

TREATISE ON THE CIRCLE

Volume II — Calculus, Complex Numbers and Harmonic Analysis

Parametrization, Integrals, Complex Exponential, Fourier Series and Transforms



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Preface

This volume arises as the direct continuation of Volume I (dedicated to the elementary geometry of the circle: angles, chords, inscribed polygons, Archimedes' method for π) and assumes familiarity with that material, recalled when necessary. The goal is not to treat differential calculus, complex numbers and Fourier analysis in the abstract, but to show how a single geometric object — the circle — traverses areas of mathematics seemingly far apart, remaining each time recognizable: the same constant curvature that in Chapter 1 characterizes the circle as the solution of a differential equation reappears, in Chapter 4, as the unit circle $\mathbb{T}$ that marks the stability boundary of a digital filter.

Target audience. The text is designed to be used in the classroom (hence the step-by-step worked solutions for every exercise, designed for student self-checking) but also for the autonomous study of an engineer who wishes to review the mathematical foundations behind tools used daily: circular interpolation in a CNC or articulated robot, digital filters on a PLC, sampling of an analog signal. The explicit links to these applications, distributed through the four chapters, are not a digression: they are the reason why it is worth studying this mathematics carefully.

How it is organized. Each chapter follows the same pattern: definitions and theorems with complete proof (when several independent proofs are available in the literature, both are presented, to accustom the student to recognizing that the same result can hide more than one idea), observations connecting the result to the rest of the treatise or to an application, exercises, and finally the fully worked solutions. Each chapter closes with a bibliography section, deliberately international: classic Western texts and articles (Ahlfors, Rudin, Stein–Shakarchi) alongside Chinese historical sources (Liu Hui, Zu Chongzhi) and contemporary research literature, also Chinese, on the digital control topics treated in Chapter 4. The Appendix collects in tabular form the most used transforms and identities, designed as a quick reference during study or exercise solving.

Prerequisites. Differential and integral calculus in one and two variables (Chapters 1–2), elementary complex numbers (premise to Chapter 3, nevertheless recalled), no prior knowledge of Fourier analysis: Chapter 4 builds its core from scratch (periodic functions, $L^2(\mathbb{T})$, coefficients, Parseval, convolution). Convergence, the Fourier, Laplace and Z transforms, DFT and FFT, sampling, control and differential geometry are treated in Volume III, to which Chapter 4 points in its closing section "Where the theory continues".

Exercise difficulty and Prerequisites boxes. Each exercise carries next to its number a difficulty indication: $\circ\circ$ (direct application of a formula or definition), $\bullet\bullet\circ$ (requires combining several results or a non-immediate algebraic step), $\bullet\bullet\bullet$ (demanding exercise, often with multiple connected parts or a complete engineering application). The few more advanced subsections — Möbius transformations and Laurent series in Chapter 3, the Poisson summation in Chapter 4 — are preceded by a brief "Prerequisites" box indicating what one must already know and, when useful, what one can accept as stated on a first reading without compromising the understanding of the rest of the chapter.

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CHAPTER 1

Parametrisation and Differential Calculus on the Circle

In Volume I the circle was studied as a *locus*: the set of points of the plane equidistant from a centre. That viewpoint is static. It is enough for synthetic geometry and for elementary analytic geometry, but it becomes insufficient as soon as one asks questions such as: *at what speed* does a cutting tool traverse an arc? *How far* has it travelled? *How tight* is the curve it is following? *In which direction* is it going?

All these questions presuppose that the circle is not merely a set, but a set that is *travelled along*: a law of motion assigning a point to every instant. The passage from the locus to the parametrised curve is the subject of this chapter, and it is the bridge between the geometry of Volume I and the analysis of the volumes that follow.

The unifying theme is that every differential quantity attached to the circle — velocity, acceleration, arc length, curvature — can be computed in closed form, with no approximation, and that the circle is the only plane curve for which this happens so completely. The chapter culminates in two characterisations: the circle as the unique plane curve of constant non-zero curvature (Theorem 1.14), and the unit circle S^1 as the first example of a space that admits *no* continuous global coordinate (Theorem 1.19) — an elementary fact to prove, but one with deep consequences, which we shall meet again in Chapter 2 when studying non-conservative fields.

INFO

Prerequisites. Volume I, Chapters 4 (central and inscribed angles), 7 (analytic geometry of the circle) and 8 (trigonometric functions and the fundamental identities). From differential calculus we assume the chain rule, second-order Taylor expansion and the Riemann integral of a continuous function.

1.1. From the locus to the parametrised curve

Definition 1.1: Plane parametrised curve. Let $I \subseteq \mathbb{R}$ be an interval. A **parametrised curve** of class C^k in the plane is a map

$$\gamma : I \rightarrow \mathbb{R}^2, \quad \gamma(t) = (x(t), y(t)),$$

whose components x, y are k times differentiable with continuous derivatives. The set $\gamma(I) \subseteq \mathbb{R}^2$ is called the **trace** (or *support*) of the curve.

Warning. The curve *is* the map γ , not its trace. Two different curves may have the same trace: $\gamma_1(t) = (\cos t, \sin t)$ and $\gamma_2(t) = (\cos 2t, \sin 2t)$, both on $[0, 2\pi]$, have the same unit circle as their trace, but the second traverses it twice and at twice the speed. Distinguishing the two notions is essential throughout the chapter: the trace carries the *geometric* information, the parametrisation the *kinematic* one.

Remark 1.1: Circle and disc. We keep the convention of Volume I: circle (or circumference) denotes the line $\{P : |P - C_0| = R\}$; disc denotes the region $\{P : |P - C_0| \leq R\}$. The curves of this chapter have the circle as their trace; the disc will appear in Chapter 2 as the domain of integration in Green's theorem.

Before stating the central result of the section we need a lemma which, elementary as it is, carries the genuine analytic content of the parametrisation: every unit vector of the plane can be written in exactly one way as $(\cos \theta, \sin \theta)$.

Lemma 1.1: Polar representation of unit vectors. For every $u = (a, b) \in \mathbb{R}^2$ with $a^2 + b^2 = 1$ there exists exactly one $\theta \in [0, 2\pi)$ such that $a = \cos \theta$ and $b = \sin \theta$.

Proof. Existence. From $a^2 + b^2 = 1$ we get $a^2 \leq 1$, that is $a \in [-1, 1]$: the domain of arccos. Set

$$\theta_0 = \arccos a \in [0, \pi],$$

so that $\cos \theta_0 = a$. Since $\theta_0 \in [0, \pi]$ we have $\sin \theta_0 \geq 0$, and by the Pythagorean identity (Volume I, Chapter 8)

$$\sin \theta_0 = +\sqrt{1 - \cos^2 \theta_0} = \sqrt{1 - a^2} = \sqrt{b^2} = |b|.$$

Two cases arise.

- If $b \geq 0$, then $|b| = b$ and the choice $\theta = \theta_0 \in [0, \pi] \subseteq [0, 2\pi)$ satisfies $\cos \theta = a$, $\sin \theta = b$.
- If $b < 0$, set $\theta = 2\pi - \theta_0$. Note that $\theta_0 \in (0, \pi]$ (the case $\theta_0 = 0$ would force $b = 0$, excluded), hence $\theta \in [\pi, 2\pi)$. By the reduction formulas, $\cos(2\pi - \theta_0) = \cos \theta_0 = a$ and $\sin(2\pi - \theta_0) = -\sin \theta_0 = -|b| = b$, since $b < 0$.

Uniqueness. Let $\theta_1, \theta_2 \in [0, 2\pi)$ with $\cos \theta_1 = \cos \theta_2$ and $\sin \theta_1 = \sin \theta_2$. Assume without loss of generality $\theta_1 \leq \theta_2$ and set $\delta = \theta_2 - \theta_1 \in [0, 2\pi)$. By the subtraction formulas,

$$\cos \delta = \cos \theta_2 \cos \theta_1 + \sin \theta_2 \sin \theta_1 = \cos^2 \theta_1 + \sin^2 \theta_1 = 1.$$

But on $[0, 2\pi)$ the equation $\cos \delta = 1$ has the unique solution $\delta = 0$: indeed $\cos$ is strictly decreasing on $[0, \pi]$ (so $\cos \delta < 1$ for $\delta \in (0, \pi]$) and $\cos \delta \leq 0 < 1$ for $\delta \in (\pi, 3\pi/2]$, while for $\delta \in (3\pi/2, 2\pi)$ we have $\cos \delta < 1$ because $\cos$ is strictly increasing on $[\pi, 2\pi]$ and equals 1 only at the endpoint $2\pi \notin [0, 2\pi)$. Hence $\theta_1 = \theta_2$. $\square$

Theorem 1.1: Standard parametrisation of the circle. Let $C_0 = (x_c, y_c) \in \mathbb{R}^2$ and $R > 0$, and let $\Gamma = C(C_0, R) = \{P \in \mathbb{R}^2 : |P - C_0| = R\}$. The map

$$\gamma : [0, 2\pi) \rightarrow \mathbb{R}^2, \quad \gamma(\theta) = (x_c + R \cos \theta, y_c + R \sin \theta)$$

enjoys the following four properties:

- (i) (Image) $\gamma([0, 2\pi)) = \Gamma$;
- (ii) (Injectivity) γ is injective on $[0, 2\pi)$;
- (iii) (Regularity) $\gamma \in C^\infty$ and $\gamma'(\theta) \neq (0, 0)$ for every θ , with $|\gamma'(\theta)| = R$;
- (iv) (Meaning of the parameter) θ is the central angle, measured counterclockwise from the ray issuing from C_0 parallel to the positive x -axis.

Extending γ to all of $\mathbb{R}$ by the same formula, γ is 2π -periodic and surjective onto Γ .

Proof. (i) Inclusion $\subseteq$. For every θ ,

$$|\gamma(\theta) - C_0|^2 = (R \cos \theta)^2 + (R \sin \theta)^2 = R^2(\cos^2 \theta + \sin^2 \theta) = R^2,$$

by the Pythagorean identity. Since $R > 0$, $|\gamma(\theta) - C_0| = R$, i.e. $\gamma(\theta) \in \Gamma$.

(i) **Inclusion** $\supseteq$. Let $P \in \Gamma$. The vector $u = (P - C_0)/R$ has norm $|P - C_0|/R = 1$: it is a unit vector. By Lemma 1.1 there is $\theta \in [0, 2\pi)$ with $u = (\cos \theta, \sin \theta)$, and then $P = C_0 + Ru = \gamma(\theta)$.

(ii) If $\gamma(\theta_1) = \gamma(\theta_2)$ then, subtracting C_0 and dividing by $R > 0$, $(\cos \theta_1, \sin \theta_1) = (\cos \theta_2, \sin \theta_2)$; the uniqueness part of Lemma 1.1 gives $\theta_1 = \theta_2$.

(iii) Sine and cosine are C^∞ on $\mathbb{R}$, hence so is γ . Differentiating componentwise,

$$\gamma'(\theta) = (-R \sin \theta, R \cos \theta), \quad |\gamma'(\theta)| = R\sqrt{\sin^2 \theta + \cos^2 \theta} = R > 0.$$

In particular $\gamma'(\theta)$ is never the zero vector.

(iv) The vector $\gamma(\theta) - C_0 = R(\cos \theta, \sin \theta)$ makes with $(1, 0)$ an angle whose cosine is

$$\frac{R(\cos \theta, \sin \theta) \cdot (1, 0)}{R \cdot 1} = \cos \theta,$$

and whose sine (the cross-product component, which fixes its sign and hence its orientation) is $\sin \theta$. By Lemma 1.1 the oriented angle is therefore exactly θ , measured counterclockwise — the positive direction by convention (Volume I, Chapter 8).

Periodicity. $\cos$ and $\sin$ have period 2π , hence $\gamma(\theta + 2\pi) = \gamma(\theta)$; surjectivity onto Γ follows from (i). $\square$

***Remark 1.2: Why $[0, 2\pi)$ and not $[0, 2\pi]$.** The half-open interval is what makes γ injective: on $[0, 2\pi]$ one would have $\gamma(0) = \gamma(2\pi)$. The choice depends on the purpose: to parametrise (describe every point exactly once) one uses $[0, 2\pi)$; to integrate along the closed curve (Chapter 2) one uses $[0, 2\pi]$, since adding a single point does not change the value of an integral but does make the curve closed in the sense of Definition 1.7.*

Example 1.1: From the Cartesian equation to the parametrisation. Parametrise the curve of equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

Step 1 — completing the squares. Group the x and y terms separately:

$$(x^2 - 6x) + (y^2 + 4y) = 12.$$

Add to both sides the quantities that complete the squares, namely $(6/2)^2 = 9$ and $(4/2)^2 = 4$:

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4 = 25,$$

that is $(x - 3)^2 + (y + 2)^2 = 25$.

Step 2 — reading off the parameters. Comparing with the canonical equation $(x - x_c)^2 + (y - y_c)^2 = R^2$ (Volume I, Chapter 7): $C_0 = (3, -2)$ and $R = \sqrt{25} = 5$. Since $25 > 0$, this is indeed a real, non-degenerate circle.

Step 3 — parametrisation. By Theorem 1.1,

$$\gamma(\theta) = (3 + 5 \cos \theta, -2 + 5 \sin \theta), \quad \theta \in [0, 2\pi).$$

Step 4 — check. Substituting into the original equation, with $x = 3 + 5 \cos \theta$ and $y = -2 + 5 \sin \theta$:

$$(x - 3)^2 + (y + 2)^2 = 25 \cos^2 \theta + 25 \sin^2 \theta = 25,$$

which is the equation obtained in Step 1, equivalent to the initial one. $\checkmark$

Step 5 — check on one point. For $\theta = 0$: $\gamma(0) = (8, -2)$. Substituting into the original equation: $64 + 4 - 48 - 8 - 12 = 0$. $\checkmark$

1.2. Phase, orientation, angular velocity and winding number

The standard parametrisation traverses the circle once, counterclockwise, at constant speed. All three features are *choices*, not necessities: replacing the parameter θ by an arbitrary function of time yields the general class of circular laws of motion.

Definition 1.2: Phase law. Let $\theta : I \rightarrow \mathbb{R}$ be of class C^k on an interval I . The curve

$$\gamma(t) = C_0 + R(\cos \theta(t), \sin \theta(t))$$

is called a **circular motion** of radius R , centre C_0 and **phase law** θ . The derivative $\omega(t) = \theta'(t)$ is the **angular velocity**, $\alpha(t) = \theta''(t)$ the **angular acceleration**.

Theorem 1.2: Kinematics of a general circular motion. With the notation of Definition 1.2, setting

$$e_r(t) = (\cos \theta(t), \sin \theta(t)), \quad e_\theta(t) = (-\sin \theta(t), \cos \theta(t)),$$

the following hold:

- (i) $\gamma(t) - C_0 = R e_r(t)$, with $|e_r| = |e_\theta| = 1$ and $e_r \cdot e_\theta = 0$;
- (ii) $\gamma'(t) = R \omega(t) e_\theta(t)$, hence $|\gamma'(t)| = R |\omega(t)|$ and $\gamma'(t) \perp (\gamma(t) - C_0)$;
- (iii) γ is regular at t if and only if $\omega(t) \neq 0$;
- (iv) the motion is counterclockwise where $\omega > 0$ and clockwise where $\omega < 0$.

Proof. (i) This is the definition; $|e_r|^2 = \cos^2 \theta + \sin^2 \theta = 1$, likewise for e_θ , and

$$e_r \cdot e_\theta = -\cos \theta \sin \theta + \sin \theta \cos \theta = 0.$$

(ii) By the chain rule applied to each component:

$$\gamma'(t) = R(-\sin \theta(t) \theta'(t), \cos \theta(t) \theta'(t)) = R \theta'(t) (-\sin \theta(t), \cos \theta(t)) = R \omega(t) e_\theta(t).$$

Its modulus is $R |\omega(t)| |e_\theta(t)| = R |\omega(t)|$; orthogonality to $\gamma - C_0 = R e_r$ follows from $e_r \cdot e_\theta = 0$.

(iii) From (ii), $\gamma'(t) = 0 \iff R |\omega(t)| = 0 \iff \omega(t) = 0$, since $R > 0$.

(iv) By Theorem 1.1(iv), $\theta(t)$ is the central angle of the point $\gamma(t)$, measured counterclockwise. If $\omega(t) = \theta'(t) > 0$ on an interval, θ is strictly increasing there: the central angle grows, i.e. the point advances counterclockwise. If $\omega < 0$, θ decreases and the direction is clockwise. $\square$

$$v = R |\omega| \quad (\text{speed of a circular motion})$$

Remark 1.3: The velocity is always tangential. Part (ii) contains a fact of complete generality, worth isolating: if a curve γ stays at constant distance from a point C_0 , then $\gamma'(t) \perp (\gamma(t) - C_0)$ for every t . The proof is one line: differentiating the identity $|\gamma(t) - C_0|^2 = R^2$ gives

$$2(\gamma(t) - C_0) \cdot \gamma'(t) = 0.$$

The same idea, applied to $|\gamma'(t)|^2 = \text{const.}$, will show that in a uniform motion acceleration and velocity are orthogonal (Section 1.10).

Definition 1.3: Closed curve and winding number. A circular motion γ defined on $[a, b]$ is **closed** if $\gamma(a) = \gamma(b)$. In that case the **winding number** about the centre is

$$n = \frac{\theta(b) - \theta(a)}{2\pi} \in \mathbb{Z}.$$

Proposition 1.1: The winding number is an integer, and counts overlaps. Let γ be a closed circular motion on $[a, b]$ with continuous phase law θ . Then:

- (i) $n = (\theta(b) - \theta(a))/2\pi$ is an integer;
- (ii) if moreover θ is strictly monotone and $|n| \geq 2$, then γ is not injective on $[a, b]$: there exist $t_1 \neq t_2$ in $[a, b]$ with $\gamma(t_1) = \gamma(t_2)$.

Proof. (i) From $\gamma(a) = \gamma(b)$ we get $e_r(a) = e_r(b)$, i.e. $\cos \theta(a) = \cos \theta(b)$ and $\sin \theta(a) = \sin \theta(b)$. Arguing as in the uniqueness part of Lemma 1.1, with $\delta = \theta(b) - \theta(a)$ we obtain $\cos \delta = 1$, whose general solution on $\mathbb{R}$ is $\delta \in 2\pi\mathbb{Z}$. Hence $n = \delta/2\pi \in \mathbb{Z}$.

(ii) Suppose θ is strictly increasing (the decreasing case is analogous, after exchanging the roles of the endpoints) and $n \geq 2$. By continuity θ attains every value of the interval $J = [\theta(a), \theta(b)]$, of length $2\pi n \geq 4\pi$. In particular both values

$$\theta(a) \quad \text{and} \quad \theta(a) + 2\pi$$

belong to J , and by strict monotonicity they are attained at two distinct points $t_1 < t_2$; moreover $t_2 < b$, because $\theta(t_2) = \theta(a) + 2\pi < \theta(a) + 2\pi n = \theta(b)$ and θ is increasing. But

$$\gamma(t_2) = C_0 + R(\cos(\theta(t_1) + 2\pi), \sin(\theta(t_1) + 2\pi)) = C_0 + R(\cos \theta(t_1), \sin \theta(t_1)) = \gamma(t_1),$$

by periodicity of sine and cosine. Hence γ is not injective on $[a, b]$. $\square$

Remark 1.4: What the winding number does not say. The winding number is a property of the parametrisation, not of the trace: the curves $\gamma_1(t) = (\cos t, \sin t)$ and $\gamma_2(t) = (\cos 2t, \sin 2t)$ on $[0, 2\pi]$ have exactly the same trace but $n_1 = 1$, $n_2 = 2$. It is therefore a kinematic invariant. Its importance will become clear in Chapter 2: the winding number is precisely what the integral of the vortex field along the curve manages to measure, and it is the first topological invariant we encounter.

Example 1.2: Building the phase law from kinematic data. A spindle of radius $R = 0.12$ m starts from rest with constant angular acceleration $\alpha = 4.0$ rad/s²; at the initial instant the marked point is at angular position $\theta_0 = \pi/6$. Determine the phase law, the position, the speed and the number of turns completed after $t = 3.0$ s.

Step 1 — integrating the angular acceleration. From $\theta''(t) = \alpha$ constant, integrating once with $\theta'(0) = 0$ (“starts from rest”):

$$\omega(t) = \theta'(t) = \alpha t = 4.0 t \quad [\text{rad/s}].$$

Integrating a second time with $\theta(0) = \pi/6$:

$$\theta(t) = \theta_0 + \frac{1}{2}\alpha t^2 = \frac{\pi}{6} + 2.0 t^2 \quad [\text{rad}].$$

Step 2 — phase and angular velocity at $t = 3.0$ s.

$$\theta(3) = \frac{\pi}{6} + 2.0 \cdot 9 = 0.5236 + 18.000 = 18.5236 \text{ rad}, \quad \omega(3) = 12.0 \text{ rad/s}.$$

Step 3 — speed. By Theorem 1.2(ii),

$$v(3) = R|\omega(3)| = 0.12 \times 12.0 = 1.44 \text{ m/s}.$$

Step 4 — number of turns. The swept angle is

$$\Delta\theta = \theta(3) - \theta(0) = 18.000 \text{ rad}, \quad \frac{\Delta\theta}{2\pi} = \frac{18.000}{6.28319} = 2.8648.$$

The spindle has completed 2 full turns plus a fraction of 0.8648 of a turn, i.e. $0.8648 \times 360^\circ = 311.3^\circ$. Note that the number of turns is *not* an integer, because the curve is not closed: Definition 1.3 requires $\gamma(a) = \gamma(b)$, which does not hold here.

Step 5 — final position. The angular position reduced to $[0, 2\pi)$ is

$$\theta(3) \bmod 2\pi = 18.5236 - 2 \cdot 6.28319 = 5.9572 \text{ rad} = 341.3^\circ,$$

so, taking C_0 at the origin,

$$\gamma(3) = 0.12 (\cos 5.9572, \sin 5.9572) = (0.1137, -0.0383) \text{ m}.$$

Check. $|\gamma(3)| = \sqrt{0.1137^2 + 0.0383^2} = \sqrt{0.012927 + 0.001467} = \sqrt{0.014394} = 0.1200 \text{ m} = R. \checkmark$

1.3. Polar, matrix and complex form

Circular motion admits three equivalent descriptions, each suited to a different kind of computation: the *polar* form (already seen), the *matrix* form, optimal for composing rotations, and the *complex* form, optimal for differentiating and integrating. This section establishes the dictionary between the three.

Definition 1.4: Rotation matrix. For $\theta \in \mathbb{R}$ set

$$Q(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad J = Q(\pi/2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Theorem 1.3: Algebraic and differential properties of $Q(\theta)$. For all $\alpha, \beta, \theta \in \mathbb{R}$:

- (i) $Q(\alpha)Q(\beta) = Q(\alpha + \beta)$ (composition law);
- (ii) $Q(\theta)^T Q(\theta) = I$ and $\det Q(\theta) = 1$ ($Q(\theta) \in SO(2)$);
- (iii) $J^2 = -I$ and $Q(\theta)^{-1} = Q(-\theta) = Q(\theta)^T$;
- (iv) $\frac{dQ}{d\theta}(\theta) = J Q(\theta) = Q(\theta) J$ (differential equation of rotation).

Proof. (i) Multiplying explicitly:

$$Q(\alpha)Q(\beta) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} = \begin{pmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -\cos \alpha \sin \beta - \sin \alpha \cos \beta \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & -\sin \alpha \sin \beta + \cos \alpha \cos \beta \end{pmatrix}.$$

By the addition formulas (Volume I, Chapter 8), the (1, 1) entry is $\cos(\alpha + \beta)$, the (2, 1) entry is $\sin(\alpha + \beta)$, the (1, 2) entry is $-\sin(\alpha + \beta)$ and the (2, 2) entry is $\cos(\alpha + \beta)$: the resulting matrix is exactly $Q(\alpha + \beta)$.

(ii) From (i) with $\beta = -\alpha$: $Q(\alpha)Q(-\alpha) = Q(0) = I$. Since $\cos(-\alpha) = \cos \alpha$ and $\sin(-\alpha) = -\sin \alpha$, we have $Q(-\alpha) = Q(\alpha)^T$; hence $Q(\theta)Q(\theta)^T = I$, and transposing also $Q(\theta)^T Q(\theta) = I$. The determinant is $\cos^2 \theta - (-\sin \theta)(\sin \theta) = \cos^2 \theta + \sin^2 \theta = 1$.

(iii) $J = Q(\pi/2)$, hence $J^2 = Q(\pi) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$. The inverse follows from (ii).

(iv) Differentiating entrywise,

$$\frac{dQ}{d\theta} = \begin{pmatrix} -\sin \theta & -\cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix},$$

while

$$JQ(\theta) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} -\sin \theta & -\cos \theta \\ \cos \theta & -\sin \theta \end{pmatrix},$$

which coincides. For the second equality note $J = Q(\pi/2)$ and use (i): $JQ(\theta) = Q(\pi/2 + \theta) = Q(\theta + \pi/2) = Q(\theta)J$. $\square$

Corollary 1.1: $SO(2)$ is an abelian group. The set $SO(2) = \{Q(\theta) : \theta \in \mathbb{R}\}$, with matrix multiplication, is an abelian group, and $\theta \mapsto Q(\theta)$ is a surjective homomorphism of $(\mathbb{R}, +)$ onto $SO(2)$ with kernel $2\pi\mathbb{Z}$. Consequently

$$SO(2) \cong \mathbb{R}/2\pi\mathbb{Z}.$$

Proof. Closure, associativity (inherited from matrix multiplication), the identity element $Q(0) = I$ and the inverse $Q(-\theta)$ follow from Theorem 1.3(i)–(iii). Commutativity is immediate from (i): $Q(\alpha)Q(\beta) = Q(\alpha + \beta) = Q(\beta + \alpha) = Q(\beta)Q(\alpha)$. Part (i) says precisely that $\theta \mapsto Q(\theta)$ is a homomorphism; it is surjective by definition of $SO(2)$, and its kernel is $\{\theta : \cos \theta = 1, \sin \theta = 0\} = 2\pi\mathbb{Z}$. The claim follows from the first isomorphism theorem. $\square$

Theorem 1.4: Complex form of circular motion. Identifying $\mathbb{R}^2$ with $\mathbb{C}$ via $(x, y) \leftrightarrow x + iy$, and writing $z_c = x_c + iy_c$, the circular motion of Definition 1.2 reads

$$z(t) = z_c + R e^{i\theta(t)},$$

and

$$z'(t) = i\omega(t)(z(t) - z_c), \quad z''(t) = (i\alpha(t) - \omega(t)^2)(z(t) - z_c).$$

Moreover multiplication by $e^{i\alpha}$ corresponds, in matrix form, to the action of $Q(\alpha)$, and the imaginary unit i corresponds to the matrix J .

Proof. By Euler's formula $e^{i\theta} = \cos \theta + i \sin \theta$ (assumed known; it will be reproved in Chapter 3), $z_c + R e^{i\theta(t)} = (x_c + R \cos \theta(t)) + i(y_c + R \sin \theta(t))$, the complex writing of $\gamma(t)$.

Differentiating in t by the chain rule, and recalling $\frac{d}{du} e^{iu} = i e^{iu}$:

$$z'(t) = R i \theta'(t) e^{i\theta(t)} = i \omega(t) R e^{i\theta(t)} = i \omega(t) (z(t) - z_c).$$

Differentiating again by the product rule:

$$z''(t) = i \omega'(t) R e^{i\theta(t)} + i \omega(t) \cdot i \omega(t) R e^{i\theta(t)} = (i\alpha(t) - \omega(t)^2)(z(t) - z_c),$$

using $i^2 = -1$ and $\alpha = \omega'$.

For the correspondence between i and J : if $w = u + iv$, then $iw = -v + iu$, which in coordinates is $(u, v) \mapsto (-v, u)$, exactly the action of J . Since $e^{i\alpha}w$ is obtained from w by a rotation through α (write $w = \rho e^{i\varphi}$ and use $e^{i\alpha} \rho e^{i\varphi} = \rho e^{i(\varphi+\alpha)}$), multiplication by $e^{i\alpha}$ corresponds to $Q(\alpha)$. $\square$

Remark 1.5: The factor i as a 90° rotation. The three identities $J^2 = -I$, $i^2 = -1$ and “rotating twice by 90° is the same as changing sign” are one and the same statement in three languages. The formula $z' = i\omega(z - z_c)$ is the complex translation of Theorem 1.2(ii): multiplying the radius vector by i rotates it by 90° , producing the tangential direction. Chapter 3 develops this dictionary systematically.

Example 1.3: Translating between the three forms. The point $P_0 = (3, 1)$ is rotated by $\alpha = 60^\circ$ about $C_0 = (1, 1)$ counterclockwise. Compute the final position in all three forms.

Polar form. The initial radius vector is $P_0 - C_0 = (2, 0)$, of modulus $R = 2$ and phase $\theta_0 = 0$ (it lies on the positive x -axis). The final phase is $\theta_1 = 0 + \pi/3$, hence

$$P_1 = C_0 + 2(\cos 60^\circ, \sin 60^\circ) = (1, 1) + 2\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = (2, 1 + \sqrt{3}) = (2, 2.7321).$$

Matrix form.

$$P_1 = C_0 + Q(60^\circ)(P_0 - C_0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} = \begin{pmatrix} 2 \\ 1 + \sqrt{3} \end{pmatrix}.$$

Complex form. With $z_c = 1 + i$ and $z_0 = 3 + i$:

$$z_1 = z_c + e^{i\pi/3}(z_0 - z_c) = (1 + i) + \left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) \cdot 2 = 2 + i(1 + \sqrt{3}).$$

The three results agree. ✓ The matrix form is the most efficient when composing many rotations (one just multiplies matrices, or adds angles thanks to Theorem 1.3(i)); the complex form is the most compact and the easiest to differentiate.

1.4. Spirals on the circle

The circle is the constant case $r(\theta) \equiv R$ of a broader family of curves described in polar coordinates by a radius $r(\theta)$ that varies with the angle. This section studies the two classical spirals, showing that the same tools — the tangent vector, arc length, the complex form — suffice to treat them without introducing anything new.

Proposition 1.2: Polar decomposition of velocity. Let $\gamma(\theta) = r(\theta)(\cos \theta, \sin \theta)$ with r differentiable and $r(\theta) > 0$. Set $e_r(\theta) = (\cos \theta, \sin \theta)$ and $e_\vartheta(\theta) = (-\sin \theta, \cos \theta)$ (as in Section 1.1). Then

$$\gamma'(\theta) = r'(\theta) e_r(\theta) + r(\theta) e_\vartheta(\theta), \quad |\gamma'(\theta)|^2 = r(\theta)^2 + r'(\theta)^2.$$

If $\psi(\theta) \in (0, \pi)$ is the angle between the radius vector e_r and the tangent $\gamma'(\theta)$, measured so that $\psi = \pi/2$ corresponds to $\gamma' \parallel e_\vartheta$, then

$$\tan \psi(\theta) = \frac{r(\theta)}{r'(\theta)} \quad (\text{where } r'(\theta) \neq 0).$$

Proof. Differentiating $\gamma(\theta) = r(\theta)e_r(\theta)$ by the product rule,

$$\gamma'(\theta) = r'(\theta) e_r(\theta) + r(\theta) e'_r(\theta).$$

But $e'_r(\theta) = (-\sin \theta, \cos \theta) = e_\vartheta(\theta)$, giving the first identity. Since $e_r \perp e_\vartheta$ and $|e_r| = |e_\vartheta| = 1$, they form an orthonormal basis: by Pythagoras, $|\gamma'|^2 = r'^2 + r^2$. Finally γ' has component r'

along e_r and component r along e_θ ; the angle between γ' and e_r therefore has tangent equal to the ratio of the orthogonal component (r) to the component along e_r (r'), i.e. $\tan \psi = r/r'$. $\square$

Remark 1.6: The circle as a special case. For $r \equiv R$ constant, $r' = 0$ and Proposition 1.2 gives $\gamma' = R e_\theta$: this recovers exactly Theorem 1.2 with $\omega = 1$, and $\psi = \pi/2$ everywhere — the tangent is always perpendicular to the radius, the elementary fact from which this chapter started.

Definition 1.5: Archimedean spiral. For $b > 0$, the **Archimedean spiral** is the curve $r(\theta) = b\theta$, $\theta \geq 0$: the radius grows *proportionally* to the angle swept.

Definition 1.6: Logarithmic spiral. For $a > 0$ and $b \in \mathbb{R} \setminus \{0\}$, the **logarithmic spiral** (or equiangular spiral) is the curve $r(\theta) = a e^{b\theta}$, $\theta \in \mathbb{R}$.

Theorem 1.5: Constant angle and self-similarity of the logarithmic spiral. For the logarithmic spiral $r(\theta) = a e^{b\theta}$:

- (i) the angle ψ between the radius vector and the tangent is constant, $\psi = \operatorname{arccot} b$;
- (ii) for every $\varphi \in \mathbb{R}$, rotating by φ is the same as scaling by $e^{b\varphi}$: $\gamma(\theta + \varphi) = e^{b\varphi} Q(\varphi) \gamma(\theta)$, where $Q(\varphi)$ is the rotation matrix of Definition 1.4.

Proof. (i) $r'(\theta) = a b e^{b\theta} = b r(\theta)$, so by Proposition 1.2, $\tan \psi = r/r' = 1/b$ for every θ : constant, and $\psi = \operatorname{arccot} b$.

(ii) In complex form (Theorem 1.4), $\gamma(\theta) \leftrightarrow z(\theta) = a e^{b\theta} e^{i\theta} = a e^{(b+i)\theta}$. Then

$$z(\theta + \varphi) = a e^{(b+i)(\theta+\varphi)} = e^{b\varphi} e^{i\varphi} (a e^{(b+i)\theta}) = e^{b\varphi} e^{i\varphi} z(\theta).$$

Since multiplication by $e^{i\varphi}$ corresponds, by Theorem 1.4, to the action of $Q(\varphi)$, this gives exactly (ii). $\square$

Remark 1.7: The logarithmic spiral as an exponential with a complex exponent. Theorem 1.4 wrote circular motion as $z(\theta) = z_c + R e^{i\theta}$: an exponential with a purely imaginary exponent. The proof just given shows that the logarithmic spiral centred at the origin is $z(\theta) = a e^{(b+i)\theta}$, the same type of formula with a complex exponent $b+i$ in place of i : the imaginary part (1, the coefficient of θ in the exponent) still produces rotation, while the real part b also produces scaling. The circle ($b = 0$) is the degenerate case in which the real part of the exponent vanishes and only pure rotation remains. This dictionary — real part of the exponent $\leftrightarrow$ growth, imaginary part $\leftrightarrow$ rotation — is exactly that of the natural modes of RLC circuits (Chapter 5, roots $-\zeta\omega_n \pm i\omega_d$ of the characteristic equation), and will be revisited in that context.

Example 1.4: Arc length of the two spirals. Compute the arc length, for $\theta \in [\theta_0, \theta_1]$, of the logarithmic spiral $r = a e^{b\theta}$ and of the Archimedean spiral $r = b\theta$.

Logarithmic spiral. By Proposition 1.2, $|\gamma'(\theta)| = \sqrt{r^2 + r'^2} = \sqrt{r^2 + b^2 r^2} = r \sqrt{1 + b^2}$. Hence

$$L = \int_{\theta_0}^{\theta_1} r(\theta) \sqrt{1 + b^2} d\theta = \sqrt{1 + b^2} \left[\frac{r(\theta)}{b} \right]_{\theta_0}^{\theta_1} = \frac{\sqrt{1 + b^2}}{b} (r(\theta_1) - r(\theta_0)),$$

using $\int r d\theta = \int a e^{b\theta} d\theta = r/b + \text{const.}$ For $a = 1$, $b = 0.15$, $\theta_0 = 0$, $\theta_1 = 4\pi$: $r(\theta_0) = 1$, $r(\theta_1) = e^{0.15 \cdot 4\pi} = e^{1.88496} = 6.58606$, and

$$L = \frac{\sqrt{1 + 0.15^2}}{0.15} (6.58606 - 1) = \frac{1.01119}{0.15} \times 5.58606 = 37.65704.$$

The length is *finite* even though $\theta_1 \rightarrow +\infty$ would make r diverge: here, conversely, as $\theta \rightarrow -\infty$ (spiral winding into the origin) $r \rightarrow 0$ and L stays finite, a fact that will be useful in the next section.

Archimedean spiral. $r' = b$, so $|\gamma'| = b\sqrt{1 + \theta^2}$ and

$$L = b \int_{\theta_0}^{\theta_1} \sqrt{1 + \theta^2} d\theta = \frac{b}{2} \left[\theta \sqrt{1 + \theta^2} + \operatorname{arcsinh} \theta \right]_{\theta_0}^{\theta_1},$$

an elementary integral (by parts, or the substitution $\theta = \sinh u$). For $b = 0.7$, $\theta_0 = 0$, $\theta_1 = 3\pi = 9.42478$:

$$L = \frac{0.7}{2} \left[9.42478 \sqrt{1 + 9.42478^2} + \operatorname{arcsinh}(9.42478) \right] = 0.35 \times (89.32504 + 2.93929) = 32.29252.$$

Independent numerical check. Both values were recomputed by direct numerical integration of $\int \sqrt{r^2 + r'^2} d\theta$ (adaptive quadrature, script `Sorgenti/Vol2/_verifiche/vol2_cap01_spirali.py`): 37.6570372569 against the closed form 37.6570372569 (difference 7×10^{-15} , machine rounding noise) for the logarithmic spiral, 32.2925162915 against 32.2925162915 (difference 5×10^{-14}) for the Archimedean one.

1.5. Regularity, simplicity, reparametrisation

Definition 1.7: Regular, simple, closed curve. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be of class C^1 . We say γ is:

- **regular** if $\gamma'(t) \neq 0$ for every $t \in [a, b]$;
- **closed** if $\gamma(a) = \gamma(b)$;
- **simple** if γ is injective on $[a, b)$ (no self-intersections, except possibly the closing point);
- **piecewise regular** if $[a, b]$ can be split into finitely many subintervals on each of which γ is regular of class C^1 .

Remark 1.8: The circle as a Jordan curve. The standard parametrisation extended to $[0, 2\pi]$ is a closed, simple, regular curve: the prototype of a regular **Jordan curve**. This makes it the reference example for Green's theorem (Chapter 2), which requires exactly these hypotheses on the boundary of the domain of integration.

Definition 1.8: Reparametrisation. Let I, J be intervals and $\varphi : J \rightarrow I$ a bijection of class C^1 with $\varphi'(u) \neq 0$ for every $u \in J$. Given $\gamma : I \rightarrow \mathbb{R}^2$, the curve $\delta = \gamma \circ \varphi : J \rightarrow \mathbb{R}^2$ is a **reparametrisation** of γ . It **preserves orientation** if $\varphi' > 0$ and **reverses** it if $\varphi' < 0$.

Theorem 1.6: What is and what is not preserved by a reparametrisation. Let $\delta = \gamma \circ \varphi$ be a reparametrisation of a regular curve γ . Then:

- (i) $\delta(J) = \gamma(I)$: the trace is invariant;

- (ii) $\delta'(u) = \varphi'(u) \gamma'(\varphi(u))$; in particular δ is regular;
- (iii) the unit tangent $T = \gamma'/|\gamma'|$ is invariant if $\varphi' > 0$ and changes sign if $\varphi' < 0$;
- (iv) the speed $|\gamma'|$ and the acceleration vector are not invariant.

Proof. (i) φ is a bijection from J onto I , hence $\delta(J) = \gamma(\varphi(J)) = \gamma(I)$.

(ii) This is the chain rule applied componentwise. Since $\varphi'(u) \neq 0$ and $\gamma'(\varphi(u)) \neq 0$ by regularity of γ , the product is non-zero.

(iii) From (ii),

$$\frac{\delta'(u)}{|\delta'(u)|} = \frac{\varphi'(u) \gamma'(\varphi(u))}{|\varphi'(u)| |\gamma'(\varphi(u))|} = \operatorname{sgn}(\varphi'(u)) T(\varphi(u)).$$

(iv) From (ii), $|\delta'(u)| = |\varphi'(u)| \cdot |\gamma'(\varphi(u))|$: the factor $|\varphi'|$ is not in general 1. For the acceleration, differentiating (ii) by the product rule:

$$\delta''(u) = \varphi''(u) \gamma'(\varphi(u)) + \varphi'(u)^2 \gamma''(\varphi(u)),$$

which agrees with $\gamma''(\varphi(u))$ only if $\varphi'' = 0$ and $\varphi'^2 = 1$, i.e. only for the translations $\varphi(u) = \pm u + c$. $\square$

Example 1.5: Smooth trace, non-regular parametrisation. Consider $\gamma(t) = (\cos t^3, \sin t^3)$ on $[-2, 2]$.

Trace. Since $t \mapsto t^3$ is onto $\mathbb{R}$, the trace is the whole unit circle: geometrically there is nothing singular.

Regularity. Differentiating,

$$\gamma'(t) = (-3t^2 \sin t^3, 3t^2 \cos t^3) = 3t^2(-\sin t^3, \cos t^3), \quad |\gamma'(t)| = 3t^2.$$

At $t = 0$ we get $\gamma'(0) = (0, 0)$: the curve is *not* regular at 0, despite having a perfectly smooth trace. The moving point stops for an instant at $\gamma(0) = (1, 0)$ and then resumes in the same direction.

Moral. Regularity is a property of the parametrisation, not of the trace — exactly like the winding number (Observation 1.4). Note also that $\varphi(t) = t^3$ is not an admissible reparametrisation in the sense of Definition 1.8, precisely because $\varphi'(0) = 0$: Theorem 1.6(ii) is not violated, it simply does not apply.

Complementary counterexample. The curve $\eta(t) = (\cos t, \sin t)$ on $[0, 4\pi]$ is regular but not simple (it traverses the circle twice), whereas γ above is simple on its domain but not regular. The two properties are independent.

1.6. Rational parametrisation and stereographic projection

The standard parametrisation uses transcendental functions. There is an alternative parametrisation employing only *rational* functions — quotients of polynomials — at the price of omitting a single point. This construction has three lives: it is the stereographic projection of geometry, the universal substitution of integral calculus, and the source of Pythagorean triples in number theory.

Theorem 1.7: Stereographic projection of the unit circle. Let $S^1 = \{(x, y) : x^2 + y^2 = 1\}$ and $N = (-1, 0)$. The map

$$p : \mathbb{R} \rightarrow S^1 \setminus \{N\}, \quad p(u) = \left(\frac{1 - u^2}{1 + u^2}, \frac{2u}{1 + u^2} \right)$$

is a bijection of class C^∞ , with inverse

$$p^{-1}(x, y) = \frac{y}{1+x}.$$

Geometrically, $p(u)$ is the second intersection point of S^1 with the line joining N to the point $(0, u)$ of the y -axis.

Proof. Step 1 — geometric construction. The line through $N = (-1, 0)$ and $(0, u)$ is parametrised as

$$\ell(s) = (-1, 0) + s((0, u) - (-1, 0)) = (-1 + s, su), \quad s \in \mathbb{R}.$$

Imposing membership of S^1 :

$$(-1+s)^2 + s^2 u^2 = 1 \iff 1 - 2s + s^2 + s^2 u^2 = 1 \iff s^2(1+u^2) - 2s = 0 \iff s[s(1+u^2) - 2] = 0.$$

The two roots are $s = 0$, giving the point N itself, and

$$s = \frac{2}{1+u^2},$$

always defined since $1+u^2 \geq 1 > 0$. Substituting:

$$x = -1 + \frac{2}{1+u^2} = \frac{1-u^2}{1+u^2}, \quad y = \frac{2}{1+u^2} u = \frac{2u}{1+u^2},$$

which is the stated formula.

Step 2 — p takes values in S^1 . Direct verification:

$$\left(\frac{1-u^2}{1+u^2}\right)^2 + \left(\frac{2u}{1+u^2}\right)^2 = \frac{(1-u^2)^2 + 4u^2}{(1+u^2)^2} = \frac{1-2u^2+u^4+4u^2}{(1+u^2)^2} = \frac{(1+u^2)^2}{(1+u^2)^2} = 1.$$

Step 3 — N is not in the image. $p(u) = N$ would require $2u/(1+u^2) = 0$, i.e. $u = 0$; but $p(0) = (1, 0) \neq N$. Hence $N \notin p(\mathbb{R})$.

Step 4 — surjectivity onto $S^1 \setminus \{N\}$. Let $(x, y) \in S^1$ with $(x, y) \neq N$, i.e. $x \neq -1$ (if $x = -1$, then $x^2 + y^2 = 1$ forces $y = 0$, i.e. the point N). Set $u = y/(1+x)$, well defined. We check $p(u) = (x, y)$. First,

$$1+u^2 = 1 + \frac{y^2}{(1+x)^2} = \frac{(1+x)^2 + y^2}{(1+x)^2} = \frac{1+2x+x^2+y^2}{(1+x)^2} = \frac{2+2x}{(1+x)^2} = \frac{2}{1+x},$$

using $x^2 + y^2 = 1$. Similarly

$$1-u^2 = \frac{(1+x)^2 - y^2}{(1+x)^2} = \frac{1+2x+x^2-(1-x^2)}{(1+x)^2} = \frac{2x+2x^2}{(1+x)^2} = \frac{2x}{1+x}.$$

Hence

$$\frac{1-u^2}{1+u^2} = \frac{2x/(1+x)}{2/(1+x)} = x, \quad \frac{2u}{1+u^2} = \frac{2y/(1+x)}{2/(1+x)} = y. \checkmark$$

Step 5 — injectivity. If $p(u_1) = p(u_2)$, applying the formula of Step 4 gives $u_1 = y/(1+x) = u_2$. The formulas for p and p^{-1} are quotients of polynomials with nowhere-vanishing denominator, hence both C^∞ . $\square$

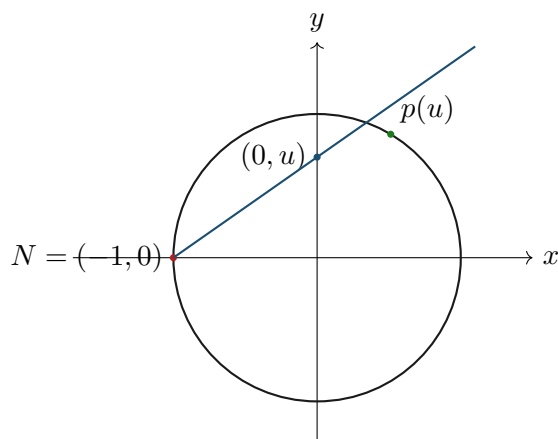


Figure 1.1: Stereographic projection from the pole $N = (-1, 0)$: the parameter u corresponds to the second intersection of the line $N(0, u)$ with the circle. As u ranges over $\mathbb{R}$ one obtains every point of S^1 except N , which intuitively corresponds to $u \rightarrow \pm\infty$.

Corollary 1.2: Relation with the half-angle tangent. For $\theta \in (-\pi, \pi)$, setting $u = \tan(\theta/2)$ we have $p(u) = (\cos \theta, \sin \theta)$, that is

$$\cos \theta = \frac{1 - u^2}{1 + u^2}, \quad \sin \theta = \frac{2u}{1 + u^2}, \quad d\theta = \frac{2 du}{1 + u^2}.$$

Proof. For $\theta \in (-\pi, \pi)$ we have $\theta/2 \in (-\pi/2, \pi/2)$, where $\cos(\theta/2) \neq 0$ and the tangent is defined. By the double-angle formulas (Volume I, Chapter 8),

$$\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}, \quad \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}.$$

Divide numerator and denominator by $\cos^2(\theta/2) \neq 0$, using in the denominator the Pythagorean identity $\cos^2(\theta/2) + \sin^2(\theta/2) = 1$:

$$\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} = \frac{1 - u^2}{1 + u^2}, \quad \sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} = \frac{2u}{1 + u^2}.$$

For the differential: from $u = \tan(\theta/2)$ we get $\theta = 2 \arctan u$ (the inverse is valid because $\theta/2 \in (-\pi/2, \pi/2)$), hence $d\theta/du = 2/(1 + u^2)$. $\square$

Remark 1.9: Three readings of the same formula.

- **Geometry.** p shows that S^1 minus a point is homeomorphic to $\mathbb{R}$: the circle is “a line plus a point at infinity”. It is the case $n = 1$ of the stereographic projection identifying $S^n \setminus \{N\}$ with $\mathbb{R}^n$.
- **Analysis.** Corollary 1.2 is the **Weierstrass substitution**, turning every integral of a rational function of $\sin \theta$ and $\cos \theta$ into an integral of a rational function of u , hence solvable by partial fractions. We shall use it in Chapter 2.
- **Number theory.** If $u = m/n \in \mathbb{Q}$, then

$$p(u) = \left(\frac{n^2 - m^2}{n^2 + m^2}, \frac{2mn}{n^2 + m^2} \right)$$

has rational coordinates; multiplying by $n^2 + m^2$ yields the Pythagorean triple $(n^2 - m^2, 2mn, n^2 + m^2)$, Euclid’s parametrisation (Volume I, Chapter 9). The bijectivity of p proves that all primitive triples arise in this way.

Warning. The rational parametrisation is not “better” than the trigonometric one: it is different. It does not cover the whole circle (the point N is missing), it does not have constant speed ($|p'(u)| = 2/(1+u^2)$, which tends to 0 as $u \rightarrow \pm\infty$) and it heavily distorts lengths near N . It is preferable when exact algebra is required (symbolic computation, rational arithmetic, NURBS curves in CAD), and inadvisable when uniform sampling is required.

1.7. Derivative, tangent vector and tangent line

Theorem 1.8: Tangent line to the circle. Let $\Gamma = C(C_0, R)$ and $P \in \Gamma$. The tangent line to Γ at P , defined as the line through P with direction γ' for any regular parametrisation, has equation

$$(P - C_0) \cdot (X - P) = 0, \quad X \in \mathbb{R}^2,$$

and it is the unique line through P meeting Γ only at P .

Proof. Step 1 — the tangent direction. Let γ be a regular parametrisation with $\gamma(t_0) = P$. By Observation 1.3, differentiating $|\gamma(t) - C_0|^2 = R^2$ gives $(\gamma(t_0) - C_0) \cdot \gamma'(t_0) = 0$, i.e. $\gamma'(t_0) \perp (P - C_0)$. Since $\gamma'(t_0) \neq 0$, the line through P with direction $\gamma'(t_0)$ is exactly the set of points X with $(X - P) \perp (P - C_0)$, that is $(P - C_0) \cdot (X - P) = 0$. The direction does not depend on the parametrisation, by Theorem 1.6(iii).

Step 2 — the intersection is P alone. Let X be a point of the tangent line, i.e. $(X - P) \cdot (P - C_0) = 0$. Writing $X - C_0 = (X - P) + (P - C_0)$ and applying Pythagoras (the two summands are orthogonal):

$$|X - C_0|^2 = |X - P|^2 + |P - C_0|^2 = |X - P|^2 + R^2 \geq R^2,$$

with equality if and only if $|X - P| = 0$, i.e. $X = P$. Hence every point of the tangent other than P lies outside Γ : the intersection reduces to $\{P\}$.

Step 3 — uniqueness. Let r be a line through P with $r \cap \Gamma = \{P\}$ and let d be the distance from C_0 to r . If $d < R$, the line would be secant and would meet Γ at two points (Volume I, Chapter 3); if $d > R$ it would not contain $P \in \Gamma$. Hence $d = R$, and since the foot of the perpendicular from C_0 to r is at distance R from C_0 and lies on r , it belongs to $\Gamma \cap r = \{P\}$: therefore $C_0P \perp r$, which is the equation of Step 1. $\square$

Corollary 1.3: The gradient as a normal. Let $F(x, y) = (x - x_c)^2 + (y - y_c)^2 - R^2$, so that $\Gamma = \{F = 0\}$. Then

$$\nabla F(P) = 2(P - C_0)$$

is orthogonal to Γ at every $P \in \Gamma$, and the tangent line at P is $\nabla F(P) \cdot (X - P) = 0$. The outward unit normal is

$$n(P) = \frac{\nabla F(P)}{|\nabla F(P)|} = \frac{P - C_0}{R}.$$

Proof. The gradient computation is immediate: $\partial_x F = 2(x - x_c)$, $\partial_y F = 2(y - y_c)$. For orthogonality, let γ be a regular parametrisation of Γ : then $F(\gamma(t)) \equiv 0$, and differentiating by the chain rule

$$\nabla F(\gamma(t)) \cdot \gamma'(t) = 0,$$

so ∇F is orthogonal to every tangent vector. Its modulus is $|\nabla F(P)| = 2|P - C_0| = 2R$, whence the unit vector. That it is *outward* follows from the fact that F increases away from the centre, and the gradient points in the direction of increase. $\square$

Example 1.6: Tangent by parametric, implicit and geometric means. Find the tangent line to $(x - 2)^2 + (y - 1)^2 = 25$ at $P = (5, 5)$.

Preliminary check. $(5 - 2)^2 + (5 - 1)^2 = 9 + 16 = 25$. $\checkmark P \in \Gamma$, with $C_0 = (2, 1)$, $R = 5$.

Method 1 — parametric. The radius vector is $P - C_0 = (3, 4)$, of modulus 5; its phase satisfies $\cos \theta_0 = 3/5$, $\sin \theta_0 = 4/5$, i.e. $\theta_0 = 0.92730$ rad. By Theorem 1.2(ii) with $\omega = 1$,

$$\gamma'(\theta_0) = 5(-\sin \theta_0, \cos \theta_0) = 5\left(-\frac{4}{5}, \frac{3}{5}\right) = (-4, 3).$$

The line through $P = (5, 5)$ with direction $(-4, 3)$ is $\{(5 - 4s, 5 + 3s)\}$; eliminating s gives $3x + 4y = 35$.

Method 2 — implicit. By Corollary 1.3, $\nabla F(P) = 2(3, 4) = (6, 8)$, and $\nabla F(P) \cdot (X - P) = 0$ gives

$$6(x - 5) + 8(y - 5) = 0 \implies 6x + 8y = 70 \implies 3x + 4y = 35. \checkmark$$

Method 3 — elementary geometry. The radius C_0P has slope $m_r = (5 - 1)/(5 - 2) = 4/3$; the tangent, being perpendicular, has $m_t = -3/4$, hence $y - 5 = -\frac{3}{4}(x - 5)$, i.e. $4y - 20 = -3x + 15$, i.e. $3x + 4y = 35$. $\checkmark$

Distance check. The distance from $C_0 = (2, 1)$ to the line $3x + 4y - 35 = 0$ is

$$\frac{|3 \cdot 2 + 4 \cdot 1 - 35|}{\sqrt{9 + 16}} = \frac{25}{5} = 5 = R,$$

as it must be for a tangent. $\checkmark$

1.8. Arc length and the natural parameter

Definition 1.9: Length of a curve. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be of class C^1 . Its **length** is

$$L(\gamma) = \int_a^b |\gamma'(t)| dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt.$$

Remark 1.10: Consistency with Volume I. In Volume I the length of a curve was defined as the supremum of the lengths of inscribed polygons, and the rectifiability of the circle was proved along those lines, independently of integral calculus (thereby avoiding any circularity in the definition of π). It is a standard result of analysis that for C^1 curves the two definitions agree. Here we adopt the integral formula as the operative definition, with the guarantee that the value obtained for the circle, $2\pi R$, coincides with the one from Volume I: consistency is checked in Theorem 1.9 below.

Theorem 1.9: Arc length of the circle. For the circular motion $\gamma(t) = C_0 + R(\cos \theta(t), \sin \theta(t))$ on $[a, b]$ with θ of class C^1 and monotone,

$$L(\gamma) = R|\theta(b) - \theta(a)|.$$

In particular, for the standard parametrisation on $[0, 2\pi]$ we get $L = 2\pi R$, and the arc subtending a central angle $\Delta\theta$ has length $R\Delta\theta$.

Proof. By Theorem 1.2(ii), $|\gamma'(t)| = R|\theta'(t)|$. Since θ is monotone, θ' has constant sign and $|\theta'| = \varepsilon\theta'$ with $\varepsilon = \pm 1$. Hence

$$L(\gamma) = \int_a^b R|\theta'(t)| dt = \varepsilon R \int_a^b \theta'(t) dt = \varepsilon R[\theta(b) - \theta(a)] = R|\theta(b) - \theta(a)|,$$

using the fundamental theorem of calculus in the third step and the definition of ε in the fourth. For $\theta(t) = t$ on $[0, 2\pi]$: $L = 2\pi R$. $\square$

$$L = R \Delta\theta \quad (\Delta\theta \text{ in radians})$$

Theorem 1.10: Length is invariant under reparametrisation. Let $\delta = \gamma \circ \varphi$ be a reparametrisation of $\gamma : [a, b] \rightarrow \mathbb{R}^2$ of class C^1 , with $\varphi : [c, d] \rightarrow [a, b]$. Then $L(\delta) = L(\gamma)$.

Proof. We treat the case $\varphi' > 0$, so that $\varphi(c) = a$ and $\varphi(d) = b$ (the case $\varphi' < 0$ is identical up to swapping the endpoints and absorbing the sign). By Theorem 1.6(ii),

$$L(\delta) = \int_c^d |\delta'(u)| du = \int_c^d |\varphi'(u)| \cdot |\gamma'(\varphi(u))| du = \int_c^d |\gamma'(\varphi(u))| \varphi'(u) du.$$

Substituting $t = \varphi(u)$, $dt = \varphi'(u) du$, the limits become $t = a$ and $t = b$:

$$L(\delta) = \int_a^b |\gamma'(t)| dt = L(\gamma).$$

$\square$

Theorem 1.11: Existence and uniqueness of the natural parameter. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be a regular curve of class C^k , $k \geq 1$. Setting

$$s(t) = \int_a^t |\gamma'(\tau)| d\tau,$$

the function $s : [a, b] \rightarrow [0, L]$ is an increasing diffeomorphism of class C^k , and the curve

$$\tilde{\gamma} = \gamma \circ s^{-1} : [0, L] \rightarrow \mathbb{R}^2$$

is a reparametrisation of γ with $|\tilde{\gamma}'(s)| = 1$ for every s (**arc-length**, or natural, parametrisation). It is unique up to a translation of the parameter and a reversal of orientation.

Proof. Step 1 — s is a diffeomorphism. The integrand $|\gamma'|$ is continuous (since $\gamma \in C^1$) and strictly positive (since γ is regular). By the fundamental theorem of calculus, s is differentiable with

$$s'(t) = |\gamma'(t)| > 0,$$

hence s is strictly increasing and therefore injective; it is continuous with $s(a) = 0$ and $s(b) = L$, so by the intermediate value theorem it is onto $[0, L]$. Its inverse s^{-1} is differentiable by the inverse function theorem, with

$$(s^{-1})'(\sigma) = \frac{1}{s'(s^{-1}(\sigma))} = \frac{1}{|\gamma'(s^{-1}(\sigma))|} > 0.$$

Step 2 — unit speed. By the chain rule,

$$\tilde{\gamma}'(\sigma) = \gamma'(s^{-1}(\sigma)) (s^{-1})'(\sigma) = \frac{\gamma'(s^{-1}(\sigma))}{|\gamma'(s^{-1}(\sigma))|},$$

a unit vector: $|\tilde{\gamma}'(\sigma)| = 1$. Note that $\tilde{\gamma}'$ coincides with the unit tangent T .

Step 3 — uniqueness. Let $\tilde{\gamma}_1, \tilde{\gamma}_2$ be two unit-speed reparametrisations of γ , related by $\tilde{\gamma}_2 = \tilde{\gamma}_1 \circ \psi$ for some admissible ψ . Differentiating and taking moduli,

$$1 = |\tilde{\gamma}'_2(\sigma)| = |\psi'(\sigma)| \cdot |\tilde{\gamma}'_1(\psi(\sigma))| = |\psi'(\sigma)|,$$

hence $\psi' = \pm 1$ identically (the sign is constant by continuity and because ψ' never vanishes), so $\psi(\sigma) = \pm\sigma + c$: a translation, possibly composed with a reversal of orientation. $\square$

Corollary 1.4: Natural parameter of the circle. The natural parametrisation of $C(C_0, R)$ is

$$\tilde{\gamma}(s) = C_0 + R \left(\cos \frac{s}{R}, \sin \frac{s}{R} \right), \quad s \in [0, 2\pi R],$$

and the parameter s is the arc length travelled from $\tilde{\gamma}(0)$.

Proof. With the standard parametrisation $\gamma(t) = C_0 + R(\cos t, \sin t)$ we have $|\gamma'| = R$, so $s(t) = \int_0^t R d\tau = Rt$, whence $t = s/R$. Substituting gives the formula. Direct check:

$$\tilde{\gamma}'(s) = \left(-\sin \frac{s}{R}, \cos \frac{s}{R} \right), \quad |\tilde{\gamma}'(s)| = 1. \checkmark$$

$\square$

1.9. Normal, curvature and the Frenet formulas

Definition 1.10: Frenet frame and signed curvature. Let γ be a regular plane curve. Set

$$T = \frac{\gamma'}{|\gamma'|} \quad (\text{unit tangent}), \quad N = J T \quad (\text{unit normal, } T \text{ rotated by } +90^\circ),$$

and define the **signed curvature**

$$\kappa = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}}.$$

The value $|\kappa|$ is the (unsigned) curvature; where $\kappa \neq 0$, the number $\rho = 1/|\kappa|$ is the **radius of curvature** and the point $\gamma + (1/\kappa)N$ is the **centre of curvature**.

Theorem 1.12: Frenet formulas in the plane. Let γ be regular and parametrised by arc length. Then

$$\frac{dT}{ds} = \kappa N, \quad \frac{dN}{ds} = -\kappa T,$$

and $\kappa = d\varphi/ds$, where $\varphi(s)$ is a continuous determination of the angle that $T(s)$ makes with the positive x -axis.

Proof. Step 1 — existence of φ . $T(s)$ is a unit vector, so by Lemma 1.1 at every point there is an angle $\varphi(s)$ with $T(s) = (\cos \varphi(s), \sin \varphi(s))$. That it can be chosen *continuously* (indeed C^1) along the curve is the content of the continuous lifting of the angle, which we prove here: writing $T = (T_1, T_2)$, in a neighbourhood of s_0 where $T_1 \neq 0$ one may take $\varphi = \arctan(T_2/T_1) + k\pi$ with suitable k ; where $T_2 \neq 0$ one uses $\varphi = \operatorname{arccot}(T_1/T_2) + k\pi$. Since $T_1^2 + T_2^2 = 1$, the two cases cover the whole curve, and the determinations glue uniquely once

$\varphi(s_0)$ is fixed.

Step 2 — $T' = \varphi' N$. Differentiating $T(s) = (\cos \varphi(s), \sin \varphi(s))$:

$$T'(s) = \varphi'(s) (-\sin \varphi(s), \cos \varphi(s)) = \varphi'(s) J T(s) = \varphi'(s) N(s),$$

using $J(\cos \varphi, \sin \varphi) = (-\sin \varphi, \cos \varphi)$.

Step 3 — $\varphi' = \kappa$. With unit speed, $x' = \cos \varphi$, $y' = \sin \varphi$, hence $x'' = -\varphi' \sin \varphi$, $y'' = \varphi' \cos \varphi$, and the denominator in Definition 1.10 equals 1. Therefore

$$\kappa = x'y'' - y'x'' = \varphi'(\cos^2 \varphi + \sin^2 \varphi) = \varphi'.$$

Step 4 — **the second formula.** From $N = JT$ with J constant,

$$N'(s) = J T'(s) = J(\kappa N(s)) = \kappa J^2 T(s) = -\kappa T(s),$$

using $N = JT$ and $J^2 = -I$ (Theorem 1.3(iii)). □

Theorem 1.13: Curvature of the circle. For every regular parametrisation of the circle $C(C_0, R)$ the curvature equals

$$\kappa = \frac{\varepsilon}{R}, \quad \varepsilon = +1 \text{ if the motion is counterclockwise, } \varepsilon = -1 \text{ if clockwise.}$$

In particular $|\kappa| = 1/R$ is independent of the parametrisation, and the radius of curvature coincides with the radius.

Proof. Let $\gamma(t) = C_0 + R(\cos \theta(t), \sin \theta(t))$ with $\theta' = \omega \neq 0$ (regularity, Theorem 1.2(iii)). Writing $c = \cos \theta(t)$ and $\sigma = \sin \theta(t)$:

$$x' = -R\omega\sigma, \quad y' = R\omega c,$$

$$x'' = -R\alpha\sigma - R\omega^2 c, \quad y'' = R\alpha c - R\omega^2 \sigma,$$

where $\alpha = \omega'$, using the product rule together with $c' = -\omega\sigma$ and $\sigma' = \omega c$. The numerator is then

$$x'y'' - y'x'' = -R^2\omega\alpha\sigma c + R^2\omega^3\sigma^2 + R^2\omega\alpha c\sigma + R^2\omega^3 c^2 = R^2\omega^3.$$

The terms in α cancel: curvature does not depend on the angular acceleration, as it must for a purely geometric quantity. The denominator is

$$(x'^2 + y'^2)^{3/2} = (R^2\omega^2)^{3/2} = R^3|\omega|^3.$$

Hence

$$\kappa = \frac{R^2\omega^3}{R^3|\omega|^3} = \frac{1}{R} \operatorname{sgn}(\omega),$$

and by Theorem 1.2(iv) the sign of ω is the direction of travel. □

Theorem 1.14: The circle is the only plane curve of constant non-zero curvature.

Let $\gamma : I \rightarrow \mathbb{R}^2$ be a regular curve of class C^2 , parametrised by arc length, with constant curvature $\kappa \equiv \kappa_0 \neq 0$. Then the trace of γ is contained in a circle of radius $1/|\kappa_0|$; if I has length $\geq 2\pi/|\kappa_0|$, the trace is the whole circle.

Proof. **Step 1** — **the centre of curvature is fixed.** Define the candidate centre

$$c(s) = \gamma(s) + \frac{1}{\kappa_0} N(s).$$

Differentiating and using the Frenet formulas (Theorem 1.12), with $\gamma' = T$ and $N' = -\kappa_0 T$:

$$c'(s) = T(s) + \frac{1}{\kappa_0} (-\kappa_0 T(s)) = 0.$$

The derivative vanishes identically on an interval, hence $c(s) \equiv C_0$ is constant. (This is where $\kappa_0 \neq 0$ is used: for $\kappa_0 = 0$ the centre of curvature is undefined, and indeed the conclusion is false — the curve is a straight line.)

Step 2 — the distance from the centre is constant. For every s ,

$$|\gamma(s) - C_0| = \left| -\frac{1}{\kappa_0} N(s) \right| = \frac{1}{|\kappa_0|} |N(s)| = \frac{1}{|\kappa_0|},$$

since N is a unit vector. Hence $\gamma(I) \subseteq C(C_0, 1/|\kappa_0|)$.

Step 3 — when the trace is the whole circle. Setting $R = 1/|\kappa_0|$, by Theorem 1.12 the tangent angle satisfies $\varphi' = \kappa_0$, hence $\varphi(s) = \varphi(s_0) + \kappa_0(s - s_0)$. The point $\gamma(s)$ has phase (relative to C_0) equal to $\varphi(s) \mp \pi/2$, varying at rate $|\kappa_0| = 1/R$; if I has length $\Delta s \geq 2\pi R$, the phase varies by at least 2π and by Theorem 1.1 the trace covers the whole circle. $\square$

. Second proof, via a differential equation. With unit speed, $\gamma'' = T' = \kappa_0 N = \kappa_0 J T = \kappa_0 J \gamma'$. Setting $w = \gamma'$, we get the linear system $w' = \kappa_0 J w$, whose solution with initial datum $w(0) = w_0$ is

$$w(s) = e^{\kappa_0 J s} w_0 = Q(\kappa_0 s) w_0,$$

the last equality following from Theorem 1.3(iv): the matrix $Q(\kappa_0 s)$ solves $\frac{d}{ds} Q = \kappa_0 J Q$ with $Q(0) = I$, and the solution of a constant-coefficient linear system is unique. Writing $w_0 = (\cos \varphi_0, \sin \varphi_0)$ (a unit vector, since $|w| = 1$),

$$\gamma'(s) = (\cos(\kappa_0 s + \varphi_0), \sin(\kappa_0 s + \varphi_0)).$$

Integrating componentwise,

$$\gamma(s) = \left(a + \frac{1}{\kappa_0} \sin(\kappa_0 s + \varphi_0), b - \frac{1}{\kappa_0} \cos(\kappa_0 s + \varphi_0) \right)$$

for suitable constants a, b . Setting $C_0 = (a, b)$ one checks directly $|\gamma(s) - C_0| = 1/|\kappa_0|$: the same conclusion as Step 2. $\square$

Remark 1.11: The osculating circle. Theorem 1.14 explains why, for an arbitrary curve, the **osculating circle** at a point is defined as the circle with centre $\gamma + (1/\kappa)N$ and radius $1/|\kappa|$: it is the unique circle sharing with the curve, at that point, position, tangent and curvature, and hence approximating it to second order. The characterisation proved above is the special case of constant κ of the fundamental theorem of plane curves (curvature determines the curve up to isometries), which will be treated in Volume III.

Remark 1.12: The locus of centres: evolutes, the cycloid, the involute. As P varies along the curve, the centre $\gamma + (1/\kappa)N$ of the osculating circle itself traces out a curve, the **evolute**. Two cases in which the circle plays the lead role have their own names here: the cycloid (the path traced by a point on the rim of a circle rolling without slipping along a line) and the involute of the circle (the curve traced by unwinding a taut string from a circle, used for the profile of gear teeth). These curves, with their complete proofs (including the historical evolute/involute result due to Huygens) and worked exercises, are treated in full in Volume III, Chapter 2 (Plane Curves: Curvature, the Osculating Circle,

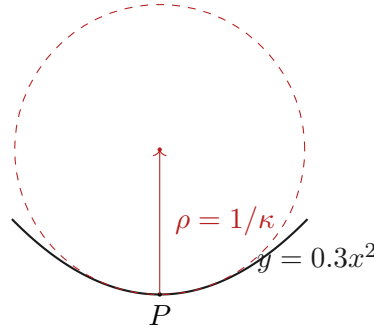


Figure 1.2: The osculating circle at $P = (0, 0)$ to the parabola $y = 0.3x^2$. Here $x' = 1$, $y' = 0.6x$, $x'' = 0$, $y'' = 0.6$, hence $\kappa(0) = 0.6$ and $\rho = 1/0.6 = 1.\bar{6}$.

Evolutes); we do not repeat them here so as not to duplicate that material.

1.10. Circular motion: decomposition of the acceleration

Theorem 1.15: Tangential–centripetal decomposition. For the circular motion $\gamma(t) = C_0 + R e_r(t)$ of Definition 1.2, with $\omega = \theta'$ and $\alpha = \theta''$:

$$\gamma''(t) = \underbrace{R\alpha(t) e_\vartheta(t)}_{\text{tangential}} - \underbrace{R\omega(t)^2 e_r(t)}_{\text{centripetal}}$$

In particular:

- (i) the centripetal component has modulus $a_c = R\omega^2 = v^2/R$, where $v = R|\omega|$;
- (ii) the tangential component has modulus $a_t = R|\alpha| = |v'|$;
- (iii) the motion is uniform (v constant) if and only if $\alpha \equiv 0$, and in that case $\gamma'' \perp \gamma'$;
- (iv) in general $\gamma'' = a_t T + \frac{v^2}{\rho} N_{\text{in}}$, with $\rho = R$ the radius of curvature: the formula holds for every regular curve, not only for the circle.

Proof. Start from $\gamma'(t) = R\omega(t) e_\vartheta(t)$ (Theorem 1.2(ii)) and differentiate by the product rule:

$$\gamma''(t) = R\omega'(t) e_\vartheta(t) + R\omega(t) e'_\vartheta(t).$$

We compute e'_ϑ . From $e_\vartheta = (-\sin \theta, \cos \theta)$, by the chain rule

$$e'_\vartheta(t) = (-\cos \theta(t) \theta'(t), -\sin \theta(t) \theta'(t)) = -\omega(t) e_r(t).$$

Substituting, and recalling $\omega' = \alpha$:

$$\gamma''(t) = R\alpha(t) e_\vartheta(t) - R\omega(t)^2 e_r(t).$$

Since $e_r \perp e_\vartheta$ and both are unit vectors, the moduli of the two components are $R|\alpha|$ and $R\omega^2$ (the latter without absolute value, as $\omega^2 \geq 0$).

(i) $R\omega^2 = R \cdot \frac{(R|\omega|)^2}{R^2} = \frac{v^2}{R}$.

(ii) $v(t) = R|\omega(t)|$; where ω does not change sign, $v' = R\omega' \operatorname{sgn} \omega$, hence $|v'| = R|\alpha|$.

(iii) $v = R|\omega|$ is constant $\iff |\omega|$ is constant $\iff \alpha = \omega' = 0$ (for ω continuous and never zero). In that case $\gamma'' = -R\omega^2 e_r$, orthogonal to $\gamma' = R\omega e_\vartheta$. Alternatively, as announced in Observation 1.3: differentiating $|\gamma'|^2 = v^2 = \text{const.}$ gives $2\gamma' \cdot \gamma'' = 0$.

(iv) The unit tangent is $T = \gamma'/|\gamma'| = \text{sgn}(\omega) e_\theta$, and $-e_r$ is the unit normal pointing towards the centre. The tangential component is $R\alpha e_\theta = v'T$, the normal one $\frac{v^2}{R}(-e_r)$. By Theorem 1.13, $R = 1/|\kappa| = \rho$. $\square$

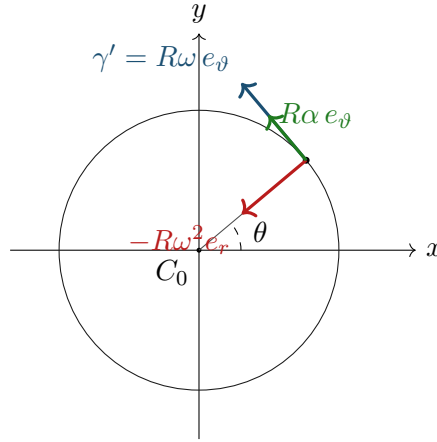


Figure 1.3: Decomposition of the acceleration in a non-uniform circular motion: the tangential component $R\alpha e_\theta$ changes the magnitude of the velocity, the centripetal component $-R\omega^2 e_r$ changes its direction.

Example 1.7: Rotor during start-up. A rotor of radius $R = 0.25$ m accelerates according to $\theta(t) = 3t^2 + 2t$ (rad, with t in seconds). Determine, at $t = 2.0$ s: the speed, the tangential acceleration, the centripetal acceleration, and the magnitude and direction of the total acceleration.

Step 1 — derivatives of the phase.

$$\omega(t) = 6t + 2, \quad \alpha(t) = 6 \text{ (constant).}$$

At $t = 2.0$: $\omega = 14.0$ rad/s, $\alpha = 6.0$ rad/s².

Step 2 — speed.

$$v = R|\omega| = 0.25 \times 14.0 = 3.50 \text{ m/s.}$$

Step 3 — tangential acceleration.

$$a_t = R|\alpha| = 0.25 \times 6.0 = 1.50 \text{ m/s}^2.$$

Step 4 — centripetal acceleration.

$$a_c = R\omega^2 = 0.25 \times 14.0^2 = 0.25 \times 196 = 49.0 \text{ m/s}^2.$$

Cross-check with the alternative formula: $v^2/R = 3.50^2/0.25 = 12.25/0.25 = 49.0$ m/s². $\checkmark$

Step 5 — total acceleration. The two components are orthogonal, so by Pythagoras

$$|a| = \sqrt{a_t^2 + a_c^2} = \sqrt{1.50^2 + 49.0^2} = \sqrt{2.25 + 2401} = \sqrt{2403.25} = 49.02 \text{ m/s}^2.$$

The angle between a and the centripetal direction is

$$\beta = \arctan \frac{a_t}{a_c} = \arctan \frac{1.50}{49.0} = \arctan(0.03061) = 1.75^\circ.$$

Engineering interpretation. Already at 14 rad/s (≈ 134 rpm) the centripetal component dominates by a factor of 33: bearing sizing is governed almost entirely by $R\omega^2$, while the tangential component matters only for the choice of drive torque. The ratio $a_c/a_t = \omega^2/\alpha$ grows quadratically with speed: at steady state ($\alpha \rightarrow 0$) the tangential component vanishes altogether.

1.11. Functions on the circle, tangential derivative and constrained optimisation

So far we have studied the curve. We now study *functions defined on the curve*: a temperature field along a ring, a cost function on a circular constraint, a potential on a boundary. The question is: what does it mean to “differentiate” a function along the circle, and where are its maxima and minima?

Definition 1.11: Restriction to the circle and tangential derivative. Let $f : A \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ be of class C^1 on an open set $A \supseteq \Gamma = C(C_0, R)$. The **restriction** of f to Γ is the one-variable function

$$g(s) = f(\tilde{\gamma}(s)),$$

where $\tilde{\gamma}$ is the natural parametrisation. Its derivative $g'(s)$ is the **tangential derivative** of f along Γ , written $\partial_T f$.

Theorem 1.16: Formulas for the tangential derivative. With the notation of Definition 1.11, and with $n = (P - C_0)/R$ the outward unit normal at $P = \tilde{\gamma}(s)$:

- (i) $\partial_T f(P) = \nabla f(P) \cdot T(P)$;
- (ii) setting $\Pi = I - nn^\top$, the matrix Π is the orthogonal projection onto the tangent line: $\Pi^2 = \Pi$, $\Pi^\top = \Pi$, $\Pi n = 0$, $\Pi T = T$;
- (iii) the **tangential gradient**

$$\nabla_\Gamma f(P) = \Pi \nabla f(P) = \nabla f(P) - (\nabla f(P) \cdot n) n$$

satisfies $\nabla_\Gamma f(P) = (\partial_T f(P)) T(P)$;

- (iv) using the angular parametrisation $f(C_0 + R(\cos \theta, \sin \theta)) = h(\theta)$, one has $\partial_T f = h'(\theta)/R$.

Proof. (i) By the multivariable chain rule,

$$g'(s) = \nabla f(\tilde{\gamma}(s)) \cdot \tilde{\gamma}'(s) = \nabla f(P) \cdot T(P),$$

since $\tilde{\gamma}' = T$ by Theorem 1.11.

- (ii) Symmetry: $\Pi^\top = I - (nn^\top)^\top = \Pi$. Idempotence, using $n^\top n = |n|^2 = 1$:

$$\Pi^2 = I - 2nn^\top + n(n^\top n)n^\top = I - 2nn^\top + nn^\top = \Pi.$$

Action on the frame: $\Pi n = n - n(n^\top n) = 0$; and since $T \perp n$ (Theorem 1.8) we have $n^\top T = 0$, hence $\Pi T = T - n(n^\top T) = T$.

- (iii) In the plane $\{T, n\}$ is an orthonormal basis, hence $\nabla f = (\nabla f \cdot T)T + (\nabla f \cdot n)n$. Applying Π and using (ii):

$$\Pi \nabla f = (\nabla f \cdot T)T = (\partial_T f)T,$$

by (i). The second expression for $\Pi \nabla f$ is the explicit form of $(I - nn^\top) \nabla f$.

- (iv) With $\gamma(\theta) = C_0 + R(\cos \theta, \sin \theta)$ we have $s = R\theta$ (Corollary 1.4), so by the chain rule $h'(\theta) = g'(s) ds/d\theta = R \partial_T f$. $\square$

Theorem 1.17: Constrained critical points and Lagrange multipliers. Let $f \in C^1$ on an open set containing $\Gamma = C(C_0, R)$ and let $P \in \Gamma$. The following are equivalent:

- (a) P is a critical point of the restriction $f|_\Gamma$, i.e. $\partial_T f(P) = 0$;
- (b) $\nabla_\Gamma f(P) = 0$;

(c) $\nabla f(P)$ is parallel to $n(P)$, i.e. there is $\lambda \in \mathbb{R}$ with $\nabla f(P) = \lambda \nabla G(P)$, where $G(x, y) = (x - x_c)^2 + (y - y_c)^2 - R^2$.

Moreover, since Γ is compact and f continuous, $f|_\Gamma$ attains an absolute maximum and an absolute minimum, both at points satisfying (a).

Proof. (a) $\iff$ (b). By Theorem 1.16(iii), $\nabla_\Gamma f = (\partial_T f)T$ with $|T| = 1$, hence $|\nabla_\Gamma f| = |\partial_T f|$: one vanishes iff the other does.

(a) $\iff$ (c). In the orthonormal basis $\{T, n\}$, $\nabla f = (\nabla f \cdot T)T + (\nabla f \cdot n)n$. Thus $\partial_T f = \nabla f \cdot T = 0$ means the tangential component vanishes, i.e. $\nabla f = (\nabla f \cdot n)n$: ∇f is parallel to n . By Corollary 1.3, $\nabla G(P) = 2(P - C_0) = 2Rn$, which is parallel to n and non-zero (as $R > 0$); hence “ $\nabla f \parallel n$ ” is equivalent to “ $\exists \lambda : \nabla f = \lambda \nabla G$ ”, with $\lambda = (\nabla f \cdot n)/(2R)$.

Existence of the extrema. Γ is closed (preimage of the closed set $\{R^2\}$ under the continuous map $P \mapsto |P - C_0|^2$) and bounded, hence compact by Heine–Borel. By Weierstrass’ theorem $f|_\Gamma$ attains a maximum and a minimum. Since Γ has no boundary (it is a closed curve), those extrema are interior points for the periodically extended function $g(s)$, where the derivative $g' = \partial_T f$ vanishes: (a) holds. $\square$

Example 1.8: Optimisation on a circular constraint, two ways. Find the maximum and minimum of $f(x, y) = xy$ subject to $x^2 + y^2 = 1$.

Method 1 — parametrisation (reduction to one variable).

Step 1. Substituting $x = \cos \theta$, $y = \sin \theta$:

$$h(\theta) = \cos \theta \sin \theta = \frac{1}{2} \sin 2\theta.$$

Step 2. Since $\sin$ ranges over $[-1, 1]$, $h \in [-\frac{1}{2}, \frac{1}{2}]$, with maximum $\frac{1}{2}$ when $2\theta = \pi/2 + 2k\pi$, i.e. $\theta \in \{\pi/4, 5\pi/4\}$, and minimum $-\frac{1}{2}$ when $\theta \in \{3\pi/4, 7\pi/4\}$.

Step 3. The corresponding points are

$$\text{max: } \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right); \quad \text{min: } \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

Method 2 — Lagrange multipliers.

Step 1. $\nabla f = (y, x)$, $G(x, y) = x^2 + y^2 - 1$, $\nabla G = (2x, 2y)$. Condition (c) of Theorem 1.17 gives the system

$$\begin{cases} y = 2\lambda x \\ x = 2\lambda y \\ x^2 + y^2 = 1. \end{cases}$$

Step 2. Substituting the first into the second: $x = 4\lambda^2 x$, i.e. $x(1 - 4\lambda^2) = 0$. If $x = 0$, the first equation would give $y = 0$, incompatible with $x^2 + y^2 = 1$. Hence $4\lambda^2 = 1$, $\lambda = \pm \frac{1}{2}$.

Step 3. For $\lambda = \frac{1}{2}$: $y = x$ and $2x^2 = 1$, so $x = \pm \frac{\sqrt{2}}{2}$, with $f = \frac{1}{2}$. For $\lambda = -\frac{1}{2}$: $y = -x$, with $f = -\frac{1}{2}$.

Step 4 — conclusion. Maximum $\frac{1}{2}$, minimum $-\frac{1}{2}$: the two methods agree. $\checkmark$

Check with the tangential gradient. At $P = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$: $n = P$ (since $R = 1$, $C_0 = 0$), $T = Jn = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$, $\nabla f(P) = P$. Hence

$$\partial_T f = \nabla f \cdot T = -\frac{1}{2} + \frac{1}{2} = 0. \checkmark$$

Best practice. When the constraint is a circle, *parametrisation* is almost always faster than Lagrange multipliers: it eliminates one variable and one auxiliary unknown, reduces the problem to a periodic function of one variable, and makes the discussion of absolute maxima and minima immediate (they always exist, by compactness). Multipliers become preferable when there is more than one constraint, or when the objective function is naturally expressed in Cartesian coordinates and the trigonometric substitution would complicate it.

1.12. The circle as a manifold, a group and a periodic space

Definition 1.12: The unit circle S^1 . $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$, with the topology induced from $\mathbb{R}^2$. Under the identification $\mathbb{R}^2 \cong \mathbb{C}$, $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, which is closed under complex multiplication.

Theorem 1.18: S^1 is a group, and $\mathbb{R}/2\pi\mathbb{Z} \cong S^1$.

- (i) $(S^1, \cdot)$ is an abelian group, with identity 1 and inverse $z^{-1} = \bar{z}$.
- (ii) The map $E : \mathbb{R} \rightarrow S^1$, $E(\theta) = e^{i\theta}$, is a surjective homomorphism of $(\mathbb{R}, +)$ onto $(S^1, \cdot)$ with kernel $2\pi\mathbb{Z}$; hence

$$\mathbb{R}/2\pi\mathbb{Z} \cong S^1 \cong SO(2).$$

- (iii) A function $F : S^1 \rightarrow \mathbb{R}$ corresponds bijectively to a 2π -periodic function $f : \mathbb{R} \rightarrow \mathbb{R}$, via $f = F \circ E$.

Proof. (i) If $|z| = |w| = 1$ then $|zw| = 1$: the product is internal. Associativity and commutativity are inherited from $\mathbb{C}$. The identity 1 has modulus 1. Finally, for $|z| = 1$ we have $z\bar{z} = |z|^2 = 1$, so $z^{-1} = \bar{z} \in S^1$.

(ii) Homomorphism: $E(\theta_1 + \theta_2) = e^{i\theta_1}e^{i\theta_2} = E(\theta_1)E(\theta_2)$ (a property of the complex exponential, equivalent to the addition formulas). Surjectivity: given $z \in S^1$, written $z = (a, b)$ with $a^2 + b^2 = 1$, Lemma 1.1 yields θ with $z = E(\theta)$. Kernel: $E(\theta) = 1 \iff \cos \theta = 1$ and $\sin \theta = 0 \iff \theta \in 2\pi\mathbb{Z}$. The first isomorphism theorem gives $\mathbb{R}/2\pi\mathbb{Z} \cong S^1$; the isomorphism with $SO(2)$ is Corollary 1.1.

(iii) Given F , the composite $f = F \circ E$ satisfies $f(\theta + 2\pi) = f(\theta)$: it is 2π -periodic. Conversely, given a periodic f , set $F(z) = f(\theta)$ for any θ with $E(\theta) = z$: this is well defined because two such θ differ by a multiple of 2π , on which f does not change. The two constructions are mutually inverse. $\square$

Remark 1.13: Why this matters. Part (iii) is the reason why all analysis on S^1 is the analysis of periodic functions: Fourier series (Chapter 4) are nothing but harmonic analysis on the group S^1 , and the trapezoidal quadrature of Chapter 2 converges exponentially precisely because it exploits periodicity. In this sense the circle is the simplest non-trivial compact group, and the reference model of abstract harmonic analysis.

Theorem 1.19: There is no continuous global angle on S^1 . There is no continuous function $\Theta : S^1 \rightarrow \mathbb{R}$ such that

$$(\cos \Theta(P), \sin \Theta(P)) = P \quad \text{for every } P \in S^1.$$

Proof. Suppose, for contradiction, that such a Θ exists. Consider the path $\gamma : [0, 2\pi] \rightarrow S^1$, $\gamma(t) = (\cos t, \sin t)$, and define

$$\Phi(t) = \Theta(\gamma(t)) - t, \quad t \in [0, 2\pi].$$

Φ is continuous, being a composition and difference of continuous functions.

Step 1 — Φ takes values only in $2\pi\mathbb{Z}$. By hypothesis

$$(\cos \Theta(\gamma(t)), \sin \Theta(\gamma(t))) = \gamma(t) = (\cos t, \sin t).$$

Arguing as in the uniqueness part of Lemma 1.1 (without the restriction to $[0, 2\pi]$), from $\cos \Theta(\gamma(t)) = \cos t$ and $\sin \Theta(\gamma(t)) = \sin t$ we get $\cos(\Theta(\gamma(t)) - t) = 1$, hence

$$\Phi(t) \in 2\pi\mathbb{Z} \quad \text{for every } t.$$

Step 2 — Φ is constant. Φ is continuous on the connected set $[0, 2\pi]$, so $\Phi([0, 2\pi])$ is a connected subset of $\mathbb{R}$ contained in the discrete set $2\pi\mathbb{Z}$. A connected subset of a discrete set with more than one point would be disconnected; hence $\Phi([0, 2\pi])$ reduces to a single point: $\Phi(t) \equiv 2\pi k$ for some fixed $k \in \mathbb{Z}$.

Step 3 — the contradiction. By Step 2, $\Theta(\gamma(t)) = t + 2\pi k$ for every t . But $\gamma(0) = (1, 0) = \gamma(2\pi)$, hence $\Theta(\gamma(0)) = \Theta(\gamma(2\pi))$, i.e.

$$0 + 2\pi k = 2\pi + 2\pi k \implies 0 = 2\pi,$$

absurd. □

Remark 1.14: Practical and theoretical consequences. Theorem 1.19 is the reason why:

- every `atan2` implementation returns values in $(-\pi, \pi]$ and exhibits a jump of 2π across the negative x -axis: the jump is not an implementation defect, it is a mathematical necessity;
- incremental angular encoders must count turns in a separate variable, and phase unwrapping algorithms are indispensable;
- on S^1 at least **two** charts are needed for continuous coordinates everywhere: S^1 is a one-dimensional differentiable manifold that cannot be covered by a single chart. The two charts may be taken as two stereographic projections, from the poles $(-1, 0)$ and $(1, 0)$ (Theorem 1.7);
- in Chapter 2 the vortex field will fail to be conservative on $\mathbb{R}^2 \setminus \{0\}$ precisely because its “natural” potential would be the angle, which by this theorem does not exist globally.

Proposition 1.3: Degree of the map $z \mapsto z^n$. For $n \in \mathbb{Z}$, the map $\Psi_n : S^1 \rightarrow S^1$, $\Psi_n(z) = z^n$, is a group homomorphism, and the path $t \mapsto \Psi_n(e^{it})$, $t \in [0, 2\pi]$, has winding number n .

Proof. $\Psi_n(zw) = (zw)^n = z^n w^n = \Psi_n(z)\Psi_n(w)$: it is a homomorphism. The path is $t \mapsto e^{int}$, with phase law $\theta(t) = nt$; it is closed, since $e^0 = 1 = e^{2\pi in}$, and by Definition 1.3 the winding number is

$$\frac{\theta(2\pi) - \theta(0)}{2\pi} = \frac{2\pi n}{2\pi} = n. \quad \square$$

1.13. Pursuit curves: the bugs-on-a-circle problem

This section applies the tangent vector and the polar decomposition of velocity (Proposition 1.2) to a classical pursuit problem, in its circular variant: n bugs at the vertices of a regular polygon, each chasing the next. The resulting trajectory is, surprisingly, a logarithmic spiral of the previous section.

Definition 1.13: The n -bugs problem. Let $n \geq 3$ point particles (**bugs**) occupy, at time $t = 0$, the vertices of a regular n -gon inscribed in a circle of radius R_0 and centre O . Each bug k moves with constant speed $v > 0$, directed at every instant towards the instantaneous position of bug $k + 1$ (indices mod n).

Theorem 1.20: Symmetry and reduction to a single bug. For every t before the meeting instant, the n bugs occupy the vertices of a regular n -gon, centred at O , that rotates and shrinks. In particular, if $z_k(t)$ is the complex position of bug k ,

$$z_k(t) = e^{2\pi i k/n} z_0(t) \quad \text{for every } k = 0, \dots, n-1.$$

Proof. The system of equations of motion is

$$\dot{z}_k(t) = v \frac{z_{k+1}(t) - z_k(t)}{|z_{k+1}(t) - z_k(t)|}, \quad k = 0, \dots, n-1,$$

with initial data $z_k(0) = R_0 e^{2\pi i k/n}$: a system of Lipschitz differential equations as long as no two bugs have collided (the denominator does not vanish), hence with a unique solution on any collision-free interval $[0, T)$ (Cauchy–Lipschitz theorem, imported: its proof belongs to ordinary differential equations courses and is not reproved here).

Let $\zeta = e^{2\pi i/n}$ and set $w_k(t) = \zeta z_k(t)$. Then $w_k(0) = \zeta R_0 e^{2\pi i k/n} = R_0 e^{2\pi i (k+1)/n} = z_{k+1}(0)$, and

$$\dot{w}_k(t) = \zeta \dot{z}_k(t) = \zeta v \frac{z_{k+1}(t) - z_k(t)}{|z_{k+1}(t) - z_k(t)|} = v \frac{w_{k+1}(t) - w_k(t)}{|w_{k+1}(t) - w_k(t)|},$$

using $\zeta z_{k+1} = w_{k+1}$, $\zeta z_k = w_k$ and $|\zeta| = 1$: the family (w_k) satisfies *the same* system of equations with *the same* initial data $w_k(0) = z_{k+1}(0)$ (indices mod n) as the family $(z_{k+1})_k$. By uniqueness, $w_k(t) = z_{k+1}(t)$ for every t , i.e. $z_{k+1}(t) = \zeta z_k(t)$ for every k . Iterating, $z_k(t) = \zeta^k z_0(t) = e^{2\pi i k/n} z_0(t)$. $\square$

Theorem 1.20 reduces the problem to a single bug: write $z_0(t) = r(t) e^{i\theta(t)}$ in polar coordinates, with $r(0) = R_0$, $\theta(0) = 0$.

Theorem 1.21: Trajectory of the pursuing bug. In the previous notation, before the meeting:

$$\dot{r}(t) = -v \sin \frac{\pi}{n} \quad (\text{constant}), \quad r(t) \dot{\theta}(t) = v \cos \frac{\pi}{n} \quad (\text{constant in magnitude}),$$

and consequently:

- (i) the meeting occurs at time $T = \frac{R_0}{v \sin(\pi/n)}$;
- (ii) the total distance travelled by each bug is $L = vT = \frac{R_0}{\sin(\pi/n)}$;
- (iii) the trajectory is an arc of a logarithmic spiral, $r(\theta) = R_0 e^{-\theta \tan(\pi/n)}$.

Proof. Fix t and let $A = z_0(t)$, $B = z_1(t)$ be the positions of bugs 0 and 1. By Theorem 1.20, $|OA| = |OB| = r(t)$ and the angle $\angle AOB = 2\pi/n$ (the bugs remain at the vertices of a regular

n -gon). Triangle OAB is therefore isosceles, and its base angles measure

$$\widehat{OAB} = \widehat{OBA} = \frac{\pi - 2\pi/n}{2} = \frac{\pi}{2} - \frac{\pi}{n}.$$

The bug at A moves with speed v directed along $\overrightarrow{AB}$ (by definition, it chases the bug at B). The angle between $\overrightarrow{AB}$ and the *outward* direction $\overrightarrow{OA}$ is the supplement of $\widehat{OAB}$, i.e. $\pi/2 + \pi/n$; the radial component of the velocity (along the outward e_r) is therefore

$$\dot{r}(t) = v \cos\left(\frac{\pi}{2} + \frac{\pi}{n}\right) = -v \sin \frac{\pi}{n},$$

constant, which proves the first identity and already (i)–(ii): since $\dot{r}$ is constant, $r(t) = R_0 - vt \sin(\pi/n)$ vanishes at time $T = R_0/(v \sin(\pi/n))$, and since the velocity has constant magnitude v , the distance travelled is $vT = R_0/\sin(\pi/n)$.

For the tangential component: the angle between $\overrightarrow{AB}$ and e_r has sine $\sin(\pi/2 + \pi/n) = \cos(\pi/n)$, so the component of the velocity along e_θ has magnitude $v \cos(\pi/n)$; by Proposition 1.2 (applied to $z_0(t) = \gamma(\theta(t))$ with $\theta = \theta(t)$ a function of t , so that $\dot{z}_0 = \dot{\theta} \gamma'(\theta) = \dot{\theta} r'(\theta) e_r + \dot{\theta} r(\theta) e_\theta$ and $\dot{r} = \dot{\theta} r'(\theta)$), the tangential component of $\dot{z}_0$ is exactly $r\dot{\theta}$; from the consistent sense of rotation (bug 1 is ahead in the direction of increasing θ), $r\dot{\theta} = v \cos(\pi/n)$.

(iii) From the two identities,

$$\frac{dr}{d\theta} = \frac{\dot{r}}{\dot{\theta}} = \frac{-v \sin(\pi/n)}{v \cos(\pi/n)/r} = -r \tan \frac{\pi}{n},$$

a first-order separable differential equation: $dr/r = -\tan(\pi/n) d\theta$, hence $\ln r = -\theta \tan(\pi/n) + \text{const.}$, and imposing $r(0) = R_0$, $r(\theta) = R_0 e^{-\theta \tan(\pi/n)}$: this is the logarithmic spiral of Definition 1.6 with $a = R_0$ and $b = -\tan(\pi/n)$. $\square$

Remark 1.15: Why the spiral has finite length. The bug travels infinitely many turns around O (as $r \rightarrow 0$, $\theta \rightarrow +\infty$: the spiral winds infinitely often), and yet its trajectory has finite length $L = R_0/\sin(\pi/n)$: this is the same phenomenon observed in Example 1.4 as $\theta \rightarrow -\infty$. An object that completes infinitely many revolutions in finite time, covering a finite distance, is possible only because the angular velocity $\dot{\theta} = v \cos(\pi/n)/r(t)$ diverges as $r(t) \rightarrow 0$: the turns become instantaneously tighter and faster, but the radius shrinks just as quickly, making the arc-length integral converge.

Example 1.9: The case $n = 3$: three bugs at the vertices of an equilateral triangle. For $R_0 = 1$ and $v = 1$: $T = 1/\sin(60^\circ) = 1/(\sqrt{3}/2) = 2/\sqrt{3} = 1.154701$, and $L = T = 1.154701$ (distance travelled equal to time, since $v = 1$). The spiral has $b = -\tan(60^\circ) = -\sqrt{3}$, a very tight wind: after just one complete turn ($\theta = 2\pi$) the radius is already reduced to $r = e^{-2\pi\sqrt{3}} \approx 1.88 \times 10^{-5}$ of its initial size.

Independent numerical check. A direct simulation of the system of n differential equations (explicit time step $\Delta t = 2 \times 10^{-5}$, script `Sorgenti/Vol2/_verifich/vol2_cap01_inseguimento.py`) gives, for $n = 3$, a simulated meeting time of 1.154680 and a travelled distance of 1.154680 against the theoretical values 1.154701 (difference 2×10^{-5} , consistent with the truncation error of the finite step); for $n = 4, 5, 6$ the agreement is of the same order (theoretical $T = 1.414214/1.701302/2.000000$ against simulated 1.414220/1.701340/2.000100). In every case, independently integrating the length of the spiral $r(\theta) = R_0 e^{-\theta \tan(\pi/n)}$ down to a residual radius $\varepsilon = 10^{-3}$ gives a value that approaches L , exactly minus the residual tail $\varepsilon/\sin(\pi/n)$ not yet travelled, as expected.

1.14. Computational laboratory

The formulas of this chapter are exact; their translation into code is not. This section collects the numerical precautions that separate a correct implementation from one that fails in the limiting cases.

Best practice. 1. Always use `atan2(y,x)`, never `arctan(y/x)`. The latter loses quadrant information (it cannot distinguish the first quadrant from the third, nor the second from the fourth) and diverges for $x = 0$. `atan2` returns the correct angle in $(-\pi, \pi]$ using the signs of both arguments.

2. Avoid `arccos` near ± 1 . If $u \cdot v \approx |u||v|$, the derivative of `arccos` blows up and the relative error is amplified like $1/\sqrt{1-c^2}$. For the angle between two vectors it is more stable to use

$$\Delta\varphi = \text{atan2}(u_x v_y - u_y v_x, u_x v_x + u_y v_y),$$

which uses the dot product and the cross product simultaneously and stays accurate over the whole range $[-\pi, \pi]$.

3. Unwrap the phase before differentiating numerically: a sampled sequence of angles must be corrected by adding $\pm 2\pi$ whenever two consecutive samples differ by more than π , otherwise the estimated angular velocity shows spurious spikes (a consequence of Theorem 1.19).

4. Renormalise unit vectors after composing rotations. Multiplying many matrices $Q(\theta_i)$ causes rounding error to destroy orthogonality; it is better to add the angles (Theorem 1.3(i)) rather than multiply the matrices, or else to re-orthonormalise periodically.

5. For curvature from sampled data, the formula $\kappa = (x'y'' - y'x'')/(x'^2 + y'^2)^{3/2}$ requires second derivatives: it is ill-conditioned on noisy data. It is better to estimate curvature by fitting a circle to a window of points (algebraic least squares, Volume I, Chapter 11) and take $\kappa = 1/R$.

Listing 1.1: Numerical verification of the chapter's formulas: curvature, arc length and the decomposition of acceleration, on a non-uniform circular motion.

```
import numpy as np

R, C0 = 2.5, np.array([1.0, -0.5])

# non-uniform phase law: theta(t) = 0.4 t^2 + 0.9 t
th = lambda t: 0.4*t**2 + 0.9*t
om = lambda t: 0.8*t + 0.9          # theta'
al = lambda t: 0.8 + 0.0*t          # theta''

def gamma(t):
    return C0[:, None] + R*np.array([np.cos(th(t)), np.sin(th(t))])

# --- curvature by centred finite differences, compared with 1/R ---
t0, h = 1.3, 1e-4
t = np.array([t0-h, t0, t0+h])
P = gamma(t)
d1 = (P[:, 2] - P[:, 0]) / (2*h)          # gamma'
d2 = (P[:, 2] - 2*P[:, 1] + P[:, 0]) / h**2 # gamma''
kap = (d1[0]*d2[1] - d1[1]*d2[0]) / np.linalg.norm(d1)**3
print(f"numerical kappa = {kap:.10f}    expected = {1/R:.10f}")

# --- arc length: quadrature vs exact formula R*|dtheta| ---
a, b = 0.0, 2.0
tt = np.linspace(a, b, 200001)
```

```

sp = R*np.abs(om(tt))
L_num = np.trapezoid(sp, tt)
L_exa = R*abs(th(b) - th(a))
print(f"numerical L = {L_num:.10f}    exact = {L_exa:.10f}")

# --- decomposition of the acceleration ---
er = np.array([np.cos(th(t0)), np.sin(th(t0))])
et = np.array([-np.sin(th(t0)), np.cos(th(t0))])
a_theory = R*al(t0)*et - R*om(t0)**2*er
print("decomposition error:", np.linalg.norm(d2 - a_theory))

```

***Remark 1.16: What to expect.** Running the code, the numerical curvature agrees with $1/R = 0.4$ to about 8 digits: centred differences have $O(h^2)$ error for the first derivative and $O(h^2)$ for the second, but the latter also suffers cancellation, with rounding error $O(\epsilon/h^2)$; the optimal step is $h \approx \epsilon^{1/4} \approx 1.2 \times 10^{-4}$, which is the choice made. The arc length, by contrast, agrees to full machine precision: the integrand is smooth and an exact formula is available.*

1.15. Application: circular interpolation in CNC and robotics

A CNC controller or an articulated robot never stores the parametrisation $\gamma(\theta) = C_0 + R(\cos \theta, \sin \theta)$ of Theorem 1.1: it receives only a start point P_0 , an end point P_1 and a minimal piece of information about the circle they belong to. The three industrial standards for specifying that minimal information are three different ways of determining the centre and radius — and it is instructive to see that all three reduce to results of elementary geometry from Volume I.

1.15.1 IJK method: centre by incremental offset

In the most widespread G-code dialect, the commands G02 (clockwise circular interpolation) and G03 (counterclockwise) specify the centre through incremental offsets I, J (and K for the third axis): the centre is $C_0 = P_0 + (I, J)$. The radius is $R = |C_0 - P_0| = \sqrt{I^2 + J^2}$, and one must verify, within tolerance, that $|C_0 - P_1| = R$.

Step-by-step proof. Step 1. Given $P_0 = (x_0, y_0)$ and I, J : compute $C_0 = (x_0 + I, y_0 + J)$.

Step 2. Compute $R = \sqrt{I^2 + J^2}$.

Step 3. Verify $||C_0 - P_1| - R| < \text{tol}$; if this fails, the program does not describe a valid arc and the controller raises an alarm.

Step 4. Compute the phases $\varphi_0 = \text{atan2}(P_0 - C_0)$ and $\varphi_1 = \text{atan2}(P_1 - C_0)$; the swept angle is

$$\Delta\varphi = \begin{cases} (\varphi_1 - \varphi_0) \bmod 2\pi & \text{for G03 (counterclockwise),} \\ (\varphi_0 - \varphi_1) \bmod 2\pi & \text{for G02 (clockwise),} \end{cases}$$

always reduced to $[0, 2\pi)$.

Step 5. The arc length is $L = R\Delta\varphi$ (Theorem 1.9), and the machining time at feed rate F is $t = L/F$. $\square$

1.15.2 R method: centre by signed radius

Theorem 1.22: Centre of a circle given two points and the radius. Let $P_0, P_1 \in \mathbb{R}^2$ be distinct, $M = \frac{1}{2}(P_0 + P_1)$, $d = |P_1 - P_0|$, and let $\hat{n}$ be a unit vector orthogonal to $P_1 - P_0$. For $|R| \geq d/2$ the centres of the circles of radius $|R|$ through P_0 and P_1 are exactly

$$C_{\pm} = M \pm h \hat{n}, \quad h = \sqrt{R^2 - (d/2)^2},$$

and they are two and distinct if $|R| > d/2$, coincident with M if $|R| = d/2$. For $|R| < d/2$ no such circle exists.

Proof. A point C is the centre of a circle through P_0 and P_1 if and only if $|C - P_0| = |C - P_1|$, i.e. if it lies on the perpendicular bisector of P_0P_1 : the line through M orthogonal to $P_1 - P_0$, parametrised as $C = M + h\hat{n}$, $h \in \mathbb{R}$. Now impose $|C - P_0| = |R|$. The vectors $M - P_0$ and $h\hat{n}$ are orthogonal, and $|M - P_0| = d/2$; by Pythagoras

$$|C - P_0|^2 = h^2 + \frac{d^2}{4}.$$

The equation $h^2 + d^2/4 = R^2$ has real solutions iff $R^2 \geq d^2/4$, i.e. $|R| \geq d/2$, and then $h = \pm\sqrt{R^2 - d^2/4}$: two distinct values if $|R| > d/2$, only $h = 0$ if $|R| = d/2$. $\square$

Remark 1.17: The sign of R in G-code. Once the direction ($G02$ or $G03$) is fixed, the two centres C_+, C_- determine two complementary arcs: one of angular width $\leq \pi$, the other $\geq \pi$. Industrial convention uses the sign of R to disambiguate: $R > 0$ selects the arc $\leq \pi$ (minor), $R < 0$ the one $\geq \pi$ (major). The R method is compact but numerically fragile when $|R| \approx d/2$: then $h = \sqrt{R^2 - d^2/4}$ is the square root of a difference of nearly equal numbers, and small data errors produce large displacements of the centre. For nearly semicircular arcs the IJK method is preferable.

1.15.3 Three-point method

Corollary 1.5: Uniqueness of the circle through three non-collinear points. Three non-collinear points $A, B, C \in \mathbb{R}^2$ determine a unique circle, whose centre is the intersection of the perpendicular bisectors of AB and BC .

Proof. The centre must be equidistant from A and B (hence on the bisector a_{AB}) and from B and C (bisector a_{BC}). The bisectors are orthogonal to $B - A$ and $C - B$ respectively; they are parallel if and only if $B - A$ and $C - B$ are parallel, i.e. if and only if A, B, C are collinear — excluded by hypothesis. Two non-parallel lines in the plane meet at a unique point O , which is therefore the only possible centre; the common distance $|O - A| = |O - B| = |O - C|$ is the radius. $\square$

Example 1.10: Three-point interpolation, Yaskawa case. A robot must traverse the arc through $A = (0, 0)$, $B = (4, 0)$, $C = (0, 4)$ (mm). Determine the centre, the radius, the swept angle and the path length.

Step 1 — centre by synthetic means. The angle at A between AB and AC is a right angle (both sides lie on the axes). By the inscribed-angle theorem in a semicircle (Volume I, Chapter 4), an inscribed right angle subtends a diameter: hence BC is a diameter and

$$O = \frac{B + C}{2} = (2, 2), \quad R = \frac{|BC|}{2} = \frac{\sqrt{32}}{2} = 2\sqrt{2} \approx 2.828 \text{ mm}.$$

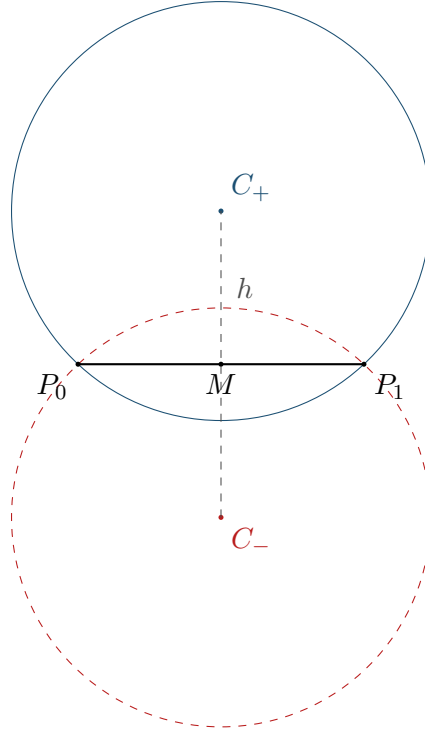


Figure 1.4: R method: the two admissible centres lie on the perpendicular bisector of P_0P_1 , symmetric about the midpoint M , at distance $h = \sqrt{R^2 - (d/2)^2}$ from it.

Step 2 — check with the bisectors (Corollary 1.5). Bisector of AB : midpoint $(2, 0)$, AB horizontal, hence the line $x = 2$. Bisector of AC : midpoint $(0, 2)$, AC vertical, hence $y = 2$. Intersection: $(2, 2)$. ✓ Radius check: $|O - A| = \sqrt{4 + 4} = 2\sqrt{2}$. ✓

Step 3 — phases of the three points relative to O .

$$A-O = (-2, -2) \Rightarrow \varphi_A = 225^\circ, \quad B-O = (2, -2) \Rightarrow \varphi_B = 315^\circ, \quad C-O = (-2, 2) \Rightarrow \varphi_C = 135^\circ.$$

Step 4 — direction and swept angle. From A (225°) to B (315°) the angle increases: the direction is counterclockwise. Continuing from 315° towards $\varphi_C = 135^\circ \equiv 495^\circ$, it keeps increasing. The total angle is

$$\Delta\varphi = 495^\circ - 225^\circ = 270^\circ = \frac{3\pi}{2} \approx 4.712 \text{ rad.}$$

Step 5 — path length. By Theorem 1.9,

$$L = R \Delta\varphi = 2\sqrt{2} \times \frac{3\pi}{2} = 3\sqrt{2}\pi \approx 13.329 \text{ mm.}$$

Order-of-magnitude check: the full circle measures $2\pi R = 17.77$ mm, and 270° is three quarters of it: $0.75 \times 17.77 = 13.33$ mm. ✓

Step 6 — admissible centripetal acceleration. If the controller limits the acceleration to $a_{\max} = 2000 \text{ mm/s}^2$, the maximum speed on the arc is, by Theorem 1.15(i),

$$v_{\max} = \sqrt{a_{\max} R} = \sqrt{2000 \times 2.828} = \sqrt{5657} = 75.2 \text{ mm/s,}$$

and the minimum traversal time is $t = L/v_{\max} = 13.329/75.2 = 0.177$ s.

Listing 1.2: INFORM job (Yaskawa Motoman) for the three-point arc of Example 1.10, simplified for teaching purposes.

```

/JOB
//NAME THREE_POINT_ARC
//POS
///NPOS 3
///RECTAN
P001=(0.00,0.00,0.00,0.00,0.00,0.00) 'point A: start of the arc
P002=(4.00,0.00,0.00,0.00,0.00,0.00) 'point B: intermediate via point
P003=(0.00,4.00,0.00,0.00,0.00,0.00) 'point C: end of the arc
//INST
NOP
MOVJ VJ=25.00 ' approach in joint coordinates
MOVL P001 V=50.0 ' place the TCP exactly at A
MOVC P002 V=75.0 PL=0 ' 1st circular segment: towards B
MOVC P003 V=75.0 PL=0 ' 2nd circular segment: towards C
MOVJ VJ=25.00 ' retract
END

```

Remark 1.18: Why two *MOVC* instructions are needed. The controller determines the circle, as in Corollary 1.5, from the current TCP position (tool centre point, here A) and the two points of the two subsequent *MOVC* instructions. Two *MOVC* commands in sequence are therefore always required: a single one would supply only two points, insufficient to determine a circle — exactly as in the *R* method, where the ambiguity is instead resolved by hand, with the sign of *R*. The *PL* parameter (position level) controls path rounding near the point: with *PL=0* the robot stops exactly at *C*.

1.16. Numerical exercises

All the exercises below are worked out in full in Section 1.18. It is advisable to attempt them before reading the solution, and in any case to read the solution step by step: the sequence of steps is as much part of the didactic content as the final answer.

Exercise 1.1: N1 — From the general equation to the parametrisation. Given the curve $2x^2 + 2y^2 - 12x + 16y - 6 = 0$, recognise that it is a circle, determine its centre and radius, write its standard parametrisation and compute the point corresponding to $\theta = 2\pi/3$.

Exercise 1.2: N2 — Phase law and kinematics. A disc of radius $R = 0.40$ m rotates with phase law $\theta(t) = 0.5t^3 - 2t$ (rad, t in s). Determine the instants at which the motion is *not* regular, the speed and the components of the acceleration at $t = 2.0$ s, and the direction of travel before and after the instant of reversal.

Exercise 1.3: N3 — Winding number. The motion $\gamma(t) = (3 \cos(2t + \pi/4), 3 \sin(2t + \pi/4))$ is considered on $[0, 3\pi]$. Verify that it is closed, compute its winding number and the total length travelled, and exhibit two distinct instants in $[0, 3\pi)$ with the same image.

Exercise 1.4: N4 — Tangent line. Determine the tangent line to the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ at the point $P = (2, 6)$, by two independent methods, and check the result by computing the distance from the centre to the line.

Exercise 1.5: N5 — Rational parametrisation. Using Theorem 1.7, compute $p(2)$ and $p(-1/3)$, and determine the parameter u corresponding to the point $(3/5, 4/5)$. Verify moreover that $u = \tan(\theta/2)$, with θ the phase of that point.

Exercise 1.6: N6 — Arc length and natural parameter. An arc of the circle of centre $(2, -1)$ and radius $R = 6$ runs from the point of phase $\theta_0 = 20^\circ$ to the point of phase $\theta_1 = 155^\circ$, counterclockwise. Compute its length, write the natural parametrisation of the arc and determine the point lying 9.00 units of length from its start.

Exercise 1.7: N7 — Curvature from a non-standard parametrisation. Verify by direct computation that the rational parametrisation $p(u) = ((1-u^2)/(1+u^2), 2u/(1+u^2))$ has curvature $\kappa = 1$ at $u = 1$, in agreement with Theorem 1.13.

Exercise 1.8: N8 — Acceleration in a non-uniform motion. A robotic arm rotates with $\theta(t) = \pi t^2/8$ on a radius $R = 0.75$ m. At what instant do the tangential and the centripetal accelerations have equal magnitude? What is the magnitude of the total acceleration at that instant?

Exercise 1.9: N9 — Optimisation on a circular constraint. Determine the maximum and minimum of $f(x, y) = 3x + 4y$ subject to $(x-1)^2 + (y+2)^2 = 25$, by both methods (parametrisation and multipliers), and give the points at which they are attained.

Exercise 1.10: N10 — CNC circular interpolation. A G-code program contains, with current position $P_0 = (10, 5)$:

G03 X22 Y17 I12 J0 F300

Verify the consistency of the data, compute the centre, the radius, the swept angle, the arc length and the machining time. Determine also which value of R would have produced the same arc in the R method.

1.17. Symbolic exercises

These exercises require no numerical computation but proofs: they are, to all intents and purposes, small theorems. Complete solutions are again in Section 1.18.

Exercise 1.11: S1 — Constant distance implies orthogonality. Let $\gamma : I \rightarrow \mathbb{R}^2$ be of class C^1 and $C_0 \in \mathbb{R}^2$ fixed. Prove that $|\gamma(t) - C_0|$ is constant if and only if $(\gamma(t) - C_0) \cdot \gamma'(t) = 0$ for every $t \in I$. Deduce that the velocity of a circular motion is always tangential.

Exercise 1.12: S2 — Invariance of curvature. Let γ be a regular plane curve of class C^2 and $\delta = \gamma \circ \varphi$ a reparametrisation. Prove that

$$\kappa_\delta(u) = \operatorname{sgn}(\varphi'(u)) \kappa_\gamma(\varphi(u)).$$

In particular $|\kappa|$ is invariant, and κ changes sign exactly when the orientation is reversed.

Exercise 1.13: S3 — Decomposition of the acceleration for an arbitrary curve. Let γ be regular of class C^2 , $v(t) = |\gamma'(t)|$, T and N the Frenet frame, κ the curvature.

Prove that

$$\gamma''(t) = v'(t) T(t) + \kappa(t) v(t)^2 N(t),$$

and check that for the circle one recovers Theorem 1.15.

Exercise 1.14: S4 — Uniqueness of the rotation matrix. Prove that the matrix Cauchy problem

$$\frac{dM}{d\theta}(\theta) = J M(\theta), \quad M(0) = I$$

has one and only one solution, and that it is $M(\theta) = Q(\theta)$. (Hint: for uniqueness consider $W(\theta) = Q(-\theta)M(\theta)$ and compute its derivative.)

Exercise 1.15: S5 — Pythagorean triples from stereographic projection. Prove that the map $u \mapsto p(u)$ of Theorem 1.7 induces a bijection between $\mathbb{Q}$ and the points of S^1 with rational coordinates other than $(-1, 0)$, and deduce that every primitive Pythagorean triple (a, b, c) with b even can be written uniquely as $a = n^2 - m^2$, $b = 2mn$, $c = n^2 + m^2$ with $n > m \geq 1$ coprime integers of opposite parity.

Exercise 1.16: S6 — Invariance of length under isometries. Let $\Phi(P) = QP + b$ be an isometry of the plane (Q orthogonal, $b \in \mathbb{R}^2$) and γ a C^1 curve. Prove that $L(\Phi \circ \gamma) = L(\gamma)$, and that the curvature satisfies $\kappa_{\Phi \circ \gamma} = (\det Q) \kappa_\gamma$.

Exercise 1.17: S7 — The integral of the tangential derivative vanishes. Let $f \in C^1$ on an open set containing $\Gamma = C(C_0, R)$. Prove that

$$\oint_{\Gamma} \partial_T f \, ds = 0,$$

and interpret the result in terms of Theorem 1.17. Deduce that $\partial_T f$ vanishes at at least two points of Γ .

Exercise 1.18: S8 — Constant speed minimises the energy. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be regular of length L . Prove the inequality

$$L^2 \leq (b - a) \int_a^b |\gamma'(t)|^2 dt,$$

with equality if and only if $|\gamma'|$ is constant. (Hint: Cauchy–Schwarz.) Conclude that, among all parametrisations of an arc on a given interval, the constant-speed one is the unique minimiser of the energy $\int |\gamma'|^2$.

Exercise 1.19: S9 — The tangential gradient does not depend on the extension. Let $f_1, f_2 \in C^1$ be defined on an open set $A \supseteq \Gamma$ and coincide on Γ . Prove that $\nabla_\Gamma f_1 = \nabla_\Gamma f_2$ at every point of Γ , whereas in general $\nabla f_1 \neq \nabla f_2$. Exhibit an explicit example of two such extensions with different gradients.

Exercise 1.20: S10 — Local existence of the angle. Theorem 1.19 denies the existence of a continuous angle on all of S^1 . Prove that, on the contrary, on every open arc $U \subsetneq S^1$ (that is, the image under $\theta \mapsto (\cos \theta, \sin \theta)$ of an open interval of length $\leq 2\pi$) there exists a continuous $\Theta : U \rightarrow \mathbb{R}$ with $(\cos \Theta(P), \sin \Theta(P)) = P$, unique up to an additive constant

in $2\pi\mathbb{Z}$. Exhibit explicitly two arcs covering S^1 and the corresponding Θ .

1.18. Solutions

1.18.1 Solutions to the numerical exercises

Solution. N1. *Step 1 — normalisation.* The coefficients of x^2 and y^2 must be 1: divide the whole equation by 2,

$$x^2 + y^2 - 6x + 8y - 3 = 0.$$

Since the coefficients of x^2 and y^2 are now equal and there is no xy term, this is a circle (possibly degenerate).

Step 2 — completing the squares.

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 3 + 9 + 16 = 28,$$

that is $(x - 3)^2 + (y + 4)^2 = 28$. Since $28 > 0$ the circle is real and non-degenerate:

$$C_0 = (3, -4), \quad R = \sqrt{28} = 2\sqrt{7} \approx 5.2915.$$

Step 3 — parametrisation.

$$\gamma(\theta) = (3 + 2\sqrt{7} \cos \theta, -4 + 2\sqrt{7} \sin \theta), \quad \theta \in [0, 2\pi).$$

Step 4 — the point for $\theta = 2\pi/3$. We have $\cos(2\pi/3) = -\frac{1}{2}$ and $\sin(2\pi/3) = \frac{\sqrt{3}}{2}$, hence

$$x = 3 - \sqrt{7} \approx 0.3543, \quad y = -4 + \sqrt{21} \approx 0.5826.$$

Step 5 — check.

$$(x - 3)^2 + (y + 4)^2 = 7 + 21 = 28. \checkmark$$

Solution. N2. *Step 1 — derivatives of the phase.* From $\theta(t) = 0.5t^3 - 2t$:

$$\omega(t) = 1.5t^2 - 2, \quad \alpha(t) = 3t.$$

Step 2 — non-regular instants. By Theorem 1.2(iii) the curve fails to be regular where $\omega = 0$:

$$1.5t^2 = 2 \implies t^2 = \frac{4}{3} \implies t = \pm \frac{2}{\sqrt{3}} = \pm 1.1547 \text{ s.}$$

For $t \geq 0$ the only instant is $t^* = 1.1547$ s: there the point momentarily stops.

Step 3 — kinematics at $t = 2.0$ s.

$$\omega(2) = 1.5 \cdot 4 - 2 = 4.00 \text{ rad/s}, \quad \alpha(2) = 6.00 \text{ rad/s}^2.$$

By Theorems 1.2(ii) and 1.15:

$$v = R|\omega| = 0.40 \times 4.00 = 1.60 \text{ m/s},$$

$$a_t = R|\alpha| = 0.40 \times 6.00 = 2.40 \text{ m/s}^2, \quad a_c = R\omega^2 = 0.40 \times 16.0 = 6.40 \text{ m/s}^2.$$

The total magnitude, the components being orthogonal, is

$$|a| = \sqrt{2.40^2 + 6.40^2} = \sqrt{5.76 + 40.96} = \sqrt{46.72} = 6.835 \text{ m/s}^2.$$

Step 4 — direction of travel. The sign of $\omega(t) = 1.5t^2 - 2$ is negative for $0 \leq t < 1.1547$ and positive for $t > 1.1547$. By Theorem 1.2(iv): the disc rotates **clockwise** before t^* , stops at t^* , and resumes **counterclockwise** afterwards. The instant t^* is a reversal of the motion, not a geometric singularity: the trace remains the same smooth circle (cf. Example 1.5).

Solution. N3. *Step 1 — phase law.* $\theta(t) = 2t + \pi/4$, hence $\theta(0) = \pi/4$ and $\theta(3\pi) = 6\pi + \pi/4$.
Step 2 — the curve is closed. $\Delta\theta = 6\pi$, an integer multiple of 2π ; by periodicity of sine and cosine, $\gamma(3\pi) = \gamma(0)$. ✓
Step 3 — winding number. By Definition 1.3, $n = 6\pi/2\pi = 3$.
Step 4 — length travelled. By Theorem 1.9,

$$L = R|\Delta\theta| = 3 \times 6\pi = 18\pi \approx 56.549,$$

consistent with intuition: three turns of a circle of radius 3, i.e. $3 \times 2\pi \times 3 = 18\pi$. ✓

Step 5 — two instants with the same image. We need $\theta(t_2) - \theta(t_1) = 2\pi$, i.e. $t_2 = t_1 + \pi$. Taking $t_1 = 0$ and $t_2 = \pi$, both in $[0, 3\pi)$:

$$\gamma(0) = \left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2}\right) = \gamma(\pi). \checkmark$$

This was guaranteed by Proposition 1.1(ii), since $|n| = 3 \geq 2$ and θ is strictly increasing.

Solution. N4. *Step 1 — canonical form.*

$$x^2 + y^2 + 4x - 6y - 12 = 0 \implies (x+2)^2 + (y-3)^2 = 25,$$

hence $C_0 = (-2, 3)$ and $R = 5$.

Step 2 — P lies on the circle. $(2+2)^2 + (6-3)^2 = 16 + 9 = 25$. ✓

Method 1 — gradient. By Corollary 1.3, $\nabla F(P) = 2(4, 3) = (8, 6)$, and the tangent is

$$8(x-2) + 6(y-6) = 0 \implies 4x + 3y = 26.$$

Method 2 — slopes. The radius C_0P has slope $m_r = \frac{3}{4}$. The tangent is perpendicular to the radius (Theorem 1.8), so $m_t = -\frac{4}{3}$, and

$$y-6 = -\frac{4}{3}(x-2) \implies 4x + 3y = 26. \checkmark$$

Step 3 — distance check. The distance from $C_0 = (-2, 3)$ to the line $4x + 3y - 26 = 0$ is

$$\frac{|-8 + 9 - 26|}{\sqrt{16 + 9}} = \frac{25}{5} = 5 = R.$$

A line at distance exactly R from the centre is tangent (Volume I, Chapter 3). ✓

Solution. N5. *Step 1 — computing $p(2)$.* With $u = 2$: $1 + u^2 = 5$, $1 - u^2 = -3$, $2u = 4$, hence

$$p(2) = \left(-\frac{3}{5}, \frac{4}{5}\right),$$

and $\frac{9+16}{25} = 1$. ✓

Step 2 — computing $p(-1/3)$. With $u = -\frac{1}{3}$: $1 + u^2 = \frac{10}{9}$, $1 - u^2 = \frac{8}{9}$, $2u = -\frac{2}{3}$, hence

$$x = \frac{8/9}{10/9} = \frac{4}{5}, \quad y = -\frac{2}{3} \cdot \frac{9}{10} = -\frac{3}{5},$$

so $p(-1/3) = \left(\frac{4}{5}, -\frac{3}{5}\right)$, and $\frac{16+9}{25} = 1$. ✓

Step 3 — parameter of the point $(3/5, 4/5)$. By the inverse formula of Theorem 1.7,

$$u = \frac{y}{1+x} = \frac{4/5}{8/5} = \frac{1}{2}.$$

Direct check: $p(1/2)$ has $1+u^2 = \frac{5}{4}$, $1-u^2 = \frac{3}{4}$, $2u = 1$, hence $p(1/2) = \left(\frac{3}{5}, \frac{4}{5}\right)$. ✓

Step 4 — verifying $u = \tan(\theta/2)$. The phase of the point satisfies $\cos \theta = \frac{3}{5}$, $\sin \theta = \frac{4}{5}$, i.e. $\theta = 0.92730$ rad (53.13°). Using the half-angle formula in the form $\tan(\theta/2) = \sin \theta / (1 + \cos \theta)$ (equivalent to Corollary 1.2, as one sees by solving for u):

$$\tan \frac{\theta}{2} = \frac{4/5}{8/5} = \frac{1}{2} = u. \checkmark$$

Numerically: $\tan(0.46365) = 0.50000$. ✓

Solution. N6. *Step 1 — conversion to radians.*

$$\theta_0 = 20^\circ = 0.349066 \text{ rad}, \quad \theta_1 = 155^\circ = 2.705260 \text{ rad}, \quad \Delta\theta = 135^\circ = \frac{3\pi}{4} = 2.356194 \text{ rad}.$$

Step 2 — arc length. By Theorem 1.9,

$$L = R \Delta\theta = 6 \times \frac{3\pi}{4} = \frac{9\pi}{2} \approx 14.1372.$$

Step 3 — natural parametrisation. By Corollary 1.4, adapted to an arc starting at θ_0 :

$$\tilde{\gamma}(s) = \left(2 + 6 \cos\left(\theta_0 + \frac{s}{6}\right), -1 + 6 \sin\left(\theta_0 + \frac{s}{6}\right) \right), \quad s \in \left[0, \frac{9\pi}{2}\right],$$

and indeed $|\tilde{\gamma}'(s)| = 6 \cdot \frac{1}{6} = 1$. ✓

Step 4 — the point at $s = 9.00$. The phase is

$$\theta = \theta_0 + \frac{9.00}{6} = 0.349066 + 1.500000 = 1.849066 \text{ rad} = 105.94^\circ.$$

Since $1.849066 < 2.705260$, the point indeed lies on the arc. We compute

$$\cos(1.849066) = -0.274692, \quad \sin(1.849066) = 0.961532,$$

hence

$$x = 2 + 6(-0.274692) = 2 - 1.648152 = 0.35185,$$

$$y = -1 + 6(0.961532) = -1 + 5.769192 = 4.76919.$$

Step 5 — check.

$$(x-2)^2 + (y+1)^2 = (-1.648152)^2 + (5.769192)^2 = 2.71640 + 33.28360 = 36.0000 = R^2. \checkmark$$

Solution. N7. *Step 1 — first derivatives.* Set $w = 1 + u^2$. From $x(u) = (1 - u^2)w^{-1}$, by the quotient rule:

$$x'(u) = \frac{(-2u)w - (1 - u^2)(2u)}{w^2} = \frac{-2u[w + (1 - u^2)]}{w^2} = \frac{-4u}{w^2},$$

using $w + (1 - u^2) = 2$. Similarly, from $y(u) = 2u w^{-1}$:

$$y'(u) = \frac{2w - 4u^2}{w^2} = \frac{2(1 - u^2)}{w^2}.$$

At $u = 1$, where $w = 2$ and $w^2 = 4$: $x'(1) = -1$, $y'(1) = 0$.

Step 2 — second derivatives. Writing $x'(u) = -4u w^{-2}$ and differentiating by the product rule, with $\frac{d}{du}w^{-2} = -4u w^{-3}$:

$$x''(u) = -4w^{-2} + 16u^2 w^{-3}, \quad x''(1) = -1 + 2 = 1.$$

Similarly, from $y'(u) = 2(1 - u^2)w^{-2}$:

$$y''(u) = -4u w^{-2} - 8u(1 - u^2)w^{-3}, \quad y''(1) = -1 - 0 = -1.$$

Step 3 — curvature. By Definition 1.10:

$$\kappa(1) = \frac{(-1)(-1) - (0)(1)}{(1 + 0)^{3/2}} = 1.$$

Step 4 — consistency. The rational parametrisation has S^1 as its trace, of radius $R = 1$; Theorem 1.13 predicts $|\kappa| = 1/R = 1$. ✓ The positive sign indicates counterclockwise travel: indeed $p(u)$ goes from $(1, 0)$ at $u = 0$ to $(0, 1)$ at $u = 1$ to $(-1, 0)$ as $u \rightarrow +\infty$. Note that the parametrisation is *not* by arc length — at $u = 1$ one happens to have $|p'(1)| = 1$, while in general $|p'(u)| = 2/(1 + u^2)$ varies — and yet the curvature comes out right: this is the invariance proved in Exercise S2.

Solution. N8. *Step 1 — derivatives of the phase.* From $\theta(t) = \pi t^2/8$:

$$\omega(t) = \frac{\pi t}{4}, \quad \alpha(t) = \frac{\pi}{4} \text{ (constant)}.$$

Step 2 — the two components. By Theorem 1.15, with $R = 0.75$ m:

$$a_t = R\alpha = 0.75 \cdot \frac{\pi}{4} = \frac{3\pi}{16} = 0.589049 \text{ m/s}^2,$$

$$a_c(t) = R\omega(t)^2 = 0.75 \cdot \frac{\pi^2 t^2}{16} = \frac{3\pi^2 t^2}{64}.$$

Step 3 — equality. Imposing $a_c(t) = a_t$:

$$\frac{3\pi^2 t^2}{64} = \frac{3\pi}{16} \implies \pi t^2 = 4 \implies t^2 = \frac{4}{\pi} \implies t = \frac{2}{\sqrt{\pi}} = 1.12838 \text{ s}.$$

Note that R cancels: the instant of balance between the two components does not depend on the radius, only on the phase law. Check: at that instant $\omega = \pi t/4 = 0.886227$ rad/s, so $\omega^2 = 0.785398 = \pi/4$, and $a_c = 0.75 \times 0.785398 = 0.589049 \text{ m/s}^2 = a_t$. ✓

Step 4 — total acceleration. The components are orthogonal and of equal magnitude, hence

$$|a| = \sqrt{a_t^2 + a_c^2} = a_t \sqrt{2} = 0.589049 \times 1.414214 = 0.833025 \text{ m/s}^2,$$

and the acceleration makes an angle of exactly 45° with the radial direction.

Solution. N9. *Method 1 — parametrisation.*

Step 1. The constraint is the circle of centre $(1, -2)$ and radius 5; parametrise it as $x = 1 + 5 \cos \theta$, $y = -2 + 5 \sin \theta$.

Step 2. Substituting into the objective function:

$$h(\theta) = 3 + 15 \cos \theta - 8 + 20 \sin \theta = -5 + 15 \cos \theta + 20 \sin \theta.$$

Step 3 — reduction to a single sinusoid. An expression $A \cos \theta + B \sin \theta$ can be written as $\sqrt{A^2 + B^2} \cos(\theta - \phi)$ with $\cos \phi = A/\sqrt{A^2 + B^2}$ and $\sin \phi = B/\sqrt{A^2 + B^2}$ (Volume I, Chapter 8). Here $\sqrt{225 + 400} = 25$, so $h(\theta) = -5 + 25 \cos(\theta - \phi)$ with $\cos \phi = \frac{3}{5}$, $\sin \phi = \frac{4}{5}$.

Step 4 — extrema. Since $\cos$ ranges over $[-1, 1]$,

$$h_{\min} = -30, \quad h_{\max} = 20.$$

The maximum occurs at $\theta = \phi$, i.e. $\cos \theta = \frac{3}{5}$, $\sin \theta = \frac{4}{5}$:

$$P_{\max} = (1 + 3, -2 + 4) = (4, 2),$$

and the minimum at $\theta = \phi + \pi$: $P_{\min} = (-2, -6)$.

Method 2 — Lagrange multipliers.

Step 1. $\nabla f = (3, 4)$; $G(x, y) = (x - 1)^2 + (y + 2)^2 - 25$, $\nabla G = (2(x - 1), 2(y + 2))$. The condition $\nabla f = \lambda \nabla G$ of Theorem 1.17 gives

$$3 = 2\lambda(x - 1), \quad 4 = 2\lambda(y + 2),$$

so necessarily $\lambda \neq 0$ and

$$x - 1 = \frac{3}{2\lambda}, \quad y + 2 = \frac{2}{\lambda}.$$

Step 2 — imposing the constraint.

$$\frac{9}{4\lambda^2} + \frac{4}{\lambda^2} = 25 \implies \frac{25}{4\lambda^2} = 25 \implies \lambda = \pm \frac{1}{2}.$$

Step 3 — the two points. For $\lambda = \frac{1}{2}$: $(4, 2)$, with $f = 20$. For $\lambda = -\frac{1}{2}$: $(-2, -6)$, with $f = -30$.

Step 4 — conclusion. The two methods give the same points and the same values: maximum 20 at $(4, 2)$, minimum -30 at $(-2, -6)$. ✓

Geometric remark. The level lines of f are the parallel lines $3x + 4y = c$; the maximum and minimum occur where one of them is *tangent* to the circle. Consistently, the distance from the centre $(1, -2)$ to the line $3x + 4y = 20$ is $|3 - 8 - 20|/5 = 5 = R$. ✓

Solution. N10. *Step 1 — reading the block.* G03 means **counterclockwise** circular interpolation; X22 Y17 is the end point $P_1 = (22, 17)$; I12 J0 are the offsets of the centre relative to $P_0 = (10, 5)$; F300 is the feed rate in mm/min.

Step 2 — centre and radius.

$$C_0 = (22, 5), \quad R = \sqrt{144} = 12 \text{ mm}.$$

Step 3 — consistency check. $|C_0 - P_1| = |(0, -12)| = 12 = R$. ✓ The data are consistent: no alarm.

Step 4 — phases of the two endpoints.

$$P_0 - C_0 = (-12, 0) \implies \varphi_0 = 180^\circ, \quad P_1 - C_0 = (0, 12) \implies \varphi_1 = 90^\circ.$$

Step 5 — swept angle. Since this is **G03** (counterclockwise), take $(\varphi_1 - \varphi_0)$ reduced to $[0, 2\pi)$:

$$\Delta\varphi = (90^\circ - 180^\circ) \bmod 360^\circ = 270^\circ = \frac{3\pi}{2} = 4.712389 \text{ rad.}$$

The direction is right: starting from 180° and increasing, one passes 270° , $360^\circ \equiv 0^\circ$ and reaches 90° , for a total of 270° .

Step 6 — arc length.

$$L = R \Delta\varphi = 12 \times \frac{3\pi}{2} = 18\pi \approx 56.549 \text{ mm.}$$

Check: the full circle is $2\pi \cdot 12 = 75.398$ mm, and 270° is three quarters of it: $0.75 \times 75.398 = 56.549$ mm. ✓

Step 7 — machining time. With $F = 300$ mm/min = 5 mm/s:

$$t = \frac{56.549}{300} \text{ min} = 0.18850 \text{ min} = 11.31 \text{ s.}$$

Step 8 — equivalent in the R method. The distance between the endpoints is

$$d = |(12, 12)| = 12\sqrt{2} = 16.9706 \text{ mm,}$$

and the existence condition $|R| \geq d/2 = 8.4853$ is satisfied by $|R| = 12$. Since the arc sweeps $270^\circ > 180^\circ$, it is the *major* arc: by the convention of Observation 1.17 a negative sign is required. The equivalent block is

G03 X22 Y17 R-12 F300

Centre check with Theorem 1.22: $M = \frac{1}{2}(P_0 + P_1) = (16, 11)$, $h = \sqrt{144 - (8.4853)^2} = \sqrt{144 - 72} = \sqrt{72} = 8.4853$; a unit vector orthogonal to $P_1 - P_0 = (12, 12)$ is $\hat{n} = \frac{1}{\sqrt{2}}(1, -1)$. Hence

$$C_+ = (16, 11) + 8.4853 \cdot \frac{1}{\sqrt{2}}(1, -1) = (16 + 6, 11 - 6) = (22, 5). \checkmark$$

1.18.2 Solutions to the symbolic exercises

Solution. S1. Set $\psi(t) = |\gamma(t) - C_0|^2 = (\gamma(t) - C_0) \cdot (\gamma(t) - C_0)$. Since $\gamma \in C^1$, so is ψ , and differentiating the (bilinear, symmetric) dot product:

$$\psi'(t) = 2(\gamma(t) - C_0) \cdot \gamma'(t).$$

($\Rightarrow$) If $|\gamma - C_0|$ is constant, so is its square ψ ; the derivative of a constant is zero, hence $\psi' \equiv 0$ and, dividing by 2, $(\gamma - C_0) \cdot \gamma' \equiv 0$.

($\Leftarrow$) If $(\gamma - C_0) \cdot \gamma' \equiv 0$ then $\psi' \equiv 0$ on I ; a differentiable function with zero derivative on an interval is constant (by the mean value theorem). Hence $\psi \equiv k \geq 0$, and since the square root is injective on the non-negatives, $|\gamma - C_0| \equiv \sqrt{k}$ is constant.

Consequence. For a circular motion $|\gamma(t) - C_0| = R$ is constant by construction, hence $\gamma'(t)$ is orthogonal to the radius vector $\gamma(t) - C_0$; the plane being two-dimensional, the direction orthogonal to the radius is exactly the direction of the tangent line (Theorem 1.8). The velocity is therefore always tangential.

Solution. S2. Write $\delta(u) = \gamma(\varphi(u))$, omitting the argument $\varphi(u)$ where unambiguous. By the chain and product rules,

$$\delta' = \varphi' \gamma'(\varphi), \quad \delta'' = \varphi'' \gamma'(\varphi) + (\varphi')^2 \gamma''(\varphi).$$

Numerator. Denoting by x, y the components of γ ,

$$\delta_1' \delta_2'' - \delta_2' \delta_1'' = \varphi' \varphi'' [x' y' - y' x'] + (\varphi')^3 [x' y'' - y' x''] = (\varphi')^3 [x' y'' - y' x''],$$

since the first bracket vanishes identically. This is the crucial point: the term in φ'' — the only one that would depend on how the traversal speed varies — cancels.

Denominator. From $|\delta'| = |\varphi'| |\gamma'(\varphi)|$,

$$(\delta_1'^2 + \delta_2'^2)^{3/2} = |\varphi'|^3 |\gamma'(\varphi)|^3.$$

Ratio.

$$\kappa_\delta(u) = \frac{(\varphi')^3}{|\varphi'|^3} \kappa_\gamma(\varphi) = \operatorname{sgn}(\varphi') \kappa_\gamma(\varphi),$$

since $t^3/|t|^3 = \operatorname{sgn}(t)$ for $t \neq 0$. In particular $|\kappa_\delta| = |\kappa_\gamma \circ \varphi|$: the unsigned curvature is a geometric invariant of the trace, while its sign depends on the chosen orientation. This justifies a posteriori the computation of Exercise N7.

Solution. S3. By definition of the unit tangent, $\gamma'(t) = v(t)T(t)$ with $v = |\gamma'| > 0$. Differentiating by the product rule:

$$\gamma''(t) = v'(t)T(t) + v(t)\frac{dT}{dt}(t).$$

To compute dT/dt pass to the natural parameter: since $ds/dt = |\gamma'| = v$, the chain rule gives

$$\frac{dT}{dt} = \frac{dT}{ds} \cdot \frac{ds}{dt} = (\kappa N)v,$$

using the first Frenet formula (Theorem 1.12). Substituting:

$$\gamma''(t) = v'(t)T(t) + \kappa(t)v(t)^2N(t).$$

The two components are orthogonal, hence $|\gamma''|^2 = (v')^2 + \kappa^2 v^4$.

Check on the circle. Take $\omega > 0$ (counterclockwise); by Theorem 1.2, $\gamma' = R\omega e_\vartheta$, so $v = R\omega$ and $T = e_\vartheta$. Then

$$N = JT = J(-\sin \theta, \cos \theta) = (-\cos \theta, -\sin \theta) = -e_r,$$

and $\kappa = 1/R$ by Theorem 1.13. Substituting, with $v' = R\alpha$:

$$\gamma'' = R\alpha e_\vartheta + \frac{(R\omega)^2}{R}(-e_r) = R\alpha e_\vartheta - R\omega^2 e_r,$$

exactly Theorem 1.15. ✓ The term $\kappa v^2 = v^2/\rho$ is the centripetal acceleration in the general form valid for any curve: the radius of the osculating circle takes the place of the radius of the circle.

Solution. S4. *Existence.* By Theorem 1.3(iv), $M(\theta) = Q(\theta)$ satisfies $M' = JM$, and $Q(0) = I$.

Uniqueness. Let M be any solution of the Cauchy problem and set, as suggested,

$$W(\theta) = Q(-\theta) M(\theta).$$

Differentiate by the product rule for matrices. By the chain rule and Theorem 1.3(iv),

$$\frac{d}{d\theta}Q(-\theta) = -JQ(-\theta).$$

Hence

$$W'(\theta) = -JQ(-\theta)M(\theta) + Q(-\theta)JM(\theta).$$

Theorem 1.3(iv) also states that J commutes with every Q : $JQ(\alpha) = Q(\alpha)J$. Applying this to the second term, $Q(-\theta)J = JQ(-\theta)$, and the two terms cancel:

$$W'(\theta) = 0.$$

A matrix-valued function with identically zero derivative on an interval is constant (each entry is), hence

$$W(\theta) \equiv W(0) = Q(0)M(0) = I.$$

Multiplying on the left by $Q(\theta)$ and using $Q(\theta)Q(-\theta) = I$ (Theorem 1.3(i)):

$$M(\theta) = Q(\theta).$$

The solution is therefore unique and equals the rotation matrix. This is the rigorous justification of the notation $Q(\theta) = e^{J\theta}$ used in the variant proof of Theorem 1.14.

Solution. S5. *Step 1 — p maps rationals to rational points.* The components of $p(u)$ are rational functions with integer coefficients and nowhere-vanishing denominator: if $u \in \mathbb{Q}$, both coordinates are rational.

Step 2 — and conversely. If $(x, y) \in S^1$ has rational coordinates and $(x, y) \neq (-1, 0)$, then $u = y/(1+x)$ is rational, and by Theorem 1.7 $p(u) = (x, y)$. Injectivity of p completes the bijection between $\mathbb{Q}$ and the rational points of $S^1 \setminus \{(-1, 0)\}$.

Step 3 — from the triple to the parameter. Let (a, b, c) be a primitive Pythagorean triple ($a^2 + b^2 = c^2$, $\gcd(a, b, c) = 1$, all positive) with b even. Then a and c are odd: if a were even, $c^2 = a^2 + b^2$ would be even, hence c even, and 2 would divide all three, contradicting primitivity. The point $(a/c, b/c)$ lies on S^1 , has rational coordinates and differs from $(-1, 0)$ (since $a/c > 0$). Its parameter is

$$u = \frac{b/c}{1 + a/c} = \frac{b}{c + a}.$$

Write $u = m/n$ in lowest terms, with $n > 0$ and $\gcd(m, n) = 1$; since $u > 0$ we have $m \geq 1$, and since $u = \tan(\theta/2)$ with θ in the first quadrant ($a, b > 0$) we have $\theta < \pi/2$, hence $u < 1$, i.e. $m < n$.

Step 4 — parity. Suppose for contradiction that m, n are both odd (they cannot both be even, being coprime). Then $n^2 + m^2 \equiv 2 \pmod{4}$ and $n^2 - m^2 \equiv 0 \pmod{8}$. From

$$\frac{a}{c} = \frac{n^2 - m^2}{n^2 + m^2}$$

one could cancel a factor 2, obtaining a fraction with even numerator and odd denominator; but a/c is already in lowest terms (since $\gcd(a, c) = 1$: a common prime would also divide $b^2 = c^2 - a^2$, against primitivity) and has odd numerator a . Contradiction: m and n have opposite parity.

Step 5 — the fraction is in lowest terms. With m, n coprime of opposite parity, $n^2 + m^2$ is odd. Let $d = \gcd(n^2 - m^2, n^2 + m^2)$: then d divides the sum $2n^2$ and the difference $2m^2$,

hence $d \mid 2\gcd(n, m)^2 = 2$; being odd, $d = 1$. Similarly $\gcd(2mn, n^2 + m^2) = 1$: an odd prime q dividing $n^2 + m^2$ and mn would divide one of m, n and hence the other, against coprimality.

Step 6 — conclusion. Since $a/c = (n^2 - m^2)/(n^2 + m^2)$ and $b/c = 2mn/(n^2 + m^2)$ are equalities between fractions both in lowest terms with positive denominators,

$$a = n^2 - m^2, \quad b = 2mn, \quad c = n^2 + m^2,$$

with $n > m \geq 1$ coprime of opposite parity. Uniqueness of the representation follows from the injectivity of p (Theorem 1.7) and the uniqueness of the reduced form of a positive rational. This recovers, by purely geometric means, Euclid's parametrisation (*Elements*, Book X, Lemma 1; cf. Volume I, Chapter 9).

Solution. S6. *Length.* Setting $\delta(t) = \Phi(\gamma(t)) = Q\gamma(t) + b$, the derivative is $\delta'(t) = Q\gamma'(t)$ (the constant translation disappears, the linear part remains). Since Q is orthogonal, $Q^\top Q = I$, and for every vector w

$$|Qw|^2 = w^\top Q^\top Q w = |w|^2.$$

Hence $|\delta'(t)| = |\gamma'(t)|$ for every t , and integrating, $L(\delta) = L(\gamma)$.

Curvature. Note that the numerator $x'y'' - y'x''$ is the determinant of the 2×2 matrix with columns γ' and γ'' :

$$x'y'' - y'x'' = \det [\gamma' \mid \gamma''].$$

From $\delta' = Q\gamma'$ and $\delta'' = Q\gamma''$ we get $[\delta' \mid \delta''] = Q[\gamma' \mid \gamma'']$, and by the Binet–Cauchy formula

$$\det [\delta' \mid \delta''] = \det Q \cdot \det [\gamma' \mid \gamma''].$$

The denominator $|\delta'|^3 = |\gamma'|^3$ is unchanged. Hence

$$\kappa_{\Phi \circ \gamma} = (\det Q) \kappa_\gamma.$$

Since Q is orthogonal, $\det Q = \pm 1$: **direct isometries** (rotations and translations, $\det Q = +1$) preserve the signed curvature; **indirect isometries** (reflections, $\det Q = -1$) reverse its sign, consistently with the fact that they exchange clockwise and counterclockwise. In either case $|\kappa|$ is invariant: it is a *metric* quantity.

Solution. S7. *Step 1 — the integral vanishes.* Let $\tilde{\gamma} : [0, L] \rightarrow \Gamma$ be the natural parametrisation, with $L = 2\pi R$, and $g(s) = f(\tilde{\gamma}(s))$. By Definition 1.11, $\partial_T f(\tilde{\gamma}(s)) = g'(s)$, hence by the fundamental theorem of calculus

$$\oint_{\Gamma} \partial_T f \, ds = \int_0^L g'(s) \, ds = g(L) - g(0).$$

But the curve is closed: $\tilde{\gamma}(L) = \tilde{\gamma}(0)$, so $g(L) = g(0)$ and the integral is 0.

Step 2 — interpretation. The result says that the tangential derivative of a function has *zero mean* on the circle: going once around, everything gained is lost again. It is the simplest instance of the fact that a gradient field has zero circulation along every closed curve, which in Chapter 2 becomes the fundamental theorem of line integrals.

Step 3 — at least two zeros. The function g is continuous on the compact $[0, L]$, so by Weierstrass' theorem it attains an absolute maximum at some s_M and an absolute minimum at some s_m . Extend g periodically to $\mathbb{R}$ (legitimate since $g(L) = g(0)$; the extension is still C^1 because $f \in C^1$ and Γ is smooth): then s_M and s_m are *interior* extremum points for the extended function, and by Fermat's theorem $g'(s_M) = g'(s_m) = 0$.

If g is not constant, then $g(s_M) \neq g(s_m)$, hence $s_M \neq s_m$ and the two points give two distinct zeros of $\partial_T f$ on Γ . If g is constant, $\partial_T f \equiv 0$ and there are infinitely many zeros. In either case there are at least two points where $\partial_T f = 0$, i.e. — by Theorem 1.17 — at least two constrained critical points: the maximum and the minimum of f on Γ .

Solution. S8. *Step 1 — Cauchy–Schwarz.* In the space of continuous functions on $[a, b]$ with the inner product $\langle \phi, \psi \rangle = \int_a^b \phi \psi \, dt$, the Cauchy–Schwarz inequality reads

$$\left(\int_a^b \phi \psi \, dt \right)^2 \leq \left(\int_a^b \phi^2 \, dt \right) \left(\int_a^b \psi^2 \, dt \right).$$

Apply it with $\phi \equiv 1$ and $\psi = |\gamma'|$ (both continuous):

$$L^2 = \left(\int_a^b |\gamma'(t)| \, dt \right)^2 \leq (b-a) \int_a^b |\gamma'(t)|^2 \, dt.$$

Step 2 — case of equality. Equality holds in Cauchy–Schwarz iff ϕ and ψ are linearly dependent, i.e. iff $|\gamma'(t)| = c$ for every t (the case $\phi = 0$ is excluded, as $\phi \equiv 1$). Since γ is regular, $c > 0$. Hence equality holds exactly for parametrisations of constant speed.

Step 3 — variational interpretation. Fix the arc (trace with its orientation) and the interval $[a, b]$. By Theorem 1.10 the length L is the same for all regular parametrisations of the arc on $[a, b]$. The inequality then reads

$$\int_a^b |\gamma'(t)|^2 \, dt \geq \frac{L^2}{b-a},$$

where the right-hand side does not depend on the parametrisation: the **energy** $\int |\gamma'|^2$ is minimised, and the minimum $L^2/(b-a)$ is attained *only* by the constant-speed parametrisation $|\gamma'| \equiv L/(b-a)$. For the circle this is the parametrisation with constant angular velocity — which explains why in robotics constant-speed profiles are the most economical for a given path and time, and why the natural parametrisation is *the* canonical choice in differential geometry.

Solution. S9. *Step 1 — the tangential gradient depends only on the restriction.* By Theorem 1.16(iii),

$$\nabla_\Gamma f(P) = (\partial_T f(P)) T(P),$$

and by Definition 1.11 $\partial_T f(\tilde{\gamma}(s)) = \frac{d}{ds} f(\tilde{\gamma}(s))$: this derivative is computed from the values that f takes *on* Γ alone. If $f_1 = f_2$ on Γ , then $f_1 \circ \tilde{\gamma} = f_2 \circ \tilde{\gamma}$ as functions of s , hence they have the same derivative, hence $\partial_T f_1 = \partial_T f_2$ and finally $\nabla_\Gamma f_1 = \nabla_\Gamma f_2$ at every point of Γ .

Step 2 — alternative proof, by projection. The difference $h = f_1 - f_2$ vanishes on Γ . Differentiating $h(\tilde{\gamma}(s)) \equiv 0$ gives $\nabla h(P) \cdot T(P) = 0$, i.e. $\nabla h(P)$ is parallel to $n(P)$: $\nabla h = \mu n$ for some scalar μ . But then by Theorem 1.16(ii),

$$\nabla_\Gamma h = \Pi \nabla h = \mu \Pi n = 0,$$

and by linearity $\nabla_\Gamma f_1 - \nabla_\Gamma f_2 = 0$.

Step 3 — explicit example. Take $\Gamma = S^1$ (so $C_0 = 0$, $R = 1$, $n(x, y) = (x, y)$) and

$$f_1(x, y) = 0, \quad f_2(x, y) = x^2 + y^2 - 1.$$

On S^1 we have $x^2 + y^2 = 1$, so $f_2 \equiv 0 = f_1$: the two extensions agree on the circle. However

$$\nabla f_1 = (0, 0), \quad \nabla f_2(x, y) = (2x, 2y) = 2n,$$

which on S^1 has modulus 2 $\neq 0$: the *ambient* gradients differ at every point of Γ . The *tangential* gradients, on the other hand, agree:

$$\nabla_{\Gamma} f_2 = \Pi(2n) = 0 = \nabla_{\Gamma} f_1. \checkmark$$

Moral. The tangential gradient is the intrinsic object — the one that “lives” on the curve — whereas the ordinary gradient also encodes the behaviour of f in the transverse direction, which depends on the chosen extension and has no meaning for the geometry of Γ . This is exactly why Theorem 1.17 characterises constrained critical points through $\nabla_{\Gamma} f = 0$ and not through $\nabla f = 0$ (a condition which, on a constraint, is in general never satisfied).

Solution. S10. Let $E : \mathbb{R} \rightarrow S^1$, $E(\theta) = (\cos \theta, \sin \theta)$, and let $U = E((\alpha, \beta))$ with $\beta - \alpha \leq 2\pi$.

Step 1 — E is injective on (α, β) . If $E(\theta_1) = E(\theta_2)$ with $\theta_1, \theta_2 \in (\alpha, \beta)$, then (as in Proposition 1.1(i)) $\theta_2 - \theta_1 \in 2\pi\mathbb{Z}$; but $|\theta_2 - \theta_1| < \beta - \alpha \leq 2\pi$, and the only multiple of 2π of modulus strictly less than 2π is 0. Hence $\theta_1 = \theta_2$.

Step 2 — definition of Θ . Set $\Theta = (E|_{(\alpha, \beta)})^{-1} : U \rightarrow (\alpha, \beta)$, well defined by Step 1. By construction $E(\Theta(P)) = P$, i.e. $(\cos \Theta(P), \sin \Theta(P)) = P$.

Step 3 — continuity of Θ . Let $P_0 \in U$ and $\theta_0 = \Theta(P_0) \in (\alpha, \beta)$. Choose $\varepsilon > 0$ with $K = [\theta_0 - \varepsilon, \theta_0 + \varepsilon] \subset (\alpha, \beta)$. The restriction $E|_K$ is a continuous bijection from the compact K into the Hausdorff space S^1 ; such a map is a homeomorphism onto its image (it is closed, because closed subsets of a compact set are compact, and continuous images of compact sets are compact, hence closed in a Hausdorff space). Therefore $\Theta = (E|_K)^{-1}$ is continuous on $E(K)$, a neighbourhood of P_0 in U . Since P_0 was arbitrary, Θ is continuous on U .

Step 4 — uniqueness up to $2\pi\mathbb{Z}$. Let $\Theta_1, \Theta_2 : U \rightarrow \mathbb{R}$ be continuous with $E \circ \Theta_i = \text{id}_U$. Then, by the same argument as in Step 1, $\Theta_1(P) - \Theta_2(P) \in 2\pi\mathbb{Z}$ for every P . The continuous function $\Theta_1 - \Theta_2$ therefore has connected image (since U , the continuous image of the interval (α, β) , is connected) contained in the discrete set $2\pi\mathbb{Z}$: it is constant. Hence $\Theta_1 = \Theta_2 + 2\pi k$ for a fixed $k \in \mathbb{Z}$.

Step 5 — two charts covering S^1 . Take

$$U_1 = E((-\pi, \pi)) = S^1 \setminus \{(-1, 0)\}, \quad \Theta_1(x, y) = \text{atan2}(y, x) \in (-\pi, \pi),$$

$$U_2 = E((0, 2\pi)) = S^1 \setminus \{(1, 0)\}, \quad \Theta_2(x, y) = \text{atan2}(y, x) \bmod 2\pi \in (0, 2\pi).$$

Since $(-1, 0) \neq (1, 0)$, we have $U_1 \cup U_2 = S^1$: two charts suffice, and by Theorem 1.19 one would never be enough. On their intersection $U_1 \cap U_2 = S^1 \setminus \{(1, 0), (-1, 0)\}$, which has *two* connected components (the upper and the lower half-plane), the two charts differ by 0 on the upper component and by 2π on the lower: the constant of Step 4 is the same only component by component, as it must be.

1.19. Self-assessment quiz

Twenty multiple-choice questions covering all sections of the chapter. The answer key, with a one-line justification for each, is at the end of the section: it is advisable to answer all questions before consulting it.

1. Which of the following is *not* a parametrisation of the unit circle?

A) $(\cos t, \sin t)$

B) $(\sin t, \cos t)$

-
- C) $(\cos t, \sin 2t)$
D) $(\cos(t + \pi/3), \sin(t + \pi/3))$
2. The *trace* of a parametrised curve is
A) the map γ itself
B) the image set $\gamma(I)$
C) the graph of γ in $\mathbb{R}^3$
D) the tangent vector
3. For the circular motion $\gamma = C_0 + R(\cos \theta(t), \sin \theta(t))$, the speed is
A) $R\theta(t)$
B) $R|\theta'(t)|$
C) $|\theta'(t)|$
D) $R^2\theta'(t)$
4. The circular motion is regular at t_0 if and only if
A) $\theta(t_0) \neq 0$
B) $\theta'(t_0) \neq 0$
C) $\theta''(t_0) \neq 0$
D) $R \neq 1$
5. The winding number of a closed circular curve is
A) always 1
B) always an integer
C) an invariant of the trace
D) equal to $L/(2\pi)$
6. For the rotation matrix, $Q'(\theta) = ?$
A) $Q(\theta)^T$
B) $-Q(\theta)$
C) $JQ(\theta)$, with $J^2 = -I$
D) $Q(\theta)^2$
7. In the complex form $z(t) = z_c + Re^{i\theta(t)}$, the derivative $z'(t)$ equals
A) $\omega(z - z_c)$
B) $i\omega(z - z_c)$
C) $i(z - z_c)$
D) $-\omega^2(z - z_c)$
8. An admissible reparametrisation preserves
A) the trace but not necessarily the speed

-
- B) the speed but not the trace
 C) both
 D) neither
9. The curve $\gamma(t) = (\cos t^3, \sin t^3)$ on $[-2, 2]$
 A) has a singular trace at $t = 0$
 B) is not regular at $t = 0$, although its trace is smooth
 C) is undefined at $t = 0$
 D) is regular everywhere
10. The stereographic projection $p(u) = ((1 - u^2)/(1 + u^2), 2u/(1 + u^2))$ from the pole $N = (-1, 0)$
 A) covers all of S^1
 B) covers S^1 except N
 C) covers only the upper semicircle
 D) does not take values in S^1
11. The Weierstrass substitution uses
 A) $u = \tan \theta$
 B) $u = \tan(\theta/2)$
 C) $u = \sin(\theta/2)$
 D) $u = \cos \theta$
12. The equation of the tangent line at P to $C(C_0, R)$ is
 A) $(P - C_0) \cdot (X - P) = 0$
 B) $(P - C_0) \cdot (X - C_0) = 0$
 C) $|X - P| = R$
 D) $(X - P) \cdot (X - C_0) = R^2$
13. The length of an arc of central angle $\Delta\theta$ (in radians) on a circle of radius R is
 A) $R^2 \Delta\theta/2$
 B) $R \Delta\theta$
 C) $2\pi R \Delta\theta$
 D) $\Delta\theta/R$
14. The natural parametrisation of $C(C_0, R)$ is
 A) $C_0 + R(\cos s, \sin s)$
 B) $C_0 + R(\cos(s/R), \sin(s/R))$
 C) $C_0 + R(\cos(Rs), \sin(Rs))$
 D) $C_0 + (\cos s, \sin s)$
15. The signed curvature of a circle of radius R traversed clockwise equals
-

-
- A) $1/R$
 B) $-1/R$
 C) R
 D) 0
16. A regular C^2 plane curve with constant curvature $\kappa_0 \neq 0$ is necessarily
 A) a straight line
 B) an arc of a circle of radius $1/|\kappa_0|$
 C) a spiral
 D) a parabola
17. In a non-uniform circular motion, the centripetal component of the acceleration equals
 A) $R\alpha$
 B) $R\omega^2 = v^2/R$
 C) $R\omega$
 D) $v^2\alpha$
18. The tangential gradient $\nabla_\Gamma f = (I - nn^\top)\nabla f$
 A) depends on the extension of f off Γ
 B) depends only on the values of f on Γ
 C) always coincides with ∇f
 D) is always zero
19. Theorem 1.19 states that
 A) no continuous function exists on S^1
 B) no globally defined continuous angle exists on S^1
 C) S^1 is not connected
 D) `atan2` is ill defined
20. In the G-code R method, if the arc sweeps more than 180° one must
 A) specify R positive
 B) specify R negative
 C) double R
 D) necessarily use G02

BEST PRACTICE

Answer key.

1-C: $(\cos t, \sin 2t)$ does not satisfy $x^2 + y^2 = 1$ (it is a Lissajous figure). **2-B:** Definition 1.1.
3-B: Theorem 1.2(ii). **4-B:** Theorem 1.2(iii): $\gamma' = R\omega e_\theta$ vanishes only with ω . **5-B:**
 Proposition 1.1(i); it is *not* an invariant of the trace (Observation 1.4), which rules out
 C. **6-C:** Theorem 1.3(iii)–(iv). **7-B:** Theorem 1.4; the factor i rotates by 90° . **8-A:**
 Theorem 1.6(i) and (iv). **9-B:** Example 1.5: $\gamma'(0) = (0, 0)$ but the trace is S^1 . **10-B:**

Theorem 1.7, Step 3. **11-B:** Corollary 1.2. **12-A:** Theorem 1.8; B is the line through C_0 orthogonal to the radius, which does not pass through P . **13-B:** Theorem 1.9. **14-B:** Corollary 1.4; only this choice gives $|\tilde{\gamma}'| = 1$. **15-B:** Theorem 1.13: the sign follows the direction. **16-B:** Theorem 1.14; A holds for $\kappa_0 = 0$. **17-B:** Theorem 1.15(i); A is the *tangential* component. **18-B:** Exercise S9, Step 1: it depends only on $\partial_T f$. **19-B:** the statement of Theorem 1.19; continuous functions on S^1 certainly exist, and S^1 is connected. **20-B:** Observation 1.17.

1.20. Chapter formulary

Parametrisations

$$\gamma(\theta) = C_0 + R(\cos \theta, \sin \theta), \quad \theta \in [0, 2\pi)$$

$$\tilde{\gamma}(s) = C_0 + R\left(\cos \frac{s}{R}, \sin \frac{s}{R}\right) \quad (\text{natural, } |\tilde{\gamma}'| = 1)$$

$$p(u) = \left(\frac{1-u^2}{1+u^2}, \frac{2u}{1+u^2} \right), \quad u = \tan \frac{\theta}{2}, \quad d\theta = \frac{2 du}{1+u^2}$$

Kinematics $(e_r = (\cos \theta, \sin \theta), e_\vartheta = (-\sin \theta, \cos \theta), \omega = \theta', \alpha = \theta'')$

$$\gamma' = R\omega e_\vartheta, \quad v = R|\omega|$$

$$\gamma'' = R\alpha e_\vartheta - R\omega^2 e_r, \quad a_t = R|\alpha| = |v'|, \quad a_c = R\omega^2 = \frac{v^2}{R}$$

$$\gamma'' = v' T + \kappa v^2 N \quad (\text{arbitrary curve})$$

Length and curvature

$$L = \int_a^b |\gamma'| dt = R |\Delta\theta|, \quad \kappa = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}} = \frac{\varepsilon}{R}$$

$$T' = \kappa N, \quad N' = -\kappa T, \quad \kappa = \frac{d\varphi}{ds}$$

Rotations

$$Q(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad Q(\alpha)Q(\beta) = Q(\alpha + \beta), \quad Q' = JQ, \quad J^2 = -I$$

$$z = z_c + Re^{i\theta}, \quad z' = i\omega(z - z_c)$$

Functions on the circle

$$\partial_T f = \nabla f \cdot T = \frac{1}{R} \frac{dh}{d\theta}, \quad \nabla_\Gamma f = (I - nn^\top) \nabla f, \quad n = \frac{P - C_0}{R}$$

$$\text{constrained critical point} \iff \nabla f \parallel n \iff \nabla f = \lambda \nabla G$$

Circular interpolation

$$\text{IJK: } C_0 = P_0 + (I, J), \quad R = \sqrt{I^2 + J^2}; \quad \text{R: } C_\pm = M \pm \sqrt{R^2 - \frac{d^2}{4}} \hat{n}$$

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CHAPTER 2

Integral Calculus on the Circle

Chapter 1 viewed the circle as an object of *differential* calculus: velocity, curvature, tangent. This chapter views it as a *domain of integration*, and does so in two distinct ways which are worth keeping apart from the outset:

- the circle as a **curve** over which one integrates — line integrals, work, circulation, flux;
- the circle as the **boundary** of a disc — Green’s theorem, which converts an integral over the boundary into an integral over the enclosed region, and vice versa.

The most important result of the chapter, however, is not a formula but a *counterexample*: the vortex field of Section 2.5, whose curl vanishes identically and yet whose circulation about the origin is 2π . That field shows that the conservativeness of a vector field depends not only on the field, but on the *shape of the domain*, and it closes the circle (so to speak) with Theorem 1.19 of the previous chapter: the missing potential is precisely the angle that cannot be defined continuously on the whole circle. It is the first point in the treatise where topology enters a computation in an essential way.

INFO

Prerequisites. Chapter 1 of this volume (parametrisation, arc length, unit tangent and normal); from integral calculus we assume the Riemann integral in one variable, the double integral over simple domains and Fubini’s theorem.

2.1. Differential elements on the circle

Every integral over a curve reduces, via a parametrisation, to an ordinary integral over an interval. The translation goes through three *differential elements*, which it is worth fixing once and for all.

Definition 2.1: Differential elements of a plane curve. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be regular, $\gamma(t) = (x(t), y(t))$. We set

$$\begin{aligned} \text{element of length:} \quad ds &= |\gamma'(t)| dt = \sqrt{x'(t)^2 + y'(t)^2} dt, \\ \text{tangential vector element:} \quad d\mathbf{r} &= \gamma'(t) dt = (dx, dy), \\ \text{normal vector element:} \quad \mathbf{n} ds &= (y'(t), -x'(t)) dt = (dy, -dx). \end{aligned}$$

Theorem 2.1: Consistency of the differential elements. With the above notation, and with $T = \gamma'/|\gamma'|$ the unit tangent and $\mathbf{n} = (y', -x')/|\gamma'|$:

- (i) $\mathbf{n}$ is a unit vector orthogonal to T , obtained by rotating T by -90° ;
- (ii) $d\mathbf{r} = T ds$, and both $d\mathbf{r}$ and $\mathbf{n} ds$ have modulus ds ;
- (iii) if γ is a Jordan curve traversed counterclockwise, then $\mathbf{n}$ is the **outward** unit normal;

(iv) for the standard parametrisation of $C(C_0, R)$,

$$ds = R d\theta, \quad d\mathbf{r} = R(-\sin \theta, \cos \theta) d\theta, \quad \mathbf{n} ds = R(\cos \theta, \sin \theta) d\theta,$$

and in particular $\mathbf{n} = (P - C_0)/R$.

Proof. (i) $|\mathbf{n}| = \sqrt{y'^2 + x'^2}/|\gamma'| = 1$; moreover

$$\mathbf{n} \cdot T = \frac{y'x' - x'y'}{|\gamma'|^2} = 0.$$

Finally, recalling $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ from Chapter 1, one checks $J(y', -x') = (x', y')$, hence $(y', -x') = J^{-1}(x', y')$: $\mathbf{n}$ is T rotated by -90° .

(ii) $d\mathbf{r} = \gamma' dt = \frac{\gamma'}{|\gamma'|} |\gamma'| dt = T ds$; similarly $(y', -x') dt = \mathbf{n} |\gamma'| dt = \mathbf{n} ds$.

(iii) For the circle this is checked directly in (iv); the general case follows from the fact that for a Jordan curve traversed counterclockwise the enclosed region always lies *to the left* of the tangent vector, so the normal pointing to the right — i.e. T rotated by -90° — points outwards.

(iv) With $\gamma(\theta) = C_0 + R(\cos \theta, \sin \theta)$ we have $\gamma' = R(-\sin \theta, \cos \theta)$ and $|\gamma'| = R$, hence $ds = R d\theta$ and $d\mathbf{r} = R(-\sin \theta, \cos \theta) d\theta$. For the normal, with $x' = -R \sin \theta$ and $y' = R \cos \theta$:

$$(y', -x') d\theta = (R \cos \theta, R \sin \theta) d\theta,$$

and dividing by $ds = R d\theta$ we get $\mathbf{n} = (\cos \theta, \sin \theta) = (P - C_0)/R$, the outward radial unit vector (Chapter 1, Corollary 1.3). $\square$

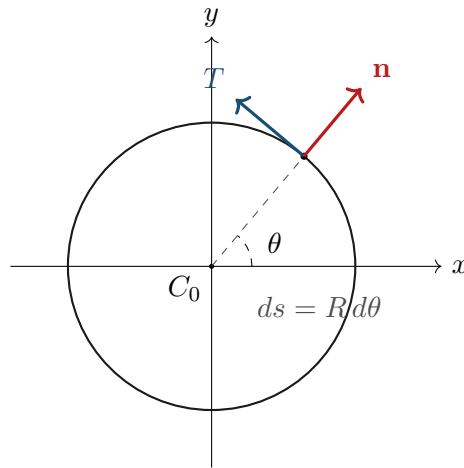


Figure 2.1: The differential elements on the circle traversed counterclockwise: the unit tangent T , the outward unit normal $\mathbf{n}$ (which coincides with the radial unit vector) and the element of length $ds = R d\theta$.

Warning. Orienting $\mathbf{n}$ outwards is a convention, but it is the one that makes every formula in this chapter true with no extra signs (divergence theorem, flux formula, Green's theorem in normal form). Reversing the direction of travel reverses both T and $\mathbf{n}$ relative to the formula $(dy, -dx)$: this is why the orientation must always be stated explicitly before writing a flux integral.

2.2. Line integrals of scalar functions

Definition 2.2: Line integral of a scalar function. Let Γ be a piecewise regular curve with parametrisation $\gamma : [a, b] \rightarrow \mathbb{R}^2$ and let f be continuous on Γ . We set

$$\int_{\Gamma} f \, ds = \int_a^b f(\gamma(t)) |\gamma'(t)| \, dt.$$

If $f \equiv 1$ one recovers the length; if $f = \lambda$ is a linear mass density, the integral is the **total mass** of the wire Γ .

Theorem 2.2: Independence of the parametrisation. The integral $\int_{\Gamma} f \, ds$ does not depend on the chosen regular parametrisation, nor on the direction of travel.

Proof. Let $\delta = \gamma \circ \varphi$ be a reparametrisation, $\varphi : [c, d] \rightarrow [a, b]$. By Chapter 1, Theorem 1.6(ii), $|\delta'(u)| = |\varphi'(u)| |\gamma'(\varphi(u))|$, hence

$$\int_c^d f(\delta(u)) |\delta'(u)| \, du = \int_c^d f(\gamma(\varphi(u))) |\gamma'(\varphi(u))| |\varphi'(u)| \, du.$$

Case $\varphi' > 0$. Then $|\varphi'| = \varphi'$, $\varphi(c) = a$, $\varphi(d) = b$; the substitution $t = \varphi(u)$ turns the integral into $\int_a^b f(\gamma(t)) |\gamma'(t)| \, dt$.

Case $\varphi' < 0$. Then $|\varphi'| = -\varphi'$, $\varphi(c) = b$, $\varphi(d) = a$; the same substitution gives

$$- \int_b^a f(\gamma(t)) |\gamma'(t)| \, dt = + \int_a^b f(\gamma(t)) |\gamma'(t)| \, dt,$$

where the minus sign from swapping the limits cancels the one from $|\varphi'| = -\varphi'$. In both cases the value is the same. $\square$

Remark 2.1: Why ds has no sign. The fact that the scalar integral is unchanged by reversing direction is the essential difference from the integrals of vector fields in the next section: ds is a length, always positive, whereas $d\mathbf{r}$ is a vector, which reverses with the orientation. A mass does not change if one walks along the wire backwards; a work does.

Theorem 2.3: Mass, mean value and centroid of a circular wire. Let Γ be an arc of the circle $C(C_0, R)$ corresponding to $\theta \in [\theta_0, \theta_1]$, and λ a continuous linear density. Then, setting $\Delta\theta = \theta_1 - \theta_0$ and $L = R \Delta\theta$:

(i) the mass is $m = R \int_{\theta_0}^{\theta_1} \lambda(\gamma(\theta)) \, d\theta$;

(ii) the mean value of f on Γ is $\bar{f} = \frac{1}{L} \int_{\Gamma} f \, ds = \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_1} f(\gamma(\theta)) \, d\theta$;

(iii) if λ is constant, the centroid is

$$G = C_0 + \frac{R}{\Delta\theta} (\sin \theta_1 - \sin \theta_0, \cos \theta_0 - \cos \theta_1);$$

(iv) for the whole circle ($\Delta\theta = 2\pi$) with λ constant, $G = C_0$.

Proof. (i) and (ii) follow at once from Definition 2.2 with $ds = R \, d\theta$ (Theorem 2.1(iv)) and from $L = R \Delta\theta$ (Chapter 1, Theorem 1.9).

(iii) The coordinates of the centroid are the mean values of the coordinates. With $C_0 = (x_c, y_c)$ and $x(\theta) = x_c + R \cos \theta$:

$$\bar{x} = \frac{1}{\Delta\theta} \int_{\theta_0}^{\theta_1} (x_c + R \cos \theta) d\theta = x_c + \frac{R(\sin \theta_1 - \sin \theta_0)}{\Delta\theta}.$$

Similarly, with $y(\theta) = y_c + R \sin \theta$ and $\int \sin = -\cos$:

$$\bar{y} = y_c + \frac{R(\cos \theta_0 - \cos \theta_1)}{\Delta\theta}.$$

(iv) For $\theta_1 = \theta_0 + 2\pi$, sine and cosine take the same values at the endpoints, so both differences vanish and $G = C_0$. This agrees with the symmetry: the centre is the only point invariant under all rotations preserving the circle. $\square$

Example 2.1: Centroid of a semicircular wire. Find the centroid of a homogeneous wire in the shape of a semicircle of radius R centred at the origin, corresponding to $\theta \in [0, \pi]$.

Step 1 — length. $L = R \Delta\theta = R\pi$.

Step 2 — abscissa of the centroid. By Theorem 2.3(iii) with $C_0 = (0, 0)$, $\theta_0 = 0$, $\theta_1 = \pi$:

$$\bar{x} = \frac{R}{\pi} (\sin \pi - \sin 0) = 0,$$

as expected by symmetry about the y -axis.

Step 3 — ordinate of the centroid.

$$\bar{y} = \frac{R}{\pi} (\cos 0 - \cos \pi) = \frac{R}{\pi} (1 + 1) = \frac{2R}{\pi} \approx 0.6366 R.$$

Step 4 — direct verification.

$$\bar{y} = \frac{1}{L} \int_{\Gamma} y \, ds = \frac{1}{\pi R} \int_0^{\pi} (R \sin \theta) R \, d\theta = \frac{R}{\pi} [-\cos \theta]_0^{\pi} = \frac{2R}{\pi}. \checkmark$$

Step 5 — plausibility check. The centroid must lie inside the disc but *off* the wire: indeed $0 < 2R/\pi \approx 0.637R < R$. $\checkmark$ Note the contrast with the centroid of the solid *half-disc*, which is $4R/(3\pi) \approx 0.424R$: all the mass of the wire is concentrated on the boundary, so its centroid is further from the centre.

2.3. Work and circulation of a vector field

Definition 2.3: Line integral of a vector field. Let $\mathbf{F} = (P, Q)$ be a continuous vector field on an open set containing the piecewise regular curve Γ , oriented by $\gamma : [a, b] \rightarrow \mathbb{R}^2$. We set

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\gamma(t)) \cdot \gamma'(t) \, dt = \int_{\Gamma} (P \, dx + Q \, dy).$$

If $\mathbf{F}$ is a force field, the integral is the **work** done along Γ ; if Γ is closed it is called the **circulation**, written $\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r}$.

Theorem 2.4: Equivalent forms and behaviour under orientation.

(i) $\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_{\Gamma} (\mathbf{F} \cdot \mathbf{T}) \, ds$: the line integral of a field is the scalar integral of its tangential component;

- (ii) the integral is invariant under orientation-preserving reparametrisations;
 (iii) denoting by $-\Gamma$ the same curve traversed in the opposite direction,

$$\int_{-\Gamma} \mathbf{F} \cdot d\mathbf{r} = - \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}.$$

Proof. (i) By Theorem 2.1(ii), $d\mathbf{r} = T ds$, hence $\mathbf{F} \cdot d\mathbf{r} = (\mathbf{F} \cdot T) ds$. Explicitly,

$$\int_a^b \mathbf{F}(\gamma) \cdot \gamma' dt = \int_a^b \mathbf{F}(\gamma) \cdot \frac{\gamma'}{|\gamma'|} |\gamma'| dt = \int_{\Gamma} (\mathbf{F} \cdot T) ds.$$

(ii) Let $\delta = \gamma \circ \varphi$ with $\varphi' > 0$. Then

$$\int_c^d \mathbf{F}(\delta(u)) \cdot \delta'(u) du = \int_c^d \mathbf{F}(\gamma(\varphi(u))) \cdot \gamma'(\varphi(u)) \varphi'(u) du = \int_a^b \mathbf{F}(\gamma(t)) \cdot \gamma'(t) dt,$$

by the substitution $t = \varphi(u)$.

(iii) Immediate from (i): reversing the direction changes the sign of T but not of ds (Observation 2.1), hence $\mathbf{F} \cdot T$ changes sign and so does the integral. $\square$

Example 2.2: Three fields on the same circle. Compute the circulation of the three fields below along $C(O, R)$ traversed counterclockwise:

$$\mathbf{F}_1(x, y) = (x, y) \text{ (radial)}, \quad \mathbf{F}_2(x, y) = (-y, x) \text{ (rotational)}, \quad \mathbf{F}_3(x, y) = (1, 0) \text{ (uniform)}.$$

Common setup. With $\gamma(\theta) = R(\cos \theta, \sin \theta)$, $\theta \in [0, 2\pi]$, we have $\gamma'(\theta) = R(-\sin \theta, \cos \theta)$.

Radial field $\mathbf{F}_1$. On the curve, $\mathbf{F}_1(\gamma) = R(\cos \theta, \sin \theta)$, hence

$$\mathbf{F}_1 \cdot \gamma' = R^2 [-\cos \theta \sin \theta + \sin \theta \cos \theta] = 0,$$

and the circulation is 0. *Interpretation:* the field is everywhere orthogonal to the tangent (it is radial, and the radius is orthogonal to the tangent by Chapter 1, Theorem 1.8); it does no work.

Rotational field $\mathbf{F}_2$. On the curve, $\mathbf{F}_2(\gamma) = R(-\sin \theta, \cos \theta) = \gamma'(\theta)$, hence

$$\oint \mathbf{F}_2 \cdot d\mathbf{r} = \int_0^{2\pi} R^2 d\theta = 2\pi R^2.$$

Interpretation: the field is everywhere parallel to and concordant with the tangent, of modulus R ; the work is modulus $\times$ length $= R \cdot 2\pi R$. $\checkmark$

Uniform field $\mathbf{F}_3$.

$$\oint \mathbf{F}_3 \cdot d\mathbf{r} = -R \int_0^{2\pi} \sin \theta d\theta = 0.$$

Interpretation: a constant field is the gradient of $\varphi(x, y) = x$, hence conservative, and its circulation on every closed curve vanishes (Theorem 2.5 below).

Remark. The two zeros have entirely different causes: $\mathbf{F}_1$ is orthogonal to the curve *pointwise*; $\mathbf{F}_3$ is not orthogonal at all, but the positive and negative contributions cancel over the loop. Only the second kind of vanishing is related to conservativeness.

2.4. Reparametrisation, orientation and concatenated curves

Proposition 2.1: Additivity under concatenation. If the curve Γ is the concatenation of $\Gamma_1, \dots, \Gamma_k$ (the final endpoint of each coinciding with the initial point of the next), then for every continuous field $\mathbf{F}$ and every continuous function f

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \sum_{j=1}^k \int_{\Gamma_j} \mathbf{F} \cdot d\mathbf{r}, \quad \int_{\Gamma} f \, ds = \sum_{j=1}^k \int_{\Gamma_j} f \, ds.$$

Proof. This is the additivity of the Riemann integral with respect to the interval: if $[a, b] = [a, t_1] \cup \dots \cup [t_{k-1}, b]$ is the subdivision corresponding to the concatenation, then $\int_a^b = \sum_j \int_{t_{j-1}}^{t_j}$ for the integrand $\mathbf{F}(\gamma) \cdot \gamma'$ (respectively $f(\gamma)|\gamma'|$), which is continuous on each subinterval by the piecewise regularity assumption. $\square$

Best practice. Before computing any line integral, fix and *write down* three things:

1. the parametrisation with its interval;
2. the direction of travel (counterclockwise is the positive convention for closed curves);
3. which of the three differential elements is needed — ds for masses and mean values, $d\mathbf{r}$ for work and circulation, $\mathbf{n} \, ds$ for fluxes.

Most errors in this chapter arise from confusing the second with the third, that is from computing a flux with the circulation formula or vice versa: the two integrals differ by a 90° rotation of the field, and on the circle they give completely different answers (Example 2.2 versus Example 2.3).

2.5. Conservative fields and the topology of the domain

This is the central section of the chapter. The question is: when does a field have zero circulation along *every* closed curve? The answer involves two independent ingredients — a local differential condition and a global topological one — and the circle is precisely the object that allows one to tell them apart.

Definition 2.4: Conservative field, potential, irrotational field. Let $\mathbf{F} = (P, Q)$ be a field of class C^1 on an open set $A \subseteq \mathbb{R}^2$.

- $\mathbf{F}$ is **conservative** on A if there exists $\phi \in C^1(A)$, called a **potential**, with $\mathbf{F} = \nabla \phi$ throughout A .
- $\mathbf{F}$ is **irrotational** (or *closed*) on A if

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \quad \text{at every point of } A.$$

The quantity $\text{curl } \mathbf{F} = \partial_x Q - \partial_y P$ is the **scalar curl** of $\mathbf{F}$.

Theorem 2.5: Fundamental theorem of line integrals. Let $\mathbf{F} = \nabla \phi$ be conservative on A and let $\Gamma \subseteq A$ be a piecewise regular curve from $\mathbf{a}$ to $\mathbf{b}$. Then

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \phi(\mathbf{b}) - \phi(\mathbf{a}).$$

In particular the integral depends only on the endpoints, and $\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve $\Gamma \subseteq A$.

Proof. Let $\gamma : [a, b] \rightarrow A$ be a parametrisation with $\gamma(a) = \mathbf{a}$, $\gamma(b) = \mathbf{b}$. Consider $g(t) = \phi(\gamma(t))$. By the chain rule,

$$g'(t) = \nabla \phi(\gamma(t)) \cdot \gamma'(t) = \mathbf{F}(\gamma(t)) \cdot \gamma'(t),$$

which is exactly the integrand of Definition 2.3. By the fundamental theorem of calculus (applicable since g' is continuous on each regular piece, the contributions adding up by Proposition 2.1):

$$\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_a^b g'(t) dt = g(b) - g(a) = \phi(\mathbf{b}) - \phi(\mathbf{a}).$$

If Γ is closed, $\mathbf{a} = \mathbf{b}$ and the difference vanishes. □

Theorem 2.6: Conservative implies irrotational. If $\mathbf{F} \in C^1(A)$ is conservative, then it is irrotational.

Proof. Let $\mathbf{F} = \nabla \phi = (\phi_x, \phi_y)$. Since $\mathbf{F} \in C^1$, the potential ϕ is of class C^2 , and by Schwarz's theorem on the interchange of the order of differentiation

$$\frac{\partial Q}{\partial x} = \frac{\partial^2 \phi}{\partial x \partial y} = \frac{\partial^2 \phi}{\partial y \partial x} = \frac{\partial P}{\partial y}.$$

□

The converse implication does *not* hold in general: this is where the shape of the domain enters.

Definition 2.5: Star-shaped domain. An open set $A \subseteq \mathbb{R}^2$ is **star-shaped** with respect to a point $\mathbf{p}_0 \in A$ if for every $\mathbf{p} \in A$ the segment from $\mathbf{p}_0$ to $\mathbf{p}$ is entirely contained in A . Every convex set is star-shaped with respect to each of its points; in particular $\mathbb{R}^2$, every disc and every rectangle are.

Theorem 2.7: Poincaré lemma on star-shaped domains. Let A be star-shaped with respect to the origin and let $\mathbf{F} = (P, Q) \in C^1(A)$ be irrotational. Then $\mathbf{F}$ is conservative on A , and an explicit potential is

$$\phi(x, y) = \int_0^1 [P(tx, ty)x + Q(tx, ty)y] dt.$$

Proof. Step 0 — the formula makes sense. Since A is star-shaped about the origin, for every $(x, y) \in A$ the point (tx, ty) lies in A for all $t \in [0, 1]$: the integrand is defined. It is continuous in t and, being a composition of C^1 functions, may be differentiated under the integral sign in x and y (the integrand and its partial derivatives are continuous on the compact $[0, 1]$ for each fixed (x, y) , and locally uniformly bounded).

Step 1 — derivative with respect to x . Differentiating under the sign,

$$\frac{\partial \phi}{\partial x}(x, y) = \int_0^1 [P(tx, ty) + tx P_x(tx, ty) + ty Q_x(tx, ty)] dt,$$

where the first term comes from the explicit factor x , the second from the dependence of P on x through tx , the third from the dependence of Q on x .

Step 2 — using irrotationality. Substitute $Q_x = P_y$ in the third term:

$$\frac{\partial \phi}{\partial x}(x, y) = \int_0^1 [P(tx, ty) + tx P_x(tx, ty) + ty P_y(tx, ty)] dt.$$

Step 3 — recognising a total derivative. Consider the one-variable function $u(t) =$

$t P(tx, ty)$. By the product and chain rules,

$$u'(t) = P(tx, ty) + t [P_x(tx, ty) x + P_y(tx, ty) y],$$

which is exactly the integrand of Step 2. Hence

$$\frac{\partial \phi}{\partial x}(x, y) = \int_0^1 u'(t) dt = u(1) - u(0) = P(x, y).$$

Step 4 — derivative with respect to y . Symmetrically, differentiating under the sign and using $P_y = Q_x$:

$$\frac{\partial \phi}{\partial y} = \int_0^1 [Q(tx, ty) + t x Q_x(tx, ty) + t y Q_y(tx, ty)] dt,$$

which is the derivative of $v(t) = t Q(tx, ty)$; integrating, $\partial_y \phi = Q(x, y)$.

Conclusion. $\nabla \phi = (P, Q) = \mathbf{F}$ on all of A : the field is conservative. $\square$

***Remark 2.2: The potential is the integral along the ray.** The formula of Theorem 2.7 is no artifice: parametrising the segment from $\mathbf{0}$ to (x, y) as $\sigma(t) = (tx, ty)$, $t \in [0, 1]$, we have $\sigma'(t) = (x, y)$ and therefore*

$$\phi(x, y) = \int_{\sigma} \mathbf{F} \cdot d\mathbf{r}.$$

The potential is simply the work done along the ray joining the centre of the star to the point. The star-shapedness hypothesis guarantees that such a ray exists inside A ; irrotationality guarantees that the result does not change if a different path is chosen.

2.5.1 The vortex field: irrotational but not conservative

Theorem 2.8: The vortex field. Let $A = \mathbb{R}^2 \setminus \{(0, 0)\}$ and

$$\mathbf{V}(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right).$$

Then:

- (i) $\mathbf{V} \in C^\infty(A)$ and is irrotational on A ;
- (ii) for every $R > 0$, denoting by C_R the circle of centre the origin and radius R traversed counterclockwise,

$$\oint_{C_R} \mathbf{V} \cdot d\mathbf{r} = 2\pi,$$

independently of R ;

- (iii) $\mathbf{V}$ is not conservative on A ;
- (iv) $\mathbf{V}$ is conservative on every open half-plane whose boundary passes through the origin — for instance on $\{x > 0\}$, where a potential is $\phi(x, y) = \arctan(y/x)$.

Proof. (i) The two components are quotients of polynomials with denominator $x^2 + y^2$, which does not vanish on A : they are C^∞ . Setting $\rho^2 = x^2 + y^2$ and using the quotient rule,

$$\frac{\partial P}{\partial y} = \frac{-\rho^2 + 2y^2}{\rho^4} = \frac{y^2 - x^2}{\rho^4}, \quad \frac{\partial Q}{\partial x} = \frac{\rho^2 - 2x^2}{\rho^4} = \frac{y^2 - x^2}{\rho^4}.$$

The two expressions agree at every point of A : $\text{curl } \mathbf{V} \equiv 0$.

(ii) Parametrise C_R by $\gamma(\theta) = R(\cos \theta, \sin \theta)$, $\theta \in [0, 2\pi]$. On the curve $\rho^2 = R^2$, hence

$$\mathbf{V}(\gamma(\theta)) = \frac{1}{R}(-\sin \theta, \cos \theta),$$

while $\gamma'(\theta) = R(-\sin \theta, \cos \theta)$. The dot product is

$$\mathbf{V}(\gamma) \cdot \gamma' = \frac{R}{R}(\sin^2 \theta + \cos^2 \theta) = 1,$$

constant. Hence

$$\oint_{C_R} \mathbf{V} \cdot d\mathbf{r} = \int_0^{2\pi} 1 \, d\theta = 2\pi,$$

and the value does not depend on R : the factors R from the field and from the velocity cancel exactly.

(iii) If $\mathbf{V}$ were conservative on A , by Theorem 2.5 its circulation along every closed curve contained in A would vanish. But $C_R \subset A$ (the origin does not belong to it) and the circulation is $2\pi \neq 0$: contradiction.

(iv) On the half-plane $\{x > 0\}$, which is convex hence star-shaped, set $\phi(x, y) = \arctan(y/x)$. Differentiating,

$$\frac{\partial \phi}{\partial x} = \frac{x^2}{x^2 + y^2} \cdot \left(-\frac{y}{x^2}\right) = \frac{-y}{x^2 + y^2} = P, \quad \frac{\partial \phi}{\partial y} = \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2} = Q.$$

Hence $\nabla \phi = \mathbf{V}$ on $\{x > 0\}$. (Alternatively, conservativeness would already follow from Theorem 2.7, the half-plane being star-shaped and $\mathbf{V}$ irrotational; the explicit computation additionally supplies the potential.) $\square$

Remark 2.3: The missing potential is the angle. The potential found in (iv), $\arctan(y/x)$, is the polar angle θ . The general formula, valid on any open set where a continuous determination of the angle can be chosen, is

$$\mathbf{V} = \nabla \theta.$$

The vortex field fails to be conservative on all of $\mathbb{R}^2 \setminus \{0\}$ for the very same reason that no continuous global angle exists on S^1 (Theorem 1.19): the candidate potential exists only locally, and going once around one accumulates a jump of 2π — exactly the value of the circulation computed in (ii). The analytic and the topological obstruction are one and the same phenomenon.

Corollary 2.1: Circulation counts the turns. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2 \setminus \{0\}$ be a closed circular motion about the origin, of radius R and phase law $\theta(t)$. Then

$$\frac{1}{2\pi} \oint_{\gamma} \mathbf{V} \cdot d\mathbf{r} = n,$$

where n is the winding number of Definition 1.3.

Proof. With $\gamma(t) = R(\cos \theta(t), \sin \theta(t))$ we have $\gamma'(t) = R\theta'(t)(-\sin \theta, \cos \theta)$ (Chapter 1, Theorem 1.2) and, as in step (ii) above, $\mathbf{V}(\gamma) = \frac{1}{R}(-\sin \theta, \cos \theta)$. Hence

$$\mathbf{V}(\gamma) \cdot \gamma' = \theta'(t),$$

and integrating

$$\oint_{\gamma} \mathbf{V} \cdot d\mathbf{r} = \theta(b) - \theta(a) = 2\pi n$$

by Definition 1.3. Dividing by 2π gives the claim. $\square$

Remark 2.4: A topological invariant computed by an integral. Corollary 2.1 is remarkable: an integer — hence insensitive to any continuous deformation — is produced by an integral, an object continuous by nature. It is the first instance of a vast family of results (indices, degrees, winding numbers, characteristic classes) in which an integral computes a discrete invariant. In Chapter 3 the same formula will reappear as $\frac{1}{2\pi i} \oint \frac{dz}{z}$, Cauchy’s integral formula in its simplest case.

Warning. The commonest error is to conclude “zero curl, therefore conservative”. The implication is correct *only* with an additional hypothesis on the domain (star-shaped, or more generally simply connected). The vortex field on $\mathbb{R}^2 \setminus \{0\}$ is the minimal counterexample, and it should be kept in mind whenever the domain has a “hole”.

2.6. Flux through a circle

Definition 2.6: Flux. Let $\mathbf{F}$ be continuous on an open set containing the closed piecewise regular curve Γ , oriented counterclockwise, and let $\mathbf{n}$ be the outward unit normal. The **outward flux** of $\mathbf{F}$ through Γ is

$$\Phi_{\Gamma}(\mathbf{F}) = \oint_{\Gamma} \mathbf{F} \cdot \mathbf{n} \, ds = \oint_{\Gamma} (P \, dy - Q \, dx),$$

using $\mathbf{n} \, ds = (dy, -dx)$ (Definition 2.1).

Theorem 2.9: Flux through a circle. Let $C_R = C(C_0, R)$ be oriented counterclockwise. Then

$$\Phi_{C_R}(\mathbf{F}) = R \int_0^{2\pi} \mathbf{F}(C_0 + R e_r(\theta)) \cdot e_r(\theta) \, d\theta, \quad e_r(\theta) = (\cos \theta, \sin \theta).$$

In particular:

- (i) for the radial field $\mathbf{F}(\mathbf{p}) = \mathbf{p} - C_0$, $\Phi = 2\pi R^2$;
- (ii) for the normalised radial field $\mathbf{F} = (\mathbf{p} - C_0)/|\mathbf{p} - C_0|$, $\Phi = 2\pi R$ (the length of the circle);
- (iii) for the source field $\mathbf{F} = (\mathbf{p} - C_0)/|\mathbf{p} - C_0|^2$, $\Phi = 2\pi$, independently of R ;
- (iv) for the rotational field $\mathbf{F}(x, y) = (-y, x)$ centred at $C_0 = O$, $\Phi = 0$.

Proof. By Theorem 2.1(iv), $\mathbf{n} \, ds = R e_r(\theta) \, d\theta$, which gives the general formula.

(i) On the curve $\mathbf{F} = R e_r$, hence $\mathbf{F} \cdot e_r = R$ and $\Phi = R \int_0^{2\pi} R \, d\theta = 2\pi R^2$.

(ii) On the curve $\mathbf{F} = e_r$, hence $\mathbf{F} \cdot e_r = 1$ and $\Phi = 2\pi R$.

(iii) On the curve $|\mathbf{p} - C_0| = R$, hence $\mathbf{F} = e_r/R$ and $\mathbf{F} \cdot e_r = 1/R$; therefore

$$\Phi = R \int_0^{2\pi} \frac{1}{R} \, d\theta = 2\pi.$$

The flux does not depend on the radius: this is the phenomenon known, in three dimensions, as Gauss’s law.

(iv) On the curve with $C_0 = O$, $\mathbf{F}(\gamma(\theta)) = R(-\sin \theta, \cos \theta)$, which is orthogonal to e_r : $\mathbf{F} \cdot e_r = 0$, hence $\Phi = 0$. The rotational field circulates but does not cross. $\square$

Example 2.3: Flux and circulation compared. Take again the fields of Example 2.2 on $C(O, R)$ counterclockwise and compute their *flux*, comparing it with the circulation already found.

Field	$\oint \mathbf{F} \cdot d\mathbf{r}$	$\oint \mathbf{F} \cdot \mathbf{n} \, ds$
$\mathbf{F}_1 = (x, y)$ (radial)	0	$2\pi R^2$
$\mathbf{F}_2 = (-y, x)$ (rotational)	$2\pi R^2$	0
$\mathbf{F}_3 = (1, 0)$ (uniform)	0	0

The first two fields swap roles exactly: $\mathbf{F}_2 = J\mathbf{F}_1$ is $\mathbf{F}_1$ rotated by 90° , and rotating the field by 90° swaps the tangential and normal components. The third field gives zero in both columns, but for yet another reason: it is conservative (zero circulation) and divergence-free (zero flux).

Check of the flux of $\mathbf{F}_3$.

$$\oint (P \, dy - Q \, dx) = \int_0^{2\pi} R \cos \theta \, d\theta = 0. \checkmark$$

Interpretation: a uniform flow enters through one side of the circle and leaves through the other, in exactly equal amounts.

2.7. Green's theorem

Theorem 2.10: Green's theorem (tangential form). Let $D \subset \mathbb{R}^2$ be a bounded closed region whose boundary ∂D is a piecewise regular Jordan curve oriented counterclockwise, and let P, Q be of class C^1 on an open set containing D . Then

$$\oint_{\partial D} (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

Proof. We prove the theorem for a region D that is *simultaneously* of type I and of type II, i.e. expressible in both forms

$$D = \{(x, y) : a \leq x \leq b, \, g_1(x) \leq y \leq g_2(x)\} = \{(x, y) : c \leq y \leq d, \, h_1(y) \leq x \leq h_2(y)\}$$

with g_i, h_i continuous and piecewise C^1 . The disc belongs to this class — and the disc is what this chapter needs. The general case follows by decomposing D into finitely many such regions: adding the identities, the line integrals along the internal cuts appear twice with opposite orientations and cancel by Theorem 2.4(iii), leaving only the contributions of ∂D .

Step 1 — the identity for P , using the type-I structure. Compute the double integral by integrating first in y , using Fubini:

$$\iint_D \frac{\partial P}{\partial y} \, dA = \int_a^b \left(\int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}(x, y) \, dy \right) dx = \int_a^b [P(x, g_2(x)) - P(x, g_1(x))] dx,$$

by the fundamental theorem of calculus in the variable y at fixed x .

Step 2 — the line integral of $P \, dx$. The boundary ∂D , traversed counterclockwise, splits into four pieces: the lower graph $\Gamma_1 : y = g_1(x)$ traversed from $x = a$ to $x = b$; the vertical segment Γ_2 at $x = b$; the upper graph $\Gamma_3 : y = g_2(x)$ traversed from $x = b$ to $x = a$; the vertical segment Γ_4 at $x = a$. On the two vertical pieces x is constant, hence $dx = 0$ and their contribution to $\oint P \, dx$ vanishes. There remains

$$\oint_{\partial D} P \, dx = \int_a^b [P(x, g_1(x)) - P(x, g_2(x))] dx = - \iint_D \frac{\partial P}{\partial y} \, dA.$$

Step 3 — the identity for Q , using the type-II structure. Symmetrically, integrating first in x :

$$\iint_D \frac{\partial Q}{\partial x} dA = \int_c^d \left[Q(h_2(y), y) - Q(h_1(y), y) \right] dy.$$

Traversing ∂D counterclockwise, the right graph $x = h_2(y)$ is traversed from $y = c$ to $y = d$ and the left graph $x = h_1(y)$ from $y = d$ to $y = c$; on the horizontal pieces $dy = 0$. Hence

$$\oint_{\partial D} Q dy = \iint_D \frac{\partial Q}{\partial x} dA.$$

Step 4 — sum. Adding the identities of Steps 2 and 3 gives the statement. $\square$

Corollary 2.2: Divergence theorem in the plane (normal form of Green). Under the same hypotheses, for every field $\mathbf{F} = (F_1, F_2)$ of class C^1 ,

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \iint_D \operatorname{div} \mathbf{F} dA, \quad \operatorname{div} \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}.$$

Proof. By Definition 2.6,

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \oint_{\partial D} (F_1 dy - F_2 dx),$$

which is the line integral of Theorem 2.10 with $P = -F_2$ and $Q = F_1$. Applying the theorem,

$$\oint_{\partial D} (-F_2 dx + F_1 dy) = \iint_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \right) dA.$$

$\square$

Corollary 2.3: Area formulas. Under the hypotheses of Theorem 2.10,

$$\operatorname{Area}(D) = \oint_{\partial D} x dy = - \oint_{\partial D} y dx = \frac{1}{2} \oint_{\partial D} (x dy - y dx).$$

Proof. Apply Theorem 2.10 three times.

- With $P = 0$, $Q = x$: $\partial_x Q - \partial_y P = 1$, hence $\oint x dy = \operatorname{Area}(D)$.
- With $P = -y$, $Q = 0$: $\partial_x Q - \partial_y P = 1$, hence $-\oint y dx = \operatorname{Area}(D)$.
- With $P = -y/2$, $Q = x/2$: $\partial_x Q - \partial_y P = 1$, hence $\frac{1}{2} \oint (x dy - y dx) = \operatorname{Area}(D)$.

The three formulas are arithmetic means of one another and naturally give the same value. $\square$

Example 2.4: The area of the disc by four methods. Compute the area of the disc of radius R in four independent ways.

Method 1 — symmetric Green formula. With $\gamma(\theta) = R(\cos \theta, \sin \theta)$: $dx = -R \sin \theta d\theta$, $dy = R \cos \theta d\theta$, hence

$$A = \frac{1}{2} \oint (x dy - y dx) = \frac{R^2}{2} \int_0^{2\pi} (\cos^2 \theta + \sin^2 \theta) d\theta = \pi R^2.$$

Method 2 — asymmetric formula $\oint x dy$.

$$A = R^2 \int_0^{2\pi} \cos^2 \theta d\theta = R^2 \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} d\theta = \pi R^2. \checkmark$$

(Here the half-angle formula $\cos^2 u = (1 + \cos 2u)/2$ was used, together with $\int_0^{2\pi} \cos 2\theta d\theta = 0$.)

Method 3 — vertical slices.

$$A = 2 \int_{-R}^R \sqrt{R^2 - x^2} \, dx.$$

With the substitution $x = R \sin u$, $dx = R \cos u \, du$, $u \in [-\pi/2, \pi/2]$ (where $\cos u \geq 0$, so $\sqrt{R^2 - x^2} = R \cos u$):

$$A = 2R^2 \int_{-\pi/2}^{\pi/2} \cos^2 u \, du = 2R^2 \cdot \frac{\pi}{2} = \pi R^2. \checkmark$$

Method 4 — polar coordinates. Anticipating Theorem 2.11 of the next section,

$$A = \int_0^{2\pi} \int_0^R \rho \, d\rho \, d\varphi = \pi R^2. \checkmark$$

The four methods use, respectively: the boundary with a symmetric field, the boundary with an asymmetric field, the region with Cartesian slices, the region with adapted coordinates. That they all give the same answer is, in essence, the content of Green's theorem.

2.8. Polar coordinates and integrals over circular domains

Theorem 2.11: Change of variables to polar coordinates. For the transformation $x = \rho \cos \varphi$, $y = \rho \sin \varphi$ with $\rho > 0$, $\varphi \in [0, 2\pi)$, the Jacobian equals $J = \rho$, hence

$$dA = dx \, dy = \rho \, d\rho \, d\varphi.$$

Proof. The four partial derivatives are

$$\frac{\partial x}{\partial \rho} = \cos \varphi, \quad \frac{\partial x}{\partial \varphi} = -\rho \sin \varphi, \quad \frac{\partial y}{\partial \rho} = \sin \varphi, \quad \frac{\partial y}{\partial \varphi} = \rho \cos \varphi,$$

and the Jacobian is the determinant of the matrix collecting them:

$$J = \det \begin{pmatrix} \cos \varphi & -\rho \sin \varphi \\ \sin \varphi & \rho \cos \varphi \end{pmatrix} = \rho \cos^2 \varphi + \rho \sin^2 \varphi = \rho(\cos^2 \varphi + \sin^2 \varphi) = \rho.$$

Since $\rho > 0$, $|J| = \rho$, and the change-of-variables formula for double integrals gives $dA = \rho \, d\rho \, d\varphi$. $\square$

Remark 2.5: Geometric reading of the factor ρ . The factor ρ is not a computational artefact: the area element in polar coordinates is a curvilinear rectangle with sides $d\rho$ (radial) and $\rho \, d\varphi$ (circumferential, by Theorem 1.9 of Chapter 1). Their product is $\rho \, d\rho \, d\varphi$. Put differently: for a given $d\varphi$, a sector far from the centre is wider than one nearby, in exact proportion to ρ .

Strategy for an integral over a circular domain Step 1. Recognise the circular symmetry of the domain or of the integrand.

Step 2. Pass to polar coordinates centred at the right point (the centre of the disc, not necessarily the origin): $x = x_c + \rho \cos \varphi$, $y = y_c + \rho \sin \varphi$, $dA = \rho \, d\rho \, d\varphi$.

Step 3. Determine the limits. For the disc $D(C_0, R)$: $\rho \in [0, R]$, $\varphi \in [0, 2\pi)$ — constant limits, so the integral factorises if the integrand is separable.

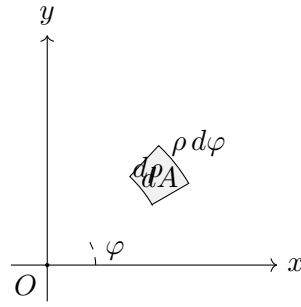


Figure 2.2: The area element in polar coordinates: a curvilinear rectangle with sides $d\rho$ and $\rho d\varphi$, whence $dA = \rho d\rho d\varphi$.

Step 4. Compute the two resulting single integrals, checking that the Jacobian factor ρ has been included. $\square$

Example 2.5: Polar moment of inertia of a disc. Compute $J_O = \iint_{D(O,R)} (x^2 + y^2) dA$, and deduce the moment of inertia of a homogeneous disc of mass m .

Step 1 — passing to polars. Integrand and domain are both circularly symmetric: polar coordinates are the natural choice, with $x^2 + y^2 = \rho^2$.

Step 2 — setting up.

$$J_O = \int_0^{2\pi} \int_0^R \rho^2 \cdot \rho d\rho d\varphi = \int_0^{2\pi} \int_0^R \rho^3 d\rho d\varphi.$$

Step 3 — radial integration. $\int_0^R \rho^3 d\rho = \frac{R^4}{4}$.

Step 4 — angular integration. $J_O = 2\pi \cdot \frac{R^4}{4} = \frac{\pi R^4}{2}$.

Step 5 — physical interpretation. With constant surface density σ , the mass is $m = \sigma\pi R^2$ and the moment of inertia about the axis through O perpendicular to the plane is

$$I_O = \sigma J_O = (\sigma\pi R^2) \frac{R^2}{2} = \frac{1}{2} m R^2,$$

the familiar formula from rigid-body mechanics. $\checkmark$

Comparison with a ring. For a ring (all the mass on the boundary) one would get $I = mR^2$: twice as much. The difference is reasonable, since in the solid disc half the mass lies closer to the axis.

2.9. Applications to engineering and physics

2.9.1 Sizing a circular clarifier

A direct application of the area and perimeter of the circle, in daily use in water treatment engineering, is the sizing of a **circular clarifier** (secondary sedimentation tank): a cylindrical basin in which the water enters at the centre, slows down, and the suspended solids settle to the bottom while the clarified water overflows a peripheral weir. The two fundamental design parameters depend, respectively, on the *area* and on the *perimeter* of the circular cross-section.

Definition 2.7: Surface overflow rate and weir loading rate. Given a flow rate Q and a circular clarifier of radius R :

$$\text{SOR} = \frac{Q}{A} = \frac{Q}{\pi R^2}, \quad \text{WLR} = \frac{Q}{P} = \frac{Q}{2\pi R}.$$

***Remark 2.6: Why both matter, and why SOR wins.** The SOR governs settling: by Hazen's classical (1904) theory of ideal tanks, a particle settles effectively if its fall velocity exceeds Q/A — which is why the SOR has the dimensions of a velocity ($\text{m}^3/(\text{m}^2 \text{h}) = \text{m/h}$). The WLR governs the risk of carry-over: if the clarified water must leave through too short a weir length, the local velocity at the edge increases and re-suspends the already settled floc. The mathematical point is that $\text{SOR} \propto R^{-2}$ whereas $\text{WLR} \propto R^{-1}$: as the radius grows, the SOR improves quadratically and the WLR only linearly. Consequently, for increasing radii the SOR constraint is satisfied first and the WLR one remains comfortably within margin: in practice it is almost always the area, not the perimeter, that sizes the tank.*

Example 2.6: Sizing a secondary clarifier. A plant treats $Q = 2000 \text{ m}^3/\text{d}$ with the design limit $\text{SOR} \leq 1.2 \text{ m/h}$.

Step 1 — flow conversion. $Q = 2000/24 = 83.33 \text{ m}^3/\text{h}$.

Step 2 — minimum area from the SOR constraint.

$$A \geq \frac{83.33}{1.2} = 69.44 \text{ m}^2.$$

Step 3 — radius and diameter.

$$R \geq \sqrt{\frac{69.44}{\pi}} = \sqrt{22.10} = 4.70 \text{ m}, \quad D = 9.40 \text{ m}.$$

Step 4 — checking the weir loading. With $P = 2\pi R = 29.53 \text{ m}$,

$$\text{WLR} = \frac{83.33}{29.53} = 2.82 \text{ m}^3/(\text{m h}) = 67.7 \text{ m}^3/(\text{m d}),$$

well below the typical literature limits ($125\text{--}375 \text{ m}^3/(\text{m d})$): the active constraint is the SOR, as predicted by Observation 2.6.

Step 5 — detention time. With a working depth $h = 3.5 \text{ m}$,

$$V = 69.44 \times 3.5 = 243.0 \text{ m}^3, \quad \tau = \frac{243.0}{83.33} = 2.92 \text{ h},$$

within the typical 2–4 hour range for a secondary clarifier. The design is consistent on all three criteria.

2.9.2 The Gaussian integral

Theorem 2.12: Gaussian integral.

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \pi, \quad \text{whence} \quad \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

Proof. (Poisson, polar coordinates.) The integrand is positive, so the integral over $\mathbb{R}^2$ is the increasing limit of the integrals over the discs $D(O, \rho_0)$ as $\rho_0 \rightarrow +\infty$. In polar coordinates

(Theorem 2.11), $x^2 + y^2 = \rho^2$ and $dA = \rho d\rho d\varphi$:

$$\iint_{D(O, \rho_0)} e^{-(x^2+y^2)} dA = \int_0^{2\pi} \int_0^{\rho_0} e^{-\rho^2} \rho d\rho d\varphi.$$

The radial integral is computed by the substitution $s = \rho^2$, $ds = 2\rho d\rho$:

$$\int_0^{\rho_0} e^{-\rho^2} \rho d\rho = \frac{1}{2} \left(1 - e^{-\rho_0^2}\right) \xrightarrow{\rho_0 \rightarrow \infty} \frac{1}{2}.$$

Hence the integral over $\mathbb{R}^2$ equals $2\pi \cdot \frac{1}{2} = \pi$.

Descent to one dimension. Since $e^{-(x^2+y^2)} = e^{-x^2}e^{-y^2}$ and the integrand is positive, Tonelli's theorem gives

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)} dA = \left(\int_{\mathbb{R}} e^{-x^2} dx \right)^2 = \pi,$$

and since the integral is positive, $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$. $\square$

. Second proof, via the Gamma and Beta functions (Legendre–Laplace). Recall $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$, $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and the relation $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$.

Step A. With the substitution $t = x^2$, $dt = 2x dx$ (for $x > 0$):

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-1/2} e^{-t} dt = 2 \int_0^\infty e^{-x^2} dx = \int_{\mathbb{R}} e^{-x^2} dx,$$

using the evenness of the integrand. So it suffices to compute $\Gamma(1/2)$.

Step B. Compute $B(1/2, 1/2)$ with $t = \sin^2 \varphi$, $dt = 2 \sin \varphi \cos \varphi d\varphi$, $\varphi \in [0, \pi/2]$ (where sine and cosine are non-negative, so $t^{-1/2} = 1/\sin \varphi$ and $(1-t)^{-1/2} = 1/\cos \varphi$):

$$B\left(\frac{1}{2}, \frac{1}{2}\right) = \int_0^{\pi/2} 2 d\varphi = \pi.$$

Step C. On the other hand $B(1/2, 1/2) = \Gamma(1/2)^2/\Gamma(1) = \Gamma(1/2)^2$, since $\Gamma(1) = 1$. Comparing, $\Gamma(1/2)^2 = \pi$, hence $\Gamma(1/2) = \sqrt{\pi}$ (positive), and by Step A $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$. $\square$

Remark 2.7: Why circular symmetry solves the problem. The function e^{-x^2} has no elementary antiderivative (Liouville's theorem rules it out), yet its integral over $\mathbb{R}$ is computable in closed form. Poisson's trick consists in going up in dimension: the square of the integral becomes a double integral whose integrand, $e^{-\rho^2}$, depends only on the distance from the origin. The circular symmetry of the problem, combined with the Jacobian factor ρ (which is exactly what is needed to make $e^{-\rho^2} \rho$ elementarily integrable), does the rest. It is the prototype of a technique used systematically in mathematical physics and in probability.

2.9.3 Volume of the sphere: a preview of Volume III

Remark 2.8: From the disc to the ball. Rotating the semicircle $y = \sqrt{R^2 - x^2}$, $x \in [-R, R]$, about the x -axis, the method of discs gives

$$\text{Vol} = \pi \int_{-R}^R (R^2 - x^2) dx = \pi \left[R^2 x - \frac{x^3}{3} \right]_{-R}^R = \frac{4}{3} \pi R^3.$$

This computation will be taken up and generalised to $\mathbb{R}^n$ in Volume III, where the volume of the n -dimensional ball will be expressed through the Γ function met above.

2.9.4 When the circle is not enough: the pendulum and the elliptic integral

The chapter has so far exploited a piece of good fortune of the circle: its maximal symmetry makes almost every integral computable in closed form. The simple pendulum — whose bob moves precisely along a circular arc — shows where that good fortune ends.

A point mass m is attached to a rigid rod of length L , of negligible mass, free to swing in a vertical plane under gravity g . Writing θ for the angle between the rod and the downward vertical, the tangential component of Newton's second law gives the equation of motion

$$\theta''(t) + \frac{g}{L} \sin \theta(t) = 0.$$

Remark 2.9: The familiar limit: small oscillations. For small θ , $\sin \theta \approx \theta$ and the equation becomes that of the harmonic oscillator $\theta'' + (g/L)\theta = 0$, with period $T_0 = 2\pi\sqrt{L/g}$, **independent of the amplitude**: this is the *isochronism* of the pendulum (Galileo, 1602). What follows quantifies the error made in using T_0 even for amplitudes that are not small.

Theorem 2.13: Exact period of the pendulum. Let $\theta_0 \in (0, \pi)$ be the maximum amplitude (the pendulum starts at rest at $\theta = \theta_0$). The exact period of the oscillations is

$$T = 4\sqrt{\frac{L}{g}} K(k), \quad k = \sin \frac{\theta_0}{2},$$

where K is the **complete elliptic integral of the first kind**

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}.$$

Proof. Step 1 — conservation of energy. Multiplying the equation of motion by θ' and integrating in t ,

$$\frac{1}{2}(\theta')^2 - \frac{g}{L} \cos \theta = \text{const.}$$

At the turning point $\theta = \theta_0$ one has $\theta' = 0$, so the constant equals $-\frac{g}{L} \cos \theta_0$, and

$$(\theta')^2 = \frac{2g}{L} (\cos \theta - \cos \theta_0).$$

Step 2 — time for a quarter period. By symmetry of the motion, the time to go from $\theta = 0$ (maximum speed) to $\theta = \theta_0$ (zero speed) is a quarter of the period. Over this stretch θ' has constant sign, so

$$dt = \frac{d\theta}{\theta'} = \sqrt{\frac{L}{2g}} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}}, \quad \frac{T}{4} = \sqrt{\frac{L}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}}.$$

Step 3 — half angle. With $\cos \theta = 1 - 2\sin^2(\theta/2)$,

$$\cos \theta - \cos \theta_0 = 2 \left[\sin^2 \frac{\theta}{2} - \sin^2 \frac{\theta_0}{2} \right],$$

hence

$$\frac{T}{4} = \frac{1}{2} \sqrt{\frac{L}{g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\sin^2(\theta/2) - \sin^2(\theta_0/2)}}.$$

Step 4 — Legendre's substitution. Set $\sin(\theta/2) = k \sin \varphi$ with $k = \sin(\theta_0/2)$: as θ runs from 0 to θ_0 , φ runs from 0 to $\pi/2$. Differentiating, $\frac{1}{2} \cos(\theta/2) d\theta = k \cos \varphi d\varphi$, so, with $\cos(\theta/2) = \sqrt{1 - k^2 \sin^2 \varphi}$ (positive for $\theta/2 \in [0, \pi/2)$),

$$d\theta = \frac{2k \cos \varphi}{\sqrt{1 - k^2 \sin^2 \varphi}} d\varphi.$$

Moreover $\sin^2(\theta_0/2) - \sin^2(\theta/2) = k^2 - k^2 \sin^2 \varphi = k^2 \cos^2 \varphi$, so the square root in the denominator equals $k \cos \varphi$ (positive on $[0, \pi/2)$), and

$$\frac{d\theta}{\sqrt{\sin^2(\theta_0/2) - \sin^2(\theta/2)}} = \frac{2k \cos \varphi / \sqrt{1 - k^2 \sin^2 \varphi}}{k \cos \varphi} d\varphi = \frac{2 d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}.$$

Step 5 — conclusion. Substituting into Step 3,

$$\frac{T}{4} = \sqrt{\frac{L}{g}} \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}} = \sqrt{\frac{L}{g}} K(k),$$

and multiplying by 4 gives the claim. $\square$

Theorem 2.14: Series expansion of K . For $|k| < 1$,

$$K(k) = \frac{\pi}{2} \sum_{n=0}^{\infty} \left[\frac{(2n-1)!!}{(2n)!!} \right]^2 k^{2n} = \frac{\pi}{2} \left(1 + \frac{1}{4}k^2 + \frac{9}{64}k^4 + \frac{25}{256}k^6 + \cdots \right).$$

Proof. **Step 0 — the binomial coefficient.** In Newton's expansion $(1-x)^{-1/2} = \sum_n \binom{-1/2}{n} (-x)^n$, the coefficient of x^n is

$$\binom{-1/2}{n} (-1)^n = \frac{(-\frac{1}{2})(-\frac{3}{2}) \cdots (-\frac{2n-1}{2})}{n!} (-1)^n = \frac{\frac{1}{2} \cdot \frac{3}{2} \cdots \frac{2n-1}{2}}{n!} = \frac{(2n-1)!!}{2^n n!} = \frac{(2n-1)!!}{(2n)!!},$$

having multiplied and divided by $2^n n! = (2n)!!$.

Step 1 — term-by-term integration. With $x = k^2 \sin^2 \varphi \in [0, k^2]$ and $k^2 < 1$, the binomial series converges uniformly in $\varphi \in [0, \pi/2]$ (its terms are bounded by those, summable, of the series in $x = k^2$), so it may be integrated term by term:

$$K(k) = \int_0^{\pi/2} \sum_{n \geq 0} \frac{(2n-1)!!}{(2n)!!} k^{2n} \sin^{2n} \varphi d\varphi = \sum_{n \geq 0} \frac{(2n-1)!!}{(2n)!!} k^{2n} \int_0^{\pi/2} \sin^{2n} \varphi d\varphi.$$

Step 2 — Wallis's integrals. By Theorem 3.26 (Chapter 3), $\int_0^{\pi/2} \sin^{2n} \varphi d\varphi = \frac{\pi}{2} \cdot \frac{(2n-1)!!}{(2n)!!}$. Substituting,

$$K(k) = \sum_{n \geq 0} \frac{(2n-1)!!}{(2n)!!} k^{2n} \cdot \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!} = \frac{\pi}{2} \sum_{n \geq 0} \left[\frac{(2n-1)!!}{(2n)!!} \right]^2 k^{2n},$$

which is the claim. $\square$

Example 2.7: The pendulum at $\theta_0 = 90^\circ$. A pendulum starts at $\theta_0 = 90^\circ = \pi/2$. Compare the exact period with that of small oscillations.

Step 1 — the parameter k . $k = \sin 45^\circ = \frac{\sqrt{2}}{2} \approx 0.707107$, $k^2 = 0.5$.

Step 2 — $K(k)$ by series, term by term (Theorem 2.14). The first eight terms give

the partial sums

1.570796; 1.767146; 1.822369; 1.841544; 1.848884; 1.851857; 1.853106; 1.853645,

approaching — slowly (the series converges, but not fast for $k^2 = 0.5$) — the value $K(k) = 1.854075$, obtained by direct numerical quadrature of the definition. This last value coincides, to the ten digits computed, with the value known in the literature $K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$: a third independent confirmation.

Step 3 — ratio of the periods.

$$\frac{T}{T_0} = \frac{4\sqrt{L/g} K(k)}{2\pi\sqrt{L/g}} = \frac{2K(k)}{\pi} \approx \frac{2 \times 1.854075}{3.141593} = 1.180341,$$

that is, the true period is about 18% longer than predicted by the small-oscillation approximation — an error far from negligible for an amplitude of only 90° .

Independent numerical check. The value of $K(k)$ by direct quadrature, the period given by the closed formula, and the period obtained by numerically integrating the original equation of motion $\theta'' = -(g/L) \sin \theta$ (fourth-order Runge–Kutta) agree with one another to one part in 10^{11} . Script `Sorgenti/Vol2/_verifiche/vol2_cap02_pendolo.py`.

***Remark 2.10: Why the series is not used in practice.** The series of Theorem 2.14 always converges, but ever more slowly as $k \rightarrow 1^-$ (that is, as $\theta_0 \rightarrow \pi$): for $\theta_0 = 150^\circ$ ($k^2 \approx 0.933$) many dozens of terms, even close to a hundred, are needed for the same precision obtained here with eight. The method used in practice to compute K is Gauss's arithmetic-geometric mean, which converges quadratically (the number of correct digits doubles at every iteration) for any $k \in (0, 1)$; we do not prove this here, but the verification script includes it as a further independent check.*

2.10. Trigonometric integrals and cross-references

Theorem 2.15: Fundamental trigonometric integrals over a period. For all integers $m, n \geq 1$:

$$\begin{aligned} \int_0^{2\pi} \cos m\theta \, d\theta &= \int_0^{2\pi} \sin m\theta \, d\theta = 0, \\ \int_0^{2\pi} \cos m\theta \cos n\theta \, d\theta &= \int_0^{2\pi} \sin m\theta \sin n\theta \, d\theta = \pi \delta_{mn}, \quad \int_0^{2\pi} \cos m\theta \sin n\theta \, d\theta = 0, \end{aligned}$$

where δ_{mn} is 1 if $m = n$ and 0 otherwise.

Proof. **First group.** For $m \geq 1$, $\int_0^{2\pi} \cos m\theta \, d\theta = [\sin(m\theta)/m]_0^{2\pi} = 0$, and likewise for the sine.

Second group. Use the product-to-sum formulas (Volume I, Chapter 8):

$$\cos m\theta \cos n\theta = \frac{1}{2} [\cos((m-n)\theta) + \cos((m+n)\theta)].$$

If $m \neq n$, both indices $m-n$ and $m+n$ are non-zero integers, so by the first group both integrals vanish and the result is 0. If $m = n$, the first summand becomes $\cos 0 = 1$ and the second $\cos(2m\theta)$ with $2m \geq 1$; hence

$$\int_0^{2\pi} \cos^2 m\theta \, d\theta = \frac{1}{2} \cdot 2\pi = \pi.$$

The sine case is identical, using $\sin m\theta \sin n\theta = \frac{1}{2}[\cos((m-n)\theta) - \cos((m+n)\theta)]$.

Third group. $\cos m\theta \sin n\theta = \frac{1}{2}[\sin((n+m)\theta) + \sin((n-m)\theta)]$; if $n \neq m$ both integrals vanish by the first group, and if $n = m$ the second summand is $\sin 0 = 0$ while the first is $\sin(2m\theta)$, of zero integral. $\square$

Remark 2.11: What these formulas mean. Theorem 2.15 says that the functions $\{1, \cos \theta, \sin \theta, \cos 2\theta, \sin 2\theta, \dots\}$ are pairwise orthogonal with respect to the inner product $\langle f, g \rangle = \int_0^{2\pi} fg \, d\theta$. This is the fact on which the whole of Fourier analysis rests: the coefficients of a trigonometric series are extracted by projecting onto the basis, exactly as the components of a vector are extracted by inner products. The systematic development is in **Chapter 4** of this volume, to which we refer.

Remark 2.12: Complex integration: see Chapter 3. Many of the integrals of this chapter can be written far more compactly by setting $z = e^{i\theta}$, whence $dz = iz \, d\theta$ and therefore $d\theta = dz/(iz)$, with

$$\cos \theta = \frac{z + z^{-1}}{2}, \quad \sin \theta = \frac{z - z^{-1}}{2i}.$$

A trigonometric integral over $[0, 2\pi]$ thus becomes a contour integral over the unit circle, computable by the residue theorem. In this light, Corollary 2.1 on the vortex field is the formula

$$\frac{1}{2\pi i} \oint_{C_R} \frac{dz}{z} = 1,$$

the elementary building block of Cauchy's theory. The full development — holomorphic functions, Cauchy's theorem, residues — is in **Chapter 3**, to which we refer in order not to duplicate the treatment.

2.11. Periodic quadrature: the exponentially convergent trapezoidal rule

An integral over the whole circle is an integral of a *periodic* function. This apparently harmless fact produces the most surprising result of the chapter from the numerical point of view: on the circle, the simplest of all quadrature formulas — the trapezoidal rule — becomes the most accurate one could wish for.

Definition 2.8: Periodic trapezoidal rule. Let f be continuous and 2π -periodic. For $N \geq 1$ set

$$T_N[f] = \frac{2\pi}{N} \sum_{k=0}^{N-1} f\left(\frac{2\pi k}{N}\right),$$

an approximation of $I[f] = \int_0^{2\pi} f(\theta) \, d\theta$.

Remark 2.13: Trapezoids, rectangles or midpoints: on the circle they coincide. For a periodic function sampled at equispaced nodes the trapezoidal rule coincides with the rectangle rule: the two half-weights at the endpoints, $\frac{1}{2}f(0)$ and $\frac{1}{2}f(2\pi)$, add up to $f(0)$ because $f(2\pi) = f(0)$. On the circle there are no “endpoints”: every node is interior. It is exactly this absence of boundary that structurally explains the theorem below.

Theorem 2.16: Spectral convergence for smooth periodic functions. Let f be of class C^m on $\mathbb{R}$ and 2π -periodic, $m \geq 2$. Then

$$|T_N[f] - I[f]| = O(N^{-m}) \quad (N \rightarrow \infty).$$

Proof. The Euler–Maclaurin formula for the trapezoidal rule on $[a, b]$ with step $h = (b - a)/N$ states

$$T_N[f] - I[f] = \sum_{j=1}^{\lfloor m/2 \rfloor} \frac{B_{2j}}{(2j)!} h^{2j} [f^{(2j-1)}(b) - f^{(2j-1)}(a)] + R_m,$$

where B_{2j} are the Bernoulli numbers and $|R_m| \leq C_m h^m \int_a^b |f^{(m)}|$.

Here $a = 0$, $b = 2\pi$ and f is 2π -periodic of class C^m : hence *all* derivatives up to order $m - 1$ take the same value at the endpoints,

$$f^{(2j-1)}(2\pi) = f^{(2j-1)}(0) \quad \text{for every } j,$$

and every term of the sum vanishes. Only the remainder survives:

$$|T_N[f] - I[f]| = |R_m| \leq C_m h^m \int_0^{2\pi} |f^{(m)}| = C'_m N^{-m}.$$

□

Remark 2.14: Accuracy that depends only on smoothness. Theorem 2.16 is remarkable: the order of convergence is not fixed by the method (like the 2 of the ordinary trapezoidal rule or the 4 of Simpson's), but by the smoothness of the function. The smoother f is, the faster the convergence — and if $f \in C^\infty$ the convergence is faster than any power of $1/N$. The next theorem shows that if f is in fact analytic, the convergence is geometric.

Theorem 2.17: Exponential convergence (Trefethen–Weideman). Let u be analytic in the disc $\{|z| < r\}$ with $r > 1$, and bounded there by $|u(z)| \leq M$. Setting

$$I = \int_0^{2\pi} u(e^{i\theta}) d\theta, \quad I_N = \frac{2\pi}{N} \sum_{k=1}^N u(z_k), \quad z_k = e^{2\pi i k/N},$$

one has, for every $N \geq 1$,

$$|I_N - I| \leq \frac{2\pi M}{r^N - 1},$$

and the constant 2π cannot be improved.

Proof. Step 1 — Taylor expansion. Being analytic in $\{|z| < r\}$, u has there a convergent power series

$$u(z) = \sum_{j=0}^{\infty} c_j z^j,$$

uniformly on every compact subset of the disc — in particular on the unit circle, which is compact and contained in $\{|z| < r\}$ since $r > 1$. This licenses the interchange of series with integral (and with finite sum) in the following steps.

Step 2 — computing I . Substituting $z = e^{i\theta}$,

$$I = \sum_{j \geq 0} c_j \int_0^{2\pi} e^{ij\theta} d\theta = 2\pi c_0,$$

since $\int_0^{2\pi} e^{ij\theta} d\theta = 0$ for $j \geq 1$ and $= 2\pi$ for $j = 0$ (Theorem 2.15).

Step 3 — computing I_N , and aliasing.

$$I_N = 2\pi \sum_{j \geq 0} c_j \left(\frac{1}{N} \sum_{k=1}^N e^{2\pi i j k / N} \right).$$

The inner sum is a geometric progression of ratio $q = e^{2\pi i j / N}$. If N divides j , then $q = 1$ and the sum equals N , i.e. 1 after division; otherwise $q \neq 1$ and

$$\sum_{k=1}^N q^k = q \frac{q^N - 1}{q - 1} = 0,$$

because $q^N = e^{2\pi i j} = 1$. Hence only the indices that are multiples of N survive:

$$I_N = 2\pi \sum_{m \geq 0} c_{mN} = 2\pi (c_0 + c_N + c_{2N} + \cdots).$$

Sampling at N points cannot distinguish z^{mN} from z^0 : this is *aliasing*, and it is the sole source of error.

Step 4 — the error. Subtracting Step 2 from Step 3,

$$I_N - I = 2\pi \sum_{m \geq 1} c_{mN}.$$

Step 5 — Cauchy estimate on the coefficients. For every ρ with $1 < \rho < r$, Cauchy's integral formula for the coefficients gives

$$|c_j| = \left| \frac{1}{2\pi i} \oint_{|z|=\rho} \frac{u(z)}{z^{j+1}} dz \right| \leq \frac{1}{2\pi} \cdot 2\pi\rho \cdot \frac{M}{\rho^{j+1}} = \frac{M}{\rho^j}.$$

Letting $\rho \rightarrow r^-$ we obtain $|c_j| \leq Mr^{-j}$.

Step 6 — geometric sum. Combining Steps 4 and 5,

$$|I_N - I| \leq 2\pi M \sum_{m \geq 1} r^{-mN} = 2\pi M \frac{r^{-N}}{1 - r^{-N}} = \frac{2\pi M}{r^N - 1},$$

using the geometric series with ratio $r^{-N} < 1$.

Step 7 — optimality of the constant. Consider $u(z) = Mz^N/r^N$, which satisfies $|u(z)| \leq M$ for $|z| < r$. For it $c_0 = 0$, hence $I = 0$, while the only non-zero coefficient is $c_N = M/r^N$, so $I_N = 2\pi M/r^N$. The actual error is then $2\pi M/r^N$, which compared with the estimate $2\pi M/(r^N - 1)$ gives

$$\frac{r^N - 1}{r^N} \xrightarrow{N \rightarrow \infty} 1 :$$

the estimate is asymptotically sharp and the constant 2π cannot be reduced. $\square$

Remark 2.15: The annulus variant. If u is analytic and bounded by M in the annulus $r^{-1} < |z| < r$ rather than in the disc (a frequent situation, since the substitution $z = e^{i\theta}$ often produces functions with poles both inside and outside the unit circle), the same argument applied to the Laurent series — which also has negative-index coefficients — doubles the number of surviving terms and the estimate becomes

$$|I_N - I| \leq \frac{4\pi M}{r^N - 1},$$

again with optimal constant. The qualitative conclusion is unchanged: the error decays like r^{-N} , where r measures how far from the unit circle the analyticity extends.

Example 2.8: Geometric convergence in practice. Approximate $I = \int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$ by the periodic trapezoidal rule, and compare with the theoretical prediction.

Step 1 — exact value. It is known (and follows from the Weierstrass substitution of Chapter 1, Corollary 1.2) that $\int_0^{2\pi} \frac{d\theta}{a + \cos \theta} = \frac{2\pi}{\sqrt{a^2 - 1}}$ for $a > 1$. With $a = 2$:

$$I = \frac{2\pi}{\sqrt{3}} = 3.6275987 \dots$$

Step 2 — identifying r . Setting $z = e^{i\theta}$, so that $\cos \theta = (z + z^{-1})/2$,

$$\frac{1}{2 + \cos \theta} = \frac{2z}{z^2 + 4z + 1}.$$

The poles are the roots of $z^2 + 4z + 1 = 0$, namely $z = -2 \pm \sqrt{3}$:

$$z_1 = -0.26795, \quad z_2 = -3.73205.$$

The function is therefore analytic in the annulus $|z_1| < |z| < |z_2|$, i.e. $r^{-1} < |z| < r$ with $r = 2 + \sqrt{3} = 3.73205$ (note $z_1 z_2 = 1$: the annulus is exactly symmetric, as required by Observation 2.15). The error must therefore decay like r^{-N} , gaining a factor of 3.73 for each added node.

Step 3 — computation for $N = 4$. The nodes are $\theta = 0, \pi/2, \pi, 3\pi/2$, with $\cos \theta = 1, 0, -1, 0$:

$$f = \frac{1}{3}, \frac{1}{2}, 1, \frac{1}{2}, \quad \sum f = 2.333333,$$

$$T_4 = \frac{2\pi}{4} \cdot 2.333333 = 1.5707963 \times 2.333333 = 3.6651914.$$

Error: $|T_4 - I| = 3.7593 \times 10^{-2}$.

Step 4 — computation for $N = 8$. The nodes are $\theta = k\pi/4$, $k = 0, \dots, 7$, with

$$\cos \theta = 1, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2}.$$

The values of f are

$$0.333333, 0.369399, 0.500000, 0.773459, 1.000000, 0.773459, 0.500000, 0.369399,$$

summing to 4.619048. Hence

$$T_8 = \frac{2\pi}{8} \times 4.619048 = 0.7853982 \times 4.619048 = 3.6277915,$$

with error $|T_8 - I| = 1.928 \times 10^{-4}$.

Step 5 — checking the rate. The ratio of the two errors is

$$\frac{3.7593 \times 10^{-2}}{1.928 \times 10^{-4}} = 195.0,$$

to be compared with the prediction $r^4 = (2 + \sqrt{3})^4 = 193.99$. The agreement is excellent (the small discrepancy is the effect of the -1 in the denominator of the estimate and of the constant M). Extrapolating, for $N = 16$ one expects an error of order $1.928 \times 10^{-4}/194 \approx 9.9 \times 10^{-7}$, and for $N = 32$ of order 5×10^{-12} .

Comparison. A fixed-order method such as Simpson's ($O(h^4)$) would gain a factor $2^4 = 16$ in going from $N = 4$ to $N = 8$, not 195. The difference — fixed order versus geometric rate — amplifies dramatically as N grows.

Best practice. If the integrand is periodic and smooth, **use the trapezoidal rule with equispaced nodes**: it is simultaneously the simplest method to implement and the most accurate available. More sophisticated formulas (Simpson, Gauss–Legendre) not only fail to improve the result: they make it worse, because they break the equispacing that is the very source of the accuracy. The advantage of Gauss–Legendre exists only for *non*-periodic integrands on intervals with endpoints.

2.12. Gauss–Legendre quadrature

The good practice just stated points to a method not yet defined. This section introduces it, showing that its strength is precisely complementary to that of periodic trapezoids: it is built exactly for intervals *with* endpoints.

Definition 2.9: Quadrature formula with n nodes. A quadrature formula with n nodes on $[-1, 1]$ is a pair of lists $(x_1, \dots, x_n)$, $(w_1, \dots, w_n)$ such that

$$\int_{-1}^1 f(x) dx \approx \sum_{i=1}^n w_i f(x_i).$$

It is called **exact of degree d** if the approximation is an equality for every polynomial of degree $\leq d$.

Theorem 2.18: Two-node Gauss–Legendre quadrature. *The formula with nodes $x_{1,2} = \mp 1/\sqrt{3}$ and weights $w_1 = w_2 = 1$ is exact of degree 3, and is the unique symmetric two-node formula with this property.*

Proof. We look for a symmetric formula, $x_{1,2} = \mp a$, $w_1 = w_2 = w$, exact on the monomials $1, x, x^2, x^3$ (the method of undetermined coefficients). For $f(x) = x$ and $f(x) = x^3$ exactness is automatic: both the exact value $\int_{-1}^1 x dx = \int_{-1}^1 x^3 dx = 0$ and the sum $w(-a) + wa = 0$ (resp. $w(-a)^3 + wa^3 = 0$) vanish by symmetry, whatever a, w are. Two conditions remain:

$$f = 1: \quad 2w = \int_{-1}^1 1 dx = 2 \Rightarrow w = 1; \quad f = x^2: \quad 2wa^2 = \int_{-1}^1 x^2 dx = \frac{2}{3} \Rightarrow a^2 = \frac{1}{3}.$$

Hence $w = 1$, $a = 1/\sqrt{3}$, uniquely determined. The formula is therefore exact on the four monomials $1, x, x^2, x^3$, hence (by linearity of the integral and of the sum) on every linear combination of them, that is, on every polynomial of degree ≤ 3 . $\square$

Theorem 2.19: Three-node Gauss–Legendre quadrature. *The formula with nodes $x_1 = -\sqrt{3/5}$, $x_2 = 0$, $x_3 = \sqrt{3/5}$ and weights $w_1 = w_3 = 5/9$, $w_2 = 8/9$ is exact of degree 5.*

Proof. Again by symmetry ($x_{1,3} = \mp b$, $x_2 = 0$, $w_1 = w_3 = w$, $w_2 = w_0$), exactness on the odd monomials x, x^3, x^5 is automatic. Three conditions remain, on the even monomials $1, x^2, x^4$:

$$f = 1: \quad w_0 + 2w = 2; \quad f = x^2: \quad 2wb^2 = \frac{2}{3}; \quad f = x^4: \quad 2wb^4 = \frac{2}{5}.$$

Dividing the third equation by the second, $b^2 = (2/5)/(2/3) = 3/5$, so $b = \sqrt{3/5}$. Substituting into the second, $w = \frac{1/3}{b^2} = \frac{1/3}{3/5} = \frac{5}{9}$. Substituting into the first, $w_0 = 2 - 2 \cdot \frac{5}{9} = \frac{8}{9}$. The formula is exact on the six monomials $1, x, \dots, x^5$, hence on every polynomial of degree ≤ 5 . $\square$

Remark 2.16: The general pattern, cited. The pattern is not accidental: it is a theorem (the general theory of Gaussian quadrature, not reproduced here in full) that the optimal n nodes are the roots of the Legendre polynomial P_n , and that the resulting formula is always exact of degree $2n - 1$ — twice what would suffice for interpolation alone ($n - 1$). This is the theoretical maximum achievable with n free nodes: no choice of n nodes and n weights can do better, because a polynomial of degree $2n$ has $2n + 1$ coefficients against the $2n$ unknowns available. The cases $n = 2, 3$ above are the direct verification of the first two steps of this general scheme.

Example 2.9: Gauss–Legendre versus the trapezoidal rule on a smooth non-periodic integrand. Compare the preceding formulas with the (non-periodic) trapezoidal rule, with halved weights at the endpoints, on the integral $I = \int_{-1}^1 e^x dx = e - e^{-1} = 2.350402$.

Gauss–Legendre, 2 evaluations. $e^{-1/\sqrt{3}} + e^{1/\sqrt{3}} = 2.342696$ (summed with weights 1): error 7.71×10^{-3} .

Gauss–Legendre, 3 evaluations. $\frac{5}{9}e^{-\sqrt{3/5}} + \frac{8}{9}e^0 + \frac{5}{9}e^{\sqrt{3/5}} = 2.350337$: error 6.55×10^{-5} , nearly two orders of magnitude better with just one more evaluation.

Trapezoidal rule, same number of evaluations. With 2 nodes (the endpoints ± 1): error 7.36×10^{-1} . With 9 equispaced nodes (four times the evaluations of the three-node Gauss–Legendre rule): error 1.22×10^{-2} — still two orders of magnitude worse than the three-node formula.

Checking the earlier good practice, on the periodic integrand from the chapter (Example 2.8). Applying Gauss–Legendre with 8 nodes (remapped from $[-1, 1]$ to $[0, 2\pi]$) to the integrand $1/(2 + \cos \theta)$ gives an error of 6.99×10^{-3} , while the periodic trapezoidal rule with the same number of nodes ($N = 8$, already computed in the cited example) has error 1.93×10^{-4} : the periodic trapezoidal rule is here 36 times more accurate, exactly the situation anticipated in the earlier good practice. On a smooth but *non*-periodic integrand the opposite holds, as just seen above: neither formula is universally better, each wins in its own domain.

Independent numerical check. All the values in this example, the nodes and weights of the preceding Definitions (also checked against the roots of the Legendre polynomial computed independently), and the check that exactness fails beyond the stated degree (degree 4 for $n = 2$, degree 6 for $n = 3$) are in the script `Sorgenti/Vol2/_verifiche/vol2_cap02_gauss_legendre.py`.

2.13. Numerical laboratory

The circle, for which exact formulas for length and area are known, is the ideal test bench for a numerical method: the approximate result can be compared with the exact value and the error measured directly.

2.13.1 When standard quadrature fails: the boundary of the circle

Definition 2.10: Composite Simpson rule. Dividing $[a, b]$ into n (even) subintervals of width $h = (b - a)/n$,

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}, 0 < i < n} f(x_i) + f(x_n) \right].$$

Remark 2.17: Simpson's error and its limitation on the circle. If $f \in C^4([a, b])$, the error of the composite rule is $-\frac{(b-a)h^4}{180}f^{(4)}(\xi)$ for some $\xi \in [a, b]$: it decreases like h^4 , much faster than the ordinary trapezoidal rule ($O(h^2)$). Applied, however, to $f(x) = \sqrt{R^2 - x^2}$ on $[-R, R]$ in order to estimate the area of the half-disc, the hypothesis $f \in C^4$ fails at the endpoints: the tangent is vertical at $x = \pm R$ and the derivatives blow up. The theoretical $O(h^4)$ order is not attained, and the effective convergence is far slower. This is the exact opposite of the situation in Section 2.11: there the circle, parametrised in θ , has no boundary and quadrature is essentially perfect; here the circle, described in x , has a singular boundary and quadrature degrades.

Best practice. Operational moral: **integrate over the circle using the angle, not the abscissa.** The change of variable $x = R \sin u$ (or directly the angular parametrisation) removes the boundary singularity and returns the problem to the periodic-analytic class, where the trapezoidal rule is exact up to exponentially small errors.

2.13.2 The Monte Carlo method

Theorem 2.20: Monte Carlo estimator of π . Let (X_i, Y_i) , $i = 1, \dots, N$, be independent points uniform in $[-1, 1]^2$, and let $S = \#\{i : X_i^2 + Y_i^2 \leq 1\}$. Then $\hat{\pi} = 4S/N$ is an unbiased estimator of π , with

$$\sigma(\hat{\pi}) = 4\sqrt{\frac{p(1-p)}{N}}, \quad p = \frac{\pi}{4}.$$

Proof. Each point falls in the unit disc with probability equal to the ratio of the areas:

$$p = \frac{\text{Area}(D(O, 1))}{\text{Area}([-1, 1]^2)} = \frac{\pi}{4},$$

using $\text{Area}(D) = \pi$ (Example 2.4 with $R = 1$) and $\text{Area}([-1, 1]^2) = 4$. Since the trials are independent with the same success probability, $S \sim \text{Bin}(N, p)$, so $E[S] = Np$ and

$$E[\hat{\pi}] = \frac{4}{N} Np = 4p = \pi :$$

the estimator is unbiased. The variance of a binomial is $Np(1-p)$, hence

$$\text{Var}(\hat{\pi}) = \frac{16p(1-p)}{N},$$

whence the stated standard deviation. □

Remark 2.18: Why $1/\sqrt{N}$, and when it is nonetheless worth it. The typical error decreases like $N^{-1/2}$: gaining one decimal digit requires 100 times as many samples. This is incomparably slower than the geometric rate of Section 2.11. The merit of Monte Carlo is therefore not speed in low dimension — where it is the worst choice — but the fact that the exponent $1/2$ does not depend on the dimension of the space, whereas deterministic quadratures degrade dramatically as it grows (“curse of dimensionality”). For an integral in $\mathbb{R}^2$ such as this one, quadrature remains vastly superior; Monte Carlo becomes competitive, and often the only option, from a few tens of dimensions onwards.

Listing 2.1: Numerical verification of the chapter’s three key claims: geometric convergence of the periodic trapezoidal rule, circulation of the vortex field, Green’s area formula.

```
import numpy as np

# --- 1. Periodic trapezoidal rule: geometric convergence (Trefethen-
#         Weideman) ---
I_exact = 2*np.pi/np.sqrt(3.0)
print("N      T_N      error      ratio")
prev = None
for N in (4, 8, 16, 32, 64):
    th = 2*np.pi*np.arange(N)/N
    T = (2*np.pi/N)*np.sum(1.0/(2.0 + np.cos(th)))
    err = abs(T - I_exact)
    rat = "-" if prev is None else f"{prev/err:10.1f}"
    print(f"{N:3d} {T:.14f} {err:.3e} {rat}")
    prev = err
print("theoretical ratio (N -> 2N):", (2+np.sqrt(3))*4)

# --- 2. Vortex field: circulation 2*pi for every radius ---
def vortex_circulation(R, N=20000):
    th = 2*np.pi*np.arange(N)/N
    x, y = R*np.cos(th), R*np.sin(th)
    r2 = x**2 + y**2
    Vx, Vy = -y/r2, x/r2          # vortex field
    dx, dy = -R*np.sin(th), R*np.cos(th)  # gamma'(theta)
    return (2*np.pi/N)*np.sum(Vx*dx + Vy*dy)

for R in (0.1, 1.0, 37.0):
    print(f"R = {R:6.2f} -> circulation = {vortex_circulation(R):.12f}")

# --- 3. Green's area formula on an ellipse (a=3, b=2): expected pi*a*b
#         ---
a, b, N = 3.0, 2.0, 200000
th = 2*np.pi*np.arange(N)/N
x, y = a*np.cos(th), b*np.sin(th)
dx, dy = -a*np.sin(th), b*np.cos(th)
area = 0.5*(2*np.pi/N)*np.sum(x*dy - y*dx)
print(f"area (Green) = {area:.12f}, expected = {np.pi*a*b:.12f}")
```

Remark 2.19: What to expect. The first block reproduces the table of Example 2.8: the ratio of successive errors settles around $194 = (2 + \sqrt{3})^4$ until the error reaches machine epsilon ($\approx 10^{-16}$), after which the ratios become erratic — not a defect of the method, but the floor of finite arithmetic. The second block returns 2π to full precision for all three radii, confirming Theorem 2.8(ii). The third returns $\pi ab = 18.849556\dots$: Green’s formula holds for every regular Jordan curve, not only for the circle.

Remark 2.20: Historical note: Liu Hui and Zu Chongzhi. Archimedes’ method (Volume I) — approximating the circle by inscribed and circumscribed regular polygons and repeatedly doubling the number of sides — has an independent counterpart in Chinese mathematics. In 263 CE Liu Hui developed an algorithm of dissection of inscribed polygons (ge yuan shu, “circle-cutting method”), reaching a 3072-sided polygon and the estimate $\pi \approx 3.1416$. About two centuries later, Zu Chongzhi (429–501) extended the computation to a 24576-sided polygon, obtaining $3.1415926 < \pi < 3.1415927$ — seven correct decimal places

— and the remarkable rational approximation $355/113$ (Milü), accurate to six decimals. The record stood for over nine hundred years.

Liu Hui, Zu Chongzhi and Archimedes share the same paradigm: a deterministic algorithm converging in a certain, monotone way. Monte Carlo represents the opposite, stochastic paradigm: no single run guarantees a small error, but the law of large numbers guarantees convergence in probability. The two paradigms coexist in modern numerical computation, and the choice between them — seen here in the simplest possible case — recurs identically in problems of far higher dimension.

2.14. Numerical exercises

All exercises are worked out in full in Section 2.16.

Exercise 2.1: N1 — Centroid of a ring of variable density. A circular wire of radius R centred at the origin has linear density $\lambda(\theta) = \lambda_0(1 + \cos \theta)$, with $\lambda_0 > 0$. Compute its total mass and the coordinates of its centroid.

Exercise 2.2: N2 — Mean value on an arc. On the arc of the circle of centre $(1, 2)$ and radius $R = 4$ between $\theta_0 = 0$ and $\theta_1 = \pi/2$, compute the length, the mean value of the ordinate y and the coordinates of the centroid (constant density).

Exercise 2.3: N3 — Circulation directly and by Green. Compute $\oint_C [(x^2 - y) dx + (x + y^2) dy]$, where C is the circle of centre the origin and radius 2 traversed counterclockwise, first by parametrising and then using Green's theorem.

Exercise 2.4: N4 — Flux by the divergence theorem. Compute the outward flux of the field $\mathbf{F}(x, y) = (x^3, y^3)$ through the circle $C(O, R)$, using Corollary 2.2, and verify the result for $R = 1$ by direct computation.

Exercise 2.5: N5 — Area by Green's formula. Use Green's formula to compute the area enclosed by the ellipse $\gamma(t) = (a \cos t, b \sin t)$, $t \in [0, 2\pi]$, and evaluate it for $a = 5$, $b = 3$. Check that for $a = b = R$ one recovers πR^2 .

Exercise 2.6: N6 — The vortex field on two different circles. Compute the circulation of the vortex field $\mathbf{V}$ of Theorem 2.8 along (a) the circle of centre the origin and radius 5; (b) the circle of centre $(3, 0)$ and radius 1. Explain the difference between the two results without explicitly computing the second integral.

Exercise 2.7: N7 — Weierstrass substitution. Compute $\int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta}$ by the substitution $u = \tan(\theta/2)$, justifying the treatment of the point $\theta = \pi$, and compare with the general formula $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}}$ for $a > |b|$.

Exercise 2.8: N8 — Periodic quadrature of an analytic integrand. Approximate $\int_0^{2\pi} e^{\cos \theta} d\theta$ by the periodic trapezoidal rule for $N = 4$ and $N = 8$, given that the exact value is $2\pi I_0(1) = 7.9549265$ (I_0 is the modified Bessel function of the first kind). Comment

on the rate of convergence in the light of Theorem 2.17.

Exercise 2.9: N9 — Sizing a Monte Carlo sample. Using Theorem 2.20, determine how many samples N are needed for the standard deviation of the estimator $\hat{\pi}$ to fall below 0.01, and how many to fall below 0.001.

Exercise 2.10: N10 — Circular clarifier. A plant must treat $Q = 3600 \text{ m}^3/\text{d}$ with the limit $\text{SOR} \leq 1.0 \text{ m/h}$. Determine the minimum radius of the clarifier and check whether the resulting weir loading rate respects the limit of $100 \text{ m}^3/(\text{m d})$.

2.15. Symbolic exercises

Exercise 2.11: S1 — The mean value property on small circles. Let f be continuous in a neighbourhood of the point P_0 . Prove that

$$\lim_{R \rightarrow 0^+} \frac{1}{2\pi R} \oint_{C(P_0, R)} f \, ds = f(P_0).$$

Exercise 2.12: S2 — Characterisation of conservative fields. Let $A \subseteq \mathbb{R}^2$ be open and connected and $\mathbf{F}$ continuous on A . Prove that $\mathbf{F}$ is conservative on A if and only if $\oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed piecewise regular curve $\Gamma \subseteq A$. (The “only if” is Theorem 2.5; prove the “if”.)

Exercise 2.13: S3 — From Green’s formula to the shoelace formula. Prove that applying Corollary 2.3 to a simple polygon with vertices $P_1, \dots, P_n, P_{n+1} = P_1$ traversed counterclockwise yields the *shoelace formula*

$$\text{Area} = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i).$$

Exercise 2.14: S4 — The two forms of Green’s theorem are equivalent. Prove that Corollary 2.2 (normal form) and Theorem 2.10 (tangential form) follow from each other by replacing $\mathbf{F}$ with $J\mathbf{F}$, where J is the 90° rotation of Chapter 1. Verify in particular that $\text{div}(J\mathbf{F}) = -\text{curl } \mathbf{F}$ and $\text{curl}(J\mathbf{F}) = \text{div } \mathbf{F}$.

Exercise 2.15: S5 — The vortex circulation does not depend on the curve. Let Γ be a piecewise regular Jordan curve, oriented counterclockwise, enclosing the origin. Prove that

$$\oint_{\Gamma} \mathbf{V} \cdot d\mathbf{r} = 2\pi,$$

with $\mathbf{V}$ the vortex field, independently of the shape of Γ . (Hint: apply Green’s theorem to the region between Γ and a small circle centred at the origin.)

Exercise 2.16: S6 — Exactness of the trapezoidal rule on trigonometric polynomials. Prove that the rule T_N of Definition 2.8 is *exact* (zero error) for every trigonometric polynomial of degree strictly less than N , i.e. for every function of the form $f(\theta) = \sum_{|j| < N} c_j e^{ij\theta}$, and that it is not exact for $e^{iN\theta}$.

Exercise 2.17: S7 — Area from the support function. Prove that for every region D satisfying the hypotheses of Theorem 2.10

$$\oint_{\partial D} (\mathbf{p} \cdot \mathbf{n}) \, ds = 2 \operatorname{Area}(D),$$

where $\mathbf{p} = (x, y)$ is the radius vector. Verify it on the disc $D(O, R)$ and interpret it geometrically.

Exercise 2.18: S8 — Green's first identity and harmonic functions. Let ϕ be of class C^2 on an open set containing $\overline{D}$, with D satisfying the hypotheses of Theorem 2.10. Prove the identity

$$\oint_{\partial D} \frac{\partial \phi}{\partial n} \, ds = \iint_D \Delta \phi \, dA, \quad \frac{\partial \phi}{\partial n} = \nabla \phi \cdot \mathbf{n}, \quad \Delta \phi = \phi_{xx} + \phi_{yy},$$

and deduce that if ϕ is harmonic ($\Delta \phi = 0$) the flux of its gradient through every circle contained in the domain vanishes.

Exercise 2.19: S9 — The integral of the normal vanishes. Prove that for every closed piecewise regular curve Γ

$$\oint_{\Gamma} \mathbf{n} \, ds = \mathbf{0}$$

(an equality between vectors), and deduce that a uniform pressure applied to the boundary of a body produces no resultant force.

Exercise 2.20: S10 — The fundamental trigonometric integral. Prove that for $a > |b|$

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}},$$

using the Weierstrass substitution and carefully justifying the limiting procedure at the point $\theta = \pi$, where the substitution is undefined.

2.16. Solutions

2.16.1 Solutions to the numerical exercises

Solution. N1. *Step 1 — setting up.* With $\gamma(\theta) = R(\cos \theta, \sin \theta)$ and $ds = R \, d\theta$ (Theorem 2.1(iv)), the mass is

$$m = \oint_{\Gamma} \lambda \, ds = \int_0^{2\pi} \lambda_0(1 + \cos \theta) R \, d\theta.$$

Step 2 — the mass.

$$m = \lambda_0 R [2\pi + 0] = 2\pi \lambda_0 R,$$

using $\int_0^{2\pi} \cos \theta \, d\theta = 0$ (Theorem 2.15). Note that the mass is the same as that of a homogeneous wire of density λ_0 : the modulation $\cos \theta$ redistributes mass, it does not add any.

Step 3 — moment about the y-axis.

$$\oint x \lambda \, ds = \lambda_0 R^2 \left[\int_0^{2\pi} \cos \theta \, d\theta + \int_0^{2\pi} \cos^2 \theta \, d\theta \right] = \lambda_0 R^2 [0 + \pi] = \pi \lambda_0 R^2,$$

the second integral being π by Theorem 2.15 (case $m = n = 1$).

Step 4 — moment about the x -axis.

$$\oint y \lambda \, ds = \lambda_0 R^2 \left[\int_0^{2\pi} \sin \theta \, d\theta + \int_0^{2\pi} \sin \theta \cos \theta \, d\theta \right] = 0,$$

the second integral vanishing by the last relation of Theorem 2.15 with $m = n = 1$.

Step 5 — centroid.

$$\bar{x} = \frac{\pi \lambda_0 R^2}{2\pi \lambda_0 R} = \frac{R}{2}, \quad \bar{y} = 0, \quad \text{that is } G = \left(\frac{R}{2}, 0 \right).$$

Step 6 — plausibility. The density is maximal at $\theta = 0$ (the point $(R, 0)$, where $\lambda = 2\lambda_0$) and zero at $\theta = \pi$ (the point $(-R, 0)$): the centroid must shift to the right of the centre, and indeed $\bar{x} = R/2 > 0$. The symmetry of the density about the x -axis ($\lambda(-\theta) = \lambda(\theta)$) forces $\bar{y} = 0$. ✓

Solution. N2. *Step 1 — length.* $\Delta\theta = \pi/2$, hence

$$L = R \Delta\theta = 4 \cdot \frac{\pi}{2} = 2\pi \approx 6.2832.$$

Step 2 — mean value of y . On the arc, $y(\theta) = 2 + 4 \sin \theta$. By Theorem 2.3(ii),

$$\bar{y} = \frac{2}{\pi} \left[2\theta - 4 \cos \theta \right]_0^{\pi/2} = \frac{2}{\pi} \left[(\pi - 0) - (0 - 4) \right] = \frac{2}{\pi} (\pi + 4) = 2 + \frac{8}{\pi} = 2 + 2.5465 = 4.5465.$$

Step 3 — mean value of x . With $x(\theta) = 1 + 4 \cos \theta$:

$$\bar{x} = \frac{2}{\pi} \left[\theta + 4 \sin \theta \right]_0^{\pi/2} = \frac{2}{\pi} \left(\frac{\pi}{2} + 4 \right) = 1 + \frac{8}{\pi} = 3.5465.$$

Step 4 — check with the general formula. Theorem 2.3(iii) gives

$$G = (1, 2) + \frac{4}{\pi/2} (1 - 0, 1 - 0) = (1, 2) + \frac{8}{\pi} (1, 1) = (3.5465, 4.5465). \checkmark$$

Step 5 — geometric check. The arc runs from $(5, 2)$ (at $\theta = 0$) to $(1, 6)$ (at $\theta = \pi/2$) along the quarter-circle in the first quadrant relative to the centre. The centroid must lie inside the triangle formed by the centre and the two endpoints, and closer to the arc than to the centre: indeed $|G - C_0| = \sqrt{2 \cdot 2.5465^2} = 3.601 < 4 = R$. ✓

Solution. N3. *Method 1 — direct computation.* With $\gamma(t) = (2 \cos t, 2 \sin t)$, $t \in [0, 2\pi]$: $dx = -2 \sin t \, dt$, $dy = 2 \cos t \, dt$, and on the curve

$$P = 4 \cos^2 t - 2 \sin t, \quad Q = 2 \cos t + 4 \sin^2 t.$$

Hence

$$P \, dx = (-8 \cos^2 t \sin t + 4 \sin^2 t) \, dt, \quad Q \, dy = (4 \cos^2 t + 8 \sin^2 t \cos t) \, dt.$$

Adding and integrating over $[0, 2\pi]$, the two “mixed” terms vanish:

$$\int_0^{2\pi} \cos^2 t \sin t \, dt = \left[-\frac{\cos^3 t}{3} \right]_0^{2\pi} = 0, \quad \int_0^{2\pi} \sin^2 t \cos t \, dt = 0,$$

while $\int_0^{2\pi} \sin^2 t \, dt = \int_0^{2\pi} \cos^2 t \, dt = \pi$ (Theorem 2.15). There remains

$$\oint = 4\pi + 4\pi = 8\pi \approx 25.133.$$

Method 2 — Green's theorem. With $P = x^2 - y$ and $Q = x + y^2$:

$$\frac{\partial Q}{\partial x} = 1, \quad \frac{\partial P}{\partial y} = -1, \quad \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2.$$

By Theorem 2.10,

$$\oint_C (P \, dx + Q \, dy) = \iint_{D(O,2)} 2 \, dA = 2\pi \cdot 2^2 = 8\pi. \checkmark$$

Remark. The second method takes three lines against the ten of the first: this is the practical reason why Green's theorem is nearly always used from right to left. Note also that the terms x^2 and y^2 , visually dominant in the integrand, contribute nothing: they are the gradients of $x^3/3$ and $y^3/3$, hence conservative, and by Theorem 2.5 their circulation vanishes.

Solution. N4. *Step 1 — divergence.* $\operatorname{div} \mathbf{F} = 3x^2 + 3y^2 = 3\rho^2$.

Step 2 — divergence theorem. By Corollary 2.2 and the change to polar coordinates (Theorem 2.11),

$$\Phi = \iint_{D(O,R)} 3\rho^2 \, dA = 3 \int_0^{2\pi} \int_0^R \rho^3 \, d\rho \, d\varphi.$$

Step 3 — computation.

$$\int_0^R \rho^3 \, d\rho = \frac{R^4}{4}, \quad \Phi = 3 \cdot 2\pi \cdot \frac{R^4}{4} = \frac{3\pi R^4}{2}.$$

Step 4 — direct verification for $R = 1$. On the unit circle $\mathbf{n} = (\cos \theta, \sin \theta)$ and $ds = d\theta$, hence

$$\Phi = \int_0^{2\pi} (\cos^4 \theta + \sin^4 \theta) \, d\theta.$$

To compute $\int_0^{2\pi} \cos^4 \theta \, d\theta$, apply the half-angle formula twice:

$$\cos^4 \theta = \frac{1 + 2 \cos 2\theta + \frac{1 + \cos 4\theta}{2}}{4},$$

whose integral over a period is $\frac{1}{4}(2\pi + 0 + \pi) = \frac{3\pi}{4}$; likewise for $\sin^4$. Hence $\Phi = \frac{3\pi}{2}$, agreeing with Step 3 for $R = 1$. $\checkmark$

Solution. N5. *Step 1 — differentials.* From $\gamma(t) = (a \cos t, b \sin t)$: $dx = -a \sin t \, dt$, $dy = b \cos t \, dt$.

Step 2 — the symmetric formula. By Corollary 2.3,

$$\text{Area} = \frac{1}{2} \int_0^{2\pi} [ab \cos^2 t + ab \sin^2 t] \, dt.$$

Step 3 — simplification. The integrand is ab , constant by the Pythagorean identity, hence

$$\text{Area} = \frac{ab}{2} \cdot 2\pi = \pi ab.$$

Step 4 — numerical value. For $a = 5$, $b = 3$: $\text{Area} = 15\pi \approx 47.124$.

Step 5 — circular case. For $a = b = R$ one obtains πR^2 , consistent with Example 2.4. ✓
Remark. The formula πab can also be obtained by noting that the ellipse is the image of the unit circle under the linear map $\text{diag}(a, b)$, of determinant ab : areas are multiplied by the determinant, hence $\pi \cdot 1^2 \cdot ab = \pi ab$. The Green computation is the direct verification of this general principle.

Solution. N6. (a) *Circle of centre the origin and radius 5.* This is exactly the case of Theorem 2.8(ii) with $R = 5$:

$$\oint_{C(O,5)} \mathbf{V} \cdot d\mathbf{r} = 2\pi \approx 6.2832,$$

and the value does not depend on the radius.

(b) *Circle of centre (3, 0) and radius 1.* The result is 0, obtained *without* computing any integral, by the following argument.

Step 1 — the curve lies in a half-plane. Every point of $C((3, 0), 1)$ has abscissa between 2 and 4, hence strictly positive: the curve is entirely contained in the open half-plane $A = \{x > 0\}$.

Step 2 — on A the field is conservative. By Theorem 2.8(iv), on A we have $\mathbf{V} = \nabla\phi$ with $\phi(x, y) = \arctan(y/x)$. (Alternatively: A is convex, hence star-shaped, and $\mathbf{V}$ is irrotational by (i); Theorem 2.7 concludes.)

Step 3 — conclusion. The curve is closed and contained in A , where the field is conservative: by Theorem 2.5 the circulation vanishes.

Step 4 — why the two results differ. The field $\mathbf{V}$ is irrotational in *both* cases — the curl vanishes identically on all of $\mathbb{R}^2 \setminus \{0\}$. The difference lies not in the field but in the *position of the curve relative to the hole*: the first circle encloses the origin, the second does not. This is precisely the situation described by Corollary 2.1: the circulation equals $2\pi n$ with n the winding number about the origin, which is 1 in the first case and 0 in the second.

Solution. N7. *Step 1 — reduction to a symmetric interval.* The integrand is 2π -periodic and continuous (the denominator is at least $5 - 4 = 1 > 0$), hence its integral over any interval of length 2π is the same:

$$\int_0^{2\pi} \frac{d\theta}{5 + 4 \cos \theta} = \int_{-\pi}^{\pi} \frac{d\theta}{5 + 4 \cos \theta}.$$

Step 2 — the point $\theta = \pi$. The substitution $u = \tan(\theta/2)$ is defined and bijective from $(-\pi, \pi)$ onto $\mathbb{R}$, but not at the two endpoints. Since the integrand is bounded (between $1/9$ and 1) and the two endpoints form a null set, removing them does not change the value of the integral: we may integrate over the open interval.

Step 3 — substitution. By Chapter 1, Corollary 1.2:

$$\cos \theta = \frac{1 - u^2}{1 + u^2}, \quad d\theta = \frac{2 du}{1 + u^2},$$

and the endpoints $\theta \rightarrow \mp\pi^\pm$ become $u \rightarrow \mp\infty$.

Step 4 — simplifying the denominator.

$$5 + 4 \frac{1 - u^2}{1 + u^2} = \frac{5 + 5u^2 + 4 - 4u^2}{1 + u^2} = \frac{9 + u^2}{1 + u^2}.$$

Step 5 — the transformed integral.

$$\int_{-\infty}^{+\infty} \frac{1 + u^2}{9 + u^2} \cdot \frac{2 du}{1 + u^2} = \int_{-\infty}^{+\infty} \frac{2 du}{9 + u^2}.$$

The factor $1 + u^2$ cancels exactly: this is why the Weierstrass substitution works.

Step 6 — computation. Recalling $\int \frac{du}{u^2+c^2} = \frac{1}{c} \arctan \frac{u}{c}$:

$$\int_{-\infty}^{+\infty} \frac{2 du}{9 + u^2} = \frac{2}{3} \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \frac{2\pi}{3} \approx 2.0944.$$

Step 7 — comparison with the general formula. With $a = 5$, $b = 4$: $2\pi/\sqrt{25 - 16} = 2\pi/3$.

✓The proof of the general formula is the object of Exercise S10.

Solution. N8. *Step 1 — $N = 4$.* The nodes are $\theta_k = k\pi/2$, $k = 0, 1, 2, 3$, with $\cos \theta_k = 1, 0, -1, 0$. Hence

$$f(\theta_k) = 2.7182818, 1, 0.3678794, 1,$$

summing to 5.0861612. Therefore

$$T_4 = \frac{2\pi}{4} \times 5.0861612 = 1.5707963 \times 5.0861612 = 7.9893225,$$

with error $|T_4 - I| = 7.9893225 - 7.9549265 = 3.4396 \times 10^{-2}$.

Step 2 — $N = 8$. The nodes are $\theta_k = k\pi/4$, with

$$\cos \theta_k = 1, \frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}, -1, -\frac{\sqrt{2}}{2}, 0, \frac{\sqrt{2}}{2},$$

where $\sqrt{2}/2 = 0.7071068$. The values of $e^{\cos \theta_k}$ are

2.7182818, 2.0281142, 1.0000000, 0.4930693, 0.3678794, 0.4930693, 1.0000000, 2.0281142,

summing to 10.1285282. Therefore

$$T_8 = \frac{2\pi}{8} \times 10.1285282 = 0.7853982 \times 10.1285282 = 7.9549278,$$

with error $|T_8 - I| \approx 1.3 \times 10^{-6}$.

Step 3 — rate of convergence. The ratio of the two errors is

$$\frac{3.44 \times 10^{-2}}{1.3 \times 10^{-6}} \approx 2.7 \times 10^4,$$

enormously larger than the factor 16 a fourth-order method would give, and indeed larger than any constant ratio: the convergence is *faster than geometric*.

Step 4 — theoretical explanation. Setting $z = e^{i\theta}$, so that $\cos \theta = (z + z^{-1})/2$,

$$u(z) = \exp\left(\frac{z + z^{-1}}{2}\right),$$

which is analytic in the whole annulus $0 < |z| < \infty$: the radius r of Observation 2.15 may be taken *arbitrarily large*. The price is that the constant M grows: on the annulus $r^{-1} < |z| < r$ one has $M \approx e^{r/2}$. The estimate becomes

$$|T_N - I| \lesssim \frac{4\pi e^{r/2}}{r^N},$$

and one may *optimise over r* : setting the derivative of $\frac{r}{2} - N \ln r$ to zero gives $r = 2N$, which substituted back yields an error of order

$$\left(\frac{e}{2N}\right)^N.$$

For $N = 8$: $(e/16)^8 = (0.16995)^8 \approx 7 \times 10^{-7}$, of the same order of magnitude as the observed error (1.3×10^{-6} ; the factor of ~ 2 difference is the effect of the approximate estimate of the constant M). ✓

Moral. When the integrand is *entire* in z and $1/z$, the convergence of the periodic trapezoidal rule is super-geometric — of factorial type. This is the regime in which an integral is computed to full machine precision with about ten evaluations.

Solution. N9. *Step 1 — the formula.* From Theorem 2.20, $\sigma(\hat{\pi}) = 4\sqrt{p(1-p)/N}$ with $p = \pi/4 = 0.7853982$, hence

$$1 - p = 0.2146018, \quad p(1 - p) = 0.7853982 \times 0.2146018 = 0.1685577.$$

Step 2 — threshold 0.01. Squaring the condition $\sigma(\hat{\pi}) \leq \varepsilon$:

$$N \geq \frac{16p(1-p)}{\varepsilon^2} = \frac{2.696923}{\varepsilon^2}.$$

For $\varepsilon = 0.01$:

$$N \geq \frac{2.696923}{10^{-4}} = 26\,969.$$

Step 3 — threshold 0.001. For $\varepsilon = 0.001$:

$$N \geq \frac{2.696923}{10^{-6}} = 2\,696\,923 \approx 2.70 \times 10^6.$$

Step 4 — comment. About 27 000 samples are needed for the second decimal digit and about 2.7 million for the third: a factor of 100 per digit gained, as predicted by the $N^{-1/2}$ rate (Observation 2.18). By comparison, the periodic trapezoidal rule of Exercise N8 reaches six correct decimals with *eight* evaluations. The gap between the two methods, in two dimensions, is about six orders of magnitude.

Solution. N10. *Step 1 — flow conversion.* $Q = 3600/24 = 150 \text{ m}^3/\text{h}$.

Step 2 — minimum area. From the SOR constraint (Definition 2.7),

$$A \geq \frac{150}{1.0} = 150 \text{ m}^2.$$

Step 3 — minimum radius.

$$R \geq \sqrt{\frac{150}{\pi}} = \sqrt{47.746} = 6.910 \text{ m}, \quad D = 13.82 \text{ m}.$$

Step 4 — weir loading rate. The perimeter is $P = 2\pi R = 43.42 \text{ m}$, hence

$$\text{WLR} = \frac{150}{43.42} = 3.455 \text{ m}^3/(\text{m h}) = 3.455 \times 24 = 82.9 \text{ m}^3/(\text{m d}).$$

Step 5 — checking the constraint. $82.9 < 100$: the limit is met, with a margin of 17%. ✓

Step 6 — which constraint is active. The SOR is at its limit (by construction), the WLR is not: it is the SOR — that is, the *area* — that sizes the tank, as predicted by Observation 2.6. To verify this: the radius that would saturate the WLR constraint is

$$R_{\text{WLR}} = \frac{150}{2\pi \times (100/24)} = \frac{150}{26.18} = 5.73 \text{ m},$$

smaller than 6.910 m: the binding constraint is indeed the one on area.

2.16.2 Solutions to the symbolic exercises

Solution. S1. *Step 1 — normalisation.* By Definition 2.2 with $f \equiv 1$, and by Theorem 1.9, $\oint_{C(P_0, R)} ds = 2\pi R$. Hence the constant $f(P_0)$ has mean exactly $f(P_0)$.

Step 2 — estimating the difference. By linearity and the triangle inequality for integrals,

$$\left| \frac{1}{2\pi R} \oint f \, ds - f(P_0) \right| = \left| \frac{1}{2\pi R} \oint [f - f(P_0)] \, ds \right| \leq \frac{1}{2\pi R} \oint |f - f(P_0)| \, ds.$$

Step 3 — bounding by the oscillation. Setting

$$\omega(R) = \max_{|Q - P_0| = R} |f(Q) - f(P_0)|$$

(the maximum exists because f is continuous and the circle is compact), we have $|f - f(P_0)| \leq \omega(R)$ along the whole curve, hence

$$\frac{1}{2\pi R} \oint |f - f(P_0)| \, ds \leq \frac{\omega(R)}{2\pi R} \cdot 2\pi R = \omega(R).$$

Step 4 — passing to the limit. Let $\varepsilon > 0$. By continuity of f at P_0 there is $\delta > 0$ such that $|Q - P_0| < \delta$ implies $|f(Q) - f(P_0)| < \varepsilon$. Hence for every $R < \delta$ we get $\omega(R) \leq \varepsilon$, and therefore

$$\left| \frac{1}{2\pi R} \oint_{C(P_0, R)} f \, ds - f(P_0) \right| \leq \varepsilon.$$

Since ε was arbitrary, the limit is $f(P_0)$.

Remark. The result says that the mean of a continuous function over ever smaller circles reproduces the central value. For *harmonic* functions much more is true — the mean equals $f(P_0)$ for every R , not only in the limit — which is the mean value property, to be taken up in Chapter 3 as a consequence of Cauchy’s integral formula.

Solution. S2. The implication “conservative $\Rightarrow$ zero circulation” is Theorem 2.5. We prove the converse.

Step 1 — path independence. Let σ_1, σ_2 be two piecewise regular paths in A from the same point $\mathbf{p}_0$ to the same point $\mathbf{p}$. The concatenation of σ_1 with $-\sigma_2$ (σ_2 traversed backwards) is a *closed* curve contained in A . By hypothesis its circulation vanishes, hence by Proposition 2.1 and Theorem 2.4(iii)

$$0 = \int_{\sigma_1} \mathbf{F} \cdot d\mathbf{r} - \int_{\sigma_2} \mathbf{F} \cdot d\mathbf{r},$$

so the two integrals coincide.

Step 2 — defining the potential. Fix $\mathbf{p}_0 \in A$. Since A is open and connected, it is connected by piecewise regular paths (every point has a convex neighbourhood, and the relation “joinable” has open equivalence classes, so by connectedness there is a single class). Set

$$\phi(\mathbf{p}) = \int_{\sigma} \mathbf{F} \cdot d\mathbf{r},$$

where σ is any path from $\mathbf{p}_0$ to $\mathbf{p}$ in A : by Step 1 this is well defined.

Step 3 — computing the partial derivative. Let $\mathbf{p} \in A$ and let $r > 0$ be such that the disc $B(\mathbf{p}, r) \subseteq A$. For $|h| < r$, a path from $\mathbf{p}_0$ to $\mathbf{p} + h\mathbf{e}_1$ is obtained by concatenating a path to $\mathbf{p}$ with the horizontal segment from $\mathbf{p}$ to $\mathbf{p} + h\mathbf{e}_1$, which lies in the disc and hence in A . By additivity,

$$\phi(\mathbf{p} + h\mathbf{e}_1) - \phi(\mathbf{p}) = \int_0^h F_1(\mathbf{p} + t\mathbf{e}_1) \, dt,$$

having parametrised the segment as $t \mapsto \mathbf{p} + t\mathbf{e}_1$, whose derivative is $\mathbf{e}_1$, so that $\mathbf{F} \cdot d\mathbf{r} = F_1 dt$.
Step 4 — limit. Dividing by h and using the mean value theorem for integrals (applicable since F_1 is continuous): there is t_h between 0 and h with

$$\frac{\phi(\mathbf{p} + h\mathbf{e}_1) - \phi(\mathbf{p})}{h} = F_1(\mathbf{p} + t_h\mathbf{e}_1) \xrightarrow{h \rightarrow 0} F_1(\mathbf{p}),$$

by continuity of F_1 . Hence $\partial\phi/\partial x = F_1$; similarly $\partial\phi/\partial y = F_2$.

Step 5 — conclusion. $\nabla\phi = \mathbf{F}$, and since $\mathbf{F}$ is continuous, ϕ is of class C^1 : the field is conservative.

Remark. Connectedness is essential: on a disconnected open set the potential is built on each component, but the additive constants are independent.

Solution. S3. *Step 1 — decomposing the boundary.* The boundary of the polygon is the concatenation of the n edges Γ_i , from $P_i = (x_i, y_i)$ to $P_{i+1} = (x_{i+1}, y_{i+1})$. By Proposition 2.1 and Corollary 2.3,

$$\text{Area} = \frac{1}{2} \sum_{i=1}^n \int_{\Gamma_i} (x dy - y dx).$$

Step 2 — parametrising an edge. Set

$$\gamma_i(t) = P_i + t(P_{i+1} - P_i), \quad t \in [0, 1],$$

i.e. $x(t) = x_i + t\Delta x_i$ and $y(t) = y_i + t\Delta y_i$ with $\Delta x_i = x_{i+1} - x_i$, $\Delta y_i = y_{i+1} - y_i$. Then $dx = \Delta x_i dt$ and $dy = \Delta y_i dt$.

Step 3 — the integrand on one edge.

$$x dy - y dx = \left[(x_i \Delta y_i - y_i \Delta x_i) + t(\Delta x_i \Delta y_i - \Delta y_i \Delta x_i) \right] dt = (x_i \Delta y_i - y_i \Delta x_i) dt.$$

The term in t vanishes identically: the integrand is *constant* on each edge. This is the key fact that makes the formula so simple.

Step 4 — integration. Being constant,

$$\int_{\Gamma_i} (x dy - y dx) = x_i(y_{i+1} - y_i) - y_i(x_{i+1} - x_i) = x_i y_{i+1} - x_{i+1} y_i,$$

the terms $-x_i y_i$ and $+y_i x_i$ cancelling.

Step 5 — conclusion.

$$\text{Area} = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i),$$

with indices understood cyclically ($P_{n+1} = P_1$). The name “shoelace” comes from the arrangement of the cross products, reminiscent of lacing a shoe.

Check on a triangle. For $P_1 = (0, 0)$, $P_2 = (4, 0)$, $P_3 = (0, 3)$:

$$\text{Area} = \frac{1}{2} [0 + 12 + 0] = 6,$$

which agrees with $\frac{1}{2} \cdot 4 \cdot 3 = 6$. ✓ The positive sign confirms that the vertices were listed counterclockwise.

Solution. S4. *Step 1 — the two differential identities.* With $J\mathbf{F} = (-F_2, F_1)$:

$$\operatorname{div}(J\mathbf{F}) = -\frac{\partial F_2}{\partial x} + \frac{\partial F_1}{\partial y} = -\operatorname{curl} \mathbf{F},$$

$$\operatorname{curl}(J\mathbf{F}) = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} = \operatorname{div} \mathbf{F}.$$

Step 2 — the geometric identity $(J\mathbf{F}) \cdot T = \mathbf{F} \cdot \mathbf{n}$. Recall $T = (x', y')/|\gamma'|$ and $\mathbf{n} = (y', -x')/|\gamma'|$ (Theorem 2.1), hence

$$JT = \frac{1}{|\gamma'|} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} = \frac{(-y', x')}{|\gamma'|} = -\mathbf{n}.$$

Since $J^T = -J$ (J is antisymmetric),

$$(J\mathbf{F}) \cdot T = \mathbf{F}^T J^T T = -\mathbf{F}^T JT = -\mathbf{F} \cdot (-\mathbf{n}) = \mathbf{F} \cdot \mathbf{n}.$$

Step 3 — deducing the normal form from the tangential one. Apply Theorem 2.10 to the field $J\mathbf{F}$, in the form $\oint (\mathbf{G} \cdot T) ds = \iint \operatorname{curl} \mathbf{G} dA$ (equivalent by Theorem 2.4(i)), with $\mathbf{G} = J\mathbf{F}$:

$$\oint_{\partial D} (J\mathbf{F}) \cdot T ds = \iint_D \operatorname{curl}(J\mathbf{F}) dA.$$

Substituting Step 2 on the left and Step 1 on the right:

$$\oint_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \iint_D \operatorname{div} \mathbf{F} dA,$$

which is Corollary 2.2.

Step 4 — and conversely. Applying the normal form to $J\mathbf{F}$ and using $(J\mathbf{F}) \cdot \mathbf{n} = -\mathbf{F} \cdot T$ (checked as in Step 2, or by noting $J^2 = -I$) together with $\operatorname{div}(J\mathbf{F}) = -\operatorname{curl} \mathbf{F}$, one recovers the tangential form. The two formulations are therefore *the same identity*, read after rotating the field by 90° : what is circulation for one field is flux for the rotated one.

Solution. S5. *Step 1 — constructing the annular region.* Since Γ is a Jordan curve enclosing the origin and the interior region is open, there is $\varepsilon > 0$ such that the closed disc $\overline{D}(O, \varepsilon)$ lies strictly inside Γ . Let

$$\Omega = \{\text{region inside } \Gamma\} \setminus D(O, \varepsilon)$$

be the (not necessarily circular) annular region between the two curves.

Step 2 — the field is smooth on Ω . The only singularity of $\mathbf{V}$ is the origin, which has been removed: $\mathbf{V} \in C^\infty$ on an open set containing $\overline{\Omega}$, and by Theorem 2.8(i) it is irrotational there.

Step 3 — the oriented boundary of Ω . For Green's theorem the boundary must be traversed leaving the region on the left: hence Γ counterclockwise and $C_\varepsilon = C(O, \varepsilon)$ clockwise. Formally, $\partial\Omega = \Gamma \cup (-C_\varepsilon)$.

Step 4 — applying Green's theorem. Theorem 2.10 also holds for regions with a hole: one makes two cuts joining Γ to C_ε , obtaining two simply connected regions to which the theorem applies; adding them, the contributions of the cuts appear twice with opposite orientations and cancel by Theorem 2.4(iii). One obtains

$$\oint_{\Gamma} \mathbf{V} \cdot d\mathbf{r} - \oint_{C_\varepsilon} \mathbf{V} \cdot d\mathbf{r} = \iint_{\Omega} \operatorname{curl} \mathbf{V} dA = 0.$$

Step 5 — conclusion. Hence

$$\oint_{\Gamma} \mathbf{V} \cdot d\mathbf{r} = \oint_{C_\varepsilon} \mathbf{V} \cdot d\mathbf{r} = 2\pi,$$

by Theorem 2.8(ii), and the value depends neither on ε nor on the shape of Γ .

Remark. This is the prototype of the “contour deformation principle”: the integral of an irrotational field does not change if the curve is continuously deformed *without crossing the singularities*. In Chapter 3 it will reappear as Cauchy’s theorem for multiply connected domains.

Solution. S6. *Step 1 — exact integral of the exponentials.* For $j \in \mathbb{Z}$,

$$I[e^{ij\theta}] = \begin{cases} 2\pi & j = 0, \\ 0 & j \neq 0, \end{cases}$$

since for $j \neq 0$ an antiderivative is $e^{ij\theta}/(ij)$, which takes the same value at both endpoints by periodicity.

Step 2 — the trapezoidal rule on the exponentials. With $\theta_k = 2\pi k/N$,

$$T_N[e^{ij\theta}] = 2\pi \left(\frac{1}{N} \sum_{k=0}^{N-1} q^k \right), \quad q = e^{2\pi ij/N}.$$

If $N \mid j$ then $q = 1$ and the sum equals N , i.e. 1 after division; otherwise $q \neq 1$ and the geometric sum gives $(q^N - 1)/(q - 1) = 0$, since $q^N = e^{2\pi ij} = 1$. Hence

$$T_N[e^{ij\theta}] = \begin{cases} 2\pi & N \mid j, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3 — comparison. For $|j| < N$ the two expressions coincide:

- if $j = 0$: both equal 2π (and $N \mid 0$);
- if $0 < |j| < N$: N does not divide j (the only multiple of N of modulus less than N is 0), so both vanish.

Hence $T_N[e^{ij\theta}] = I[e^{ij\theta}]$ for every $|j| < N$, and by *linearity* of T_N and I the equality extends to every linear combination $f(\theta) = \sum_{|j| < N} c_j e^{ij\theta}$: the rule is exact on all trigonometric polynomials of degree $< N$.

Step 4 — and no further. For $f(\theta) = e^{iN\theta}$ we have $I[f] = 0$ (since $N \neq 0$) but $T_N[f] = 2\pi$ (since $N \mid N$): the error equals 2π , the largest possible. This is precisely the aliasing of Theorem 2.17, Step 3: sampling at N nodes confuses $e^{iN\theta}$ with the constant 1.

Remark. The rule T_N uses N evaluations and is exact on a space of dimension $2N - 1$ (the exponentials with $|j| < N$). This is the maximum possible for equispaced nodes, and it explains why no “better” formula exists on the circle: T_N already has optimal degree of exactness in its class.

Solution. S7. *Step 1 — divergence of the radius vector.* With $\mathbf{p} = (x, y)$, $\operatorname{div} \mathbf{p} = 1 + 1 = 2$.

Step 2 — divergence theorem. By Corollary 2.2,

$$\oint_{\partial D} (\mathbf{p} \cdot \mathbf{n}) \, ds = \iint_D 2 \, dA = 2 \operatorname{Area}(D).$$

Step 3 — check on the disc. On $\partial D(O, R)$ we have $\mathbf{n} = \mathbf{p}/R$ (Theorem 2.1(iv) with $C_0 = O$), hence $\mathbf{p} \cdot \mathbf{n} = R^2/R = R$, constant. Therefore

$$\oint_{\partial D} (\mathbf{p} \cdot \mathbf{n}) \, ds = R \cdot 2\pi R = 2\pi R^2 = 2 \text{Area}(D(O, R)). \checkmark$$

Step 4 — geometric interpretation. The quantity $h(P) = \mathbf{p} \cdot \mathbf{n}$ is the **support function**: the signed distance from the origin to the tangent line at P (the projection of the radius vector onto the normal). The formula can be rewritten

$$\text{Area}(D) = \frac{1}{2} \oint_{\partial D} h \, ds,$$

which has an elementary reading: decompose D into infinitesimal triangles with apex at the origin, base ds on the boundary and height h ; each has area $\frac{1}{2}h \, ds$, and the sum is the total area. For the circle, $h \equiv R$ and one recovers $\frac{1}{2}R \cdot 2\pi R = \pi R^2$: it is the “pie-slice” proof of the area of the circle, made rigorous by the divergence theorem. Note that the formula holds with respect to *any* origin, provided it is held fixed: changing origin changes h but not the integral.

Solution. S8. *Step 1 — choice of field.* Apply Corollary 2.2 to the field $\mathbf{F} = \nabla\phi = (\phi_x, \phi_y)$, which is of class C^1 because $\phi \in C^2$.

Step 2 — the divergence is the Laplacian.

$$\text{div}(\nabla\phi) = \phi_{xx} + \phi_{yy} = \Delta\phi.$$

Step 3 — the flux is the normal derivative. By definition of directional derivative, $\mathbf{F} \cdot \mathbf{n} = \nabla\phi \cdot \mathbf{n} = \partial\phi/\partial n$.

Step 4 — conclusion. Substituting into the corollary:

$$\oint_{\partial D} \frac{\partial\phi}{\partial n} \, ds = \iint_D \Delta\phi \, dA.$$

Step 5 — harmonic case. If $\Delta\phi \equiv 0$ on an open set A , then for every disc $\overline{D(P_0, R)} \subset A$

$$\oint_{C(P_0, R)} \frac{\partial\phi}{\partial n} \, ds = 0 :$$

the flux of the gradient of a harmonic function through every circle contained in the domain vanishes.

Physical interpretation. If ϕ is a temperature potential in steady state with no sources (Laplace’s equation), the result states that the heat entering a disc exactly equals the heat leaving it: there is no accumulation. It is the integral formulation of the conservation principle, and the reason why harmonic functions cannot have interior maxima or minima — a fact to be proved in Chapter 3 as the maximum principle.

Solution. S9. *Step 1 — writing componentwise.* By Definition 2.1, $\mathbf{n} \, ds = (dy, -dx)$. Hence the vector identity to be proved is equivalent to the two scalar identities

$$\oint_{\Gamma} dy = 0, \quad \oint_{\Gamma} dx = 0.$$

Step 2 — computation. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be a piecewise regular parametrisation of Γ . By the fundamental theorem of calculus applied on each piece and summing (Proposition 2.1),

$$\oint_{\Gamma} dy = \int_a^b y'(t) dt = y(b) - y(a) = 0,$$

since the curve is closed, i.e. $\gamma(b) = \gamma(a)$ and in particular $y(b) = y(a)$. Identically $\oint dx = 0$.

Step 3 — conclusion. $\oint_{\Gamma} \mathbf{n} ds = \mathbf{0}$.

Step 4 — alternative proof, by the divergence theorem. Let $\mathbf{c}$ be an arbitrary constant vector. Applying Corollary 2.2 to the constant field $\mathbf{F} \equiv \mathbf{c}$, whose divergence vanishes,

$$\mathbf{c} \cdot \oint_{\Gamma} \mathbf{n} ds = \iint_D 0 dA = 0.$$

Since the dot product with $\oint \mathbf{n} ds$ vanishes for *every* $\mathbf{c}$, the vector $\oint \mathbf{n} ds$ is zero (take $\mathbf{c}$ equal to the vector itself).

Step 5 — physical consequence. A fluid at rest exerts on the boundary of an immersed body a uniform pressure p , directed inwards: the infinitesimal force is $d\mathbf{F} = -p \mathbf{n} ds$. The resultant is

$$\mathbf{F}_{\text{tot}} = -p \oint_{\Gamma} \mathbf{n} ds = \mathbf{0} :$$

a *uniform* pressure produces no resultant force, whatever the shape of the body. Archimedes' buoyancy indeed arises from the *non-uniformity* of pressure with depth, not from pressure as such.

Solution. S10. *Step 0 — trivial case.* If $b = 0$ the integral equals $2\pi/a = 2\pi/\sqrt{a^2}$, consistent with the formula. Assume then $b \neq 0$.

Step 1 — the integrand is well defined. From $a > |b|$ it follows that

$$a + b \cos \theta \geq a - |b| > 0 \quad \text{for every } \theta,$$

so the integrand is continuous and bounded on $\mathbb{R}$, and 2π -periodic: its integral over $[0, 2\pi]$ equals that over $[-\pi, \pi]$.

Step 2 — removing the endpoints. The substitution $u = \tan(\theta/2)$ is an increasing diffeomorphism from $(-\pi, \pi)$ onto $\mathbb{R}$, but it is undefined at $\theta = \pm\pi$. Since the integrand is bounded and $\{-\pi, \pi\}$ is a null set,

$$\int_{-\pi}^{\pi} = \int_{(-\pi, \pi)},$$

and the substitution is legitimate on the open interval. (Alternatively, and entirely elementarily: integrate over $[-\pi + \delta, \pi - \delta]$ and let $\delta \rightarrow 0^+$, which is legitimate because the integrand is bounded.)

Step 3 — substitution. By Chapter 1, Corollary 1.2: $\cos \theta = \frac{1 - u^2}{1 + u^2}$ and $d\theta = \frac{2 du}{1 + u^2}$. The denominator becomes

$$a + b \frac{1 - u^2}{1 + u^2} = \frac{(a + b) + (a - b)u^2}{1 + u^2}.$$

Note that $a > |b|$ guarantees $a + b > 0$ and $a - b > 0$.

Step 4 — simplification.

$$\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \int_{-\infty}^{+\infty} \frac{2 du}{(a + b) + (a - b)u^2},$$

the factor $1 + u^2$ cancelling exactly.

Step 5 — reduction to canonical form. Factoring out $a - b > 0$ and setting

$$c = \sqrt{\frac{a+b}{a-b}} > 0,$$

we have $(a+b) + (a-b)u^2 = (a-b)(u^2 + c^2)$, hence

$$\frac{2}{a-b} \int_{-\infty}^{+\infty} \frac{du}{u^2 + c^2} = \frac{2}{a-b} \cdot \frac{\pi}{c}.$$

Step 6 — final simplification.

$$\frac{2\pi}{(a-b)c} = \frac{2\pi}{a-b} \sqrt{\frac{a-b}{a+b}} = \frac{2\pi}{\sqrt{(a-b)(a+b)}} = \frac{2\pi}{\sqrt{a^2 - b^2}},$$

using $a - b > 0$ to write $a - b = \sqrt{(a-b)^2}$.

Step 7 — checks. As $b \rightarrow 0^+$ one obtains $2\pi/a$, consistent with Step 0. As $b \rightarrow a^-$ the integral diverges, consistently with the denominator $a + b \cos \theta$ vanishing at $\theta = \pi$. For $a = 5$, $b = 4$ one recovers $2\pi/3$, as in Exercise N7. ✓

2.17. Self-assessment quiz

- For the standard parametrisation of $C(C_0, R)$, the element of length is
A) $d\theta$ B) $R d\theta$ C) $R^2 d\theta$ D) dR
- The normal vector element $\mathbf{n} ds$ is written
A) (dx, dy) B) $(dy, -dx)$ C) $(-dy, dx)$ D) $(dx, -dy)$
- Does $\int_{\Gamma} f ds$ change sign when the direction of travel is reversed?
A) Yes, always B) No, never C) Only if $f < 0$ D) Only for closed curves
- Does $\int_{\Gamma} \mathbf{F} \cdot d\mathbf{r}$ change sign when the direction is reversed?
A) Yes, always B) No, never C) Only for conservative fields D) Only for closed curves
- The centroid of a homogeneous semicircular wire of radius R lies, from the centre, at distance
A) $R/2$ B) $2R/\pi$ C) $4R/(3\pi)$ D) R
- The circulation of the radial field (x, y) along $C(O, R)$ equals
A) 0 B) $2\pi R^2$ C) πR^2 D) $2\pi R$
- If $\mathbf{F} = \nabla\phi$ on an open set A , then for every closed curve $\Gamma \subseteq A$
A) $\oint \mathbf{F} \cdot d\mathbf{r} = 0$
B) $\oint \mathbf{F} \cdot d\mathbf{r} = \phi$
C) $\oint \mathbf{F} \cdot d\mathbf{r} = 2\pi$
D) nothing can be said
- An irrotational field on a *star-shaped* open set is

-
- A) always conservative B) never conservative C) conservative only if bounded D) conservative only if C^∞
9. The vortex field $\mathbf{V} = (-y/(x^2 + y^2), x/(x^2 + y^2))$ on $\mathbb{R}^2 \setminus \{0\}$
- A) has non-zero curl
 B) is irrotational but not conservative
 C) is conservative
 D) is undefined on the unit circle
10. The circulation of $\mathbf{V}$ along a Jordan curve enclosing the origin equals
- A) 0 B) 2π C) π D) it depends on the shape of the curve
11. The “missing” potential of the vortex field is
- A) the radius ρ B) the angle θ C) $\log \rho$ D) $x^2 + y^2$
12. Green’s theorem in tangential form states
- A) $\oint (P dx + Q dy) = \iint_D (\partial_x Q - \partial_y P) dA$
 B) $\oint (P dx + Q dy) = \iint_D (\partial_x P + \partial_y Q) dA$
 C) $\oint \mathbf{F} \cdot \mathbf{n} ds = \iint_D \text{curl } \mathbf{F} dA$
 D) $\oint f ds = \iint_D f dA$
13. The area enclosed by a simple closed curve is computed as
- A) $\oint (x dx + y dy)$ B) $\frac{1}{2} \oint (x dy - y dx)$ C) $\oint \rho d\theta$ D) $\frac{1}{2} \oint \rho ds$
14. The Jacobian of the change to polar coordinates equals
- A) 1 B) ρ C) ρ^2 D) $1/\rho$
15. The value of $\int_{\mathbb{R}} e^{-x^2} dx$ is
- A) π B) $\sqrt{\pi}$ C) 2π D) $\pi/2$
16. For a 2π -periodic integrand of class C^m , the error of the trapezoidal rule is
- A) $O(N^{-2})$ always B) $O(N^{-4})$ always C) $O(N^{-m})$ D) independent of N
17. For a periodic *analytic* integrand, the trapezoidal rule converges
- A) linearly B) like N^{-4} C) geometrically, like r^{-N} D) not at all
18. The rule T_N on N nodes is exact for trigonometric polynomials of degree
- A) $< N$ B) $< N/2$ C) $\leq 2N$ D) any
19. The error of the Monte Carlo method for π decreases like
- A) $1/N$ B) $1/\sqrt{N}$ C) $1/N^2$ D) $\log N/N$
20. In sizing a circular clarifier, the typically active constraint is
- A) the SOR, because $\propto R^{-2}$ B) the WLR, because $\propto R^{-1}$ C) always both D) neither
-

BEST PRACTICE

Answer key.

1-B: Theorem 2.1(iv). **2-B:** Definition 2.1; $\mathbf{C}$ would be the *inward* normal. **3-B:** Theorem 2.2; ds is a length, always positive. **4-A:** Theorem 2.4(iii); $d\mathbf{r}$ is a vector and reverses. **5-B:** Example 2.1; $\mathbf{C}$ is the centroid of the solid *half-disc*. **6-A:** Example 2.2: the field is everywhere orthogonal to the tangent. $\mathbf{B}$ is its *flux*. **7-A:** Theorem 2.5. **8-A:** Theorem 2.7. **9-B:** Theorem 2.8(i) and (iii): the central counterexample of the chapter. **10-B:** Exercise S5 (contour deformation principle): the value does not depend on the shape. **11-B:** Observation 2.3; $\mathbf{C}$ is the potential of the source field, not of the vortex. **12-A:** Theorem 2.10; $\mathbf{C}$ confuses the two forms. **13-B:** Corollary 2.3. **14-B:** Theorem 2.11. **15-B:** Theorem 2.12; π is the value of the *double* integral. **16-C:** Theorem 2.16: the order is set by smoothness, not by the method. **17-C:** Theorem 2.17. **18-A:** Exercise S6. **19-B:** Theorem 2.20. **20-A:** Observation 2.6: the SOR improves quadratically with the radius, so it saturates first.

2.18. Chapter formulary

Differential elements (counterclockwise curve)

$$ds = |\gamma'| dt = R d\theta, \quad d\mathbf{r} = T ds = (dx, dy), \quad \mathbf{n} ds = (dy, -dx), \quad \mathbf{n} = \frac{P - C_0}{R}$$

Line integrals

$$\int_{\Gamma} f ds = \int_a^b f(\gamma) |\gamma'| dt, \quad \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\gamma) \cdot \gamma' dt = \int_{\Gamma} (\mathbf{F} \cdot T) ds$$

$$m = \oint \lambda ds, \quad \bar{f} = \frac{1}{L} \oint f ds, \quad G = C_0 + \frac{R}{\Delta\theta} (\sin \theta_1 - \sin \theta_0, \cos \theta_0 - \cos \theta_1)$$

Conservative fields

$$\mathbf{F} = \nabla \phi \implies \int_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = \phi(\mathbf{b}) - \phi(\mathbf{a}), \quad \oint_{\Gamma} \mathbf{F} \cdot d\mathbf{r} = 0$$

conservative $\implies \operatorname{curl} \mathbf{F} = 0$; $\operatorname{curl} \mathbf{F} = 0$ and A star-shaped $\implies$ conservative

$$\mathbf{V} = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right) = \nabla \theta, \quad \frac{1}{2\pi} \oint_{\Gamma} \mathbf{V} \cdot d\mathbf{r} = n \text{ (winding number)}$$

Green and divergence

$$\oint_{\partial D} (P dx + Q dy) = \iint_D (\partial_x Q - \partial_y P) dA, \quad \oint_{\partial D} \mathbf{F} \cdot \mathbf{n} ds = \iint_D \operatorname{div} \mathbf{F} dA$$

$$\operatorname{Area}(D) = \oint x dy = - \oint y dx = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \oint (\mathbf{p} \cdot \mathbf{n}) ds$$

Polar coordinates and notable integrals

$$dA = \rho d\rho d\varphi, \quad \int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}, \quad \int_0^{2\pi} \frac{d\theta}{a + b \cos \theta} = \frac{2\pi}{\sqrt{a^2 - b^2}} \quad (a > |b|)$$

$$\int_0^{2\pi} \cos m\theta \cos n\theta d\theta = \pi \delta_{mn} \quad (m, n \geq 1), \quad \int_0^{2\pi} \cos m\theta \sin n\theta d\theta = 0$$

Periodic quadrature

$$T_N[f] = \frac{2\pi}{N} \sum_{k=0}^{N-1} f\left(\frac{2\pi k}{N}\right), \quad |T_N - I| = O(N^{-m}) \quad (f \in C^m), \quad |I_N - I| \leq \frac{2\pi M}{r^N - 1} \quad (f \text{ analytic})$$

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CHAPTER 3

Complex Numbers and the Circle

With complex numbers the circle ceases to be merely a locus and becomes an *algebraic* object: the set $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ of numbers of unit modulus, closed under multiplication. This identification, made possible by Euler's formula, is probably the most fruitful connection in the whole treatise: it turns geometric questions (regular polygons, rotations, inversions) into purely algebraic ones (the equations $z^n = 1$, multiplication in $\mathbb{C}$), and prepares the ground for the harmonic analysis of Chapter 4, where $\mathbb{T}$ will use exactly this group structure.

The chapter follows a four-stage path. First the vocabulary is built: complex plane, modulus, argument, conjugation, and the choice — *declared*, never implicit — of the interval for the argument. Then loci and plane transformations are translated into complex language, revealing that a single equation describes circles and lines together. Next comes the heart of the matter: Euler's formula, de Moivre's theorem, powers and n -th roots. Finally the roots of unity reveal their double nature — vertices of a regular polygon and orthogonal basis — which is precisely the bridge to the harmonic modes.

Warning. **Convention for the argument, valid throughout the chapter.** We write $\text{Arg } z$ for the **principal value** of the argument, with

$$\text{Arg } z \in (-\pi, \pi],$$

and $\arg z$ for the *set* of all values $\text{Arg } z + 2k\pi$, $k \in \mathbb{Z}$. Angles are always in radians unless degrees are explicitly indicated. This is the convention of the *NIST Digital Library of Mathematical Functions*, §4.2, and it is the only one used in the chapter: confusing the principal value with the set of values is the commonest source of error in computing powers and roots, and Section 3.7 returns to the point explicitly.

INFO

Prerequisites. Volume I, Chapters 4 (angles), 7 (analytic geometry of the circle) and 8 (trigonometry); from this volume, Chapter 1 (parametrisation of the circle, Lemma 1.1 on the polar representation of unit vectors) and, for (Volume III, Chapter 15), Chapter 2 (line integrals and the contour deformation principle).

3.1. The complex plane: point, vector, modulus, argument

Definition 3.1: The complex plane. We identify $\mathbb{C}$ with $\mathbb{R}^2$ by setting

$$z = x + iy \longleftrightarrow (x, y), \quad x = \text{Re } z, \quad y = \text{Im } z,$$

where i satisfies $i^2 = -1$. The plane so coordinatised is the **complex plane** (or *Argand-Gauss plane*). The **conjugate** of z is $\bar{z} = x - iy$ and the **modulus** is $|z| = \sqrt{x^2 + y^2}$.

Remark 3.1: Point or vector? The same symbol z denotes, depending on context, a point of the plane (when position matters) or a vector based at the origin (when sums and differences matter). The sum $z+w$ is the parallelogram rule; the difference $z-w$ is the vector from w to z , and its modulus $|z-w|$ is the distance between the two points (Theorem 3.2). Multiplication, by contrast, has no elementary vector analogue: it is precisely this extra operation that makes $\mathbb{C}$ so much more powerful than $\mathbb{R}^2$ for the geometry of the circle.

Theorem 3.1: Properties of modulus and conjugation. For all $z, w \in \mathbb{C}$:

- (i) $z\bar{z} = |z|^2$, and $|z| = 0 \iff z = 0$;
- (ii) $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\bar{w}$, $\bar{\bar{z}} = z$;
- (iii) $|zw| = |z||w|$ (multiplicativity);
- (iv) $|z+w| \leq |z| + |w|$, with equality if and only if z and w are aligned and concordant, i.e. $z = 0$ or $w = tz$ with $t \geq 0$ (triangle inequality);
- (v) $||z| - |w|| \leq |z - w|$;
- (vi) $\operatorname{Re} z = \frac{z + \bar{z}}{2}$, $\operatorname{Im} z = \frac{z - \bar{z}}{2i}$, and $|\operatorname{Re} z| \leq |z|$.

Proof. (i) $z\bar{z} = (x+iy)(x-iy) = x^2 - (iy)^2 = x^2 + y^2 = |z|^2$, using $i^2 = -1$. If $|z| = 0$ then $x^2 + y^2 = 0$ with x, y real, hence $x = y = 0$.

(ii) Direct verification on components: if $z = x+iy$, $w = u+iv$, then $\overline{z+w} = (x+u) - i(y+v) = \bar{z} + \bar{w}$. For the product,

$$zw = (xu - yv) + i(xv + yu) \implies \overline{zw} = (xu - yv) - i(xv + yu),$$

while $\bar{z}\bar{w} = (x-iy)(u-iv) = (xu - yv) - i(xv + yu)$, which agrees.

(iii) Using (i) and (ii),

$$|zw|^2 = (zw)\overline{(zw)} = zw\bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = |z|^2|w|^2,$$

and taking square roots of two non-negative numbers gives the claim.

(iv) By (i) and (ii),

$$|z+w|^2 = (z+w)(\bar{z} + \bar{w}) = |z|^2 + |w|^2 + (z\bar{w} + \overline{z\bar{w}}) = |z|^2 + |w|^2 + 2\operatorname{Re}(z\bar{w}),$$

using (vi). Now $\operatorname{Re}(z\bar{w}) \leq |z\bar{w}| = |z||w|$, hence

$$|z+w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

Equality holds iff $\operatorname{Re}(z\bar{w}) = |z||w|$, i.e. iff $z\bar{w}$ is real and non-negative. If $z \neq 0$, multiplying by $z/|z|^2$ gives $w = tz$ with $t = z\bar{w}/|z|^2 \geq 0$.

(v) From (iv), $|z| = |(z-w) + w| \leq |z-w| + |w|$, i.e. $|z| - |w| \leq |z-w|$. Swapping the roles of z and w gives $|w| - |z| \leq |z-w|$; together they give the claim.

(vi) $z + \bar{z} = 2x$ and $z - \bar{z} = 2iy$, whence the two formulas. Finally $|\operatorname{Re} z| = |x| \leq \sqrt{x^2 + y^2} = |z|$. $\square$

Theorem 3.2: The modulus of the difference is the Euclidean distance. For $z = x+iy$ and $w = u+iv$,

$$|z-w| = \sqrt{(x-u)^2 + (y-v)^2},$$

that is, $|z-w|$ is exactly the Euclidean distance between the corresponding points of $\mathbb{R}^2$. Consequently $(\mathbb{C}, d)$ with $d(z, w) = |z-w|$ is a metric space, and the topological notions (disc, open set, convergence) coincide with those of $\mathbb{R}^2$.

Proof. $z - w = (x - u) + i(y - v)$, so by Definition 3.1 $|z - w| = \sqrt{(x - u)^2 + (y - v)^2}$, the Euclidean distance formula (Volume I, Chapter 7). The metric properties follow from Theorem 3.1: positivity and vanishing from (i), symmetry from $|\zeta| = |-\zeta|$, the triangle inequality from (iv) applied to $z - w = (z - \zeta) + (\zeta - w)$. $\square$

Corollary 3.1: Circles and discs in complex form. For $c \in \mathbb{C}$ and $r > 0$:

$$C(c, r) = \{z \in \mathbb{C} : |z - c| = r\}, \quad D(c, r) = \{z \in \mathbb{C} : |z - c| < r\}.$$

In particular the **unit circle** is $\mathbb{T} = C(0, 1)$, and by property (iii) of Theorem 3.1 it is closed under multiplication: if $|z| = |w| = 1$ then $|zw| = 1$.

Proof. Immediate from Theorem 3.2: the condition $|z - c| = r$ says exactly that z is at distance r from c , which is the definition of a circle (Volume I, Chapter 2). Multiplicative closure is the multiplicativity of the modulus. $\square$

3.1.1 Polar form and argument

Theorem 3.3: Polar form and its uniqueness. Every $z \in \mathbb{C} \setminus \{0\}$ can be written in exactly one way as

$$z = \rho(\cos \theta + i \sin \theta), \quad \rho = |z| > 0, \quad \theta = \operatorname{Arg} z \in (-\pi, \pi].$$

The set of all arguments of z is $\arg z = \{\operatorname{Arg} z + 2k\pi : k \in \mathbb{Z}\}$. For $z = 0$ the modulus is 0 and the argument is undefined.

Proof. Existence. Let $z \neq 0$ and $\rho = |z| > 0$. The number $u = z/\rho$ has modulus 1 by Theorem 3.1(iii); identifying it with the vector $(x/\rho, y/\rho) \in \mathbb{R}^2$, it is a unit vector, so by Lemma 1.1 of Chapter 1 there is exactly one $\theta_0 \in [0, 2\pi)$ with $u = \cos \theta_0 + i \sin \theta_0$. We move θ_0 into the chosen interval: if $\theta_0 \in [0, \pi]$ set $\theta = \theta_0$; if $\theta_0 \in (\pi, 2\pi)$ set $\theta = \theta_0 - 2\pi \in (-\pi, 0)$. In both cases $\theta \in (-\pi, \pi]$ and, by periodicity, $\cos \theta + i \sin \theta = u$.

Uniqueness. If $\rho_1(\cos \theta_1 + i \sin \theta_1) = \rho_2(\cos \theta_2 + i \sin \theta_2)$ with $\rho_j > 0$ and $\theta_j \in (-\pi, \pi]$, taking moduli gives $\rho_1 = \rho_2$, and cancelling leaves two equal unit vectors; by the uniqueness part of Lemma 1.1, applied after moving both angles into $[0, 2\pi)$, we get $\theta_1 = \theta_2$.

Set of arguments. If $\cos \theta + i \sin \theta = \cos \theta' + i \sin \theta'$, then (again by Lemma 1.1, without the interval restriction) $\theta - \theta' \in 2\pi\mathbb{Z}$; conversely, translation by 2π leaves sine and cosine unchanged. $\square$

Best practice. How to compute $\operatorname{Arg} z$ without getting the quadrant wrong. The formula $\theta = \arctan(y/x)$ is *wrong* in two quadrants out of four and undefined for $x = 0$: $\arctan$ always returns a value in $(-\pi/2, \pi/2)$, i.e. in the right half-plane. The correct formula is

$$\operatorname{Arg} z = \operatorname{atan2}(y, x),$$

the same function already recommended in the laboratory of Chapter 1. If only $\arctan$ is

available, the explicit rule is

$$\operatorname{Arg}(x + iy) = \begin{cases} \arctan(y/x) & x > 0, \\ \arctan(y/x) + \pi & x < 0, y \geq 0, \\ \arctan(y/x) - \pi & x < 0, y < 0, \\ +\pi/2 & x = 0, y > 0, \\ -\pi/2 & x = 0, y < 0, \\ \text{undefined} & x = 0, y = 0. \end{cases}$$

Theorem 3.4: Multiplication, division and the argument. For $z, w \in \mathbb{C} \setminus \{0\}$:

$$|zw| = |z||w|, \quad \arg(zw) = \arg z + \arg w,$$

$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|}, \quad \arg\left(\frac{z}{w}\right) = \arg z - \arg w,$$

where equalities between arguments are equalities of sets of values, i.e. modulo 2π .

Proof. The modulus part is Theorem 3.1(iii) (and, for the quotient, $|z| = |(z/w)w| = |z/w||w|$). For the arguments, write $z = \rho(\cos \alpha + i \sin \alpha)$ and $w = \sigma(\cos \beta + i \sin \beta)$ and multiply:

$$zw = \rho\sigma \left[(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta) \right].$$

By the addition formulas (Volume I, Chapter 8) the two brackets are $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$, so

$$zw = \rho\sigma [\cos(\alpha + \beta) + i \sin(\alpha + \beta)],$$

showing that $\alpha + \beta$ is an argument of zw ; by Theorem 3.3 all others differ by multiples of 2π . The quotient formula follows by applying the above to $z = (z/w) \cdot w$. $\square$

Warning. The identity $\arg(zw) = \arg z + \arg w$ does *not* hold in general for principal values: with $z = w = -1$ we have $\operatorname{Arg} z = \operatorname{Arg} w = \pi$, while $zw = 1$ and $\operatorname{Arg}(zw) = 0 \neq 2\pi$. This is the minimal example of why the principal value must be treated as a *branch choice* and not as a function with the algebraic properties of the real logarithm.

Example 3.1: From Cartesian to polar form and back. Write $z_1 = -3 + 3i$ and $z_2 = -2 - 2\sqrt{3}i$ in polar form, compute their product and quotient, and check the results in Cartesian form.

Step 1 — moduli.

$$|z_1| = \sqrt{9 + 9} = 3\sqrt{2} \approx 4.2426, \quad |z_2| = \sqrt{4 + 12} = 4.$$

Step 2 — arguments (minding the quadrant). The point $z_1 = (-3, 3)$ is in the *second* quadrant ($x < 0, y > 0$): the rule gives $\operatorname{Arg} z_1 = \arctan(3/(-3)) + \pi = -\pi/4 + \pi = 3\pi/4$ (135°). The point $z_2 = (-2, -2\sqrt{3})$ is in the *third* quadrant ($x < 0, y < 0$):

$$\operatorname{Arg} z_2 = \arctan(\sqrt{3}) - \pi = \frac{\pi}{3} - \pi = -\frac{2\pi}{3} \quad (-120^\circ).$$

Note that naive use of $\arctan(y/x)$ would have given $-\pi/4$ for z_1 and $+\pi/3$ for z_2 : both wrong by π , i.e. diametrically opposite.

Step 3 — polar form.

$$z_1 = 3\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right), \quad z_2 = 4 \left(\cos \left(-\frac{2\pi}{3} \right) + i \sin \left(-\frac{2\pi}{3} \right) \right).$$

Step 4 — product in polar form. By Theorem 3.4,

$$|z_1 z_2| = 12\sqrt{2}, \quad \arg(z_1 z_2) = \frac{3\pi}{4} - \frac{2\pi}{3} = \frac{9\pi - 8\pi}{12} = \frac{\pi}{12},$$

which already lies in $(-\pi, \pi]$: it is the principal value (15°).

Step 5 — check in Cartesian form.

$$z_1 z_2 = (-3 + 3i)(-2 - 2\sqrt{3}i) = 6 + 6\sqrt{3}i - 6i - 6\sqrt{3}i^2 = (6 + 6\sqrt{3}) + i(6\sqrt{3} - 6).$$

Numerically: $z_1 z_2 = 16.392 + 4.392i$. Checking modulus and argument:

$$|z_1 z_2| = \sqrt{16.392^2 + 4.392^2} = \sqrt{268.70 + 19.29} = \sqrt{287.99} = 16.97 = 12\sqrt{2} \checkmark$$

$$\text{Arg}(z_1 z_2) = \arctan \frac{4.392}{16.392} = \arctan(0.26794) = 0.26180 = \frac{\pi}{12} \checkmark$$

(the point is in the first quadrant, so arctan is legitimate here).

Step 6 — quotient.

$$\left| \frac{z_1}{z_2} \right| = \frac{3\sqrt{2}}{4} = 1.0607, \quad \arg \frac{z_1}{z_2} = \frac{3\pi}{4} + \frac{2\pi}{3} = \frac{17\pi}{12}.$$

This value does *not* lie in $(-\pi, \pi]$: to obtain the principal value we subtract 2π ,

$$\text{Arg} \frac{z_1}{z_2} = \frac{17\pi}{12} - 2\pi = -\frac{7\pi}{12} \quad (-105^\circ).$$

This is the step that Theorem 3.4 makes necessary: the sum of principal arguments is not, in general, the principal argument of the product.

3.2. Euler's formula and the exponential form

Theorem 3.5: Euler's formula. For every $\theta \in \mathbb{R}$:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Proof. [power series] The complex exponential is defined by the series $e^w = \sum_{n \geq 0} w^n/n!$, absolutely convergent for every $w \in \mathbb{C}$ (the ratio test gives infinite radius of convergence). Putting $w = i\theta$ and separating even from odd indices — legitimate by absolute convergence, which permits rearrangement —

$$e^{i\theta} = \sum_{k=0}^{\infty} \frac{(i\theta)^{2k}}{(2k)!} + \sum_{k=0}^{\infty} \frac{(i\theta)^{2k+1}}{(2k+1)!}.$$

Now $i^{2k} = (-1)^k$ and $i^{2k+1} = i(-1)^k$, hence

$$e^{i\theta} = \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k \theta^{2k+1}}{(2k+1)!} = \cos \theta + i \sin \theta,$$

recognising the Taylor series of cosine and sine. □

. Second proof — differential equation. Put $g(\theta) = (\cos \theta + i \sin \theta)e^{-i\theta}$. Termwise differentiation of power series is legitimate inside the radius of convergence and gives $\frac{d}{d\theta}e^{i\theta} = ie^{i\theta}$, hence $\frac{d}{d\theta}e^{-i\theta} = -ie^{-i\theta}$. By the product rule,

$$g'(\theta) = (-\sin \theta + i \cos \theta)e^{-i\theta} + (\cos \theta + i \sin \theta)(-i)e^{-i\theta}.$$

The second term equals $e^{-i\theta}(-i \cos \theta + \sin \theta)$, using $-i \cdot i \sin \theta = \sin \theta$. Adding inside the bracket,

$$(-\sin \theta + i \cos \theta) + (-i \cos \theta + \sin \theta) = 0,$$

so $g' \equiv 0$ and g is constant on $\mathbb{R}$. Since $g(0) = 1$, we get $g \equiv 1$, i.e. $\cos \theta + i \sin \theta = e^{i\theta}$.

Third proof — definition by a limit. Historically close to Euler's compound-interest intuition, this route starts from $e^{i\theta} := \lim_{n \rightarrow \infty} (1 + i\theta/n)^n$. Write $1 + i\theta/n$ in polar form: its modulus is $(1 + \theta^2/n^2)^{1/2}$ and its principal argument is $\arctan(\theta/n)$ (the point is in the right half-plane for large n , so $\arctan$ is legitimate). By Theorem 3.4, raising to the n -th power raises the modulus to the n -th power and multiplies the argument by n :

$$\left| \left(1 + \frac{i\theta}{n}\right)^n \right| = \exp \left[\frac{n}{2} \ln \left(1 + \frac{\theta^2}{n^2}\right) \right] \xrightarrow{n \rightarrow \infty} \exp(0) = 1,$$

since $\ln(1 + u) \sim u$ gives $\frac{\theta^2}{2n} \rightarrow 0$. For the argument, $n \arctan(\theta/n) \rightarrow \theta$, using $\arctan u \sim u$. Modulus $\rightarrow 1$ and argument $\rightarrow \theta$: the limit is the point of $\mathbb{T}$ with argument θ , i.e. $\cos \theta + i \sin \theta$. The fully rigorous version of this argument, with the uniform estimates in n that justify passing to the limit in modulus and argument simultaneously, is in Rudin, *Principles of Mathematical Analysis*, Chapter 8. $\square$

Corollary 3.2: Exponential parametrisation of the circle.

$$\mathbb{T} = \{e^{i\theta} : \theta \in (-\pi, \pi]\}, \quad C(0, r) = \{re^{i\theta}\},$$

and multiplication $z \mapsto e^{i\alpha}z$ is rotation by α about the origin. Moreover $(\mathbb{T}, \cdot)$ is an abelian group, with $(e^{i\theta})^{-1} = e^{-i\theta} = \overline{e^{i\theta}}$.

Proof. From Euler's formula, $|e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$, and by Theorem 3.3 every element of $\mathbb{T}$ is so written for a unique $\theta \in (-\pi, \pi]$. For the rotation:

$$e^{i\alpha} \cdot re^{i\theta} = re^{i(\alpha+\theta)},$$

where $e^{i\alpha}e^{i\theta} = e^{i(\alpha+\theta)}$ is Theorem 3.4 read in exponential form: modulus unchanged, argument shifted by α . The group structure is Corollary 3.1 (closure) together with the inverse $e^{-i\theta}$, of modulus 1; the identity $e^{-i\theta} = \overline{e^{i\theta}}$ follows from $\cos(-\theta) = \cos \theta$ and $\sin(-\theta) = -\sin \theta$. $\square$

Theorem 3.6: Inverse Euler formulas. For every $\theta \in \mathbb{R}$:

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Proof. From Euler's formula applied to θ and to $-\theta$:

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad e^{-i\theta} = \cos \theta - i \sin \theta,$$

the second because cosine is even and sine is odd. Adding eliminates the sine and gives

$e^{i\theta} + e^{-i\theta} = 2 \cos \theta$; subtracting eliminates the cosine and gives $e^{i\theta} - e^{-i\theta} = 2i \sin \theta$. Dividing by 2 and by $2i$ respectively concludes. $\square$

Remark 3.2: Why these two formulas matter so much. Theorem 3.6 is the dictionary allowing every trigonometric function to be replaced by complex exponentials. We already used it implicitly in Chapter 2 (Observation 2.12) to turn trigonometric integrals into contour integrals, and we shall use it systematically in Chapter 4: the complex Fourier coefficients are nothing but the real ones rewritten with these two identities. It is also the reason why multiplying exponentials “already knows” all the trigonometry of addition: the addition formulas are a corollary of $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$, as Exercise S1 shows.

Remark 3.3: Euler’s identity. For $\theta = \pi$ Euler’s formula gives $e^{i\pi} = -1$, that is

$$e^{i\pi} + 1 = 0,$$

a relation linking the five constants $0, 1, i, \pi, e$ in a single equality. Geometrically the content is modest and clear: the point of $\mathbb{T}$ with argument π is $(-1, 0)$. The historical context of the formula — published by Euler in the *Introductio in analysin infinitorum* (1748) — is documented in the MacTutor biography cited in the bibliography.

3.3. Hyperbolic functions and the unit circle

Euler’s formula (Theorem 3.5) links the exponential with *purely imaginary* argument $i\theta$ to cosine and sine. What happens for a *real* argument? The answer introduces the hyperbolic functions, the analogue of the unit hyperbola $x^2 - y^2 = 1$ for $(\cos t, \sin t)$ on the unit circle $x^2 + y^2 = 1$.

Definition 3.2: Hyperbolic cosine and sine. For every $x \in \mathbb{R}$,

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}.$$

Theorem 3.7: Hyperbolic Pythagorean identity. For every $x \in \mathbb{R}$, $\cosh^2 x - \sinh^2 x = 1$.

Proof.

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 = \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{4}.$$

The numerator is a difference of squares, $(A + B)^2 - (A - B)^2 = 4AB$ with $A = e^x$, $B = e^{-x}$, hence equals $4e^x e^{-x} = 4e^0 = 4$: the ratio is 1. $\square$

Theorem 3.8: Cosine and sine of an imaginary argument. For every $x \in \mathbb{R}$: $\cos(ix) = \cosh x$ and $\sin(ix) = i \sinh x$.

Proof. By Theorem 3.6 (Euler’s inverse formulas, valid for every complex argument, not only real, being identities between power series/entire holomorphic functions),

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Setting $\theta = ix$ ($x \in \mathbb{R}$): $e^{i\theta} = e^{i \cdot ix} = e^{-x}$ and $e^{-i\theta} = e^x$, hence

$$\cos(ix) = \frac{e^{-x} + e^x}{2} = \cosh x, \quad \sin(ix) = \frac{e^{-x} - e^x}{2i} = \frac{-(e^x - e^{-x})}{2i} = i \frac{e^x - e^{-x}}{2} = i \sinh x,$$

having used $-1/i = i$ (since $i \cdot i = -1$) in the last step. $\square$

Remark 3.4: The circle and the hyperbola: the same structure, two signs. This comparison is the pedagogical heart of this section. The parametrisation $\gamma(t) = (\cos t, \sin t)$ traces the unit circle $x^2 + y^2 = 1$ (Chapter 1) because $\cos^2 t + \sin^2 t = 1$ (ordinary Pythagorean identity). The parametrisation $\eta(t) = (\cosh t, \sinh t)$ traces, by Theorem 3.7, the right branch of the **unit hyperbola** $x^2 - y^2 = 1$ (since $\cosh t \geq 1 > 0$ for every t , being, by the Definition, a sum of two positive exponentials divided by 2, minimal at $t = 0$ where it equals 1): as t ranges over $\mathbb{R}$, $\eta(t)$ traces all of the right branch, because $\sinh$ is strictly increasing and bijective from $\mathbb{R}$ to $\mathbb{R}$ (derivative $\cosh t > 0$ always) and $\cosh t = \sqrt{1 + \sinh^2 t}$ is determined by $\sinh t$ on the right branch. Theorem 3.8 makes precise in what sense the two families are "the same formula": rotating the argument by 90° in the complex plane ($\theta \rightarrow i\theta$) turns the trigonometric circle into the hyperbola, flipping the sign in the Pythagorean identity.

Example 3.2: Numerical check. For $x = 1$: $\cosh 1 = \frac{e+e^{-1}}{2} = 1.5430806$, $\sinh 1 = \frac{e-e^{-1}}{2} = 1.1752012$, and $\cosh^2 1 - \sinh^2 1 = 2.3810978 - 1.3810978 = 1.0000000$ (also verified at full precision by the script). Independent check of the other identity: $\cos(i) = 1.5430806 - 0i = \cosh 1$ ✓ and $\sin(i) = 0 + 1.1752012i = i \sinh 1$ ✓ (script `vol2_cap03_iperboliche.py`, direct evaluation of $\cos, \sin$ on a complex argument and comparison at machine precision).

3.4. Loci in complex form

Complex notation is not merely a compact way of writing coordinates: it is a language in which certain loci take a *single* form. The central result of this section is that circles and lines, which in Cartesian coordinates have equations of different degree, obey one and the same equation in complex form.

Theorem 3.9: General equation of circles and lines. Consider the equation

$$A z \bar{z} + \bar{B} z + B \bar{z} + C = 0, \quad A, C \in \mathbb{R}, \quad B \in \mathbb{C}, \quad |B|^2 > AC. \quad (\star)$$

Then:

- (i) if $A \neq 0$, the solution set is the circle of centre $c = -B/A$ and radius $r = \sqrt{|B|^2 - AC} / |A|$;
- (ii) if $A = 0$ (and then necessarily $B \neq 0$), the set is a line;
- (iii) conversely, every circle and every line of the plane can be written in the form $(\star)$;
- (iv) the locus passes through the origin if and only if $C = 0$.

Proof. Put $z = x + iy$ and $B = \beta_1 + i\beta_2$ with $\beta_1, \beta_2 \in \mathbb{R}$. By Theorem 3.1(i), $z\bar{z} = x^2 + y^2$. Moreover, since $\overline{\bar{B}z} = B\bar{z}$, Theorem 3.1(vi) gives

$$\bar{B}z + B\bar{z} = \bar{B}z + \overline{\bar{B}z} = 2 \operatorname{Re}(\bar{B}z),$$

and

$$\bar{B}z = (\beta_1 - i\beta_2)(x + iy) = (\beta_1x + \beta_2y) + i(\beta_1y - \beta_2x),$$

so $\bar{B}z + B\bar{z} = 2(\beta_1x + \beta_2y)$. Equation $(\star)$ therefore becomes

$$A(x^2 + y^2) + 2\beta_1x + 2\beta_2y + C = 0. \quad (\star\star)$$

(i) **Case** $A \neq 0$. Dividing by A and completing the squares,

$$\left(x + \frac{\beta_1}{A}\right)^2 + \left(y + \frac{\beta_2}{A}\right)^2 = \frac{|B|^2 - AC}{A^2},$$

using $|B|^2 = \beta_1^2 + \beta_2^2$. Since $|B|^2 - AC > 0$ by hypothesis, the right-hand side is positive: this is the circle of radius $\sqrt{|B|^2 - AC} / |A|$ centred at the point with coordinates $(-\beta_1/A, -\beta_2/A)$, that is, in complex notation,

$$c = -\frac{\beta_1}{A} + i\left(-\frac{\beta_2}{A}\right) = -\frac{\beta_1 + i\beta_2}{A} = -\frac{B}{A}.$$

(ii) **Case** $A = 0$. The hypothesis $|B|^2 > AC$ becomes $|B|^2 > 0$, i.e. $B \neq 0$, and $(\star\star)$ reduces to $2\beta_1x + 2\beta_2y + C = 0$ with $(\beta_1, \beta_2) \neq (0, 0)$: the equation of a line.

(iii) **Conversely**. A circle of centre c and radius $r > 0$ has equation $|z - c| = r$, i.e., squaring and using Theorem 3.1(i),

$$z\bar{z} - \bar{c}z - c\bar{z} + |c|^2 - r^2 = 0,$$

which is $(\star)$ with $A = 1$, $B = -c$, $C = |c|^2 - r^2$; indeed $|B|^2 - AC = r^2 > 0$. A line $ax + by + d = 0$ with $(a, b) \neq (0, 0)$ is $(\star\star)$ with $A = 0$, $\beta_1 = a/2$, $\beta_2 = b/2$, $C = d$, i.e. $(\star)$ with $B = (a + ib)/2 \neq 0$.

(iv) Setting $z = 0$ in $(\star)$ leaves $C = 0$; conversely if $C = 0$ the point $z = 0$ satisfies the equation. $\square$

Warning. Step (i) of the proof shows how easy it is to make a sign error in a conjugate. The correct form of the centre is

$$c = -\frac{B}{A} \quad \text{when the equation is written as } Az\bar{z} + \bar{B}z + B\bar{z} + C = 0,$$

and the safe way to check it is always the same: go back to real coordinates with $z\bar{z} = x^2 + y^2$ and $\bar{B}z + B\bar{z} = 2\operatorname{Re}(B\bar{z})$, complete the squares, and compare. The radius is $r = \sqrt{|B|^2 - AC} / |A|$ in every case.

Theorem 3.10: The perpendicular bisector of a segment. Let $a, b \in \mathbb{C}$ be distinct. The locus

$$\{z \in \mathbb{C} : |z - a| = |z - b|\}$$

is the line through the midpoint of ab perpendicular to it (the perpendicular bisector), of equation

$$2\operatorname{Re}[(\bar{b} - \bar{a})z] = |b|^2 - |a|^2.$$

Proof. Squaring (legitimate: both sides are non-negative) and using $|w|^2 = w\bar{w}$:

$$(z - a)(\bar{z} - \bar{a}) = (z - b)(\bar{z} - \bar{b}),$$

that is

$$z\bar{z} - \bar{a}z - a\bar{z} + |a|^2 = z\bar{z} - \bar{b}z - b\bar{z} + |b|^2.$$

The term $z\bar{z}$ cancels — and this is the point: the resulting equation has degree one, hence a line. Rearranging,

$$(\bar{b} - \bar{a})z + (b - a)\bar{z} = |b|^2 - |a|^2,$$

whose left-hand side is $w + \bar{w}$ with $w = (\bar{b} - \bar{a})z$, i.e. $2 \operatorname{Re}[(\bar{b} - \bar{a})z]$ by Theorem 3.1(vi). That this is indeed the perpendicular bisector follows in two steps. First: the midpoint $m = (a + b)/2$ belongs to the locus, since $|m - a| = |b - a|/2 = |m - b|$. Second: the direction of the line is orthogonal to $b - a$. Indeed, if z_1, z_2 are two points of the locus, subtracting their equations gives

$$(\bar{b} - \bar{a})(z_1 - z_2) + (b - a)\overline{(z_1 - z_2)} = 0,$$

i.e. $\operatorname{Re}[(\bar{b} - \bar{a})(z_1 - z_2)] = 0$: the scalar product of the direction vector $z_1 - z_2$ with $b - a$ vanishes. (Recall that for $u, v \in \mathbb{C}$ the Euclidean scalar product is $\operatorname{Re}(\bar{u}v)$.) $\square$

Theorem 3.11: The Apollonius circle in complex form. Let $a, b \in \mathbb{C}$ be distinct and $k > 0$. The locus

$$\mathcal{A}_k = \left\{ z \in \mathbb{C} : \frac{|z - a|}{|z - b|} = k \right\}$$

is:

- (i) for $k = 1$, the perpendicular bisector of ab (Theorem 3.10);
- (ii) for $k \neq 1$, the circle of centre and radius

$$c = \frac{a - k^2 b}{1 - k^2}, \quad r = \frac{k|a - b|}{|1 - k^2|}.$$

Proof. The condition $|z - a| = k|z - b|$ squares to

$$z\bar{z} - \bar{a}z - a\bar{z} + |a|^2 = k^2(z\bar{z} - \bar{b}z - b\bar{z} + |b|^2),$$

that is, collecting terms,

$$(1 - k^2)z\bar{z} - (\bar{a} - k^2\bar{b})z - (a - k^2b)\bar{z} + (|a|^2 - k^2|b|^2) = 0. \quad (\dagger)$$

(i) **Case $k = 1$.** The coefficient of $z\bar{z}$ vanishes and $(\dagger)$ becomes the first-degree equation of Theorem 3.10.

(ii) **Case $k \neq 1$.** Divide $(\dagger)$ by $1 - k^2 \neq 0$ and set

$$c = \frac{a - k^2 b}{1 - k^2}, \quad \gamma = \frac{|a|^2 - k^2|b|^2}{1 - k^2}.$$

The equation takes the form $z\bar{z} - \bar{c}z - c\bar{z} + \gamma = 0$, i.e., completing the square as in the proof of Theorem 3.9(iii),

$$|z - c|^2 = |c|^2 - \gamma.$$

It remains to compute $|c|^2 - \gamma$. Writing $\lambda = 1 - k^2$,

$$|c|^2 = \frac{|a|^2 - k^2(\bar{a}\bar{b} + \bar{a}b) + k^4|b|^2}{\lambda^2}, \quad \gamma = \frac{|a|^2 - k^2|b|^2 - k^2|a|^2 + k^4|b|^2}{\lambda^2}.$$

Subtracting, the terms $|a|^2$ and $k^4|b|^2$ cancel:

$$|c|^2 - \gamma = \frac{k^2(|a|^2 - (\bar{a}\bar{b} + \bar{a}b) + |b|^2)}{\lambda^2}.$$

The bracket is exactly $|a - b|^2 = (a - b)(\bar{a} - \bar{b})$, hence

$$|c|^2 - \gamma = \frac{k^2|a - b|^2}{\lambda^2} = \left(\frac{k|a - b|}{|1 - k^2|} \right)^2 = r^2,$$

positive since $a \neq b$ and $k > 0$: the locus is the circle $C(c, r)$ as claimed. $\square$

Remark 3.5: Consistency with Volume I. Theorem 3.11 is the complex version of the Apollonius theorem already proved analytically in Volume I, Chapter 7. The comparison is instructive: there one had to expand in coordinates and complete two squares separately; here the quadratic term $z\bar{z}$ appears once, and the whole proof reduces to factoring out $1 - k^2$. It is an example of the concrete gain of complex notation: the circular symmetry of the problem is already built into the writing.

Example 3.3: An Apollonius locus, with test points. Determine the locus of points z with $|z - 2| = 2|z + 1|$, and verify it on test points.

Step 1 — identifying the data. Comparing with Theorem 3.11: $a = 2$, $b = -1$, $k = 2$. Since $k \neq 1$, the locus is a circle.

Step 2 — centre.

$$c = \frac{2 - 4(-1)}{1 - 4} = \frac{6}{-3} = -2.$$

Step 3 — radius.

$$r = \frac{2|2 - (-1)|}{3} = \frac{6}{3} = 2.$$

The locus is $C(-2, 2)$: the circle of centre $(-2, 0)$ and radius 2.

Step 4 — verification on three points of the locus.

- $z = 0$ (on $C(-2, 2)$, since $|0 + 2| = 2$): $|0 - 2| = 2$ and $2|0 + 1| = 2$. ✓
- $z = -4$ (since $|-4 + 2| = 2$): $|-4 - 2| = 6$ and $2|-4 + 1| = 6$. ✓
- $z = -2 + 2i$ (since $|(-2 + 2i) + 2| = 2$): $|-4 + 2i| = \sqrt{20}$ and $2|-1 + 2i| = 2\sqrt{5} = \sqrt{20}$. ✓

Step 5 — check on an external point. The point $z = 1$ is not on $C(-2, 2)$ (it is at distance 3 from the centre): indeed $|1 - 2| = 1$ while $2|1 + 1| = 4 \neq 1$. ✓

Typical error. Squaring $|z - 2| = 2|z + 1|$ and forgetting the factor $k^2 = 4$ (not 2) on the right. The check of Step 4 on at least two points catches this immediately, because with $k^2 = 2$ one would obtain a different centre and radius that none of the test points satisfies.

3.5. Plane transformations in complex form

Theorem 3.12: Direct similarities. Let $f(z) = \alpha z + \beta$ with $\alpha, \beta \in \mathbb{C}$, $\alpha \neq 0$. Then:

- (i) f is a **direct similarity** of ratio $|\alpha|$: for all z, w , $|f(z) - f(w)| = |\alpha||z - w|$;
- (ii) f maps circles to circles and lines to lines; the image of $C(c, r)$ is $C(f(c), |\alpha|r)$;
- (iii) if $\alpha \neq 1$, f has a unique fixed point $z^* = \beta/(1 - \alpha)$ and can be written

$$f(z) = z^* + \alpha(z - z^*),$$

i.e. it is the composition of a homothety of ratio $|\alpha|$ and a rotation by $\text{Arg } \alpha$, both centred at z^* ;

- (iv) if $\alpha = 1$ and $\beta \neq 0$, f is the translation by β and has no fixed point.

Proof. (i) $f(z) - f(w) = \alpha(z - w)$, and by Theorem 3.1(iii), $|f(z) - f(w)| = |\alpha||z - w|$.

(ii) From (i), $|z - c| = r \iff |f(z) - f(c)| = |\alpha|r$: the image of $C(c, r)$ is contained in $C(f(c), |\alpha|r)$, and equals it because f is bijective (with inverse $w \mapsto (w - \beta)/\alpha$, of the same form). For lines: if $2\text{Re}(\bar{\nu}z) = d$ is the equation of a line, substituting $z = (w - \beta)/\alpha$ still yields a first-degree equation in $\text{Re } w$ and $\text{Im } w$, hence a line.

(iii) The equation $f(z) = z$ reads $(1 - \alpha)z = -\beta$; for $\alpha \neq 1$ it has the unique solution $z^* = \beta/(1 - \alpha)$ — one checks directly $\alpha z^* + \beta = \frac{\alpha\beta + \beta(1 - \alpha)}{1 - \alpha} = z^*$. ✓ Rewriting,

$$f(z) - z^* = \alpha z + \beta - z^* = \alpha(z - z^*),$$

using $\beta = z^*(1 - \alpha)$. Writing $\alpha = |\alpha|e^{i\varphi}$ with $\varphi = \text{Arg } \alpha$, the map $z \mapsto z^* + \alpha(z - z^*)$ is: translation taking z^* to the origin, then rotation by φ (by Corollary 3.2) composed with homothety of ratio $|\alpha|$, then the inverse translation.

(iv) If $\alpha = 1$, $f(z) = z + \beta$ and $f(z) = z$ would require $\beta = 0$. $\square$

Theorem 3.13: Isometries fixing the unit circle. Every isometry of the plane fixing the origin has the form

$$z \mapsto e^{i\alpha}z \quad (\text{rotation}) \quad \text{or} \quad z \mapsto e^{i\alpha}\bar{z} \quad (\text{reflection}),$$

for some $\alpha \in \mathbb{R}$; both map $\mathbb{T}$ onto itself.

Proof. An isometry of the plane fixing the origin is linear (a classical fact: it preserves distances and the origin, hence preserves scalar products by the polarisation identity, and a map preserving scalar products and fixing 0 is linear) and is represented by a 2×2 orthogonal matrix, of determinant $+1$ (rotation) or -1 (reflection).

In the first case the matrix is $Q(\alpha)$ of Chapter 1, Definition 1.4, and by Corollary 3.2 its action in complex notation is $z \mapsto e^{i\alpha}z$.

In the second case, composing with the reflection in the real axis — which in complex notation is conjugation $z \mapsto \bar{z}$, since $\overline{x + iy} = x - iy$ — yields a matrix of determinant $+1$, i.e. a rotation: hence the original reflection is $z \mapsto e^{i\alpha}\bar{z}$ for some α .

Both preserve the modulus ($|e^{i\alpha}z| = |z|$ and $|\bar{z}| = |z|$), hence map $\mathbb{T}$ onto itself. $\square$

Definition 3.3: Circular inversion in complex form. The **inversion** in $C(0, r)$ is the map

$$\iota(z) = \frac{r^2}{\bar{z}}, \quad z \in \mathbb{C} \setminus \{0\},$$

extended to $\mathbb{C} \cup \{\infty\}$ by $\iota(0) = \infty$ and $\iota(\infty) = 0$. With respect to $\mathbb{T}$ ($r = 1$) it is $\iota(z) = 1/\bar{z}$.

Proposition 3.1: Properties of inversion. With $\iota(z) = r^2/\bar{z}$:

- (i) ι is involutive: $\iota(\iota(z)) = z$;
- (ii) $|z| \cdot |\iota(z)| = r^2$, and $z, \iota(z)$ lie on the same ray from the origin;
- (iii) the points of $C(0, r)$ are exactly the fixed points of ι ;
- (iv) ι interchanges the interior and the exterior of the disc.

Proof. (i) $\iota(\iota(z)) = \frac{r^2}{\overline{r^2/\bar{z}}} = \frac{r^2}{r^2/z} = z$, since r^2 is real so $\overline{r^2/\bar{z}} = r^2/z$.

(ii) $|\iota(z)| = r^2/|z|$, whence the product. For the direction,

$$\iota(z) = \frac{r^2}{\bar{z}} = \frac{r^2 z}{\bar{z}z} = \frac{r^2}{|z|^2} z,$$

i.e. $\iota(z)$ is a *positive real* multiple of z : same argument, same ray.

(iii) $\iota(z) = z \iff r^2/|z|^2 = 1 \iff |z| = r$.

(iv) From (ii), $|z| < r \iff |\iota(z)| = r^2/|z| > r$. $\square$

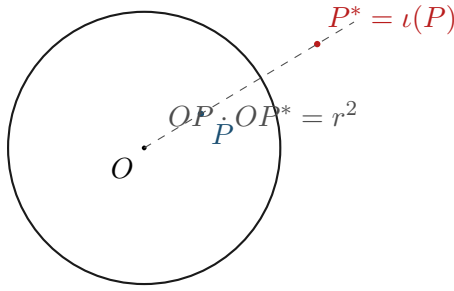


Figure 3.1: Inversion in $C(O, r)$: P and P^* lie on the same ray from O , with $OP \cdot OP^* = r^2$; the points of the circle are fixed (Proposition 3.1).

3.5.1 Möbius transformations

Remark 3.6: Prerequisites. This subsection presupposes the extended complex plane $\mathbb{C} \cup \{\infty\}$ — the plane with a single “point at infinity” adjoined, already met in Chapter 1 in connection with stereographic projection (Theorem 1.7) — and Theorem 3.9 on the single equation of circles and lines.

Definition 3.4: Möbius transformation. A Möbius transformation is a map $f : \mathbb{C} \cup \{\infty\} \rightarrow \mathbb{C} \cup \{\infty\}$ of the form

$$f(z) = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc \neq 0,$$

with the conventions $f(-d/c) = \infty$ and $f(\infty) = a/c$ when $c \neq 0$, and $f(\infty) = \infty$ when $c = 0$. Translations, rotations, homotheties and the inversion $z \mapsto 1/z$ are particular cases.

Theorem 3.14: Möbius transformations map circles and lines to circles and lines. If f is a Möbius transformation, the image of every circle or line of the extended plane is again a circle or a line. More precisely, a circle or line through the pole $z = -d/c$ has a line as image; if it does not pass through the pole, the image is a circle.

Proof. Step 1 — decomposition. If $c = 0$ (and then $d \neq 0$ since $ad - bc \neq 0$), $f(z) = \frac{a}{d}z + \frac{b}{d}$ is a direct similarity and the claim is Theorem 3.12(ii). If $c \neq 0$, Euclidean division gives

$$f(z) = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d},$$

as one verifies by putting the right-hand side over a common denominator:

$$\frac{a(cz + d) + bc - ad}{c(cz + d)} = \frac{acz + bc}{c(cz + d)} = \frac{az + b}{cz + d}. \checkmark$$

Hence f is the composition of the chain

$$z \mapsto cz \mapsto cz + d \mapsto \frac{1}{cz + d} \mapsto \frac{bc - ad}{c} \cdot \frac{1}{cz + d} + \frac{a}{c},$$

that is: two similarities, two translations and *one single* inversion $w \mapsto 1/w$. Since similarities preserve the family “circles and lines” (Theorem 3.12(ii)), it suffices to prove the claim for $w = 1/z$.

Step 2 — invariance of the general equation. By Theorem 3.9, every circle or line has equation

$$Az\bar{z} + \bar{B}z + B\bar{z} + C = 0, \quad A, C \in \mathbb{R}, \quad B \in \mathbb{C}, \quad |B|^2 > AC.$$

Substitute $z = 1/w$, $\bar{z} = 1/\bar{w}$ (legitimate for $w \neq 0$) and multiply by $w\bar{w} = |w|^2 \neq 0$:

$$\frac{A}{w\bar{w}} + \frac{\bar{B}}{w} + \frac{B}{\bar{w}} + C = 0 \xrightarrow{\times w\bar{w}} A + \bar{B}\bar{w} + Bw + Cw\bar{w} = 0.$$

Rearranging and renaming,

$$A'w\bar{w} + \bar{B}'w + B'\bar{w} + C' = 0, \quad A' = C, \quad B' = \bar{B}, \quad C' = A,$$

an equation of the same form, with $A', C' \in \mathbb{R}$ and $B' \in \mathbb{C}$; and the non-degeneracy condition is preserved:

$$|B'|^2 - A'C' = |B|^2 - AC > 0.$$

Step 3 — when the image is a line. By Theorem 3.9(ii), the image is a line exactly when $A' = 0$, i.e. when $C = 0$, i.e. — by part (iv) of the same theorem — when the original curve passes through the origin, which is the pole of the inversion $w = 1/z$. Returning to the chain of Step 1, the pole of f is the point sent to the origin by the first part of the chain, namely $z = -d/c$: whence the precise statement. $\square$

***Remark 3.7: Why a circle can become a line.** A circle is compact, a line is not: apparently the two cannot correspond. The contradiction dissolves in the extended plane: the image line is really a line plus the point at infinity, and the point of the circle that goes there is precisely the pole, i.e. the point f sends to ∞ . In the extended plane, lines and circles are the same thing — both are circles on the Riemann sphere, and lines are those passing through the north pole.*

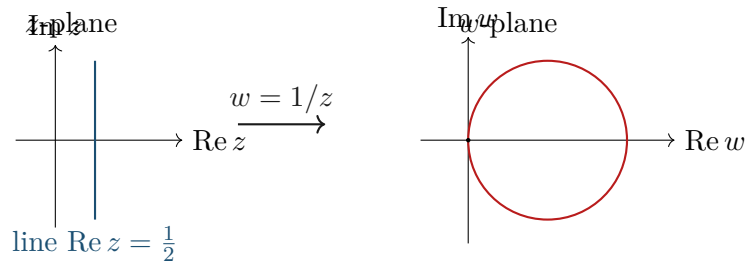


Figure 3.2: The line $\operatorname{Re} z = \frac{1}{2}$ (which does *not* pass through the origin) is mapped by the inversion $w = 1/z$ to the circle of centre 1 and radius 1, which *does* pass through the origin: this is the rule $A = 0 \Rightarrow C' = A = 0$ of Step 3, read backwards.

***Corollary 3.3: The unit circle becomes the imaginary axis.** The Möbius transformation $w = f(z) = \frac{z-1}{z+1}$ maps $\mathbb{T} \setminus \{-1\}$ onto the imaginary axis of the w -plane; precisely $f(e^{i\theta}) = i \tan(\theta/2)$ for $\theta \in (-\pi, \pi)$.*

Proof. For $z = e^{i\theta}$ with $\theta \neq \pi$, multiply numerator and denominator by $e^{-i\theta/2}$:

$$w = \frac{e^{i\theta/2} - e^{-i\theta/2}}{e^{i\theta/2} + e^{-i\theta/2}}.$$

By Theorem 3.6 the numerator is $2i \sin(\theta/2)$ and the denominator $2 \cos(\theta/2)$, non-zero for $\theta \in (-\pi, \pi)$. Hence

$$w = i \tan \frac{\theta}{2},$$

purely imaginary. As θ ranges over $(-\pi, \pi)$, $\tan(\theta/2)$ ranges over all of $\mathbb{R}$: the image is the whole imaginary axis. That $z = -1$ goes to ∞ is consistent with Theorem 3.14: $\mathbb{T}$ passes through the pole $z = -1$ of f , hence its image is a line. $\square$

Remark 3.8: Why it matters: the bilinear transform. Corollary 3.3 is not an isolated curiosity: it is essentially the **bilinear** (or Tustin) **transform** of digital control, which in the form

$$s = \frac{2}{T} \cdot \frac{z - 1}{z + 1}$$

(T = sampling period) links the discrete z -domain to the continuous s -domain. That $\mathbb{T}$ corresponds exactly to the imaginary axis is no accident: they are, respectively, the locus of marginal stability for discrete-time and for continuous-time systems. A pole inside $\mathbb{T}$ corresponds to a pole in the left half-plane, i.e. to a stable system; a pole outside, to an unstable one. We shall meet this again in Chapter 4, where the Z -transform makes it operational (Theorem, Volume III, Chapter 21).

3.6. Powers: de Moivre's theorem

Theorem 3.15: De Moivre's theorem. For every $\theta \in \mathbb{R}$ and every $n \in \mathbb{Z}$:

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta,$$

that is, in exponential form, $(e^{i\theta})^n = e^{in\theta}$. More generally, for $z = \rho e^{i\theta}$ with $\rho > 0$,

$$z^n = \rho^n e^{in\theta}, \quad \text{i.e.} \quad |z^n| = |z|^n \quad \text{and} \quad \arg(z^n) = n \arg z.$$

Proof. [by induction] **Case** $n \geq 0$. For $n = 0$ both sides equal 1. Assume the formula for some $n \geq 0$ and prove it for $n + 1$:

$$(\cos \theta + i \sin \theta)^{n+1} = (\cos n\theta + i \sin n\theta)(\cos \theta + i \sin \theta),$$

by the inductive hypothesis. Expanding,

$$= (\cos n\theta \cos \theta - \sin n\theta \sin \theta) + i(\sin n\theta \cos \theta + \cos n\theta \sin \theta),$$

and by the addition formulas (Volume I, Chapter 8) the two brackets are $\cos((n+1)\theta)$ and $\sin((n+1)\theta)$. Induction concludes.

Case $n < 0$. Put $n = -m$ with $m > 0$. Since $(\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta) = 1$, the inverse of $\cos \theta + i \sin \theta$ is $\cos(-\theta) + i \sin(-\theta)$. Applying the positive case to $-\theta$:

$$(\cos \theta + i \sin \theta)^{-m} = \cos(-m\theta) + i \sin(-m\theta) = \cos n\theta + i \sin n\theta.$$

$\square$

. Second proof — via Euler's formula. By Theorem 3.5, $\cos \theta + i \sin \theta = e^{i\theta}$. The property $e^u e^v = e^{u+v}$ of the complex exponential — which for purely imaginary exponents is Theorem 3.4 — gives at once, by induction, $(e^{i\theta})^n = e^{in\theta}$ for $n \geq 0$, and for $n < 0$ by passing to

the inverse $(e^{i\theta})^{-1} = e^{-i\theta}$. Re-applying Euler's formula to the result gives $\cos n\theta + i \sin n\theta$. The two proofs are not logically equivalent: the first uses only the addition formulas and is therefore available to a reader who does not yet have Euler's formula (it is de Moivre's historical route, 1707–1730); the second is shorter but presupposes the complex exponential. $\square$

Corollary 3.4: Multiple-angle formulas and Chebyshev polynomials. For every $n \geq 0$ there is a polynomial T_n with integer coefficients, of degree n , such that

$$\cos n\theta = T_n(\cos \theta) \quad \text{for every } \theta \in \mathbb{R}.$$

These polynomials (the **Chebyshev polynomials of the first kind**) satisfy

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

Proof. From the product-to-sum formulas (Volume I, Chapter 8),

$$\cos((n+1)\theta) = 2 \cos \theta \cos n\theta - \cos((n-1)\theta).$$

We argue by strong induction on n . For $n = 0$ and $n = 1$ the identities $\cos 0 = 1$ and $\cos \theta = \cos \theta$ give $T_0 = 1$ and $T_1(x) = x$. Assume polynomials T_{n-1}, T_n with integer coefficients exist with $\cos((n-1)\theta) = T_{n-1}(\cos \theta)$ and $\cos n\theta = T_n(\cos \theta)$. Substituting in the relation just written,

$$\cos((n+1)\theta) = 2 \cos \theta T_n(\cos \theta) - T_{n-1}(\cos \theta) = T_{n+1}(\cos \theta),$$

where $T_{n+1}(x) := 2xT_n(x) - T_{n-1}(x)$ still has integer coefficients. The degree increases by one at each step (the term $2xT_n$ has degree $n+1$ with leading coefficient 2^n , which T_{n-1} , of degree $n-1$, cannot cancel). $\square$

Example 3.4: The triple-angle formulas from de Moivre. Derive $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos \theta$ and $\sin \theta$.

Step 1 — de Moivre with $n = 3$.

$$\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3.$$

Step 2 — binomial expansion. Writing $c = \cos \theta$, $s = \sin \theta$:

$$(c + is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3,$$

using $i^2 = -1$ and $i^3 = -i$.

Step 3 — separating real and imaginary parts.

$$\cos 3\theta = c^3 - 3cs^2, \quad \sin 3\theta = 3c^2s - s^3.$$

Step 4 — reduction to a single function. With $s^2 = 1 - c^2$:

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta,$$

and with $c^2 = 1 - s^2$:

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta.$$

The first identity shows $T_3(x) = 4x^3 - 3x$, in agreement with the recurrence of Corollary 3.4: $T_3 = 2x(2x^2 - 1) - x = 4x^3 - 3x$. $\checkmark$

Step 5 — numerical check. For $\theta = \pi/9$ (20°), $\cos \theta = 0.9396926$:

$$4(0.9396926)^3 - 3(0.9396926) = 4(0.8297694) - 2.8190779 = 3.3190776 - 2.8190779 = 0.4999997,$$

while $\cos(3\theta) = \cos(60^\circ) = 0.5$. ✓

Historical note. The identity $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ is the one reducing angle trisection to a cubic equation, and it is by this route that the impossibility of trisection with straightedge and compass is proved: the topic is treated in Volume I, Chapters 8-9.

3.7. n -th roots

Theorem 3.16: n -th roots of a complex number. Let $w \in \mathbb{C} \setminus \{0\}$, $w = \rho e^{i\varphi}$ with $\rho = |w| > 0$ and $\varphi = \text{Arg } w$, and let $n \geq 1$ be an integer. The equation $z^n = w$ has exactly n distinct solutions in $\mathbb{C}$:

$$z_k = \rho^{1/n} \exp\left(i \frac{\varphi + 2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1,$$

where $\rho^{1/n}$ is the unique positive real n -th root of ρ . They are the vertices of a regular n -gon inscribed in the circle $C(0, \rho^{1/n})$.

Proof. **Step 1 — each z_k is a solution.** By Theorem 3.15,

$$z_k^n = \rho e^{i(\varphi + 2k\pi)} = \rho e^{i\varphi} e^{2k\pi i} = w,$$

since $e^{2k\pi i} = 1$ for every $k \in \mathbb{Z}$.

Step 2 — there are no others. Let z be any solution. First $z \neq 0$ (otherwise $w = 0$), so $z = re^{i\alpha}$ with $r > 0$. From $z^n = w$ and Theorem 3.15, $r^n e^{in\alpha} = \rho e^{i\varphi}$. Equating moduli (Theorem 3.1(iii)) gives $r^n = \rho$, and since $t \mapsto t^n$ is strictly increasing and bijective from $(0, \infty)$ onto itself, $r = \rho^{1/n}$ is uniquely determined. Cancelling the moduli leaves $e^{in\alpha} = e^{i\varphi}$, i.e., by Theorem 3.3 (set of arguments),

$$n\alpha = \varphi + 2m\pi \quad \text{for some } m \in \mathbb{Z}, \quad \text{i.e.} \quad \alpha = \frac{\varphi + 2m\pi}{n}.$$

Step 3 — only n distinct values. Writing $m = qn + k$ with $k \in \{0, \dots, n-1\}$ (Euclidean division),

$$\frac{\varphi + 2m\pi}{n} = \frac{\varphi + 2k\pi}{n} + 2q\pi,$$

and since the exponential has period $2\pi i$, $z_m = z_k$. So at most the n values $z_0, \dots, z_{n-1}$ remain. That they are *distinct* follows because their arguments

$$\frac{\varphi}{n}, \frac{\varphi}{n} + \frac{2\pi}{n}, \dots, \frac{\varphi}{n} + \frac{2(n-1)\pi}{n}$$

lie in an interval of length $\frac{2(n-1)\pi}{n} < 2\pi$ and are pairwise different: no two of them can differ by a non-zero multiple of 2π , hence by Theorem 3.3 the corresponding complex numbers are different.

Step 4 — geometric configuration. All the z_k have the same modulus $\rho^{1/n}$, hence lie on $C(0, \rho^{1/n})$; consecutive arguments differ by exactly $2\pi/n$, which is the regularity condition for an inscribed polygon (equal central angles, by definition of a regular polygon). $\square$

Warning. The symbol $\sqrt[n]{w}$ is *ambiguous* over $\mathbb{C}$: it denotes n values, not one. Writing $\sqrt[n]{w} = \rho^{1/n} e^{i\varphi/n}$ means choosing the **principal value** (the one with $k = 0$), and the choice must be declared. With this convention $\sqrt{zw} = \sqrt{z}\sqrt{w}$ does *not* hold in general: for $z = w = -1$ the left side is $\sqrt{1} = 1$ and the right side $i \cdot i = -1$. It is the same phenomenon as the principal value of the argument, and the reason why this chapter always speaks of “the n -th roots”, in the plural.

Definition 3.5: n -th roots of unity. For an integer $n \geq 1$ set

$$\zeta_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1,$$

and write $\mu_n = \{\zeta_0, \dots, \zeta_{n-1}\}$. By Theorem 3.16 (case $w = 1$) these are exactly the n solutions of $z^n = 1$.

Theorem 3.17: The roots of unity form a regular polygon. The points $\zeta_0, \dots, \zeta_{n-1}$ are the vertices of a regular n -gon inscribed in $\mathbb{T}$, with one vertex at 1. The side of the polygon measures $2 \sin(\pi/n)$ and the perimeter $2n \sin(\pi/n)$.

Proof. The polygonal configuration is Step 4 of the proof of Theorem 3.16 with $\rho = 1$, and $\zeta_0 = 1$.

For the side length, compute $|\zeta_{k+1} - \zeta_k|$. Factoring out ζ_k and using $|\zeta_k| = 1$,

$$|\zeta_{k+1} - \zeta_k| = |e^{2\pi i/n} - 1|,$$

independent of k : the polygon is equilateral. Now, with $\alpha = 2\pi/n$ and using $e^{i\alpha} - 1 = e^{i\alpha/2}(e^{i\alpha/2} - e^{-i\alpha/2})$ together with Theorem 3.6,

$$|e^{i\alpha} - 1| = 2|\sin(\alpha/2)| = 2 \sin \frac{\pi}{n},$$

positive since $0 < \pi/n \leq \pi$ for $n \geq 1$. The perimeter is n times the side. $\square$

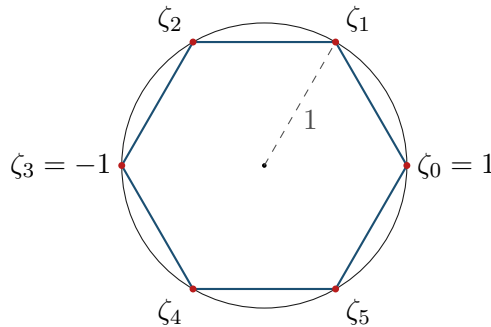


Figure 3.3: The sixth roots of unity: vertices of a regular hexagon inscribed in $\mathbb{T}$, with one vertex at 1. The side measures $2 \sin(\pi/6) = 1$, equal to the radius — the characteristic property of the regular hexagon already noted in Volume I.

Theorem 3.18: μ_n is a cyclic group of order n . μ_n is a subgroup of $(\mathbb{T}, \cdot)$ of order n , cyclic, generated by $\zeta_1 = e^{2\pi i/n}$; it is the kernel of the homomorphism $\mathbb{T} \rightarrow \mathbb{T}$, $z \mapsto z^n$.

Proof. Subgroup. If $z^n = 1$ and $w^n = 1$ then $(zw)^n = 1$ and $(z^{-1})^n = 1$; moreover $1 \in \mu_n$. The order is n by Theorem 3.16.

Cyclicity. By Theorem 3.15, $\zeta_1^k = e^{2\pi i k/n} = \zeta_k$: the powers of ζ_1 exhaust μ_n , so ζ_1 generates it.

Kernel. The map $\pi_n : \mathbb{T} \rightarrow \mathbb{T}$, $\pi_n(z) = z^n$, is a homomorphism (since $(zw)^n = z^n w^n$) and its kernel is by definition $\{z \in \mathbb{T} : z^n = 1\} = \mu_n$, since every solution of $z^n = 1$ has modulus 1 (Step 2 of the proof of Theorem 3.16 with $\rho = 1$). $\square$

Definition 3.6: Primitive root. A root $\zeta \in \mu_n$ is **primitive** if it generates μ_n , i.e. if the least integer $d \geq 1$ with $\zeta^d = 1$ is exactly n .

Theorem 3.19: Characterisation of primitive roots. For $k \in \{0, \dots, n-1\}$, the order of $\zeta_k = e^{2\pi i k/n}$ in the group μ_n is

$$\text{ord}(\zeta_k) = \frac{n}{\gcd(k, n)}.$$

In particular ζ_k is primitive if and only if $\gcd(k, n) = 1$, and the number of primitive n -th roots of unity is $\varphi(n)$, Euler's totient function.

Proof. Put $g = \gcd(k, n)$, $k = gk'$, $n = gn'$ with $\gcd(k', n') = 1$. By Theorem 3.15,

$$\zeta_k^d = e^{2\pi i k d/n} = 1 \iff \frac{kd}{n} \in \mathbb{Z} \iff n \mid kd.$$

Substituting $k = gk'$, $n = gn'$: $n' \mid k'd$. Since $\gcd(k', n') = 1$, Euclid's lemma gives $n' \mid d$. The least $d \geq 1$ with this property is $d = n' = n/g$, which is therefore the order of ζ_k .

Consequently ζ_k is primitive ($\text{ord} = n$) iff $g = 1$, and the number of such k in $\{0, \dots, n-1\}$ is by definition $\varphi(n)$. $\square$

Example 3.5: The twelfth roots of unity and their primitives. List the twelfth roots of unity, identify the primitive ones and check the order of some of them.

Step 1 — the twelve roots. $\zeta_k = e^{i\pi k/6}$, $k = 0, \dots, 11$: the vertices of the regular dodecagon inscribed in $\mathbb{T}$ with one vertex at 1. For instance

$$\zeta_0 = 1, \quad \zeta_1 = \frac{\sqrt{3}}{2} + \frac{i}{2}, \quad \zeta_3 = i, \quad \zeta_6 = -1, \quad \zeta_9 = -i.$$

Step 2 — orders. By Theorem 3.19, $\text{ord}(\zeta_k) = 12/\gcd(k, 12)$:

k	0	1	2	3	4	5	6	7	8	9	10	11
$\gcd(k, 12)$	12	1	2	3	4	1	6	1	4	3	2	1
ord	1	12	6	4	3	12	2	12	3	4	6	12

(for $k = 0$ we read $\gcd(0, 12) = 12$, and indeed $\zeta_0 = 1$ has order 1).

Step 3 — the primitive roots. They are those of order 12, i.e. $k \in \{1, 5, 7, 11\}$: four of them, $= \varphi(12)$. $\checkmark$ (Indeed $12 = 2^2 \cdot 3$ and $\varphi(12) = 12(1 - \frac{1}{2})(1 - \frac{1}{3}) = 4$.)

Step 4 — direct check on $\zeta_3 = i$. The predicted order is 4: indeed $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and no smaller power equals 1. $\checkmark$ The powers of ζ_3 generate $\{1, i, -1, -i\} = \mu_4 \subset \mu_{12}$: a proper subgroup, consistently with ζ_3 not being primitive.

Step 5 — check on ζ_5 . Since $\gcd(5, 12) = 1$, ζ_5 should be primitive. Indeed the powers ζ_5^j correspond, for $j = 0, \dots, 11$, to the indices $0, 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7$: all twelve indices, each exactly once. $\checkmark$

3.8. Sums, products and the master orthogonality identity

Theorem 3.20: Sum of powers of the roots of unity. Let $\zeta_1 = e^{2\pi i/n}$ and $m \in \mathbb{Z}$. Then

$$\sum_{k=0}^{n-1} \zeta_1^{km} = \sum_{k=0}^{n-1} e^{2\pi i km/n} = \begin{cases} n & \text{if } n \mid m, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Put $q = \zeta_1^m = e^{2\pi im/n}$; the sum is $\sum_{k=0}^{n-1} q^k$.

Case $n \mid m$. Then $m/n \in \mathbb{Z}$ and $q = 1$: the sum is n .

Case $n \nmid m$. Then $m/n \notin \mathbb{Z}$, so $q \neq 1$ by Theorem 3.3. The finite geometric sum formula gives

$$\sum_{k=0}^{n-1} q^k = \frac{q^n - 1}{q - 1},$$

and $q^n = e^{2\pi im} = 1$ because m is an integer. The numerator vanishes while the denominator does not: the sum is 0. $\square$

Remark 3.9: The master identity. Theorem 3.20 is, without exaggeration, the most reused identity of the whole volume. It appears, always the same, under at least five disguises:

- as the sum of the roots of unity ($m = 1$), Corollary 3.5;
- as the orthogonality of the columns of the DFT matrix, Theorem 3.22;
- as the integral of powers over a circle, Lemma (Volume III, Chapter 15), its continuous version;
- as the orthogonality of complex exponentials in harmonic analysis (Chapter 4, Theorem 4.2);
- as the aliasing mechanism in the periodic trapezoidal rule (Chapter 2, Theorem 2.17, Step 3).

In every case the content is the same: summing (or integrating) a non-trivial character over the whole group gives zero. It is the orthogonality of characters, which Chapter 4 will state in its general form.

Corollary 3.5: Sum of the n -th roots of unity. For $n \geq 2$: $\sum_{k=0}^{n-1} \zeta_k = 0$.

Proof. This is Theorem 3.20 with $m = 1$: since $n \geq 2$, $n \nmid 1$ and the sum vanishes. $\square$

Remark 3.10: Physical reading of the corollary. Interpreting each ζ_k as a vector from the centre to a vertex of the regular polygon, Corollary 3.5 says that their resultant vanishes. This is exactly why n equal masses at the vertices of a regular polygon have their centre of mass at the centre, and why a symmetric n -phase system (with $n \geq 2$) has zero instantaneous sum of voltages — the principle of the neutral conductor in three-phase systems, taken up in Section 3.10.

Theorem 3.21: Cyclotomic factorisation and product of the roots. For every $n \geq 1$:

(i) $z^n - 1 = \prod_{k=0}^{n-1} (z - \zeta_k)$ for all $z \in \mathbb{C}$;

(ii) $\prod_{k=0}^{n-1} \zeta_k = (-1)^{n-1}$;

$$(iii) \prod_{k=1}^{n-1} (1 - \zeta_k) = n \quad (\text{for } n \geq 2).$$

Proof. (i) The polynomial $p(z) = z^n - 1$ has degree n and leading coefficient 1; by Theorem 3.16 its roots are exactly $\zeta_0, \dots, \zeta_{n-1}$, which are n and distinct. A monic polynomial of degree n with n distinct roots factors as the product of the corresponding linear factors (repeated application of the factor theorem).

(ii) Evaluating (i) at $z = 0$:

$$-1 = \prod_{k=0}^{n-1} (-\zeta_k) = (-1)^n \prod_{k=0}^{n-1} \zeta_k,$$

whence $\prod_k \zeta_k = (-1)^{1-n} = (-1)^{n-1}$, the last step because $1 - n$ and $n - 1$ have the same parity.

(iii) From (i), isolating the factor $k = 0$:

$$z^n - 1 = (z - 1) \prod_{k=1}^{n-1} (z - \zeta_k).$$

On the other hand $z^n - 1 = (z - 1)(1 + z + \dots + z^{n-1})$. Comparing and cancelling $z - 1$ (legitimate for $z \neq 1$),

$$\prod_{k=1}^{n-1} (z - \zeta_k) = 1 + z + \dots + z^{n-1} \quad (z \neq 1).$$

Both sides are polynomials, hence continuous: the identity extends to $z = 1$. Evaluating there gives $\prod_{k=1}^{n-1} (1 - \zeta_k) = n$. $\square$

Corollary 3.6: The product of the diagonals of a regular polygon. In a regular n -gon inscribed in the unit circle, the product of the distances from one fixed vertex to all the other $n - 1$ vertices equals exactly n :

$$\prod_{k=1}^{n-1} |1 - \zeta_k| = n.$$

Proof. Take moduli in Theorem 3.21(iii) and use multiplicativity (Theorem 3.1(iii)): $\prod_k |1 - \zeta_k| = |n| = n$. The fixed vertex is $\zeta_0 = 1$; by symmetry (multiplying everything by ζ_j , a rotation, which preserves distances) the same holds for every other vertex. $\square$

Example 3.6: Checking the product of the diagonals for $n = 4$ and $n = 6$. **Case $n = 4$.** The fourth roots are $1, i, -1, -i$. The distances from the vertex 1 to the other three:

$$|1 - i| = \sqrt{2}, \quad |1 - (-1)| = 2, \quad |1 - (-i)| = \sqrt{2}.$$

Product: $\sqrt{2} \cdot 2 \cdot \sqrt{2} = 4 = n$. $\checkmark$

Case $n = 6$. The chord subtending a central angle α on $\mathbb{T}$ measures $2 \sin(\alpha/2)$ (Theorem 3.17), so the five distances from the vertex 1 are

$$2 \sin 30^\circ = 1, \quad 2 \sin 60^\circ = \sqrt{3}, \quad 2 \sin 90^\circ = 2, \quad 2 \sin 120^\circ = \sqrt{3}, \quad 2 \sin 150^\circ = 1.$$

Product: $1 \cdot \sqrt{3} \cdot 2 \cdot \sqrt{3} \cdot 1 = 6 = n$. $\checkmark$

Remark. The result is far from obvious a priori: as n grows the individual distances become tiny (the nearest are $\approx 2\pi/n$), yet their product grows *linearly* in n . It is an example of how the algebraic factorisation of Theorem 3.21 says things that direct geometric computation would not make evident.

Remark 3.11: Connection with Archimedes' method. The n -th roots of unity are exactly the vertices of the regular polygon used by Archimedes (Volume I, Chapter 1) to approximate π . By Theorem 3.17 the perimeter of the polygon inscribed in $\mathbb{T}$ is

$$P_n = 2n \sin \frac{\pi}{n},$$

consistent with $P_n = 2nr \sin(\pi/n)$ from Volume I (here $r = 1$). The limit $P_n \rightarrow 2\pi$ as $n \rightarrow \infty$ is the notable limit $\sin x/x \rightarrow 1$: the very relation which, in Volume I, required the whole anti-circularity architecture to be established without presupposing π .

3.9. The roots of unity as a basis: the DFT matrix

The n -th roots of unity are not only a geometric object: seen as numbers, they are the basis on which the **discrete Fourier transform** (DFT) is built — the finite-dimensional version of the Fourier series that Chapter 4 will treat at length. It is worth constructing it here, while the roots of unity are at the centre of attention.

Definition 3.7: DFT matrix. Let $\zeta = e^{-2\pi i/n}$. The **DFT matrix** of order n is

$$W_n = (\zeta^{jk})_{j,k=0}^{n-1}.$$

For a finite sequence $x = (x_0, \dots, x_{n-1})$, the discrete Fourier transform is $X = W_n x$, i.e.

$$X_j = \sum_{k=0}^{n-1} x_k \zeta^{jk} = \sum_{k=0}^{n-1} x_k e^{-2\pi i j k / n}.$$

Theorem 3.22: Orthogonality and unitarity of the DFT matrix. The columns of W_n are pairwise orthogonal with respect to the Hermitian inner product:

$$\sum_{j=0}^{n-1} \zeta^{jk} \overline{\zeta^{jk'}} = n \delta_{kk'}.$$

Consequently $U_n := \frac{1}{\sqrt{n}} W_n$ is unitary ($U_n^* U_n = I$), W_n is invertible with

$$W_n^{-1} = \frac{1}{n} \overline{W_n} = \frac{1}{n} (e^{+2\pi i j k / n})_{j,k},$$

and the **discrete Parseval identity** holds:

$$\sum_{k=0}^{n-1} |x_k|^2 = \frac{1}{n} \sum_{j=0}^{n-1} |X_j|^2.$$

Proof. Orthogonality. Since $|\zeta| = 1$, $\overline{\zeta^{jk'}} = \zeta^{-jk'}$, hence

$$\sum_{j=0}^{n-1} \zeta^{jk} \overline{\zeta^{jk'}} = \sum_{j=0}^{n-1} e^{-2\pi i j(k-k')/n}.$$

Setting $m = -(k - k')$, this is exactly the sum of Theorem 3.20, which equals n if $n \mid (k - k')$ and 0 otherwise. Since $k, k' \in \{0, \dots, n-1\}$, $n \mid (k - k')$ is equivalent to $k = k'$.

Unitarity. The (k, k') entry of $W_n^* W_n$ is precisely the sum just computed, hence $W_n^* W_n = nI$ and $U_n^* U_n = I$.

Inverse. From $W_n^* W_n = nI$ we get $W_n^{-1} = \frac{1}{n} W_n^*$. But W_n is symmetric ($\zeta^{jk} = \zeta^{kj}$), so $W_n^* = \overline{W_n}$, with entries $e^{+2\pi i jk/n}$.

Discrete Parseval. With $X = W_n x$,

$$\sum_j |X_j|^2 = X^* X = x^* W_n^* W_n x = n x^* x = n \sum_k |x_k|^2.$$

□

Remark 3.12: The same identity, two disguises. Theorem 3.22 is not a new result: it is Theorem 3.20 disguised as a fact of linear algebra, as announced in Observation 3.9. The same phenomenon will reappear, in continuous form, as the orthogonality of complex exponentials (Chapter 4, Theorem 4.2): the DFT is, in a precise sense that Chapter 4 will make exact, the Fourier series restricted to n equispaced frequencies on $\mathbb{T}$.

Example 3.7: The four-point DFT, computed by hand. Construct W_4 and compute the DFT of three elementary signals.

Step 1 — the matrix. For $n = 4$, $\zeta = e^{-i\pi/2} = -i$. The exponents jk are reduced modulo 4 because $\zeta^4 = 1$:

$$W_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$

Step 2 — constant signal $x = (1, 1, 1, 1)$.

$$X_0 = 4, \quad X_1 = 1 - i - 1 + i = 0, \quad X_2 = 1 - 1 + 1 - 1 = 0, \quad X_3 = 0.$$

So $X = (4, 0, 0, 0)$: all the energy is in the DC component $j = 0$, as it must be for a signal that does not vary.

Step 3 — alternating signal $x = (1, -1, 1, -1)$.

$$X_0 = 0, \quad X_1 = 0, \quad X_2 = 1 + 1 + 1 + 1 = 4, \quad X_3 = 0.$$

So $X = (0, 0, 4, 0)$: all the energy in the component $j = 2$, which with $n = 4$ samples is the highest representable frequency — the discrete Nyquist frequency (Chapter 4, Theorem, Volume III, Chapter 24).

Step 4 — impulsive signal $x = (1, 0, 0, 0)$.

$$X_j = x_0 \zeta^0 = 1 \quad \text{for every } j,$$

i.e. $X = (1, 1, 1, 1)$: an impulse concentrated in time has a perfectly flat spectrum. This is the time–frequency uncertainty principle in its most elementary form, and together with Step 2 it shows the duality: constant $\leftrightarrow$ impulse, in both directions.

Step 5 — check with discrete Parseval. For the signal of Step 3: $\sum_k |x_k|^2 = 4$ and $\frac{1}{4} \sum_j |X_j|^2 = \frac{16}{4} = 4$. ✓

Listing 3.1: Constructing the DFT matrix and comparing with NumPy's FFT.

```
import numpy as np

def dft_matrix(n):
    j, k = np.meshgrid(np.arange(n), np.arange(n), indexing='ij')
    zeta = np.exp(-2j * np.pi / n)
    return zeta ** (j * k)

n = 4
W = dft_matrix(n)

# unitarity check: W* W = n I (orthogonality theorem)
print(np.allclose(W.conj().T @ W, n * np.eye(n))) # True

for x in ([1, 1, 1, 1], [1, -1, 1, -1], [1, 0, 0, 0]):
    x = np.array(x, dtype=complex)
    print(x.real, "->", np.round(W @ x, 12).real,
          "| FFT:", np.round(np.fft.fft(x), 12).real)
```

3.10. Phasors: from the circle to oscillations

This section is the explicit bridge to Chapter 4. The idea is simple: a quantity oscillating sinusoidally in time is the *projection* of a point going round the circle at constant angular speed — the uniform circular motion of Chapter 1. The complex number describing that point is called a phasor, and it turns the differential equations of circuits into algebraic ones.

Definition 3.8: Phasor. Let $v(t) = V_m \cos(\omega t + \phi)$ be a sinusoidal quantity of amplitude $V_m > 0$, angular frequency $\omega > 0$ and initial phase ϕ . The associated **phasor** (in the peak-amplitude convention) is the complex number

$$\underline{V} = V_m e^{i\phi},$$

and $v(t) = \operatorname{Re} [\underline{V} e^{i\omega t}]$.

Warning. Conventions that must always be declared. Three conventions circulate in engineering practice: phasors in *peak amplitude* ($\underline{V} = V_m e^{i\phi}$, adopted here), in *RMS value* ($\underline{V} = \frac{V_m}{\sqrt{2}} e^{i\phi}$, common in electrical engineering), and with *sine* rather than cosine as reference. Electrical engineering also often writes j for the imaginary unit, to avoid confusion with current. In this treatise: peak amplitude, cosine reference, imaginary unit i . Comparing two results obtained with different conventions without converting them is among the commonest and hardest-to-spot errors.

Theorem 3.23: Fundamental properties of phasors. Let v_1, v_2 be sinusoidal of the same angular frequency ω , with phasors $\underline{V}_1, \underline{V}_2$. Then:

- (i) (linearity) $v_1 + v_2$ is sinusoidal of the same frequency, with phasor $\underline{V}_1 + \underline{V}_2$;
- (ii) (differentiation) $\frac{dv_1}{dt}$ is sinusoidal of the same frequency, with phasor $i\omega \underline{V}_1$;
- (iii) (integration) a primitive of v_1 with zero mean has phasor $\underline{V}_1/(i\omega)$;
- (iv) the correspondence $v \leftrightarrow \underline{V}$ is a bijection between sinusoids of frequency ω and $\mathbb{C}$.

Proof. (i) By Definition 3.8 and linearity of Re ,

$$v_1(t) + v_2(t) = \text{Re}[(\underline{V}_1 + \underline{V}_2)e^{i\omega t}].$$

Writing $\underline{V}_1 + \underline{V}_2 = Ae^{i\psi}$ in polar form (Theorem 3.3), the right-hand side is $A \cos(\omega t + \psi)$: a sinusoid of frequency ω .

(ii) Differentiating and exchanging derivative and real part (legitimate because Re is $\mathbb{R}$ -linear and continuous):

$$\frac{dv_1}{dt} = \text{Re}\left[\underline{V}_1 \frac{d}{dt}e^{i\omega t}\right] = \text{Re}[i\omega \underline{V}_1 e^{i\omega t}],$$

which by Definition 3.8 has phasor $i\omega \underline{V}_1$.

(iii) Follows from (ii): if u has phasor $\underline{U}$ and $u' = v_1$, then $i\omega \underline{U} = \underline{V}_1$, i.e. $\underline{U} = \underline{V}_1/(i\omega)$ (well defined since $\omega > 0$). The zero-mean condition removes the additive constant, which is not representable as a phasor of frequency ω .

(iv) If $\underline{V} = V_m e^{i\phi}$ with $V_m > 0$, then $\text{Re}[\underline{V}e^{i\omega t}] = V_m \cos(\omega t + \phi)$ by Theorem 3.5; conversely each sinusoid $V_m \cos(\omega t + \phi)$ determines $V_m = |\underline{V}|$ and ϕ modulo 2π , hence a unique $\underline{V} \in \mathbb{C}$. $\square$

Remark 3.13: Why phasors work. Theorem 3.23(ii) is the heart of the method: differentiating in time becomes multiplying by $i\omega$. A linear constant-coefficient differential equation, in sinusoidal steady state, thus becomes an algebraic equation in the phasors. Geometrically, multiplying by $i\omega$ means rotating by 90° and scaling by ω : the derivative of an oscillation is in quadrature with it, exactly as in Chapter 1 the velocity of a circular motion was orthogonal to the radius (Theorem 1.2(ii)). Phasors and uniform circular motion are the same object read in two languages.

Example 3.8: Sum of two phase-shifted sinusoids. Add $v_1(t) = 10 \cos(\omega t)$ and $v_2(t) = 6 \cos(\omega t + 60^\circ)$, expressing the result as a single sinusoid.

Step 1 — phasors. By Definition 3.8,

$$\underline{V}_1 = 10 e^{i0} = 10, \quad \underline{V}_2 = 6 e^{i\pi/3} = 6(\cos 60^\circ + i \sin 60^\circ) = 3 + 3\sqrt{3}i.$$

Numerically $\underline{V}_2 = 3 + 5.19615i$.

Step 2 — sum (Theorem 3.23(i)).

$$\underline{V} = \underline{V}_1 + \underline{V}_2 = 13 + 5.19615i.$$

Step 3 — back to polar form.

$$|\underline{V}| = \sqrt{13^2 + 5.19615^2} = \sqrt{169 + 27.0} = \sqrt{196.0} = 14.000,$$

$$\text{Arg } \underline{V} = \arctan \frac{5.19615}{13} = \arctan(0.399704) = 0.380251 \text{ rad} = 21.787^\circ,$$

legitimate with $\arctan$ since the point is in the first quadrant.

Step 4 — result.

$$v_1(t) + v_2(t) = 14.00 \cos(\omega t + 21.79^\circ).$$

Step 5 — independent trigonometric check. Expanding v_2 by the addition formula:

$$v_2(t) = 6(\cos \omega t \cos 60^\circ - \sin \omega t \sin 60^\circ) = 3 \cos \omega t - 3\sqrt{3} \sin \omega t.$$

Hence

$$v_1 + v_2 = 13 \cos \omega t - 5.19615 \sin \omega t.$$

An expression $A \cos \omega t - B \sin \omega t$ equals $R \cos(\omega t + \psi)$ with $R = \sqrt{A^2 + B^2}$ and $\tan \psi = B/A$ (Volume I, Chapter 8): here $R = \sqrt{169 + 27} = 14$ and $\psi = \arctan(5.19615/13) = 21.79^\circ$. $\checkmark$

Typical error. Adding the *amplitudes* ($10 + 6 = 16$), ignoring the phase shift. The correct result, 14, is smaller: the two sinusoids partly cancel. Only for zero phase shift would adding amplitudes be correct.

Example 3.9: The neutral of a symmetric three-phase system. A symmetric three-phase system has three voltages of equal amplitude V_m shifted by 120° :

$$v_k(t) = V_m \cos\left(\omega t - \frac{2\pi k}{3}\right), \quad k = 0, 1, 2.$$

Show that their instantaneous sum is identically zero.

Step 1 — phasors. $\underline{V}_k = V_m e^{-2\pi i k/3} = V_m \bar{\zeta}_1^k$, where $\zeta_1 = e^{2\pi i/3}$.

Step 2 — sum of the phasors. Since $\zeta_1^3 = 1$, we have $\zeta_1^{-1} = \zeta_1^2$ and $\zeta_1^{-2} = \zeta_1$: the set $\{1, \zeta_1^{-1}, \zeta_1^{-2}\}$ coincides with $\{1, \zeta_1, \zeta_1^2\} = \mu_3$, i.e. with all the cube roots of unity. By Theorem 3.23(i) and Corollary 3.5 with $n = 3$,

$$\underline{V}_0 + \underline{V}_1 + \underline{V}_2 = V_m(1 + \zeta_1 + \zeta_1^2) = V_m \cdot 0 = 0.$$

Step 3 — conclusion in the time domain. Since the phasor–sinusoid correspondence is linear and bijective (Theorem 3.23(i),(iv)), a zero phasor means an identically zero sinusoid:

$$v_0(t) + v_1(t) + v_2(t) \equiv 0 \quad \text{for every } t.$$

Step 4 — direct check at one instant. For $\omega t = 0$:

$$V_m \left[1 - \frac{1}{2} - \frac{1}{2}\right] = 0. \checkmark$$

Interpretation. This is why, in a balanced three-phase load, the neutral conductor carries no current: the instantaneous sum vanishes by the algebraic identity of Corollary 3.5, i.e. because the three phasors are the vertices of an equilateral triangle centred at the origin. The same argument applies to an n -phase system with $n \geq 2$.

3.11. Application: rotary encoders and angular quantisation

A rotary encoder — the sensor measuring the angular position of a motor, a robot joint or a shaft — does nothing but *sample* the circle at a finite number N of equispaced positions: exactly the partition of the circle into the N N -th roots of unity of Section 3.7, here reinterpreted as physical positions rather than as solutions of an algebraic equation.

Definition 3.9: Angular resolution. An encoder with N positions per revolution has **angular resolution**

$$\Delta\theta = \frac{2\pi}{N} \text{ rad} = \frac{360^\circ}{N}.$$

The N positions correspond exactly to the points $\zeta_k = e^{2\pi i k/N}$, $k = 0, \dots, N-1$, of Definition 3.5.

Remark 3.14: Two families: incremental and absolute. An **incremental** encoder produces two pulse trains in quadrature (channels A and B , shifted by 90° , i.e. a quarter of a position): the direction of rotation is read from the order in which A and B switch, and counting all four edges (rise and fall of both channels) gives an effective resolution $4N$, four times the nominal one. An **absolute** encoder instead codes each position by a fixed binary

word, read optically or magnetically on concentric tracks: it needs no homing at power-up, but how the position is coded is far from trivial, as the next observation shows.

Definition 3.10: Gray code. The **Gray code** (or reflected binary) on b bits is an enumeration of the 2^b binary words in which two consecutive words — including the last-to-first pair, cyclically — differ in *exactly one bit*. It is obtained from an integer n by

$$g(n) = n \oplus (n \gg 1),$$

the bitwise XOR of n with n shifted one place right.

Proposition 3.2: The Gray code is genuinely reflected. For every $n \geq 0$, the words $g(n)$ and $g(n+1)$ differ in exactly one bit.

Proof. Write n in binary and let $t \geq 0$ be the number of consecutive 1 bits at its tail, so that $n = m \underbrace{0 \ 1 \cdots 1}_t$ for a suitable prefix m (if n is even, $t = 0$). Then $n+1 = m \underbrace{1 \ 0 \cdots 0}_t$: the increment propagates the carry through the t trailing ones and flips the 0 preceding them. Now $g(n) \oplus g(n+1) = [n \oplus (n+1)] \oplus [(n \gg 1) \oplus ((n+1) \gg 1)]$. From the two binary expansions, $n \oplus (n+1)$ has exactly $t+1$ bits set, all in the $t+1$ least significant positions: it is the word $\underbrace{1 \cdots 1}_{t+1}$. Since the right shift commutes with XOR, the second term is $[n \oplus (n+1)] \gg 1 = \underbrace{1 \cdots 1}_t$. The XOR of the two words leaves only the bit in position t set and clears all the others. Hence $g(n)$ and $g(n+1)$ differ in exactly one bit, the one in position t . $\square$

Remark 3.15: Why it is needed: the ambiguity of pure binary. With pure binary coding, moving between two adjacent angular positions may require changing several bits at once: from 011 to 100 (from 3 to 4) all three change. If the sensors reading the individual tracks do not switch perfectly synchronously — inevitable given mechanical tolerances — the encoder may return, for an instant, a completely wrong intermediate word (for instance 000 or 111), corresponding to an angular position very far from the true one. With Gray code, by Proposition 3.2 every transition changes a single bit: the worst possible transient error is reading the adjacent position, never an arbitrarily distant one. This is an essential reliability gain when the encoder feeds a position control loop.

Example 3.10: A 12-bit absolute encoder on a robot joint. A 12-bit absolute encoder is mounted on a joint with arm $r = 0.5$ m. Determine the angular and linear resolution, and construct the 3-bit Gray code.

Step 1 — number of positions. $N = 2^{12} = 4096$.

Step 2 — angular resolution.

$$\Delta\theta = \frac{360^\circ}{4096} = 0.087890625^\circ = 1.53398 \times 10^{-3} \text{ rad.}$$

Step 3 — linear resolution at the arm tip.

$$r \Delta\theta = 0.5 \times 1.53398 \times 10^{-3} = 7.670 \times 10^{-4} \text{ m} = 0.767 \text{ mm,}$$

a typical value, sufficient for most industrial positioning.

Step 4 — 3-bit Gray code. Applying $g(n) = n \oplus (n \gg 1)$:

Position k	Pure binary	$k \gg 1$	Gray $g(k)$	bits changed
0	000	000	000	—
1	001	000	001	1
2	010	001	011	1
3	011	001	010	1
4	100	010	110	1
5	101	010	111	1
6	110	011	101	1
7	111	011	100	1

Three-bit Gray code for an absolute encoder with $N = 8$ positions. The last column confirms Proposition 3.2; note that the cyclic return from 7 to 0 ($100 \rightarrow 000$) also changes a single bit.

Remark 3.16: A look ahead: vibration diagnostics. If one records, revolution after revolution, the instantaneous angular velocity derived from an encoder (rather than position alone), the resulting signal is periodic with period one revolution: a natural candidate for the frequency analysis of Chapter 4. A mechanical unbalance typically produces a dominant component at the rotation frequency (once-per-revolution), while a bearing defect produces components at higher characteristic frequencies: exactly the kind of diagnosis for which the FFT and the spectral windows of Chapter 4 are everyday tools in predictive maintenance.

3.12. The Basel problem

De Moivre's theorem (Theorem 3.15) is not only useful for computing powers and roots: applied with the odd exponent $m = 2n + 1$ and compared with the binomial expansion, it yields a polynomial whose zeros are $\cot^2(k\pi/m)$, $k = 1, \dots, n$. This section uses that polynomial to prove a result Euler announced in 1735: the sum of the reciprocal squares equals $\pi^2/6$. The proof is due to Tom Apostol (1975) and is entirely elementary: it uses only de Moivre, the binomial theorem, Vieta's formulas and the squeeze theorem.

Theorem 3.24: Apostol's identity. For every integer $n \geq 1$, setting $m = 2n + 1$,

$$\sum_{k=1}^n \cot^2 \frac{k\pi}{m} = \frac{n(2n-1)}{3}.$$

Proof. By Theorem 3.15 and the binomial theorem, for every $\theta \in \mathbb{R}$,

$$\cos(m\theta) + i \sin(m\theta) = (\cos \theta + i \sin \theta)^m = \sum_{j=0}^m \binom{m}{j} \cos^{m-j} \theta (i \sin \theta)^j.$$

Isolating the imaginary part (the terms with j odd, $j = 2\ell + 1$, $\ell = 0, \dots, n$, since $m = 2n + 1$ is odd; $i^{2\ell+1} = i \cdot (-1)^\ell$):

$$\sin(m\theta) = \sum_{\ell=0}^n (-1)^\ell \binom{m}{2\ell+1} \cos^{m-2\ell-1} \theta \sin^{2\ell+1} \theta.$$

For $\theta \in (0, \pi/2)$, divide both sides by $\sin^m \theta \neq 0$; since $m - 2\ell - 1 = 2(n - \ell)$ is even,

$$\frac{\sin(m\theta)}{\sin^m \theta} = \sum_{\ell=0}^n (-1)^\ell \binom{m}{2\ell+1} \cot^{2(n-\ell)} \theta =: P_n(x), \quad x := \cot^2 \theta,$$

a polynomial of degree n in x , with leading coefficient (term $\ell = 0$) $\binom{m}{1} = m$ and coefficient of x^{n-1} (term $\ell = 1$) equal to $-\binom{m}{3}$.

For $k = 1, \dots, n$ set $\theta_k = k\pi/m \in (0, \pi/2)$ (because $k \leq n < m/2$). Then $\sin(m\theta_k) = \sin(k\pi) = 0$ while $\sin \theta_k \neq 0$, so $P_n(\cot^2 \theta_k) = 0$: the n numbers $x_k = \cot^2 \theta_k$ are roots of P_n . They are *distinct*, because $\cot^2$ is strictly decreasing and injective on $(0, \pi/2)$ and the θ_k are distinct; since P_n has degree n , these are *all* the zeros of P_n .

By Vieta's formulas, the sum of the roots of a polynomial $c_n x^n + c_{n-1} x^{n-1} + \dots$ is $-c_{n-1}/c_n$. Here $c_n = \binom{m}{1} = m$ and $c_{n-1} = -\binom{m}{3}$, so

$$\sum_{k=1}^n \cot^2 \theta_k = \frac{\binom{m}{3}}{\binom{m}{1}} = \frac{m(m-1)(m-2)/6}{m} = \frac{(m-1)(m-2)}{6}.$$

With $m = 2n + 1$: $(m-1)(m-2) = 2n(2n-1)$, and dividing by 6 gives $n(2n-1)/3$. $\square$

Theorem 3.25: The Basel problem (Euler, 1735; proof by Apostol).

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

Proof. For $0 < \theta < \pi/2$ the elementary inequalities $\sin \theta < \theta < \tan \theta$ hold (Volume I, Chapter 8: the first compares the arc with the chord, the second the arc with the geometric tangent to the unit circle). From the second, $\theta \cos \theta < \sin \theta$, i.e. $\cot \theta < 1/\theta$; from the first, $1/\theta < 1/\sin \theta = \csc \theta$. Hence

$$\cot \theta < \frac{1}{\theta} < \csc \theta.$$

Squaring (valid, since all terms are positive),

$$\cot^2 \theta < \frac{1}{\theta^2} < \csc^2 \theta = 1 + \cot^2 \theta.$$

Apply this with $\theta_k = k\pi/m$, $k = 1, \dots, n$, $m = 2n + 1$, and sum over k :

$$\sum_{k=1}^n \cot^2 \theta_k < \sum_{k=1}^n \frac{1}{\theta_k^2} < \sum_{k=1}^n \cot^2 \theta_k + n.$$

The middle term is $\sum_k m^2/(k^2 \pi^2) = (m^2/\pi^2) \sum_{k=1}^n 1/k^2$. By Theorem 3.24, the sum of cotangents equals $n(2n-1)/3$; substituting and multiplying by π^2/m^2 :

$$\frac{\pi^2}{m^2} \cdot \frac{n(2n-1)}{3} < \sum_{k=1}^n \frac{1}{k^2} < \frac{\pi^2}{m^2} \left(\frac{n(2n-1)}{3} + n \right).$$

With $m = 2n + 1$, both bounds have the same limit as $n \rightarrow \infty$: dividing numerator and denominator by n^2 ,

$$\frac{\pi^2}{m^2} \cdot \frac{n(2n-1)}{3} = \pi^2 \cdot \frac{2n^2 - n}{3(2n+1)^2} \xrightarrow{n \rightarrow \infty} \pi^2 \cdot \frac{2}{3 \cdot 4} = \frac{\pi^2}{6},$$

and the upper bound differs from the lower one by an extra term $\pi^2 n/m^2 = \pi^2 n/(2n+1)^2 \rightarrow 0$, since the numerator is linear and the denominator quadratic. By the squeeze theorem, the sequence of partial sums $\sum_{k=1}^n 1/k^2$ converges to $\pi^2/6$. $\square$

Remark 3.17: A second proof, already carried out: via Parseval. The Basel problem also admits a completely independent proof, with opposite tools (Fourier analysis rather than finite trigonometric algebra): it is Exercise 4.6 of Chapter 4, fully solved in Section 4.12 of that chapter, which applies Parseval's identity (Theorem 4.15) to the function $f(\theta) = \theta$ on $(-\pi, \pi]$. The two proofs do not share a single step — one is finite trigonometric algebra followed by a limiting argument, the other is measure theory applied to a single function — yet they produce the same number. This is exactly the kind of cross-check this treatise favours: neither proof depends on the other, so an error in either one would produce a disagreement, not the same value from two opposite routes.

Remark 3.18: Independent numerical verification. For $n = 1, 2, 3, 5, 10, 20$ Apostol's identity was recomputed directly (sum of $\cot^2$ against $n(2n-1)/3$: agreement to machine precision, script `Sorgenti/Vol2/_verifich/vol2_cap03_basilea.py`); the binomial coefficient $\binom{m}{3}/\binom{m}{1}$ was independently verified to equal $n(2n-1)/3$ for the same values of n ; the bounds of the squeeze inequality were computed for n up to 1000, narrowing from $[0.366, 1.462]$ ($n = 1$) to $[1.64247, 1.64493]$ ($n = 1000$), around $\pi^2/6 = 1.64493407\dots$; finally the direct sum $\sum_{k=1}^{2 \times 10^6} 1/k^2 = 1.64493357$ confirms the value, with the slow convergence (error $\sim 1/N$) typical of this series.

3.13. The Wallis product

A second classical way to connect the circle with the integers, independent of the previous one, is Wallis's infinite product (1656), which expresses $\pi/2$ as the limit of a product of rational fractions. The proof uses the Wallis integrals, defined here for the first time, and the reduction formula obtained by integration by parts.

Definition 3.11: Wallis integrals. For integer $n \geq 0$, $I_n := \int_0^{\pi/2} \sin^n x \, dx$.

Theorem 3.26: Reduction formula and closed values. $I_0 = \pi/2$, $I_1 = 1$, and for $n \geq 2$,

$$I_n = \frac{n-1}{n} I_{n-2}.$$

Consequently, for $m \geq 1$,

$$I_{2m} = \frac{\pi}{2} \cdot \frac{(2m-1)!!}{(2m)!!}, \quad I_{2m+1} = \frac{(2m)!!}{(2m+1)!!},$$

where $k!! = k(k-2)(k-4)\cdots$ (down to 2 or 1) is the double factorial.

Proof. $I_0 = \int_0^{\pi/2} dx = \pi/2$ and $I_1 = \int_0^{\pi/2} \sin x \, dx = [-\cos x]_0^{\pi/2} = 1$ are immediate. For $n \geq 2$, integrate by parts with $u = \sin^{n-1} x$, $dv = \sin x \, dx$ (so $du = (n-1)\sin^{n-2} x \cos x \, dx$, $v = -\cos x$):

$$I_n = [-\sin^{n-1} x \cos x]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x \, dx.$$

The boundary term vanishes (at $x = \pi/2$, $\cos x = 0$; at $x = 0$, $\sin^{n-1} x = 0$ for $n \geq 2$). With $\cos^2 x = 1 - \sin^2 x$,

$$I_n = (n-1) \int_0^{\pi/2} \sin^{n-2} x \, dx - (n-1) \int_0^{\pi/2} \sin^n x \, dx = (n-1)I_{n-2} - (n-1)I_n,$$

so $nI_n = (n-1)I_{n-2}$, the reduction formula. Applying it repeatedly starting from I_0 or I_1 gives the two closed formulas by immediate induction. $\square$

Theorem 3.27: Wallis product.

$$\frac{\pi}{2} = \prod_{k=1}^{\infty} \frac{2k}{2k-1} \cdot \frac{2k}{2k+1} = \lim_{m \rightarrow \infty} \prod_{k=1}^m \frac{2k}{2k-1} \cdot \frac{2k}{2k+1}.$$

Proof. On $[0, \pi/2]$ we have $0 \leq \sin x \leq 1$, so $\sin^{n+1} x \leq \sin^n x$ for every $n \geq 0$; integrating, $I_{n+1} \leq I_n$: the sequence (I_n) is decreasing. In particular, for every $m \geq 1$,

$$I_{2m+1} \leq I_{2m} \leq I_{2m-1}.$$

By the reduction formula, $I_{2m+1} = \frac{2m}{2m+1} I_{2m-1}$, so $I_{2m-1}/I_{2m+1} = (2m+1)/(2m)$. Dividing the double inequality by $I_{2m+1} > 0$,

$$1 \leq \frac{I_{2m}}{I_{2m+1}} \leq \frac{2m+1}{2m} \xrightarrow{m \rightarrow \infty} 1,$$

hence by the squeeze theorem $I_{2m}/I_{2m+1} \rightarrow 1$.

Define $P_m := \prod_{k=1}^m \frac{2k}{2k-1} \cdot \frac{2k}{2k+1}$. Since $\prod_{k=1}^m \frac{2k}{2k-1} = \frac{(2m)!!}{(2m-1)!!}$ and $\prod_{k=1}^m \frac{2k}{2k+1} = \frac{(2m)!!}{(2m+1)!!}$ (each even factor $2k$ of the double factorial cancels progressively with the odd ones in the denominator),

$$P_m = \frac{(2m)!!}{(2m-1)!!} \cdot \frac{(2m)!!}{(2m+1)!!} = \frac{[(2m)!!]^2}{(2m-1)!!(2m+1)!!}.$$

Compare with the closed formulas of Theorem 3.26:

$$\frac{I_{2m}}{I_{2m+1}} = \frac{\frac{\pi}{2} \cdot \frac{(2m-1)!!}{(2m)!!}}{\frac{(2m)!!}{(2m+1)!!}} = \frac{\pi}{2} \cdot \frac{(2m-1)!!(2m+1)!!}{[(2m)!!]^2} = \frac{\pi}{2} \cdot \frac{1}{P_m}.$$

Hence $P_m = \frac{\pi}{2} \cdot \frac{I_{2m+1}}{I_{2m}}$. Since $I_{2m}/I_{2m+1} \rightarrow 1$, its reciprocal $I_{2m+1}/I_{2m} \rightarrow 1$ as well, and therefore

$$\lim_{m \rightarrow \infty} P_m = \frac{\pi}{2},$$

which is the claim. $\square$

Remark 3.19: Independent numerical verification. The reduction formula was rechecked by direct numerical integration of I_n ($n = 2, \dots, 9$): agreement to machine precision. The ratio I_{2m}/I_{2m+1} was computed for $m = 1, 5, 20, 100$, decreasing towards 1 ($1.17810 \rightarrow 1.00249$) as expected. The partial product P_m was computed directly up to $m = 2 \times 10^5$: $P_{200000} = 1.57079436$ against $\pi/2 = 1.57079633$ (difference

2×10^{-6} , consistent with the $O(1/m)$ convergence rate typical of this product). Script `Sorgenti/Vol2/_verifiche/vol2_cap03_wallis.py`.

3.14. Numerical exercises

All exercises are worked out in full in Section 3.16, with an independent check and a note on the typical error. It is advisable to attempt them before reading the solution.

Exercise 3.1: N1 — Polar form, product and quotient. Write $z_1 = -5 + 5\sqrt{3}i$ and $z_2 = 4 - 4i$ in polar form, stating the quadrant of each. Then compute $z_1 z_2$ and z_1/z_2 in polar form, reducing every argument to the principal value in $(-\pi, \pi]$, and check the product in Cartesian form.

Exercise 3.2: N2 — An Apollonius locus. Determine the locus of points $z \in \mathbb{C}$ with $|z - 3i| = 2|z + 3i|$, finding its centre and radius by Theorem 3.11, and check the result on at least two points of the locus and one external point.

Exercise 3.3: N3 — A high power by de Moivre. Compute $(1 + i\sqrt{3})^{10}$ in exact Cartesian form using Theorem 3.15, and state the principal argument of the result.

Exercise 3.4: N4 — All the fourth roots. Determine all solutions of $z^4 = -8 + 8\sqrt{3}i$ in exponential and exact Cartesian form, and verify at least one by direct substitution. Describe the geometric configuration of the four solutions.

Exercise 3.5: N5 — Orders and primitive roots. For $n = 18$, compute the order of ζ_4 , ζ_9 and ζ_{15} in the group μ_{18} using Theorem 3.19, list all indices k for which ζ_k is primitive, and check that their number is $\varphi(18)$.

Exercise 3.6: N6 — A four-point DFT. Compute by hand the DFT of the signal $x = (2, 1, -1, 0)$ with $n = 4$ (Definition 3.7), and check the result with the discrete Parseval identity of Theorem 3.22.

Exercise 3.7: N7 — Adding two sinusoids with phasors. Add $v_1(t) = 8 \cos(\omega t + 30^\circ)$ and $v_2(t) = 6 \cos(\omega t - 60^\circ)$ using phasors, expressing the result as a single sinusoid $V \cos(\omega t + \psi)$ with $V > 0$ and $\psi \in (-180^\circ, 180^\circ]$. Explain why the resulting amplitude is a “round” number.

Exercise 3.8: N8 — Images under inversion. Using Theorem 3.14 (Step 2), determine the image under $w = 1/z$ of (a) the circle $|z| = 2$; (b) the line $\operatorname{Re} z = 1$. In each case decide whether the image is a circle or a line *before* computing it, using the through-the-origin criterion.

Exercise 3.9: N9 — A 14-bit encoder and the Gray code. A 14-bit absolute encoder is mounted on an arm of length $r = 0.8$ m. (a) Compute N , the angular resolution in degrees and radians, and the linear resolution at the tip. (b) The positions $k = 8191$ and $k = 8192$ are adjacent on the circle: write their 14-bit pure binary representations and the corresponding Gray words. (c) Count how many bits change in each case and comment in

the light of Proposition 3.2.

3.15. Symbolic exercises

These exercises call for proofs, not numerical computation: they are small theorems. Complete solutions are again in Section 3.16.

Exercise 3.10: S1 — The trigonometry inside complex multiplication. Deduce both addition formulas for sine and cosine from the identity $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$, and derive from them the double-angle formulas. Show moreover that the inverse formulas of Theorem 3.6 yield the product-to-sum formulas in two lines of computation.

Exercise 3.11: S2 — The unit circle and the reciprocal. Prove that for $z \neq 0$, $|z| = 1 \iff 1/z = \bar{z}$, and deduce that $\mathbb{T}$ is closed under taking reciprocals. Prove further that for $z, w \in \mathbb{T}$

$$|z - w| = |1 - \bar{z}w|,$$

and interpret this identity geometrically.

Exercise 3.12: S3 — The n -th power as a covering of $\mathbb{T}$. Let $\pi_n : \mathbb{T} \rightarrow \mathbb{T}$, $\pi_n(z) = z^n$. Prove that π_n is a surjective group homomorphism with kernel μ_n , that every $w \in \mathbb{T}$ has exactly n preimages, and deduce the isomorphism $\mathbb{T}/\mu_n \cong \mathbb{T}$. Connect the result with the winding number of Chapter 1 (Proposition 1.3).

Exercise 3.13: S4 — Discrete orthogonality and the inversion formula. Starting from Theorem 3.20, prove directly the DFT inversion formula

$$x_k = \frac{1}{n} \sum_{j=0}^{n-1} X_j e^{+2\pi i j k / n},$$

without invoking invertibility of the matrix, but computing the double sum explicitly.

Exercise 3.14: S5 — Cross-ratio and determination of a Möbius map. The **cross-ratio** of four distinct points is

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}.$$

Prove that the cross-ratio is invariant under every Möbius transformation, and deduce that a Möbius transformation is uniquely determined by the images of three distinct points.

Exercise 3.15: S6 — Ptolemy's identity in complex form. Prove the algebraic identity

$$(z_1 - z_3)(z_2 - z_4) = (z_1 - z_2)(z_3 - z_4) + (z_2 - z_3)(z_1 - z_4)$$

valid for all $z_1, z_2, z_3, z_4 \in \mathbb{C}$, and deduce *Ptolemy's inequality*

$$|z_1 - z_3| |z_2 - z_4| \leq |z_1 - z_2| |z_3 - z_4| + |z_2 - z_3| |z_1 - z_4|.$$

Compare with Ptolemy's theorem proved in Volume I, Chapter 5.

Exercise 3.16: S7 — Sum of the squared diagonals. Prove that in a regular n -gon inscribed in $\mathbb{T}$ the sum of the squared distances from one fixed vertex to all the others equals

$$\sum_{k=1}^{n-1} |1 - \zeta_k|^2 = 2n.$$

Compare with Corollary 3.6 (the *product* of the same distances equals n) and comment on the difference between the two behaviours.

Exercise 3.17: S8 — Negative frequencies and polyphase systems. (a) Prove that every real sinusoid is the sum of two conjugate phasors:

$$A \cos(\omega t + \phi) = \frac{1}{2} \underline{V} e^{i\omega t} + \frac{1}{2} \overline{\underline{V}} e^{-i\omega t}, \quad \underline{V} = A e^{i\phi},$$

and explain in what sense this justifies the “two-sided” spectrum of Chapter 4. (b) Prove that for every $n \geq 2$ a symmetric n -phase system has identically zero instantaneous sum, generalising Example 3.9.

3.16. Solutions

3.16.1 Solutions to the numerical exercises

Solution. N1. *Step 1 — moduli.*

$$|z_1| = \sqrt{25 + 75} = 10, \quad |z_2| = \sqrt{32} = 4\sqrt{2} \approx 5.6569.$$

Step 2 — arguments, with the quadrant. The point $z_1 = (-5, 5\sqrt{3})$ has $x < 0$, $y > 0$: **second quadrant**. By the rule in the *good practice* box of Section 3.1,

$$\text{Arg } z_1 = \arctan(-\sqrt{3}) + \pi = -\frac{\pi}{3} + \pi = \frac{2\pi}{3} \quad (120^\circ).$$

The point $z_2 = (4, -4)$ has $x > 0$: **fourth quadrant**, and $\arctan$ applies directly,

$$\text{Arg } z_2 = \arctan(-1) = -\frac{\pi}{4} \quad (-45^\circ).$$

Hence $z_1 = 10e^{i2\pi/3}$ and $z_2 = 4\sqrt{2}e^{-i\pi/4}$.

Step 3 — product. By Theorem 3.4,

$$|z_1 z_2| = 40\sqrt{2} \approx 56.569, \quad \arg(z_1 z_2) = \frac{2\pi}{3} - \frac{\pi}{4} = \frac{5\pi}{12} \quad (75^\circ),$$

already the principal value.

Step 4 — quotient.

$$\left| \frac{z_1}{z_2} \right| = \frac{10}{4\sqrt{2}} = \frac{5\sqrt{2}}{4} \approx 1.7678, \quad \arg \frac{z_1}{z_2} = \frac{2\pi}{3} + \frac{\pi}{4} = \frac{11\pi}{12} \quad (165^\circ),$$

also already in $(-\pi, \pi]$.

Step 5 — Cartesian check of the product.

$$z_1 z_2 = (-5 + 5\sqrt{3}i)(4 - 4i) = -20 + 20i + 20\sqrt{3}i - 20\sqrt{3}i^2 = (-20 + 20\sqrt{3}) + i(20 + 20\sqrt{3}).$$

Numerically: $z_1 z_2 = 14.641 + 54.641i$. Check:

$$|z_1 z_2| = \sqrt{214.36 + 2985.6} = \sqrt{3200.0} = 56.569 = 40\sqrt{2} \quad \checkmark$$

$$\text{Arg}(z_1 z_2) = \arctan \frac{54.641}{14.641} = \arctan(3.7321) = 1.30900 \text{ rad} = 75.000^\circ = \frac{5\pi}{12} \quad \checkmark$$

(first quadrant, so arctan is legitimate).

Typical error. Using $\arctan(y/x)$ for z_1 without correcting the quadrant: one would get -60° instead of 120° , i.e. the diametrically opposite point. Checking the signs of x and y before applying arctan is the only effective defence.

Solution. N2. *Step 1 — identifying the parameters.* Comparing $|z - 3i| = 2|z + 3i|$ with $|z - a| = k|z - b|$ of Theorem 3.11: $a = 3i$, $b = -3i$, $k = 2$. Since $k \neq 1$, the locus is a circle.
Step 2 — centre.

$$c = \frac{3i - 4(-3i)}{1 - 4} = \frac{15i}{-3} = -5i,$$

i.e. the point $(0, -5)$.

Step 3 — radius.

$$r = \frac{2|3i - (-3i)|}{3} = \frac{2 \cdot 6}{3} = 4.$$

The locus is $C(-5i, 4)$.

Step 4 — check on two points of the locus.

- $z = -i$ (at distance $|-i + 5i| = 4$ from the centre $\checkmark$): $|-i - 3i| = 4$ and $2|-i + 3i| = 4$. $\checkmark$
- $z = -9i$ (at distance $|-9i + 5i| = 4$ $\checkmark$): $|-9i - 3i| = 12$ and $2|-9i + 3i| = 12$. $\checkmark$

Step 5 — check on an external point. The point $z = 0$ is at distance $5 \neq 4$ from the centre, hence not on the locus: indeed $|0 - 3i| = 3$ while $2|0 + 3i| = 6 \neq 3$. $\checkmark$

Typical error. Writing $c = (a - kb)/(1 - k)$, forgetting the squares: one would get $c = -9i$ and a wrong radius, which none of the test points would satisfy. Step 4 on two distinct points catches the error immediately.

Solution. N3. *Step 1 — polar form of the base.*

$$|1 + i\sqrt{3}| = 2, \quad \text{Arg}(1 + i\sqrt{3}) = \frac{\pi}{3}$$

(first quadrant, arctan legitimate). Hence $1 + i\sqrt{3} = 2e^{i\pi/3}$.

Step 2 — de Moivre. By Theorem 3.15,

$$(1 + i\sqrt{3})^{10} = 2^{10} e^{i10\pi/3} = 1024 e^{i10\pi/3}.$$

Step 3 — reducing the argument to the principal value. We have $10\pi/3 = 3.3\bar{3}\pi$, which is not in $(-\pi, \pi]$. Subtracting 2π :

$$\frac{10\pi}{3} - 2\pi = \frac{10\pi - 6\pi}{3} = \frac{4\pi}{3},$$

still out of range ($4\pi/3 > \pi$). Subtracting 2π again:

$$\frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3} \in (-\pi, \pi]. \quad \checkmark$$

Hence $\text{Arg}[(1 + i\sqrt{3})^{10}] = -2\pi/3$ (-120°).

Step 4 — Cartesian form.

$$1024 e^{-i2\pi/3} = 1024 \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) = -512 - 512\sqrt{3}i = -512 - 886.810i.$$

Step 5 — independent check by successive squarings. $(1 + i\sqrt{3})^2 = -2 + 2\sqrt{3}i$; $(1 + i\sqrt{3})^4 = (-2 + 2\sqrt{3}i)^2 = -8 - 8\sqrt{3}i$; $(1 + i\sqrt{3})^8 = (-8 - 8\sqrt{3}i)^2 = -128 + 128\sqrt{3}i$. Finally

$$(1 + i\sqrt{3})^{10} = (-128 + 128\sqrt{3}i)(-2 + 2\sqrt{3}i) = 256 - 768 - 512\sqrt{3}i = -512 - 512\sqrt{3}i. \checkmark$$

Typical error. Stopping after the first subtraction of 2π and declaring $\text{Arg} = 4\pi/3$. The principal value requires $\theta \in (-\pi, \pi]$: one must subtract 2π as many times as needed.

Solution. N4. Step 1 — polar form of the right-hand side.

$$|-8 + 8\sqrt{3}i| = \sqrt{64 + 192} = 16,$$

and the point $(-8, 8\sqrt{3})$ is in the second quadrant, so

$$\varphi = \arctan(-\sqrt{3}) + \pi = \frac{2\pi}{3}, \quad w = 16 e^{i2\pi/3}.$$

Step 2 — applying Theorem 3.16. With $n = 4$:

$$\rho^{1/4} = 2, \quad \theta_k = \frac{2\pi/3 + 2k\pi}{4} = \frac{\pi}{6} + \frac{k\pi}{2}, \quad k = 0, 1, 2, 3.$$

Step 3 — the four roots.

$$\begin{aligned} z_0 &= 2e^{i\pi/6} &= \sqrt{3} + i, \\ z_1 &= 2e^{i2\pi/3} &= -1 + \sqrt{3}i, \\ z_2 &= 2e^{i7\pi/6} &= -\sqrt{3} - i, \\ z_3 &= 2e^{i5\pi/3} &= 1 - \sqrt{3}i. \end{aligned}$$

Reduced to principal values, the arguments are $\pi/6, 2\pi/3, -5\pi/6, -\pi/3$.

Step 4 — check by substitution on z_0 .

$$(\sqrt{3} + i)^2 = 2 + 2\sqrt{3}i, \quad (\sqrt{3} + i)^4 = (2 + 2\sqrt{3}i)^2 = -8 + 8\sqrt{3}i. \checkmark$$

Step 5 — geometric configuration. The four roots all have modulus 2 and arguments differing by $\pi/2$: they are the vertices of a **square** inscribed in $C(0, 2)$, rotated by 30° relative to the axes. Note that $z_2 = -z_0$ and $z_3 = -z_1$: the roots come in opposite pairs, as they must for even n .

Typical error. Giving a single root (the “principal value”) and declaring the equation solved. Over $\mathbb{C}$ the equation $z^n = w$ *always* has exactly n solutions for $w \neq 0$: omitting some is the error flagged in the *warning* of Section 3.7.

Solution. N5. Step 1 — the order formula. By Theorem 3.19, $\text{ord}(\zeta_k) = 18/\text{gcd}(k, 18)$.

Step 2 — the three required cases.

$$\text{gcd}(4, 18) = 2 \implies \text{ord}(\zeta_4) = 9,$$

$$\text{gcd}(9, 18) = 9 \implies \text{ord}(\zeta_9) = 2,$$

$$\text{gcd}(15, 18) = 3 \implies \text{ord}(\zeta_{15}) = 6.$$

Check on $\zeta_9 = e^{i\pi} = -1$: indeed $(-1)^2 = 1$ and $(-1)^1 \neq 1$, order 2. ✓

Step 3 — the primitive roots. They are those with $\gcd(k, 18) = 1$. Since $18 = 2 \cdot 3^2$, the integers in $\{0, \dots, 17\}$ coprime with 18 are those divisible neither by 2 nor by 3:

$$k \in \{1, 5, 7, 11, 13, 17\}.$$

Step 4 — count. There are 6. On the other hand

$$\varphi(18) = 18(1 - \frac{1}{2})(1 - \frac{1}{3}) = 18 \cdot \frac{1}{2} \cdot \frac{2}{3} = 6. \checkmark$$

Step 5 — structural check. The possible orders are exactly the divisors of 18, namely 1, 2, 3, 6, 9, 18. Each divisor d is the order of exactly $\varphi(d)$ elements, and $\sum_{d|18} \varphi(d) = 1 + 1 + 2 + 2 + 6 + 6 = 18$. ✓

Solution. N6. *Step 1 — the matrix.* For $n = 4$, $\zeta = -i$ and (as in Example 3.7)

$$W_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$

Step 2 — the four components. With $x = (2, 1, -1, 0)$:

$$X_0 = 2 + 1 - 1 + 0 = 2, \quad X_1 = 2 - i + 1 + 0 = 3 - i,$$

$$X_2 = 2 - 1 - 1 + 0 = 0, \quad X_3 = 2 + i + 1 + 0 = 3 + i.$$

Hence $X = (2, 3 - i, 0, 3 + i)$.

Step 3 — Hermitian symmetry check. The signal x is *real*: for it we must have $X_{n-j} = \overline{X_j}$. Indeed $X_3 = 3 + i = \overline{X_1}$ ✓, and X_0, X_2 are real ✓. This symmetry is the discrete version of the one Chapter 4 will prove for the Fourier coefficients of a real function.

Step 4 — check with discrete Parseval. By Theorem 3.22,

$$\sum_k |x_k|^2 = 4 + 1 + 1 + 0 = 6, \quad \frac{1}{4} \sum_j |X_j|^2 = \frac{1}{4}(4 + 10 + 0 + 10) = 6. \checkmark$$

Typical error. Forgetting the factor $1/n$ in the discrete Parseval identity and concluding that “it does not work out”. The asymmetric normalisation (1 forward, $1/n$ backward) is the one adopted here and in most numerical libraries, but it is not universal: it must always be declared.

Solution. N7. *Step 1 — phasors.* By Definition 3.8,

$$\underline{V}_1 = 8e^{i30^\circ} = 4\sqrt{3} + 4i \approx 6.92820 + 4.00000i,$$

$$\underline{V}_2 = 6e^{-i60^\circ} = 3 - 3\sqrt{3}i \approx 3.00000 - 5.19615i.$$

Step 2 — sum. By Theorem 3.23(i),

$$\underline{V} = (4\sqrt{3} + 3) + i(4 - 3\sqrt{3}) \approx 9.92820 - 1.19615i.$$

Step 3 — modulus.

$$|\underline{V}| = \sqrt{9.92820^2 + 1.19615^2} = \sqrt{98.5691 + 1.4308} = \sqrt{99.9999} = 10.0000.$$

Step 4 — argument. The point is in the fourth quadrant ($x > 0$, $y < 0$), so arctan applies directly:

$$\psi = \arctan \frac{-1.19615}{9.92820} = \arctan(-0.120480) = -0.119903 \text{ rad} = -6.8699^\circ.$$

Step 5 — result.

$$v_1(t) + v_2(t) = 10.00 \cos(\omega t - 6.87^\circ).$$

Step 6 — why the amplitude is “round”. The phase difference between the two sinusoids is

$$30^\circ - (-60^\circ) = 90^\circ,$$

so the two phasors are *orthogonal* in the complex plane. By Pythagoras the modulus of the sum is then

$$\sqrt{8^2 + 6^2} = 10$$

exactly, independently of the common phase: it is the Pythagorean triple (6, 8, 10). This explains the round value and also provides the required independent check.

Typical error. Adding the amplitudes ($8 + 6 = 14$) or subtracting them (2). These are correct only for phase differences of 0° and 180° respectively; in all other cases the vector sum is needed.

Solution. N8. *Preliminary prediction.* By Theorem 3.14 the image under $w = 1/z$ is a line if and only if the original curve passes through the origin. The circle $|z| = 2$ does *not* pass through the origin $\Rightarrow$ image: a circle. The line $\operatorname{Re} z = 1$ does *not* pass through the origin $\Rightarrow$ image: a circle (through the origin, since the line passes through ∞).

(a) *The circle $|z| = 2$.*

Step 1 — equation in standard form. $|z| = 2 \iff z\bar{z} - 4 = 0$: this is $(\star)$ with $A = 1$, $B = 0$, $C = -4$. Check: $|B|^2 - AC = 4 > 0 \checkmark$.

Step 2 — transforming the coefficients. By Step 2 of the proof of Theorem 3.14: $A' = C = -4$, $B' = 0$, $C' = A = 1$. The image equation is

$$-4|w|^2 + 1 = 0 \iff |w| = \frac{1}{2}.$$

Step 3 — conclusion. The image is the circle $C(0, \frac{1}{2})$: consistent with the prediction and with Proposition 3.1(ii), since $|z| \cdot |1/z| = 1$ and here $|z| = 2$.

(b) *The line $\operatorname{Re} z = 1$.*

Step 1 — equation in standard form. $\operatorname{Re} z = 1$ means $z + \bar{z} - 2 = 0$: this is $(\star)$ with $A = 0$, $B = 1$, $C = -2$. Check: $|B|^2 - AC = 1 > 0 \checkmark$.

Step 2 — transforming the coefficients. $A' = C = -2$, $B' = \bar{B} = 1$, $C' = A = 0$. The image equation is

$$-2|w|^2 + w + \bar{w} = 0.$$

Step 3 — canonical form. Dividing by -2 : $|w|^2 - \operatorname{Re} w = 0$, i.e. with $w = u + iv$,

$$\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}.$$

Step 4 — conclusion. The image is the circle of centre $\frac{1}{2}$ and radius $\frac{1}{2}$, passing through the origin ($C' = 0$): consistent with the prediction and with Figure 3.2.

Check on points. The point $z = 1$ is on the line; its image is $w = 1$, which lies on the circle found ($|1 - \frac{1}{2}| = \frac{1}{2} \checkmark$). The point $z = 1 + i$ is on the line; $w = (1 - i)/2$, and $|\frac{1-i}{2} - \frac{1}{2}| = \frac{1}{2} \checkmark$.

Solution. N9. (a) *Resolutions.*

Step 1. $N = 2^{14} = 16\,384$ positions per revolution.

Step 2 — *angular resolution.* By Definition 3.9,

$$\Delta\theta = \frac{360^\circ}{16\,384} = 0.0219727^\circ = 3.83495 \times 10^{-4} \text{ rad.}$$

Step 3 — *linear resolution.* With $r = 0.8$ m,

$$r \Delta\theta = 0.8 \times 3.83495 \times 10^{-4} = 3.0680 \times 10^{-4} \text{ m} = 0.3068 \text{ mm.}$$

(b) *Coding of the two positions.*

Step 4 — *14-bit pure binary.* Since $8191 = 2^{13} - 1$ and $8192 = 2^{13}$:

$$8191 = 01\,1111\,1111\,1111_2, \quad 8192 = 10\,0000\,0000\,0000_2.$$

Step 5 — *Gray code.* Applying $g(n) = n \oplus (n \gg 1)$:

$$g(8191) = 01\,1111\,1111\,1111_2 \oplus 00\,1111\,1111\,1111_2 = 01\,0000\,0000\,0000_2 = 4096,$$

$$g(8192) = 10\,0000\,0000\,0000_2 \oplus 01\,0000\,0000\,0000_2 = 11\,0000\,0000\,0000_2 = 12\,288.$$

(c) *Counting the changed bits.*

Step 6 — *pure binary.*

$$8191 \oplus 8192 = 11\,1111\,1111\,1111_2,$$

which has **fourteen** bits set: *all* the bits change.

Step 7 — *Gray code.*

$$g(8191) \oplus g(8192) = 10\,0000\,0000\,0000_2,$$

which has a **single** bit set. ✓

Step 8 — *comment.* This pair is the worst possible case for pure binary on 14 bits, and it is exactly the scenario described in Observation 3.15: if the fourteen tracks did not switch simultaneously, the encoder could return, for an instant, any word between 0 and 16 383, i.e. an arbitrary angular position over the whole revolution. With Gray code, Proposition 3.2 guarantees that the worst transient error is a single position, i.e. 0.022° — irrelevant for the control loop.

3.16.2 Solutions to the symbolic exercises

Solution. S1. Step 1 — *addition formulas.* Apply Euler's formula (Theorem 3.5) to both sides of $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$. On the right: $\cos(\alpha + \beta) + i\sin(\alpha + \beta)$. On the left:

$$(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + i(\sin \alpha \cos \beta + \cos \alpha \sin \beta).$$

Two complex numbers are equal iff their real and imaginary parts agree (Definition 3.1), so *both* addition formulas follow from a single computation:

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta, \quad \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$$

Step 2 — *doubling.* Setting $\beta = \alpha$:

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha, \quad \sin 2\alpha = 2 \sin \alpha \cos \alpha.$$

Step 3 — *product-to-sum formulas.* By Theorem 3.6,

$$\cos \alpha \cos \beta = \frac{e^{i(\alpha+\beta)} + e^{i(\alpha-\beta)} + e^{-i(\alpha-\beta)} + e^{-i(\alpha+\beta)}}{4}.$$

Grouping the two conjugate pairs and re-applying the inverse formulas,

$$\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha + \beta) + \cos(\alpha - \beta)].$$

Similarly,

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)], \quad \sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)].$$

Remark. The trigonometric content of angle addition is *entirely* contained in the multiplication rule of $\mathbb{C}$: not a coincidence, but the reason why complex notation so simplifies trigonometric computation. The product-to-sum formulas are the ones used in Chapter 4 to prove the orthogonality of the real modes.

Solution. S2. *Step 1 — the equivalence.* By Theorem 3.1(i), $z\bar{z} = |z|^2$. For $z \neq 0$, dividing by z :

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}, \quad \text{hence} \quad \frac{1}{z} = \bar{z} \iff \bar{z} \left(\frac{1}{|z|^2} - 1 \right) = 0.$$

Since $z \neq 0$ implies $\bar{z} \neq 0$, this is equivalent to $|z| = 1$.

Step 2 — closure of $\mathbb{T}$ under reciprocals. If $z \in \mathbb{T}$ then $1/z = \bar{z}$ has modulus $|z| = 1$, so $1/z \in \mathbb{T}$. (Together with multiplicative closure from Corollary 3.1 this again gives the group structure of Corollary 3.2.)

Step 3 — the identity $|z - w| = |1 - \bar{z}w|$. Let $z, w \in \mathbb{T}$. Compute both squares using $|\zeta|^2 = \zeta\bar{\zeta}$ and $|z|^2 = |w|^2 = 1$:

$$|z - w|^2 = |z|^2 - z\bar{w} - \bar{z}w + |w|^2 = 2 - (z\bar{w} + \bar{z}w),$$

$$|1 - \bar{z}w|^2 = (1 - \bar{z}w)(1 - z\bar{w}) = 1 - z\bar{w} - \bar{z}w + |z|^2|w|^2 = 2 - (z\bar{w} + \bar{z}w).$$

The two results agree; since both moduli are non-negative, $|z - w| = |1 - \bar{z}w|$.

Step 4 — geometric interpretation. By Corollary 3.2, the map $\zeta \mapsto \bar{z}\zeta$ is a rotation by $-\text{Arg } z$; it sends z to 1 and w to $\bar{z}w$. The identity therefore says simply that *rotation preserves distances*: the distance between z and w is the same as between their images 1 and $\bar{z}w$. It is the one-line translation of Theorem 3.13.

Note. This identity is the key to computations in circle geometry with complex coordinates: it lets one reduce every pair of points of $\mathbb{T}$ to a pair in which one of them is 1, drastically simplifying the expressions.

Solution. S3. *Step 1 — homomorphism.* For $z, w \in \mathbb{T}$, $\pi_n(zw) = z^n w^n = \pi_n(z)\pi_n(w)$; moreover $|\pi_n(z)| = 1$, so the image really lies in $\mathbb{T}$.

Step 2 — surjectivity. Let $w = e^{i\varphi} \in \mathbb{T}$. The number $z = e^{i\varphi/n}$ has modulus 1 and, by Theorem 3.15, $z^n = w$.

Step 3 — kernel. By definition $\ker \pi_n = \{z \in \mathbb{T} : z^n = 1\}$. Every solution of $z^n = 1$ automatically has modulus 1 (taking moduli: $|z|^n = 1$ with $|z| > 0$ real gives $|z| = 1$), so $\ker \pi_n = \mu_n$, of order n by Theorem 3.18.

Step 4 — preimages. Let $w \in \mathbb{T}$ and let z_0 be a preimage (it exists by Step 2). Then

$$\pi_n^{-1}(w) = \{z \in \mathbb{T} : z^n = w\} = z_0 \mu_n,$$

because $z^n = z_0^n \iff (z/z_0)^n = 1 \iff z/z_0 \in \mu_n$. The coset $z_0 \mu_n$ has exactly $|\mu_n| = n$ elements: every w has n preimages, in agreement with Theorem 3.16.

Step 5 — isomorphism. By the first isomorphism theorem applied to the surjective homomorphism π_n with kernel μ_n , $\mathbb{T}/\mu_n \cong \mathbb{T}$.

Step 6 — link with the winding number. Composing π_n with the parametrisation $t \mapsto e^{it}$ gives the path $t \mapsto e^{int}$, which by Proposition 1.3 of Chapter 1 has winding number n . The fact that every point has n preimages is precisely the set-theoretic reading of that winding number: traversing the domain once, the image traverses the circle n times, and every point is reached n times.

Solution. S4. *Step 1 — setting up.* With $X_j = \sum_{k=0}^{n-1} x_k e^{-2\pi ijk/n}$, we must show that

$$\frac{1}{n} \sum_{j=0}^{n-1} X_j e^{+2\pi i j \ell / n} = x_\ell \quad \text{for every } \ell \in \{0, \dots, n-1\}.$$

Step 2 — substitution and exchange of sums. Substituting the definition of X_j and exchanging the two finite sums:

$$\frac{1}{n} \sum_{k=0}^{n-1} x_k \left(\sum_{j=0}^{n-1} e^{2\pi i j (\ell - k) / n} \right).$$

Step 3 — the master identity. The inner sum is exactly that of Theorem 3.20 with $m = \ell - k$:

$$\sum_{j=0}^{n-1} e^{2\pi i j (\ell - k) / n} = \begin{cases} n & \text{if } n \mid (\ell - k), \\ 0 & \text{otherwise.} \end{cases}$$

Since $k, \ell \in \{0, \dots, n-1\}$, we have $|\ell - k| < n$, so $n \mid (\ell - k)$ iff $\ell = k$.

Step 4 — conclusion. Only the term $k = \ell$ survives:

$$\frac{1}{n} \sum_k x_k n \delta_{k\ell} = x_\ell.$$

Remark. The proof uses no linear algebra: only the geometric sum of Theorem 3.20. It is the cheapest way to see that the DFT is invertible, and it makes clear that the only ingredient is the orthogonality of the characters of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ — the same ingredient that in Chapter 4 will make the Fourier series invertible.

Solution. S5. *Step 1 — it suffices to check it on the generators.* By Step 1 of the proof of Theorem 3.14, every Möbius transformation is a composition of maps of the three types

$$(a) \ z \mapsto z + b, \quad (b) \ z \mapsto az \ (a \neq 0), \quad (c) \ z \mapsto 1/z.$$

Since the cross-ratio is invariant under each of these, it is invariant under any composition.

Step 2 — translations. If $w_k = z_k + b$, every difference is unchanged: $w_j - w_k = z_j - z_k$. The cross-ratio, a quotient of differences, does not change.

Step 3 — homotheties-rotations. If $w_k = az_k$, then $w_j - w_k = a(z_j - z_k)$. Two differences appear in the numerator and two in the denominator, so the factor a^2 cancels.

Step 4 — inversion. If $w_k = 1/z_k$ (with $z_k \neq 0$),

$$w_j - w_k = \frac{z_k - z_j}{z_j z_k}.$$

Substituting into the cross-ratio,

$$\frac{\frac{z_3 - z_1}{z_1 z_3} \cdot \frac{z_4 - z_2}{z_2 z_4}}{\frac{z_4 - z_1}{z_1 z_4} \cdot \frac{z_3 - z_2}{z_2 z_3}} = \frac{(z_3 - z_1)(z_4 - z_2)}{(z_4 - z_1)(z_3 - z_2)} \cdot \frac{z_1 z_2 z_3 z_4}{z_1 z_2 z_3 z_4}.$$

The second factor is 1; the first equals the original cross-ratio because the two minus signs in the numerator and the two in the denominator compensate.

Step 5 — determination by three points. Let z_2, z_3, z_4 be three distinct points and w_2, w_3, w_4 three assigned distinct points. The map

$$f(z) := \text{the unique } w \text{ with } (w, w_2; w_3, w_4) = (z, z_2; z_3, z_4)$$

is well defined: with w_2, w_3, w_4 fixed, the equation $(w, w_2; w_3, w_4) = \lambda$ is of first degree in w (numerator and denominator are linear in w) and has a unique solution in $\mathbb{C} \cup \{\infty\}$; moreover f is of the form $(aw + b)/(cw + d)$, i.e. a Möbius map. By construction $f(z_k) = w_k$ for $k = 2, 3, 4$.

For uniqueness: if f, g are two Möbius maps with $f(z_k) = g(z_k) = w_k$, then $h = g^{-1} \circ f$ is a Möbius map fixing the three points z_2, z_3, z_4 . But a Möbius map $h(z) = (az + b)/(cz + d)$ fixing a point satisfies $cz^2 + (d - a)z - b = 0$: if $h \neq \text{id}$, this equation of degree at most 2 has at most two solutions and cannot have three fixed points. Hence $h = \text{id}$ and $f = g$.

Solution. S6. *Step 1 — the algebraic identity.* Expand the right-hand side:

$$(z_1 - z_2)(z_3 - z_4) = z_1 z_3 - z_1 z_4 - z_2 z_3 + z_2 z_4,$$

$$(z_2 - z_3)(z_1 - z_4) = z_1 z_2 - z_2 z_4 - z_1 z_3 + z_3 z_4.$$

Adding, the terms $z_1 z_3$ and $z_2 z_4$ cancel in pairs:

$$(z_1 - z_2)(z_3 - z_4) + (z_2 - z_3)(z_1 - z_4) = -z_1 z_4 - z_2 z_3 + z_1 z_2 + z_3 z_4.$$

Now expand the left-hand side:

$$(z_1 - z_3)(z_2 - z_4) = z_1 z_2 - z_1 z_4 - z_2 z_3 + z_3 z_4,$$

which agrees term by term. The identity is proved, and holds for *any* quadruple of complex numbers, with no geometric hypothesis.

Step 2 — the inequality. Taking moduli and applying the triangle inequality (Theorem 3.1(iv)) to the right-hand side, then multiplicativity (part (iii) of the same theorem):

$$|z_1 - z_3||z_2 - z_4| \leq |z_1 - z_2||z_3 - z_4| + |z_2 - z_3||z_1 - z_4|.$$

Step 3 — geometric reading. Interpreting z_1, z_2, z_3, z_4 as the vertices of a quadrilateral (in this order), the two sides are:

- on the left, the product of the *diagonals*;
- on the right, the sum of the products of *opposite sides*.

Ptolemy's inequality therefore states that the product of the diagonals never exceeds the sum of the products of the opposite sides.

Step 4 — the equality case. By Theorem 3.1(iv), equality in the triangle inequality holds iff the two summands $(z_1 - z_2)(z_3 - z_4)$ and $(z_2 - z_3)(z_1 - z_4)$ are aligned and concordant, i.e. their ratio is real and positive. That ratio is, up to sign, precisely the cross-ratio of Exercise S5: the condition amounts to saying that the four points are *conyclic* and arranged

in the order z_1, z_2, z_3, z_4 along the circle. One thus recovers **Ptolemy's theorem** of Volume I, Chapter 5, which already gives an equivalent fourth proof there ("Proof 4 — via complex numbers"): here we present it again with this chapter's notation, additionally obtaining the equality case (concyclicity) as a direct reading of the cross-ratio from Exercise S5.

Solution. S7. *Step 1 — a single square.* For each k ,

$$|1 - \zeta_k|^2 = (1 - \zeta_k)(1 - \bar{\zeta}_k) = 2 - \zeta_k - \bar{\zeta}_k,$$

using $|\zeta_k| = 1$.

Step 2 — summing over $k = 1, \dots, n-1$.

$$\sum_{k=1}^{n-1} |1 - \zeta_k|^2 = 2(n-1) - \sum_{k=1}^{n-1} \zeta_k - \sum_{k=1}^{n-1} \bar{\zeta}_k.$$

Step 3 — the two partial sums. By Corollary 3.5, $\sum_{k=0}^{n-1} \zeta_k = 0$; isolating $\zeta_0 = 1$,

$$\sum_{k=1}^{n-1} \zeta_k = -1.$$

Conjugating this equality (conjugation is additive by Theorem 3.1(ii)), also $\sum_{k=1}^{n-1} \bar{\zeta}_k = -1$.

Step 4 — conclusion.

$$\sum_{k=1}^{n-1} |1 - \zeta_k|^2 = 2(n-1) + 1 + 1 = 2n.$$

Step 5 — check for $n = 4$. The distances from the vertex 1 are $\sqrt{2}$, 2, $\sqrt{2}$; the sum of squares is $2 + 4 + 2 = 8 = 2 \cdot 4$. ✓

Step 6 — comparison with the product. Corollary 3.6 gives $\prod_{k=1}^{n-1} |1 - \zeta_k| = n$, while here $\sum_{k=1}^{n-1} |1 - \zeta_k|^2 = 2n$. Both behaviours are *linear* in n , which is remarkable: the individual distances do not grow (all are ≤ 2) and the smallest tend to zero like $2\pi/n$, yet neither the sum of squares nor the product degenerates. The structural reason is the same in both cases — the vanishing of the sum of the roots — but applied to two different symmetric functions.

Solution. S8. (a) *Decomposition into two conjugate phasors.*

Step 1. By Theorem 3.6 applied to the argument $\omega t + \phi$:

$$A \cos(\omega t + \phi) = \frac{A}{2} e^{i\phi} e^{i\omega t} + \frac{A}{2} e^{-i\phi} e^{-i\omega t}.$$

Step 2. Setting $\underline{V} = Ae^{i\phi}$ we have $\overline{\underline{V}} = Ae^{-i\phi}$, hence

$$A \cos(\omega t + \phi) = \frac{1}{2} \underline{V} e^{i\omega t} + \frac{1}{2} \overline{\underline{V}} e^{-i\omega t}.$$

Note that the two summands are *conjugate to each other*, which is why their sum is real: a complex number added to its conjugate gives $2\operatorname{Re}$, a real number.

Step 3 — interpretation. The decomposition shows that a single real oscillation of frequency ω contains *two* complex exponential modes, of frequencies $+\omega$ and $-\omega$, with conjugate complex amplitudes. "Negative frequencies" have no autonomous physical existence: they are the half needed to make the result real. This is exactly why the complex spectrum of Chapter 4 is indexed by $n \in \mathbb{Z}$ and not only by $n \geq 0$, and why for a real signal $c_{-n} = \overline{c_n}$ (already observed in the solution to Exercise N6 in discrete form).

(b) *n-phase system.*

Step 4. The n voltages are $v_k(t) = V_m \cos(\omega t - \frac{2\pi k}{n})$, with phasors $\underline{V}_k = V_m \bar{\zeta}_1^k$, where $\zeta_1 = e^{2\pi i/n}$.

Step 5. Since $\bar{\zeta}_1 = \zeta_1^{-1} = \zeta_1^{n-1}$ is again a primitive n -th root of unity (by Theorem 3.19, $\gcd(n-1, n) = 1$), its powers $\bar{\zeta}_1^k$, $k = 0, \dots, n-1$, run through *all* the n -th roots of unity, each once. Hence, by Corollary 3.5 (which requires $n \geq 2$),

$$\sum_{k=0}^{n-1} \underline{V}_k = V_m \sum_{j=0}^{n-1} \zeta_j = 0.$$

Step 6 — conclusion in the time domain. The phasor–sinusoid correspondence is linear and bijective (Theorem 3.23(i),(iv)): a zero resultant phasor means an identically zero resultant sinusoid,

$$\sum_{k=0}^{n-1} v_k(t) \equiv 0 \quad \text{for every } t.$$

The case $n = 3$ is Example 3.9; the case $n = 2$ gives the trivial $\cos \omega t + \cos(\omega t - \pi) = 0$.

3.17. Self-assessment quiz

Seventeen questions covering all sections. The key, with one line of justification for each, is at the end: it is advisable to answer all of them before consulting it.

- The principal value $\text{Arg } z$ adopted in this chapter lies in
A) $[0, 2\pi)$ B) $(-\pi, \pi]$ C) $[-\pi, \pi)$ D) $(0, 2\pi]$
- For $z = -1 + i$, $\text{Arg } z$ equals
A) $\pi/4$ B) $3\pi/4$ C) $-\pi/4$ D) $-3\pi/4$
- The identity $\arg(zw) = \arg z + \arg w$
A) always holds for principal values
B) holds as an equality of sets of values, i.e. modulo 2π
C) is false in general
D) holds only if $|z| = |w| = 1$
- The equation $Az\bar{z} + \bar{B}z + B\bar{z} + C = 0$ with $A \neq 0$ and $|B|^2 > AC$ represents
A) always a line B) a circle of centre $-B/A$ C) an ellipse D) a parabola
- The locus $|z - a| = |z - b|$ with $a \neq b$ is
A) a circle B) the perpendicular bisector of ab C) the empty set D) a ray
- The locus $|z - a|/|z - b| = k$ with $k \neq 1$ is
A) a line B) a circle (of Apollonius) C) a hyperbola D) depends on the sign of $k - 1$
- Möbius transformations map circles and lines
A) always to circles B) always to lines C) to circles or lines D) to ellipses
- Under $w = 1/z$, the image is a *line* exactly when the original curve

- A) is a circle B) passes through the origin C) has radius 1 D) is symmetric about the real axis
9. De Moivre's theorem states that $(\cos \theta + i \sin \theta)^n$ equals
 A) $\cos^n \theta + i \sin^n \theta$ B) $\cos n\theta + i \sin n\theta$ C) $n(\cos \theta + i \sin \theta)$ D) $\cos(\theta^n) + i \sin(\theta^n)$
10. The equation $z^n = w$ with $w \neq 0$ has
 A) one solution B) exactly n distinct solutions C) infinitely many solutions D) n solutions only if $|w| = 1$
11. The order of $\zeta_k = e^{2\pi i k/n}$ in the group μ_n is
 A) always n B) k C) $n/\gcd(k, n)$ D) $\gcd(k, n)$
12. The number of *primitive* n -th roots of unity is
 A) n B) $n - 1$ C) $\varphi(n)$ D) 1
13. The sum $\sum_{k=0}^{n-1} e^{2\pi i k m/n}$ equals
 A) always 0 B) n if $n \mid m$, otherwise 0 C) always n D) m
14. The product of the distances from one vertex to all the others, in a regular n -gon inscribed in $\mathbb{T}$, equals
 A) 1 B) n C) $2n$ D) 2^{n-1}
15. The columns of the DFT matrix W_n are
 A) parallel B) orthogonal, of norm $\sqrt{n}$ C) identical D) unrelated
16. In the phasor formalism, differentiating with respect to time corresponds to
 A) multiplying by ω B) multiplying by $i\omega$ C) dividing by $i\omega$ D) conjugating
17. In an absolute encoder, Gray code is preferred to pure binary because
 A) it is more compact B) between adjacent positions a single bit changes C) it is faster to read D) it uses fewer tracks

BEST PRACTICE

Answer key.

1-B: the convention declared at the start of the chapter, consistent with NIST DLMF §4.2.
2-B: $(-1, 1)$ is in the second quadrant: $\arctan(-1) + \pi = 3\pi/4$; answer C is the classic error from naive use of $\arctan$. **3-B:** Theorem 3.4 and the accompanying *warning*: for principal values the identity may fail by 2π . **4-B:** Theorem 3.9(i). **5-B:** Theorem 3.10: the term $z\bar{z}$ cancels and a first-degree equation remains. **6-B:** Theorem 3.11(ii); for $k = 1$ one falls back on B of the previous question. **7-C:** Theorem 3.14. **8-B:** same theorem, Step 3: the image is a line iff $C = 0$, i.e. iff the curve passes through the pole of the inversion. **9-B:** Theorem 3.15; A is the error of raising the components separately. **10-B:** Theorem 3.16.
11-C: Theorem 3.19. **12-C:** same theorem, second part. **13-B:** Theorem 3.20, the master identity. **14-B:** Corollary 3.6; C is the *sum of the squares* of the same distances (Exercise S7). **15-B:** Theorem 3.22: $W_n^* W_n = nI$. **16-B:** Theorem 3.23(ii). **17-B:** Proposition 3.2 and Observation 3.15.

3.18. Chapter formulary

Modulus, conjugation, argument

$$z\bar{z} = |z|^2, \quad |zw| = |z||w|, \quad |z+w| \leq |z| + |w|,$$
$$\operatorname{Re} z = \frac{z + \bar{z}}{2}, \quad \operatorname{Im} z = \frac{z - \bar{z}}{2i}, \quad \operatorname{Arg} z = \operatorname{atan2}(y, x) \in (-\pi, \pi]$$

Polar and exponential form

$$z = \rho e^{i\theta}, \quad e^{i\theta} = \cos \theta + i \sin \theta, \quad e^{i\pi} + 1 = 0$$
$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Loci

$$Az\bar{z} + \bar{B}z + B\bar{z} + C = 0 \quad (A, C \in \mathbb{R}, |B|^2 > AC) :$$
$$A \neq 0 \Rightarrow \text{circle } C\left(-\frac{B}{A}, \frac{\sqrt{|B|^2 - AC}}{|A|}\right), \quad A = 0 \Rightarrow \text{line}$$
$$\frac{|z-a|}{|z-b|} = k \quad (k \neq 1) \Rightarrow C\left(\frac{a - k^2b}{1 - k^2}, \frac{k|a-b|}{|1 - k^2|}\right)$$

Powers and roots

$$(\rho e^{i\theta})^n = \rho^n e^{in\theta}, \quad z^n = \rho e^{i\varphi} \Rightarrow z_k = \rho^{1/n} e^{i(\varphi + 2k\pi)/n}, \quad k = 0, \dots, n-1$$
$$\cos n\theta = T_n(\cos \theta), \quad T_{n+1} = 2xT_n - T_{n-1}$$

Roots of unity

 ($\zeta_k = e^{2\pi i k/n}$)

$$\sum_{k=0}^{n-1} \zeta_k^m = n [n \mid m], \quad \sum_{k=0}^{n-1} \zeta_k = 0, \quad \prod_{k=0}^{n-1} \zeta_k = (-1)^{n-1}$$
$$\prod_{k=1}^{n-1} (1 - \zeta_k) = n, \quad \sum_{k=1}^{n-1} |1 - \zeta_k|^2 = 2n, \quad \operatorname{ord}(\zeta_k) = \frac{n}{\gcd(k, n)}$$

DFT and phasors

$$X_j = \sum_{k=0}^{n-1} x_k e^{-2\pi i j k/n}, \quad x_k = \frac{1}{n} \sum_{j=0}^{n-1} X_j e^{+2\pi i j k/n}, \quad \sum_k |x_k|^2 = \frac{1}{n} \sum_j |X_j|^2$$
$$v(t) = \operatorname{Re} [\underline{V} e^{i\omega t}], \quad \underline{V} = V_m e^{i\phi}, \quad \frac{d}{dt} \longleftrightarrow \times i\omega$$

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CHAPTER 4

Harmonic Analysis and Transforms on the Circle

This chapter concludes the volume by returning, with the more powerful tools built in previous chapters, to the question that motivated them all: how does one decompose a periodic function into its fundamental components? The answer — Fourier series — lives naturally on $\mathbb{T}$, the complex unit circle of Chapter 3. Volume III generalizes this idea to the Fourier, Laplace and Zeta transforms, which are not a digression: they are exactly the tools with which an automation engineer analyzes the stability of a control loop, designs a digital filter or chooses the scan time of a PLC. Not by chance, as we will see there, the unit circle $\mathbb{T}$ reappears as the critical stability locus for digital systems: the geometry of Chapter 3 and the analysis of this chapter are, in the end, the same story.

4.1. Periodic functions and the circle as natural domain

Remark 4.1: Identification $\mathbb{T} \cong \mathbb{R}/2\pi\mathbb{Z}$. Every 2π -periodic function $f : \mathbb{R} \rightarrow \mathbb{C}$ can be viewed as a function $\tilde{f} : \mathbb{T} \rightarrow \mathbb{C}$ defined by $\tilde{f}(e^{i\theta}) = f(\theta)$, well-defined precisely because $f(\theta + 2\pi) = f(\theta)$. The circle $\mathbb{T}$ is therefore the natural domain of classical Fourier analysis: periodicity in $\theta \equiv$ function on $\mathbb{T}$.

Warning. Conventions card — valid throughout the chapter. Harmonic analysis is the territory where conventions proliferate and comparison between texts becomes impossible unless one's own is declared. This is ours, and every formula in the chapter refers back to it.

Object	Convention adopted
Domain	$\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, represented by $\theta \in [0, 2\pi)$ or $(-\pi, \pi]$
Measure	normalised: $d\mu = \frac{d\theta}{2\pi}$, so that $\mu(\mathbb{T}) = 1$
Inner product	$\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \overline{g(\theta)} d\theta$
Modes	$e_n(\theta) = e^{in\theta}$, $n \in \mathbb{Z}$; they are <i>orthonormal</i>
Coefficients	$\hat{f}(n) = \langle f, e_n \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$
Sign in the exponent	– in analysis, + in synthesis $f \sim \sum_n \hat{f}(n) e^{in\theta}$
Transform on $\mathbb{R}$	$\hat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ (no residual 2π factors)
DFT	$X_j = \sum_{k=0}^{N-1} x_k e^{-2\pi i j k / N}$, $x_k = \frac{1}{N} \sum_j X_j e^{+2\pi i j k / N}$

The normalised measure is the choice that makes the e_n *orthonormal* rather than merely orthogonal, removes the 2π factors from Parseval and from the convolution theorem, and makes the analogy with the DFT literal (where the factor $1/N$ appears for the same reason).

A reader comparing with texts using the unnormalised measure $d\theta$ must remember that their coefficients differ by a factor 2π and their Parseval identities by a factor 2π or $1/2\pi$ depending on direction.

4.2. The space $L^2(\mathbb{T})$ and the characters of the circle

Definition 4.1: The space $L^2(\mathbb{T})$. $L^2(\mathbb{T})$ is the space of measurable functions $f : \mathbb{T} \rightarrow \mathbb{C}$ with $\int_0^{2\pi} |f|^2 d\theta < \infty$, identified when they agree almost everywhere, equipped with the inner product and norm

$$\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \overline{g(\theta)} d\theta, \quad \|f\|_2 = \left(\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta \right)^{1/2}.$$

With this structure $L^2(\mathbb{T})$ is a (complete) Hilbert space; completeness — far from obvious — is proved later, without circularity, in Theorem 4.12 (Riesz–Fischer).

Remark 4.2: Why the mean and not the integral. The normalisation $\frac{1}{2\pi}$ turns the integral into a mean: $\|1\|_2 = 1$, $\langle f, 1 \rangle$ is the mean value of f , and $\|f\|_2^2$ is the mean square value — in engineering, the power of the signal. This energetic reading is the one used systematically from Theorem 4.15 onwards.

Definition 4.2: Character of the circle. A **character** of $\mathbb{T}$ is a continuous group homomorphism $\chi : \mathbb{T} \rightarrow \mathbb{T}$, i.e. a continuous map with $\chi(zw) = \chi(z)\chi(w)$ for all $z, w \in \mathbb{T}$ and $|\chi(z)| = 1$.

Theorem 4.1: Classification of the characters of $\mathbb{T}$. The characters of $\mathbb{T}$ are exactly the power maps

$$\chi_n(z) = z^n, \quad n \in \mathbb{Z},$$

that is, in angular coordinates, $\chi_n(e^{i\theta}) = e^{in\theta} = e_n(\theta)$. In particular the exponential modes of the conventions card are not an arbitrary choice: they are all and only the continuous homomorphisms of the group $\mathbb{T}$ into itself.

Proof. Step 1 — power maps are characters. $\chi_n(zw) = (zw)^n = z^n w^n$ by commutativity of $\mathbb{C}$, and $|z^n| = 1$: they are continuous homomorphisms of $\mathbb{T}$ into itself (Chapter 1, Proposition 1.3).

Step 2 — reduction to a homomorphism of $\mathbb{R}$. Let χ be a character and set

$$\varphi(t) = \chi(e^{it}), \quad t \in \mathbb{R}.$$

Then φ is continuous, $|\varphi| = 1$, $\varphi(0) = \chi(1) = 1$, $\varphi(s+t) = \varphi(s)\varphi(t)$ and φ is 2π -periodic.

Step 3 — φ is differentiable. By continuity and $\varphi(0) = 1$, there is $\delta > 0$ with

$$c := \int_0^\delta \varphi(t) dt \neq 0$$

(indeed $\frac{1}{\delta} \int_0^\delta \varphi \rightarrow \varphi(0) = 1$ as $\delta \rightarrow 0^+$). Then, for all s ,

$$c\varphi(s) = \int_0^\delta \varphi(s)\varphi(t) dt = \int_0^\delta \varphi(s+t) dt = \int_s^{s+\delta} \varphi(u) du,$$

using $u = s + t$. The right-hand side is differentiable in s (integral of a continuous function with variable endpoints), hence so is φ , and differentiating,

$$c\varphi'(s) = \varphi(s + \delta) - \varphi(s) = \varphi(s)(\varphi(\delta) - 1).$$

Step 4 — solving the differential equation. With $a = (\varphi(\delta) - 1)/c$, the equation is $\varphi' = a\varphi$ with $\varphi(0) = 1$, whose unique solution is $\varphi(t) = e^{at}$ (one checks that $\frac{d}{dt}[\varphi(t)e^{-at}] = 0$, so $\varphi(t)e^{-at}$ is constant and equals 1 at $t = 0$).

Step 5 — a is purely imaginary. From $|\varphi(t)| = 1$ for all t we get $e^{t\operatorname{Re} a} = 1$ for all t , forcing $\operatorname{Re} a = 0$: hence $a = i\alpha$ with $\alpha \in \mathbb{R}$.

Step 6 — periodicity quantises α . The 2π -periodicity gives $e^{2\pi i\alpha} = \varphi(2\pi) = 1$, i.e. $2\pi\alpha \in 2\pi\mathbb{Z}$ by Theorem 3.3 of Chapter 3: hence $\alpha = n \in \mathbb{Z}$.

Conclusion. $\chi(e^{i\theta}) = e^{in\theta} = (e^{i\theta})^n$, i.e. $\chi = \chi_n$. $\square$

Remark 4.3: Why this theorem matters. Theorem 4.1 answers the question “why sinusoids?”. It is not a matter of convenience, nor of physical intuition: the exponentials $e^{in\theta}$ are the only continuous homomorphisms of the group $\mathbb{T}$ into itself, i.e. the only functions turning the additive structure of angles into the multiplicative structure of numbers of modulus 1. Every operator commuting with rotations of the circle — and such is, for instance, every linear time-invariant system — necessarily has these as eigenfunctions. This is the observation from which abstract harmonic analysis is born, recalled in (Volume III, Chapter 13): on every locally compact abelian group the characters play the same role, and the Fourier transform is the decomposition over them.

Remark 4.4: Three transforms, three domains: how not to confuse them. The chapter treats three tools that must not be conflated. We summarise the distinction here; every later section refers back to it.

	<i>Fourier series</i>	<i>Fourier transform</i>	<i>DFT</i>
<i>Domain</i>	$\mathbb{T}$ (compact)	$\mathbb{R}$ (non-compact)	$\mathbb{Z}/N\mathbb{Z}$ (finite)
<i>Function</i>	periodic	non-periodic, integrable	N samples
<i>Spectrum</i>	discrete, $n \in \mathbb{Z}$	continuous, $\xi \in \mathbb{R}$	discrete and finite
<i>Formula</i>	$\hat{f}(n) = \langle f, e_n \rangle$	$\hat{f}(\xi) = \int_{\mathbb{R}} f e^{-2\pi i x \xi} dx$	$X_j = \sum_k x_k e^{-2\pi i j k / N}$
<i>Inversion</i>	series	integral	finite sum

The practical rule: compact domain $\Rightarrow$ discrete spectrum; discrete domain $\Rightarrow$ periodic spectrum. The circle is compact, so its modes are countable; a grid of N points is discrete and finite, so its spectrum is discrete and periodic of period N — which is exactly the reason for aliasing (Volume III, Chapter 24). Using the continuous transform on periodic data, or comparing energies computed with different normalisations, are the commonest and hardest-to-spot errors downstream.

4.3. Fourier series: coefficients and orthogonality

Theorem 4.2: Orthogonality of complex exponentials. For $n, m \in \mathbb{Z}$:

$$\frac{1}{2\pi} \int_0^{2\pi} e^{in\theta} \overline{e^{im\theta}} d\theta = \delta_{nm} = \begin{cases} 1 & n = m \\ 0 & n \neq m. \end{cases}$$

Proof. $\overline{e^{im\theta}} = e^{-im\theta}$, so the integral is $\frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)\theta} d\theta$. If $n = m$ the integrand is 1, integral $= 2\pi/2\pi = 1$. If $n \neq m$, let $k = n - m \neq 0$:

$$\int_0^{2\pi} e^{ik\theta} d\theta = \left[\frac{e^{ik\theta}}{ik} \right]_0^{2\pi} = \frac{e^{2\pi ik} - 1}{ik} = \frac{1 - 1}{ik} = 0,$$

since $e^{2\pi ik} = \cos(2\pi k) + i \sin(2\pi k) = 1$ for $k \in \mathbb{Z}$, by periodicity of sine and cosine. $\square$

Definition 4.3: Fourier coefficients. For $f \in L^2(\mathbb{T})$ (or $f : \mathbb{R} \rightarrow \mathbb{C}$ 2π -periodic, sufficiently regular):

$$\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta, \quad n \in \mathbb{Z},$$

and formally $f(\theta) \sim \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta}$ (Fourier series).

Theorem 4.3: Trigonometric form and complex form. Let $f \in L^2(\mathbb{T})$ with complex coefficients $c_n = \hat{f}(n)$. Setting, for $n \geq 1$,

$$a_n = c_n + c_{-n}, \quad b_n = i(c_n - c_{-n}), \quad a_0 = 2c_0,$$

one has

$$c_n e^{in\theta} + c_{-n} e^{-in\theta} = a_n \cos n\theta + b_n \sin n\theta,$$

and hence the **trigonometric form**

$$f(\theta) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\theta + b_n \sin n\theta).$$

The direct formulas for the real coefficients are

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos n\theta d\theta, \quad b_n = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin n\theta d\theta,$$

and the inverse formulas

$$c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \frac{a_n + ib_n}{2} \quad (n \geq 1), \quad c_0 = \frac{a_0}{2}.$$

Proof. Step 1 — inverting the definitions. From $a_n = c_n + c_{-n}$ and $b_n = i(c_n - c_{-n})$ we get $c_n - c_{-n} = -ib_n$, and adding or subtracting with the first:

$$c_n = \frac{a_n - ib_n}{2}, \quad c_{-n} = \frac{a_n + ib_n}{2}.$$

Step 2 — recombining the two modes. Using Euler's formula (Chapter 3, Theorem 3.5),

$$c_n e^{in\theta} = \frac{1}{2} [a_n \cos n\theta + b_n \sin n\theta] + \frac{i}{2} [a_n \sin n\theta - b_n \cos n\theta],$$

$$c_{-n} e^{-in\theta} = \frac{1}{2} [a_n \cos n\theta + b_n \sin n\theta] - \frac{i}{2} [a_n \sin n\theta - b_n \cos n\theta].$$

Adding, the imaginary parts cancel and $a_n \cos n\theta + b_n \sin n\theta$ remains.

Step 3 — direct formulas. By Definition 4.3,

$$a_n = c_n + c_{-n} = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) 2 \cos n\theta d\theta,$$

by Theorem 3.6 of Chapter 3, whence the formula for a_n . Similarly

$$b_n = i(c_n - c_{-n}) = \frac{i}{2\pi} \int_0^{2\pi} f(\theta) (-2i \sin n\theta) d\theta,$$

and since $i \cdot (-2i) = 2$ we obtain the formula for b_n . Finally $a_0 = 2c_0 = \frac{1}{\pi} \int_0^{2\pi} f$, consistent with the general formula for $n = 0$: this is the reason for the factor $a_0/2$ in the series. $\square$

Theorem 4.4: Symmetries and coefficients. Let $f \in L^2(\mathbb{T})$ with coefficients $c_n = \hat{f}(n)$. Then:

- (i) f is real-valued $\iff c_{-n} = \overline{c_n}$ for every n ;
- (ii) f is even ($f(-\theta) = f(\theta)$) $\iff c_{-n} = c_n$ for every n ; if moreover f is real, all the b_n vanish and the series is a cosine series;
- (iii) f is odd ($f(-\theta) = -f(\theta)$) $\iff c_{-n} = -c_n$ for every n ; if moreover f is real, all the a_n vanish (including a_0) and the series is a sine series;
- (iv) if f is real, $|c_{-n}| = |c_n|$: the amplitude spectrum is even.

Proof. (i) Conjugating the definition of c_n and using that conjugation commutes with the integral,

$$\overline{c_n} = \frac{1}{2\pi} \int_0^{2\pi} \overline{f(\theta)} e^{in\theta} d\theta.$$

If f is real, $\overline{f} = f$ and the right-hand side is by definition c_{-n} . Conversely, if $c_{-n} = \overline{c_n}$ for all n , then f and $\overline{f}$ have the same coefficients (since $\widehat{\overline{f}}(n) = \overline{\widehat{f}(-n)} = \overline{c_{-n}} = c_n$), hence they agree by uniqueness (Corollary, Volume III, Chapter 17).

(ii) In the definition of c_{-n} substitute $\theta \mapsto -\theta$, legitimate because the integrand is 2π -periodic and the integral over a period does not depend on the chosen interval:

$$c_{-n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(-u) e^{-inu} du.$$

If f is even, $f(-u) = f(u)$ and the right-hand side is c_n . The converse again follows from uniqueness. If moreover f is real and even, then by (i) and (ii) $c_n = c_{-n} = \overline{c_n}$, so $c_n \in \mathbb{R}$ and $b_n = i(c_n - c_{-n}) = 0$.

(iii) Identical to (ii) with $f(-u) = -f(u)$, giving $c_{-n} = -c_n$. If f is real and odd, $a_n = c_n + c_{-n} = 0$ for every $n \geq 0$; in particular $a_0 = 2c_0 = 0$, i.e. f has zero mean (as it must for an odd function).

(iv) From (i), $|c_{-n}| = |\overline{c_n}| = |c_n|$. $\square$

Best practice. Theorem 4.4 should be used *before* computing, not after: recognising that a function is even or odd halves the work, because it kills half the coefficients without any integral. Likewise, for a real function it suffices to compute the c_n with $n \geq 0$: the negative ones follow by conjugation. Forgetting this is the commonest error in hand computations, and it produces the typical symptom of “doubled” coefficients or of an energy that fails to balance in Parseval.

Theorem 4.5: Bessel’s inequality and best approximation. Let $f \in L^2(\mathbb{T})$, $c_n = \hat{f}(n)$, and let

$$S_N f(\theta) = \sum_{|n| \leq N} c_n e^{in\theta}$$

be its partial sum of order N . Then, for every trigonometric polynomial $P(\theta) = \sum_{|n| \leq N} d_n e^{in\theta}$ of degree $\leq N$:

- (i) $\|f - P\|_2^2 = \|f - S_N f\|_2^2 + \sum_{|n| \leq N} |c_n - d_n|^2$;
(ii) consequently $\|f - S_N f\|_2 \leq \|f - P\|_2$, with equality iff $P = S_N f$: the partial sum is the best approximation in mean square;
(iii) $\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |c_n|^2$;
(iv) (Bessel's inequality) $\sum_{n \in \mathbb{Z}} |c_n|^2 \leq \|f\|_2^2 < \infty$.

Proof. Write $e_n(\theta) = e^{in\theta}$; by Theorem 4.2 the system $\{e_n\}$ is orthonormal in $L^2(\mathbb{T})$, and by definition $c_n = \langle f, e_n \rangle$.

Step 1 — expanding the norm. For $P = \sum_{|n| \leq N} d_n e_n$,

$$\|f - P\|_2^2 = \|f\|_2^2 - \langle f, P \rangle - \langle P, f \rangle + \|P\|_2^2.$$

By sesquilinearity and orthonormality, $\langle f, P \rangle = \sum_{|n| \leq N} \overline{d_n} c_n$, $\|P\|_2^2 = \sum_{|n| \leq N} |d_n|^2$, and $\langle P, f \rangle = \sum_{|n| \leq N} d_n \overline{c_n}$. Hence

$$\|f - P\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} (\overline{d_n} c_n + d_n \overline{c_n}) + \sum_{|n| \leq N} |d_n|^2.$$

Step 2 — completing the square. Adding and subtracting $\sum_{|n| \leq N} |c_n|^2$ and recognising $|c_n|^2 - \overline{d_n} c_n - d_n \overline{c_n} + |d_n|^2 = |c_n - d_n|^2$:

$$\|f - P\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |c_n|^2 + \sum_{|n| \leq N} |c_n - d_n|^2. \quad (*)$$

Step 3 — (iii). Setting $d_n = c_n$ (i.e. $P = S_N f$) the last sum vanishes.

Step 4 — (i) and (ii). Substituting Step 3 into (*) gives (i). Since the last sum in (*) is non-negative and vanishes only if $d_n = c_n$ for every n , the minimum is attained exactly at $P = S_N f$: this is (ii).

Step 5 — (iv). The left-hand side of Step 3 is ≥ 0 , hence $\sum_{|n| \leq N} |c_n|^2 \leq \|f\|_2^2$ for every N . The partial sums of the non-negative series $\sum_n |c_n|^2$ are bounded by $\|f\|_2^2$: the series converges and its sum does not exceed $\|f\|_2^2$. $\square$

Corollary 4.1: Riemann–Lebesgue lemma on $\mathbb{T}$. If $f \in L^2(\mathbb{T})$, then $\hat{f}(n) \rightarrow 0$ as $|n| \rightarrow \infty$. The same holds, more generally, for $f \in L^1(\mathbb{T})$.

Proof. Case L^2 . By Bessel's inequality (Theorem 4.5(iv)) the series $\sum_n |\hat{f}(n)|^2$ converges; the general term of a convergent series tends to zero, so $\hat{f}(n) \rightarrow 0$.

Case L^1 . Given $\varepsilon > 0$, by density of trigonometric polynomials in $L^1(\mathbb{T})$ — a consequence of Fejér's theorem (Corollary, Volume III, Chapter 17) — there is a trigonometric polynomial P of degree M with $\|f - P\|_1 < \varepsilon$. For $|n| > M$ we have $\hat{P}(n) = 0$ (Theorem 4.2), hence

$$|\hat{f}(n)| = |\widehat{f - P}(n)| \leq \|f - P\|_1 < \varepsilon.$$

Since ε was arbitrary, the claim follows. $\square$

Remark 4.5: What Riemann–Lebesgue means. The lemma says that every integrable function has decaying coefficients: no function “contains” an arbitrarily high-frequency mode with non-negligible amplitude. The speed at which the coefficients tend to zero measures the

smoothness of f : (Volume III, Chapter 17) will show that greater smoothness gives faster decay, and that it is precisely this decay that governs the type of convergence of the series. It is the same principle already met in Chapter 2 with the periodic trapezoidal rule, where the smoothness of the integrand dictated the order of the error (Theorem 2.16).

Definition 4.4: Amplitude and phase spectrum. For $f \in L^2(\mathbb{T})$ with coefficients $c_n = \hat{f}(n)$ one calls

$$\text{amplitude spectrum: } n \mapsto |c_n|, \quad \text{phase spectrum: } n \mapsto \text{Arg } c_n.$$

In the trigonometric form, the n -th mode of a real function is written

$$a_n \cos n\theta + b_n \sin n\theta = A_n \cos(n\theta - \phi_n), \quad A_n = \sqrt{a_n^2 + b_n^2} = 2|c_n|, \quad \phi_n = \text{atan2}(b_n, a_n).$$

4.4. From the Riemann integral to Lebesgue measure

INFO

Why this section. The previous sections built the Fourier coefficients and their orthogonality using only the Riemann integral. But Parseval's identity, stated further on, requires knowing that the trigonometric system is **complete** in $L^2(\mathbb{T})$ — a fact the Riemann integral alone is not flexible enough to establish rigorously (the interchange of limit and integral needed to prove it fails in general for Riemann). This section builds the tools — Lebesgue measure and integral, completeness of L^2 — needed to fill that gap, proving in full what is within reach at this level and honestly stating, with a source, what goes beyond it. The prerequisites are completeness of the real numbers, elementary properties of sequences and series, compactness of closed intervals, and the Riemann calculus of the preceding chapters: not an axiomatic construction of $\mathbb{R}$.

Theorem 4.6: A pointwise limit that is not Riemann integrable. There are Riemann-integrable functions $f_n : [0, 1] \rightarrow \mathbb{R}$, all with integral zero, whose pointwise limit is not Riemann integrable.

Proof. Step 1 — construction. Enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ by ordering fractions p/q by $|p| + q$ and removing repetitions. Set $f_n = \mathbf{1}_{\{q_1, \dots, q_n\}}$ and $f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}$. Every rational occurs from some index onward, while no irrational ever occurs. Hence $f_n \uparrow f$ pointwise.

Step 2 — Darboux sums. Every lower Darboux sum of f_n is zero. Given $\varepsilon > 0$, surround its finitely many nonzero points by intervals of total length less than ε ; a partition containing their endpoints has upper sum less than ε (at the endpoints 0, 1 use one-sided intervals). Hence $\int_0^1 f_n = 0$.

Step 3 — failure for the limit. Every nondegenerate interval contains a rational: choose m with $m(b - a) > 1$ and an integer k with $ma < k < mb$. It also contains an irrational, obtained by perturbing an interior rational by $\sqrt{2}/N$ for sufficiently large N . For f , therefore, every lower sum equals 0 and every upper sum equals 1. The two Darboux integrals differ. $\square$

Remark 4.6: What this example really says. The limit function $f = \mathbf{1}_{\mathbb{Q} \cap [0, 1]}$ is the classical Dirichlet function. The example shows that the class of Riemann-integrable functions is not closed under monotone pointwise limits — even in the simplest possible case, indicators of finite sets. A more robust integration theory is needed with respect to this kind

of limit: that is exactly what follows.

4.4.1 Outer measure and measurable sets

Definition 4.5: Outer measure. For $E \subseteq \mathbb{R}$, using bounded open intervals I_j , define

$$m^*(E) = \inf \left\{ \sum_{j=1}^{\infty} |I_j| : E \subseteq \bigcup_{j=1}^{\infty} I_j \right\}.$$

The infimum is $+\infty$ when all the sums are infinite. The empty cover is allowed for $E = \emptyset$.

Lemma 4.1: Properties of outer measure. Outer measure is zero on the empty set, monotone, countably subadditive and translation invariant. Every bounded interval has outer measure equal to its length; every countable set has outer measure zero.

Proof. **Step 1 — basic properties and subadditivity.** A cover of F also covers $E \subset F$, so $m^*(E) \leq m^*(F)$. For subadditivity only $\sum_j m^*(E_j) < \infty$ needs proof: for each j choose a cover costing less than $m^*(E_j) + \varepsilon 2^{-j}$. Combining them gives $m^*(\bigcup_j E_j) \leq \sum_j m^*(E_j) + \varepsilon$; let ε decrease to zero. Translation preserves interval lengths, so $m^*(E + a) = m^*(E)$, using the inverse translation for the reverse inequality.

Step 2 — intervals. The cover $(a - \varepsilon, b + \varepsilon)$ gives $m^*([a, b]) \leq b - a$. Compactness extracts a finite subcover from any open cover of $[a, b]$. Its lengths sum to at least $b - a$: ordering and merging overlapping intervals cannot lower total length, and their union contains $[a, b]$. This proves the reverse inequality. For open or half-open intervals use $[a + \varepsilon, b - \varepsilon] \subset I \subset [a, b]$ and let ε tend to zero.

Step 3 — countable sets. A point admits interval covers of arbitrarily small length and therefore has outer measure zero. Countable subadditivity gives $m^*({x_1, x_2, \dots}) \leq \sum_j m^*({x_j}) = 0$. $\square$

Definition 4.6: Carathéodory criterion. A set $E \subseteq \mathbb{R}$ is Lebesgue measurable when, for every $A \subseteq \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \setminus E).$$

Write $\mathcal{L}$ for this family and m for the restriction of m^* to $\mathcal{L}$.

Theorem 4.7: Construction of Lebesgue measure. The family $\mathcal{L}$ is a σ -algebra: it contains $\mathbb{R}$ and is closed under complements and countable unions. It contains the Borel sets and all subsets of null sets. The function m is a complete, countably additive, translation-invariant measure.

Proof. **Step 1 — finite operations.** The criterion is immediate for $\mathbb{R}$ and unchanged on replacing E by E^c . For $E, F \in \mathcal{L}$, split A first by E , then split both pieces by F . This expresses $m^*(A)$ as the sum of the outer measures of four disjoint pieces. Combining three of them and using subadditivity gives $m^*(A) \geq m^*(A \cap E \cap F) + m^*(A \setminus (E \cap F))$. The reverse inequality always holds by subadditivity. Thus $E \cap F \in \mathcal{L}$, and finite unions and differences follow.

Step 2 — countable unions. Let $E_j \in \mathcal{L}$ be disjoint and $E = \bigcup_j E_j$. Successively splitting off the first N sets gives, for every A :

$$m^*(A) \geq \sum_{j=1}^N m^*(A \cap E_j) + m^*(A \setminus E).$$

Take the supremum over N and use $m^*(A \cap E) \leq \sum_j m^*(A \cap E_j)$ to obtain the criterion for E . Reduce an arbitrary union to the disjoint case by replacing E_j by $E_j \setminus \bigcup_{k < j} E_k$, which is measurable by Step 1.

Step 3 — additivity and completeness. Set $A = E$ in Step 2: $m(E) \geq \sum_j m(E_j)$, and subadditivity gives the opposite inequality. If $m^*(N) = 0$, monotonicity gives $m^*(B) = 0$ for every $B \subseteq N$. For any A , $m^*(A \cap B) = 0$ and $m^*(A \setminus B) \leq m^*(A)$. Together with subadditivity this proves the criterion for B . Thus every subset of a null set is measurable: this is the completeness of the measure.

Step 4 — half-lines and Borel sets. Fix $H = (-\infty, a]$. Split each covering interval of A at a . This gives covers of $A \cap H$ and $A \setminus H$ whose combined cost is at most the original cost plus ε ; enlarge half-open pieces by amounts with sum less than ε . Taking the infimum and then $\varepsilon \downarrow 0$ gives $m^*(A \cap H) + m^*(A \setminus H) \leq m^*(A)$. Thus H is measurable. Half-lines generate intervals, and every open set is the countable union of its rational-endpoint intervals. Hence the Borel σ -algebra, generated by open sets, is contained in $\mathcal{L}$. Translating the criterion and using invariance of m^* proves invariance of $\mathcal{L}$ and m . $\square$

Lemma 4.2: Continuity of measure from below. If E_n are measurable and $E_n \uparrow E$, then $m(E_n) \uparrow m(E)$.

Proof. Put $D_1 = E_1$ and $D_n = E_n \setminus E_{n-1}$ for $n \geq 2$. These sets are disjoint, $E = \bigcup_n D_n$ and $E_N = \bigcup_{n \leq N} D_n$. Countable additivity gives $m(E) = \sum_n m(D_n) = \lim_N \sum_{n \leq N} m(D_n) = \lim_N m(E_N)$, including an infinite limit. $\square$

Remark 4.7: Measure on the circle. On the circle choose the representative $\theta \in [0, 2\pi)$ and set $\mu(E) = m(E)/(2\pi)$. Endpoints have measure zero. A rotation is two translations on the arcs separated by the cut; additivity and translation invariance of m give rotation invariance of μ and $\mu(\mathbb{T}) = 1$. From now on circle integrals mean $\int_{\mathbb{T}} d\mu$. The following construction also applies to $[0, 1]$ with measure m .

4.4.2 Measurable functions and the integral

Definition 4.7: Measurability and simple functions. An extended real function $f : X \rightarrow [-\infty, +\infty]$ is measurable when $\{f > a\}$ is measurable for every $a \in \mathbb{R}$. A complex function is measurable when its real and imaginary parts are. A nonnegative simple function has the form $s = \sum_{j=1}^r a_j \mathbf{1}_{E_j}$, with $a_j \geq 0$ and disjoint measurable E_j . Define

$$\int s \, d\mu = \sum_{j=1}^r a_j \mu(E_j), \quad \int f \, d\mu = \sup \left\{ \int s \, d\mu : 0 \leq s \leq f, s \in \mathcal{S}_+ \right\} \quad (f \geq 0).$$

In the formula, $\mathcal{S}_+$ denotes the set of nonnegative simple functions. Refining two representations of s by intersecting their sets, additivity of the measure shows that the integral is independent of the representation. The same refinement proves additivity and homogeneity for simple functions; monotonicity follows by comparing values on each common cell. For nonnegative functions monotonicity follows directly from the supremum definition.

Lemma 4.3: Increasing measurable approximation. Pointwise limits and finite algebraic operations, wherever defined, preserve measurability. Every nonnegative measurable function is an increasing limit of nonnegative simple functions.

Proof. Step 1 — operations. For finite real functions, $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g > a - r\})$. Also $\{f \geq a\} = \bigcap_k \{f > a - 1/k\}$; complements give sublevel sets. Preimages of open intervals are measurable, and every open set is a countable union of rational intervals, so composition with a continuous function preserves measurability. This covers squares, absolute values and products through $fg = ((f + g)^2 - f^2 - g^2)/2$; use real components for complex functions. Finally $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$, and changing signs gives infima. Hence $\liminf_n f_n = \sup_N \inf_{n \geq N} f_n$ and existing pointwise limits are measurable.

Step 2 — explicit approximation.

$$s_n(x) = 2^{-n} \lfloor 2^n \min\{f(x), n\} \rfloor.$$

The level sets of s_n are intersections of measurable level sets, and it takes finitely many values. Refining the mesh from 2^{-n} to $2^{-(n+1)}$ and raising the cutoff from n to $n + 1$ cannot lower its value: $s_n \leq s_{n+1} \leq f$. If $f(x) < \infty$, then $0 \leq f(x) - s_n(x) < 2^{-n}$ for $n > f(x)$; if $f(x) = \infty$, then $s_n(x) = n$. Thus $s_n \uparrow f$. $\square$

Theorem 4.8: Monotone convergence (Beppo Levi). If $0 \leq f_n \uparrow f$ with f_n measurable, then $\int f_n \uparrow \int f$, allowing an infinite value.

Proof. Step 1 — the easy inequality. Monotonicity gives the existence of $L = \lim_n \int f_n$ in $[0, \infty]$ and $L \leq \int f$.

Step 2 — comparison with every simple function. Fix a simple $0 \leq s \leq f$ and $0 < c < 1$. The sets $A_n = \{f_n \geq cs\}$ increase to X : membership is immediate where $s = 0$, and where $s > 0$ we have $f \geq s > cs$. Since $f_n \geq cs \mathbf{1}_{A_n}$, we get $\int f_n \geq c \int s \mathbf{1}_{A_n}$. Write $s = \sum_j a_j \mathbf{1}_{E_j}$ and apply continuity of measure from below to $E_j \cap A_n$, obtaining $L \geq c \sum_j a_j \mu(E_j) = c \int s$. Let $c \uparrow 1$ and take the supremum over simple $s \leq f$: $L \geq \int f$. No interchange theorem has been assumed in this argument. $\square$

Corollary 4.2: Additivity and nonnegative series. For measurable $f, g \geq 0$ and $a \geq 0$, $\int(f + g) = \int f + \int g$ and $\int af = a \int f$ (with $0 \cdot \infty = 0$). For a measurable sequence $u_j \geq 0$, $\int \sum_j u_j = \sum_j \int u_j$.

Proof. Choose simple $s_n \uparrow f$ and $t_n \uparrow g$. Then $s_n + t_n \uparrow f + g$: monotone convergence and simple-function additivity give $\int(f + g) = \lim_n (\int s_n + \int t_n) = \int f + \int g$. For $a > 0$ use $as_n \uparrow af$; for $a = 0$ the identity is immediate. Partial sums $\sum_{j=1}^N u_j$ increase to $\sum_j u_j$: apply monotone convergence again, and finite additivity to each partial sum, to obtain the final identity. $\square$

Lemma 4.4: Fatou. For measurable $f_n \geq 0$, $\int \liminf_n f_n \leq \liminf_n \int f_n$.

Proof. Set $g_N = \inf_{n \geq N} f_n$. These measurable functions increase to $\liminf_n f_n$. Since $g_N \leq f_n$ for every $n \geq N$, $\int g_N \leq \inf_{n \geq N} \int f_n$. Letting $N \rightarrow \infty$, monotone convergence on the left and the definition of the lower limit on the right prove the claim. $\square$

Definition 4.8: Integrability and equality almost everywhere. A measurable complex function f is integrable, $f \in L^1$, when $\int |f| < \infty$. For real f , define $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$ and $\int f = \int f^+ - \int f^-$. For a complex function, integrate real and imaginary parts separately. Two functions agree *almost everywhere* if they differ only on a set of measure zero.

Lemma 4.5: Linearity, absolute value and null sets. The integral on L^1 is linear and $|\int f| \leq \int |f|$. For $h \geq 0$, $\int h = 0$ exactly when $h = 0$ almost everywhere. Modifications on

a null set preserve measurability and the integral.

Proof. Step 1 — linearity. For integrable real f, g , $|f + g| \leq |f| + |g|$ ensures finiteness of all integrals in $(f + g)^+ + f^- + g^- = (f + g)^- + f^+ + g^+$. Integrate by nonnegative additivity and subtract finite quantities to get $\int(f + g) = \int f + \int g$. Positive homogeneity is already proved; negative scalars exchange positive and negative parts. Real and imaginary components give complex linearity.

Step 2 — inequality and null sets. If $z = \int f \neq 0$, choose $c = \bar{z}/|z|$. Then $|z| = \Re \int cf = \int \Re(cf) \leq \int |f|$; $z = 0$ is immediate. If $\int h = 0$, $h \geq k^{-1} \mathbf{1}_{\{h \geq 1/k\}}$ shows that each set $\{h \geq 1/k\}$ is null; their union is $\{h > 0\}$. Conversely, every simple $0 \leq s \leq h$ supported on a null set has zero integral, hence so does h . Completeness of measure makes null-set modifications measurable; their difference has zero absolute integral, so the integral is unchanged. $\square$

Theorem 4.9: Lebesgue dominated convergence. Let f_n be measurable complex functions, $f_n \rightarrow f$ almost everywhere, and $|f_n| \leq G$ almost everywhere for every n , with integrable $G \geq 0$. Then $f \in L^1$, $\int |f_n - f| \rightarrow 0$ and $\int f_n \rightarrow \int f$.

Proof. Step 1 — exceptions and integrability. The countable union of exceptional sets is null. Set all functions to zero there. The limit and bounds now hold everywhere: f is measurable and $|f| \leq G$, hence integrable.

Step 2 — application of Fatou. Apply Fatou to the nonnegative functions $2G - h_n$ with $h_n = |f_n - f|$: $0 \leq h_n \leq 2G$ and $h_n \rightarrow 0$. This gives $2 \int G \leq \liminf_n \int (2G - h_n) = 2 \int G - \limsup_n \int h_n$. Since $\int G$ is finite, subtraction is legitimate, yielding $\limsup_n \int h_n \leq 0$. These integrals are nonnegative, hence converge to zero. Finally $|\int f_n - \int f| \leq \int |f_n - f| \rightarrow 0$. $\square$

Theorem 4.10: Compatibility with Riemann. Every bounded Riemann-integrable function on $[a, b]$ is Lebesgue measurable and has the same integral.

Proof. For real f , choose partitions whose upper and lower sums differ by less than 2^{-n} and refine them cumulatively. Their lower step functions l_n increase and upper step functions u_n decrease, with $l_n \leq f \leq u_n$ away from the countable set of endpoints. Their limits l, u are measurable. Fatou gives $\int(u - l) \leq \liminf_n \int(u_n - l_n) = 0$. Thus $u = l$ almost everywhere and $f = l$ almost everywhere: completeness of measure ensures measurability of f . The inequalities $\int l_n \leq \int f \leq \int u_n$ and Darboux's criterion show equality of the Lebesgue and Riemann integrals. Apply this argument to both components for complex functions. $\square$

Example 4.1: What changes with Lebesgue. For the finite rational-set indicators of the opening example, $|f_n| \leq 1$ and $f_n \rightarrow \mathbf{1}_{\mathbb{Q} \cap [0,1]}$. Dominated convergence justifies the interchange: the limit has integral zero. The function was not Riemann integrable. A common majorant is essential: $g_n = n \mathbf{1}_{(0,1/n)}$ tends to zero everywhere but $\int g_n = 1$. No common integrable majorant can exist, or the theorem just proved would imply $1 \rightarrow 0$.

4.4.3 Geometry and completeness of L^2

From now on we use $\mu(\mathbb{T}) = 1$. The space $L^2(\mathbb{T})$ consists of classes of measurable functions satisfying $\int |f|^2 d\mu < \infty$, identified when equal almost everywhere. The inner product is linear in its first argument:

$$\langle f, g \rangle = \int f \bar{g} d\mu, \quad \|f\|_2 = \sqrt{\langle f, f \rangle}.$$

The product is integrable because $2|f\bar{g}| \leq |f|^2 + |g|^2$. The sum remains in L^2 since $|f + g|^2 \leq 2|f|^2 + 2|g|^2$. Lemma 4.5 ensures independence from the chosen representatives.

Theorem 4.11: Cauchy–Schwarz, triangle and parallelogram.

$$|\langle f, g \rangle| \leq \|f\|_2 \|g\|_2, \quad \|f + g\|_2 \leq \|f\|_2 + \|g\|_2, \quad \|f\|_1 \leq \|f\|_2,$$

$$\|f + g\|_2^2 + \|f - g\|_2^2 = 2\|f\|_2^2 + 2\|g\|_2^2.$$

Equality in Cauchy–Schwarz holds exactly when f, g are linearly dependent as almost-everywhere classes.

Proof. Step 1 — Cauchy–Schwarz. If $\|g\|_2 = 0$, then $g = 0$ almost everywhere and the result is immediate. Otherwise put $c = \langle f, g \rangle / \|g\|_2^2$. Expanding the square using linearity in the first argument,

$$0 \leq \|f - cg\|_2^2 = \|f\|_2^2 - \frac{|\langle f, g \rangle|^2}{\|g\|_2^2}.$$

Multiply by $\|g\|_2^2$ to obtain the inequality. Equality is equivalent to $f = cg$ almost everywhere; if either vector is zero, dependence is automatic.

Step 2 — the remaining identities. Expand $\|f + g\|_2^2 = \|f\|_2^2 + 2\Re\langle f, g \rangle + \|g\|_2^2 \leq (\|f\|_2 + \|g\|_2)^2$ and take square roots. For $f - g$ the mixed term changes sign: adding the two expansions proves the parallelogram identity. Finally apply Cauchy–Schwarz to $|f|$ and 1: $\int |f| \leq \|f\|_2 \|1\|_2 = \|f\|_2$ since $\mu(\mathbb{T}) = 1$. $\square$

Theorem 4.12: Riesz–Fischer: completeness of L^2 . Every Cauchy sequence in the $L^2(\mathbb{T})$ norm converges in that norm to an element of $L^2(\mathbb{T})$. Consequently $L^2(\mathbb{T})$ is a Hilbert space.

Proof. Step 1 — a fast subsequence. The Cauchy condition gives increasing indices n_k with $\|f_{n_{k+1}} - f_{n_k}\|_2 \leq 2^{-k}$. Choose measurable representatives and set $d_k = f_{n_{k+1}} - f_{n_k}$, $H_N = \sum_{k=1}^N |d_k|$. The triangle inequality gives $\|H_N\|_2 \leq \sum_{k=1}^N 2^{-k} \leq 1$.

Step 2 — construction of the limit. For $H = \sum_k |d_k|$, we have $H_N^2 \uparrow H^2$. Monotone convergence gives $\int H^2 \leq 1$, hence $H < \infty$ almost everywhere. Indeed, if $H = \infty$ on a positive-measure set, its squared integral would exceed M times that measure for every M . Outside a null set, $\sum_k d_k$ converges absolutely. Define $f = f_{n_1} + \sum_k d_k$ there and zero on the exception. It is measurable and belongs to L^2 , because $|f| \leq |f_{n_1}| + H$.

Step 3 — norm convergence.

$$\|f - f_{n_k}\|_2^2 \leq \liminf_{M \rightarrow \infty} \left\| \sum_{j=k}^M d_j \right\|_2^2 \leq \left(\sum_{j=k}^{\infty} 2^{-j} \right)^2 = 2^{2-2k}.$$

The first inequality is Fatou applied to squared absolute values of finite tail sums; the second is the triangle inequality. Thus the subsequence converges to f . Given $\varepsilon > 0$, choose N such that $\|f_n - f_m\|_2 < \varepsilon/2$ for $n, m \geq N$, then k with $n_k \geq N$ and $\|f_{n_k} - f\|_2 < \varepsilon/2$. For $n \geq N$, the triangle inequality gives $\|f_n - f\|_2 < \varepsilon$. The inner product is positive definite on almost-everywhere classes by the nullity lemma, and sesquilinear by linearity of the integral. A complete inner-product space is, by definition, a Hilbert space. $\square$

4.4.4 Worked exercises and self-assessment

Exercise 4.1: A limit and an inequality. Compute $\lim_n \int_0^1 x^n dx$, justifying the interchange. Check Cauchy–Schwarz for $f = 1$, $g = x$ on $[0, 1]$.

Solution. For $x < 1$, $x^n \rightarrow 0$; the exception $x = 1$ is null and the majorant 1 is integrable. Dominated convergence gives limit zero, confirmed by $\int_0^1 x^n dx = 1/(n+1)$. Also $|\int_0^1 x dx|^2 = 1/4 < 1/3 = (\int_0^1 1 dx)(\int_0^1 x^2 dx)$. Equality fails because x is not almost-everywhere constant.

Exercise 4.2: A series in L^2 . If $\|u_k\|_2 \leq 3^{-k}$ for $k \geq 1$, prove convergence of $\sum_k u_k$ in L^2 and bound its tail after N terms.

Solution. For $M > N$, $\|\sum_{k=N+1}^M u_k\|_2 \leq \sum_{k=N+1}^M 3^{-k} \leq 3^{-N}/2$. Partial sums are Cauchy and converge by Riesz–Fischer. Letting $M \rightarrow \infty$, the tail norm is at most $3^{-N}/2$. The passage is legitimate because $||a|_2 - |b|_2| \leq \|a - b\|_2$, obtained by applying the triangle inequality twice.

Warning. Quick self-check (true or false, justify).

1. A dense set always has positive measure.
2. $\mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ has zero Lebesgue integral.
3. Pointwise convergence suffices for interchanging limit and integral.
4. The dominated-convergence majorant may depend on n .
5. A function nonzero at one point may have zero L^2 norm.
6. Completeness implies that all sequences converge.
7. Dominated convergence and Riesz–Fischer have been proved here.
8. Orthogonality alone proves Parseval for every $f \in L^2$.

Teacher’s key. 1 F: rationals are dense and null. 2 T: a null-set indicator. 3 F: counterexample $n\mathbf{1}_{(0,1/n)}$. 4 F: one common integrable majorant is required. 5 T: classes are defined almost everywhere. 6 F: only Cauchy sequences. 7 T: the proofs are in this section. 8 F: completeness of the trigonometric system is still needed and is proved later.

4.5. From orthogonality to completeness: Fejér’s bridge

Orthogonality alone (Theorem 4.2) does not guarantee that every function in $L^2(\mathbb{T})$ can be reconstructed as a sum of the modes $e^{in\theta}$: for that we must prove that trigonometric polynomials are **dense** in $L^2(\mathbb{T})$. We do so without using Parseval in the density argument — otherwise the argument would be circular, since it is precisely density (together with the completeness of L^2 , just proved) that justifies Parseval.

Recall $e_k(\theta) = e^{ik\theta}$ and $\hat{f}(k) = \langle f, e_k \rangle = \int_{\mathbb{T}} f(\theta) e^{-ik\theta} d\mu$. For $f \in L^2$, these integrals exist because $\|f\|_1 \leq \|f\|_2$ (Theorem 4.11). The inner product is linear in its first argument.

Lemma 4.6: Density of continuous periodic functions. For every $f \in L^2(\mathbb{T})$ and $\varepsilon > 0$ there is a continuous periodic g with $\|f - g\|_2 < \varepsilon$.

Proof. Step 1 — reduction to simple functions. Set $f_M = f\mathbf{1}_{\{|f| \leq M\}}$. Since $|f - f_M|^2 \rightarrow 0$ almost everywhere and is dominated by $|f|^2$, dominated convergence gives $\|f - f_M\|_2 \rightarrow 0$. Rounding the real and imaginary parts of f_M to multiples of $1/r$, we obtain a simple s with $|s - f_M| \leq \sqrt{2}/r$, hence $\|s - f_M\|_2 \leq \sqrt{2}/r$. It therefore suffices to approximate measurable

indicators by continuous functions.

Step 2 — approximating a set by arcs. Let $E \subset \mathbb{T}$ be measurable. The definition of outer measure gives a countable union of open arcs $O \supset E$ with $\mu(O) < \mu(E) + \eta$, hence $\mu(O \setminus E) < \eta$. By continuity of measure from below, a finite union J of the first arcs has $\mu(O \setminus J) < \eta$. Then $E \Delta J \subset (O \setminus E) \cup (O \setminus J)$ and $\mu(E \Delta J) < 2\eta$. Consequently $\|\mathbf{1}_E - \mathbf{1}_J\|_2 < \sqrt{2\eta}$. Passing from intervals on the line to arcs cut at 0 changes at most null endpoints.

Step 3 — continuous transitions. The boundary of a finite union of arcs is finite. Replace every jump of $\mathbf{1}_J$ by a linear transition in disjoint small neighborhoods of total measure less than η . The resulting periodic continuous h satisfies $0 \leq h \leq 1$, equals $\mathbf{1}_J$ outside these neighborhoods, and $\|h - \mathbf{1}_J\|_2 \leq \sqrt{\eta}$. For empty J or the whole circle, a constant suffices. The triangle inequality now approximates $\mathbf{1}_E$. For $s = \sum_{j=1}^r a_j \mathbf{1}_{E_j}$ choose h_j with errors below $\varepsilon/[3r(1 + |a_j|)]$; then $g = \sum_j a_j h_j$ is continuous and $\|s - g\|_2 < \varepsilon/3$. Combine with the two errors of Step 1, each below $\varepsilon/3$. $\square$

Lemma 4.7: The Fejér kernel in this volume's normalization. For $N \geq 0$ define:

$$K_N(t) = \frac{1}{N+1} \left| \sum_{j=0}^N e^{ijt} \right|^2 = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1} \right) e^{ikt}.$$

Then $K_N \geq 0$, $\int_{\mathbb{T}} K_N d\mu = 1$, and for $0 < \delta \leq |t| \leq \pi$,

$$K_N(t) = \frac{\sin^2((N+1)t/2)}{(N+1)\sin^2(t/2)} \leq \frac{1}{(N+1)\sin^2(\delta/2)}.$$

Proof. Nonnegativity follows from the squared absolute value. In the expansion $\sum_{j,l=0}^N e^{i(j-l)t}$, the index $k = j - l$ occurs $N+1 - |k|$ times; dividing by $N+1$ gives the second expression. By the orthogonality already proved, all nonconstant exponentials integrate to zero, leaving the constant coefficient 1. If $t \notin 2\pi\mathbb{Z}$, the geometric sum is $(1 - e^{i(N+1)t})/(1 - e^{it})$; use $|1 - e^{ix}| = 2|\sin(x/2)|$. The numerator is at most 1, and $\sin(|t|/2) \geq \sin(\delta/2)$ on the stated interval. By continuity, $K_N(0) = N+1$. In this normalization the mass is integrated over one period, not over the whole real line. $\square$

Theorem 4.13: Fejér for continuous functions. For continuous f on $\mathbb{T}$, the polynomials

$$\sigma_N f(x) = \int_{\mathbb{T}} f(x-t) K_N(t) d\mu(t) = \sum_{|k| \leq N} \left(1 - \frac{|k|}{N+1} \right) \hat{f}(k) e^{ikx}$$

converge uniformly to f .

Proof. Step 1 — polynomial form. Insert the finite expansion of K_N into the integral. In term k set $u = x - t$; periodicity and change of variable for continuous functions give $\int f(x-t) e^{ikt} d\mu(t) = e^{ikx} \hat{f}(k)$. No infinite sum is being interchanged with integration.

Step 2 — uniform estimate. Given $\varepsilon > 0$, uniform continuity on $\mathbb{T}$ gives $\delta > 0$ with $|f(x-t) - f(x)| < \varepsilon/2$ for $|t| < \delta$, uniformly in x . Since the kernel is positive and has mass 1, splitting the integral into $|t| < \delta$ and its complement gives

$$|\sigma_N f(x) - f(x)| \leq \frac{\varepsilon}{2} + \frac{2\|f\|_{\infty}}{(N+1)\sin^2(\delta/2)}.$$

The second term is less than $\varepsilon/2$ for sufficiently large N , independently of x . This proves uniform convergence. The coefficient formula also shows $\sigma_N f = (N+1)^{-1} \sum_{j=0}^N S_j f$, because frequency k occurs in $N+1 - |k|$ partial sums. These are Cesàro means, not the partial sums $S_N f$ themselves. $\square$

Theorem 4.14: Trigonometric completeness and mean-square convergence. Trigonometric polynomials are dense in $L^2(\mathbb{T})$. For every $f \in L^2(\mathbb{T})$, the sums $S_N f = \sum_{|k| \leq N} \hat{f}(k) e^{ikx}$ converge to f in L^2 norm.

Proof. Step 1 — density. Choose continuous g with $\|f - g\|_2 < \varepsilon/2$ (Lemma 4.6). Fejér's theorem gives a polynomial $p = \sigma_M g$ with $\|g - p\|_\infty < \varepsilon/2$ (Theorem 4.13). Since $\mu(\mathbb{T}) = 1$, $\|g - p\|_2 \leq \|g - p\|_\infty$, hence $\|f - p\|_2 < \varepsilon$.

Step 2 — optimal projection. For $|k| \leq N$, orthogonality gives $\langle f - S_N f, e_k \rangle = 0$. If p has degree at most N , then $S_N f - p$ is a combination of these modes, so mixed terms vanish in the following expansion:

$$\|f - p\|_2^2 = \|f - S_N f\|_2^2 + \|S_N f - p\|_2^2 \geq \|f - S_N f\|_2^2.$$

Fix the polynomial of Step 1. For every $N \geq M$, $\|f - S_N f\|_2 \leq \|f - p\|_2 < \varepsilon$, proving convergence. In particular, a function orthogonal to every mode has $S_N f = 0$ for every N and therefore has norm zero: the system is complete. $\square$

Corollary 4.3: Existence for square-summable coefficients. For every family $(a_k)_{k \in \mathbb{Z}}$ with $\sum_k |a_k|^2 < \infty$, there is a unique class $f \in L^2(\mathbb{T})$ satisfying $\hat{f}(k) = a_k$ for every k .

Proof. The sums $p_N = \sum_{|k| \leq N} a_k e_k$ are Cauchy: for $M > N$, orthogonality gives $\|p_M - p_N\|_2^2 = \sum_{N < |k| \leq M} |a_k|^2 \rightarrow 0$ because the numerical series converges. Completeness of L^2 (Theorem 4.12) gives $f = \lim p_N$. For fixed k , $|\hat{f}(k) - \langle p_N, e_k \rangle| \leq \|f - p_N\|_2 \|e_k\|_2 \rightarrow 0$; for $N \geq |k|$ the second coefficient is a_k . Two solutions differ by a function with all coefficients zero, hence agree almost everywhere by the preceding theorem. $\square$

Remark 4.8: An import that is now unnecessary. The property "equal coefficients imply the same class in L^2 ", used in point (i) of the proof of Theorem 4.4 by importing it from Volume III, is actually a special case of this corollary (applied to the sequence $a_k = \hat{f}(k) = \hat{g}(k)$, to deduce $f = g$ almost everywhere from equal coefficients). The import was needed only because, at that point in the chapter, completeness of L^2 had not yet been proved: from here on the fact is self-contained, internal to this chapter, with no need for Volume III.

4.5.1 Two learning checks

Exercise 4.3: Partial sums and means are different. For $f(x) = 1 + 2 \cos x + 3 \sin(2x)$, compute $S_2 f$, $\sigma_2 f$ and $\|f - \sigma_2 f\|_2^2$.

Solution. The degree is 2, so $S_2 f = f$. Fejér weights at frequencies 0, 1, 2 are 1, 2/3, 1/3, giving $\sigma_2 f = 1 + (4/3) \cos x + \sin(2x)$. The error is $(2/3) \cos x + 2 \sin(2x)$. Its mixed product integrates to zero by orthogonality, while $\int \cos^2 x = \int \sin^2(2x) = 1/2$. The squared error norm is therefore $(4/9)/2 + 4/2 = 20/9$. Finite integrals suffice here, without Parseval.

Exercise 4.4: Mean-square convergence without pointwise convergence. On $[0, 1]$ define $u_{2^k+j} = \mathbf{1}_{[2^{-k}, (j+1)2^{-k})}$ for $k \geq 0$ and $0 \leq j < 2^k$. Prove $u_n \rightarrow 0$ in L^2 but failure of pointwise convergence at every point.

Solution. Every function in block k has squared norm 2^{-k} ; as $n \rightarrow \infty$, also $k \rightarrow \infty$, hence $\|u_n\|_2 \rightarrow 0$. For fixed x , exactly one half-open interval in each block contains x , giving a value 1; for $k \geq 1$ there are values 0 too. Two subsequences have different limits. Completeness of L^2 concerns the norm and does not establish pointwise convergence of the whole sequence.

Remark 4.9: Link to Volume III. The construction of measure, dominated convergence and completeness of L^2 are now available with proofs. Orthogonality and density have been distinguished. Volume III's Fourier-convergence chapter can therefore develop pointwise convergence, kernels and boundary phenomena without implicitly assuming these foundations. The numbers of the automation chapters of Volume III must instead be checked against its updated table of contents, not inferred from this volume.

4.6. Parseval's identity and the isoperimetric inequality: corrected proofs

Theorem 4.15: Parseval's identity on $\mathbb{T}$. For $f \in L^2(\mathbb{T})$:

$$\frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2.$$

Proof. Step 1 — finite identity. By definition $S_N f = \sum_{|n| \leq N} \hat{f}(n) e_n$, with $e_n(\theta) = e^{in\theta}$ and $\hat{f}(n) = \langle f, e_n \rangle$. Orthogonality (Theorem 4.2) gives $\|S_N f\|_2^2 = \sum_{|n| \leq N} |\hat{f}(n)|^2$. Also $\langle f - S_N f, e_n \rangle = 0$ for $|n| \leq N$, so $f - S_N f$ is orthogonal to $S_N f$. Expanding the square,

$$\|f\|_2^2 = \|f - S_N f\|_2^2 + \sum_{|n| \leq N} |\hat{f}(n)|^2.$$

Step 2 — passing to the limit. Theorem 4.14, proved by density and Fejér *without* using Parseval, ensures $\|f - S_N f\|_2 \rightarrow 0$. In the finite identity the first term therefore tends to zero. The nonnegative sums on the right tend, by definition, to the series over $\mathbb{Z}$, proving the formula. The required completeness was proved separately: it does not follow from orthogonality alone. $\square$

Example 4.2: Square wave and sawtooth wave. Expand in Fourier series the two 2π -periodic functions defined on $(-\pi, \pi)$ by

$$f(\theta) = \operatorname{sgn}(\theta) = \begin{cases} +1 & 0 < \theta < \pi \\ -1 & -\pi < \theta < 0 \end{cases} \quad (\text{square wave}), \quad g(\theta) = \theta \quad (\text{sawtooth}),$$

and use Parseval to derive two notable sums.

Step 1 — exploiting the symmetries. Both functions are *real* and *odd*. By Theorem 4.4(iii) all the a_n vanish and only sines remain: half the work is done with no computation at all.

Step 2 — coefficients of the square wave. By Theorem 4.3, using that $f \sin n\theta$ is even:

$$b_n = \frac{2}{\pi} \int_0^\pi \sin n\theta d\theta = \frac{2}{\pi} \left[\frac{-\cos n\theta}{n} \right]_0^\pi = \frac{2(1 - (-1)^n)}{n\pi},$$

using $\cos n\pi = (-1)^n$. Hence

$$b_n = \begin{cases} \frac{4}{n\pi} & n \text{ odd,} \\ 0 & n \text{ even,} \end{cases} \quad f(\theta) \sim \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)\theta)}{2k+1}.$$

Step 3 — coefficients of the sawtooth. Integrating by parts,

$$\int_0^\pi \theta \sin n\theta \, d\theta = \left[\frac{-\theta \cos n\theta}{n} \right]_0^\pi + \frac{1}{n} \int_0^\pi \cos n\theta \, d\theta = -\frac{\pi(-1)^n}{n},$$

the last term vanishing since $\sin n\pi = 0$. Hence

$$b_n = \frac{2}{\pi} \cdot \left(-\frac{\pi(-1)^n}{n} \right) = \frac{2(-1)^{n+1}}{n}, \quad g(\theta) \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin n\theta.$$

Step 4 — check with Parseval on the square wave. Since $|f| = 1$ almost everywhere, $\|f\|_2^2 = 1$. For a real odd function, $c_n = -ib_n/2$ and $c_{-n} = ib_n/2$, so $|c_n|^2 = |c_{-n}|^2 = b_n^2/4$ and

$$\sum_{n \in \mathbb{Z}} |c_n|^2 = \frac{1}{2} \sum_{n \geq 1} b_n^2.$$

By Theorem 4.15 this equals $\|f\|_2^2 = 1$:

$$\frac{8}{\pi^2} \sum_{k \geq 0} \frac{1}{(2k+1)^2} = 1 \implies \sum_{k \geq 0} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}.$$

Step 5 — check with Parseval on the sawtooth.

$$\|g\|_2^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 \, d\theta = \frac{\pi^2}{3},$$

while

$$\frac{1}{2} \sum_{n \geq 1} b_n^2 = 2 \sum_{n \geq 1} \frac{1}{n^2},$$

so $2 \sum_{n \geq 1} n^{-2} = \pi^2/3$, that is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Step 6 — consistency of the two results. Separating even from odd terms,

$$\sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{8} + \frac{1}{4} \sum_{m \geq 1} \frac{1}{m^2},$$

whence $\frac{3}{4} \sum n^{-2} = \pi^2/8$, i.e. $\sum n^{-2} = \pi^2/6$. ✓

Remark. Step 5 yields a proof of the *Basel problem* — the value of $\sum 1/n^2$, solved by Euler in 1735 — as a simple corollary of Parseval applied to the most elementary function imaginable. It is perhaps the cheapest route to that result, and a good example of how harmonic analysis makes accessible identities that are hard to obtain directly.

One of the most surprising applications of Parseval's identity is a completely analytic proof of the isoperimetric inequality, due to Adolf Hurwitz (1901): among all simple closed curves of given length L , the circle encloses the maximum area. The following proof uses exclusively tools already built in this treatise: Green's formula (Chapter 2) and Parseval.

Theorem 4.16: Isoperimetric inequality (Hurwitz, 1901). Let γ be a plane closed, simple, regular curve of length L , enclosing an area A . Then

$$L^2 \geq 4\pi A,$$

with equality if and only if γ is a circumference.

Proof. We reparametrize γ so that the parameter $\theta \in [0, 2\pi]$ is proportional to arc length: $s = \frac{L}{2\pi}\theta$. Since the arc-length parametrization satisfies $|dz/ds| = 1$ (Theorem 1.11, Chapter 1, with $z(\theta) = x(\theta) + iy(\theta)$ the position in complex notation), one has $|dz/d\theta| = |dz/ds| \cdot |ds/d\theta| = L/(2\pi)$, constant. Hence

$$\int_0^{2\pi} |z'(\theta)|^2 d\theta = 2\pi \left(\frac{L}{2\pi} \right)^2 = \frac{L^2}{2\pi}. \quad (i)$$

We write the Fourier expansion $z(\theta) = \sum_{n \in \mathbb{Z}} c_n e^{in\theta}$; then $z'(\theta) = \sum_{n \in \mathbb{Z}} (inc_n) e^{in\theta}$, and by Parseval (Theorem 4.15 applied to z' , whose n -th Fourier frequency is inc_n):

$$\int_0^{2\pi} |z'(\theta)|^2 d\theta = 2\pi \sum_{n \in \mathbb{Z}} n^2 |c_n|^2.$$

Comparing with (i): $\sum_{n \in \mathbb{Z}} n^2 |c_n|^2 = L^2/(4\pi^2)$. (*)

For the area, we use Green's formula in complex notation: $A = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \int_0^{2\pi} \text{Im}(\bar{z}(\theta) z'(\theta)) d\theta$ (one verifies that $\text{Im}(\bar{z}z') = xy' - yx'$ expanding $\bar{z} = x - iy$, $z' = x' + iy'$). Substituting the Fourier series and using orthogonality again (only terms with equal indices survive integration over $[0, 2\pi]$):

$$\int_0^{2\pi} \bar{z}(\theta) z'(\theta) d\theta = 2\pi i \sum_{n \in \mathbb{Z}} n |c_n|^2,$$

a purely imaginary number multiplied by i , hence with imaginary part $2\pi \sum_{n \in \mathbb{Z}} n |c_n|^2$. Therefore

$$A = \frac{1}{2} \cdot 2\pi \sum_{n \in \mathbb{Z}} n |c_n|^2 = \pi \sum_{n \in \mathbb{Z}} n |c_n|^2. \quad (**)$$

We now compare (*) and (**). For every integer n we have $n^2 \geq n$ (equivalent to $n(n-1) \geq 0$, true for every $n \in \mathbb{Z}$: for $n \leq 0$ it is obvious, for $n \geq 1$ it follows from $n-1 \geq 0$). Hence $n^2 |c_n|^2 \geq n |c_n|^2$ term by term, and summing:

$$\frac{L^2}{4\pi^2} = \sum_{n \in \mathbb{Z}} n^2 |c_n|^2 \geq \sum_{n \in \mathbb{Z}} n |c_n|^2 = \frac{A}{\pi}.$$

Multiplying by $4\pi^2$: $L^2 \geq 4\pi A$, as desired.

Equality case. Equality holds if and only if $n^2 |c_n|^2 = n |c_n|^2$ for every n , i.e. $n(n-1) |c_n|^2 = 0$ for every n : this forces $c_n = 0$ for every $n \notin \{0, 1\}$. Hence $z(\theta) = c_0 + c_1 e^{i\theta}$, the exponential parametrization (Corollary 3.2, Chapter 3) of a circle of center c_0 and radius $|c_1|$. $\square$

Remark 4.10: Why this proof is remarkable. Unlike classical proofs (variational, or based on Brunn-Minkowski inequalities), Hurwitz's proof requires no calculus of variations: it reduces entirely the geometric problem to a comparison between two sums of Fourier coefficients, made possible only by Parseval's identity. It is a paradigmatic example of how harmonic analysis translates geometric problems into algebraic problems on coefficients — exactly the guiding theme of this chapter.

4.7. The Gibbs phenomenon

INFO

Why this section. Theorem 4.14 guarantees that $S_N f \rightarrow f$ in L^2 norm even for discontinuous functions such as the square wave of Example 4.2. But convergence in quadratic mean says nothing about the *pointwise* behaviour of the partial sums near a jump. This section shows that, there, the partial sums behave in a way that surprised even Gibbs: they produce an *overshoot* that never shrinks, however large N becomes.

Recall the square wave of Example 4.2, $f(\theta) = \text{sgn}(\theta)$ on $(-\pi, \pi)$, with Fourier series $f(\theta) \sim \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin((2k+1)\theta)}{2k+1}$. Set

$$T_n(\theta) = \frac{4}{\pi} \sum_{k=0}^n \frac{\sin((2k+1)\theta)}{2k+1},$$

the partial sum through frequency $2n+1$ (the sums S_N with N odd coincide with T_n for $N = 2n+1$, since all even coefficients vanish).

Lemma 4.8: Closed sum of the odd cosine harmonics. For every $n \geq 0$ and $\theta \not\equiv 0 \pmod{\pi}$:

$$\sum_{k=0}^n \cos((2k+1)\theta) = \frac{\sin(2(n+1)\theta)}{2 \sin \theta}.$$

Proof. From the product-to-sum formula $2 \sin \theta \cos((2k+1)\theta) = \sin((2k+2)\theta) - \sin(2k\theta)$. Summing over $k = 0, \dots, n$ the sum telescopes:

$$2 \sin \theta \sum_{k=0}^n \cos((2k+1)\theta) = \sin(2(n+1)\theta) - \sin(0) = \sin(2(n+1)\theta).$$

Dividing by $2 \sin \theta \neq 0$ gives the statement. $\square$

Theorem 4.17: Gibbs phenomenon for the square wave. The first positive local maximum of T_n is at $\theta_n = \frac{\pi}{2(n+1)}$, and

$$T_n(\theta_n) \xrightarrow{n \rightarrow \infty} \frac{2}{\pi} \text{Si}(\pi) = 1.1789797445 \dots,$$

where $\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt$ is the **sine integral function**. Since $f(0^+) = 1$, the **overshoot** $T_n(\theta_n) - 1$ does not vanish: in the limit it equals ≈ 0.1789797 , about 8.949% of the total jump amplitude (which is $f(0^+) - f(0^-) = 2$).

Proof. Step 1 — derivative of T_n . By Lemma 4.8,

$$T'_n(\theta) = \frac{4}{\pi} \sum_{k=0}^n \cos((2k+1)\theta) = \frac{2}{\pi} \cdot \frac{\sin(2(n+1)\theta)}{\sin \theta}.$$

Step 2 — the first maximum. For $0 < \theta < \pi/(2(n+1))$ one has $0 < 2(n+1)\theta < \pi$ and $0 < \theta < \pi$, hence $\sin(2(n+1)\theta) > 0$ and $\sin \theta > 0$: $T'_n(\theta) > 0$ and T_n is strictly increasing on this interval. At $\theta_n = \pi/(2(n+1))$ one has $T'_n(\theta_n) = 0$, and right after $\sin(2(n+1)\theta)$ turns negative: θ_n is thus the first local maximum. Since $T_n(0) = 0$ (a series of pure sines evaluated

at 0),

$$T_n(\theta_n) = \int_0^{\theta_n} T_n'(t) dt = \frac{2}{\pi} \int_0^{\theta_n} \frac{\sin(2(n+1)t)}{\sin t} dt.$$

Step 3 — change of variable. Put $u = 2(n+1)t$, $t = u/(2(n+1))$, $dt = du/(2(n+1))$; as t runs from 0 to θ_n , u runs from 0 to π :

$$T_n(\theta_n) = \frac{2}{\pi} \int_0^\pi \frac{\sin u}{\sin(u/(2(n+1)))} \cdot \frac{du}{2(n+1)} = \frac{2}{\pi} \int_0^\pi \frac{\sin u}{u} \cdot g_n(u) du,$$

$$g_n(u) := \frac{u/(2(n+1))}{\sin(u/(2(n+1)))}.$$

Step 4 — pointwise limit and domination. As $x \rightarrow 0$, $x/\sin x \rightarrow 1$; hence $g_n(u) \rightarrow 1$ for every fixed $u \in (0, \pi]$ as $n \rightarrow \infty$. By Jordan's inequality $\sin x \geq \frac{2}{\pi}x$ on $[0, \pi/2]$ (concavity of $\sin$ on that interval: the graph lies above the chord joining $(0, 0)$ and $(\pi/2, 1)$), and since $u/(2(n+1)) \leq \pi/2$ for every $u \in [0, \pi]$ and $n \geq 0$, we get $g_n(u) \leq \pi/2$ for every n, u . Hence

$$\left| \frac{\sin u}{u} g_n(u) \right| \leq \frac{\pi}{2} \cdot \frac{|\sin u|}{u} \leq \frac{\pi}{2},$$

an integrable bound on $[0, \pi]$, independent of n . By the dominated convergence theorem (Theorem 4.9, applicable because $[0, \pi]$ has finite Lebesgue measure and the integrand is continuous, hence measurable):

$$T_n(\theta_n) \longrightarrow \frac{2}{\pi} \int_0^\pi \frac{\sin u}{u} du = \frac{2}{\pi} \text{Si}(\pi).$$

□

Proposition 4.1: Cesàro means produce no overshoot. Let $f \in L^1(\mathbb{T})$ with $m \leq f \leq M$ almost everywhere. Then, for every $N \geq 0$ and every θ , $m \leq \sigma_N f(\theta) \leq M$, where $\sigma_N f$ is the Cesàro mean (convolution with the Fejér kernel) of Theorem 4.13, defined by the same integral formula also for f merely integrable.

Proof. By Lemma 4.7, $K_N \geq 0$ and $\int_{\mathbb{T}} K_N d\mu = 1$. Then

$$M - \sigma_N f(\theta) = \int_{\mathbb{T}} (M - f(\theta - t)) K_N(t) d\mu(t) \geq 0,$$

since $M - f \geq 0$ almost everywhere and $K_N \geq 0$: the integral of a product of non-negative functions is non-negative. Likewise $\sigma_N f(\theta) - m \geq 0$ by integrating $f(\theta - t) - m \geq 0$ against K_N . □

Corollary 4.4: Application to the square wave: no overshoot for Fejér. For the square wave ($m = -1$, $M = 1$): $-1 \leq \sigma_N f(\theta) \leq 1$ for every N and every θ .

Proof. Immediate from Proposition 4.1 with $m = -1$, $M = 1$, since $-1 \leq f \leq 1$ everywhere for the square wave. □

Remark 4.11: A defect of the sums, not of the series. Unlike the partial sums $S_N = T_{\lfloor (N-1)/2 \rfloor}$, Cesàro means never exceed the amplitude of the function: the Gibbs phenomenon is a defect of the partial sums, not of the Fourier series itself, and it is exactly the kind of pathology that the Fejér mean, by construction, eliminates.

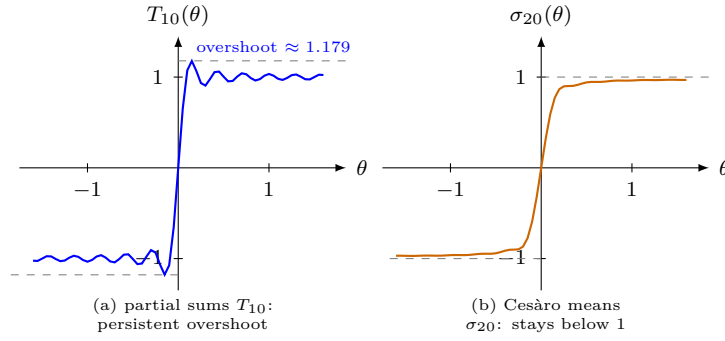


Figure 4.1: Comparison between the partial sums T_{10} (a), which stably exceed the amplitude 1 of the function near the jump (Theorem 4.17), and the Cesàro means σ_{20} (b), which never exceed it (Proposition 4.1) — same data as the script `vol2_cap04_gibbs.py`.

Remark 4.12: Why there is no contradiction with Riemann–Lebesgue. Corollary 4.1 guarantees $\hat{f}(n) \rightarrow 0$: why, then, does the overshoot not disappear? There is no contradiction, because Riemann–Lebesgue controls the amplitude of the individual modes, not the pointwise behaviour of the sum. What truly shrinks with n is the **location** of the maximum, $\theta_n = \pi/(2(n+1)) \rightarrow 0$: the overshoot peak moves ever closer to the jump, narrowing in width, while its height stays ≈ 1.179 . It is exactly this narrowing base (against a constant height) that makes the area of the excess “bump” ever smaller, consistent with the L^2 convergence of Theorem 4.14 (which requires $\int |f - S_N f|^2 \rightarrow 0$, not uniform convergence): a narrow, tall peak contributes little to the integral of the square if its width shrinks fast enough.

Remark 4.13: Historical notes and sources. The phenomenon was first observed by Henry Wilbraham, “On a certain periodic function”, Cambridge and Dublin Mathematical Journal 3 (1848), 198–201, but went unnoticed for about half a century. It was rediscovered by J. Willard Gibbs, who described it in two letters to Nature: vol. 59, p. 200 (1898) and — with the quantitatively correct account, the one reported above — p. 606 (1899). The rigorous mathematical treatment, and the name by which the phenomenon is known today, are due to Maxime Bôcher, “Introduction to the theory of Fourier’s series”, Annals of Mathematics (2nd series) 7 (1905–1906), 81–152.

Example 4.3: Convergence of the overshoot: numerical check. The script `Sorgenti/Vol2/_verifiche/vol2_cap04_gibbs.py` computes $T_n(\theta_n)$ for increasing n , using two independent methods: direct summation of the truncated series, and adaptive quadrature of $\frac{2}{\pi} \text{Si}(\pi)$ (independent of the formula in Theorem 4.17). The results:

n	$T_n(\theta_n)$
1	1.2004217549
10	1.1796690928
100	1.1789879137
1000	1.1789798276
20000	1.1789797447

against the limiting value $\frac{2}{\pi} \text{Si}(\pi) = 1.1789797445$ (adaptive quadrature, estimated error 2×10^{-14}): the difference for $n = 20000$ is 2.1×10^{-10} . The same script verifies the identity of Lemma 4.8 for $n = 3, 7, 15$ (agreement to 10^{-15} – 10^{-16}) and confirms that the maxima of

the Cesàro means σ_N stay below 1 and converge to it from below (0.891 for $N = 5$; 0.971 for $N = 20$; 0.994 for $N = 100$), consistent with Proposition 4.1.

Exercise 4.5: Computing $\text{Si}(\pi)$ and the overshoot at finite n . (a) Approximate $\text{Si}(\pi)$ by summing the first four terms of the power series $\text{Si}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)(2k+1)!}$ for $x = \pi$, and compare with the reference value $\text{Si}(\pi) = 1.851937052$. (b) Compute $T_{10}(\theta_{10})$ using the closed formula of Lemma 4.8 and compare it with the limit of Theorem 4.17.

Solution. (a) The first four terms ($k = 0, 1, 2, 3$):

$$\begin{aligned} \frac{\pi}{1} &= 3.14159265, & -\frac{\pi^3}{3 \cdot 3!} &= -1.72257093, \\ \frac{\pi^5}{5 \cdot 5!} &= 0.51003281, & -\frac{\pi^7}{7 \cdot 7!} &= -0.08560922. \end{aligned}$$

Partial sum: $3.14159265 - 1.72257093 + 0.51003281 - 0.08560922 = 1.84344531$. Compared with 1.851937052, the discrepancy is 8.5×10^{-3} (about 0.46%): the series converges, but a few more terms are needed for the precision reported in the theorem (with 6 terms one reaches 1.8519026, discrepancy 3.4×10^{-5}) — which is exactly why the script uses an independent adaptive quadrature for the high-precision comparison.

(b) With $n = 10$, $\theta_{10} = \pi/22 = 0.142799666$. Summing directly $T_{10}(\theta_{10}) = \frac{4}{\pi} \sum_{k=0}^{10} \frac{\sin((2k+1)\theta_{10})}{2k+1}$ (eleven terms) gives $T_{10}(\theta_{10}) = 1.1796690928$. Compared with the limit 1.1789797445, the discrepancy is 6.9×10^{-4} : already with eleven harmonics the overshoot has reached the limiting value to within an error of order 10^{-4} , confirming the rapid convergence predicted by Theorem 4.17.

4.8. Circular convolution and filtering

Definition 4.9: Circular convolution. For $f, g \in L^1(\mathbb{T})$ set

$$(f * g)(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta - t) g(t) dt.$$

Theorem 4.18: Convolution theorem. For $f, g \in L^1(\mathbb{T})$ the convolution $f * g$ is well defined almost everywhere, belongs to $L^1(\mathbb{T})$, and is commutative, associative and bilinear. Moreover

$$\widehat{f * g}(n) = \hat{f}(n) \hat{g}(n) \quad \text{for every } n \in \mathbb{Z}.$$

Proof. Step 1 — well-posedness. By Fubini–Tonelli applied to the non-negative function $(\theta, t) \mapsto |f(\theta - t)| |g(t)|$,

$$\frac{1}{2\pi} \int \left(\frac{1}{2\pi} \int |f(\theta - t)| |g(t)| dt \right) d\theta = \|f\|_1 \|g\|_1 < \infty,$$

the inner integral equalling $\|f\|_1$ by translation invariance of the measure on $\mathbb{T}$. Hence the defining integral converges absolutely for almost every θ and $\|f * g\|_1 \leq \|f\|_1 \|g\|_1$.

Step 2 — commutativity. With the substitution $s = \theta - t$, $(f * g)(\theta) = (g * f)(\theta)$.

Step 3 — the coefficient of the product. Applying the definition and exchanging the order of integration (legitimate by Step 1):

$$\widehat{f * g}(n) = \frac{1}{2\pi} \int g(t) \left(\frac{1}{2\pi} \int f(\theta - t) e^{-in\theta} d\theta \right) dt.$$

In the inner integral substitute $s = \theta - t$, so that $e^{-in\theta} = e^{-ins} e^{-int}$: the inner integral equals $\hat{f}(n) e^{-int}$. Hence

$$\widehat{f * g}(n) = \hat{f}(n) \cdot \frac{1}{2\pi} \int g(t) e^{-int} dt = \hat{f}(n) \hat{g}(n).$$

Step 4 — associativity and bilinearity. Bilinearity is immediate from linearity of the integral. Associativity follows from Step 3 and uniqueness (Corollary 4.3): $(f * g) * h$ and $f * (g * h)$ have the same coefficients $\hat{f}\hat{g}\hat{h}$. $\square$

***Remark 4.14: The convolution theorem is the foundation of filtering.** Theorem 4.18 states that convolving in the time domain is multiplying in the frequency domain. This is the statement on which the whole theory of linear filters rests: a linear time-invariant system acts on a periodic signal by convolving it with its own impulse response h , and its effect on the spectrum is multiplication by $\hat{h}(n)$, the frequency response. Designing a filter therefore means choosing the desired sequence $\hat{h}(n)$ — letting some frequencies through ($\hat{h}(n) \approx 1$) and suppressing others ($\hat{h}(n) \approx 0$) — and deriving h from it. That is exactly what is done in (Volume III, Chapter 21), and the reason why the three kernels of the previous section are, literally, three low-pass filters: D_N is the ideal low-pass (sharp cut at $|n| \leq N$), while F_N and P_r are low-pass filters with gradual transition.*

***Corollary 4.5: Parseval's identity in polarised form (Plancherel).** For $f, g \in L^2(\mathbb{T})$:*

$$\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f \bar{g} d\theta = \sum_{n \in \mathbb{Z}} \hat{f}(n) \overline{\hat{g}(n)}.$$

Proof. By Theorem 4.14 the partial sums $S_N f$ and $S_N g$ converge to f and g in L^2 ; since the inner product is continuous with respect to the norm (Cauchy–Schwarz), $\langle f, g \rangle = \lim_N \langle S_N f, S_N g \rangle$. By orthonormality (Theorem 4.2),

$$\langle S_N f, S_N g \rangle = \sum_{|n| \leq N} \hat{f}(n) \overline{\hat{g}(n)},$$

and letting $N \rightarrow \infty$ concludes. The case $g = f$ returns Parseval's identity of Theorem 4.15. $\square$

4.9. Where the theory continues

This chapter stops on the threshold of four developments. Each of them is taken up and completed, with full proofs, in Volume III.

- **Pointwise convergence of Fourier series.** Lebesgue measure, completeness of L^2 , the Fejér kernel, trigonometric completeness and Parseval's identity are now proved in full in this chapter (Sections 4.4 and 4.5); the Gibbs phenomenon is proved in full later in this same chapter (Section 4.7). Still to be developed is *pointwise* convergence in general (distinct from the L^2 -norm convergence proved here): Dirichlet–Jordan conditions and the Kolmogorov–Carleson panorama: Volume III, Chapter 17.
- **The circle as a group.** Compact topological group, invariant measure $d\theta/2\pi$, continuous characters, duality with $\mathbb{Z}$ and uniqueness of the Haar measure: Volume III, Chapter 13.

- **Partial differential equations.** The heat equation on $\mathbb{T}$, the Laplacian in polar coordinates, the Dirichlet problem in the disc and the circular membrane: Volume III, Chapters 16 and 18.
- **Transforms and signals.** Fourier transform (Volume III, Chapter 19), Laplace transform (Chapter 20), Z-transform and DTFT (Chapter 21), discrete transform (Chapter 22), FFT algorithm (Chapter 23), sampling and linear systems (Chapter 24), control and transmission lines (Chapter 25).

What remains in this chapter, because it is the minimum language needed to read all the rest: periodic functions, the space $L^2(\mathbb{T})$ with its characters, Fourier coefficients, the construction of Lebesgue measure with its convergence theorems, completeness of L^2 , Fejér's bridge to trigonometric completeness, Parseval's identity (now proved in full, no longer taken for granted) with the isoperimetric inequality, and circular convolution.

4.10. Numerical exercises

The three exercises that follow are computational in nature and concern the Fourier coefficients and their symmetries. They are worked out in full in Section 4.12; the *symbolic* exercises on the theoretical part (characters, symmetries, convolution) are in Section 4.11.

Warm-up 4.1. Is the function $f(t) = \cos(3t)$ even or odd? Which Fourier coefficients (a_n or b_n in the real form) do you expect to vanish, and why?

Exercise 4.6. Compute the Fourier coefficients of $f(\theta) = \theta$ on $(-\pi, \pi]$ (extended periodically) and verify, using Parseval, that $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$ (Basel problem).

Exercise 4.7. Prove that if f is a real even function, then $b_n = 0$ for every n in the real form of the series (only cosines survive).

Exercise 4.8. Verify directly, by integration by parts, that the Fourier coefficients $\hat{f}(n)$ of a function $f \in C^1(\mathbb{T})$ satisfy $\hat{f}'(n) = in\hat{f}(n)$, and deduce why smoother functions have coefficients that decay faster.

4.11. Symbolic exercises

These five exercises require proofs and concern the theoretical part of the chapter: characters, symmetries, convolution and Wirtinger's inequality. The solutions are in Section 4.13.

Exercise 4.9: S1 — Translation, modulation and reflection. Let $f \in L^1(\mathbb{T})$, $\alpha \in \mathbb{R}$ and $m \in \mathbb{Z}$. Prove that, writing $f_\alpha(\theta) = f(\theta - \alpha)$ (translate), $g(\theta) = e^{im\theta}f(\theta)$ (modulate) and $\check{f}(\theta) = f(-\theta)$ (reflection),

$$\widehat{f_\alpha}(n) = e^{-in\alpha}\hat{f}(n), \quad \hat{g}(n) = \hat{f}(n - m), \quad \widehat{\check{f}}(n) = \hat{f}(-n).$$

Interpret the first as “translating in time does not change the amplitude spectrum, only the phase spectrum”.

Exercise 4.10: S2 — Differentiation and antiderivative in the spectral domain. Let f be continuous, 2π -periodic and piecewise C^1 . Prove that $\hat{f}'(n) = in\hat{f}(n)$. Deduce

that f admits a 2π -periodic antiderivative if and only if $\hat{f}(0) = 0$, and that in that case the coefficients of the antiderivative are $\hat{f}(n)/(in)$ for $n \neq 0$.

Exercise 4.11: S3 — A non-continuous character does not exist. Prove that every measurable group homomorphism $\chi : \mathbb{T} \rightarrow \mathbb{T}$ not identically 1 is automatically continuous, and deduce that the classification of Theorem 4.1 holds even without assuming continuity. (Hint: adapt Step 3 of that proof, which uses only local integrability.)

Exercise 4.12: S4 — Convolution and smoothness. Prove that if $f \in L^1(\mathbb{T})$ and $g \in C^k(\mathbb{T})$, then $f * g \in C^k(\mathbb{T})$ and $(f * g)^{(j)} = f * g^{(j)}$ for $j \leq k$. Interpret the result by saying that convolution “cannot worsen” smoothness, and connect it with the fact that the Fejér and Poisson kernels are C^∞ .

Exercise 4.13: S5 — Weighted Parseval and Wirtinger’s inequality. Let f be continuous, 2π -periodic, piecewise C^1 , with zero mean ($\hat{f}(0) = 0$). Prove **Wirtinger’s inequality**

$$\int_{-\pi}^{\pi} |f(\theta)|^2 d\theta \leq \int_{-\pi}^{\pi} |f'(\theta)|^2 d\theta,$$

with equality if and only if $f(\theta) = c_1 e^{i\theta} + c_{-1} e^{-i\theta}$. Observe that this is exactly the key step in Hurwitz’s proof of the isoperimetric inequality (Theorem 4.16).

4.12. Solutions to the numerical exercises

Solution Warm-up 4.1. $\cos(3(-t)) = \cos(-3t) = \cos(3t)$: the function is even. For a real even function, the b_n coefficients (those of the sines, odd) all vanish, and only the a_n coefficients of the cosines survive (here $a_3 \neq 0$, all others zero).

Solution. For $n = 0$: $\hat{f}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta d\theta = 0$ (odd function on a symmetric interval). For $n \neq 0$, integrating by parts with $u = \theta$, $dv = e^{-in\theta} d\theta$, $v = e^{-in\theta}/(-in)$:

$$\int_{-\pi}^{\pi} \theta e^{-in\theta} d\theta = \left[\frac{\theta e^{-in\theta}}{-in} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{e^{-in\theta}}{-in} d\theta.$$

The second integral is zero (complete period of a non-constant complex exponential, Theorem 4.2 with $m = 0$). The boundary term:

$$\frac{\pi e^{-in\pi} - (-\pi) e^{in\pi}}{-in} = \frac{\pi(e^{-in\pi} + e^{in\pi})}{-in} = \frac{2\pi \cos(n\pi)}{-in} = \frac{2\pi(-1)^n}{-in} = \frac{2\pi i(-1)^n}{n}$$

(rationalizing $1/(-in) = i/n$). Hence $\hat{f}(n) = \frac{1}{2\pi} \cdot \frac{2\pi i(-1)^n}{n} = \frac{i(-1)^n}{n}$ for $n \neq 0$, and $|\hat{f}(n)|^2 = 1/n^2$.

Applying Parseval:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \theta^2 d\theta = \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2.$$

Left side: $\frac{1}{2\pi} \left[\frac{\theta^3}{3} \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \cdot \frac{2\pi^3}{3} = \frac{\pi^2}{3}$. Right side: $\sum_{n \neq 0} \frac{1}{n^2} = 2 \sum_{n=1}^{\infty} \frac{1}{n^2}$ (the terms n and $-n$ contribute equally). Equating:

$$\frac{\pi^2}{3} = 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

Euler's celebrated result on the Basel problem, here obtained as an elementary consequence of Parseval.

Solution. If f is even, $f(-\theta) = f(\theta)$. We compute $\hat{f}(-n)$ with the substitution $\theta \mapsto -\theta$:

$$\hat{f}(-n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{in\theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(-\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(\theta) e^{-in\theta} d\theta = \hat{f}(n),$$

having used the parity of f and then renamed the integration variable. Hence $\hat{f}(n) = \hat{f}(-n)$ for every n . By Theorem 4.3, $b_n = i(\hat{f}(n) - \hat{f}(-n)) = i \cdot 0 = 0$: in the real form only cosines survive, as expected for an even function (sine is odd, it cannot appear in the expansion of an even function).

Solution. Integrating by parts with $u = e^{-in\theta}$, $dv = f'(\theta)d\theta$, $v = f(\theta)$:

$$\int_0^{2\pi} f'(\theta) e^{-in\theta} d\theta = \left[f(\theta) e^{-in\theta} \right]_0^{2\pi} + in \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

The boundary term vanishes: $f(2\pi)e^{-in2\pi} - f(0)e^0 = f(2\pi) - f(0) = 0$ by periodicity of f (and $e^{-in2\pi} = 1$). Dividing by 2π :

$$\widehat{f'}(n) = in\hat{f}(n).$$

Applying this relation k times to a function $f \in C^k(\mathbb{T})$ one obtains $\widehat{f^{(k)}}(n) = (in)^k \hat{f}(n)$. Since $f^{(k)} \in L^2(\mathbb{T})$, by Parseval $\sum_n |\widehat{f^{(k)}}(n)|^2 = \sum_n n^{2k} |\hat{f}(n)|^2 < \infty$: the sum converges, so $n^{2k} |\hat{f}(n)|^2 \rightarrow 0$, i.e. $\hat{f}(n)$ decays faster than n^{-k} . In summary: the more continuous derivatives f has, the faster its Fourier coefficients decay — regularity in the physical domain translates to decay in the frequency domain.

4.13. Solutions to the symbolic exercises

Solution. S1. *Translation.* By definition, with the substitution $s = \theta - \alpha$, legitimate because the integrand is 2π -periodic:

$$\widehat{f_\alpha}(n) = \frac{1}{2\pi} \int f(\theta - \alpha) e^{-in\theta} d\theta = \frac{1}{2\pi} \int f(s) e^{-in(s+\alpha)} ds = e^{-in\alpha} \hat{f}(n).$$

Modulation.

$$\hat{g}(n) = \frac{1}{2\pi} \int e^{im\theta} f(\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \int f(\theta) e^{-i(n-m)\theta} d\theta = \hat{f}(n-m).$$

Reflection. With $u = -\theta$:

$$\widehat{\tilde{f}}(n) = \frac{1}{2\pi} \int f(-\theta) e^{-in\theta} d\theta = \frac{1}{2\pi} \int f(u) e^{inu} du = \hat{f}(-n).$$

Interpretation. Since $|e^{-in\alpha}| = 1$, translation leaves the amplitude spectrum $|\hat{f}(n)|$ unchanged and adds to the phase spectrum the term $-n\alpha$, which is *linear* in n . This is why in signal processing a pure delay is recognised by a linear phase, and why the amplitude spectrum says nothing about *when* an event happens — that information lies entirely in the phase.

Solution. S2. *Step 1 — the differentiation formula.* Integrating by parts over a period, with f continuous and 2π -periodic so that the boundary term vanishes:

$$\hat{f}'(n) = \frac{1}{2\pi} \int_0^{2\pi} f'(\theta) e^{-in\theta} d\theta = \underbrace{\frac{1}{2\pi} [f(\theta) e^{-in\theta}]_0^{2\pi}}_{=0} + \frac{in}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta = in \hat{f}(n).$$

(The integration by parts is legitimate on each smoothness piece; the contributions at the junction points cancel because f is continuous.)

Step 2 — existence of a periodic antiderivative. Put $F(\theta) = \int_0^\theta f(s) ds$. Then $F' = f$ and

$$F(\theta + 2\pi) - F(\theta) = \int_\theta^{\theta+2\pi} f(s) ds = \int_0^{2\pi} f(s) ds = 2\pi \hat{f}(0),$$

the second equality holding because the integral of a periodic function over a period does not depend on the starting endpoint. Hence F is 2π -periodic if and only if $\hat{f}(0) = 0$; every other antiderivative differs from F by a constant, which does not affect periodicity.

Step 3 — coefficients of the antiderivative. If $\hat{f}(0) = 0$, applying Step 1 to F (continuous, periodic, piecewise C^1) gives $\hat{f}(n) = \widehat{F'}(n) = in \hat{F}(n)$, hence

$$\hat{F}(n) = \frac{\hat{f}(n)}{in} \quad (n \neq 0),$$

while $\hat{F}(0)$ remains free: it corresponds to the constant of integration.

Remark. Step 1 explains why differentiation “amplifies high frequencies” (multiplying by $|n|$) and integration damps them (dividing by $|n|$): this is why differentiating a noisy signal is an ill-conditioned operation while integrating it is stabilising — the same phenomenon already observed numerically in Chapter 1 (Observation 1.16).

Solution. S3. *Step 1 — reduction.* As in Step 2 of the proof of Theorem 4.1, put $\varphi(t) = \chi(e^{it})$: it is measurable, $|\varphi| = 1$ (hence bounded and locally integrable), $\varphi(0) = 1$, $\varphi(s+t) = \varphi(s)\varphi(t)$ and φ is 2π -periodic.

Step 2 — there is a δ with non-zero integral. Suppose, for contradiction, that $\int_0^\delta \varphi(t) dt = 0$ for every $\delta > 0$. Then $\Phi(\delta) = \int_0^\delta \varphi$ is identically zero, and by Lebesgue’s differentiation theorem $\varphi(t) = \Phi'(t) = 0$ for almost every t — contradicting $|\varphi| = 1$. Hence there is $\delta > 0$ with $c = \int_0^\delta \varphi \neq 0$.

Step 3 — automatic continuity. Exactly as in Step 3 of that proof, from the homomorphism property,

$$c\varphi(s) = \int_0^\delta \varphi(s+t) dt = \int_s^{s+\delta} \varphi(u) du.$$

The right-hand side is a *continuous* function of s : indeed, for $\varphi \in L^1_{\text{loc}}$,

$$\left| \int_{s'}^{s'+\delta} \varphi - \int_s^{s+\delta} \varphi \right| \leq \int_{\min}^{\max} |\varphi| \leq |s' - s|,$$

using $|\varphi| = 1$. Hence $\varphi = \frac{1}{c} \int_s^{s+\delta} \varphi$ is continuous.

Step 4 — conclusion. Once continuity is established, Steps 3–6 of the proof of Theorem 4.1 apply unchanged and give $\chi = \chi_n$ for some $n \in \mathbb{Z}$.

Remark. The result is an instance of *automatic regularity*: for homomorphisms of topological groups, very weak hypotheses (measurability) imply very strong ones (continuity, and here even analyticity). Without *any* regularity hypothesis the statement is false: there exist non-measurable, pathological homomorphisms $\mathbb{R} \rightarrow \mathbb{R}$, whose construction requires the axiom of choice.

Solution. S4. *Step 1 — choosing the form of the convolution.* By commutativity (Theorem 4.18, Step 2) write

$$(f * g)(\theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) g(\theta - t) dt,$$

so that the dependence on θ sits entirely in g , which is smooth.

Step 2 — differentiating under the integral sign. Consider the difference quotient:

$$\frac{(f * g)(\theta + h) - (f * g)(\theta)}{h} = \frac{1}{2\pi} \int f(t) \frac{g(\theta + h - t) - g(\theta - t)}{h} dt.$$

By the mean value theorem applied to g , the inner difference quotient equals $g'(\xi)$ for a suitable ξ , hence is bounded in modulus by $\|g'\|_{\infty}$, finite because $g \in C^1$ and $\mathbb{T}$ is compact. The dominating function $|f(t)| \cdot \|g'\|_{\infty}$ is integrable, and the difference quotient converges pointwise to $g'(\theta - t)$ as $h \rightarrow 0$: by dominated convergence one may pass to the limit under the integral sign, obtaining

$$(f * g)'(\theta) = \frac{1}{2\pi} \int f(t) g'(\theta - t) dt = (f * g')(\theta).$$

Step 3 — iteration and continuity. Applying Step 2 k times gives $(f * g)^{(j)} = f * g^{(j)}$ for $j \leq k$. Each of these is continuous: indeed, since $g^{(j)}$ is uniformly continuous on $\mathbb{T}$,

$$|(f * g^{(j)})'(\theta') - (f * g^{(j)})'(\theta)| \leq \|f\|_1 \cdot \sup_t |g^{(j)}(\theta' - t) - g^{(j)}(\theta - t)| \rightarrow 0$$

as $\theta' \rightarrow \theta$. Hence $f * g \in C^k(\mathbb{T})$.

Step 4 — interpretation. Convolution inherits the smoothness of the *smoother* of the two factors: convolving cannot worsen smoothness, and typically improves it. Since the Fejér and Poisson kernels are trigonometric polynomials (the former) or analytic functions (the latter), the means $\sigma_N f$ and $f * P_r$ are C^{∞} even when f is merely integrable: this is precisely the *regularisation* mechanism that makes those means so much better behaved than the partial sums.

Solution. S5. *Step 1 — Parseval for f and for f' .* By Theorem 4.15 (Parseval's identity) and the hypothesis $\hat{f}(0) = 0$:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f|^2 d\theta = \sum_{n \neq 0} |\hat{f}(n)|^2.$$

By Exercise S2, $\hat{f}'(n) = in\hat{f}(n)$, hence $|\hat{f}'(n)|^2 = n^2|\hat{f}(n)|^2$ and

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f'|^2 d\theta = \sum_{n \neq 0} n^2 |\hat{f}(n)|^2.$$

Step 2 — termwise comparison. For every non-zero integer n we have $n^2 \geq 1$, hence

$$|\hat{f}(n)|^2 \leq n^2 |\hat{f}(n)|^2,$$

and summing over $n \neq 0$:

$$\sum_{n \neq 0} |\hat{f}(n)|^2 \leq \sum_{n \neq 0} n^2 |\hat{f}(n)|^2.$$

Multiplying by 2π gives Wirtinger's inequality.

Step 3 — equality case. Equality holds if and only if $(n^2 - 1)|\hat{f}(n)|^2 = 0$ for every $n \neq 0$, i.e. $\hat{f}(n) = 0$ for $|n| \geq 2$. With $\hat{f}(0) = 0$, only the modes $n = \pm 1$ remain:

$$f(\theta) = c_1 e^{i\theta} + c_{-1} e^{-i\theta}.$$

Step 4 — connection with Hurwitz. In the proof of Theorem 4.16 one compares $\sum n^2 |c_n|^2$ (linked to the length) and $\sum n |c_n|^2$ (linked to the area), using $n^2 \geq n$. Here one compares $\sum n^2 |c_n|^2$ and $\sum |c_n|^2$ using $n^2 \geq 1$. The two inequalities are the same idea — comparing two symmetric functions of the coefficients termwise — and in both cases the equality case isolates only the modes $n \in \{0, \pm 1\}$, i.e. the circle.

4.14. Self-assessment quiz

One multiple-choice question per main section of the chapter, designed for a quick self-check (5 minutes) before moving on to the exercises. The answer key is at the end.

- The measure adopted on $\mathbb{T}$ in this chapter is
A) $d\theta$ B) $d\theta/2\pi$ (normalised) C) $2\pi d\theta$ D) it depends on the section
- The characters of the group $\mathbb{T}$ are
A) all continuous maps $\mathbb{T} \rightarrow \mathbb{T}$
B) exactly the maps $z \mapsto z^n$, $n \in \mathbb{Z}$
C) only $z \mapsto z$ and $z \mapsto \bar{z}$
D) infinitely many and not classifiable
- If f is real and *odd*, then in the trigonometric form
A) all the b_n vanish B) all the a_n vanish, including a_0 C) nothing vanishes D) only a_0 vanishes
- Bessel's inequality states that
A) $\sum_n |\hat{f}(n)|^2 \leq \|f\|_2^2$ B) $\sum_n |\hat{f}(n)|^2 \geq \|f\|_2^2$ C) $\sum_n |\hat{f}(n)| \leq \|f\|_2$ D) $\hat{f}(n) \rightarrow 0$
- Among all trigonometric polynomials of degree $\leq N$, the one minimising $\|f - P\|_2$ is
A) the Fejér mean $\sigma_N f$ B) the partial sum $S_N f$ C) the polynomial interpolating at the nodes D) it depends on f
- The convolution theorem states that
A) $\widehat{f * g} = \hat{f} + \hat{g}$ B) $\widehat{f * g} = \hat{f} \cdot \hat{g}$ C) $\widehat{fg} = \hat{f} * \hat{g}$ only D) $\widehat{f * g} = \hat{f} / \hat{g}$

BEST PRACTICE

Answer key.

1-B: conventions card: $\mu(\mathbb{T}) = 1$ makes the e_n orthonormal. **2-B:** Theorem 4.1; A confuses homomorphisms with arbitrary functions. **3-B:** Theorem 4.4(iii); A holds for *even* functions. **4-A:** Theorem 4.5(iv); it becomes an equality (Parseval) by Theorem 4.15. **5-B:** Theorem 4.5(ii): the partial sum is the orthogonal projection. **6-B:** Theorem 4.18: convolving in time is multiplying in frequency.

4.15. Chapter formulary

Conventions $(d\mu = d\theta/2\pi, e_n(\theta) = e^{in\theta})$

$$\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f \bar{g} d\theta, \quad \hat{f}(n) = \langle f, e_n \rangle, \quad f \sim \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta}$$

Real and complex forms

$$a_n = \hat{f}(n) + \hat{f}(-n), \quad b_n = i(\hat{f}(n) - \hat{f}(-n)), \quad \hat{f}(\pm n) = \frac{a_n \mp ib_n}{2}$$

$$f \sim \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos n\theta + b_n \sin n\theta), \quad A_n = \sqrt{a_n^2 + b_n^2} = 2|\hat{f}(n)|$$

Symmetries $(f \text{ real} \Rightarrow \hat{f}(-n) = \overline{\hat{f}(n)})$

$$f \text{ even} \Rightarrow b_n = 0, \quad f \text{ odd} \Rightarrow a_n = 0$$

Orthogonality, Bessel, Parseval

$$\langle e_n, e_m \rangle = \delta_{nm}, \quad \sum_{|n| \leq N} |\hat{f}(n)|^2 \leq \|f\|_2^2, \quad \sum_{n \in \mathbb{Z}} |\hat{f}(n)|^2 = \|f\|_2^2$$

$$\|f - S_N f\|_2^2 = \|f\|_2^2 - \sum_{|n| \leq N} |\hat{f}(n)|^2, \quad \langle f, g \rangle = \sum_n \hat{f}(n) \overline{\hat{g}(n)}$$

Operations and coefficients

$$\widehat{f(\cdot - \alpha)}(n) = e^{-in\alpha} \hat{f}(n), \quad \widehat{e^{im\theta} f}(n) = \hat{f}(n - m), \quad \widehat{f'}(n) = in \hat{f}(n)$$

$$(f * g)(\theta) = \frac{1}{2\pi} \int f(\theta - t) g(t) dt, \quad \widehat{f * g}(n) = \hat{f}(n) \hat{g}(n)$$

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Volume III, Chapter 24) and its applications.

CHAPTER 5

AC circuit theory: the circle and the phasor

Chapter 3 introduced the phasor as the complex number attached to a sinusoidal oscillation (Section 3.10); Chapter 4 showed that every periodic signal is a sum of such oscillations. This chapter gathers the two threads and takes them where they were born: *electrical engineering*. There is a single idea, and it is worth stating at once in its barest form.

INFO

The idea in one line. An alternating current is the *projection* of a point travelling around a circle at constant angular velocity. The point is the phasor; the circle is the locus of its possible values; the sinusoid one reads on the oscilloscope is the shadow of the point on the vertical axis. Everything that follows — impedances, power, power-factor correction, resonance, three-phase systems — is an algebraic consequence of this single picture.

The practical gain is enormous and worth quantifying. An RLC circuit in the sinusoidal steady state is described by an integro-differential equation; passing to phasors turns it into an *algebraic linear equation with complex coefficients*, that is, an object solved by the same arithmetic as Chapter 3. Differentiation becomes multiplication by $i\omega$ (Theorem 3.23(ii)), integration division by $i\omega$, and Kirchhoff's laws remain formally unchanged. It was Charles Proteus Steinmetz who, in 1893, systematically proposed this method to the community of electrical engineers; his memoir is cited in the bibliography.

Warning. Conventions of the chapter, stated once and for all.

- **Imaginary unit.** In this chapter, as in the rest of the treatise, we use i . Electrical engineering almost always writes j , to avoid confusion with instantaneous current: the formulas are identical, one simply substitutes.
- **Reference.** Cosine: $v(t) = V_m \cos(\omega t + \phi)$, as in Definition 3.8.
- **Phasor magnitude.** *RMS value*, not peak value: $\underline{V} = V e^{i\phi}$ with $V = V_m/\sqrt{2}$. This is the dominant convention in power engineering and the one that makes $S = \underline{V} \underline{I}^*$ the complex power with no factor of 1/2. *Note that it differs from the peak-amplitude convention of Chapter 3:* the results are the same, the numerical factors are not.
- **Sign convention.** Passive (load) sign convention: current enters the terminal at the higher potential, and absorbed power is positive.
- **Frequency.** $\omega = 2\pi f$ in rad/s; $f = 50$ Hz unless otherwise stated.

Comparing results obtained under different conventions without converting them is, in this field, the most frequent and the hardest error to spot.

INFO

Prerequisites. Chapter 1 (uniform circular motion, Theorem 1.2); Chapter 3 (complex numbers, polar form, phasors, roots of unity, Möbius transformations); Chapter 4 (only for references to transforms). No prior knowledge of electrical engineering is assumed: the physical quantities are defined where they are needed.

5.1. From circular motion to the sinusoid: the phasor

Definition 5.1: Sinusoidal quantity. A **sinusoidal quantity** is a function of time of the form

$$a(t) = A_m \cos(\omega t + \alpha), \quad A_m > 0, \omega > 0, \alpha \in (-\pi, \pi],$$

where A_m is the *amplitude* (or peak value), ω the *angular frequency*, α the *initial phase* and $\omega t + \alpha$ the *instantaneous phase*. The period is $T = 2\pi/\omega$ and the frequency $f = 1/T = \omega/2\pi$.

Theorem 5.1: The sinusoid is the projection of uniform circular motion. Let $P(t)$ be the point travelling around the circle $C(0, A_m)$ with uniform circular motion of angular velocity ω and initial position of argument α , that is,

$$P(t) = A_m e^{i(\omega t + \alpha)} \in \mathbb{C}.$$

Then:

- (i) $\operatorname{Re} P(t) = A_m \cos(\omega t + \alpha)$ and $\operatorname{Im} P(t) = A_m \sin(\omega t + \alpha)$: the two Cartesian projections of P are sinusoidal quantities of equal amplitude and angular frequency, $\pi/2$ apart in phase;
- (ii) $P(t) = \underline{A}_m e^{i\omega t}$ with $\underline{A}_m = A_m e^{i\alpha}$ constant in time: all the information about the sinusoidal quantity is contained in the complex number $\underline{A}_m$, once ω is fixed;
- (iii) the velocity $\dot{P}(t)$ has modulus ωA_m and is orthogonal to $P(t)$, rotated by $+\pi/2$;
- (iv) two sinusoidal quantities of equal ω keep a constant phase difference in time, equal to the difference of the arguments of the corresponding points.

Proof. (i) By Euler's formula (Theorem 3.5), $e^{i(\omega t + \alpha)} = \cos(\omega t + \alpha) + i \sin(\omega t + \alpha)$; multiplying by $A_m > 0$ and separating real and imaginary parts gives the claim. The phase difference between the two is $\pi/2$ because $\sin \theta = \cos(\theta - \pi/2)$.

(ii) By Theorem 3.15 (property of the exponential), $e^{i(\omega t + \alpha)} = e^{i\alpha} e^{i\omega t}$, hence $P(t) = (A_m e^{i\alpha}) e^{i\omega t}$. The factor $\underline{A}_m = A_m e^{i\alpha}$ does not depend on t , while $e^{i\omega t}$ is the same for all quantities of angular frequency ω : it is therefore a common factor, which may be left understood.

(iii) Differentiating,

$$\dot{P}(t) = i\omega A_m e^{i(\omega t + \alpha)} = i\omega P(t).$$

By Theorem 3.1(iii), $|\dot{P}| = |i||\omega||P| = \omega A_m$; and by Corollary 3.2 multiplication by $i = e^{i\pi/2}$ is the rotation by $+\pi/2$, which sends a vector to an orthogonal one. This recovers, in complex language, Theorem 1.2(ii) of Chapter 1.

(iv) Let $P_1(t) = \underline{A}_1 e^{i\omega t}$ and $P_2(t) = \underline{A}_2 e^{i\omega t}$. Their ratio

$$\frac{P_1(t)}{P_2(t)} = \frac{\underline{A}_1}{\underline{A}_2}$$

is constant in time; by Theorem 3.4 its argument is $\arg \underline{A}_1 - \arg \underline{A}_2$, which is therefore the phase difference, independent of t . $\square$

Remark 5.1: Why the circle and not an ellipse. Figure 5.1 is a circle, and not some arbitrary curve, for a precise reason: the motion is uniform, i.e. $|\dot{P}|$ is constant and so is $|P|$. If the amplitude varied in time — as in a transient, or in a modulated signal — the point would describe a spiral, and the notion of phasor would have to be extended (this is what the Laplace transform of Chapter 4 does, replacing $i\omega$ by $s = \sigma + i\omega$: a non-zero

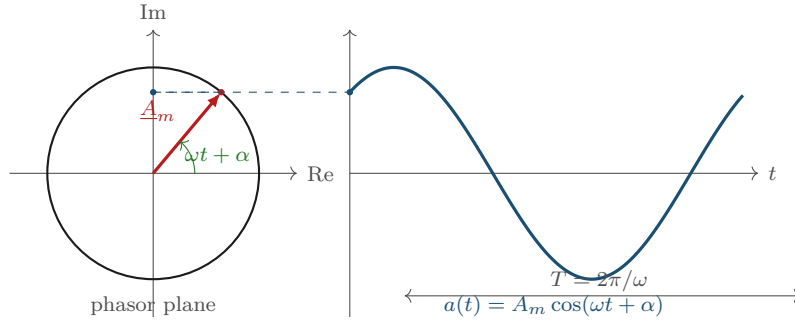


Figure 5.1: The phasor and its shadow. On the left, the point $\underline{A}_m e^{i\omega t}$ travelling around the circle of radius A_m with angular velocity ω ; on the right, the sinusoid generated by its projection, read as a function of time. The two figures are the *same* object: the circle is its instantaneous photograph, the sinusoid its recording (Theorem 5.1).

real part means a growing or decaying radius). The sinusoidal steady state is exactly the hypothesis that makes the circle, and not the spiral, the right figure.

Definition 5.2: RMS value. The **RMS value** (root mean square, also *effective value*) of a periodic quantity $a(t)$ of period T is

$$A = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt}.$$

Theorem 5.2: RMS value of a sinusoid and its energetic meaning. For $a(t) = A_m \cos(\omega t + \alpha)$:

- (i) $A = \frac{A_m}{\sqrt{2}}$;
- (ii) a sinusoidal current of RMS value I dissipates in a resistor R the same average power as a direct current of intensity I ;
- (iii) the mean value of a over a period is zero, whereas the RMS value vanishes only if $A_m = 0$.

Proof. (i) Using the half-angle formula $\cos^2 u = \frac{1}{2}(1 + \cos 2u)$ (Volume I, Chapter 8), with $u = \omega t + \alpha$:

$$\frac{1}{T} \int_0^T A_m^2 \cos^2(\omega t + \alpha) dt = \frac{A_m^2}{2T} \int_0^T (1 + \cos(2\omega t + 2\alpha)) dt.$$

The first term gives $A_m^2/2$. The second vanishes: an antiderivative of $\cos(2\omega t + 2\alpha)$ is $\sin(2\omega t + 2\alpha)/(2\omega)$, which takes the same value at 0 and at $T = 2\pi/\omega$, since $2\omega T = 4\pi$ and the sine has period 2π . Hence $A^2 = A_m^2/2$ and, taking the positive root, $A = A_m/\sqrt{2}$.

(ii) The instantaneous power dissipated by R carrying $i(t)$ is $p(t) = Ri^2(t)$ (Joule's law). Its average over a period is

$$P = \frac{1}{T} \int_0^T Ri^2(t) dt = R \cdot \frac{1}{T} \int_0^T i^2(t) dt = RI^2,$$

by Definition 5.2. In DC with current I the power is RI^2 : identical. It is precisely this property that justifies the name “effective” and makes the RMS value, not the peak value, the quantity read by instruments.

(iii) The mean of $\cos(\omega t + \alpha)$ over a period is zero by the same periodicity argument used

in (i). The RMS value is instead the root of the mean of a square: it vanishes only if $a \equiv 0$ almost everywhere, i.e. if $A_m = 0$. $\square$

Example 5.1: The 230 V mains: what it really means. The nominal voltage of the European low-voltage distribution network is 230 V at 50 Hz (IEC 60038). What exactly does this number identify?

Step 1 — it is an RMS value. 230 V is the RMS value V , not the amplitude. By Theorem 5.2(i),

$$V_m = V\sqrt{2} = 230 \times 1.4142136 = 325.27 \text{ V}.$$

The instantaneous voltage therefore swings between -325.27 V and $+325.27$ V.

Step 2 — the period.

$$T = \frac{1}{f} = \frac{1}{50} = 20 \text{ ms}, \quad \omega = 2\pi \cdot 50 = 314.159 \text{ rad/s}.$$

The phasor therefore completes fifty full turns per second: Figure 5.1, taken literally, is a point going round the circle every 20 ms.

Step 3 — energetic check. A 100 W lamp on 230 V draws an RMS current $I = P/V = 100/230 = 0.4348$ A. The peak current is $0.4348 \times \sqrt{2} = 0.6149$ A, and the *instantaneous* power swings between 0 and $V_m I_m = 325.27 \times 0.6149 = 200$ W, with mean 100 W. $\checkmark$ The peak instantaneous power is therefore *twice* the average one: a fact that Theorem 5.10 will make general.

Typical error. Sizing the insulation of a component for 230 V instead of 325 V. The voltage the insulation must withstand is the peak one (in fact more, because of transient overvoltages), not the RMS one.

Definition 5.3: Phasor in RMS value. To the quantity $a(t) = A_m \cos(\omega t + \alpha)$ one associates the **phasor**

$$\underline{A} = A e^{i\alpha} = \frac{A_m}{\sqrt{2}} e^{i\alpha},$$

and one writes $a(t) = \operatorname{Re} [\sqrt{2} \underline{A} e^{i\omega t}]$. The modulus of $\underline{A}$ is the RMS value, its argument the initial phase.

Best practice. When drawing a phasor diagram it always pays to choose *one* quantity as the phase reference and place it on the positive real axis: the choice is free because multiplying all phasors by the same $e^{-i\alpha_0}$ is a rigid rotation (Theorem 3.13) and alters neither moduli nor relative phase differences — that is, it alters nothing physically measurable. In series circuits one takes the *current* as reference (common to all elements); in parallel circuits the *voltage*.

5.2. Complex impedance and Kirchhoff's laws in symbolic form

Definition 5.4: Impedance and admittance. For a linear one-port in the sinusoidal steady state of angular frequency ω , carrying the current of phasor $\underline{I} \neq 0$ and subject to the voltage of phasor $\underline{V}$ (load sign convention), the **impedance** is defined as

$$\underline{Z} = \frac{\underline{V}}{\underline{I}} = R + iX \quad [\Omega],$$

where $R = \operatorname{Re} \underline{Z}$ is the *resistance* and $X = \operatorname{Im} \underline{Z}$ the *reactance*. The **admittance** is

$\underline{Y} = 1/\underline{Z} = G + iB$, with G the *conductance* and B the *susceptance* [S].

Theorem 5.3: Impedance of the elementary one-ports. In the conventions fixed above:

$$\underline{Z}_R = R, \quad \underline{Z}_L = i\omega L, \quad \underline{Z}_C = \frac{1}{i\omega C} = -\frac{i}{\omega C}.$$

Consequently: in the resistor voltage and current are in phase; in the inductor the voltage leads the current by $\pi/2$; in the capacitor it lags by $\pi/2$.

Proof. Resistor. Ohm's law $v(t) = Ri(t)$ is pointwise in time; by the linearity of the phasor–sinusoid correspondence (Theorem 3.23(i),(iv)) it translates into $\underline{V} = R\underline{I}$, i.e. $\underline{Z}_R = R$, real and positive: zero argument, no phase shift.

Inductor. The constitutive law is $v(t) = L \frac{di}{dt}$. By Theorem 3.23(ii) time differentiation corresponds to multiplication by $i\omega$, hence $\underline{V} = i\omega L \underline{I}$ and $\underline{Z}_L = i\omega L$. Since $i = e^{i\pi/2}$, the argument of $\underline{Z}_L$ is $+\pi/2$: by Theorem 3.4, $\arg \underline{V} = \arg \underline{I} + \pi/2$, i.e. the voltage leads.

Capacitor. The constitutive law is $i(t) = C \frac{dv}{dt}$, which by the same argument gives $\underline{I} = i\omega C \underline{V}$, whence $\underline{Z}_C = 1/(i\omega C)$. Rationalising, $1/i = -i$, hence $\underline{Z}_C = -i/(\omega C)$, of argument $-\pi/2$: the voltage lags. $\square$

Remark 5.2: A mnemonic that does not betray. The pair $\underline{Z}_L = i\omega L$, $\underline{Z}_C = -i/(\omega C)$ is remembered in English by the acronym ELI THE ICE MAN (E before I in L , I before E in C). But the reason is structural and need not be memorised: differentiating multiplies by $i\omega$, i.e. rotates by $+90^\circ$; integrating divides, i.e. rotates by -90° . The inductor differentiates the current, the capacitor integrates it.

Theorem 5.4: Symbolic Ohm's law and Kirchhoff's laws. In the sinusoidal steady state of angular frequency ω :

- (i) (symbolic Ohm) $\underline{V} = \underline{Z} \underline{I}$ for every linear one-port;
- (ii) (Kirchhoff's current law) at every node, $\sum_k \underline{I}_k = 0$;
- (iii) (Kirchhoff's voltage law) around every loop, $\sum_k \underline{V}_k = 0$;
- (iv) (series and parallel) $\underline{Z}_{\text{series}} = \sum_k \underline{Z}_k$ and $\underline{Y}_{\text{par}} = \sum_k \underline{Y}_k$.

Proof. (i) It is Definition 5.4 read as a relation, together with the existence of the impedance for the elementary one-ports (Theorem 5.3) and their composition, given in (iv).

(ii) Kirchhoff's current law in the time domain states $\sum_k i_k(t) = 0$ for every t . All the i_k are sinusoidal of the same angular frequency ω (in a linear circuit at steady state, driven by sources of angular frequency ω); by Theorem 3.23(i) the sum has phasor $\sum_k \underline{I}_k$, and by (iv) of the same theorem the correspondence is bijective: identically zero sinusoid $\iff$ zero phasor. Hence $\sum_k \underline{I}_k = 0$.

(iii) Identical to (ii) applied to loop voltages.

(iv) Series: the elements carry the same current $\underline{I}$; by (iii) the total voltage is $\sum_k \underline{V}_k = \sum_k \underline{Z}_k \underline{I} = (\sum_k \underline{Z}_k) \underline{I}$, and by Definition 5.4 the equivalent impedance is the sum. **Parallel:** the elements are subject to the same voltage $\underline{V}$; by (ii) the total current is $\sum_k \underline{I}_k = \sum_k \underline{Y}_k \underline{V}$, whence the sum of the admittances. $\square$

Remark 5.3: What has really been gained. Theorem 5.4 says that all of DC circuit theory — dividers, Thévenin, Norton, superposition, node and mesh analysis — carries over unchanged to the sinusoidal steady state, provided the real numbers R are replaced by the complex numbers $\underline{Z}$. This is not an analogy: it is literally the same linear system, solved over $\mathbb{C}$ instead of $\mathbb{R}$. The price paid is that the solutions are complex and must be reinterpreted; the price avoided is solving a system of integro-differential equations.

Theorem 5.5: Impedance triangle. For a one-port of impedance $\underline{Z} = R + iX$ with $\underline{Z} \neq 0$, setting $Z = |\underline{Z}|$ and $\varphi = \text{Arg } \underline{Z}$:

$$Z = \sqrt{R^2 + X^2}, \quad \varphi = \text{atan2}(X, R), \quad R = Z \cos \varphi, \quad X = Z \sin \varphi,$$

and φ is the phase of the voltage relative to the current. Moreover $\varphi \in [-\pi/2, \pi/2]$ for every passive one-port, with $\varphi > 0$ (inductive behaviour) if $X > 0$ and $\varphi < 0$ (capacitive) if $X < 0$.

Proof. The first four relations are the polar form of $\underline{Z}$ (Theorem 3.3) and the operational rule of the *good practice* box of Section 3.1 for computing the argument. That φ is the phase difference follows from $\underline{V} = \underline{Z}\underline{I}$ and Theorem 3.4: $\arg \underline{V} = \arg \underline{Z} + \arg \underline{I}$.

For the bound: a *passive* one-port delivers no energy, hence the average power absorbed is non-negative; by Theorem 5.11, which we shall prove, $P = ZI^2 \cos \varphi \geq 0$ with $Z, I > 0$, whence $\cos \varphi \geq 0$, i.e. $\varphi \in [-\pi/2, \pi/2]$, that is $R \geq 0$. The sign of φ coincides with that of $X = Z \sin \varphi$ because $Z > 0$ and the sine is positive exactly on $(0, \pi)$. $\square$

Example 5.2: Series RLC circuit: complete calculation. A series branch with $R = 30 \, \Omega$, $L = 0.12 \, \text{H}$ and $C = 40 \, \mu\text{F}$ is supplied at $V = 230 \, \text{V}$, $f = 50 \, \text{Hz}$. Find the impedance, the current, the voltages across the individual elements and verify Kirchhoff's law.

Step 1 — reactances. With $\omega = 2\pi \cdot 50 = 314.159 \, \text{rad/s}$:

$$X_L = \omega L = 314.159 \times 0.12 = 37.699 \, \Omega,$$

$$X_C = \frac{1}{\omega C} = \frac{1}{314.159 \times 40 \times 10^{-6}} = \frac{1}{0.01256637} = 79.577 \, \Omega.$$

Step 2 — impedance. By Theorem 5.4(iv),

$$\underline{Z} = R + i(X_L - X_C) = 30 + i(37.699 - 79.577) = 30 - i 41.878 \, \Omega.$$

Modulus and argument (Theorem 5.5):

$$Z = \sqrt{900 + 1753.8} = \sqrt{2653.8} = 51.515 \, \Omega,$$

$$\varphi = \arctan \frac{-41.878}{30} = \arctan(-1.39593) = -0.94908 \, \text{rad} = -54.377^\circ,$$

legitimate with $\arctan$ because $R > 0$ (fourth quadrant). The circuit is *capacitive*: the current leads the voltage.

Step 3 — current. Taking $\underline{V} = 230 \angle 0^\circ$ as reference,

$$\underline{I} = \frac{\underline{V}}{\underline{Z}} = \frac{230 \angle 0^\circ}{51.515 \angle -54.377^\circ} = 4.4647 \angle 54.377^\circ \, \text{A}.$$

Step 4 — voltages across the individual elements.

$$\underline{V}_R = R\underline{I} = 30 \times 4.4647 \angle 54.377^\circ = 133.94 \angle 54.377^\circ \, \text{V},$$

$$\underline{V}_L = iX_L\underline{I} = 37.699 \times 4.4647 \angle (54.377^\circ + 90^\circ) = 168.31 \angle 144.377^\circ \, \text{V},$$

$$\underline{V}_C = -iX_C \underline{I} = 79.577 \times 4.4647 \angle (54.377^\circ - 90^\circ) = 355.30 \angle -35.623^\circ \text{ V.}$$

Step 5 — Kirchhoff check. In Cartesian components:

$$\underline{V}_R = 78.00 + i108.86, \quad \underline{V}_L = -136.97 + i98.03, \quad \underline{V}_C = 288.97 - i206.89.$$

Sum: $(78.00 - 136.97 + 288.97) + i(108.86 + 98.03 - 206.89) = 230.00 + i0.00$. ✓

Step 6 — physical remark. The voltage across the capacitor, 355 V, is *greater* than the supply voltage, 230 V. Kirchhoff's law is not violated: voltages add as *vectors*, not as numbers, and $\underline{V}_L$ and $\underline{V}_C$ are in antiphase, partly cancelling. The phenomenon is the prelude to resonance (Section 5.6), where it becomes spectacular.

Typical error. Adding the moduli: $133.94 + 168.31 + 355.30 = 657.6 \neq 230$. It is the same error as in Example 3.8 of Chapter 3, here with practical consequences: it leads to badly sized components.

5.3. RC and RLC circuits as dynamical systems

INFO

What changes with respect to the previous sections. Up to now the phasor has described only the *sinusoidal steady state*, not the full transient. This section fixes input, output and initial conditions, and derives the responses directly from the circuit's differential equation, without requiring the Laplace transform. We write i for the imaginary unit, as throughout this volume; electrical engineering often writes j . Components are ideal and linear.

Theorem 5.6: RC low-pass: transient, magnitude and phase. For a series RC with $R, C > 0$, input $u(t)$ across the series branch and output $v(t)$ across the capacitor, put $\tau = RC$. Then

$$\tau v'(t) + v(t) = u(t), \quad v(t) = v(0)e^{-t/\tau} + \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds.$$

For a constant step U with $v(0) = 0$, $v(t) = U(1 - e^{-t/\tau})$ for $t \geq 0$. In sinusoidal steady state:

$$\underline{H}_C(i\omega) = \frac{1}{1 + i\omega\tau}, \quad |\underline{H}_C(i\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}, \quad \arg \underline{H}_C(i\omega) = -\arctan(\omega\tau) \quad (\omega \geq 0).$$

Proof. Step 1 — circuit laws. The capacitor current is $I(t) = Cv'(t)$ and the resistor voltage is $RI(t)$. Kirchhoff's law gives $u = RI + v = RCv' + v$. Divide by τ and multiply by $e^{t/\tau}$: $(e^{t/\tau}v)' = e^{t/\tau}u/\tau$. Integrate from 0 to t and isolate v to obtain the formula. The difference of two solutions with the same initial value satisfies $(e^{t/\tau}w)' = 0$, $w(0) = 0$, hence vanishes: the solution is unique. The formula holds for continuous inputs, and piecewise between finitely many jumps.

Step 2 — step and phasor. For $u = U$, the integral evaluates to $U(1 - e^{-t/\tau})$. In sinusoidal steady state substitute $u(t) = \Re(\underline{U}e^{i\omega t})$ and $v(t) = \Re(\underline{V}e^{i\omega t})$ into the equation: $(1 + i\omega\tau)\underline{V} = \underline{U}$. The magnitude of the denominator is $\sqrt{1 + (\omega\tau)^2}$ and, having positive real part, its argument is $\arctan(\omega\tau)$. The reciprocal has opposite argument. The homogeneous transient decays, because $\tau > 0$. □

Corollary 5.1: Cutoff and decibels: the exact value. The cutoff angular frequency of the low-pass is $\omega_c = 1/\tau$. There the magnitude is $1/\sqrt{2}$, the phase is $-\pi/4$ and the gain in decibels is $-10 \log_{10} 2$, not exactly -3 .

Proof. The zero-frequency value is 1. Imposing that the squared magnitude be halved means $1/(1 + (\omega\tau)^2) = 1/2$, hence $(\omega\tau)^2 = 1$; for $\omega > 0$ this gives $\omega = 1/\tau$. Substituting into the phase gives $-\arctan 1 = -\pi/4$. By definition the amplitude gain in decibels is $20 \log_{10} |H|$: $20 \log_{10}(2^{-1/2}) = -10 \log_{10} 2$.

For $R = 1000 \Omega$ and $C = 1 \mu\text{F}$: $\tau = 1 \text{ ms}$, $\omega_c = 1000 \text{ rad/s}$, $f_c \simeq 159.154943 \text{ Hz}$ and the gain at cutoff is about -3.010299957 dB (verified). The usual phrase “ -3 dB ” is a rounding. $\square$

Theorem 5.7: Series RLC and damping classification. For a series RLC with $L, C > 0$, $R \geq 0$, input u across the whole branch and output v across the capacitor, the equation and parameters are

$$LCv'' + RCv' + v = u, \quad \omega_n = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{R}{2} \sqrt{\frac{C}{L}}, \quad v'' + 2\zeta\omega_n v' + \omega_n^2 v = \omega_n^2 u.$$

The characteristic roots are $r_{\pm} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$. For $0 \leq \zeta < 1$ the regime is underdamped, for $\zeta = 1$ critically damped, for $\zeta > 1$ overdamped. For $\zeta = 0$ it is undamped. If $R > 0$, the branch quality factor satisfies $Q = 1/(2\zeta)$.

Proof. Step 1 — equation and dimensions. With $I = Cv'$, Kirchhoff's law gives $u = RI + LI' + v = RCv' + LCv'' + v$. Dividing by LC identifies $\omega_n^2 = 1/(LC)$ and $2\zeta\omega_n = R/L$, giving the formula for ζ . The product LC has dimension time squared and R/L has dimension inverse time, and ζ is dimensionless.

Step 2 — roots and modes. For $u = 0$ try $v = e^{rt}$, obtaining $r^2 + 2\zeta\omega_n r + \omega_n^2 = 0$, with discriminant $4\omega_n^2(\zeta^2 - 1)$. If $0 \leq \zeta < 1$, put $\omega_d = \omega_n\sqrt{1 - \zeta^2}$: the real modes are $e^{-\zeta\omega_n t} \cos(\omega_d t)$ and $e^{-\zeta\omega_n t} \sin(\omega_d t)$. If $\zeta = 1$, the modes are $e^{-\omega_n t}$ and $te^{-\omega_n t}$; substituting $v = e^{-\omega_n t} w$ into the equation reduces it to $w'' = 0$. If $\zeta > 1$, the modes are $e^{r_+ t}, e^{r_- t}$, both decaying because $\sqrt{\zeta^2 - 1} < \zeta$. Two independent modes can reproduce any pair $(v(0), v'(0))$; uniqueness is proved in the next theorem. Explicitly, the two-mode initial-data matrices have determinant ω_d in the oscillatory case, 1 in the critical case and $r_- - r_+$ in the overdamped case: all nonzero in their respective cases, so the linear system for the two coefficients has exactly one solution.

Step 3 — quality factor. For the series branch, $Q = \omega_n L/R$. Substituting $\omega_n = 1/\sqrt{LC}$, $Q = R^{-1} \sqrt{L/C} = 1/(2\zeta)$. At $R = 0$ we must not divide by zero: the ideal circuit has no dissipation and no finite quality factor. $\square$

Theorem 5.8: Step response and uniqueness. For a step $U \neq 0$ applied at $t = 0$, with $v(0) = 0$ and initial current $I(0) = 0$, put $y = v/U$. The solution for $t \geq 0$ is:

$$\begin{aligned} 0 \leq \zeta < 1: \quad y &= 1 - e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta\omega_n}{\omega_d} \sin \omega_d t \right), \\ \zeta = 1: \quad y &= 1 - e^{-\omega_n t} (1 + \omega_n t), \quad \zeta > 1: \quad y = 1 + \frac{r_- e^{r_+ t} - r_+ e^{r_- t}}{r_+ - r_-}. \end{aligned}$$

Proof. Step 1 — initial data. Since $I = Cv'$, the data become $y(0) = y'(0) = 0$. The equation is $y'' + 2\zeta\omega_n y' + \omega_n^2 y = \omega_n^2$ and a particular solution is $y = 1$. Subtracting it reduces the problem to the homogeneous modes of the previous theorem.

Step 2 — verification for the oscillatory regime. Write $\alpha = \zeta\omega_n$, so $\omega_d^2 + \alpha^2 = \omega_n^2$.

Differentiating the first formula:

$$y' = \frac{\omega_n^2}{\omega_d} e^{-\alpha t} \sin \omega_d t, \quad y'' = \omega_n^2 e^{-\alpha t} \left(\cos \omega_d t - \frac{\alpha}{\omega_d} \sin \omega_d t \right).$$

Substituting into $y'' + 2\alpha y' + \omega_n^2 y$, the cosines cancel and the coefficients of the sines are $-\alpha + 2\alpha - \alpha = 0$: what remains is ω_n^2 . This includes $\zeta = 0$, for which $y = 1 - \cos \omega_n t$.

Step 3 — critical and overdamped cases. In the critical case $y' = \omega_n^2 t e^{-\omega_n t}$ and $y'' = \omega_n^2(1 - \omega_n t)e^{-\omega_n t}$: substitution cancels the exponential terms and leaves ω_n^2 ; the initial data vanish. In the overdamped case write $y = 1 + Ae^{r_+ t} + Be^{r_- t}$. The initial data require $A+B = -1$ and $r_+ A + r_- B = 0$, so $A = r_-/(r_+ - r_-)$ and $B = -r_+/(r_+ - r_-)$. Each exponential solves the homogeneous equation because $r_{\pm}^2 + 2\zeta\omega_n r_{\pm} + \omega_n^2 = 0$. This proves the third formula.

Step 4 — uniqueness without an external ODE theorem. The difference w of two solutions with identical initial data satisfies $w'' + 2\zeta\omega_n w' + \omega_n^2 w = 0$ and $w(0) = w'(0) = 0$. For $E(t) = ((w')^2 + \omega_n^2 w^2)/2$, $E' = w'(w'' + \omega_n^2 w) = -2\zeta\omega_n(w')^2 \leq 0$. Thus $0 \leq E(t) \leq E(0) = 0$ for $t \geq 0$, forcing $w = w' = 0$. The verified formulas are therefore the unique solutions. $\square$

Corollary 5.2: First peak and overshoot. If $0 < \zeta < 1$, the first maximum of the normalized response is at $t_p = \pi/\omega_d$, and the relative overshoot is

$$M_p = y(t_p) - 1 = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right).$$

Proof. The preceding formula for y' has the sign of $\sin \omega_d t$: positive on $(0, \pi/\omega_d)$ and negative just afterwards. Hence t_p is the first maximum. There the sine is zero and the cosine is -1 , so $y(t_p) = 1 + e^{-\zeta\omega_n\pi/\omega_d}$. Substitute $\omega_d = \omega_n\sqrt{1-\zeta^2}$ to obtain the formula. $\square$

Theorem 5.9: Three angular frequencies not to confuse. For $R > 0$, the current of the series RLC is maximal at ω_n . The angular frequency of the free oscillations is $\omega_d = \omega_n\sqrt{1-\zeta^2}$ when $0 < \zeta < 1$. By contrast, the magnitude of the ratio capacitor voltage / input voltage has a positive-frequency maximum only if $0 < \zeta < 1/\sqrt{2}$, and then

$$\omega_r = \omega_n\sqrt{1-2\zeta^2}, \quad |\underline{H}_C(i\omega_r)| = \frac{1}{2\zeta\sqrt{1-\zeta^2}}.$$

Proof. The series impedance has squared magnitude $R^2 + (\omega L - 1/(\omega C))^2$, minimal when $\omega L = 1/(\omega C)$, i.e. at ω_n . For the output across the capacitor, instead, the differential equation gives $\underline{H}_C(i\omega) = 1/(1-x^2+2i\zeta x)$, with $x = \omega/\omega_n$. The denominator of the squared magnitude is $D(x) = (1-x^2)^2 + 4\zeta^2 x^2$ and $D'(x) = 4x(x^2 + 2\zeta^2 - 1)$. For $x > 0$ this has an interior minimum precisely when $\zeta < 1/\sqrt{2}$, at $x_r = \sqrt{1-2\zeta^2}$; the derivative is negative before and positive afterwards. Substitution gives $D(x_r) = 4\zeta^2(1-\zeta^2)$. If $\zeta \geq 1/\sqrt{2}$, D increases for $x > 0$ and the maximum gain is the zero-frequency value 1. The formula for ω_d was already derived from the characteristic roots. $\square$

5.3.1 Reproducible example: formula and numerical integration

Example 5.3: Four regimes on the same circuit. Take again $L = 50 \text{ mH}$, $C = 20 \text{ }\mu\text{F}$ and a step $U = 100 \text{ V}$. The output is the capacitor voltage, initially uncharged, with zero initial current. We vary only R to compare regimes.

Step 1 — exact parameters.

$$\omega_n = (LC)^{-1/2} = 1000 \text{ rad/s}, \quad R_{\text{crit}} = 2\sqrt{L/C} = 100 \text{ }\Omega, \quad \zeta = R/(100 \text{ }\Omega).$$

Step 2 — two independent routes. The first route evaluates the closed formulas of Theorem 5.8. The second integrates the system $y' = w$, $w' = \omega_n^2(1 - y) - 2\zeta\omega_n w$ with $y(0) = w(0) = 0$, with a Python script (`vol2_cap05_rlc.py`, method DOP853, relative tolerance 10^{-11} , absolute tolerance 10^{-13} , maximum step $5 \cdot 10^{-5}$ s).

$R \ (\Omega)$	ζ	$y(5 \text{ ms}), \text{ formula}$	$y(5 \text{ ms}), \text{ numerical}$
0	0	0.716337815	0.716337815
10	0.1	0.901449332	0.901449332
100	1	0.959572318	0.959572318
250	2.5	0.631760871	0.631760871

Step 3 — reading the comparison. The four cases are, in order, undamped, underdamped, critical and overdamped ($R = 0, 10, 100, 250 \ \Omega$ corresponding to $\zeta = 0; 0.1; 1; 2.5$). In the recorded run the discrepancy between the closed formula and the numerical integration is below 10^{-9} on all four values (verified). For $R = 0$ the regime never damps (pure oscillation at ω_n); for $R = 10 \ \Omega$ ($\zeta = 0.1$, underdamped), $\omega_d = 1000\sqrt{0.99} \text{ rad/s}$ and $\omega_r = 1000\sqrt{0.98} \text{ rad/s}$: the two values do not coincide, consistently with Theorem 5.9. The simulation is an independent check, not the proof of the formulas.

Exercise 5.1. For the same L, C , put $R = 80 \ \Omega$. Classify the transient and decide whether $|\underline{H}_C|$ has a maximum at positive frequency.

Solution. We have $\zeta = 0.8 < 1$: the roots are complex and $\omega_d = 1000\sqrt{1 - 0.64} = 600 \text{ rad/s}$. However $0.8 > 1/\sqrt{2}$, so $D'(x) > 0$ for $x > 0$: the gain decreases without a positive peak. The transient can therefore oscillate even when the magnitude response shows no resonance. These are two different properties, not equivalent definitions.

Warning. Quick self-check (true or false, justify).

1. For RC, the time constant is RC .
2. At the RC cutoff, the gain magnitude is $1/2$.
3. The exact cutoff value is -3 dB .
4. For series RLC, $R = 0$ implies no damping.
5. For $R > 0$, $Q = 1/(2\zeta)$.
6. The current peak and the capacitor-voltage peak always occur at the same frequency.
7. The critical case has terms $te^{-\omega_n t}$.
8. An agreeing simulation replaces the proof of a theorem.

Teacher's key. 1 T: the coefficient of the derivative in the normalized equation. 2 F: the magnitude is $1/\sqrt{2}$, its square is $1/2$. 3 F: $-10 \log_{10} 2$. 4 T: the roots are purely imaginary. 5 T: substitute the definitions. 6 F: it depends on the output and on ζ . 7 T: a repeated root. 8 F: it only checks numerical cases.

Remark 5.4: Connection with Volume III. The parameters ω_n, ζ and the form of the step response are the same tools used to describe a vibrational mode and design a command that reduces its excitation. Here they were derived from circuit equations, without introducing a Laplace transform that has not yet been developed. Feedback and complete control design remain subsequent topics. An ideal model excludes saturation, component tolerances and real-plant safety limits.

5.4. Power in the sinusoidal steady state

Definition 5.5: Instantaneous power. For a one-port in the load sign convention, the **instantaneous power** absorbed is

$$p(t) = v(t) i(t) \quad [\text{W}],$$

and the **active power** (or mean power) is $P = \frac{1}{T} \int_0^T p(t) dt$.

Theorem 5.10: Decomposition of the instantaneous power. Let $v(t) = V\sqrt{2}\cos(\omega t)$ and $i(t) = I\sqrt{2}\cos(\omega t - \varphi)$, with $V, I > 0$ RMS values. Then

$$p(t) = \underbrace{VI \cos \varphi}_P (1 + \cos 2\omega t) + \underbrace{VI \sin \varphi}_Q \sin 2\omega t.$$

In particular:

- (i) the instantaneous power oscillates at twice the mains frequency;
- (ii) $P = VI \cos \varphi$ is the mean value of p ;
- (iii) the term in $Q = VI \sin \varphi$ has zero mean: it represents energy sloshing back and forth between source and one-port without being converted;
- (iv) $p(t) \geq 0$ for every t if and only if $\varphi = 0$.

Proof. **Step 1 — the product.** Multiplying,

$$p(t) = 2VI \cos(\omega t) \cos(\omega t - \varphi).$$

By the product-to-sum formula $2 \cos A \cos B = \cos(A - B) + \cos(A + B)$ (Volume I, Chapter 8; or Exercise S1 of Chapter 3), with $A = \omega t$ and $B = \omega t - \varphi$:

$$p(t) = VI [\cos \varphi + \cos(2\omega t - \varphi)].$$

Step 2 — expanding the second cosine. By the addition formula,

$$\cos(2\omega t - \varphi) = \cos 2\omega t \cos \varphi + \sin 2\omega t \sin \varphi.$$

Substituting and collecting:

$$p(t) = VI \cos \varphi (1 + \cos 2\omega t) + VI \sin \varphi \sin 2\omega t,$$

which is the claim.

Step 3 — (i) and (ii). Both oscillating terms have angular frequency 2ω , hence frequency $2f$. The mean over a period of $\cos 2\omega t$ and $\sin 2\omega t$ is zero (same periodicity argument as in Theorem 5.2(i)), so $P = VI \cos \varphi$.

Step 4 — (iii). The term $Q \sin 2\omega t$ has zero mean for the same reason: for half a period energy flows towards the one-port, for the other half it flows back. The balance is exactly zero.

Step 5 — (iv). The first term is ≥ 0 if $\cos \varphi \geq 0$, and vanishes twice per period. If $\sin \varphi \neq 0$, the second term changes sign and near the instants where the first vanishes makes p negative; if instead $\varphi = 0$ we are left with $p(t) = VI(1 + \cos 2\omega t) = 2VI \cos^2 \omega t \geq 0$. Conversely, if $p \geq 0$ everywhere, at the instants t^* with $\cos 2\omega t^* = -1$ we have $\sin 2\omega t^* = 0$ and $p(t^*) = 0$; differentiating, $p'(t^*) = 2\omega Q \cos 2\omega t^* = -2\omega Q$ must vanish because t^* is an interior minimum, hence $Q = 0$, i.e. $\varphi = 0$ (since $VI > 0$). $\square$

Definition 5.6: Active, reactive, apparent and complex power. With φ the phase of the voltage relative to the current:

$$P = VI \cos \varphi \text{ [W]}, \quad Q = VI \sin \varphi \text{ [var]}, \quad S = VI \text{ [VA]},$$

called respectively **active**, **reactive** and **apparent** power. The **complex power** is

$$\underline{S} = \underline{V} \underline{I}^* = P + iQ,$$

and the **power factor** is $\cos \varphi = P/S$.

Theorem 5.11: Complex power and the power triangle. With the previous definitions:

- (i) $\underline{S} = \underline{V} \underline{I}^* = P + iQ$, i.e. $P = \operatorname{Re} \underline{S}$ and $Q = \operatorname{Im} \underline{S}$;
- (ii) $\underline{S} = \underline{Z} |\underline{I}|^2 = \underline{Y}^* |\underline{V}|^2$; in particular $P = RI^2$ and $Q = XI^2$;
- (iii) $S^2 = P^2 + Q^2$ (power triangle);
- (iv) $Q > 0$ for an inductive load, $Q < 0$ for a capacitive load, $Q = 0$ for a purely resistive one;
- (v) $0 \leq \cos \varphi \leq 1$ for every passive one-port, with $\cos \varphi = 1$ if and only if the one-port is purely resistive.

Proof. (i) Write $\underline{V} = V e^{i\alpha}$ and $\underline{I} = I e^{i\beta}$, so that $\varphi = \alpha - \beta$. Then, since conjugation reverses the sign of the argument (Theorem 3.1(ii)),

$$\underline{V} \underline{I}^* = V e^{i\alpha} \cdot I e^{-i\beta} = VI e^{i(\alpha-\beta)} = VI e^{i\varphi} = VI \cos \varphi + i VI \sin \varphi = P + iQ,$$

by Euler's formula and Definition 5.6.

Why the conjugate. Writing $\underline{V} \underline{I}$ would give the argument $\alpha + \beta$, which depends on the arbitrary choice of phase reference and is therefore not a physical quantity. Conjugation is what makes $\underline{S}$ invariant under a rigid rotation of the phasors: multiplying $\underline{V}$ and $\underline{I}$ by the same $e^{i\theta_0}$ leaves $\underline{V} \underline{I}^*$ unchanged. It is the only choice compatible with the arbitrariness of the reference.

(ii) Substituting $\underline{V} = \underline{Z} \underline{I}$:

$$\underline{S} = \underline{Z} \underline{I} \underline{I}^* = \underline{Z} |\underline{I}|^2 = (R + iX) I^2,$$

using $\underline{I} \underline{I}^* = |\underline{I}|^2$ (Theorem 3.1(i)). Separating real and imaginary parts gives $P = RI^2$ and $Q = XI^2$. The other form follows from $\underline{I} = \underline{Y} \underline{V}$, giving $\underline{S} = \underline{V} \underline{Y}^* \underline{V}^* = \underline{Y}^* |\underline{V}|^2$.

(iii) $S^2 = |\underline{S}|^2 = P^2 + Q^2$ by the definition of modulus.

(iv) From (ii), $Q = XI^2$ with $I^2 > 0$: the sign of Q is that of X , and by Theorem 5.5 $X > 0$ means inductive behaviour.

(v) By Theorem 5.5 a passive one-port has $R \geq 0$, hence $P = RI^2 \geq 0$ and $\cos \varphi = P/S \geq 0$; moreover $\cos \varphi \leq 1$ because $P^2 \leq P^2 + Q^2 = S^2$. Equality $\cos \varphi = 1$ requires $Q = 0$, i.e. $X = 0$. $\square$

Remark 5.5: What reactive power “costs”. Reactive power does no work: by Theorem 5.10(iii) its energy balance over a period is exactly zero. Yet it costs, for a purely geometric reason. The current flowing in the line has modulus $I = S/V = \sqrt{P^2 + Q^2}/V$: for the same active power P demanded by the load, a non-zero Q increases I , and with it the Joule losses in the line, which equal $R_{\text{line}} I^2$. It is the power triangle read as a right triangle: the hypotenuse S is what the line must carry, the leg P is what the load uses. Hence the

practice of power-factor correction (Section 5.5) and the billing of reactive power to large customers.

Theorem 5.12: Boucherot's theorem (conservation of power). In a linear network in the sinusoidal steady state, the complex power delivered by the sources equals the sum of the complex powers absorbed by the one-ports:

$$\sum_k \underline{S}_k = 0,$$

the sum ranging over all branches in the load sign convention. In particular active and reactive power are conserved separately: $\sum_k P_k = 0$ and $\sum_k Q_k = 0$. The apparent power, by contrast, is not conserved.

Proof. Step 1 — Tellegen's theorem in phasor form. Let the network have b branches and n nodes. Denote by $\underline{V}_k$ the voltage of branch k and by $\underline{I}_k$ its current. By Kirchhoff's voltage law (Theorem 5.4(iii)) there exist node potentials $\underline{E}_1, \dots, \underline{E}_n$ with $\underline{V}_k = \underline{E}_{p(k)} - \underline{E}_{q(k)}$, where $p(k)$ and $q(k)$ are the terminal nodes of branch k . Then

$$\sum_{k=1}^b \underline{V}_k \underline{I}_k^* = \sum_{k=1}^b (\underline{E}_{p(k)} - \underline{E}_{q(k)}) \underline{I}_k^* = \sum_{j=1}^n \underline{E}_j \left(\sum_{k \text{ incident at } j} \pm \underline{I}_k^* \right),$$

rearranging the sum by nodes. The inner sum is the conjugate of the sum of the currents leaving node j , which vanishes by Kirchhoff's current law (Theorem 5.4(ii)); the conjugate of zero is zero. Hence $\sum_k \underline{V}_k \underline{I}_k^* = 0$.

Step 2 — conclusion. By Definition 5.6, $\underline{S}_k = \underline{V}_k \underline{I}_k^*$, hence $\sum_k \underline{S}_k = 0$. Separating real and imaginary parts gives the two conservation laws.

Step 3 — why S is not conserved. The apparent power is $S_k = |\underline{S}_k|$, and the modulus is not additive: in general $|\sum_k \underline{S}_k| \neq \sum_k |\underline{S}_k|$, with equality only if all the $\underline{S}_k$ have the same argument (Theorem 3.1(iv)), i.e. only if all loads have the same power factor. $\square$

Example 5.4: Power balance of the series RLC circuit. Take up again Example 5.2: $R = 30 \, \Omega$, $X_L = 37.699 \, \Omega$, $X_C = 79.577 \, \Omega$, $V = 230 \, \text{V}$, $I = 4.4647 \, \text{A}$, $\varphi = -54.377^\circ$.

Step 1 — powers from the source.

$$S = VI = 230 \times 4.4647 = 1026.9 \, \text{VA},$$

$$P = S \cos \varphi = 1026.9 \times 0.582452 = 598.1 \, \text{W},$$

$$Q = S \sin \varphi = 1026.9 \times (-0.812865) = -834.7 \, \text{var}.$$

Step 2 — powers of the individual elements. With $I^2 = 19.9335 \, \text{A}^2$ and Theorem 5.11(ii):

$$P_R = RI^2 = 30 \times 19.9335 = 598.0 \, \text{W}, \quad P_L = P_C = 0,$$

$$Q_L = X_L I^2 = 37.699 \times 19.9335 = 751.4 \, \text{var},$$

$$Q_C = -X_C I^2 = -79.577 \times 19.9335 = -1586.1 \, \text{var}.$$

Step 3 — Boucherot check.

$$P_R + P_L + P_C = 598.0 \, \text{W} = P \quad \checkmark$$

$$Q_L + Q_C = 751.4 - 1586.1 = -834.7 \, \text{var} = Q \quad \checkmark$$

Step 4 — and the apparent power? The apparent powers of the three elements are 598.0, 751.4 and 1586.1 VA, summing to 2935.5 VA, quite different from $S = 1026.9$ VA. ✓ Apparent power, as announced in Theorem 5.12, is *not* conserved: adding it is a classic mistake and produces gross oversizing.

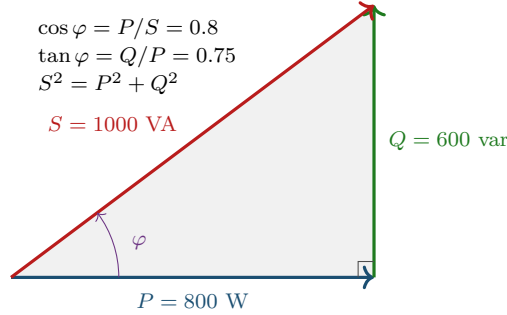


Figure 5.2: The power triangle. The complex apparent power $\underline{S} = \underline{V} \underline{I}^* = P + jQ$ has as real part the active power P (the one that does work and that the meter bills) and as imaginary part the reactive power Q (the one that swings back and forth between source and magnetic or electric field, doing no net work). Since P and Q are orthogonal, $S = \sqrt{P^2 + Q^2}$ and the power factor is $\cos \varphi = P/S$. In the example $P = 800 \text{ W}$ and $Q = 600 \text{ var}$ give $S = 1000 \text{ VA}$ and $\cos \varphi = 0.8$ lagging: the line must carry 1000 VA in order to deliver only 800 W of useful power.

5.5. Power-factor correction

Theorem 5.13: Correction capacitance. A single-phase load absorbs active power P with power factor $\cos \varphi_1$ under RMS voltage V and angular frequency ω . Placing a capacitor of capacitance C in parallel with the load, the overall power factor becomes $\cos \varphi_2$ with

$$C = \frac{P(\tan \varphi_1 - \tan \varphi_2)}{\omega V^2},$$

provided $\varphi_2 \leq \varphi_1$. The capacitor does not change the active power absorbed, and the line current goes from $I_1 = P/(V \cos \varphi_1)$ to $I_2 = P/(V \cos \varphi_2)$.

Proof. Step 1 — the capacitor consumes no active power. By Theorem 5.3 the impedance of the capacitor is purely imaginary, so $R_C = 0$ and, by Theorem 5.11(ii), $P_C = 0$: the active power of the whole remains P .

Step 2 — reactive power of the capacitor. The capacitor is in parallel, hence subject to the same voltage V . By Theorem 5.11(ii) in admittance form, with $\underline{Y}_C = i\omega C$:

$$\underline{S}_C = \underline{Y}_C^* V^2 = -i\omega C V^2 \implies Q_C = -\omega C V^2 < 0.$$

The capacitor *supplies* reactive power, consistently with Theorem 5.11(iv).

Step 3 — reactive balance. By Theorem 5.12 reactive powers add:

$$Q_2 = Q_1 + Q_C = P \tan \varphi_1 - \omega C V^2,$$

where $Q_1 = V I_1 \sin \varphi_1 = P \tan \varphi_1$, a consequence of $P = V I_1 \cos \varphi_1$.

Step 4 — imposing the target. We want $Q_2 = P \tan \varphi_2$. Substituting and solving for C :

$$P \tan \varphi_2 = P \tan \varphi_1 - \omega C V^2 \implies C = \frac{P(\tan \varphi_1 - \tan \varphi_2)}{\omega V^2}.$$

The condition $\varphi_2 \leq \varphi_1$ guarantees $C \geq 0$, i.e. a physically realisable capacitor.

Step 5 — line current. From $P = VI \cos \varphi$ we get $I = P/(V \cos \varphi)$, which gives the two expressions; since $\cos \varphi_2 > \cos \varphi_1$, we have $I_2 < I_1$. $\square$

***Remark 5.6: Why in parallel and not in series.** The correction capacitor must be placed in parallel with the load. In series it would change the voltage across the load and hence its operation; in parallel the load voltage remains V and the load “notices” nothing, while the line current is reduced. It is an intervention on the transport of energy, not on its use: the active power delivered to the load is identical before and after.*

Example 5.5: Power-factor correction of an industrial load. A single-phase load absorbs $P = 15$ kW with $\cos \varphi_1 = 0.75$ (inductive) at $V = 400$ V, $f = 50$ Hz. Bring it to $\cos \varphi_2 = 0.95$ and quantify the benefit.

Step 1 — angles and tangents.

$$\varphi_1 = \arccos 0.75 = 41.4096^\circ, \quad \tan \varphi_1 = 0.881917,$$

$$\varphi_2 = \arccos 0.95 = 18.1949^\circ, \quad \tan \varphi_2 = 0.328684.$$

(Check: $\sin \varphi_1 = \sqrt{1 - 0.75^2} = \sqrt{0.4375} = 0.661438$, and $0.661438/0.75 = 0.881917$ ✓.)

Step 2 — reactive power to be compensated.

$$Q_C = P(\tan \varphi_1 - \tan \varphi_2) = 15\,000 \times 0.553233 = 8298.5 \text{ var.}$$

Step 3 — capacitance. With $\omega = 314.159$ rad/s and $V^2 = 160\,000$ V²:

$$C = \frac{8298.5}{314.159 \times 160\,000} = \frac{8298.5}{50\,265\,440} = 1.651 \times 10^{-4} \text{ F} = 165.1 \text{ }\mu\text{F}.$$

Step 4 — line currents.

$$I_1 = \frac{15\,000}{400 \times 0.75} = 50.00 \text{ A}, \quad I_2 = \frac{15\,000}{400 \times 0.95} = 39.47 \text{ A}.$$

Current reduction: 21.1%.

Step 5 — reduction of line losses. The losses are $R_{\text{line}} I^2$, so their ratio is

$$\left(\frac{I_2}{I_1}\right)^2 = \left(\frac{39.47}{50.00}\right)^2 = 0.7894^2 = 0.6232,$$

i.e. a 37.7% reduction. On a line dissipating 500 W before correction, 188 W are saved permanently.

Step 6 — check with apparent powers. $S_1 = 15\,000/0.75 = 20\,000$ VA and $S_2 = 15\,000/0.95 = 15\,789$ VA; the ratio $15\,789/20\,000 = 0.7895$ matches I_2/I_1 . ✓

Typical error. Correcting to $\cos \varphi = 1$ exactly. Besides requiring a larger capacitance ($C = 15\,000 \times 0.881917/50\,265\,440 = 263.2$ μF , i.e. 59% more for a further 5% of current), full correction is unstable: as the load varies one easily slips into capacitive operation, with overvoltages and possible resonance with the network inductance. Industrial practice stops at 0.95, and that is why supply contracts typically set that threshold.

5.6. Resonance

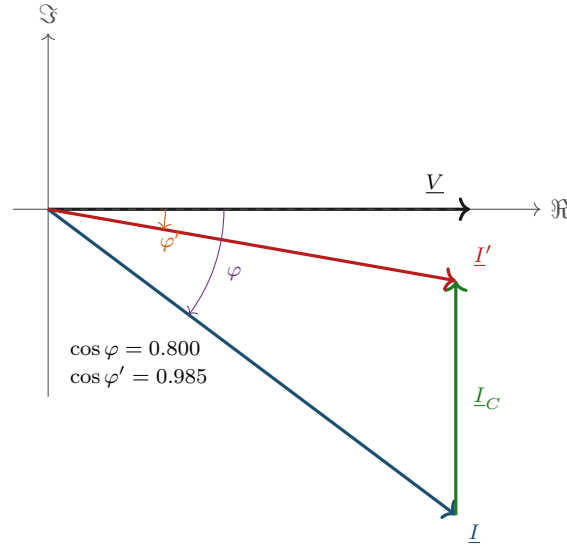


Figure 5.3: Power-factor correction seen on the phasor diagram. The inductive load draws the current $\underline{I}$ lagging the voltage $\underline{V}$ by $\varphi = 36.87^\circ$ ($\cos \varphi = 0.8$). The parallel capacitor draws a purely imaginary current $\underline{I}_C$ leading by 90° , which adds vertically and shortens the reactive component without touching the active one: the line current becomes $\underline{I}'$, smaller in modulus and out of phase by only $\varphi' = 9.94^\circ$ ($\cos \varphi' = 0.985$). The active power is unchanged ($P = \Re \underline{V} \underline{I}^*$ does not move), but the line carries 19% less current and the Joule losses, which go as I^2 , drop by 34%.

Theorem 5.14: Series resonance. For a series RLC branch of impedance $\underline{Z}(\omega) = R + i(\omega L - 1/(\omega C))$ supplied at constant RMS voltage V :

(i) there is a unique angular frequency $\omega_0 > 0$ at which $\underline{Z}$ is real, namely

$$\omega_0 = \frac{1}{\sqrt{LC}};$$

(ii) at ω_0 the modulus $|\underline{Z}|$ is minimal and equals R ; the current is maximal, $I_0 = V/R$, and is in phase with the voltage;

(iii) for $\omega < \omega_0$ the one-port is capacitive, for $\omega > \omega_0$ inductive;

(iv) at resonance the voltages across L and C are equal in modulus and opposite in phase, and their ratio to V equals $Q = \omega_0 L/R$.

Proof. (i) $\text{Im } \underline{Z} = \omega L - 1/(\omega C) = 0$ is equivalent, for $\omega > 0$, to $\omega^2 = 1/(LC)$, whose unique positive root is $\omega_0 = 1/\sqrt{LC}$.

(ii) By Theorem 5.5, $|\underline{Z}|^2 = R^2 + X^2(\omega)$ with $X(\omega) = \omega L - 1/(\omega C)$. Since R is constant, $|\underline{Z}|$ is minimal exactly when X^2 is, i.e. when $X = 0$: at ω_0 , where $|\underline{Z}| = R$. The current $I = V/|\underline{Z}|$ is therefore maximal and equals V/R ; the argument of $\underline{Z}$ vanishes, so there is no phase shift.

(iii) $X(\omega)$ is strictly increasing on $(0, \infty)$, being the sum of ωL (increasing) and $-1/(\omega C)$ (increasing); it vanishes at ω_0 , hence it is negative before and positive after. By Theorem 5.5, $X < 0$ means capacitive and $X > 0$ inductive.

(iv) At resonance $\underline{I} = V/R$ (real, taking $\underline{V}$ real) and

$$\underline{V}_L = i\omega_0 L \underline{I} = i \frac{\omega_0 L}{R} V, \quad \underline{V}_C = \frac{-i}{\omega_0 C} \underline{I} = -i \frac{1}{\omega_0 C R} V.$$

The two moduli coincide because $\omega_0 L = 1/(\omega_0 C)$, a consequence of $\omega_0^2 = 1/(LC)$; the argu-

ments differ by π , i.e. the two phasors are opposite and their sum is zero — as it must be, since at resonance the whole voltage falls across R . Their common ratio is $\omega_0 L/R$. $\square$

Definition 5.7: Quality factor and bandwidth. For the series RLC branch one sets

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R} \sqrt{\frac{L}{C}},$$

called the **quality factor**. The **cut-off angular frequencies** $\omega_1 < \omega_2$ are those at which the active power dissipated is half its value at resonance, and $B = \omega_2 - \omega_1$ is the **bandwidth**.

Theorem 5.15: Bandwidth and quality factor. With the previous notation:

$$\omega_{1,2} = \omega_0 \left(\sqrt{1 + \frac{1}{4Q^2}} \mp \frac{1}{2Q} \right), \quad \omega_1 \omega_2 = \omega_0^2, \quad B = \omega_2 - \omega_1 = \frac{\omega_0}{Q} = \frac{R}{L}.$$

Proof. Step 1 — translating the condition. The dissipated power is $P = RI^2$ with $I = V/|Z|$; halving P at constant V is equivalent to $I = I_0/\sqrt{2}$, i.e. $|Z| = R\sqrt{2}$, i.e.

$$R^2 + X^2 = 2R^2 \iff X(\omega) = \pm R.$$

Step 2 — solving. The equation $\omega L - \frac{1}{\omega C} = \pm R$, multiplied by $\omega/L > 0$, becomes

$$\omega^2 \mp \frac{R}{L} \omega - \frac{1}{LC} = 0,$$

i.e. $\omega^2 \mp \frac{\omega_0}{Q} \omega - \omega_0^2 = 0$, using $1/(LC) = \omega_0^2$ and $R/L = \omega_0/Q$ (from Definition 5.7). The positive roots are

$$\omega = \frac{\omega_0}{2Q} \pm \sqrt{\frac{\omega_0^2}{4Q^2} + \omega_0^2} = \omega_0 \left(\pm \sqrt{1 + \frac{1}{4Q^2}} + \frac{1}{2Q} \right),$$

and choosing for each sign the positive root gives the two expressions of the statement.

Step 3 — product and difference.

$$\omega_1 \omega_2 = \omega_0^2 \left[\left(\sqrt{1 + \frac{1}{4Q^2}} \right)^2 - \left(\frac{1}{2Q} \right)^2 \right] = \omega_0^2 \left[1 + \frac{1}{4Q^2} - \frac{1}{4Q^2} \right] = \omega_0^2,$$

$$\omega_2 - \omega_1 = \omega_0 \left(\frac{1}{2Q} + \frac{1}{2Q} \right) = \frac{\omega_0}{Q} = \frac{\omega_0 R}{\omega_0 L} = \frac{R}{L}.$$

Remark on the result. The relation $\omega_1 \omega_2 = \omega_0^2$ says that ω_0 is the *geometric*, not arithmetic, mean of the cut-off frequencies: the resonance curve is symmetric on a logarithmic scale, not on a linear one. $\square$

Example 5.6: Resonant circuit: design and verification. Let $R = 10 \, \Omega$, $L = 50 \, \text{mH}$, $C = 20 \, \mu\text{F}$, supply $V = 100 \, \text{V}$.

Step 1 — resonance angular frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50 \times 10^{-3} \times 20 \times 10^{-6}}} = \frac{1}{\sqrt{10^{-6}}} = 1000 \, \text{rad/s},$$

i.e. $f_0 = 1000/(2\pi) = 159.15 \, \text{Hz}$.

Step 2 — quality factor.

$$Q = \frac{\omega_0 L}{R} = \frac{1000 \times 0.05}{10} = 5.$$

Check with the alternative form: $\frac{1}{R}\sqrt{L/C} = \frac{1}{10}\sqrt{0.05/2 \times 10^{-5}} = \frac{1}{10}\sqrt{2500} = \frac{50}{10} = 5$. ✓

Step 3 — current and voltages at resonance.

$$I_0 = \frac{V}{R} = 10 \text{ A}, \quad V_L = V_C = QV = 5 \times 100 = 500 \text{ V}.$$

Direct check: $V_L = \omega_0 L I_0 = 1000 \times 0.05 \times 10 = 500 \text{ V}$ ✓; $V_C = I_0/(\omega_0 C) = 10/(1000 \times 2 \times 10^{-5}) = 10/0.02 = 500 \text{ V}$ ✓.

Step 4 — bandwidth and cut-off frequencies.

$$B = \frac{\omega_0}{Q} = 200 \text{ rad/s} \quad (31.83 \text{ Hz}),$$

$$\omega_{1,2} = 1000 \left(\sqrt{1 + \frac{1}{100}} \mp \frac{1}{10} \right) = 1000(1.0049876 \mp 0.1),$$

i.e. $\omega_1 = 904.99 \text{ rad/s}$ and $\omega_2 = 1104.99 \text{ rad/s}$. Checks: $\omega_2 - \omega_1 = 200.00$ ✓; $\omega_1 \omega_2 = 904.99 \times 1104.99 = 1.0000 \times 10^6 = \omega_0^2$ ✓.

Step 5 — check at ω_1 . $X(\omega_1) = 904.99 \times 0.05 - 1/(904.99 \times 2 \times 10^{-5}) = 45.250 - 55.250 = -10.000 \Omega = -R$ ✓, in agreement with Step 1 of the proof of Theorem 5.15.

Practical interpretation. $V_L = V_C = 500 \text{ V}$ on a 100 V supply: the *resonant overvoltage* is Q times the applied voltage. This is the principle of radio tuning circuits (where a high Q is desirable because it makes the filter selective) and at the same time a hazard in power installations, where an accidental resonance between the capacitance of correction banks and the network inductance can destroy insulation.

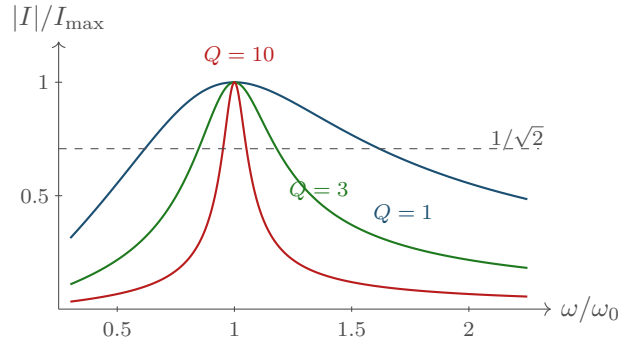


Figure 5.4: The resonance curves of a series RLC circuit, for three values of the quality factor $Q = \omega_0 L/R$. In normalised form $|I|/I_{\max} = 1/\sqrt{1 + Q^2(x - 1/x)^2}$ with $x = \omega/\omega_0$: all curves equal 1 at $x = 1$, where the reactance vanishes and the circuit is purely resistive. The -3 dB bandwidth, i.e. the interval where $|I|/I_{\max} \geq 1/\sqrt{2}$, has width $\Delta\omega = \omega_0/Q$: the quality factor is exactly the ratio of resonant frequency to bandwidth. Note the asymmetry of the curves about $x = 1$: the natural scale of the phenomenon is logarithmic, and the curve is symmetric in $\ln x$.

5.7. Butterworth poles on a circle

The selective filter just met (the observation on resonant overvoltage) selects frequencies near ω_0 while letting others through more or less abruptly. The low-pass filters used in practice — for instance to remove the harmonics of Section 5.15 — are designed by explicitly choosing the shape

of the response curve. The simplest family, Butterworth's (*maximally flat* in the passband), has a striking property: its poles lie *exactly* on a circle.

Definition 5.8: Butterworth response. A low-pass filter of order $n \in \mathbb{N}$ has a **Butterworth response** of cutoff pulsation $\omega_c > 0$ if its transfer function $H(s)$, rational, proper, with real coefficients, stable and causal (all poles with $\Re(s) < 0$), satisfies

$$|H(i\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}} \quad \text{for every } \omega \in \mathbb{R}.$$

Theorem 5.16: Butterworth poles lie on a circle. The n poles of a transfer function $H(s)$ with a Butterworth response of order n and cutoff pulsation ω_c are exactly

$$s_k = \omega_c e^{i\pi(2k+n-1)/(2n)}, \quad k = 1, \dots, n,$$

all lying on the circle $|s| = \omega_c$ of the complex s -plane, and angularly equally spaced by π/n .

Proof. Step 1 — from $|H(i\omega)|^2$ to an identity between rational functions. Since H has real coefficients, $H(-i\omega) = \overline{H(i\omega)}$ for every $\omega \in \mathbb{R}$ (conjugating a real-coefficient function evaluated at a point and at its conjugate gives conjugate values; here $\overline{i\omega} = -i\omega$). Hence, for real ω ,

$$H(i\omega) H(-i\omega) = H(i\omega) \overline{H(i\omega)} = |H(i\omega)|^2 = \frac{1}{1 + (\omega/\omega_c)^{2n}}.$$

The left side, $G(s) := H(s)H(-s)$, is a rational function of s with real coefficients (a product of two rational functions with real coefficients). The right side, as a function of ω , is rewritten in $s = i\omega$ by noting that $\omega = s/i = -is$, hence $\omega^{2n} = (-is)^{2n} = (-i)^{2n} s^{2n}$; since $(-i)^{2n} = ((-i)^2)^n = (-1)^n$, we get $\omega^{2n} = (-1)^n s^{2n}$ and

$$\frac{1}{1 + (\omega/\omega_c)^{2n}} = \frac{1}{1 + (-1)^n s^{2n}/\omega_c^{2n}} \quad (s = i\omega).$$

Two rational functions that agree on the entire imaginary axis (an infinite set of points) agree as rational functions (a rational function is determined by its values on any infinite set, in particular on a line of the complex plane): hence the identity

$$G(s) = \frac{1}{1 + (-1)^n s^{2n}/\omega_c^{2n}}$$

holds for *every* $s \in \mathbb{C}$, not only for purely imaginary s .

Step 2 — the $2n$ poles of G . The poles of G are the zeros of the denominator:

$$1 + \frac{(-1)^n s^{2n}}{\omega_c^{2n}} = 0 \quad \Longleftrightarrow \quad s^{2n} = (-1)^{n+1} \omega_c^{2n} = \omega_c^{2n} e^{i\pi(n+1)}.$$

The $2n$ $2n$ -th roots of $\omega_c^{2n} e^{i\pi(n+1)}$ are

$$\sigma_k = \omega_c e^{i\pi(n+1+2k)/(2n)}, \quad k = 0, 1, \dots, 2n-1.$$

Step 3 — symmetry and no purely imaginary pole. If σ_k is a root, so is $-\sigma_k = \sigma_k e^{i\pi}$: substituting, $-\sigma_k = \omega_c e^{i\pi(n+1+2k+2n)/(2n)} = \sigma_{k+n \bmod 2n}$, which is still in the list (index $k+n$ taken modulo $2n$). The set of $2n$ poles of G is thus closed under $s \mapsto -s$: it splits into n pairs $\{\sigma, -\sigma\}$. No σ_k is purely imaginary: if, for contradiction, $\sigma_k = i\omega$ for some $\omega \in \mathbb{R}$, then G would have a pole at $s = i\omega$, but by Step 1, at every point $s = i\omega$ of the imaginary axis the denominator of G equals $1 + (-1)^n (i\omega)^{2n}/\omega_c^{2n} = 1 + (\omega/\omega_c)^{2n} \geq 1 > 0$ (a sum of 1

and a non-negative number, $(\omega/\omega_c)^{2n}$ being an even power): it cannot vanish, a contradiction. Hence no pole of G is purely imaginary, and the n pairs $\{\sigma, -\sigma\}$ (in each of which $\Re(\sigma)$ and $\Re(-\sigma)$ have opposite sign, since $\Re(\sigma) \neq 0$) split into exactly n poles with $\Re < 0$ and n with $\Re > 0$.

Step 4 — selecting the poles of H . By hypothesis H is stable and causal: all its poles have $\Re < 0$. The poles of $H(-s)$ are then all with $\Re > 0$ (the negatives of those of H). The poles of $G = H(s)H(-s)$ are the union (with no cancellation, since H is proper and the two factors share no pole by Step 3) of the poles of H and of $H(-s)$: hence the poles of H are exactly the σ_k with $\Re(\sigma_k) < 0$, i.e. with $\pi/2 < \arg \sigma_k < 3\pi/2$. Working these out explicitly (Example 5.7 for $n = 3$) gives, after reindexing, the points $s_k = \omega_c e^{i\pi(2k+n-1)/(2n)}$, $k = 1, \dots, n$.

Step 5 — circle and equal spacing. By construction $|s_k| = \omega_c$ for every k (Step 2): all poles lie on the circle $|s| = \omega_c$. The angle of s_k is $\pi(2k + n - 1)/(2n)$, an affine function of k with increment $\pi \cdot 2/(2n) = \pi/n$ per unit of k : consecutive poles are thus separated by exactly π/n . $\square$

Example 5.7: Butterworth poles for $n = 3$. With $\omega_c = 1$ and $n = 3$, the theorem's formula gives

$$s_1 = e^{i2\pi/6} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad s_2 = e^{i\pi} = -1, \quad s_3 = e^{i4\pi/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

Independent check (script `vol2_cap05_butterworth.py`): the six poles of $G(s) = H(s)H(-s)$ for $n = 3$ were computed by solving $s^6 = (-1)^4 = 1$ rotated by $\pi \cdot 4/6$ (Step 2), finding exactly the three pairs $\{s_1, -s_1\}, \{s_2, -s_2\}, \{s_3, -s_3\}$; the three with negative real part match, digit for digit up to 10^{-15} , s_1, s_2, s_3 above. All have modulus $1 = \omega_c$ ✓, and the angles $120^\circ, 180^\circ, 240^\circ$ are equally spaced by $60^\circ = \pi/3 = \pi/n$ ✓. The script also verifies, for $n = 2, 4, 5$, the same circle and equal-spacing property, and that $|H(i\omega_c)|^2 = 1/2$ for every order (the -3 dB cutoff is, by construction, independent of n).

5.8. Circle diagrams: the admittance locus

This section is the “circular” heart of the chapter. We show that, as *one single* circuit parameter varies, the representative phasor does not wander arbitrarily in the complex plane but travels along a **circle**. The result is no lucky accident: it is a direct consequence of Theorem 3.14 of Chapter 3.

Theorem 5.17: Circular locus of the admittance. Consider the series one-port of impedance $\underline{Z} = R + iX$ with fixed reactance $X \neq 0$ and variable resistance R in $[0, +\infty)$. Then the admittance $\underline{Y} = 1/\underline{Z} = G + iB$ describes the semicircle

$$G^2 + \left(B + \frac{1}{2X}\right)^2 = \left(\frac{1}{2X}\right)^2, \quad G \geq 0,$$

of centre $-\frac{i}{2X}$ and radius $\frac{1}{2|X|}$, passing through the origin (limit $R \rightarrow \infty$) and through the point $-i/X$ (value $R = 0$). Consequently, at fixed voltage $\underline{V}$, the current phasor $\underline{I} = \underline{V}\underline{Y}$ also describes a semicircle, obtained from the previous one by a direct similarity of ratio $|\underline{V}|$ and rotation $\arg \underline{V}$.

Proof. First proof — direct algebra.

Step 1. Rationalising,

$$\underline{Y} = \frac{1}{R + iX} = \frac{R - iX}{R^2 + X^2} \implies G = \frac{R}{R^2 + X^2}, \quad B = \frac{-X}{R^2 + X^2}.$$

Step 2. Adding the squares,

$$G^2 + B^2 = \frac{R^2 + X^2}{(R^2 + X^2)^2} = \frac{1}{R^2 + X^2}.$$

On the other hand, from the second relation of Step 1, $\frac{1}{R^2 + X^2} = -\frac{B}{X}$. Hence

$$G^2 + B^2 = -\frac{B}{X} \iff G^2 + B^2 + \frac{B}{X} = 0.$$

Step 3. Completing the square in B ,

$$G^2 + \left(B + \frac{1}{2X}\right)^2 = \frac{1}{4X^2},$$

which is the stated equation. The restriction $G \geq 0$ comes from $R \geq 0$ in Step 1: only the semicircle contained in the right half-plane is traced.

Step 4 — the endpoints. For $R = 0$: $G = 0$, $B = -1/X$, i.e. the point $-i/X$, which lies on the circle ($|-1/X + 1/(2X)| = 1/(2|X|)$ ✓). For $R \rightarrow +\infty$: $G \rightarrow 0^+$ and $B \rightarrow 0^-$, i.e. the origin, also on the circle.

Second proof — geometric, via Möbius.

The map $\underline{Z} \mapsto \underline{Y} = 1/\underline{Z}$ is a Möbius transformation (Definition 3.4 with $a = 0$, $b = 1$, $c = 1$, $d = 0$; the determinant $ad - bc = -1 \neq 0$). As R ranges over $[0, \infty)$ with X fixed, the point $\underline{Z} = R + iX$ travels along the horizontal ray at height X , contained in the line $\text{Im } \underline{Z} = X$. By Theorem 3.14, the image of a line is a circle or a line, and it is a circle exactly when the line does not pass through the pole of the inversion, which here is the origin. Since $X \neq 0$, the line $\text{Im } \underline{Z} = X$ does not pass through the origin: the image is therefore a circle, and it passes through the origin because the line passes through the point at infinity (Observation 3.7). Centre and radius follow from two points and symmetry: the image of iX is $-i/X$, the image of ∞ is 0, and the circle through these two points with centre on the imaginary axis is the one of centre $-i/(2X)$ and radius $1/(2|X|)$. $\square$

Remark 5.7: Why this result matters. Before the age of numerical computation, the circle diagram was the main design tool for induction machines and transmission lines: once a circle had been drawn on graph paper, current, power, torque and efficiency could be read off as the load varied, with no equation to solve. Today the computation is entrusted to the computer, but the conceptual content remains valuable: it says that a one-parameter family of solutions does not fill the plane arbitrarily but lives on a curve of known equation. In (Volume III, Chapter 25) the same principle will produce the M and N circles and the Smith chart, which is literally the same theorem applied to the reflection coefficient.

Example 5.8: Admittance locus of an RL branch with variable resistance. Let $X_L = 10 \, \Omega$ be fixed and R variable. By Theorem 5.17 the locus is the semicircle of centre $-i/20 = -i0.05 \, \text{S}$ and radius $0.05 \, \text{S}$.

Check points.

- $R = 0$: $\underline{Y} = 1/(i10) = -i0.1 \, \text{S}$. Distance from the centre: $|-0.1 + 0.05| = 0.05$ ✓

- $R = 10 \Omega$: $\underline{Y} = \frac{1}{10 + 10i} = \frac{10 - 10i}{200} = 0.05 - i 0.05$ S. Distance: $\sqrt{0.05^2 + 0} = 0.05$ ✓
- $R = 20 \Omega$: $\underline{Y} = \frac{20 - 10i}{500} = 0.04 - i 0.02$ S. Distance: $\sqrt{0.04^2 + 0.03^2} = \sqrt{0.0016 + 0.0009} = \sqrt{0.0025} = 0.05$ ✓
- $R \rightarrow \infty$: $\underline{Y} \rightarrow 0$. Distance from the centre: 0.05 ✓

Physical reading. The point of maximum conductance is the rightmost point of the circle, $G_{\max} = 0.05$ S, reached for $R = X_L = 10 \Omega$. Since at fixed voltage the active power is $P = GV^2$ (Theorem 5.11(ii) in admittance form), *maximum active power is transferred when R equals the reactance* — a special case of the maximum power transfer theorem, read here as the simple search for the extreme point of a circle.

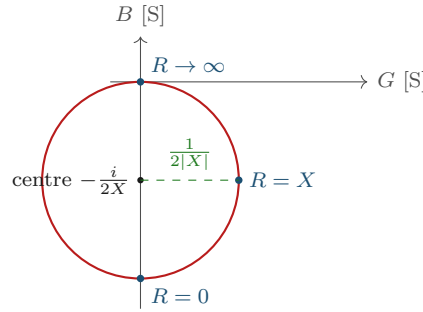


Figure 5.5: The circle diagram. As the resistance R alone varies, with fixed reactance $X > 0$, the admittance travels along the right semicircle of centre $-i/(2X)$ and radius $1/(2|X|)$ (Theorem 5.17). The point of maximum conductance, and hence of maximum active power at fixed voltage, occurs at $R = X$.

5.9. The circle diagram of the induction machine

Section 5.8 showed that the admittance locus of a branch with fixed reactance and variable resistance is a circle. That result is not an exercise: it is the foundation of the *circle diagram* of the induction motor, introduced by Alexander Heyland in 1894 and for a century the design and testing tool for induction machines. The reason is that the equivalent circuit of the motor contains exactly one variable resistance — the referred rotor resistance divided by the slip — and the whole behaviour of the machine, from no-load operation to short circuit, can be read on a single circle.

Definition 5.9: Slip and approximate equivalent circuit. Let ω_s be the angular velocity of the rotating field and ω_r that of the rotor. The pure number

$$s = \frac{\omega_s - \omega_r}{\omega_s}$$

is called the *slip*, so that $s = 1$ at standstill and $s = 0$ at synchronous speed. In the approximate (Steinmetz) equivalent circuit, with the magnetising branch moved to the terminals, the rotor branch carries the current

$$\underline{I}'_2 = \frac{\underline{V}}{R_1 + \frac{R'_2}{s} + jX}, \quad X = X_1 + X'_2,$$

where R_1 is the stator resistance, R'_2 and X'_2 are the rotor resistance and reactance referred to the stator, and the absorbed current is $\underline{I}_1 = \underline{I}_0 + \underline{I}'_2$ with $\underline{I}_0$ the no-load current, practically constant.

Theorem 5.18: The current locus is a circle (Heyland). As the slip s varies over the whole of $\mathbb{R} \setminus \{0\}$, the tip of the phasor $\underline{I}'_2$ of Definition 5.9 describes a circle through the origin, of diameter

$$d = \frac{|\underline{V}|}{X}$$

whose centre lies on the perpendicular to $\underline{V}$ drawn through the origin. The locus of the absorbed current $\underline{I}_1$ is the same circle translated by the constant vector $\underline{I}_0$.

Proof. Step 1 — reduction to one real variable. Put $u := R_1 + R'_2/s$. As s runs over $\mathbb{R} \setminus \{0\}$, the quantity R'_2/s runs over $\mathbb{R} \setminus \{0\}$ and hence u runs over the whole real line with the point R_1 removed. The denominator $u + jX$ then describes the horizontal line $L = \{u + jX : u \in \mathbb{R}\}$ of the complex plane, which is at distance X from the origin and does not contain it.

Step 2 — inversion maps lines to circles. Let $z = 1/(u + jX)$. Multiplying numerator and denominator by the conjugate,

$$z = \frac{u - jX}{u^2 + X^2}.$$

Let us compute the distance of z from the point $-j/(2X)$:

$$z + \frac{j}{2X} = \frac{u}{u^2 + X^2} + j \left(\frac{1}{2X} - \frac{X}{u^2 + X^2} \right) = \frac{u}{u^2 + X^2} + j \frac{u^2 + X^2 - 2X^2}{2X(u^2 + X^2)} = \frac{u}{u^2 + X^2} + j \frac{u^2 - X^2}{2X(u^2 + X^2)}.$$

Hence

$$\left| z + \frac{j}{2X} \right|^2 = \frac{4X^2u^2 + (u^2 - X^2)^2}{4X^2(u^2 + X^2)^2} = \frac{u^4 + 2u^2X^2 + X^4}{4X^2(u^2 + X^2)^2} = \frac{(u^2 + X^2)^2}{4X^2(u^2 + X^2)^2} = \frac{1}{4X^2},$$

that is $|z + j/(2X)| = 1/(2X)$, independently of u : the point z lies on the circle with centre $-j/(2X)$ and radius $1/(2X)$, which passes through the origin.

Step 3 — back to the current. Since $\underline{I}'_2 = \underline{V}z$, and since multiplication by the constant complex number $\underline{V}$ is a direct similarity (rotation by $\arg \underline{V}$ and homothety of ratio $|\underline{V}|$), the locus of $\underline{I}'_2$ is again a circle, of radius $|\underline{V}|/(2X)$ and therefore of diameter $|\underline{V}|/X$, passing through the origin. Its centre is $-j\underline{V}/(2X)$: if $\underline{V}$ is taken as the real reference, the centre lies on the negative imaginary axis, that is on the perpendicular to $\underline{V}$, and the diameter through the origin is perpendicular to $\underline{V}$.

Step 4 — the translation. Finally $\underline{I}_1 = \underline{I}_0 + \underline{I}'_2$ with $\underline{I}_0$ constant: adding a constant vector translates the circle without deforming it. The Heyland circle therefore has the same radius and its centre shifted to $\underline{I}_0 - j\underline{V}/(2X)$.

Step 5 — the notable points. The parameter u locates the points on the circle:

- $s \rightarrow 0$ (synchronism): $u \rightarrow \infty$, $z \rightarrow 0$, hence $\underline{I}'_2 = 0$ and $\underline{I}_1 = \underline{I}_0$: this is the *no-load operating point*;
- $s = 1$ (standstill): $u = R_1 + R'_2$, the *short-circuit*, or starting, point;
- $s \rightarrow \pm\infty$: $u \rightarrow R_1$, the ideal limit point;
- $s < 0$ (rotor driven above synchronism): $u < R_1$, and one travels the arc corresponding to operation as an *induction generator*, with active power delivered to the network.

A single circle therefore contains the three regimes of the machine — motor, generator and brake — distinguished only by the arc one happens to be on. $\square$

Remark 5.8: Why it is the same circle as in Section 5.8. Theorem 5.18 is not a new result but Theorem 5.17 applied to a concrete case: the rotor branch is a series branch with fixed reactance X and variable resistance u , and its admittance describes the circle of diameter $1/X$. What electrical engineering adds is the interpretation: the variable resistance

is not a rheostat but R'_2/s , that is the rotor speed disguised as a resistance. This is the central idea of Steinmetz's model, and it explains why the power dissipated in R'_2/s contains both the rotor Joule losses (R'_2) and the mechanical power delivered ($R'_2(1-s)/s$).

Theorem 5.19: Maximum torque and Kloss's formula. Neglecting the stator resistance ($R_1 \simeq 0$), the electromagnetic torque is

$$T(s) = \frac{3|V|^2}{\omega_s} \frac{R'_2/s}{(R'_2/s)^2 + X^2},$$

and is maximum at the slip

$$s_m = \frac{R'_2}{X}, \quad \text{with} \quad T_{\max} = \frac{3|V|^2}{2\omega_s X}$$

The maximum torque does not depend on R'_2 , whereas the slip at which it occurs is proportional to it. Moreover Kloss's formula holds:

$$\frac{T(s)}{T_{\max}} = \frac{2}{\frac{s}{s_m} + \frac{s_m}{s}}$$

Proof. Step 1 — expression of the torque. The power transmitted across the air gap is the one dissipated in the fictitious resistance R'_2/s in the three phases, $P_t = 3(R'_2/s)|\underline{I}_2|^2$, and the torque is $T = P_t/\omega_s$. With $R_1 = 0$,

$$|\underline{I}_2|^2 = \frac{|V|^2}{(R'_2/s)^2 + X^2},$$

whence the expression of the statement.

Step 2 — change of variable. Put $\xi := R'_2/s > 0$ for $s > 0$. Then

$$T = \frac{3|V|^2}{\omega_s} f(\xi), \quad f(\xi) = \frac{\xi}{\xi^2 + X^2}.$$

Step 3 — maximum. Differentiating,

$$f'(\xi) = \frac{(\xi^2 + X^2) - \xi \cdot 2\xi}{(\xi^2 + X^2)^2} = \frac{X^2 - \xi^2}{(\xi^2 + X^2)^2},$$

which vanishes for $\xi = X$, is positive for $\xi < X$ and negative for $\xi > X$: it is therefore an absolute maximum on $\xi > 0$. From $\xi = R'_2/s = X$ it follows that $s_m = R'_2/X$.

Step 4 — maximum value. Substituting $\xi = X$:

$$f(X) = \frac{X}{2X^2} = \frac{1}{2X} \implies T_{\max} = \frac{3|V|^2}{\omega_s} \cdot \frac{1}{2X} = \frac{3|V|^2}{2\omega_s X},$$

in which R'_2 does not appear: increasing the rotor resistance shifts the maximum towards larger slips without changing its value.

Step 5 — Kloss's formula. Dividing term by term,

$$\frac{T(s)}{T_{\max}} = \frac{f(\xi)}{f(X)} = \frac{\xi/(\xi^2 + X^2)}{1/(2X)} = \frac{2X\xi}{\xi^2 + X^2} = \frac{2}{\frac{\xi}{X} + \frac{X}{\xi}},$$

and since $\xi/X = (R'_2/s)/(R'_2/s_m) = s_m/s$, the formula of the statement follows. Note that the right-hand side depends on s only through the ratio s/s_m : all mechanical characteristics, for whatever value of R'_2 , are the *same* curve rescaled along the abscissa.

Step 6 — symmetry. The function $x \mapsto 2/(x + 1/x)$ is invariant under $x \mapsto 1/x$: hence $T(s)/T_{\max}$ takes the same value at $s = ks_m$ and at $s = s_m/k$. The mechanical characteristic is symmetric about s_m on a logarithmic scale, exactly like the resonance curves of Section 5.6: this is no coincidence, because both have the form $2/(x + 1/x)$, which is the same function that governs the bandwidth of a resonant circuit. $\square$

Corollary 5.3: Rotor-resistance starting. For a wound-rotor motor, inserting into the rotor circuit an additional resistance such that the total referred resistance equals $R''_2 = X$, one obtains $s_m = 1$: the maximum torque occurs at standstill, and it is the largest starting torque obtainable, equal to $T_{\max}$.

Proof. By Theorem 5.19, $s_m = R''_2/X$; imposing $s_m = 1$ gives $R''_2 = X$. Since $T_{\max}$ does not depend on R''_2 , the starting torque equals exactly $T_{\max}$, and no choice of R''_2 can exceed it because $T(s) \leq T_{\max}$ for every s . Progressively cutting out the additional resistance as the speed rises keeps the torque near its maximum throughout the start: this is the principle of the rheostatic starter. $\square$

Example 5.9: A four-pole induction motor. A three-phase 4-pole, 400 V, 50 Hz motor has $X = 1.2 \Omega$ and $R'_2 = 0.24 \Omega$ per phase (star-connected stator, R_1 neglected).

Step 1 — synchronous quantities. With $p = 2$ pole pairs, the synchronous speed is $n_s = 60f/p = 1500$ rpm, that is $\omega_s = 2\pi \cdot 1500/60 = 157.08$ rad/s. The phase voltage is $V = 400/\sqrt{3} = 230.94$ V.

Step 2 — slip and breakdown torque.

$$s_m = \frac{0.24}{1.2} = 0.20, \quad T_{\max} = \frac{3 \cdot 230.94^2}{2 \cdot 157.08 \cdot 1.2} = \frac{160\,000}{376.99} = 424.4 \text{ N} \cdot \text{m}.$$

Step 3 — rated torque. At $s = 0.03$ (that is 1455 rpm), Kloss's formula gives

$$\frac{T}{T_{\max}} = \frac{2}{\frac{0.03}{0.20} + \frac{0.20}{0.03}} = \frac{2}{0.1500 + 6.6667} = \frac{2}{6.8167} = 0.29340,$$

hence $T = 0.29340 \cdot 424.4 = 124.5$ N·m, and the mechanical power is $P = T\omega_r = 124.5 \cdot 157.08 \cdot 0.97 = 18.97$ kW.

Step 4 — starting torque. At $s = 1$,

$$\frac{T}{T_{\max}} = \frac{2}{5.0000 + 0.2000} = \frac{2}{5.2000} = 0.38462 \implies T_{\text{st}} = 163.3 \text{ N} \cdot \text{m},$$

that is 1.31 times the rated torque: a typical value, sufficient for a no-load start, insufficient for many constant-torque loads.

Step 5 — rheostatic starting. By Corollary 5.3, raising the referred rotor resistance to $R''_2 = X = 1.2 \Omega$, that is adding 0.96Ω referred, raises the starting torque to $T_{\max} = 424.4$ N·m, that is 3.41 times the rated torque and 2.60 times the starting torque with the rotor short-circuited.

Step 6 — the circle. The diameter of the Heyland circle is $|V|/X = 230.94/1.2 = 192.45$ A: it is the rotor current one would have with zero total resistance, that is the ideal short-circuit limit. The actual starting current, with $u = R'_2 = 0.24$, is

$$|I'_2| = \frac{230.94}{\sqrt{0.24^2 + 1.2^2}} = \frac{230.94}{1.22376} = 188.7 \text{ A},$$

that is about six times the rated current, which with $\cos \varphi \simeq 0.85$ and efficiency 0.90 is $19\,000/(\sqrt{3} \cdot 400 \cdot 0.85 \cdot 0.90) \simeq 35.9$ A. This is the well-known ratio $I_{\text{st}}/I_n \simeq 6$ of direct-on-line induction motors, and the reason why star-delta starting and soft starters exist.

5.10. Three-phase systems

Definition 5.10: Balanced positive-sequence set. A balanced positive-sequence set of voltages of RMS value V is

$$\underline{V}_a = V, \quad \underline{V}_b = V a^2, \quad \underline{V}_c = V a, \quad a := e^{i2\pi/3},$$

where a is the primitive cube root of unity of Definition 3.5. Equivalently, the three phasors have equal modulus and arguments $0, -120^\circ, +120^\circ$.

Theorem 5.20: Fundamental properties of the balanced set. With $a = e^{i2\pi/3}$:

- (i) $a^3 = 1$, $1 + a + a^2 = 0$, $a^* = a^2$, $|a| = 1$;
- (ii) $\underline{V}_a + \underline{V}_b + \underline{V}_c = 0$, and hence the instantaneous sum $v_a(t) + v_b(t) + v_c(t)$ is identically zero;
- (iii) the line-to-line voltages $\underline{V}_{ab} = \underline{V}_a - \underline{V}_b$ (and cyclically) again form a balanced positive-sequence set, of modulus $\sqrt{3}V$ and leading the corresponding phase voltages by 30° .

Proof. (i) $a^3 = e^{2\pi i} = 1$ by Theorem 3.15; $1 + a + a^2 = 0$ is Corollary 3.5 with $n = 3$; $a^* = e^{-2\pi i/3} = e^{4\pi i/3} = a^2$; $|a| = 1$ because $a \in \mathbb{T}$.

(ii) $\underline{V}_a + \underline{V}_b + \underline{V}_c = V(1 + a^2 + a) = 0$ by (i). By Theorem 3.23(i),(iv) the phasor–sinusoid correspondence is linear and bijective: a zero resultant phasor means an identically zero resultant sinusoid. This is Example 3.9 of Chapter 3, obtained here in one line.

(iii) We compute:

$$\underline{V}_{ab} = \underline{V}_a - \underline{V}_b = V(1 - a^2).$$

Now, by Theorem 3.6,

$$1 - a^2 = 1 - e^{-i2\pi/3} = e^{-i\pi/3}(e^{i\pi/3} - e^{-i\pi/3}) = e^{-i\pi/3} \cdot 2i \sin \frac{\pi}{3} = 2 \sin \frac{\pi}{3} e^{i(\pi/2 - \pi/3)} = \sqrt{3} e^{i\pi/6},$$

using $i = e^{i\pi/2}$ and $2 \sin(\pi/3) = \sqrt{3}$. Hence $\underline{V}_{ab} = \sqrt{3}V e^{i\pi/6}$: modulus $\sqrt{3}V$ and a 30° lead on $\underline{V}_a$. The other two follow by multiplying by a^2 and a , which preserves moduli and argument differences: the line-to-line set is therefore again balanced and of positive sequence. $\square$

Theorem 5.21: Power in a balanced three-phase system. A balanced three-phase load of impedance $\underline{Z} = Z \angle \varphi$ per phase, supplied by a balanced set of line voltage V_l and carrying line current I_l , absorbs

$$P = \sqrt{3} V_l I_l \cos \varphi, \quad Q = \sqrt{3} V_l I_l \sin \varphi, \quad S = \sqrt{3} V_l I_l,$$

both in star and in delta connection. Moreover the total instantaneous power is constant and equal to P .

Proof. Step 1 — star connection. The phase voltage is $V_f = V_l/\sqrt{3}$ by Theorem 5.20(iii), and the phase current equals the line current, $I_f = I_l$. The total active power is the sum of

the three equal phase powers:

$$P = 3V_f I_f \cos \varphi = 3 \cdot \frac{V_l}{\sqrt{3}} \cdot I_l \cos \varphi = \sqrt{3} V_l I_l \cos \varphi,$$

using $3/\sqrt{3} = \sqrt{3}$. Identical reasoning for Q and S .

Step 2 — delta connection. Now $V_f = V_l$ and $I_f = I_l/\sqrt{3}$ (by symmetry, with the same proof as Theorem 5.20(iii) applied to the currents). Hence

$$P = 3V_l I_f \cos \varphi = 3V_l \frac{I_l}{\sqrt{3}} \cos \varphi = \sqrt{3} V_l I_l \cos \varphi,$$

identical to the previous one: the formula is the same in both connections, and this is why it is used without specifying the connection.

Step 3 — constant instantaneous power. Apply Theorem 5.10 to each phase, with phase shifts $0, -2\pi/3, +2\pi/3$:

$$p_k(t) = P_f \left(1 + \cos(2\omega t - 2\gamma_k)\right) + Q_f \sin(2\omega t - 2\gamma_k), \quad \gamma_k \in \left\{0, \frac{2\pi}{3}, -\frac{2\pi}{3}\right\}.$$

Summing over the three phases, the constant terms give $3P_f = P$, while the oscillating terms require the sums

$$\sum_k \cos(2\omega t - 2\gamma_k) \quad \text{and} \quad \sum_k \sin(2\omega t - 2\gamma_k).$$

The three angles $2\gamma_k$ are $0, 4\pi/3, -4\pi/3$, i.e. modulo 2π : $0, -2\pi/3, +2\pi/3$ — again a balanced set. By (ii) of Theorem 5.20 both sums vanish identically. We are left with $p(t) \equiv P$. $\square$

Remark 5.9: The industrial reason for three phases. Step 3 is no technicality: it is the reason the world electrified in three phases. In a single-phase system the instantaneous power pulses between 0 and $2P$ (Theorem 5.10), and a motor so supplied suffers a pulsating torque, with vibration and noise. In a balanced three-phase system the instantaneous power is constant: the torque is smooth, and the machine can be built lighter for the same power. The ultimate reason is algebraic, and it is Corollary 3.5 of Chapter 3: the three cube roots of unity sum to zero.

Theorem 5.22: Aron connection (two-wattmeter method). In a three-wire three-phase system (balanced or not), two wattmeters with current coils on lines a and c and voltage coils referred to line b read W_1 and W_2 with

$$W_1 + W_2 = P.$$

If in addition the system is balanced and symmetric,

$$W_1 = V_l I_l \cos(\varphi - 30^\circ), \quad W_2 = V_l I_l \cos(\varphi + 30^\circ), \quad Q = \sqrt{3}(W_1 - W_2).$$

Proof. **Step 1 — the general case.** The two wattmeters read

$$W_1 = \operatorname{Re} [\underline{V}_{ab} \underline{I}_a^*], \quad W_2 = \operatorname{Re} [\underline{V}_{cb} \underline{I}_c^*].$$

In a three-wire system, Kirchhoff's current law (Theorem 5.4(ii)) applied to the load as a whole gives $\underline{I}_a + \underline{I}_b + \underline{I}_c = 0$, hence $\underline{I}_b = -\underline{I}_a - \underline{I}_c$. The total active power, by Theorem 5.11(i), is

$$P = \operatorname{Re} [\underline{V}_a \underline{I}_a^* + \underline{V}_b \underline{I}_b^* + \underline{V}_c \underline{I}_c^*].$$

Substituting $\underline{I}_b^* = -\underline{I}_a^* - \underline{I}_c^*$ and collecting,

$$P = \operatorname{Re} \left[(\underline{V}_a - \underline{V}_b) \underline{I}_a^* + (\underline{V}_c - \underline{V}_b) \underline{I}_c^* \right] = W_1 + W_2.$$

Note that no symmetry or balance hypothesis was used: the sum of the two wattmeters gives the total active power in *every* three-wire system.

Step 2 — balanced symmetric case. Take $\underline{V}_a = V_f$ real. By Theorem 5.20(iii), $\underline{V}_{ab} = \sqrt{3}V_f e^{i\pi/6} = V_l e^{i\pi/6}$; similarly $\underline{V}_{cb} = \underline{V}_c - \underline{V}_b = V_f(a - a^2) = V_f \cdot i\sqrt{3} = V_l e^{i\pi/2}$, using $a - a^2 = 2i \sin(2\pi/3) = i\sqrt{3}$. The currents are $\underline{I}_a = I_l e^{-i\varphi}$ and $\underline{I}_c = I_l e^{i(2\pi/3-\varphi)}$. Hence, with $\underline{I}_a^* = I_l e^{+i\varphi}$,

$$W_1 = \operatorname{Re} \left[V_l e^{i\pi/6} I_l e^{i\varphi} \right] = V_l I_l \cos(30^\circ + \varphi),$$

$$W_2 = \operatorname{Re} \left[V_l e^{i\pi/2} I_l e^{-i(2\pi/3-\varphi)} \right] = V_l I_l \cos(\varphi - 30^\circ),$$

since $90^\circ - 120^\circ = -30^\circ$. These coincide with the statement up to exchanging the names $W_1 \leftrightarrow W_2$, which depends on which line carries the current coil; from now on we adopt the numbering of the statement, i.e. $W_1 = V_l I_l \cos(\varphi - 30^\circ)$ and $W_2 = V_l I_l \cos(\varphi + 30^\circ)$.

Step 3 — sum and difference. By the sum-to-product formulas,

$$W_1 + W_2 = 2V_l I_l \cos \varphi \cos 30^\circ = \sqrt{3} V_l I_l \cos \varphi = P \quad \checkmark$$

in agreement with Theorem 5.21, and

$$W_1 - W_2 = 2V_l I_l \sin \varphi \sin 30^\circ = V_l I_l \sin \varphi = \frac{Q}{\sqrt{3}},$$

whence $Q = \sqrt{3}(W_1 - W_2)$. □

Example 5.10: Three-phase load: full calculation and Aron check. A balanced star-connected load with $\underline{Z} = 20 \angle 30^\circ \, \Omega$ per phase is supplied at $V_l = 400 \, \text{V}$, 50 Hz.

Step 1 — phase voltage and current.

$$V_f = \frac{400}{\sqrt{3}} = 230.94 \, \text{V}, \quad I_f = \frac{230.94}{20} = 11.547 \, \text{A} = I_l.$$

Step 2 — powers. By Theorem 5.21, with $\cos 30^\circ = 0.866025$ and $\sin 30^\circ = 0.5$:

$$S = \sqrt{3} \times 400 \times 11.547 = 8000.0 \, \text{VA},$$

$$P = 8000 \times 0.866025 = 6928.2 \, \text{W}, \quad Q = 8000 \times 0.5 = 4000.0 \, \text{var}.$$

Check with the power triangle: $\sqrt{6928.2^2 + 4000^2} = \sqrt{64\,000\,000} = 8000 \quad \checkmark$

Step 3 — per-phase check. $P_f = V_f I_f \cos \varphi = 230.94 \times 11.547 \times 0.866025 = 2309.4 \, \text{W}$, and $3P_f = 6928.2 \, \text{W} \quad \checkmark$

Step 4 — Aron connection. By Theorem 5.22, with $\varphi = 30^\circ$:

$$W_1 = 400 \times 11.547 \times \cos(0^\circ) = 4618.8 \, \text{W},$$

$$W_2 = 400 \times 11.547 \times \cos(60^\circ) = 4618.8 \times 0.5 = 2309.4 \, \text{W}.$$

Checks: $W_1 + W_2 = 6928.2 \, \text{W} = P \quad \checkmark$; $\sqrt{3}(W_1 - W_2) = 1.7320508 \times 2309.4 = 4000.0 \, \text{var} = Q \quad \checkmark$

Step 5 — the same load in delta. With $\underline{Z}$ unchanged and delta connection, $V_f = 400 \, \text{V}$, $I_f = 20 \, \text{A}$, $I_l = 20\sqrt{3} = 34.64 \, \text{A}$ and

$$P = \sqrt{3} \times 400 \times 34.64 \times 0.866025 = 20\,784.6 \, \text{W} = 3 \times 6928.2.$$

The delta-connected load absorbs *three times* the power of the same load in star: this is the reason for star–delta starting of induction motors, which reduces power (and torque) to one third during run-up.

Typical error. Using $P = 3V_l I_l \cos \varphi$ instead of $P = \sqrt{3}V_l I_l \cos \varphi$: an error of a factor $\sqrt{3} = 1.732$. The form with $\sqrt{3}$ *already* contains the passage from phase to line quantities.

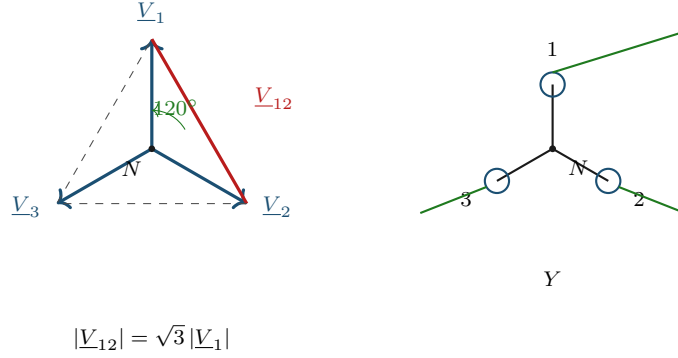


Figure 5.6: The symmetrical three-phase system. Left, the star voltage triple: three phasors of equal modulus, 120° apart; their sum vanishes, because $1 + a + a^2 = 0$ with $a = e^{j2\pi/3}$. It is the same identity that says the cube roots of unity sum to zero, and from it follows that in a balanced system the neutral conductor carries no current. The line voltage $\underline{V}_{12} = \underline{V}_1 - \underline{V}_2$ is the side of the inscribed triangle: its modulus is $\sqrt{3}$ times the phase value and it leads by 30° , whence the 230/400 V pair of the European grid, with $400 \simeq 230\sqrt{3}$. Right, the corresponding star connection.

5.11. Symmetrical components: the three-point DFT

When the three-phase system is unbalanced — through a fault, or an asymmetric load — the three quantities no longer form a balanced set and the results of the previous section fail. The method of *symmetrical components*, introduced by Charles LeGeyt Fortescue in 1918, solves the problem by decomposing an arbitrary set into three particular ones. As we shall now see, this is *exactly* the three-point discrete Fourier transform of Chapter 3.

Definition 5.11: Symmetrical components. Given an arbitrary set $(\underline{V}_a, \underline{V}_b, \underline{V}_c) \in \mathbb{C}^3$ and setting $a = e^{j2\pi/3}$, one defines the **zero-sequence** $(\underline{V}_0)$, **positive-sequence** $(\underline{V}_1)$ and **negative-sequence** $(\underline{V}_2)$ components:

$$\begin{pmatrix} \underline{V}_0 \\ \underline{V}_1 \\ \underline{V}_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{pmatrix} \begin{pmatrix} \underline{V}_a \\ \underline{V}_b \\ \underline{V}_c \end{pmatrix}.$$

Theorem 5.23: Fortescue’s theorem. The transformation of Definition 5.11 is invertible, with

$$\begin{pmatrix} \underline{V}_a \\ \underline{V}_b \\ \underline{V}_c \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{pmatrix} \begin{pmatrix} \underline{V}_0 \\ \underline{V}_1 \\ \underline{V}_2 \end{pmatrix},$$

that is, every set decomposes uniquely as the sum of

- a zero-sequence set $(\underline{V}_0, \underline{V}_0, \underline{V}_0)$, with the three quantities equal;
- a positive-sequence set $(\underline{V}_1, a^2 \underline{V}_1, a \underline{V}_1)$, balanced with sequence $a \rightarrow b \rightarrow c$;

- a negative-sequence set $(\underline{V}_2, a\underline{V}_2, a^2\underline{V}_2)$, balanced with the opposite sequence.

Moreover: the set is balanced of positive sequence if and only if $\underline{V}_0 = \underline{V}_2 = 0$; it has no zero-sequence component if and only if $\underline{V}_a + \underline{V}_b + \underline{V}_c = 0$.

Proof. Step 1 — recognising the DFT matrix. Setting $\zeta = e^{-2\pi i/3} = a^2 = a^{-1}$, the matrix of Definition 3.7 of Chapter 3 with $n = 3$ is

$$W_3 = (\zeta^{jk})_{j,k=0}^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{pmatrix},$$

using $\zeta^0 = 1$, $\zeta^1 = a^2$, $\zeta^2 = a^4 = a$. The matrix of Definition 5.11 is therefore $\frac{1}{3}\overline{W}_3$, and the inverse matrix of the statement is W_3 .

Step 2 — invertibility. By Theorem 3.22 of Chapter 3, $W_3^{-1} = \frac{1}{3}\overline{W}_3$. Hence the forward transformation $\frac{1}{3}\overline{W}_3$ and the inverse one W_3 are mutually inverse, and the theorem is proved — with no computation at all, as a consequence of the orthogonality of the roots of unity.

Step 3 — explicit check, for safety. The product $\frac{1}{3}\overline{W}_3 W_3$ has (j, k) entry

$$\frac{1}{3} \sum_{m=0}^2 \overline{\zeta^{jm}} \zeta^{mk} = \frac{1}{3} \sum_{m=0}^2 \zeta^{m(k-j)},$$

which by Theorem 3.20 equals 1 if $3 \mid (k - j)$, i.e. $j = k$ within the indices $\{0, 1, 2\}$, and 0 otherwise: it is the identity matrix. ✓

Step 4 — structure of the three sets. Expanding the inverse formula row by row:

$$\underline{V}_a = \underline{V}_0 + \underline{V}_1 + \underline{V}_2, \quad \underline{V}_b = \underline{V}_0 + a^2\underline{V}_1 + a\underline{V}_2, \quad \underline{V}_c = \underline{V}_0 + a\underline{V}_1 + a^2\underline{V}_2,$$

which is exactly the sum of the three sets described in the statement.

Step 5 — the two criteria. The set is balanced of positive sequence, i.e. of the form (V, a^2V, aV) , if and only if it coincides with the positive-sequence component alone, i.e. $\underline{V}_0 = \underline{V}_2 = 0$ (by uniqueness of the decomposition, guaranteed by invertibility). Finally, from the first row of the forward transformation, $\underline{V}_0 = \frac{1}{3}(\underline{V}_a + \underline{V}_b + \underline{V}_c)$, which vanishes exactly when the sum vanishes. □

Remark 5.10: An engineering theorem that is an algebra theorem. Theorem 5.23 is, literally, Theorem 3.22 of Chapter 3 with $n = 3$. Fortescue published it in 1918 as an analysis technique for polyphase networks; the discrete Fourier transform was known in other guises, and the fast algorithm would arrive forty-eight years later. It is an instructive example of how the same mathematical object is rediscovered in different domains: here the orthogonality of the cube roots of unity — the mother identity of Observation 3.9 — appears as a diagnostic tool for faults in power networks. The practical advantage is that, for a linear symmetric network, the three sequences propagate independently: the coupled 3×3 system decouples into three single-phase problems, exactly as a Fourier series decouples a differential equation into infinitely many algebraic ones.

Example 5.11: Decomposition of an unbalanced set. Decompose the set $\underline{V}_a = 100\angle 0^\circ$, $\underline{V}_b = 80\angle -100^\circ$, $\underline{V}_c = 120\angle 140^\circ$ (volts).

Step 1 — Cartesian form.

$$\underline{V}_a = 100, \quad \underline{V}_b = -13.892 - i78.785, \quad \underline{V}_c = -91.925 + i77.135.$$

Step 2 — zero-sequence component.

$$\underline{V}_0 = \frac{100 - 13.892 - 91.925 + i(-78.785 + 77.135)}{3} = \frac{-5.817 - i 1.650}{3} = -1.939 - i 0.550,$$

that is, $\underline{V}_0 = 2.016 \angle -164.17^\circ$ V.

Step 3 — positive-sequence component. $a\underline{V}_b = 80 \angle (-100^\circ + 120^\circ) = 80 \angle 20^\circ = 75.175 + i 27.362$; $a^2\underline{V}_c = 120 \angle (140^\circ + 240^\circ) = 120 \angle 20^\circ = 112.763 + i 41.042$. Hence

$$\underline{V}_1 = \frac{100 + 75.175 + 112.763 + i(27.362 + 41.042)}{3} = \frac{287.938 + i 68.404}{3} = 95.979 + i 22.801,$$

that is, $\underline{V}_1 = 98.65 \angle 13.36^\circ$ V.

Step 4 — negative-sequence component. $a^2\underline{V}_b = 80 \angle (-100^\circ + 240^\circ) = 80 \angle 140^\circ = -61.284 + i 51.423$; $a\underline{V}_c = 120 \angle (140^\circ + 120^\circ) = 120 \angle 260^\circ = -20.838 - i 118.177$. Hence

$$\underline{V}_2 = \frac{100 - 61.284 - 20.838 + i(51.423 - 118.177)}{3} = \frac{17.878 - i 66.754}{3} = 5.959 - i 22.251,$$

that is, $\underline{V}_2 = 23.03 \angle -75.00^\circ$ V.

Step 5 — reconstruction check.

$$\underline{V}_0 + \underline{V}_1 + \underline{V}_2 = (-1.939 + 95.979 + 5.959) + i(-0.550 + 22.801 - 22.251) = 99.999 + i 0.000 \approx \underline{V}_a \checkmark$$

Second check, on phase b : $a^2\underline{V}_1 = 98.65 \angle 253.36^\circ = -28.294 - i 94.507$ and $a\underline{V}_2 = 23.03 \angle 45^\circ = 16.285 + i 16.285$; adding $\underline{V}_0$:

$$(-1.939 - 28.294 + 16.285) + i(-0.550 - 94.507 + 16.285) = -13.948 - i 78.772 \approx \underline{V}_b \checkmark$$

(discrepancies in the last digit are rounding).

Step 6 — diagnostic reading. The *negative-sequence unbalance* is $|\underline{V}_2|/|\underline{V}_1| = 23.03/98.65 = 23.3\%$: an enormous value. Standard EN 50160 on voltage quality sets a typical limit of 2% on the negative-to-positive ratio, averaged over ten minutes: this set would therefore correspond to a fault in progress, not to a normal load unbalance. Moreover the non-zero zero-sequence component (2.02 V) shows that the three voltages do not sum to zero, which in a three-wire system without neutral cannot happen: the measured set necessarily comes from a system with a neutral or an earth return.

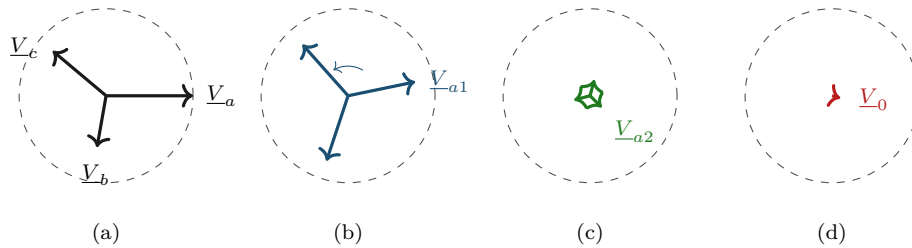


Figure 5.7: Fortescue's decomposition of an unbalanced triple. In (a) the given triple $\underline{V}_a = 1 \angle 0^\circ$, $\underline{V}_b = 0.6 \angle -100^\circ$, $\underline{V}_c = 0.8 \angle 140^\circ$. In (b), (c), (d) the three component triples: the positive sequence ($0.788 \angle 11.7^\circ$, rotating in the a - b - c sense), the negative sequence ($0.189 \angle -45^\circ$, rotating the other way) and the zero sequence ($0.098 \angle -15.2^\circ$, three identical phasors). The formulas $\underline{V}_{0,1,2} = \frac{1}{3}(\underline{V}_a + a^k \underline{V}_b + a^{2k} \underline{V}_c)$ are exactly the three-point discrete Fourier transform, and the reconstruction $\underline{V}_a = \underline{V}_0 + \underline{V}_1 + \underline{V}_2$ is its inverse.

5.12. The per-unit system

So far we have worked in volts, amperes and ohms. In power-system practice one almost always prefers a system of relative quantities, called *per unit* (p.u.), in which every quantity is expressed as a fraction of a reference quantity chosen in advance. The reason is not cosmetic. In per unit, as we shall prove, the $\sqrt{3}$ factors of three-phase systems cancel, the turns ratio of transformers disappears altogether, and the numerical values of machine impedances fall into ranges so narrow that a wrong figure is recognised at a glance.

Definition 5.12: Base quantities and per-unit values. Given a base power S_b (three-phase apparent power, in VA) and a base voltage V_b (line voltage, in V), one defines the base current and the base impedance

$$I_b = \frac{S_b}{\sqrt{3} V_b}, \quad Z_b = \frac{V_b/\sqrt{3}}{I_b} = \frac{V_b^2}{S_b}$$

and the *per-unit* value of a quantity is the ratio of its actual value to the corresponding base quantity:

$$V_{\text{pu}} = \frac{V}{V_b}, \quad I_{\text{pu}} = \frac{I}{I_b}, \quad Z_{\text{pu}} = \frac{Z}{Z_b}, \quad S_{\text{pu}} = \frac{S}{S_b}.$$

Of the four bases only two are independent: once S_b and V_b are fixed, the other two follow. The per-unit value multiplied by 100 is called the *percentage* value.

Theorem 5.24: In per unit the $\sqrt{3}$ factors disappear. With the bases of Definition 5.12:

- (i) the phase voltage and the line voltage have the same per-unit value;
- (ii) $\underline{S}_{\text{pu}} = \underline{V}_{\text{pu}} \underline{I}_{\text{pu}}^*$, with no $\sqrt{3}$ factor;
- (iii) Ohm's law reads $\underline{V}_{\text{pu}} = \underline{Z}_{\text{pu}} \underline{I}_{\text{pu}}$;
- (iv) single-phase and three-phase powers, referred to their respective bases, have the same per-unit value.

Proof. Step 1 — the phase-voltage base. For dimensional consistency with Z_b , the base for star voltages is $V_{b,\text{ph}} = V_b/\sqrt{3}$. In a symmetrical system $V_{\text{ph}} = V_{\text{line}}/\sqrt{3}$ (Theorem 5.20), hence

$$\frac{V_{\text{ph}}}{V_{b,\text{ph}}} = \frac{V_{\text{line}}/\sqrt{3}}{V_b/\sqrt{3}} = \frac{V_{\text{line}}}{V_b},$$

which is (i): one and the same number represents both. This is the first practical benefit, because it removes the commonest mistake in network calculations, namely confusing the two voltages.

Step 2 — power. For a balanced three-phase system $S = \sqrt{3} V_{\text{line}} I$. Dividing by $S_b = \sqrt{3} V_b I_b$:

$$\frac{S}{S_b} = \frac{\sqrt{3} V_{\text{line}} I}{\sqrt{3} V_b I_b} = \frac{V_{\text{line}}}{V_b} \cdot \frac{I}{I_b} = V_{\text{pu}} I_{\text{pu}}.$$

Since the argument of $\underline{S}$ is by definition $\arg \underline{V} - \arg \underline{I}$, and the bases are positive real numbers, which do not alter arguments, the identity also holds in complex form: $\underline{S}_{\text{pu}} = \underline{V}_{\text{pu}} \underline{I}_{\text{pu}}^*$. This is (ii).

Step 3 — Ohm's law. On a single phase $\underline{V}_{\text{ph}} = \underline{Z} \underline{I}$. Dividing by $V_{b,\text{ph}} = Z_b I_b$ (an identity, since Z_b was defined precisely as $V_{b,\text{ph}}/I_b$):

$$\frac{\underline{V}_{\text{ph}}}{V_{b,\text{ph}}} = \frac{\underline{Z} \underline{I}}{Z_b I_b} = \frac{\underline{Z}}{Z_b} \cdot \frac{\underline{I}}{I_b},$$

which is (iii). Note that, by Step 1, the left-hand side is also the per-unit value of the line voltage: Ohm's law in per unit does not require one to declare which of the two voltages is being used.

Step 4 — single-phase and three-phase. If one takes as single-phase base $S_{b,1} = S_b/3$ and as voltage base $V_{b,ph} = V_b/\sqrt{3}$, then the power of one phase $S_1 = V_{ph}I$ gives

$$\frac{S_1}{S_{b,1}} = \frac{V_{ph}I}{S_b/3} = \frac{(V_{line}/\sqrt{3})I}{\sqrt{3}V_bI_b/3} = \frac{V_{line}I}{\sqrt{3}V_bI_b} = \frac{S}{S_b},$$

that is (iv): in per unit the distinction between single-phase and three-phase quantities no longer exists, and single-line diagrams can be read directly as equivalent single-phase circuits. $\square$

Theorem 5.25: Invariance of the per-unit impedance across a transformer. Let an ideal transformer of ratio $n = N_1/N_2$ be given, and choose on the two sides the same base power S_b and base voltages in the turns ratio, $V_{b1}/V_{b2} = n$. Then an impedance $\underline{Z}_2$ connected to the secondary and its image referred to the primary, $\underline{Z}_1 = n^2 \underline{Z}_2$, have the same per-unit value. Consequently, in per unit the ideal transformer is a short circuit, and disappears from the diagram.

Proof. **Step 1 — referring to the primary.** For an ideal transformer $\underline{V}_1 = n\underline{V}_2$ and $\underline{I}_1 = \underline{I}_2/n$, so the impedance seen at the primary is

$$\underline{Z}_1 = \frac{\underline{V}_1}{\underline{I}_1} = \frac{n\underline{V}_2}{\underline{I}_2/n} = n^2 \underline{Z}_2.$$

Step 2 — ratio of the base impedances. From Definition 5.12, $Z_{b1} = V_{b1}^2/S_b$ and $Z_{b2} = V_{b2}^2/S_b$, with the same S_b . Hence

$$\frac{Z_{b1}}{Z_{b2}} = \left(\frac{V_{b1}}{V_{b2}} \right)^2 = n^2.$$

Step 3 — conclusion. Dividing Step 1 by Step 2:

$$\frac{\underline{Z}_1}{Z_{b1}} = \frac{n^2 \underline{Z}_2}{n^2 Z_{b2}} = \frac{\underline{Z}_2}{Z_{b2}},$$

that is $\underline{Z}_{1,pu} = \underline{Z}_{2,pu}$: the factor n^2 of the referral and the factor n^2 of the bases cancel exactly.

Step 4 — the transformer vanishes. Applying the same computation to voltage and current, $\underline{V}_{1,pu} = n\underline{V}_2/(nV_{b2}) = \underline{V}_{2,pu}$ and likewise $\underline{I}_{1,pu} = \underline{I}_{2,pu}$. A two-port having the same voltage and the same current on both sides is, in per unit, a plain connection: the ideal transformer has become invisible. What remains in the diagram is only its short-circuit impedance, which is what matters in network calculations. $\square$

Remark 5.11: Why this is the point. A real network contains dozens of voltage levels. In physical quantities, every transformer crossing forces one to refer all impedances from one side to the other by multiplying or dividing by n^2 , with one opportunity for error at every step. Theorem 5.25 says that, with bases chosen in the ratio of the rated voltages, that work has already been done once and for all: the network in per unit is a single circuit, with no transformers. This is why every power-system computation program works internally in per unit.

Theorem 5.26: Change of base. If an impedance equals Z_{pu}^o on bases (S_b^o, V_b^o) , then on the bases (S_b^n, V_b^n) it equals

$$Z_{\text{pu}}^n = Z_{\text{pu}}^o \left(\frac{V_b^o}{V_b^n} \right)^2 \frac{S_b^n}{S_b^o}$$

Proof. The physical impedance $\underline{Z}$ does not depend on the choice of bases, hence $Z_{\text{pu}}^o Z_b^o = \underline{Z} = Z_{\text{pu}}^n Z_b^n$. Substituting $Z_b = V_b^2/S_b$:

$$Z_{\text{pu}}^n = Z_{\text{pu}}^o \frac{Z_b^o}{Z_b^n} = Z_{\text{pu}}^o \frac{(V_b^o)^2/S_b^o}{(V_b^n)^2/S_b^n} = Z_{\text{pu}}^o \left(\frac{V_b^o}{V_b^n} \right)^2 \frac{S_b^n}{S_b^o}.$$

The formula reads: the per-unit impedance is *proportional* to the base power and *inversely proportional* to the square of the base voltage. $\square$

Example 5.12: A three-level network brought to a common base. Consider: a 50 MVA, 13.8 kV alternator with subtransient reactance $x_d'' = 20\%$ on its own rating; a 60 MVA, 13.8/132 kV step-up transformer with short-circuit voltage $v_{sc} = 11\%$; a 132 kV line of impedance $\underline{Z}_L = 8 + j40 \Omega$. Take as system bases $S_b = 100$ MVA and $V_b = 132$ kV on the line side.

Step 1 — bases at the various levels. The base voltages follow the rated ratio of the transformers: $V_b = 132$ kV on the line and $V_b = 13.8$ kV upstream. The base impedances are

$$Z_b^{(132)} = \frac{132^2}{100} = 174.24 \Omega, \quad Z_b^{(13.8)} = \frac{13.8^2}{100} = 1.9044 \Omega,$$

using the convenient rule $Z_b[\Omega] = (V_b[\text{kV}])^2/S_b[\text{MVA}]$, which is the same formula with the units scaled by 10^6 in numerator and denominator.

Step 2 — alternator. The datum $x_d'' = 0.20$ refers to the bases (50 MVA, 13.8 kV). By Theorem 5.26, with the base voltage unchanged,

$$x_{d,\text{pu}}'' = 0.20 \cdot \left(\frac{13.8}{13.8} \right)^2 \cdot \frac{100}{50} = 0.40 \text{ p.u.}$$

Step 3 — transformer. Likewise,

$$x_{T,\text{pu}} = 0.11 \cdot \frac{100}{60} = 0.18333 \text{ p.u.}$$

and by Theorem 5.25 this value is the same seen from either side: there is no need to decide whether to refer it to 13.8 or to 132 kV.

Step 4 — line. Here one divides by the base impedance of its own level:

$$\underline{Z}_{L,\text{pu}} = \frac{8 + j40}{174.24} = 0.045914 + j0.229568 \text{ p.u.}$$

Step 5 — total impedance. Since in per unit the transformers have vanished, the three elements are simply in series:

$$\underline{Z}_{\text{tot}} = 0.045914 + j(0.40 + 0.18333 + 0.229568) = 0.045914 + j0.812901 \text{ p.u.},$$

of modulus 0.814197. The base current on the line is $I_b = 100 \cdot 10^6 / (\sqrt{3} \cdot 132 \cdot 10^3) = 437.39$ A; hence a three-phase short circuit at the far end of the line, with 1.0 p.u. open-circuit voltage, gives

$$I_{sc} = \frac{1}{0.814197} = 1.22820 \text{ p.u.} = 1.22820 \cdot 437.39 = 537.2 \text{ A},$$

and the short-circuit power is $S_{sc} = 1.2282 \cdot 100 = 122.8$ MVA. Note that the whole computation was carried out without ever referring an impedance across a transformer.

Best practice. Per-unit values of machines fall into very narrow ranges, and this makes them an excellent plausibility check: short-circuit reactance of power transformers $4 \div 15\%$; subtransient reactance of alternators $10 \div 25\%$; synchronous reactance $100 \div 250\%$; leakage reactance of induction motors $15 \div 25\%$. A figure outside these ranges is almost always a base error, not an unusual machine: suspect number one is having forgotten the factor S_b^n/S_b^o of Theorem 5.26.

5.13. The real transformer: equivalent circuit and Kapp diagram

Theorem 5.25 showed that, in per unit, the ideal transformer disappears. The *real* transformer does not: a series impedance remains, and it is what determines the voltage drop, the short-circuit current and the efficiency. This section builds its equivalent circuit, shows how to measure its parameters with two standardised tests, and observes that the locus of the primary voltage phasor, as the power factor varies at constant current, is a *circle*: the Kapp diagram.

Definition 5.13: T equivalent circuit and Steinmetz circuit. Referring all quantities to the primary, a single-phase transformer (or one phase of a three-phase one) is represented by the **T equivalent circuit**: a primary series impedance $R_1 + jX_1$, a shunt magnetising branch $R_{fe} \parallel jX_\mu$, and a referred secondary series impedance $R'_2 + jX'_2$, followed by the ideal ratio $m = N_1/N_2$.

Since the no-load current is of the order of $1 \div 3\%$ of the rated one, one almost always adopts the **Steinmetz circuit** (or Γ circuit), in which the shunt branch is moved to the input terminals and the two series branches are added:

$$\underline{Z}_{sc} = R_{sc} + jX_{sc}, \quad R_{sc} = R_1 + R'_2, \quad X_{sc} = X_1 + X'_2.$$

Theorem 5.27: Parameters from the two standardised tests. Let S_n be the rated power, V_n the rated voltage and $I_n = S_n/V_n$ the rated current (per phase, in a three-phase machine with phase quantities). Then:

- (i) Open-circuit test (secondary open, primary at V_n): the measured power P_0 is essentially the iron loss, and

$$R_{fe} = \frac{V_n^2}{P_0}, \quad I_0 = \frac{S_0}{V_n}, \quad X_\mu = \frac{V_n^2}{\sqrt{S_0^2 - P_0^2}}.$$

- (ii) Short-circuit test (secondary shorted, primary fed at the voltage V_{sc} that makes I_n flow): the measured power P_{sc} is essentially the copper loss at rated load, and

$$u_k = \frac{V_{sc}}{V_n}, \quad Z_{sc} = \frac{V_{sc}}{I_n} = u_k Z_{base}, \quad R_{sc} = \frac{P_{sc}}{I_n^2}, \quad X_{sc} = \sqrt{Z_{sc}^2 - R_{sc}^2},$$

where $Z_{base} = V_n^2/S_n$. In per unit

$$u_r = \frac{P_{sc}}{S_n}, \quad u_x = \sqrt{u_k^2 - u_r^2}.$$

Proof. (i). At no load the current is the magnetising one, of very small amplitude compared with the rated one; the drop across $R_1 + jX_1$ is therefore negligible and the shunt branch sees practically V_n . The active power drawn is dissipated in R_{fe} alone (the magnetising component is purely reactive), hence $P_0 = V_n^2/R_{fe}$, which gives R_{fe} . The apparent power is $S_0 = V_n I_0$

and the reactive one $Q_0 = \sqrt{S_0^2 - P_0^2}$, dissipated in X_μ : $Q_0 = V_n^2/X_\mu$.

(ii). In short circuit the applied voltage is of the order of a few per cent of the rated one, so the core flux, proportional to the voltage, is reduced in the same proportion, and the iron losses, proportional to the square of the flux, are reduced by a factor $u_k^2 \approx 10^{-3}$: they are negligible, and P_{sc} is entirely Joule loss in the windings. For the same reason the shunt branch, carrying a current reduced in the same proportion, may be ignored, and the circuit reduces to $\underline{Z}_{sc}$ alone carrying I_n . Hence $Z_{sc} = V_{sc}/I_n$ and $P_{sc} = R_{sc}I_n^2$, which give R_{sc} and, by Pythagoras, X_{sc} . Dividing $Z_{sc} = V_{sc}/I_n$ by $Z_{base} = V_n/I_n$ gives $Z_{sc}/Z_{base} = V_{sc}/V_n = u_k$; and $R_{sc}/Z_{base} = P_{sc}/(I_n^2 Z_{base}) = P_{sc}/(V_n I_n) = P_{sc}/S_n$. $\square$

Remark 5.12: u_k is a normative definition, not a free design choice. The standard IEEE C57.12.00 defines the percent impedance exactly as in (ii): the voltage, expressed as a percentage of the rated one, that must be applied to one winding to circulate rated current when the other is short-circuited. The same standard fixes the manufacturing tolerance: $\pm 7.5\%$ of the declared value for two-winding transformers. This tolerance must be borne in mind in two opposite calculations: the short-circuit current, to be computed with the minimum u_k (worst case for the equipment), and the load sharing between parallel transformers, to be checked with the maximum admissible difference between their u_k values.

Theorem 5.28: Industrial voltage drop and the Kapp diagram. Let $\underline{V}_2$ be the secondary voltage referred to the primary, taken as reference, and let $\underline{I}$ be the load current of amplitude I and lagging phase angle φ . Then

$$\underline{V}_1 = \underline{V}_2 + \underline{Z}_{sc} \underline{I}.$$

- (i) (circular locus) At constant I and varying φ , the point $\underline{V}_1$ of the complex plane describes the circle of centre $\underline{V}_2$ and radius $Z_{sc}I$.
(ii) (industrial drop) In per unit, with $V_2 = 1$ and $I = 1$,

$$\Delta V\% \simeq u_r \cos \varphi + u_x \sin \varphi + \frac{(u_x \cos \varphi - u_r \sin \varphi)^2}{200},$$

where u_r, u_x are expressed as percentages.

- (iii) (maximum drop) As φ varies, ΔV has maximum u_k , attained for $\tan \varphi = X_{sc}/R_{sc}$, and minimum $-u_k$.

Proof. (i). As φ varies with $|\underline{I}| = I$ fixed, the product $\underline{Z}_{sc}\underline{I}$ has constant modulus $Z_{sc}I$ and varying argument: it therefore describes the circle of centre the origin and radius $Z_{sc}I$. Adding the constant vector $\underline{V}_2$ translates that circle, giving the circle of centre $\underline{V}_2$ and radius $Z_{sc}I$. It is the same mechanism as in Theorem 5.17: a constant modulus and a free argument always give a circle.

(iii), which is immediate from the geometric picture. The modulus $|\underline{V}_1|$ is the distance from the origin of a point running along that circle; the maximum and the minimum of that distance occur at the two points collinear with the origin and the centre, and equal

$$V_2 + Z_{sc}I \quad \text{and} \quad V_2 - Z_{sc}I.$$

In per unit, with $V_2 = 1$ and $I = 1$, the extremes of $\Delta V = |V_1| - 1$ are therefore $\pm Z_{sc} = \pm u_k$. The maximum occurs when $\underline{Z}_{sc}\underline{I}$ is in phase with $\underline{V}_2$, that is when the argument of $\underline{I}$ compensates that of $\underline{Z}_{sc}$: setting $\underline{I} = I e^{-j\varphi}$ and $\underline{Z}_{sc} = Z_{sc} e^{j\vartheta}$ with $\tan \vartheta = X_{sc}/R_{sc}$, this happens for $\varphi = \vartheta$.

(ii). With $\underline{V}_2 = 1$, $\underline{I} = \cos \varphi - j \sin \varphi$ and $\underline{Z}_{sc} = u_r + j u_x$ (in per unit),

$$\underline{Z}_{sc} \underline{I} = (u_r \cos \varphi + u_x \sin \varphi) + j(u_x \cos \varphi - u_r \sin \varphi) =: a + jb.$$

Then $|\underline{V}_1| = \sqrt{(1+a)^2 + b^2} = \sqrt{1 + 2a + a^2 + b^2}$, and since a, b are of order 10^{-2} the expansion $\sqrt{1+\varepsilon} \simeq 1 + \varepsilon/2$ applies:

$$|\underline{V}_1| \simeq 1 + a + \frac{a^2 + b^2}{2} \simeq 1 + a + \frac{b^2}{2},$$

having neglected $a^2/2$, of order 10^{-4} relative to a itself. Expressing a and b as percentages gives the stated formula: the term $b^2/200$ is the celebrated “quadratic term” of the industrial drop. $\square$

Remark 5.13: Why it is a circle, and what knowing that is good for. The Kapp diagram is the graphical translation of Theorem 5.28: draw the vector $\underline{V}_2$, set the compass on its tip with opening $\underline{Z}_{sc} \underline{I}$, and read off the arc the primary voltage needed for each power factor. A single drawing shows that:

- the drop is never greater than u_k , whatever the rated load;
- with a capacitive load ($\varphi < 0$) the drop can become negative: the no-load secondary voltage is lower than the loaded one, which is the Ferranti effect in miniature;
- the drop vanishes at one particular capacitive power factor, the one for which $\underline{Z}_{sc} \underline{I}$ is perpendicular to $\underline{V}_2$.

Once again it is the circle that carries all the information: the locus of a phasor of fixed modulus and free phase.

Example 5.13: A 630 kVA transformer, computed in full. Nameplate data: $S_n = 630$ kVA, 20/0.4 kV, Dyn11, $u_k = 4.00\%$, losses at rated load $P_{sc} = 6500$ W, no-load losses $P_0 = 800$ W. We refer everything to the LV side.

Base quantities.

$$I_n = \frac{630\,000}{\sqrt{3} \cdot 400} = 909.327 \text{ A}, \quad Z_{base} = \frac{400^2}{630\,000} = 0.253968 \, \Omega.$$

Short-circuit impedance.

$$u_r = \frac{6500}{630\,000} = 0.0103175 = 1.03175\%, \quad u_x = \sqrt{0.04^2 - 0.0103175^2} = 0.038646 = 3.8646\%,$$

$$R_{sc} = 0.0103175 \cdot 0.253968 = 2.6203 \text{ m}\Omega, \quad X_{sc} = 0.038646 \cdot 0.253968 = 9.8151 \text{ m}\Omega.$$

Check: $\sqrt{2.6203^2 + 9.8151^2} = \sqrt{6.866 + 96.336} = 10.1589 \text{ m}\Omega$, and $10.1589/253.968 = 0.0400$: it matches u_k .

Voltage drop at $\cos \varphi = 0.80$ lagging. With $\sin \varphi = 0.60$,

$$a = 1.03175 \cdot 0.80 + 3.8646 \cdot 0.60 = 0.82540 + 2.31876 = 3.14416\%,$$

$$b = 3.8646 \cdot 0.80 - 1.03175 \cdot 0.60 = 3.09168 - 0.61905 = 2.47263\%,$$

$$\Delta V\% \simeq 3.14416 + \frac{2.47263^2}{200} = 3.14416 + 0.03057 = 3.17473\%.$$

Exact value: $|\underline{V}_1| = \sqrt{1.0314416^2 + 0.0247263^2} = \sqrt{1.0644832} = 1.0317379$, that is $\Delta V = 3.17379\%$. The error of the industrial formula is $9.4 \cdot 10^{-4}$ percentage points.

Maximum drop (Theorem 5.28(iii)). One has $\tan \vartheta = 9.8151/2.6203 = 3.74578$, hence $\vartheta = 75.05^\circ$ and $\cos \varphi = 0.2579$: at that power factor the drop is exactly $u_k = 4.00\%$. Check with

the industrial formula: $1.03175 \cdot 0.2579 + 3.8646 \cdot 0.96617 = 0.26609 + 3.73385 = 3.99994\%$: it matches.

Kapp diagram. The secondary phase voltage is $400/\sqrt{3} = 230.940$ V and the radius of the circle is $Z_{sc}I_n = 10.1589 \text{ m}\Omega \cdot 909.327 \text{ A} = 9.2378 \text{ V}$: as the power factor varies, the required primary voltage ranges between 221.702 V and 240.178 V, that is between -4% and $+4\%$, as predicted.

Efficiency. At full load and $\cos \varphi = 0.80$,

$$\eta = \frac{630\,000 \cdot 0.80}{630\,000 \cdot 0.80 + 800 + 6500} = \frac{504\,000}{511\,300} = 0.985723 = 98.57\%.$$

Efficiency is maximal when the variable losses equal the constant ones, that is at the load factor

$$x = \sqrt{\frac{P_0}{P_{sc}}} = \sqrt{\frac{800}{6500}} = 0.350823,$$

at which $\eta_{\max} = \frac{176\,815}{178\,415} = 99.10\%$: the transformer works best at little more than a third of the load, which is why in a plant with strongly varying load it pays to split the power over several machines.

Short-circuit current at the LV terminals. Neglecting the upstream network impedance,

$$I_{sc} = \frac{I_n}{u_k} = \frac{909.327}{0.04} = 22.73 \text{ kA}.$$

With the IEEE C57.12.00 tolerance of -7.5% on u_k , that is $u_k = 3.70\%$, one gets $I_{sc} = 24.58 \text{ kA}$: this, and not the nominal value, is what must be used to size the breaking capacity.

5.14. Fault calculation with symmetrical components

Fortescue's Theorem 5.23 gave us a decomposition; now we cash it in. Fault current calculation is the industrial reason why symmetrical components exist, and it is what justifies teaching them: without them an unsymmetrical fault requires solving a coupled three-phase network, with them it reduces to three decoupled single-phase networks connected to one another in different ways according to the type of fault. We shall work in per unit, following Section 5.12.

5.14.1 Why the sequences decouple

Theorem 5.29: Diagonalisation of the cyclic impedance matrix. *Let a cyclically symmetric three-phase element be given, that is one described by*

$$\begin{pmatrix} V_a \\ V_b \\ V_c \end{pmatrix} = \begin{pmatrix} \underline{Z}_s & \underline{Z}_m & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_s & \underline{Z}_m \\ \underline{Z}_m & \underline{Z}_m & \underline{Z}_s \end{pmatrix} \begin{pmatrix} I_a \\ I_b \\ I_c \end{pmatrix}.$$

Then, in the sequence coordinates (V_0, V_1, V_2) of Definition 5.11, the matrix becomes diagonal:

$$\underline{Z}_0 = \underline{Z}_s + 2\underline{Z}_m, \quad \underline{Z}_1 = \underline{Z}_2 = \underline{Z}_s - \underline{Z}_m.$$

The three sequence networks are therefore independent of one another.

Proof. Step 1 — the matrix is circulant. Each row is obtained from the previous one by a cyclic shift of one position: this is precisely the structure studied in Chapter 7 in connection

with cyclic codes, that is the matrix of an operator commuting with the cyclic permutation P of the three phases. Written with the matrix P (which sends $a \rightarrow b \rightarrow c \rightarrow a$), one has $Z = \underline{Z}_s I + \underline{Z}_m (P + P^2)$.

Step 2 — the eigenvectors of P are the Fortescue triples. Since $P^3 = I$, the eigenvalues of P are the cube roots of unity $1, a, a^2$ with $a = e^{j2\pi/3}$, and the eigenvectors are

$$u_0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad u_1 = \begin{pmatrix} 1 \\ a^2 \\ a \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ a \\ a^2 \end{pmatrix},$$

which are exactly the zero-, positive- and negative-sequence triples. The check is immediate: for instance Pu_1 has components $(a^2, a, 1) = a^2(1, a^2, a) = a^2 u_1$ (using $a^3 = 1$), so u_1 is an eigenvector with eigenvalue a^2 . Likewise $Pu_0 = u_0$ and $Pu_2 = a u_2$.

Step 3 — the eigenvalues of Z . If $Pu = \lambda u$ then $P^2 u = \lambda^2 u$, and therefore

$$Zu = (\underline{Z}_s + \underline{Z}_m(\lambda + \lambda^2))u.$$

For $\lambda = 1$: $\lambda + \lambda^2 = 2$, and the eigenvalue is $\underline{Z}_s + 2\underline{Z}_m$. For $\lambda = a$ or $\lambda = a^2$: $\lambda + \lambda^2 = a + a^2 = -1$, because $1 + a + a^2 = 0$ (Theorem 5.20); the eigenvalue is therefore $\underline{Z}_s - \underline{Z}_m$ in both cases.

Step 4 — conclusion. In the basis $\{u_0, u_1, u_2\}$, which is precisely the basis in which symmetrical components are read, the matrix Z is diagonal with entries $\underline{Z}_s + 2\underline{Z}_m$, $\underline{Z}_s - \underline{Z}_m$, $\underline{Z}_s - \underline{Z}_m$. It follows that $\underline{V}_k = \underline{Z}_k \underline{I}_k$ for $k = 0, 1, 2$ separately: the three sequence networks exchange no energy.

Step 5 — what has really been proved. The content of the theorem is not electrotechnical but algebraic: every $n \times n$ circulant matrix is diagonalised by the discrete Fourier matrix, and symmetrical components are the case $n = 3$. It is the same fact that makes the DFT the natural tool for the shift-invariant systems of Chapter 4: the cyclic symmetry of the three phases plays here the role that time invariance plays there. $\square$

***Remark 5.14: Why $\underline{Z}_0$ is different, and usually smaller.** For the positive and negative sequences the sum of the three currents vanishes and the neutral conductor (or earth) carries no current: the two networks see the same geometry and $\underline{Z}_1 = \underline{Z}_2$ for passive elements. For the zero sequence, instead, the three currents are equal and in phase, and must close through earth: the zero-sequence network therefore depends on the earthing arrangement and on the transformer connections (a delta winding interrupts it, a solidly earthed neutral closes it), and its value can differ greatly from $\underline{Z}_1$. It is from this asymmetry that the industrially most important results of the next paragraph arise.*

5.14.2 The four fundamental faults

Let us fix the model. From the fault point one sees three sequence networks, each reduced to its Thévenin equivalent: the positive-sequence network contains the electromotive force $\underline{E}$ (machines generate a positive-sequence symmetrical triple, and only that) in series with $\underline{Z}_1$; the negative- and zero-sequence networks are passive, with impedances $\underline{Z}_2$ and $\underline{Z}_0$. Hence, with the currents flowing *out* of the network into the fault,

$$\underline{V}_1 = \underline{E} - \underline{Z}_1 \underline{I}_1, \quad \underline{V}_2 = -\underline{Z}_2 \underline{I}_2, \quad \underline{V}_0 = -\underline{Z}_0 \underline{I}_0. \quad (\star)$$

Theorem 5.30: Symmetrical three-phase fault. For a solid three-phase short circuit ($\underline{V}_a = \underline{V}_b = \underline{V}_c = 0$) the fault current is, in each phase,

$$\underline{I}_{sc}^{(3)} = \frac{\underline{E}}{\underline{Z}_1}$$

and the negative and zero sequences vanish.

Proof. The three voltages at the fault point form a null triple, which is positive-sequence symmetrical (of zero amplitude); by the uniqueness of the Fortescue decomposition, $\underline{V}_0 = \underline{V}_1 = \underline{V}_2 = 0$. From the second and third of $(\star)$ it follows that $\underline{I}_2 = \underline{I}_0 = 0$, and from the first $\underline{I}_1 = \underline{E}/\underline{Z}_1$. Recomposing, $\underline{I}_a = \underline{I}_1$, $\underline{I}_b = a^2 \underline{I}_1$, $\underline{I}_c = a \underline{I}_1$: the current triple is positive-sequence symmetrical, as it had to be by the symmetry of the problem. $\square$

Theorem 5.31: Single line-to-ground fault. For a solid fault of phase a alone to earth ($\underline{V}_a = 0$, $\underline{I}_b = \underline{I}_c = 0$):

$$\underline{I}_a = \frac{3\underline{E}}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_0}$$

and the three sequence networks turn out to be connected in series.

Proof. **Step 1 — the sequence currents are equal.** Applying the direct transformation with $\underline{I}_b = \underline{I}_c = 0$:

$$\underline{I}_0 = \frac{1}{3}(\underline{I}_a + 0 + 0) = \frac{1}{3}\underline{I}_a, \quad \underline{I}_1 = \frac{1}{3}(\underline{I}_a + a \cdot 0 + a^2 \cdot 0) = \frac{1}{3}\underline{I}_a, \quad \underline{I}_2 = \frac{1}{3}(\underline{I}_a + a^2 \cdot 0 + a \cdot 0) = \frac{1}{3}\underline{I}_a.$$

Hence $\underline{I}_0 = \underline{I}_1 = \underline{I}_2 = \underline{I}_a/3$: a single unknown.

Step 2 — the voltage condition. The second boundary condition, $\underline{V}_a = 0$, reads in sequence coordinates $\underline{V}_0 + \underline{V}_1 + \underline{V}_2 = 0$.

Step 3 — substitution. Inserting $(\star)$ and taking $\underline{I}_1$ as the common unknown:

$$-\underline{Z}_0 \underline{I}_1 + (\underline{E} - \underline{Z}_1 \underline{I}_1) - \underline{Z}_2 \underline{I}_1 = 0 \implies \underline{I}_1 = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_0}.$$

Step 4 — back to the phase. Since $\underline{I}_a = \underline{I}_0 + \underline{I}_1 + \underline{I}_2 = 3\underline{I}_1$, the formula of the statement follows.

Step 5 — circuit reading. The equation of Step 3 says that the same current flows in succession through the three impedances $\underline{Z}_1, \underline{Z}_2, \underline{Z}_0$ and that their total drop equals $\underline{E}$: the three sequence networks are connected *in series*. Each type of fault corresponds to a different way of connecting the three networks, and this is the operational advantage of the method. $\square$

Theorem 5.32: Line-to-line fault. For a solid short circuit between phases b and c only ($\underline{I}_a = 0$, $\underline{I}_b = -\underline{I}_c$, $\underline{V}_b = \underline{V}_c$):

$$\underline{I}_1 = -\underline{I}_2 = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_2}, \quad \underline{I}_0 = 0, \quad |\underline{I}_b| = \frac{\sqrt{3} |\underline{E}|}{|\underline{Z}_1 + \underline{Z}_2|}$$

and the positive- and negative-sequence networks turn out to be connected in parallel (in opposition), while the zero-sequence one does not intervene.

Proof. **Step 1 — no zero sequence.** $\underline{I}_0 = \frac{1}{3}(\underline{I}_a + \underline{I}_b + \underline{I}_c) = \frac{1}{3}(0 + \underline{I}_b - \underline{I}_b) = 0$: with no earth contact there is no zero-sequence current, and the zero-sequence network stays out whatever

$\underline{Z}_0$ may be.

Step 2 — the two remaining sequences. With $\underline{I}_a = 0$ and $\underline{I}_c = -\underline{I}_b$:

$$\underline{I}_1 = \frac{1}{3}(a\underline{I}_b + a^2\underline{I}_c) = \frac{1}{3}(a - a^2)\underline{I}_b, \quad \underline{I}_2 = \frac{1}{3}(a^2\underline{I}_b + a\underline{I}_c) = \frac{1}{3}(a^2 - a)\underline{I}_b = -\underline{I}_1.$$

Since $a - a^2 = 2j \sin(2\pi/3) = j\sqrt{3}$, one has $\underline{I}_1 = \frac{j\sqrt{3}}{3}\underline{I}_b = \frac{j}{\sqrt{3}}\underline{I}_b$, that is $\underline{I}_b = -j\sqrt{3}\underline{I}_1$.

Step 3 — the voltage condition. From $\underline{V}_b = \underline{V}_c$ it follows that

$$\underline{V}_1 - \underline{V}_2 = \frac{1}{3}[(a - a^2)\underline{V}_b + (a^2 - a)\underline{V}_c] = \frac{1}{3}(a - a^2)(\underline{V}_b - \underline{V}_c) = 0,$$

hence $\underline{V}_1 = \underline{V}_2$.

Step 4 — solution. With $(\star)$ and $\underline{I}_2 = -\underline{I}_1$:

$$\underline{E} - \underline{Z}_1\underline{I}_1 = -\underline{Z}_2\underline{I}_2 = \underline{Z}_2\underline{I}_1 \implies \underline{I}_1 = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_2},$$

and by Step 2 $|\underline{I}_b| = \sqrt{3}|\underline{I}_1|$, which is the formula of the statement.

Step 5 — a remarkable comparison. If $\underline{Z}_2 = \underline{Z}_1$, as happens for passive elements and, to a first approximation, for rotating machines in the subtransient regime, then

$$\frac{|\underline{I}_b|}{|\underline{I}_{sc}^{(3)}|} = \frac{\sqrt{3}|\underline{E}|/(2|\underline{Z}_1|)}{|\underline{E}|/|\underline{Z}_1|} = \frac{\sqrt{3}}{2} = 0.866 :$$

the line-to-line fault is always less severe than the three-phase one, exactly in the ratio $\sqrt{3}/2$, independently of the network data. $\square$

Theorem 5.33: Double line-to-ground fault. For a solid short circuit between phases b, c and earth ($\underline{I}_a = 0, \underline{V}_b = \underline{V}_c = 0$):

$$\underline{I}_1 = \frac{\underline{E}}{\underline{Z}_1 + \frac{\underline{Z}_0\underline{Z}_2}{\underline{Z}_0 + \underline{Z}_2}}$$

that is, the positive-sequence network closes on the parallel of the negative- and zero-sequence networks. The current returning from earth equals $3\underline{I}_0$.

Proof. **Step 1 — the sequence voltages are equal.** From $\underline{V}_b = \underline{V}_c = 0$:

$$\underline{V}_0 = \frac{1}{3}\underline{V}_a, \quad \underline{V}_1 = \frac{1}{3}(\underline{V}_a + a \cdot 0 + a^2 \cdot 0) = \frac{1}{3}\underline{V}_a, \quad \underline{V}_2 = \frac{1}{3}\underline{V}_a,$$

hence $\underline{V}_0 = \underline{V}_1 = \underline{V}_2$.

Step 2 — the current condition. From $\underline{I}_a = 0$ it follows that $\underline{I}_0 + \underline{I}_1 + \underline{I}_2 = 0$.

Step 3 — elimination. Setting $\underline{U} := \underline{V}_1 = \underline{E} - \underline{Z}_1\underline{I}_1$, the other two of $(\star)$ give $\underline{I}_2 = -\underline{U}/\underline{Z}_2$ and $\underline{I}_0 = -\underline{U}/\underline{Z}_0$. Substituting into Step 2:

$$\underline{I}_1 = \underline{U} \left(\frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_0} \right) = \frac{\underline{U}}{\underline{Z}_0 \parallel \underline{Z}_2},$$

having written $\underline{Z}_0 \parallel \underline{Z}_2 = \underline{Z}_0\underline{Z}_2/(\underline{Z}_0 + \underline{Z}_2)$.

Step 4 — solution. Substituting $\underline{U} = \underline{E} - \underline{Z}_1\underline{I}_1$ and solving for $\underline{I}_1$:

$$\underline{I}_1(\underline{Z}_0 \parallel \underline{Z}_2) = \underline{E} - \underline{Z}_1\underline{I}_1 \implies \underline{I}_1 = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_0 \parallel \underline{Z}_2},$$

which is the claim. The equivalent circuit is therefore: electromotive force $\underline{E}$, impedance $\underline{Z}_1$ in series, and at the end the parallel of $\underline{Z}_2$ and $\underline{Z}_0$.

Step 5 — earth current. The current entering the ground is $\underline{I}_b + \underline{I}_c = (\underline{I}_a + \underline{I}_b + \underline{I}_c) - \underline{I}_a = 3\underline{I}_0 - 0 = 3\underline{I}_0$, since $\underline{I}_a = 0$. This is the value that sizes the earthing grid, and in practice it is often the most severe of all. $\square$

Example 5.14: The four fault currents on one and the same busbar. A 20 kV busbar is fed from a network with, in per unit on $S_b = 100$ MVA,

$$\underline{Z}_1 = \underline{Z}_2 = j0.20, \quad \underline{Z}_0 = j0.05, \quad \underline{E} = 1.00.$$

The base current is $I_b = 100 \cdot 10^6 / (\sqrt{3} \cdot 20 \cdot 10^3) = 2886.75$ A.

Three-phase. $\underline{I}_{sc}^{(3)} = 1/(j0.20) = -j5.000$ p.u., that is $5.000 \cdot 2886.75 = 14.43$ kA. The short-circuit power is $S_{sc} = 5 \cdot 100 = 500$ MVA.

Line-to-line. $|\underline{I}_b| = \sqrt{3}/0.40 = 4.330$ p.u. = 12.50 kA, exactly 0.866 times the three-phase value, as predicted by Step 5 of Theorem 5.32.

Single line-to-ground. $\underline{I}_a = 3/(j0.45) = -j6.667$ p.u. = 19.25 kA. Note that this current is *larger* than the three-phase one: this happens whenever $|\underline{Z}_0| < |\underline{Z}_1|$, that is typically on busbars fed by transformers with solidly earthed neutral. It is the reason why the breaking capacity of switchgear is not sized on the three-phase fault alone.

Double line-to-ground. Here $\underline{Z}_0 \parallel \underline{Z}_2 = (j0.05)(j0.20)/(j0.25) = j0.04$, hence

$$\underline{I}_1 = \frac{1}{j0.24} = -j4.1667, \quad \underline{U} = 1 - (j0.20)(-j4.1667) = 1 - 0.83333 = 0.16667,$$

$$\underline{I}_2 = -\frac{0.16667}{j0.20} = +j0.83333, \quad \underline{I}_0 = -\frac{0.16667}{j0.05} = +j3.33333.$$

Check: $\underline{I}_0 + \underline{I}_1 + \underline{I}_2 = j(3.33333 - 4.16667 + 0.83333) = 0$ ✓. Recomposing, $\underline{I}_b = \underline{I}_0 + a^2 \underline{I}_1 + a \underline{I}_2 = -4.3301 + j5.0000$, of modulus 6.614 p.u. = 19.09 kA, and $\underline{I}_c$ has the same modulus. The earth current is $|3\underline{I}_0| = 10.000$ p.u. = 28.87 kA: twice the three-phase fault current.

Type of fault	Network connection	p.u.	kA
Three-phase	positive only	5.000	14.43
Line-to-line	positive + negative in series	4.330	12.50
Single line-to-ground	three networks in series	6.667	19.25
Double line-to-ground (phase)	positive + (negative $\parallel$ zero)	6.614	19.09
Double line-to-ground (earth)	$3\underline{I}_0$	10.000	28.87

Warning. The zero sequence is not a theoretical curiosity: in the table of the example it is responsible for the two highest values. Neglecting it, or assuming $\underline{Z}_0 = \underline{Z}_1$ “for simplicity”, underestimates the single line-to-ground fault current by 33% and the earth current by 100%. The zero-sequence network must be built separately, with attention to the transformer winding connections and to the neutral earthing arrangement.

5.15. Harmonics and power quality

Everything so far assumes that voltages and currents are sinusoidal. Modern loads — switch-mode power supplies, inverter drives, LED lighting — instead draw heavily distorted currents, even under a practically sinusoidal voltage. Phasor theory no longer applies directly, but it is not to be thrown away: it suffices to apply it *to each harmonic*, and the Fourier series of Chapter 4 supplies exactly the tool. The outcome is that the power factor splits into two distinct factors, one of displacement and one of distortion, and that harmonics of order a multiple of three behave as zero-sequence components — bringing us back to Fortescue’s theorem.

Definition 5.14: Non-sinusoidal periodic quantities. Let

$$v(t) = \sqrt{2} \sum_{h \geq 1} V_h \sin(h\omega t + \alpha_h), \quad i(t) = \sqrt{2} \sum_{h \geq 1} I_h \sin(h\omega t + \beta_h)$$

be the Fourier expansions of two periodic quantities with zero mean, where $V_h, I_h \geq 0$ are the rms values of the individual harmonics. One defines the total rms values and the *total harmonic distortion*

$$V = \sqrt{\sum_{h \geq 1} V_h^2}, \quad I = \sqrt{\sum_{h \geq 1} I_h^2}, \quad \text{THD}_I = \frac{\sqrt{\sum_{h \geq 2} I_h^2}}{I_1}.$$

Theorem 5.34: Active power under distorted conditions. With the notation of Definition 5.14, the active power is

$$P = \sum_{h \geq 1} V_h I_h \cos(\alpha_h - \beta_h)$$

that is, the sum of the powers of the individual harmonics: harmonics of different order exchange no active power. In particular, if the voltage is purely sinusoidal ($V_h = 0$ for $h \geq 2$), then $P = V_1 I_1 \cos \varphi_1$ with $\varphi_1 = \alpha_1 - \beta_1$: only the fundamental of the current does work.

Proof. Step 1 — definition. By definition $P = \frac{1}{T} \int_0^T v(t)i(t) dt$ with $T = 2\pi/\omega$.

Step 2 — product of the series. Substituting the expansions and interchanging series and integral (legitimate because the series converge in $L^2(0, T)$ and the inner product is continuous, Chapter 4):

$$P = 2 \sum_{h \geq 1} \sum_{k \geq 1} V_h I_k \frac{1}{T} \int_0^T \sin(h\omega t + \alpha_h) \sin(k\omega t + \beta_k) dt.$$

Step 3 — orthogonality. By the product-to-sum formulas,

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)].$$

The second term has angular frequency $(h + k)\omega \neq 0$ and zero mean over one period. The first has angular frequency $(h - k)\omega$: if $h \neq k$ its mean is zero, if $h = k$ it is the constant $\frac{1}{2} \cos(\alpha_h - \beta_h)$. Hence

$$\frac{1}{T} \int_0^T \sin(h\omega t + \alpha_h) \sin(k\omega t + \beta_k) dt = \begin{cases} \frac{1}{2} \cos(\alpha_h - \beta_h) & h = k \\ 0 & h \neq k. \end{cases}$$

Step 4 — conclusion. Only the diagonal terms survive and the factor 2 cancels the $\frac{1}{2}$: $P = \sum_h V_h I_h \cos(\alpha_h - \beta_h)$. The final statement is the case in which the sum reduces to the single term $h = 1$.

Step 5 — the meaning. The orthogonality of the trigonometric system, which in Chapter 4 was a computational device, is here a physical assertion: a fifth-harmonic current, under a pure 50 Hz voltage, carries *exactly zero* mean active power. It does no work, yet it travels along the line, heats it and occupies transformer capacity. $\square$

Theorem 5.35: Decomposition of the power factor. Under sinusoidal voltage and distorted current, the apparent power is $S = V_1 I$ and the power factor splits into the product

of a displacement factor and a distortion factor:

$$\text{PF} = \frac{P}{S} = \underbrace{\cos \varphi_1}_{\text{displacement}} \cdot \underbrace{\frac{I_1}{I}}_{\text{distortion}} = \frac{\cos \varphi_1}{\sqrt{1 + \text{THD}_I^2}}$$

In particular $\text{PF} \leq \cos \varphi_1$, with equality if and only if $\text{THD}_I = 0$: distortion always worsens the power factor, and no capacitor can correct it.

Proof. Step 1 — active power. By Theorem 5.34 with $V_h = 0$ for $h \geq 2$: $P = V_1 I_1 \cos \varphi_1$.

Step 2 — apparent power. By definition $S = V \cdot I$ with V, I the total rms values; since the voltage is sinusoidal, $V = V_1$, hence $S = V_1 I$.

Step 3 — ratio.

$$\text{PF} = \frac{P}{S} = \frac{V_1 I_1 \cos \varphi_1}{V_1 I} = \frac{I_1}{I} \cos \varphi_1.$$

Step 4 — the distortion factor in terms of the THD. From Definition 5.14,

$$I^2 = I_1^2 + \sum_{h \geq 2} I_h^2 = I_1^2 (1 + \text{THD}_I^2) \implies \frac{I_1}{I} = \frac{1}{\sqrt{1 + \text{THD}_I^2}}.$$

Step 5 — inequality. Since $\text{THD}_I \geq 0$, we have $1/\sqrt{1 + \text{THD}_I^2} \leq 1$ with equality if and only if $\text{THD}_I = 0$; hence $\text{PF} \leq \cos \varphi_1$ with the same equality condition.

Step 6 — why a capacitor is not enough. The power-factor correction of Section 5.5 acts on the reactive component of the *fundamental*, that is it can bring $\cos \varphi_1$ to 1; but it leaves the factor I_1/I untouched. With $\text{THD}_I = 0.80$, for example, the power factor cannot exceed $1/\sqrt{1.64} = 0.781$ even with perfect correction. Worse: a parallel capacitor forms with the network inductance a resonant circuit (Section 5.6), and if its resonant frequency falls near a harmonic that is present, it amplifies it instead of attenuating it. This is why power-factor correction banks in the presence of harmonics are installed *detuned*, with a series reactor bringing the resonance below the fifth harmonic. $\square$

Theorem 5.36: Sequence of the harmonics and neutral current. In a three-phase system with balanced loads and identical waveforms shifted by one third of a period, the triple of the h -th harmonics is

positive-sequence if $h \equiv 1 \pmod{3}$, negative-sequence if $h \equiv 2 \pmod{3}$, zero-sequence if $h \equiv 0 \pmod{3}$

Consequently, in the neutral conductor of a four-wire system the harmonics that are multiples of three do not cancel but add, and

$$I_N = 3 \sqrt{\sum_{h \equiv 0 \pmod{3}} I_h^2}$$

Proof. Step 1 — the model. If $i_a(t)$ is the current of phase a , the other two are $i_b(t) = i_a(t - T/3)$ and $i_c(t) = i_a(t - 2T/3)$: this is the definition of a balanced system with identical loads. For the h -th harmonic, a delay of $T/3$ corresponds to a phase shift of $h \cdot 2\pi/3$.

Step 2 — the three phasors of the h -th harmonic. Let $\underline{I}_h$ be the phasor of the h -th harmonic of phase a ; those of phases b and c are

$$\underline{I}_{h,b} = \underline{I}_h e^{-jh2\pi/3} = \underline{I}_h (a^{-1})^h = \underline{I}_h a^{-h}, \quad \underline{I}_{h,c} = \underline{I}_h a^{-2h},$$

with $a = e^{j2\pi/3}$.

Step 3 — classification. Since $a^3 = 1$, the exponent counts only modulo 3:

- $h \equiv 1$: $a^{-h} = a^{-1} = a^2$, so the triple is $(\underline{I}_h, a^2 \underline{I}_h, a \underline{I}_h)$, that is symmetrical *positive-sequence* (Definition 5.10);
- $h \equiv 2$: $a^{-h} = a^{-2} = a$, and the triple is $(\underline{I}_h, a \underline{I}_h, a^2 \underline{I}_h)$, symmetrical *negative-sequence*;
- $h \equiv 0$: $a^{-h} = 1$, and the three currents are *identical*: a zero-sequence triple.

The first harmonics therefore give: $h = 1$ positive, $h = 2$ negative, $h = 3$ zero, $h = 4$ positive, $h = 5$ negative, $h = 7$ positive, $h = 9$ zero, and so on.

Step 4 — the neutral. The neutral current is $i_N = i_a + i_b + i_c$. By Step 3, the harmonics with $h \neq 0$ form symmetrical triples, whose sum vanishes because $1 + a + a^2 = 0$ (Theorem 5.20); only the harmonics with $3 \mid h$ remain, for which the three currents are equal and the sum is 3 times the phase value. Passing to rms values, and using once more the orthogonality of the harmonics (Step 3 of Theorem 5.34), one obtains the formula of the statement.

Step 5 — sizing consequence. The neutral current can *exceed* the phase current. If the triplen harmonics weigh more than one third of the phase rms value, that is if $\sqrt{\sum_{3|h} I_h^2} > I/3$, then $I_N > I$. For a typical non-linear single-phase load (bridge rectifier with smoothing capacitor) the third harmonic reaches $60 \div 70\%$ of the fundamental, and the condition is amply satisfied: this is why in switchboards feeding many electronic loads the neutral is sized with a cross-section equal to or larger than that of the phases, contrary to traditional practice. $\square$

Example 5.15: An electronic load and its neutral. A balanced three-phase load of electronic equipment draws, per phase, the harmonic currents

$$I_1 = 100 \text{ A}, \quad I_3 = 60 \text{ A}, \quad I_5 = 25 \text{ A}, \quad I_7 = 12 \text{ A},$$

with $\cos \varphi_1 = 0.95$ and a practically sinusoidal phase voltage of 230 V.

Step 1 — total rms value.

$$I = \sqrt{100^2 + 60^2 + 25^2 + 12^2} = \sqrt{10\,000 + 3\,600 + 625 + 144} = \sqrt{14\,369} = 119.87 \text{ A}.$$

Step 2 — distortion.

$$\text{THD}_I = \frac{\sqrt{3\,600 + 625 + 144}}{100} = \frac{\sqrt{4\,369}}{100} = \frac{66.098}{100} = 66.10\%.$$

Step 3 — powers. By Theorem 5.34, $P = 3 \cdot 230 \cdot 100 \cdot 0.95 = 65.55 \text{ kW}$ (fundamental only). The apparent power is $S = 3 \cdot 230 \cdot 119.87 = 82.71 \text{ kVA}$.

Step 4 — power factor.

$$\text{PF} = \frac{65.55}{82.71} = 0.7925,$$

in agreement with Theorem 5.35: $0.95/\sqrt{1 + 0.6610^2} = 0.95/1.19871 = 0.7925 \checkmark$. Perfect correction of the fundamental would raise it at most to $1/1.19871 = 0.8342$: the remaining 17 percentage points can be recovered only by filtering the harmonics.

Step 5 — neutral current. The only triplen harmonic present is the third, hence

$$I_N = 3 \cdot 60 = 180 \text{ A},$$

that is 1.50 times the phase current 119.87 A and 1.80 times the fundamental. A neutral sized at half the phase cross-section, as is customary for linear loads, would carry three times the current for which it was chosen.

Step 6 — the standards. IEC 61000-3-2 sets the harmonic emission limits for equipment; IEC 61000-2-4 and EN 50160 set the compatibility levels and the quality of the supplied voltage; IEEE 519-2022 sets the distortion limits at the point of common coupling as a function of the ratio between short-circuit current and load current. None of them limits PF directly: they limit the *harmonics*, because it is the distorted current, not the resulting power factor, that damages the network.

INFO

The thread linking this section to the rest of the chapter is always the same circle. The orthogonality of the harmonics in Theorem 5.34 is the orthogonality of the characters $e^{jh\theta}$ of the circle; the classification of the sequences in Theorem 5.36 is the computation of a^{-h} with a a cube root of unity, that is the same arithmetic on $\mathbb{Z}/3\mathbb{Z}$ that underpins Fortescue's theorem. The most insidious industrial phenomenon of the modern grid — the overloaded neutral — is therefore a corollary of the fact that 3 divides 9 but does not divide 5.

5.16. Voltage quality beyond harmonics

Harmonics are only one chapter of voltage quality. The standard IEC 61000-4-30 lists the parameters to be measured and *how* to measure them; EN 50160 fixes the limits that the voltage supplied by a public network must satisfy. This section runs through the other parameters, with particular attention to the two that can be read directly with the tools already built in this chapter: **unbalance**, which is the ratio of two Fortescue components, and **flicker**, which is a statistic over time windows.

Definition 5.15: The parameters of voltage quality. IEC 61000-4-30 (*Testing and measurement techniques — Power quality measurement methods*) defines methods and accuracy classes (Class A for contractual and arbitration purposes, Class S for statistical surveys) for the following parameters:

1. power **frequency**;
2. supply voltage **magnitude**;
3. **flicker** (short-term severity P_{st} and long-term severity P_{lt});
4. **dips, swells and interruptions**;
5. voltage **transients**;
6. supply voltage **unbalance**;
7. **harmonics and interharmonics**;
8. **mains signalling** voltages;
9. **rapid voltage changes**.

The basic measurement interval is 10 cycles at 50 Hz (200 ms), then aggregated over 3 s, 10 min and 2 h.

Theorem 5.37: Unbalance as a ratio of symmetrical components. Let $(\underline{V}_{ab}, \underline{V}_{bc}, \underline{V}_{ca})$ be a set of line-to-line voltages, and let $\underline{V}^{(1)}$ and $\underline{V}^{(2)}$ be its positive- and negative-sequence components (Theorem 5.23). The **unbalance factor** is

$$u_2 = \frac{|\underline{V}^{(2)}|}{|\underline{V}^{(1)}|}.$$

If only the magnitudes V_{ab}, V_{bc}, V_{ca} of the three phasors are known, then

$$u_2 = \sqrt{\frac{1 - \sqrt{3 - 6\beta}}{1 + \sqrt{3 - 6\beta}}}, \quad \beta = \frac{V_{ab}^4 + V_{bc}^4 + V_{ca}^4}{(V_{ab}^2 + V_{bc}^2 + V_{ca}^2)^2}.$$

Proof. Why the magnitudes alone suffice. The three line voltages identically satisfy $\underline{V}_{ab} + \underline{V}_{bc} + \underline{V}_{ca} = 0$: they are the three sides of a closed triangle. Given the three magnitudes, the triangle is determined up to rotation and one reflection; the rotation does not change u_2 (it multiplies both components by the same unit phasor), and the reflection swaps positive and negative sequence, hence changes u_2 into $1/u_2$: with the cyclic order $a \rightarrow b \rightarrow c$ fixed, the value is

unique. The formula therefore expresses a quantity genuinely determined by the magnitudes alone.

The computation. The closure $\sum \underline{V} = 0$ says that the zero-sequence component vanishes, hence $\underline{V}_{ab} = \underline{V}^{(1)} + \underline{V}^{(2)}$ and the other two are obtained by multiplying by α^2, α and α, α^2 respectively, with $\alpha = e^{j2\pi/3}$. Setting $p = |\underline{V}^{(1)}|$, $q = |\underline{V}^{(2)}|$ and δ the phase difference between the two components, the law of cosines gives

$$V_{ab}^2 = p^2 + q^2 + 2pq \cos \delta, \quad V_{bc}^2 = p^2 + q^2 + 2pq \cos(\delta - \frac{2\pi}{3}), \quad V_{ca}^2 = p^2 + q^2 + 2pq \cos(\delta + \frac{2\pi}{3}).$$

Adding and using $\sum \cos(\delta + k\frac{2\pi}{3}) = 0$,

$$\Sigma_2 := V_{ab}^2 + V_{bc}^2 + V_{ca}^2 = 3(p^2 + q^2).$$

Squaring each and adding, with $\sum \cos^2(\delta + k\frac{2\pi}{3}) = \frac{3}{2}$,

$$\Sigma_4 := V_{ab}^4 + V_{bc}^4 + V_{ca}^4 = 3(p^2 + q^2)^2 + 4p^2q^2 \cdot \frac{3}{2} = 3(p^2 + q^2)^2 + 6p^2q^2.$$

Hence, setting $u = q/p$,

$$\beta = \frac{\Sigma_4}{\Sigma_2^2} = \frac{3(p^2 + q^2)^2 + 6p^2q^2}{9(p^2 + q^2)^2} = \frac{1}{3} + \frac{2}{3} \frac{p^2q^2}{(p^2 + q^2)^2} = \frac{1}{3} + \frac{2}{3} \frac{u^2}{(1 + u^2)^2}.$$

Solving for u : $3 - 6\beta = 1 - \frac{4u^2}{(1 + u^2)^2} = \left(\frac{1 - u^2}{1 + u^2} \right)^2$, hence, since $0 \leq u \leq 1$,

$$\sqrt{3 - 6\beta} = \frac{1 - u^2}{1 + u^2} \implies u^2 = \frac{1 - \sqrt{3 - 6\beta}}{1 + \sqrt{3 - 6\beta}},$$

which is the assertion. □

Remark 5.15: The normative limit, and why unbalance is dangerous. EN 50160 prescribes that, under normal operating conditions, the unbalance factor shall not exceed 2% for 95% of the 10-minute mean rms values of each weekly period (in some areas with a heavy presence of single-phase loads the limit rises to 3%).

The reason for the limit lies in Theorem 5.18: the negative-sequence component produces, in an induction motor, a rotating field in the opposite direction, that is a slip close to 2. At that slip the rotor impedance is very low, and a small negative-sequence voltage produces a large negative-sequence current: with the typical ratio $Z^{(1)}/Z^{(2)} \approx 6$, a 2% voltage unbalance generates a 12% current unbalance, and the extra Joule loss forces the machine to be derated. The NEMA rule of thumb is to derate a motor by 5% for a 2% unbalance and by 18% for a 5% unbalance.

Definition 5.16: Flicker severity. Flicker is the visual perception of the luminous fluctuation caused by variations of the supply voltage. The flickermeter of IEC 61000-4-15 yields the **short-term severity** P_{st} , evaluated over 10 minutes and normalised so that $P_{st} = 1$ corresponds to the irritability threshold of 50% of observers. The **long-term severity** is the cubic mean of twelve consecutive values, that is over two hours:

$$P_{lt} = \sqrt[3]{\frac{1}{12} \sum_{i=1}^{12} P_{st,i}^3}.$$

Remark 5.16: Why the mean is cubic. The cubic mean is not an arbitrary convention: it gives intense episodes far more weight than the arithmetic mean would, reproducing the experimental fact that perceived annoyance grows more than linearly with the amplitude of the fluctuation. One practical consequence must be kept in mind: only two bad ten-minute intervals out of twelve can push P_{lt} above the limit even if the other ten are amply compliant, as Example 5.16 shows. EN 50160 prescribes $P_{lt} \leq 1$ for 95% of the time over each weekly period.

Example 5.16: Unbalance and flicker on an industrial switchboard. *Unbalance.* On an LV switchboard the three line voltages are measured, as 10-minute means,

$$V_{ab} = 402 \text{ V}, \quad V_{bc} = 395 \text{ V}, \quad V_{ca} = 410 \text{ V}.$$

One has

$$\Sigma_2 = 161\,604 + 156\,025 + 168\,100 = 485\,729,$$

$$\Sigma_4 = 26\,115\,852\,816 + 24\,343\,800\,625 + 28\,257\,610\,000 = 78\,717\,263\,441,$$

$$\beta = \frac{78\,717\,263\,441}{485\,729^2} = \frac{78\,717\,263\,441}{235\,932\,661\,441} = 0.3336429.$$

Then $3 - 6\beta = 0.9981424$, $\sqrt{3 - 6\beta} = 0.9990708$, and

$$u_2 = \sqrt{\frac{1 - 0.9990708}{1 + 0.9990708}} = \sqrt{\frac{0.0009292}{1.9990708}} = \sqrt{0.00046483} = 0.021560 = 2.16\%.$$

The value *exceeds* the 2% limit of EN 50160.

Independent check with Fortescue. Closing the triangle with $\underline{V}_{ab} = 402\angle 0^\circ$ gives $\cos \theta = (168\,100 - 317\,629)/317\,580 = -0.470838$, that is $\underline{V}_{bc} = 395\angle(-118.089^\circ) = -185.98 - j348.48$ and $\underline{V}_{ca} = -216.02 + j348.48$. Applying Theorem 5.23,

$$\underline{V}^{(1)} = 402.19 + j8.67, \quad |\underline{V}^{(1)}| = 402.28, \quad \underline{V}^{(2)} = -0.19 - j8.67, \quad |\underline{V}^{(2)}| = 8.674,$$

and $u_2 = 8.674/402.28 = 0.021562 = 2.16\%$: it matches the magnitude-only formula.

Flicker. Over the same two hours the flickermeter records twelve P_{st} values: ten equal to 0.80 and two equal to 1.60 (the two starts of a welding set). Then

$$\sum_i P_{st,i}^3 = 10 \cdot 0.512 + 2 \cdot 4.096 = 5.120 + 8.192 = 13.312,$$

$$P_{lt} = \sqrt[3]{\frac{13.312}{12}} = \sqrt[3]{1.109333} = 1.0352.$$

The value *exceeds* the unit limit, even though ten readings out of twelve are at 80% of it: this is precisely the effect of the cubic mean described in Remark 5.16. The arithmetic mean would have given 0.9333, that is a false “compliant”.

Overall reading. The switchboard shows two independent non-conformities: a structural one (the unbalance, corrected by redistributing the single-phase loads over the three phases) and an intermittent one (the flicker, corrected by stiffening the point of common coupling or by installing a static compensator). Harmonics, measured separately with the methods of Section 5.15, must be assessed as a third independent parameter: compliance on harmonics says nothing about the other two.

Remark 5.17: Voltage dips: what the standard measures and what it does not promise. A **dip** is a temporary reduction of the rms voltage below 90% of the declared value, lasting between 10 ms and 1 minute; below 5% one speaks of an interruption. IEC 61000-4-30 prescribes characterising each event by two numbers only — the residual voltage (the minimum rms value over a one-cycle window updated every half cycle) and the duration — and classifying it in a two-entry table by residual-voltage class ($90 \div 80$, $80 \div 70$, $70 \div 40$, $40 \div 5$, $< 5\%$) and duration class ($10 \div 200$ ms, $200 \div 500$ ms, $0.5 \div 1$ s, $1 \div 5$ s, $5 \div 60$ s). The point to stress is that EN 50160 fixes no limit on the number of dips: it merely indicates that in a year there may be from a few tens to a few hundreds, most of them shorter than a second and with residual voltage above 40%. The reason is that dips are largely caused by faults on the transmission network, that is by weather, and are not controllable by the distributor. A user who cannot tolerate them must provide for it, and the choice is made by comparing the measured dip table with the immunity curve of one's own equipment (ITIC/CBEMA or IEC 61000-4-11): it is a problem of coordination, not of compliance.

5.17. Numerical exercises

All exercises are fully solved in Section 5.19, with an independent check and a note on the typical error. Unless stated otherwise: $f = 50$ Hz, RMS values, load sign convention.

Exercise 5.2: N1 — From sinusoid to phasor. Given $v(t) = 311 \cos(314t - 30^\circ)$ V, find the RMS value, the frequency, the period and the phasor in RMS value. State by how much the instantaneous voltage exceeds the RMS value at its maximum.

Exercise 5.3: N2 — Series RL branch. A series branch with $R = 40 \Omega$ and $L = 0.1$ H is supplied at $V = 230$ V. Compute X_L , $\underline{Z}$ in polar form, the current, the three powers and the power factor. Verify the power triangle.

Exercise 5.4: N3 — Parallel RC. A resistor $R = 100 \Omega$ and a capacitor $C = 20 \mu\text{F}$ are in parallel under $V = 230$ V. Compute the admittance, the equivalent impedance, the total current and the powers, working *in admittance* rather than reducing the parallel to a series impedance.

Exercise 5.5: N4 — Power-factor correction. A single-phase load absorbs $P = 5$ kW at $V = 230$ V with $\cos \varphi_1 = 0.6$ inductive. Find the capacitance needed to raise the power factor to 0.9, the line currents before and after, and the percentage reduction of line losses.

Exercise 5.6: N5 — Design of a resonant circuit. Design a series RLC branch resonating at $f_0 = 1$ kHz with $L = 20$ mH and $R = 5 \Omega$. Find C , the quality factor, the bandwidth in rad/s and in Hz, and the overvoltage across the capacitor for a 10 V supply.

Exercise 5.7: N6 — A circle diagram. A series branch has fixed capacitive reactance $X = -25 \Omega$ and variable resistance R . Find the centre and radius of the admittance locus, and verify that the points corresponding to $R = 0$, $R = 25 \Omega$ and $R = 50 \Omega$ lie on the circle. State for which R the active power absorbed is maximal, at fixed voltage.

Exercise 5.8: N7 — Delta-connected three-phase load. A balanced delta-connected load with $\underline{Z} = 30\angle -40^\circ \Omega$ per phase is supplied at $V_l = 400$ V. Compute the phase and line currents, the three powers, and state whether the load is inductive or capacitive.

Exercise 5.9: N8 — Aron connection on the previous load. For the load of Exercise N7, compute the readings of the two wattmeters in the Aron connection and verify that their sum gives P and that $\sqrt{3}$ times their difference gives Q . Comment on the sign of each reading.

Exercise 5.10: N9 — Symmetrical components with one phase lost. A set has $\underline{V}_a = 200\angle 0^\circ$, $\underline{V}_b = 200\angle -120^\circ$, $\underline{V}_c = 0$ (interruption of phase c). Compute the three symmetrical components, verify the reconstruction and comment on the negative-to-positive sequence ratio.

Exercise 5.11: N10 — Three-phase correction and contractual penalty. A three-phase motor absorbs $P = 30$ kW with $\cos \varphi = 0.8$ at $V_l = 400$ V, for 8 hours a day and 250 days a year. (a) Compute the annual active and reactive energy. (b) Knowing that the contract imposes a penalty if the kvarh/kWh ratio exceeds 0.5, state whether the penalty applies. (c) Size the correction bank, both in delta and in star, to bring the ratio back to exactly 0.5.

Exercise 5.12: N11 — Network in per unit and fault currents. A power station consists of a 40 MVA, 11 kV alternator with $x_d'' = 18\%$, connected through a 40 MVA, 11/66 kV transformer with $v_{sc} = 9\%$ to a 66 kV busbar. Taking $S_b = 100$ MVA and $V_b = 66$ kV, and $\underline{Z}_2 = \underline{Z}_1$, $\underline{Z}_0 = j0.06$ p.u. as seen from the busbar, compute: (a) the per-unit reactances on the system base; (b) the three-phase short-circuit current at the busbar, in p.u. and in kA; (c) the single line-to-ground fault current; (d) the ratio of the two and its theoretical upper limit.

Exercise 5.13: N12 — Mechanical characteristic of an induction motor. A three-phase 6-pole, 400 V, 50 Hz induction motor has $X = 0.9 \Omega$ and $R'_2 = 0.135 \Omega$ per phase, with R_1 negligible. Compute: (a) the synchronous speed and ω_s ; (b) the breakdown slip and torque; (c) torque and mechanical power at $s = 0.04$; (d) the starting torque; (e) the additional referred resistance that brings the maximum torque to standstill.

5.18. Symbolic exercises

These ten exercises require proofs. The solutions are in Section 5.20.

Exercise 5.14: S1 — Passivity and reflection coefficient. Let $\underline{Z} = R + iX$ be the impedance of a passive one-port and $Z_0 > 0$ a real number. Prove that

$$R \geq 0 \iff P \geq 0 \iff |\Gamma| \leq 1, \quad \Gamma := \frac{\underline{Z} - Z_0}{\underline{Z} + Z_0}.$$

Recognise Γ as a Möbius transformation and deduce, using Corollary 3.3 of Chapter 3, that the imaginary axis of the $\underline{Z}$ -plane corresponds to the unit circle of the Γ -plane.

Exercise 5.15: S2 — Maximum power transfer. A real source of electromotive force $\underline{E}$ and internal impedance $\underline{Z}_s = R_s + iX_s$ (with $R_s > 0$) feeds a load $\underline{Z}_L = R_L + iX_L$. Prove that the active power delivered to the load is maximal for $\underline{Z}_L = \underline{Z}_s^*$, and that

$$P_{\max} = \frac{|\underline{E}|^2}{4R_s}.$$

Discuss why this criterion is *not* used in power systems.

Exercise 5.16: S3 — The two counter-rotating phasors. Prove that every real sinusoidal quantity is the sum of two conjugate counter-rotating phasors,

$$a(t) = \frac{1}{\sqrt{2}} \left(\underline{A} e^{i\omega t} + \underline{A}^* e^{-i\omega t} \right),$$

and deduce, by multiplying two such expressions, Theorem 5.10 on instantaneous power. Explain in this language why power pulses at twice the frequency.

Exercise 5.17: S4 — When apparent power is conserved. Prove that, for a set of one-ports in parallel, $\sum_k S_k = S$ holds if and only if all the one-ports have the *same* power factor *and* the same sign of Q . Connect this to the equality case in the triangle inequality (Theorem 3.1(iv)).

Exercise 5.18: S5 — The dual locus. Prove that if the resistance $R > 0$ is held fixed and the reactance X varies over all of $\mathbb{R}$, the admittance describes the circle

$$\left(G - \frac{1}{2R} \right)^2 + B^2 = \left(\frac{1}{2R} \right)^2,$$

of centre $1/(2R)$ and radius $1/(2R)$. Deduce that the admittance locus of a series RLC branch, as ω ranges over $(0, \infty)$, is this same circle minus the origin, traversed exactly once.

Exercise 5.19: S6 — Where the capacitor voltage is really maximal. For the series RLC branch supplied at constant voltage V , prove that the ratio $|\underline{V}_C|/V$ is maximal not at ω_0 but at

$$\omega_m = \omega_0 \sqrt{1 - \frac{1}{2Q^2}} \quad (Q > 1/\sqrt{2}),$$

and that the maximum value is $\frac{2Q^2}{\sqrt{4Q^2 - 1}}$. Check that for $Q \gg 1$ one has $\omega_m \rightarrow \omega_0$ and maximum value $\rightarrow Q$.

Exercise 5.20: S7 — Constant instantaneous power and unbalance. Prove that, in a three-phase system with symmetric voltages, the total instantaneous power is constant if and only if the currents have no negative-sequence component. Interpret the result in the light of Theorem 5.23.

Exercise 5.21: S8 — Power in symmetrical components. Prove the *power form* of Fortescue's transformation,

$$\underline{S} = \underline{V}_a \underline{I}_a^* + \underline{V}_b \underline{I}_b^* + \underline{V}_c \underline{I}_c^* = 3 \left(\underline{V}_0 \underline{I}_0^* + \underline{V}_1 \underline{I}_1^* + \underline{V}_2 \underline{I}_2^* \right),$$

and explain the origin of the factor 3 and how a unitary normalisation removes it.

Exercise 5.22: S9 — Power with a distorted current. A purely sinusoidal voltage $v(t) = V\sqrt{2}\cos\omega t$ feeds a nonlinear load drawing a periodic current $i(t)$ of RMS value I , with Fourier series $i(t) = \sum_{n\geq 1} I_n\sqrt{2}\cos(n\omega t - \varphi_n)$. Prove that

$$P = VI_1 \cos \varphi_1, \quad \frac{P}{S} = \frac{I_1}{I} \cos \varphi_1,$$

that is, only the *fundamental* of the current produces active power, and the power factor factorises into a distortion factor times a displacement factor.

Exercise 5.23: S10 — The admittance circle as frequency varies. For the series RLC branch, prove that as ω varies the point $\underline{Y}(\omega)$ traverses the circle of Exercise S5 *clockwise*, starting from the origin as $\omega \rightarrow 0^+$, passing through the real point $1/R$ at $\omega = \omega_0$ and returning to the origin as $\omega \rightarrow +\infty$. Deduce a graphical construction of the cut-off angular frequencies.

Exercise 5.24: S11 — When the single line-to-ground fault is the severest. With $\underline{Z}_2 = \underline{Z}_1$ and purely reactive impedances $\underline{Z}_1 = jX_1$, $\underline{Z}_0 = jX_0$ ($X_1, X_0 > 0$), prove that

$$\frac{|\underline{I}_a^{(1)}|}{|\underline{I}_{sc}^{(3)}|} = \frac{3X_1}{2X_1 + X_0},$$

that this ratio exceeds 1 if and only if $X_0 < X_1$, and that its supremum is $3/2$. Comment on the meaning of the criterion $X_0/X_1 \leq 3$ used to define an effectively earthed system.

Exercise 5.25: S12 — The ceiling of the power factor in the presence of harmonics. Prove that, under sinusoidal voltage, $\text{PF} \leq 1/\sqrt{1 + \text{THD}_I^2}$ whatever φ_1 may be, and that perfect correction of the fundamental alone multiplies the power factor by $1/\cos \varphi_1$ and no more. Then determine the largest THD_I compatible with $\text{PF} \geq 0.90$ after perfect correction.

5.19. Solutions to the numerical exercises

Solution. N1. *Step 1 — reading off the parameters.* Comparing with Definition 5.1: $V_m = 311$ V, $\omega = 314$ rad/s, $\alpha = -30^\circ$.

Step 2 — RMS value. By Theorem 5.2(i),

$$V = \frac{311}{1.4142136} = 219.91 \text{ V}.$$

Step 3 — frequency and period.

$$f = \frac{314}{6.2831853} = 49.974 \text{ Hz}, \quad T = 20.011 \text{ ms}.$$

(The value $\omega = 314$ is the usual rounding of $2\pi \cdot 50 = 314.159$: hence the 0.05% discrepancy in frequency.)

Step 4 — phasor. By Definition 5.3,

$$\underline{V} = 219.91 \angle -30^\circ \text{ V} = 190.45 - j109.96 \text{ V}.$$

Step 5 — excess of the peak.

$$\frac{V_m - V}{V} = \sqrt{2} - 1 = 0.41421 = 41.42\%.$$

The maximum instantaneous value exceeds the RMS value by 41.4%, always, for every sinusoid.

Typical error. Writing the phasor as $311\angle -30^\circ$. In the RMS convention adopted in this chapter the phasor modulus is V , not V_m ; using V_m leads to powers twice the true ones.

Solution. N2. Step 1 — reactance. $X_L = \omega L = 314.159 \times 0.1 = 31.416 \, \Omega$.

Step 2 — impedance. $\underline{Z} = 40 + i 31.416 \, \Omega$, and by Theorem 5.5

$$Z = \sqrt{1600 + 986.96} = \sqrt{2586.96} = 50.862 \, \Omega,$$

$$\varphi = \arctan \frac{31.416}{40} = \arctan(0.785398) = 0.66577 \text{ rad} = 38.146^\circ,$$

legitimate with $\arctan$ because the point is in the first quadrant. *Inductive* load.

Step 3 — current.

$$I = \frac{230}{50.862} = 4.5220 \text{ A}, \quad \underline{I} = 4.5220 \angle -38.146^\circ \text{ A}$$

(taking $\underline{V} = 230\angle 0^\circ$).

Step 4 — powers. With $I^2 = 20.448 \text{ A}^2$ and Theorem 5.11(ii):

$$P = RI^2 = 40 \times 20.448 = 817.9 \text{ W}, \quad Q = X_L I^2 = 31.416 \times 20.448 = 642.4 \text{ var},$$

$$S = VI = 230 \times 4.5220 = 1040.1 \text{ VA}.$$

Step 5 — power triangle check.

$$\sqrt{P^2 + Q^2} = \sqrt{668\,960 + 412\,678} = \sqrt{1\,081\,638} = 1040.0 \text{ VA} \checkmark$$

$$\cos \varphi = \frac{P}{S} = \frac{817.9}{1040.1} = 0.78636,$$

and $\cos(38.146^\circ) = 0.78637 \checkmark$

Typical error. Computing $Q = VI \sin \varphi$ and $P = RI^2$ with two different rounded values of I . It pays to fix I to five significant digits and use it throughout.

Solution. N3. Step 1 — admittances of the two branches. By Theorem 5.3 and Definition 5.4,

$$\underline{Y}_R = \frac{1}{100} = 0.01 \text{ S}, \quad \underline{Y}_C = i\omega C = i 314.159 \times 20 \times 10^{-6} = i 0.0062832 \text{ S}.$$

Step 2 — total admittance. By Theorem 5.4(iv), admittances in parallel add:

$$\underline{Y} = 0.01 + i 0.0062832 \text{ S},$$

$$|\underline{Y}| = \sqrt{10^{-4} + 3.9478 \times 10^{-5}} = \sqrt{1.39478 \times 10^{-4}} = 0.0118101 \text{ S},$$

$$\arg \underline{Y} = \arctan \frac{0.0062832}{0.01} = \arctan(0.62832) = 32.142^\circ.$$

Step 3 — equivalent impedance.

$$Z = \frac{1}{\underline{Y}} = 84.674 \, \Omega, \quad \varphi = -32.142^\circ,$$

the sign being reversed because $\underline{Z} = 1/\underline{Y}$ (Theorem 3.4). *Capacitive* load, as it must be.

Step 4 — current and powers.

$$I = 0.0118101 \times 230 = 2.7163 \, \text{A},$$

and by Theorem 5.11(ii) in admittance form with $V^2 = 52\,900$:

$$P = 0.01 \times 52\,900 = 529.0 \, \text{W}, \quad Q = -0.0062832 \times 52\,900 = -332.4 \, \text{var}.$$

(The minus sign comes from the conjugate: $\underline{S} = \underline{Y}^* V^2$.)

Step 5 — independent check. The resistor current is $230/100 = 2.30 \, \text{A}$ and the capacitor current $230 \times 0.0062832 = 1.4451 \, \text{A}$, in quadrature with each other; the modulus of the vector sum is $\sqrt{2.30^2 + 1.4451^2} = \sqrt{5.29 + 2.0883} = \sqrt{7.3783} = 2.7163 \, \text{A} \checkmark$. Moreover $P = 230 \times 2.30 = 529 \, \text{W} \checkmark$

Typical error. Adding impedances instead of admittances in a parallel connection. The rule $\underline{Z}_{\text{eq}} = \underline{Z}_1 \underline{Z}_2 / (\underline{Z}_1 + \underline{Z}_2)$ remains valid, but the admittance computation is shorter and less prone to sign errors.

Solution. N4. Step 1 — tangents. $\cos \varphi_1 = 0.6 \Rightarrow \sin \varphi_1 = 0.8 \Rightarrow \tan \varphi_1 = 1.333333$; $\cos \varphi_2 = 0.9 \Rightarrow \sin \varphi_2 = \sqrt{1 - 0.81} = 0.435890 \Rightarrow \tan \varphi_2 = 0.484322$.

Step 2 — reactive power to compensate.

$$Q_C = 5000 \times 0.849011 = 4245.1 \, \text{var}.$$

Step 3 — capacitance. By Theorem 5.13, with $V^2 = 52\,900$:

$$C = \frac{4245.1}{314.159 \times 52\,900} = \frac{4245.1}{16\,619\,011} = 2.5544 \times 10^{-4} \, \text{F} = 255.4 \, \mu\text{F}.$$

Step 4 — currents.

$$I_1 = \frac{5000}{230 \times 0.6} = 36.232 \, \text{A}, \quad I_2 = \frac{5000}{230 \times 0.9} = 24.155 \, \text{A}.$$

Step 5 — reduction of losses.

$$\left(\frac{I_2}{I_1}\right)^2 = \left(\frac{24.155}{36.232}\right)^2 = 0.66667^2 = 0.44444,$$

i.e. a 55.6% reduction. (The ratio $I_2/I_1 = \cos \varphi_1 / \cos \varphi_2 = 0.6/0.9 = 2/3$ exactly: a useful instant check.)

Step 6 — reactive balance check. $Q_1 = 6666.7 \, \text{var}$; $Q_2 = 6666.7 - 4245.1 = 2421.6 \, \text{var}$; and $P \tan \varphi_2 = 2421.6 \, \text{var} \checkmark$

Typical error. Using the correction formula with the *line* voltage of a three-phase system in a single-phase problem, or vice versa. The formula $C = Q_C / (\omega V^2)$ requires V to be the voltage actually applied to the capacitor.

Solution. N5. Step 1 — target angular frequency. $\omega_0 = 2\pi \times 1000 = 6283.19 \, \text{rad/s}$.

Step 2 — capacitance. From $\omega_0 = 1/\sqrt{LC}$ (Theorem 5.14(i)),

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(6283.19)^2 \times 0.02} = \frac{1}{3.94784 \times 10^7 \times 0.02} = \frac{1}{789\,568} = 1.2665 \times 10^{-6} \text{ F} = 1.267 \mu\text{F}.$$

Step 3 — quality factor. By Definition 5.7,

$$Q = \frac{6283.19 \times 0.02}{5} = \frac{125.664}{5} = 25.133.$$

Check with $\frac{1}{R}\sqrt{L/C}$: $\sqrt{0.02/1.2665 \times 10^{-6}} = \sqrt{15\,791} = 125.66$, divided by 5 gives 25.13 ✓

Step 4 — bandwidth. By Theorem 5.15,

$$B = \frac{6283.19}{25.133} = 250.0 \text{ rad/s}, \quad B_f = 39.79 \text{ Hz}.$$

Alternative check: $B = R/L = 5/0.02 = 250 \text{ rad/s}$ ✓

Step 5 — overvoltage. By Theorem 5.14(iv), at resonance

$$V_C = QV = 25.133 \times 10 = 251.3 \text{ V}.$$

On a 10 V supply the capacitor sees 251 V: its rated voltage must be chosen on this value, not on the supply one.

Step 6 — cut-off frequencies. $f_{1,2} \approx f_0 \mp B_f/2$ to first order (exact to within $1/(8Q^2) = 0.0198\%$): $f_1 \approx 980.1 \text{ Hz}$, $f_2 \approx 1019.9 \text{ Hz}$. The relative selectivity is $B_f/f_0 = 1/Q = 3.98\%$.

Typical error. Computing C from $\omega_0 = 1/\sqrt{LC}$ while forgetting the square: one would get $C = 1/(\omega_0 L) = 7.96 \mu\text{F}$, six times too large, and the resonance would fall at 398 Hz.

Solution. N6. *Step 1 — centre and radius.* By Theorem 5.17 with $X = -25 \Omega$:

$$\text{centre} = -\frac{i}{-50} = +i\,0.02 \text{ S}, \quad \text{radius} = \frac{1}{50} = 0.02 \text{ S}.$$

The circle lies in the upper half-plane (positive susceptance: capacitive behaviour).

Step 2 — checking the three points.

- $R = 0$: $\underline{Y} = \frac{1}{-25i} = \frac{i}{25} = i\,0.04 \text{ S}$. Distance from centre: $|0.04 - 0.02| = 0.02$ ✓
- $R = 25 \Omega$: $\underline{Y} = \frac{1}{25 - 25i} = \frac{25 + 25i}{1250} = 0.02 + i\,0.02 \text{ S}$. Distance: $\sqrt{0.02^2 + 0} = 0.02$ ✓
- $R = 50 \Omega$: $\underline{Y} = \frac{1}{50 + 25i} = \frac{50 + 25i}{2500 + 625} = \frac{50 + 25i}{3125} = 0.016 + i\,0.008 \text{ S}$. Distance: $\sqrt{0.016^2 + 0.012^2} = \sqrt{2.56 \times 10^{-4} + 1.44 \times 10^{-4}} = \sqrt{4 \times 10^{-4}} = 0.02$ ✓

Step 3 — maximum active power. At fixed voltage $P = GV^2$, so P is maximal where the conductance is, i.e. at the rightmost point of the circle: $G_{\max} = 0.02 \text{ S}$, reached for $R = |X| = 25 \Omega$. The value is $P_{\max} = 0.02 V^2$.

Step 4 — analytic check. $G(R) = R/(R^2 + 625)$; differentiating, $G'(R) = (625 - R^2)/(R^2 + 625)^2$, which vanishes at $R = 25$, is positive before and negative after: a maximum ✓

Remark. The result $R = |X|$ is a special case of maximum power transfer (Exercise S2) when the reactive part cannot be adjusted: one can only match the modulus, not conjugate it.

Solution. N7. *Step 1 — phase voltage and current.* In delta the phase voltage equals the line voltage: $V_f = 400 \text{ V}$. Hence $I_f = 400/30 = 13.333 \text{ A}$.

Step 2 — line current. By the dual of Theorem 5.20(iii),

$$I_l = \sqrt{3} I_f = 1.7320508 \times 13.333 = 23.094 \text{ A.}$$

Step 3 — powers. By Theorem 5.21, with $\cos(-40^\circ) = 0.766044$ and $\sin(-40^\circ) = -0.642788$:

$$S = \sqrt{3} \times 400 \times 23.094 = 16\,000.0 \text{ VA,}$$

$$P = 16\,000 \times 0.766044 = 12\,256.7 \text{ W,} \quad Q = 16\,000 \times (-0.642788) = -10\,284.6 \text{ var.}$$

Step 4 — per-phase check. $S_f = 400 \times 13.333 = 5333.3 \text{ VA}$, and $3S_f = 16\,000 \text{ VA}$ ✓

Step 5 — nature of the load. The argument of the impedance is $-40^\circ < 0$, so $X < 0$: **capacitive** load, and indeed $Q < 0$ (Theorem 5.11(iv)). The current *leads* the voltage.

Typical error. Applying $I_l = \sqrt{3}I_f$ to the star connection too, where instead $I_l = I_f$. The two $\sqrt{3}$ relations are *dual*: in star they concern voltages, in delta currents.

Solution. N8. *Step 1 — applying the formula.* By Theorem 5.22 with $\varphi = -40^\circ$, $V_l = 400 \text{ V}$, $I_l = 23.094 \text{ A}$ and $V_l I_l = 9237.6 \text{ VA}$:

$$W_1 = 9237.6 \cos(-40^\circ - 30^\circ) = 9237.6 \cos(-70^\circ) = 9237.6 \times 0.342020 = 3159.4 \text{ W,}$$

$$W_2 = 9237.6 \cos(-40^\circ + 30^\circ) = 9237.6 \cos(-10^\circ) = 9237.6 \times 0.984808 = 9097.3 \text{ W.}$$

Step 2 — sum check.

$$W_1 + W_2 = 3159.4 + 9097.3 = 12\,256.7 \text{ W} = P \quad \checkmark$$

Step 3 — difference check.

$$\sqrt{3}(W_1 - W_2) = 1.7320508 \times (3159.4 - 9097.3) = 1.7320508 \times (-5937.9) = -10\,284.6 \text{ var} = Q \quad \checkmark$$

Step 4 — comment on signs. Both readings are positive because $|\varphi| < 60^\circ$: indeed $W_2 = V_l I_l \cos(\varphi + 30^\circ)$ changes sign when $|\varphi + 30^\circ| > 90^\circ$. Since a passive load has $|\varphi| \leq 90^\circ$, *one* of the two readings becomes negative exactly when $|\varphi| > 60^\circ$, i.e. when $\cos \varphi < 0.5$. This is the classic warning sign: in the Aron connection, a wattmeter driven backwards indicates a power factor below 0.5, not an instrument fault.

Step 5 — cross-check. The power factor can be recovered from the readings alone:

$$\tan \varphi = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2} = 1.7320508 \times \frac{-5937.9}{12\,256.7} = -0.83910,$$

whence $\varphi = -40.00^\circ \quad \checkmark$

Solution. N9. *Step 1 — Cartesian form.*

$$\underline{V}_a = 200, \quad \underline{V}_b = 200 \angle -120^\circ = -100 - i 173.205, \quad \underline{V}_c = 0.$$

Step 2 — zero-sequence component.

$$\underline{V}_0 = \frac{200 - 100 - i 173.205}{3} = \frac{100 - i 173.205}{3} = 33.333 - i 57.735 = 66.667 \angle -60^\circ \text{ V.}$$

Step 3 — positive-sequence component. $a \underline{V}_b = 200 \angle (-120^\circ + 120^\circ) = 200 \angle 0^\circ = 200$. Hence

$$\underline{V}_1 = \frac{200 + 200 + 0}{3} = 133.333 \angle 0^\circ \text{ V.}$$

Step 4 — negative-sequence component. $a^2 \underline{V}_b = 200 \angle (-120^\circ + 240^\circ) = 200 \angle 120^\circ = -100 + i 173.205$. Hence

$$\underline{V}_2 = \frac{200 - 100 + i 173.205}{3} = 33.333 + i 57.735 = 66.667 \angle 60^\circ \text{ V.}$$

Step 5 — reconstruction check.

$$\underline{V}_0 + \underline{V}_1 + \underline{V}_2 = (33.333 + 133.333 + 33.333) + i(-57.735 + 0 + 57.735) = 200.00 + i 0 = \underline{V}_a \checkmark$$

On phase c : $\underline{V}_0 + a \underline{V}_1 + a^2 \underline{V}_2 = 66.667 \angle -60^\circ + 133.333 \angle 120^\circ + 66.667 \angle (60^\circ + 240^\circ = 300^\circ)$. In Cartesian form: $(33.333 - i 57.735) + (-66.667 + i 115.470) + (33.333 - i 57.735) = 0.00 - i 0.00 = \underline{V}_c \checkmark$

Step 6 — comment. The negative-to-positive sequence ratio is

$$\frac{66.667}{133.333} = 0.50 = 50\%,$$

an enormous value: the loss of one phase is, in terms of symmetrical components, a 50% unbalance. This is why negative-sequence overcurrent relays are the standard protection against phase loss in three-phase motors: a large negative-sequence component induces rotor currents at nearly twice the frequency and rapid overheating. Note also that $\underline{V}_0 \neq 0$: the three voltages do not sum to zero, consistently with Theorem 5.23.

Solution. N10. (a) *Annual energies.*

Step 1. Annual operating hours: $8 \times 250 = 2000$ h, so $W_a = 30 \times 2000 = 60\,000$ kWh.

Step 2. With $\cos \varphi = 0.8$ we have $\sin \varphi = 0.6$ and $\tan \varphi = 0.75$, hence

$$Q = P \tan \varphi = 30 \times 0.75 = 22.5 \text{ kvar}, \quad W_r = 22.5 \times 2000 = 45\,000 \text{ kvarh.}$$

(b) *Penalty.*

Step 3. The ratio is $45\,000/60\,000 = 0.75 = \tan \varphi$, an identity that is no accident: the ratio of reactive to active energy equals $\tan \varphi$ when the power factor is constant in time. Since $0.75 > 0.50$, the penalty **applies**. The threshold 0.5 corresponds to $\cos \varphi = 1/\sqrt{1+0.25} = 0.894$.

(c) *Correction bank.*

Step 4 — reactive power to compensate. $Q_C = 30 \times (0.75 - 0.50) = 7.5$ kvar.

Step 5 — delta connection. Each capacitor is subject to the line voltage V_l and supplies $\omega C V_l^2$; the three together supply $3\omega C V_l^2$. Hence

$$C_\Delta = \frac{Q_C}{3\omega V_l^2} = \frac{7500}{3 \times 314.159 \times 160\,000} = \frac{7500}{150\,796\,320} = 4.974 \times 10^{-5} \text{ F} = 49.7 \text{ }\mu\text{F}.$$

Step 6 — star connection. Each capacitor sees $V_l/\sqrt{3}$ and supplies $\omega C V_l^2/3$; the three together $\omega C V_l^2$. Hence

$$C_Y = \frac{Q_C}{\omega V_l^2} = \frac{7500}{50\,265\,440} = 1.4921 \times 10^{-4} \text{ F} = 149.2 \text{ }\mu\text{F}.$$

Step 7 — comparison and choice. $C_Y = 3C_\Delta$: for the same reactive power, the delta connection needs *one third* of the capacitance. Since the cost of a capacitor grows with capacitance (at equal rated voltage), but delta capacitors must withstand 400 V instead of 231 V, the actual choice depends on prices; in low-voltage industrial practice delta prevails.

Step 8 — final check. After correction, $Q_2 = 22.5 - 7.5 = 15$ kvar, $\tan \varphi_2 = 15/30 = 0.50 \checkmark$, $\cos \varphi_2 = 0.894$, and the annual reactive energy drops to 30 000 kvarh: exactly the threshold.

Solution. (a) Reactances on the system base. By Theorem 5.26, with base voltages coinciding with the rated ones,

$$x_d'' = 0.18 \cdot \frac{100}{40} = 0.450 \text{ p.u.}, \quad x_T = 0.09 \cdot \frac{100}{40} = 0.225 \text{ p.u.}$$

From the busbar one therefore sees $\underline{Z}_1 = j(0.450 + 0.225) = j0.675 \text{ p.u.}$, and by hypothesis $\underline{Z}_2 = j0.675$, $\underline{Z}_0 = j0.06$.

(b) Three-phase fault. By Theorem 5.30,

$$|\underline{I}_{sc}^{(3)}| = \frac{1}{0.675} = 1.4815 \text{ p.u.}$$

The base current is $I_b = 100 \cdot 10^6 / (\sqrt{3} \cdot 66 \cdot 10^3) = 874.77 \text{ A}$, hence $I_{sc}^{(3)} = 1.4815 \cdot 874.77 = 1296 \text{ A} = 1.296 \text{ kA}$. The short-circuit power is $S_{sc} = 1.4815 \cdot 100 = 148.15 \text{ MVA}$.

(c) Single line-to-ground fault. By Theorem 5.31,

$$|\underline{I}_a| = \frac{3}{0.675 + 0.675 + 0.06} = \frac{3}{1.41} = 2.1277 \text{ p.u.} = 2.1277 \cdot 874.77 = 1861 \text{ A} = 1.861 \text{ kA}.$$

(d) Ratio.

$$\frac{2.1277}{1.4815} = 1.436 = \frac{3 \cdot 0.675}{2 \cdot 0.675 + 0.06} = \frac{2.025}{1.41} \checkmark$$

The single line-to-ground fault is therefore 43.6% more severe than the three-phase one. The limiting value is obtained for $X_0 \rightarrow 0$ and equals 3/2: no network configuration can make the single-phase fault exceed the three-phase one by more than 50% (Exercise S11).

Typical mistake. Forgetting the factor 100/40 in the change of base and using 0.18 and 0.09 directly: one would get $\underline{Z}_1 = j0.27$ and a short-circuit current of 3.70 p.u., that is 2.5 times the true value. This is precisely the error that the plausibility check of Section 5.12 is meant to catch.

Solution. (a) Synchronism. Six poles mean $p = 3$ pole pairs, hence

$$n_s = \frac{60f}{p} = \frac{60 \cdot 50}{3} = 1000 \text{ rpm}, \quad \omega_s = \frac{2\pi \cdot 1000}{60} = 104.72 \text{ rad/s},$$

and the phase voltage is $V = 400/\sqrt{3} = 230.94 \text{ V}$.

(b) Breakdown. By Theorem 5.19,

$$s_m = \frac{0.135}{0.9} = 0.150, \quad T_{\max} = \frac{3 \cdot 230.94^2}{2 \cdot 104.72 \cdot 0.9} = \frac{160\,000}{188.50} = 848.8 \text{ N} \cdot \text{m}.$$

(c) Operating point $s = 0.04$. Kloss's formula gives

$$\frac{T}{T_{\max}} = \frac{2}{\frac{0.04}{0.15} + \frac{0.15}{0.04}} = \frac{2}{0.26667 + 3.75000} = \frac{2}{4.01667} = 0.49793,$$

hence $T = 0.49793 \cdot 848.8 = 422.6 \text{ N} \cdot \text{m}$. Since $\omega_r = \omega_s(1 - s) = 104.72 \cdot 0.96 = 100.53 \text{ rad/s}$, the mechanical power is $P = T\omega_r = 422.6 \cdot 100.53 = 42.49 \text{ kW}$.

(d) Starting. At $s = 1$,

$$\frac{T}{T_{\max}} = \frac{2}{\frac{1}{0.15} + 0.15} = \frac{2}{6.81667} = 0.29340 \implies T_{\text{st}} = 249.1 \text{ N} \cdot \text{m},$$

that is only 0.59 times the rated torque: such a motor will not start under rated load.

(e) Additional resistance. By Corollary 5.3 one needs $R_2'' = X = 0.9 \Omega$, hence an additional referred resistance of $0.9 - 0.135 = 0.765 \Omega$. With it the starting torque rises to $848.8 \text{ N}\cdot\text{m}$, that is 3.41 times the previous value and 2.01 times the rated torque.

A coincidence that is not one. The ratio $T/T_{\max} = 0.29340$ found here at standstill is identical to the one in Example 5.9 at the rated point, although the two motors are different. The reason is Step 6 of Theorem 5.19: Kloss's formula depends only on $x = s/s_m$ and is invariant under $x \mapsto 1/x$. There $x = 0.03/0.20 = 0.15$; here $x = 1/0.15 = 6.6667$: the two values are reciprocals of each other, so the relative torque must coincide. ✓

5.20. Solutions to the symbolic exercises

Solution. S1. *Step 1* — $R \geq 0 \iff P \geq 0$. By Theorem 5.11(ii), $P = R|\underline{I}|^2$ with $|\underline{I}|^2 \geq 0$; if $\underline{I} \neq 0$ the two signs coincide.

Step 2 — *computing* $|\Gamma|^2$. With $\underline{Z} = R + iX$ and $Z_0 > 0$ real,

$$|\Gamma|^2 = \frac{(R - Z_0)^2 + X^2}{(R + Z_0)^2 + X^2},$$

well defined because the denominator vanishes only for $\underline{Z} = -Z_0$, impossible if $R \geq 0$ and $Z_0 > 0$.

Step 3 — *the inequality*. Subtracting,

$$|\underline{Z} + Z_0|^2 - |\underline{Z} - Z_0|^2 = (R + Z_0)^2 - (R - Z_0)^2 = 4RZ_0.$$

Since $Z_0 > 0$, this quantity is ≥ 0 if and only if $R \geq 0$. But it is ≥ 0 if and only if $|\underline{Z} - Z_0|^2 \leq |\underline{Z} + Z_0|^2$, i.e. if and only if $|\Gamma| \leq 1$. The three conditions are therefore equivalent, and $|\Gamma| = 1$ exactly when $R = 0$.

Step 4 — Γ is a Möbius map. It is $\Gamma(\underline{Z}) = \frac{\underline{Z} - Z_0}{\underline{Z} + Z_0}$, i.e. Definition 3.4 with $a = 1$, $b = -Z_0$, $c = 1$, $d = Z_0$ and determinant $2Z_0 \neq 0$.

Step 5 — *the case* $Z_0 = 1$ *and Chapter 3*. For $Z_0 = 1$ one gets exactly the map $w = (z - 1)/(z + 1)$ of Corollary 3.3, which sends $\mathbb{T}$ onto the imaginary axis. Here we read it the other way round: the imaginary axis of the $\underline{Z}$ -plane (purely reactive one-ports, $R = 0$) goes to the unit circle of the Γ -plane, and the half-plane $R > 0$ (passive dissipative one-ports) to the open disc $|\Gamma| < 1$. (Volume III, Chapter 25) will build the Smith chart on this correspondence.

Solution. S2. *Step 1* — *expression of the power*. The current is $\underline{I} = \underline{E}/(\underline{Z}_s + \underline{Z}_L)$, and the active power on the load, by Theorem 5.11(ii), is

$$P_L = \frac{R_L |\underline{E}|^2}{(R_s + R_L)^2 + (X_s + X_L)^2}.$$

Step 2 — *optimising in* X_L . At fixed R_L the numerator does not depend on X_L and the denominator is minimal when $(X_s + X_L)^2$ is, i.e. for $X_L = -X_s$. With this choice $P_L = R_L |\underline{E}|^2 / (R_s + R_L)^2$.

Step 3 — *optimising in* R_L . Differentiating,

$$\frac{dP_L}{dR_L} = |\underline{E}|^2 \frac{R_s - R_L}{(R_s + R_L)^3},$$

positive for $R_L < R_s$ and negative for $R_L > R_s$: the maximum is at $R_L = R_s$.

Step 4 — conclusion. The optimum is $\underline{Z}_L = \underline{Z}_s^*$ (conjugate matching), and $P_{\max} = R_s |\underline{E}|^2 / (2R_s)^2 = |\underline{E}|^2 / (4R_s)$.

Step 5 — an alternative reading, without derivatives. By the AM–GM inequality, $(R_s + R_L)^2 \geq 4R_s R_L$ with equality iff $R_L = R_s$; hence

$$P_L \leq \frac{R_L |\underline{E}|^2}{4R_s R_L} = \frac{|\underline{E}|^2}{4R_s},$$

with equality exactly in the matched case. This route yields the maximum value and the equality case in one stroke.

Step 6 — why it is not used in power systems. At conjugate matching the efficiency is

$$\eta = \frac{R_L}{R_L + R_s} = \frac{1}{2},$$

i.e. half the energy is dissipated in the generator. Acceptable in telecommunications, where what matters is maximising the received *signal* and powers are tiny; unacceptable in a power network, which is designed for $\eta \rightarrow 1$, i.e. $R_s \ll R_L$, deliberately giving up maximum transfer.

Solution. S3. *Step 1 — the decomposition.* By Theorem 3.6 applied to the argument $\omega t + \alpha$,

$$a(t) = \frac{A_m}{2} e^{i\alpha} e^{i\omega t} + \frac{A_m}{2} e^{-i\alpha} e^{-i\omega t}.$$

With $\underline{A} = \frac{A_m}{\sqrt{2}} e^{i\alpha}$ (Definition 5.3) we have $A_m/2 = \underline{A}/\sqrt{2}$ and $\frac{A_m}{2} e^{-i\alpha} = \underline{A}^*/\sqrt{2}$, whence the claim.

Step 2 — interpretation. The two terms are points travelling around the same circle of radius $A_m/2$ in opposite senses, and their sum is always real because they are conjugate. A real sinusoid is therefore the superposition of two counter-rotating circular motions, not of one: this is the physical content of the “negative frequencies” of Exercise S8 of Chapter 3.

Step 3 — product of two quantities. With $v(t) = \frac{1}{\sqrt{2}} (\underline{V} e^{i\omega t} + \underline{V}^* e^{-i\omega t})$ and similarly for $i(t)$,

$$p(t) = \frac{1}{2} (\underline{V} \underline{I} e^{2i\omega t} + \underline{V} \underline{I}^* + \underline{V}^* \underline{I} + \underline{V}^* \underline{I}^* e^{-2i\omega t}).$$

Step 4 — grouping. The first and fourth terms are conjugate to each other, as are the second and third. Using $z + z^* = 2 \operatorname{Re} z$,

$$p(t) = \operatorname{Re} [\underline{V} \underline{I}^*] + \operatorname{Re} [\underline{V} \underline{I} e^{2i\omega t}].$$

The first term is P by Theorem 5.11(i); the second oscillates at 2ω with amplitude $|\underline{V} \underline{I}| = VI = S$.

Step 5 — recovering Theorem 5.10. Taking $\underline{V} = V$ real and $\underline{I} = I e^{-i\varphi}$,

$$\operatorname{Re} [\underline{V} \underline{I} e^{2i\omega t}] = VI \cos \varphi \cos 2\omega t + VI \sin \varphi \sin 2\omega t,$$

and adding $P = VI \cos \varphi$ recovers exactly the decomposition of Theorem 5.10.

Step 6 — why twice the frequency. It is now evident: the product of two exponentials $e^{i\omega t}$ gives $e^{2i\omega t}$, while the product of $e^{i\omega t}$ by $e^{-i\omega t}$ gives a constant. A product of two quantities of angular frequency ω therefore contains *only* the angular frequencies 0 and 2ω . In the language of Chapter 4: multiplying in time is convolving in frequency, and the convolution of two spectral lines at $\pm\omega$ produces lines at 0 and $\pm 2\omega$.

Solution. S4. *Step 1 — setting up.* By Theorem 5.12, $\underline{S} = \sum_k \underline{S}_k$. Taking moduli and using the triangle inequality (Theorem 3.1(iv)) iteratively, $S \leq \sum_k S_k$: the total apparent power never exceeds the sum of the apparent powers.

Step 2 — equality for two terms. By Theorem 3.1(iv), $|\underline{S}_1 + \underline{S}_2| = |\underline{S}_1| + |\underline{S}_2|$ if and only if $\underline{S}_1 = 0$ or $\underline{S}_2 = t \underline{S}_1$ with $t \geq 0$, i.e. if and only if the two complex numbers have the *same argument*.

Step 3 — extension to n terms. By induction: splitting $\sum_{k=1}^n \underline{S}_k = \sum_{k=1}^{n-1} \underline{S}_k + \underline{S}_n$ and applying Step 1 twice,

$$\sum_{k=1}^n S_k = \left| \sum_{k=1}^{n-1} \underline{S}_k + \underline{S}_n \right| \leq \left| \sum_{k=1}^{n-1} \underline{S}_k \right| + S_n \leq \sum_{k=1}^{n-1} S_k + S_n,$$

and since the two ends coincide, both inequalities are equalities. From the first, by Step 2, $\underline{S}_n$ has the same argument as $\sum_{k < n} \underline{S}_k$; from the second, by the inductive hypothesis, all $\underline{S}_k$ with $k < n$ share one argument. Hence all n do.

Step 4 — electrical translation. The argument of $\underline{S}_k$ is φ_k , and equal arguments mean: same φ_k , hence the same $\cos \varphi_k$ and the same sign of $\sin \varphi_k$ — same power factor and same nature (all inductive or all capacitive).

Step 5 — an instructive counterexample. A load with $\underline{S}_1 = 100 + i100$ VA and one with $\underline{S}_2 = 100 - i100$ VA have the same power factor $1/\sqrt{2}$ but opposite nature. The sum is $\underline{S} = 200$ VA, whereas $S_1 + S_2 = 282.84$ VA. The 41% difference is exactly what power-factor correction exploits: placing an inductive and a capacitive load in parallel reduces the total apparent power, and with it the line current.

Solution. S5. *Step 1 — real and imaginary parts.* With $R > 0$ fixed and $X \in \mathbb{R}$,

$$G = \frac{R}{R^2 + X^2}, \quad B = \frac{-X}{R^2 + X^2}.$$

Step 2 — the key relation. $G^2 + B^2 = \frac{1}{R^2 + X^2} = \frac{G}{R}$.

Step 3 — completing the square. From $G^2 - \frac{G}{R} + B^2 = 0$,

$$\left(G - \frac{1}{2R}\right)^2 + B^2 = \frac{1}{4R^2},$$

the announced circle: centre at $1/(2R)$ on the positive real axis, radius $1/(2R)$, through the origin and through $1/R$.

Step 4 — covering. For $X = 0$, $\underline{Y} = 1/R$ (rightmost point); for $X \rightarrow \pm\infty$, $\underline{Y} \rightarrow 0$ (the origin, never attained); and $X \mapsto \underline{Y}$ is injective because $X \mapsto R + iX$ is and inversion is bijective on $\mathbb{C} \setminus \{0\}$. As X ranges over $\mathbb{R}$ the whole circle minus the origin is traversed exactly once.

Step 5 — the series RLC case. For the RLC branch, $X(\omega) = \omega L - 1/(\omega C)$ is *strictly increasing* on $(0, +\infty)$ (step (iii) of the proof of Theorem 5.14) with limits $-\infty$ and $+\infty$: it is a bijection of $(0, \infty)$ onto $\mathbb{R}$. By Step 4, $\underline{Y}(\omega)$ traverses the whole circle minus the origin exactly once.

Step 6 — alternative geometric route. As in the second proof of Theorem 5.17: the vertical line $\operatorname{Re} \underline{Z} = R$ does not pass through the origin (because $R > 0$), hence its image under the inversion is a circle through the origin (Theorem 3.14). The point R maps to $1/R$ and the point at infinity to the origin: the circle therefore has $[0, 1/R]$ as a diameter, i.e. centre $1/(2R)$ and radius $1/(2R)$.

Solution. S6. *Step 1 — normalisation.* Put $u = \omega/\omega_0$. With $Q = \omega_0 L/R$ we have $\omega L = uQR$ and $1/(\omega C) = QR/u$, hence

$$|Z| = R\sqrt{1 + Q^2\left(u - \frac{1}{u}\right)^2}.$$

Step 2 — the ratio to maximise.

$$\frac{|V_C|}{V} = \frac{Q}{u\sqrt{1 + Q^2\left(u - \frac{1}{u}\right)^2}},$$

so we must minimise $f(u) = u^2 + Q^2(u^2 - 1)^2$.

Step 3 — minimisation. With $x = u^2 > 0$, $f = x + Q^2(x - 1)^2$, a quadratic in x with positive leading coefficient. Its derivative $1 + 2Q^2(x - 1)$ vanishes at

$$x^* = 1 - \frac{1}{2Q^2},$$

an absolute minimum, admissible ($x^* > 0$) iff $Q > 1/\sqrt{2}$. Hence $\omega_m = \omega_0\sqrt{1 - 1/(2Q^2)} < \omega_0$.

Step 4 — maximum value. Substituting,

$$f(x^*) = 1 - \frac{1}{2Q^2} + \frac{1}{4Q^2} = \frac{4Q^2 - 1}{4Q^2},$$

hence

$$\left(\frac{|V_C|}{V}\right)_{\max} = \frac{Q}{\sqrt{f(x^*)}} = \frac{2Q^2}{\sqrt{4Q^2 - 1}}.$$

Step 5 — limit for large Q . For $Q \gg 1$, $\sqrt{1 - 1/(2Q^2)} = 1 - \frac{1}{4Q^2} + O(Q^{-4})$, so $\omega_m \rightarrow \omega_0$; and

$$\frac{2Q^2}{\sqrt{4Q^2 - 1}} = Q\left(1 + \frac{1}{8Q^2} + O(Q^{-4})\right) \rightarrow Q.$$

Step 6 — numerical check. For $Q = 5$ (Example 5.6): $\omega_m = 1000\sqrt{1 - 0.02} = 1000 \times 0.98995 = 989.95$ rad/s, and the maximum is $50/\sqrt{99} = 50/9.94987 = 5.0252$, against $Q = 5$ exactly at ω_0 . The discrepancy is 0.5% in amplitude and 1% in frequency: in practice they are indistinguishable, which is why Theorem 5.14(iv) is usually stated “at resonance” with no further qualification. The distinction matters only for Q close to $1/\sqrt{2} = 0.707$, where the maximum disappears altogether.

Solution. S7. *Step 1 — instantaneous power in complex form.* By Step 4 of the solution to S3, applied to each phase,

$$p_k(t) = \operatorname{Re} [\underline{V}_k \underline{I}_k^*] + \operatorname{Re} [\underline{V}_k \underline{I}_k e^{2i\omega t}].$$

Summing over the three phases, the constant part is P (Theorem 5.12) and the oscillating part is $\operatorname{Re} [e^{2i\omega t} \underline{N}]$ with $\underline{N} = \sum_k \underline{V}_k \underline{I}_k$. The instantaneous power is constant iff $\underline{N} = 0$.

Step 2 — symmetric voltages. By hypothesis $\underline{V}_a = V$, $\underline{V}_b = a^2V$, $\underline{V}_c = aV$. Write the currents in symmetrical components (Theorem 5.23).

Step 3 — computing $\underline{N}$. Collecting the coefficients of the three components:

- $\underline{I}_0$: $1 + a^2 + a = 0$;
- $\underline{I}_1$: $1 + a^4 + a^2 = 1 + a + a^2 = 0$ (since $a^4 = a$);
- $\underline{I}_2$: $1 + a^3 + a^3 = 3$ (since $a^3 = 1$).

Hence $\underline{N} = 3V \underline{I}_2$.

Step 4 — conclusion, first part. $\underline{N} = 0 \iff \underline{I}_2 = 0$: with symmetric voltages, the instantaneous power is constant if and only if the current has no *negative-sequence* component. The zero-sequence component, surprisingly, does not appear in $\underline{N}$.

Step 5 — the role of the zero sequence. The zero-sequence component causes no power oscillation *with symmetric voltages*, but it does cause neutral current: indeed $\underline{I}_a + \underline{I}_b + \underline{I}_c = 3\underline{I}_0$. In a three-wire system $\underline{I}_0 = 0$ necessarily (there is no return path), while in a four-wire system $\underline{I}_0 \neq 0$ overloads the neutral without disturbing the constancy of power. The statement of the exercise must therefore be made precise: constancy is equivalent to $\underline{I}_2 = 0$, and in three-wire systems this also entails $\underline{I}_0 = 0$.

Step 6 — interpretation. The positive sequence carries “smooth” power; the negative sequence, beating against the positive-sequence voltages, produces power pulsating at 2ω . In a rotating machine this pulsation becomes an oscillating torque at 100 Hz and rotor currents at twice the frequency: precisely the mechanism by which unbalance damages motors, and the quantitative justification of the comment in Step 6 of the solution to N9.

Solution. S8. *Step 1 — matrix notation.* Set

$$\mathbf{V} = \begin{pmatrix} V_a \\ V_b \\ V_c \end{pmatrix}, \quad \mathbf{V}_s = \begin{pmatrix} V_0 \\ V_1 \\ V_2 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{pmatrix},$$

so that $\mathbf{V} = A\mathbf{V}_s$ and likewise $\mathbf{I} = A\mathbf{I}_s$ (Theorem 5.23). The total complex power is $\underline{S} = \mathbf{V}^T \bar{\mathbf{I}} = \mathbf{I}^* \mathbf{V}$, in the notation with $\mathbf{I}^* = \bar{\mathbf{I}}^T$.

Step 2 — substitution. $\underline{S} = (A\mathbf{I}_s)^* (A\mathbf{V}_s) = \mathbf{I}_s^* (A^* A) \mathbf{V}_s$.

Step 3 — computing $A^ A$.* The matrix A is the DFT matrix W_3 of Step 1 of the proof of Theorem 5.23; by Theorem 3.22 of Chapter 3, $W_3^* W_3 = 3I$. Hence

$$\underline{S} = 3\mathbf{I}_s^* \mathbf{V}_s = 3(V_0 \underline{I}_0^* + V_1 \underline{I}_1^* + V_2 \underline{I}_2^*).$$

Step 4 — origin of the factor 3. It comes from A not being unitary but satisfying $A^* A = 3I$: its columns are orthogonal but of norm $\sqrt{3}$, not 1. It is the same factor n that in Definition 3.7 makes $W_n^{-1} = \frac{1}{n} \bar{W}_n$ rather than $\bar{W}_n$.

Step 5 — unitary normalisation. Setting $\tilde{A} = A/\sqrt{3}$ gives $\tilde{A}^* \tilde{A} = I$ and the transformation becomes *unitary*; with the components $\tilde{\mathbf{V}}_s = \sqrt{3} \mathbf{V}_s$ the power formula loses the factor 3. This is the same choice that in Chapter 3 makes $U_n = W_n/\sqrt{n}$ unitary. The engineering literature nevertheless prefers the $1/3$ normalisation because it makes $\underline{V}_1$ coincide *numerically* with the phase voltage for a balanced positive-sequence set — a practical convenience worth the price of the factor 3 in the power formulas.

Solution. S9. *Step 1 — setting up.* $P = \frac{1}{T} \int_0^T v i dt$ with $v(t) = V\sqrt{2} \cos \omega t$ and $i(t) = \sum_{n \geq 1} I_n \sqrt{2} \cos(n\omega t - \varphi_n)$.

Step 2 — orthogonality. Exchanging series and integral (legitimate by L^2 convergence, Theorem (Volume III, Chapter 17) of Chapter 4),

$$P = \sum_{n \geq 1} 2V I_n \cdot \frac{1}{T} \int_0^T \cos(\omega t) \cos(n\omega t - \varphi_n) dt.$$

By the product-to-sum formula the integrand is $\frac{1}{2} [\cos((n-1)\omega t - \varphi_n) + \cos((n+1)\omega t - \varphi_n)]$. Over a period, every cosine of argument $k\omega t + \text{const.}$ with $k \neq 0$ has zero mean (Theorem 4.2

of Chapter 4, or the direct calculation of Theorem 5.2). The only surviving term is $n = 1$, contributing $\frac{1}{2} \cos \varphi_1$.

Step 3 — conclusion on active power. $P = VI_1 \cos \varphi_1$. Current harmonics of order $n \geq 2$ produce *no* active power when the voltage is purely sinusoidal: they circulate current, and hence line losses, without transferring useful energy.

Step 4 — RMS value of the current. By Parseval's identity (Theorem 4.15 of Chapter 4), $I^2 = \sum_{n \geq 1} I_n^2 \geq I_1^2$, with equality iff the current is purely sinusoidal.

Step 5 — factorising the power factor. Since $S = VI$,

$$\frac{P}{S} = \underbrace{\frac{I_1}{I}}_{\text{distortion}} \cdot \underbrace{\cos \varphi_1}_{\text{displacement}}.$$

Introducing the total harmonic distortion $\text{THD} = \sqrt{\sum_{n \geq 2} I_n^2} / I_1$, the first factor equals $1/\sqrt{1 + \text{THD}^2}$, since $I^2 = I_1^2(1 + \text{THD}^2)$.

Step 6 — practical consequence. A rectifier drawing a heavily distorted current may have $\cos \varphi_1 = 1$ (fundamental in phase with the voltage) and yet an overall power factor of 0.6 or less, purely through distortion. Correcting with capacitors is then useless — indeed it may trigger resonances with the harmonics. This is why IEEE Std 1459 explicitly separates the two components, and why modern power supplies include an active power-factor correction stage.

Solution. S10. *Step 1 — the circle.* It is Exercise S5: as ω varies, $\underline{Y}(\omega)$ traverses the circle of centre $1/(2R)$ and radius $1/(2R)$, minus the origin.

Step 2 — direction of travel. Parametrise by $X = X(\omega)$, strictly increasing. The susceptance is $B = -X/(R^2 + X^2)$, hence

- $\omega \rightarrow 0^+$: $X \rightarrow -\infty$, $\underline{Y} \rightarrow 0$ with $B \rightarrow 0^+$ — one starts at the origin from *above*;
- $\omega < \omega_0$: $X < 0$ and $B > 0$, upper semicircle;
- $\omega = \omega_0$: $X = 0$, $\underline{Y} = 1/R$, rightmost point;
- $\omega > \omega_0$: $X > 0$ and $B < 0$, lower semicircle;
- $\omega \rightarrow +\infty$: $\underline{Y} \rightarrow 0$ with $B \rightarrow 0^-$ — one returns to the origin from *below*.

The path therefore runs from the origin rightwards via the top and returns via the bottom: **clockwise**, as stated. The reason is that B has the sign *opposite* to X .

Step 3 — graphical construction of the cut-off frequencies. By Step 1 of the proof of Theorem 5.15, the cut-off angular frequencies are those with $|X| = R$, i.e. $|\underline{Z}| = R\sqrt{2}$, i.e.

$$|\underline{Y}| = \frac{1}{R\sqrt{2}}.$$

Graphically: draw the circle centred at the origin of radius $1/(R\sqrt{2})$; it meets the admittance circle at two points, and the corresponding angular frequencies are ω_1 (upper point) and ω_2 (lower point).

Step 4 — checking the coordinates. With $X = \mp R$,

$$G = \frac{1}{2R}, \quad B = \pm \frac{1}{2R},$$

so the two points are $\frac{1}{2R}(1 \pm i)$: they lie on the admittance circle (distance $1/(2R)$ from the centre $\checkmark$) and have modulus $1/(R\sqrt{2})$ $\checkmark$. They are the endpoints of the vertical diameter, i.e. the points at $\pm 45^\circ$ from the centre: the bandwidth corresponds to a 90° arc centred on the resonance point.

Step 5 — reading. The “useful” arc of the circle is thus a quarter of the circumference, independently of R , L , C . What changes with Q is not the portion of circle but the *speed*

at which it is traversed as ω varies: a high Q means the 90° arc is crossed within a very narrow band of frequencies, i.e. a selective filter.

Solution. Step 1 — the two currents. By Theorems 5.30 and 5.31, with $\underline{Z}_2 = \underline{Z}_1$,

$$\underline{I}_{sc}^{(3)} = \frac{\underline{E}}{\underline{Z}_1}, \quad \underline{I}_a^{(1)} = \frac{3\underline{E}}{2\underline{Z}_1 + \underline{Z}_0}.$$

Step 2 — the ratio. Dividing,

$$\frac{\underline{I}_a^{(1)}}{\underline{I}_{sc}^{(3)}} = \frac{3\underline{Z}_1}{2\underline{Z}_1 + \underline{Z}_0}.$$

With $\underline{Z}_1 = jX_1$ and $\underline{Z}_0 = jX_0$ the ratio is real and positive, and equals $3X_1/(2X_1 + X_0)$: the assumption of pure reactances is exactly what guarantees that the moduli behave like the values, with no cancellation of phase.

Step 3 — when it exceeds unity.

$$\frac{3X_1}{2X_1 + X_0} > 1 \iff 3X_1 > 2X_1 + X_0 \iff X_1 > X_0,$$

having been able to multiply by $2X_1 + X_0 > 0$ without reversing the inequality.

Step 4 — supremum. The function $X_0 \mapsto 3X_1/(2X_1 + X_0)$ is strictly decreasing on $X_0 > 0$; its supremum is attained as $X_0 \rightarrow 0^+$ and equals $3X_1/(2X_1) = 3/2$, not reached because $X_0 > 0$. Hence

$$\frac{|\underline{I}_a^{(1)}|}{|\underline{I}_{sc}^{(3)}|} < \frac{3}{2} \quad \text{always.}$$

Step 5 — practical reading. A system is called *effectively* earthed when $X_0/X_1 \leq 3$ (and $R_0/X_1 \leq 1$). Substituting $X_0 = 3X_1$ in the ratio gives $3X_1/(5X_1) = 0.6$: at the upper limit of effective earthing the single-phase fault is 60% of the three-phase one. The criterion, however, arises from a different requirement: with $X_0/X_1 \leq 3$ the overvoltage on the healthy phases during the fault stays below 80% of the line voltage, which allows less costly surge arresters and insulation. The two quantities — fault current and overvoltage — move in opposite directions as X_0 varies: earthing is always a compromise.

Solution. Step 1 — the bound. By Theorem 5.35,

$$\text{PF} = \frac{\cos \varphi_1}{\sqrt{1 + \text{THD}_I^2}}.$$

Since $\cos \varphi_1 \leq 1$ for every φ_1 , it follows at once that

$$\text{PF} \leq \frac{1}{\sqrt{1 + \text{THD}_I^2}},$$

with equality if and only if $\varphi_1 = 0$.

Step 2 — what correction does. Power-factor correction acts on the reactive component of the fundamental alone, so it changes $\cos \varphi_1$ and leaves I_1 and the harmonics — that is THD_I — unchanged. Bringing $\cos \varphi_1$ from an initial value to 1, the power factor goes from $\cos \varphi_1 / \sqrt{1 + \text{THD}_I^2}$ to $1 / \sqrt{1 + \text{THD}_I^2}$: the gain is exactly the factor $1 / \cos \varphi_1$, and by Step 1 one cannot do better. In particular, if $\text{THD}_I > 0$ then unity power factor is *unattainable* with capacitors alone.

Step 3 — the admissible THD. Imposing $1/\sqrt{1 + \text{THD}_I^2} \geq 0.90$:

$$1 + \text{THD}_I^2 \leq \frac{1}{0.90^2} = \frac{1}{0.81} = 1.234568 \implies \text{THD}_I^2 \leq 0.234568 \implies \text{THD}_I \leq 0.48432.$$

The ceiling is therefore 48.4%. Note that this is a much looser bound than the standards' limits on individual harmonics: a load may satisfy $\text{PF} \geq 0.90$ and still be non-compliant on harmonic emission, because the power factor weighs the harmonics only through their sum of squares.

Step 4 — check against Example 5.15. There $\text{THD}_I = 0.6610 > 0.48432$, and indeed the power factor after perfect correction came out as $0.8342 < 0.90$, consistently. ✓

5.21. Self-assessment quiz

Twenty questions covering all sections. The key, with a one-line justification for each, is at the end.

1. The phasor associated with $v(t) = V_m \cos(\omega t + \alpha)$, in the convention of this chapter, has modulus
 - A) V_m
 - B) $V_m/\sqrt{2}$
 - C) $V_m\sqrt{2}$
 - D) $V_m/2$
2. The RMS value of a sinusoid is the number that
 - A) equals the maximum value
 - B) dissipates in a resistor the same average power as a DC current of equal value
 - C) is the arithmetic mean over a period
 - D) depends on the initial phase
3. The impedance of an inductor is
 - A) ωL
 - B) $i\omega L$
 - C) $-i\omega L$
 - D) $1/(i\omega L)$
4. In a capacitor, the current relative to the voltage is
 - A) in phase
 - B) leading by 90°
 - C) lagging by 90°
 - D) in antiphase
5. In a parallel connection one adds
 - A) impedances
 - B) admittances
 - C) resistances
 - D) power factors
6. In single phase, instantaneous power oscillates at frequency
 - A) f
 - B) $2f$
 - C) $f/2$
 - D) it is constant
7. Complex power is defined as
 - A) $\underline{V}\underline{I}$
 - B) $\underline{V}\underline{I}^*$
 - C) $\underline{V}^*\underline{I}^*$
 - D) $|\underline{V}||\underline{I}|$
8. Boucherot's theorem states that what is conserved is
 - A) P , Q and S
 - B) P and Q , but not S
 - C) only P
 - D) only S
9. Power-factor correction reduces
 - A) the active power absorbed by the load
 - B) the line current and the Joule losses

-
- C) the voltage across the load
D) the mains frequency
10. The correction capacitor must be connected
A) in series with the load B) in parallel with the load C) in series with the line D) between two phases
11. At series resonance the impedance is
A) maximal and real B) minimal and real, equal to R C) zero D) purely imaginary
12. The bandwidth of a series RLC is
A) $\omega_0 Q$ B) ω_0/Q C) Q/ω_0 D) ω_0
13. At resonance, the capacitor voltage relative to the supply voltage is
A) equal B) Q times larger C) Q times smaller D) zero
14. As the resistance R alone varies, with fixed reactance, the admittance describes
A) a line B) a semicircle C) an ellipse D) a hyperbola branch
15. In a balanced set of voltages, the instantaneous sum equals
A) $3V$ B) $\sqrt{3}V$ C) 0 D) it depends on the instant
16. The ratio of line to phase voltage in a star connection is
A) 3 B) $\sqrt{3}$ C) 1 D) $1/\sqrt{3}$
17. The active power of a balanced three-phase load is
A) $3V_l I_l \cos \varphi$ B) $\sqrt{3} V_l I_l \cos \varphi$ C) $V_l I_l \cos \varphi$ D) $\sqrt{3} V_l I_l$
18. In the Aron connection, a negative reading indicates
A) an instrument fault B) $\cos \varphi < 0.5$ C) a capacitive load D) phase inversion
19. Fortescue's transformation coincides, up to normalisation, with
A) the Laplace transform B) the three-point DFT C) the Z-transform D) the Fourier series
20. With sinusoidal voltage and distorted current, active power is produced
A) by all harmonics B) by the fundamental of the current alone C) by odd harmonics only D) it is always zero

BEST PRACTICE

Answer key.

1-B: RMS convention, stated in the opening *warning* box and in Definition 5.3. **2-B:** Theorem 5.2(ii); it is the operational definition of the RMS value. **3-B:** Theorem 5.3; differentiating corresponds to multiplying by $i\omega$. **4-B:** same theorem: in C the voltage lags, hence the current leads. **5-B:** Theorem 5.4(iv). **6-B:** Theorem 5.10(i). **7-B:** Definition 5.6; the conjugate is needed for $\underline{S}$ to be independent of the phase reference (Theorem 5.11(i)). **8-B:** Theorem 5.12; A is the commonest error, and Example 5.4 shows its numerical size. **9-B:** Observation 5.5 and Theorem 5.13; the active power is unchanged. **10-B:** Observation 5.6: in series it would alter the load voltage. **11-B:** Theorem 5.14(ii). **12-B:** Theorem 5.15. **13-B:** Theorem 5.14(iv); with $Q = 25$ a 10 V

supply produces 251 V (Exercise N5). **14-B:** Theorem 5.17; it is Theorem 3.14 of Chapter 3 applied to inversion. **15-C:** Theorem 5.20(ii), a consequence of $1 + a + a^2 = 0$. **16-B:** Theorem 5.20(iii); the factor $\sqrt{3}$ comes from $|1 - a^2| = 2 \sin(\pi/3)$. **17-B:** Theorem 5.21; A is off by a factor $\sqrt{3}$. **18-B:** Theorem 5.22 and Step 4 of the solution to N8. **19-B:** Theorem 5.23 and Observation 5.10. **20-B:** Exercise S9: current harmonics find no matching voltage harmonic with which to produce average power.

5.22. Chapter formulary

Phasors (RMS value, cosine reference)

$$a(t) = A_m \cos(\omega t + \alpha) \longleftrightarrow \underline{A} = \frac{A_m}{\sqrt{2}} e^{i\alpha}, \quad A = \frac{A_m}{\sqrt{2}}, \quad \omega = 2\pi f$$

$$\frac{d}{dt} \longleftrightarrow \times i\omega, \quad \int dt \longleftrightarrow \div i\omega$$

Impedances

$$\underline{Z}_R = R, \quad \underline{Z}_L = i\omega L, \quad \underline{Z}_C = \frac{-i}{\omega C}$$

$$\underline{Z} = R + iX = Z e^{i\varphi}, \quad Z = \sqrt{R^2 + X^2}, \quad \varphi = \text{atan2}(X, R)$$

$$\underline{Z}_{\text{series}} = \sum_k \underline{Z}_k, \quad \underline{Y}_{\text{par}} = \sum_k \underline{Y}_k$$

Power

$$\underline{S} = \underline{V} \underline{I}^* = P + iQ = \underline{Z} I^2 = \underline{Y}^* V^2$$

$$P = VI \cos \varphi = RI^2, \quad Q = VI \sin \varphi = XI^2, \quad S = VI = \sqrt{P^2 + Q^2}$$

$$p(t) = P(1 + \cos 2\omega t) + Q \sin 2\omega t, \quad \sum_k P_k = 0, \quad \sum_k Q_k = 0 \text{ (Boucherot)}$$

Power-factor correction

$$C = \frac{P(\tan \varphi_1 - \tan \varphi_2)}{\omega V^2} \text{ (single phase)}, \quad C_\Delta = \frac{Q_C}{3\omega V_l^2}, \quad C_Y = \frac{Q_C}{\omega V_l^2}$$

Series resonance

$$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R} \sqrt{\frac{L}{C}}, \quad B = \frac{\omega_0}{Q} = \frac{R}{L}$$

$$\omega_1 \omega_2 = \omega_0^2, \quad V_L = V_C = QV \text{ (at resonance)}$$

Circular loci

$$X \text{ fixed, } R \text{ variable: } G^2 + \left(B + \frac{1}{2X}\right)^2 = \left(\frac{1}{2X}\right)^2$$

$$R \text{ fixed, } X \text{ variable: } \left(G - \frac{1}{2R}\right)^2 + B^2 = \left(\frac{1}{2R}\right)^2$$

Three-phase ($a = e^{i2\pi/3}$, $1 + a + a^2 = 0$)

$$V_l = \sqrt{3} V_f \text{ (star)}, \quad I_l = \sqrt{3} I_f \text{ (delta)}$$

$$P = \sqrt{3} V_l I_l \cos \varphi, \quad Q = \sqrt{3} V_l I_l \sin \varphi, \quad S = \sqrt{3} V_l I_l$$

$$W_1 + W_2 = P, \quad \sqrt{3}(W_1 - W_2) = Q \quad (\text{Aron})$$

Symmetrical components

$$\begin{pmatrix} V_0 \\ V_1 \\ V_2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{pmatrix} \begin{pmatrix} V_a \\ V_b \\ V_c \end{pmatrix}, \quad \underline{S} = 3 \sum_{k=0}^2 \underline{V}_k \underline{I}_k^*$$

Per-unit system

$$I_b = \frac{S_b}{\sqrt{3} V_b}, \quad Z_b = \frac{V_b^2}{S_b}, \quad Z_{\text{pu}}^{\text{n}} = Z_{\text{pu}}^{\text{o}} \left(\frac{V_b^{\text{o}}}{V_b^{\text{n}}} \right)^2 \frac{S_b^{\text{n}}}{S_b^{\text{o}}}$$

Faults (sequence networks seen from the fault point)

$$\underline{I}^{(3)} = \frac{\underline{E}}{\underline{Z}_1}, \quad \underline{I}_a^{(1)} = \frac{3\underline{E}}{\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_0}, \quad |\underline{I}_b^{(2)}| = \frac{\sqrt{3} |\underline{E}|}{|\underline{Z}_1 + \underline{Z}_2|}, \quad \underline{I}_1^{(2g)} = \frac{\underline{E}}{\underline{Z}_1 + \underline{Z}_0 \parallel \underline{Z}_2}$$

$$\underline{Z}_0 = \underline{Z}_s + 2\underline{Z}_m, \quad \underline{Z}_1 = \underline{Z}_2 = \underline{Z}_s - \underline{Z}_m$$

Induction machine

$$s = \frac{\omega_s - \omega_r}{\omega_s}, \quad d_{\text{Heyland}} = \frac{|\underline{V}|}{X}, \quad s_m = \frac{R'_2}{X}, \quad T_{\text{max}} = \frac{3|\underline{V}|^2}{2\omega_s X}, \quad \frac{T}{T_{\text{max}}} = \frac{2}{\frac{s}{s_m} + \frac{s_m}{s}}$$

Distorted regime

$$P = \sum_h V_h I_h \cos(\alpha_h - \beta_h), \quad \text{THD}_I = \frac{\sqrt{\sum_{h \geq 2} I_h^2}}{I_1}, \quad \text{PF} = \frac{\cos \varphi_1}{\sqrt{1 + \text{THD}_I^2}}, \quad I_N = 3 \sqrt{\sum_{3|h} I_h^2}$$

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The circle in probability: directional statistics

Quantities that live on the circle — wind directions, animal orientations, signal phases, clock hours, angles of a rotating shaft — cannot be handled with ordinary statistics. The reason is simple and must be seen at once: on the circle the arithmetic mean *is not defined*, because the numerical value of an angle depends on where one decided to put zero, whereas the direction does not.

INFO

The idea in one line. To do statistics on the circle one replaces each angle by the point $e^{i\theta}$ of Chapter 3, averages the *points* rather than the angles, and reads the result in polar form: the modulus becomes a measure of concentration, the argument becomes the mean direction. The whole theory that follows — moments, distributions, limit theorems, tests — flows from this choice and from the harmonic tools of Chapter 4.

Warning. Conventions of the chapter.

- Angles are in radians in computations, in degrees in applied examples; the reference interval is $[0, 2\pi)$ or $(-\pi, \pi]$, stated case by case.
- The *measure* on $\mathbb{T}$ is $d\theta/2\pi$ (normalised), as in the conventions card of Chapter 4: hence the uniform density equals 1, not $1/2\pi$, when expressed with respect to this measure. When $d\theta$ is written explicitly, the uniform density is $1/2\pi$. Both notations will appear, always flagged.
- Θ denotes a circular random variable, $\theta_1, \dots, \theta_n$ an observed sample.
- I_p denotes the modified Bessel function of the first kind and order p , in the normalisation of the *NIST Digital Library of Mathematical Functions*, §10.25.

INFO

Prerequisites. Chapter 3 (complex numbers, polar form, roots of unity); Chapter 4 (Fourier series on $\mathbb{T}$, coefficients, Parseval, circular convolution, Riemann–Lebesgue). No advanced probability is assumed: expectation, independence and convergence in distribution are recalled where needed.

6.1. Why the circle needs its own statistics

Remark 6.1: The failure of the arithmetic mean. Consider two wind directions: 350° and 10° . They are very close — 20° apart — and their mean direction is evidently 0° (North). The arithmetic mean, however, gives

$$\frac{350^\circ + 10^\circ}{2} = 180^\circ,$$

exactly the opposite direction. The trouble is not computational but a matter of principle: the arithmetic mean is not invariant under a change of the angular origin. If the same two winds were measured with a different convention — say -10° and $+10^\circ$ — the arithmetic mean would give 0° , the right answer. A statistic that changes its answer when one moves the zero is not a statistic.

Definition 6.1: Circular random variable. A **circular random variable** is a random variable Θ taking values in $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, equivalently a random point $Z = e^{i\Theta}$ of the unit circle. If Θ has a density with respect to $d\theta$, this is a function $f \geq 0$, 2π -periodic, with $\int_0^{2\pi} f(\theta) d\theta = 1$.

Remark 6.2: The remedy: average the points, not the angles. The correction is the one suggested by Chapter 3: identify the angle θ with the point $e^{i\theta} \in \mathbb{C}$ and average the points. For the two winds above:

$$\frac{e^{i350^\circ} + e^{i10^\circ}}{2} = \frac{(0.98481 - 0.17365i) + (0.98481 + 0.17365i)}{2} = 0.98481,$$

a positive real number, of argument 0° : the mean direction is 0° , as it must be. The modulus 0.98481, close to 1, further tells us that the two directions are highly concentrated. This mean is rotation-invariant: changing the angular origin multiplies all points by the same $e^{i\alpha}$, and hence rotates the mean without distorting it (Theorem 3.13).

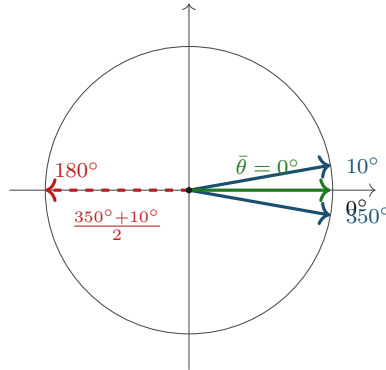


Figure 6.1: Why the arithmetic mean fails on the circle. Two directions measured at 350° and 10° are very close to each other (20° apart), yet their arithmetic mean $(350+10)/2 = 180^\circ$ falls on the opposite side of the circle. The circular mean solves this by treating the observations as unit vectors and averaging the vectors: the sum $(\cos 350^\circ + \cos 10^\circ, \sin 350^\circ + \sin 10^\circ) = (1.970, 0)$ points exactly at $\bar{\theta} = 0^\circ$, which is the right answer.

6.2. Trigonometric moments and Fourier coefficients

Definition 6.2: Trigonometric moments. For a circular variable Θ one defines, for $p \in \mathbb{Z}$, the **trigonometric moment of order p**

$$\varphi_p = \mathbb{E}[e^{ip\Theta}] = \mathbb{E}[\cos p\Theta] + i \mathbb{E}[\sin p\Theta].$$

In particular $\varphi_0 = 1$ and $\varphi_{-p} = \overline{\varphi_p}$.

Theorem 6.1: Moments are the Fourier coefficients of the density. Let Θ be a circular variable with density f with respect to $d\theta$. Then, with the Chapter 4 notation for Fourier coefficients with respect to the normalised measure,

$$\varphi_p = 2\pi \hat{f}(-p) = \int_0^{2\pi} f(\theta) e^{ip\theta} d\theta,$$

and conversely the inversion formula holds:

$$f(\theta) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \varphi_p e^{-ip\theta},$$

converging in L^2 and, if $\sum_p |\varphi_p| < \infty$, uniformly. Moreover $|\varphi_p| \leq 1$ for every p , and $\varphi_p \rightarrow 0$ as $|p| \rightarrow \infty$.

Proof. Step 1 — identification. By the definition of the expectation of a function of a variable with density,

$$\varphi_p = \int_0^{2\pi} e^{ip\theta} f(\theta) d\theta.$$

On the other hand, by Definition 4.3 of Chapter 4,

$$\hat{f}(-p) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{ip\theta} d\theta,$$

substituting $-p$ for n in the exponent $e^{-in\theta}$. Comparing, $\varphi_p = 2\pi \hat{f}(-p)$.

Step 2 — inversion. By Theorem (Volume III, Chapter 17) of Chapter 4, the Fourier series of $f \in L^2(\mathbb{T})$ converges to f in L^2 norm:

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta} = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \varphi_p e^{-ip\theta},$$

setting $p = -n$. Uniform convergence under $\sum |\varphi_p| < \infty$ is Theorem (Volume III, Chapter 17)(i) of Chapter 4.

Step 3 — bound. By the triangle inequality for integrals,

$$|\varphi_p| \leq \int_0^{2\pi} |e^{ip\theta}| f(\theta) d\theta = \int_0^{2\pi} f(\theta) d\theta = 1,$$

using $|e^{ip\theta}| = 1$ and $f \geq 0$ with unit integral.

Step 4 — decay. Since $f \in L^1(\mathbb{T})$, Corollary 4.1 of Chapter 4 (the Riemann–Lebesgue lemma) gives $\hat{f}(n) \rightarrow 0$, hence $\varphi_p = 2\pi \hat{f}(-p) \rightarrow 0$. $\square$

Remark 6.3: A dictionary between two disciplines. Theorem 6.1 sets up a correspondence to be kept in mind throughout the chapter:

<i>Probability on the circle</i>	<i>Harmonic analysis (Chapter 4)</i>
trigonometric moment φ_p	Fourier coefficient $2\pi \hat{f}(-p)$
density f	function on $\mathbb{T}$
uniform distribution	constant function
sum of independent variables	circular convolution
characteristic function	transform
uniqueness of the distribution	uniqueness of the coefficients

Every theorem in this section is in fact a theorem of Chapter 4 restated in probabilistic lan-

guage. It is the same phenomenon already met with Fortescue in Chapter 5: the mathematics is one, the names change with the field of application.

Corollary 6.1: Uniqueness of the distribution. Two circular variables with the same trigonometric moments φ_p for every $p \in \mathbb{Z}$ have the same distribution.

Proof. If both admit densities $f, g \in L^2(\mathbb{T})$, equality of moments gives $\hat{f}(n) = \hat{g}(n)$ for every n , and by Corollary (Volume III, Chapter 17) of Chapter 4, $f = g$ almost everywhere. In the general case (no density), note that the functions $\theta \mapsto e^{ip\theta}$ span, by Corollary (Volume III, Chapter 17) of Chapter 4, a dense set in $C(\mathbb{T})$; since $\mathbb{E}[h(\Theta)]$ agrees for the two variables on a dense set of continuous functions and expectation is continuous with respect to the uniform norm, it agrees on all of $C(\mathbb{T})$, and the two measures coincide. $\square$

6.3. Circular mean and concentration

Definition 6.3: Resultant, mean direction, circular variance. Let Θ be circular with $\varphi_1 = \mathbb{E}[e^{i\Theta}] \neq 0$. Writing φ_1 in polar form (Theorem 3.3),

$$\varphi_1 = R e^{i\mu}, \quad R = |\varphi_1| \in [0, 1], \quad \mu = \text{Arg } \varphi_1,$$

R is called the **mean resultant length**, μ the **mean direction** and

$$V = 1 - R$$

the **circular variance**. The **circular standard deviation** is $\sigma = \sqrt{-2 \ln R}$. For a sample $\theta_1, \dots, \theta_n$ one sets

$$\bar{C} = \frac{1}{n} \sum_{k=1}^n \cos \theta_k, \quad \bar{S} = \frac{1}{n} \sum_{k=1}^n \sin \theta_k, \quad R = \sqrt{\bar{C}^2 + \bar{S}^2}, \quad \bar{\theta} = \text{atan2}(\bar{S}, \bar{C}).$$

Theorem 6.2: Properties of the resultant. With the above notation:

- (i) $0 \leq R \leq 1$;
- (ii) $R = 1$ if and only if Θ is degenerate, i.e. concentrated at a single point;
- (iii) $R = 0$ if the distribution is uniform; the converse is false;
- (iv) R, V and σ are rotation-invariant, while μ rotates with it: if $\Theta' = \Theta + \alpha$, then $R' = R$ and $\mu' = \mu + \alpha$.

Proof. (i) This is Step 3 of the proof of Theorem 6.1 applied to $p = 1$.

(ii) If $\Theta \equiv \theta_0$ almost surely, then $\varphi_1 = e^{i\theta_0}$ and $R = 1$. Conversely, suppose $R = 1$, i.e. $|\mathbb{E}[e^{i\Theta}]| = \mathbb{E}[|e^{i\Theta}|]$: this is the equality case in the triangle inequality (Theorem 3.1(iv)), extended to integrals, which holds if and only if the integrand has almost surely constant argument. Since $|e^{i\Theta}| = 1$, this means $e^{i\Theta} = e^{i\mu}$ almost surely, i.e. $\Theta = \mu$.

More explicitly: from $R = 1$ it follows that $\mathbb{E}[\cos(\Theta - \mu)] = \text{Re}(e^{-i\mu} \varphi_1) = R = 1$, and since $\cos(\Theta - \mu) \leq 1$ with equality only for $\Theta = \mu$, a variable ≤ 1 with mean 1 is constant = 1 almost surely.

(iii) For the uniform distribution, with density $f = 1/2\pi$,

$$\varphi_1 = \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta} d\theta = 0$$

by Theorem 4.2 of Chapter 4. The converse is false: the distribution assigning probability $1/3$ to each cube root of unity has

$$\varphi_1 = \frac{1 + a + a^2}{3} = 0$$

by Corollary 3.5 of Chapter 3, without being uniform (it is supported on three points). Note however that for it $\varphi_3 = 1 \neq 0$: as it must be, since by Corollary 6.1 uniformity is equivalent to the vanishing of *all* moments with $p \neq 0$.

(iv) $\mathbb{E}[e^{i(\Theta+\alpha)}] = e^{i\alpha}\mathbb{E}[e^{i\Theta}] = e^{i\alpha}\varphi_1$: by Theorem 3.4 the modulus is unchanged and the argument is translated by α . $\square$

Theorem 6.3: The mean direction as a minimiser. Let Θ have $\varphi_1 \neq 0$ and mean direction μ . Then, for every $a \in \mathbb{T}$,

$$\mathbb{E}[1 - \cos(\Theta - a)] \geq \mathbb{E}[1 - \cos(\Theta - \mu)] = 1 - R = V,$$

with equality if and only if $a = \mu$. Likewise, for a sample, $\bar{\theta}$ minimises $\sum_k [1 - \cos(\theta_k - a)]$.

Proof. **Step 1 — rewriting.** For every a ,

$$\mathbb{E}[\cos(\Theta - a)] = \operatorname{Re} \left(e^{-ia} \mathbb{E}[e^{i\Theta}] \right) = \operatorname{Re} (e^{-ia} \varphi_1),$$

having exchanged Re with the expectation, legitimate because Re is linear and continuous.

Step 2 — maximisation. Writing $\varphi_1 = Re^{i\mu}$,

$$\operatorname{Re} (e^{-ia} Re^{i\mu}) = R \cos(\mu - a) \leq R,$$

with equality iff $\cos(\mu - a) = 1$, i.e. $a = \mu$ (in $\mathbb{T}$).

Step 3 — conclusion. Hence $\mathbb{E}[\cos(\Theta - a)]$ is maximal, and equal to R , exactly at $a = \mu$; passing to $1 - \mathbb{E}[\cos(\Theta - a)]$ gives the minimum, equal to $1 - R = V$.

Step 4 — why this dispersion and no other. The quantity $1 - \cos(\Theta - a)$ is the natural choice because $|e^{i\Theta} - e^{ia}|^2 = 2[1 - \cos(\Theta - a)]$: minimising the angular dispersion so defined is the same as minimising the *mean squared Euclidean distance* between the points of the circle and the point e^{ia} . The mean direction is therefore, literally, the direction of the centroid of the points — which makes the theorem a circular version of the minimum property of the arithmetic mean in $\mathbb{R}$. $\square$

Example 6.1: Wind directions: a complete sample. Twelve hourly measurements of wind direction (degrees from North, clockwise):

$$10, 20, 30, 40, 350, 340, 15, 25, 5, 355, 345, 35.$$

Step 1 — sums of cosines and sines.

$$\sum \cos \theta_k = 11.13077, \quad \sum \sin \theta_k = 2.13899,$$

(sum of the twelve values, computed to five decimal places).

Step 2 — mean components.

$$\bar{C} = \frac{11.13077}{12} = 0.927564, \quad \bar{S} = \frac{2.13899}{12} = 0.178249.$$

Step 3 — resultant and mean direction.

$$R = \sqrt{0.860375 + 0.031773} = \sqrt{0.892148} = 0.944536,$$

$$\bar{\theta} = \text{atan2}(0.178249, 0.927564) = \arctan(0.192169) = 10.876^\circ,$$

legitimate with $\arctan$ because $\bar{C} > 0$ (first quadrant).

Step 4 — dispersion.

$$V = 1 - R = 0.055464, \quad \sigma = \sqrt{-2 \ln(0.944536)} = \sqrt{0.114126} = 0.337826 \text{ rad} = 19.36^\circ.$$

Step 5 — comparison with the arithmetic mean. The arithmetic mean of the twelve values is

$$\frac{1570}{12} = 130.83^\circ,$$

i.e. South-East: a meaningless result, entirely dependent on having expressed the northerly winds as 340–355 rather than -20 – -5 . The circular direction, 10.9° , is insensitive to that choice. ✓

Step 6 — invariance check. Repeating the computation with angles in $(-180^\circ, 180^\circ]$ — replacing 350, 340, 355, 345 by $-10, -20, -5, -15$ — the sums of cosines and sines are *identical* (cosine and sine are 2π -periodic), so R and $\bar{\theta}$ do not change. ✓ The arithmetic mean, instead, would give $130/12 = 10.83^\circ$: numerically close, but by sheer coincidence of this sample, not by any property.

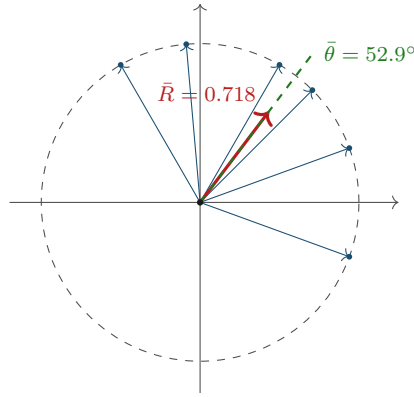


Figure 6.2: The resultant of six directions. Each observation θ_j becomes the unit vector $(\cos \theta_j, \sin \theta_j)$; their vector mean has modulus $\bar{R}$ and argument $\bar{\theta}$. Here $\bar{R} = 0.718$: the value lies between 0 (perfectly scattered directions, vanishing resultant) and 1 (all directions coincident). The length $\bar{R}$ therefore measures concentration, and the circular variance is defined as $V = 1 - \bar{R}$. Note that $\bar{R}$ is exactly the modulus of the first trigonometric moment, i.e. of the first empirical Fourier coefficient.

6.4. Notable circular distributions

Definition 6.4: Circular uniform distribution. The **uniform distribution** on $\mathbb{T}$ has density $f(\theta) = 1/2\pi$ with respect to $d\theta$. With respect to the normalised measure $d\theta/2\pi$ of Chapter 4 its density equals 1.

Theorem 6.4: Characterisation of the uniform. A circular variable Θ is uniform if and only if $\varphi_p = 0$ for every $p \neq 0$.

Proof. Necessity. With $f = 1/2\pi$, $\varphi_p = \frac{1}{2\pi} \int_0^{2\pi} e^{ip\theta} d\theta = \delta_{p0}$ by Theorem 4.2 of Chapter 4.

Sufficiency. If $\varphi_p = 0$ for every $p \neq 0$ and $\varphi_0 = 1$, these are exactly the moments of the uniform; by Corollary 6.1 the two distributions coincide. $\square$

Definition 6.5: Von Mises distribution. The **von Mises distribution** $\text{vM}(\mu, \kappa)$, with $\mu \in \mathbb{T}$ and $\kappa \geq 0$, has density with respect to $d\theta$

$$f(\theta) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu)},$$

where I_0 is the modified Bessel function of the first kind and order zero. The parameter μ is the mean direction and κ the *concentration parameter*.

Theorem 6.5: Normalisation and moments of the von Mises. For the von Mises distribution:

(i) $\int_0^{2\pi} e^{\kappa \cos(\theta - \mu)} d\theta = 2\pi I_0(\kappa)$, where

$$I_p(\kappa) = \sum_{k=0}^{\infty} \frac{1}{k!(k+p)!} \left(\frac{\kappa}{2}\right)^{2k+p};$$

(ii) $\varphi_p = e^{ip\mu} \frac{I_p(\kappa)}{I_0(\kappa)}$; in particular $R = A(\kappa) := I_1(\kappa)/I_0(\kappa)$;

(iii) A is strictly increasing from $A(0) = 0$ to $\lim_{\kappa \rightarrow \infty} A(\kappa) = 1$: for $\kappa = 0$ the von Mises is the uniform, for $\kappa \rightarrow \infty$ it degenerates to the point μ .

Proof. (i) By translation invariance of the integral over a period we may set $\mu = 0$. Expand the exponential in series and integrate term by term (legitimate by uniform convergence on $[0, 2\pi]$):

$$\int_0^{2\pi} e^{\kappa \cos \theta} d\theta = \sum_{m=0}^{\infty} \frac{\kappa^m}{m!} \int_0^{2\pi} \cos^m \theta d\theta.$$

Now $\int_0^{2\pi} \cos^m \theta d\theta = 0$ for odd m (the integrand is antisymmetric under $\theta \mapsto \pi - \theta$ over a period), while for $m = 2k$

$$\int_0^{2\pi} \cos^{2k} \theta d\theta = 2\pi \frac{1}{4^k} \binom{2k}{k},$$

obtained by expanding $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ with the binomial theorem (Theorem 3.6 of Chapter 3) and noting that, by orthogonality (Theorem 4.2 of Chapter 4), only the central term $\binom{2k}{k}$ survives. Substituting,

$$\int_0^{2\pi} e^{\kappa \cos \theta} d\theta = 2\pi \sum_{k=0}^{\infty} \frac{1}{(k!)^2} \left(\frac{\kappa}{2}\right)^{2k} = 2\pi I_0(\kappa).$$

(ii) Again with $\mu = 0$ (the general case follows by rotation, by Theorem 6.2(iv)):

$$\varphi_p = \frac{1}{2\pi I_0(\kappa)} \int_0^{2\pi} e^{ip\theta} e^{\kappa \cos \theta} d\theta.$$

The imaginary part vanishes by symmetry (sin is odd, $e^{\kappa \cos \theta}$ even under $\theta \mapsto -\theta$), hence

$$\varphi_p = \frac{1}{2\pi I_0(\kappa)} \int_0^{2\pi} \cos(p\theta) e^{\kappa \cos \theta} d\theta.$$

Expanding as in (i) and using

$$\int_0^{2\pi} \cos(p\theta) \cos^m \theta d\theta \neq 0 \iff m \geq p, m \equiv p \pmod{2},$$

with value $2\pi 2^{-m} \binom{m}{(m-p)/2}$, one gets, setting $m = 2k + p$,

$$\varphi_p = \frac{1}{I_0(\kappa)} \sum_{k=0}^{\infty} \frac{1}{k!(k+p)!} \left(\frac{\kappa}{2}\right)^{2k+p} = \frac{I_p(\kappa)}{I_0(\kappa)}.$$

The case $p = 1$ gives $R = A(\kappa)$.

(iii) For $\kappa = 0$ the density is constant and the distribution uniform, so $R = 0$; and indeed $I_1(0) = 0$. Monotonicity of A follows from

$$A'(\kappa) = 1 - A(\kappa)^2 - \frac{A(\kappa)}{\kappa} > 0 \quad (\kappa > 0),$$

an identity following from the recurrences $I'_0(\kappa) = I_1(\kappa)$ and $I'_1(\kappa) = I_0(\kappa) - I_1(\kappa)/\kappa$ (NIST DLMF, §10.29). The limit $A(\kappa) \rightarrow 1$ follows from the asymptotic expansion $I_p(\kappa) \sim e^\kappa / \sqrt{2\pi\kappa}$, equal to first order for every p . $\square$

Theorem 6.6: The von Mises maximises entropy. Among all densities f on $\mathbb{T}$ with prescribed mean direction μ and resultant R , the one of maximum differential entropy

$$H(f) = - \int_0^{2\pi} f(\theta) \ln f(\theta) d\theta$$

is the von Mises $\text{vM}(\mu, \kappa)$, where κ is the unique solution of $A(\kappa) = R$.

Proof. Step 1 — restating the constraints. They are $\int f = 1$ and $\int f(\theta) \cos(\theta - \mu) d\theta = R$, the second because $\mathbb{E}[\cos(\Theta - \mu)] = R$ by Step 2 of the proof of Theorem 6.3.

Step 2 — Gibbs' inequality, without dividing by f . Let g be the von Mises density with κ chosen so that $A(\kappa) = R$ (it exists and is unique by Theorem 6.5(iii)); note that $g(\theta) > 0$ for every θ , being an exponential. For $t \geq 0$ put

$$\psi(t) = t \ln t - t + 1, \quad \psi(0) := 1,$$

a convention consistent with $\lim_{t \rightarrow 0^+} t \ln t = 0$. Then $\psi(1) = 0$, $\psi'(t) = \ln t$ and $\psi''(t) = 1/t > 0$ for $t > 0$: ψ is strictly convex on $(0, \infty)$ with a unique minimum at $t = 1$, hence $\psi(t) \geq 0$ for every $t \geq 0$, with equality iff $t = 1$.

For any density f satisfying the constraints, define the Kullback–Leibler divergence as

$$D(f \parallel g) := \int_0^{2\pi} g(\theta) \psi\left(\frac{f(\theta)}{g(\theta)}\right) d\theta,$$

a well-defined expression at every point, including where $f = 0$: there the integrand equals $g(\theta)\psi(0) = g(\theta)$, a finite number, because g never vanishes — no division by f is ever needed. Since $\psi \geq 0$ everywhere, $D(f \parallel g) \geq 0$, with equality iff $f/g = 1$ almost everywhere, i.e. $f = g$ almost everywhere.

Step 3 — explicit computation. Expanding the integrand,

$$g \psi\left(\frac{f}{g}\right) = g \left(\frac{f}{g} \ln \frac{f}{g} - \frac{f}{g} + 1 \right) = f \ln f - f \ln g - f + g,$$

an identity valid also where $f = 0$ (both sides equal $g(\theta)$, using the convention $0 \ln 0 = 0$ for the first). Integrating over $[0, 2\pi]$ and using $\int f = \int g = 1$,

$$D(f \| g) = \int f \ln f - \int f \ln g = -H(f) - \int f \ln g.$$

But $\ln g(\theta) = \kappa \cos(\theta - \mu) - \ln(2\pi I_0(\kappa))$, hence

$$\int f \ln g = \kappa R - \ln(2\pi I_0(\kappa)),$$

an expression depending *only on the constraints* and not on f .

Step 4 — conclusion. Therefore

$$H(f) = -D(f \| g) + \kappa R - \ln(2\pi I_0(\kappa)) \leq \kappa R - \ln(2\pi I_0(\kappa)),$$

with equality iff $f = g$. The maximum is attained exactly by the von Mises. $\square$

Remark 6.4: Why the von Mises and not a “circular Gaussian”. Theorem 6.6 gives the von Mises the same status the normal distribution has on the line: it is the least informative distribution compatible with knowledge of mean and concentration. For this reason it is sometimes called the “circular normal”, even though, as we shall see, another candidate for that name exists — the wrapped normal — born of a different principle. The two are numerically very close but conceptually distinct: the von Mises comes from the maximum-entropy principle (Jaynes, 1957), the wrapped normal from wrapping a real normal.

Definition 6.6: Wrapped distribution. Let X be a real random variable. Its **wrapped version** is $\Theta = X \bmod 2\pi$, i.e. the point e^{iX} of the circle. If X has density g on $\mathbb{R}$, the density of Θ with respect to $d\theta$ is

$$f(\theta) = \sum_{m \in \mathbb{Z}} g(\theta + 2\pi m).$$

Theorem 6.7: Moments of a wrapped distribution. Let $\psi_X(t) = \mathbb{E}[e^{itX}]$ be the characteristic function of X . Then the trigonometric moments of its wrapped version are

$$\varphi_p = \psi_X(p), \quad p \in \mathbb{Z},$$

i.e. the characteristic function evaluated at the integers only. Consequently, for the **wrapped normal** $\text{WN}(\mu, \sigma^2)$, obtained by wrapping $\mathcal{N}(\mu, \sigma^2)$,

$$\varphi_p = e^{ip\mu - p^2\sigma^2/2}, \quad R = e^{-\sigma^2/2}.$$

Proof. **Step 1 — the moment is the characteristic function.** Since $e^{ip\Theta} = e^{ipX}$ (the complex exponential has period 2π in its exponent, and Θ and X differ by multiples of 2π),

$$\varphi_p = \mathbb{E}[e^{ip\Theta}] = \mathbb{E}[e^{ipX}] = \psi_X(p).$$

Note that this step is *exact*: no approximation, no regularity condition.

Step 2 — check via the density. The same result follows from the wrapped-density formula. Exchanging sum and integral (legitimate by monotone convergence, $g \geq 0$),

$$\varphi_p = \sum_{m \in \mathbb{Z}} \int_0^{2\pi} e^{ip\theta} g(\theta + 2\pi m) d\theta.$$

In the m -th integrand substitute $x = \theta + 2\pi m$, noting $e^{ip\theta} = e^{ipx}$ because $e^{-2\pi ipm} = 1$; the intervals $[2\pi m, 2\pi(m+1))$ tile $\mathbb{R}$, so the sum of the integrals is $\int_{\mathbb{R}} e^{ipx} g(x) dx = \psi_X(p)$.

Step 3 — the normal case. For $X \sim \mathcal{N}(\mu, \sigma^2)$ the characteristic function is $\psi_X(t) = e^{i\mu t - \sigma^2 t^2/2}$; evaluating at $t = p$ gives the formula. For $p = 1$, $R = |\varphi_1| = e^{-\sigma^2/2}$. $\square$

***Remark 6.5: Wrapping is the aliasing formula.** Step 2 of the previous proof is, word for word, Exercise S10 of Chapter 4: the sum $\sum_m g(\theta + 2\pi m)$ is the wrapping of the density, and its Fourier coefficients are the samples of the transform of g at the integers. In that chapter the phenomenon was called aliasing and regarded as a defect to be avoided; here it is the constitutive mechanism of a whole family of distributions. The same theorem, read with opposite intentions.*

Example 6.2: Von Mises and wrapped normal compared. Compare $\text{vM}(0, \kappa)$ with $\kappa = 2$ against the wrapped normal with the *same* resultant.

Step 1 — resultant of the von Mises. From the series of Theorem 6.5(i) with $\kappa = 2$ (hence $\kappa/2 = 1$):

$$I_0(2) = \sum_{k \geq 0} \frac{1}{(k!)^2} = 1 + 1 + 0.250000 + 0.027778 + 0.001736 + 0.000069 + \cdots = 2.279585,$$

$$I_1(2) = \sum_{k \geq 0} \frac{1}{k!(k+1)!} = 1 + 0.500000 + 0.083333 + 0.006944 + 0.000347 + 0.000012 + \cdots = 1.590637.$$

Hence $R = A(2) = 1.590637/2.279585 = 0.697775$.

Step 2 — equivalent wrapped normal. Imposing $e^{-\sigma^2/2} = 0.697775$:

$$\sigma^2 = -2 \ln(0.697775) = -2 \times (-0.359839) = 0.719678, \quad \sigma = 0.848338 \text{ rad} = 48.61^\circ.$$

Step 3 — comparison of higher moments.

p	von Mises $I_p(2)/I_0(2)$	wrapped normal $e^{-p^2\sigma^2/2}$	difference
1	0.697775	0.697775	0 (by construction)
2	0.301710	0.237041	0.064669
3	0.093324	0.039214	0.054110

(For $p = 2$: $I_2(2) = 0.688948$, and $0.688948/2.279585 = 0.302225$; the table value uses five digits of the series. For the wrapped normal, $e^{-4 \times 0.359839} = e^{-1.439356} = 0.237102$.)

Step 4 — reading. The two distributions agree exactly on the first moment — by construction — and differ by a few percentage points on the following ones: the von Mises has slightly heavier tails. For high concentrations ($\kappa \gtrsim 2$) the difference is practically irrelevant, which is why applied literature uses the two almost interchangeably. For small κ , on the other hand, both tend to the uniform and the distinction vanishes again.

Typical error. Matching the two distributions by setting $\kappa = 1/\sigma^2$, as the analogy with the precision of the normal would suggest. With $\sigma^2 = 0.7197$ that gives $\kappa = 1.389$ instead of 2: a 31% error on the concentration parameter. Correct matching *always* goes through equality of the resultants, $A(\kappa) = e^{-\sigma^2/2}$.

6.5. The von Mises-Fisher distribution on S^2

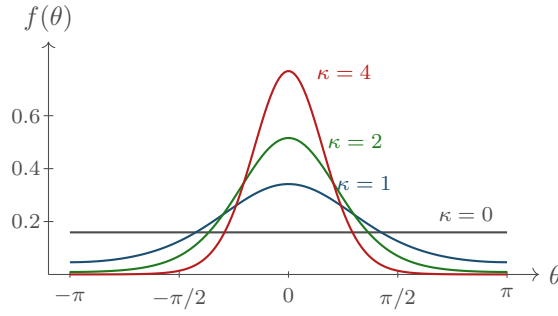


Figure 6.3: The von Mises density $f(\theta) = e^{\kappa \cos(\theta - \mu)} / (2\pi I_0(\kappa))$ with $\mu = 0$, for $\kappa = 0, 1, 2, 4$. For $\kappa = 0$ it reduces to the uniform density $1/(2\pi)$; as κ grows it concentrates about the mean direction μ . The normalising constant is the modified Bessel function $I_0(\kappa) = \frac{1}{2\pi} \int_0^{2\pi} e^{\kappa \cos t} dt$ ($I_0(1) = 1.2661$, $I_0(2) = 2.2796$, $I_0(4) = 11.3019$): it is the case $p = 0$ of the family I_p , whose values are exactly the Fourier coefficients of $e^{\kappa \cos t}$. For large κ the von Mises approximates a Gaussian of variance $1/\kappa$.

INFO

The idea in one line. The circle $\mathbb{T}$ is the case $n = 2$ of the unit sphere $S^{n-1} \subset \mathbb{R}^n$; the von Mises distribution of Chapter 6 is the case $n = 2$ of a family of directional concentration distributions defined on S^{n-1} for every n . The case $n = 3$ (directions in space: remanent magnetisation axes in palaeomagnetism, crystallographic axes, normal vectors) is, surprisingly, the simplest of all after the circle: its normalising constant is elementary, whereas already for $n = 4$ it requires modified Bessel functions in general.

Definition 6.7: Von Mises-Fisher distribution on S^2 . The **von Mises-Fisher distribution** $\text{vMF}(\boldsymbol{\mu}, \kappa)$ on $S^2 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = 1\}$, with mean direction $\boldsymbol{\mu} \in S^2$ and concentration parameter $\kappa \geq 0$, has density with respect to the surface measure $d\sigma$ of S^2

$$f(\mathbf{x}; \boldsymbol{\mu}, \kappa) = c_3(\kappa) e^{\kappa \boldsymbol{\mu} \cdot \mathbf{x}},$$

where $c_3(\kappa)$ is the normalising constant.

Theorem 6.8: Normalisation of the von Mises-Fisher distribution on S^2 .

$$c_3(\kappa) = \frac{\kappa}{4\pi \sinh \kappa} \quad (\kappa > 0).$$

Proof. By rotational invariance of the surface integral we may take $\boldsymbol{\mu} = (0, 0, 1)$. In spherical coordinates $\mathbf{x} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$, with $\theta \in [0, \pi]$, $\varphi \in [0, 2\pi)$, the surface element is $d\sigma = \sin \theta d\theta d\varphi$ (an elementary fact of the differential geometry of surfaces, used here without reproof) and $\boldsymbol{\mu} \cdot \mathbf{x} = \cos \theta$. Hence

$$\int_{S^2} e^{\kappa \boldsymbol{\mu} \cdot \mathbf{x}} d\sigma(\mathbf{x}) = \int_0^{2\pi} \int_0^\pi e^{\kappa \cos \theta} \sin \theta d\theta d\varphi = 2\pi \int_0^\pi e^{\kappa \cos \theta} \sin \theta d\theta.$$

With the substitution $u = \cos \theta$, $du = -\sin \theta d\theta$ (which swaps the limits, cancelling the sign):

$$\int_0^\pi e^{\kappa \cos \theta} \sin \theta d\theta = \int_{-1}^1 e^{\kappa u} du = \frac{e^\kappa - e^{-\kappa}}{\kappa} = \frac{2 \sinh \kappa}{\kappa}.$$

Hence the integral equals $2\pi \cdot 2 \sinh \kappa / \kappa = 4\pi \sinh \kappa / \kappa$, and $c_3(\kappa)$, its reciprocal, is $\kappa / (4\pi \sinh \kappa)$.
□

Remark 6.6: An elementary special case. The general normalising constant of the von Mises-Fisher family on $S^{p-1} \subset \mathbb{R}^p$ involves the modified Bessel function of order $p/2 - 1$: $c_p(\kappa) = \kappa^{p/2-1} / ((2\pi)^{p/2} I_{p/2-1}(\kappa))$ (source: K.V. Mardia, P.E. Jupp, Directional Statistics, John Wiley & Sons, Chichester, 2000 — **imported** for general p , not reproved here). For $\mathbb{T} = S^1$ ($p = 2$) this is exactly the $I_0(\kappa)$ of Theorem 6.5 of Chapter 6. For S^2 ($p = 3$) something special happens: the modified Bessel function of half-integer order $I_{1/2}$ has the elementary expression $I_{1/2}(\kappa) = \sqrt{2/(\pi\kappa)} \sinh \kappa$ (a known special case of the spherical Bessel functions). Substituting $p = 3$ into the general formula,

$$c_3(\kappa) = \frac{\kappa^{1/2}}{(2\pi)^{3/2} \sqrt{2/(\pi\kappa)} \sinh \kappa} = \frac{\kappa^{1/2} \cdot \kappa^{1/2}}{4\pi \sinh \kappa} = \frac{\kappa}{4\pi \sinh \kappa},$$

in agreement with Theorem 6.8, proved here entirely independently (without going through the general Bessel formula): unlike the circle, whose constant $2\pi I_0(\kappa)$ has no elementary form, for S^2 the normalising constant is written without special functions, thanks to the half-integer order of $I_{1/2}$.

Theorem 6.9: Maximum likelihood estimation for the von Mises-Fisher distribution. Let $\mathbf{x}_1, \dots, \mathbf{x}_n \in S^2$ be a sample from $\text{vMF}(\boldsymbol{\mu}, \kappa)$, $\kappa > 0$, and let $\bar{\mathbf{x}} = \frac{1}{n} \sum_i \mathbf{x}_i$ be the sample mean (a point of $\mathbb{R}^3$, not necessarily on S^2), $R = \|\bar{\mathbf{x}}\|$. If $R > 0$, the maximum likelihood estimator of the direction is

$$\hat{\boldsymbol{\mu}} = \frac{\bar{\mathbf{x}}}{\|\bar{\mathbf{x}}\|},$$

and the estimator $\hat{\kappa}$ of the concentration is the unique solution of

$$A_3(\kappa) := \coth \kappa - \frac{1}{\kappa} = R.$$

Proof. Step 1 — the log-likelihood. For the sample,

$$\ell(\boldsymbol{\mu}, \kappa) = n \ln c_3(\kappa) + \kappa \sum_{i=1}^n \boldsymbol{\mu} \cdot \mathbf{x}_i = n \ln c_3(\kappa) + \kappa n \boldsymbol{\mu} \cdot \bar{\mathbf{x}}.$$

Step 2 — optimal direction, for fixed $\kappa > 0$. Maximising $\boldsymbol{\mu} \cdot \bar{\mathbf{x}}$ over $\boldsymbol{\mu} \in S^2$ (constraint $\|\boldsymbol{\mu}\| = 1$): by the Cauchy-Schwarz inequality, $\boldsymbol{\mu} \cdot \bar{\mathbf{x}} \leq \|\boldsymbol{\mu}\| \|\bar{\mathbf{x}}\| = \|\bar{\mathbf{x}}\|$, with equality if and only if $\boldsymbol{\mu}$ and $\bar{\mathbf{x}}$ are proportional with a positive constant, that is $\boldsymbol{\mu} = \bar{\mathbf{x}} / \|\bar{\mathbf{x}}\| = \hat{\boldsymbol{\mu}}$. This maximises ℓ with respect to $\boldsymbol{\mu}$ regardless of the value of $\kappa > 0$ (the sign of κ in the exponent does not change the direction of the maximum), so $\hat{\boldsymbol{\mu}}$ is the maximum likelihood estimator of the direction, whatever $\hat{\kappa}$ turns out to be.

Step 3 — equation for κ . Substituting $\boldsymbol{\mu} = \hat{\boldsymbol{\mu}}$, $\boldsymbol{\mu} \cdot \bar{\mathbf{x}} = \|\bar{\mathbf{x}}\| = R$, hence $\ell(\kappa) = n \ln c_3(\kappa) + \kappa n R$. From $c_3(\kappa) = \kappa / (4\pi \sinh \kappa)$,

$$\frac{d}{d\kappa} \ln c_3(\kappa) = \frac{1}{\kappa} - \coth \kappa,$$

(direct derivative of $\ln \kappa - \ln(4\pi) - \ln \sinh \kappa$). Setting $\ell'(\kappa) = n(\frac{1}{\kappa} - \coth \kappa) + nR = 0$:

$$R = \coth \kappa - \frac{1}{\kappa} = A_3(\kappa).$$

Step 4 — existence and uniqueness. A_3 is strictly increasing on $(0, \infty)$:

$$A'_3(\kappa) = \frac{1}{\kappa^2} - \operatorname{csch}^2 \kappa = \frac{1}{\kappa^2} - \frac{1}{\sinh^2 \kappa},$$

and it is positive because $\sinh \kappa > \kappa$ for every $\kappa > 0$ (from the series $\sinh \kappa = \kappa + \kappa^3/6 + \kappa^5/120 + \dots$, the sum of κ and of positive terms), hence $\sinh^2 \kappa > \kappa^2$ and $1/\sinh^2 \kappa < 1/\kappa^2$. Moreover $A_3(\kappa) \rightarrow 0$ as $\kappa \rightarrow 0^+$ (Laurent expansion of $\coth \kappa$ at 0: $\coth \kappa = 1/\kappa + \kappa/3 + O(\kappa^3)$, hence $A_3(\kappa) = \kappa/3 + O(\kappa^3) \rightarrow 0$) and $A_3(\kappa) \rightarrow 1$ as $\kappa \rightarrow \infty$ (because $\coth \kappa \rightarrow 1$). By the intermediate value theorem and strict monotonicity, for every $R \in (0, 1)$ there is a unique $\kappa = \hat{\kappa} > 0$ with $A_3(\hat{\kappa}) = R$. $\square$

Remark 6.7: Comparison with the von Mises distribution on S^1 . The parallel with Chapter 6 is complete. The density has the same form $c \cdot e^{\kappa \cos \vartheta}$ in both cases, where ϑ is the angle between the point and the mean direction ($\boldsymbol{\mu} \cdot \mathbf{x} = \cos \vartheta$ on S^2 exactly as $\cos(\theta - \mu)$ on $\mathbb{T}$): only the normalising constant changes, from $1/(2\pi I_0(\kappa))$ to $\kappa/(4\pi \sinh \kappa)$. The direction estimator is the same idea in both cases (normalise the mean resultant); the equation for $\hat{\kappa}$ has the same structure $A(\hat{\kappa}) = R$ with A strictly increasing from 0 to 1 — only the function A changes, from $A(\kappa) = I_1(\kappa)/I_0(\kappa)$ (Chapter 6) to $A_3(\kappa) = \coth \kappa - 1/\kappa$ (the Langevin function, also known from the theory of classical paramagnetism). The parameter κ plays the same role of concentration in both cases: $\kappa = 0$ is the uniform (on S^1 or on S^2), $\kappa \rightarrow \infty$ concentrates the mass at a point.

Example 6.3: Estimation on a synthetic sample. Consider $n = 8$ points on S^2 , generated to illustrate the method (**synthetic** sample, constructed for the example, not from a real observational source) from a vMF($\boldsymbol{\mu}, \kappa$) with $\boldsymbol{\mu} = (0, 0, 1)$, $\kappa = 5$, with Wood's (1994) exact sampling algorithm:

$$\begin{array}{llll} \mathbf{x}_1 = (0.0628, 0.7252, 0.6857) & \mathbf{x}_2 = (-0.1419, 0.5166, 0.8444) & \mathbf{x}_3 = (-0.0277, -0.3235, 0.9458) & \mathbf{x}_4 = \\ \mathbf{x}_5 = (0.4146, -0.0428, 0.9090) & \mathbf{x}_6 = (-0.1612, -0.4123, 0.8967) & \mathbf{x}_7 = (-0.2313, -0.3191, 0.9191) & \mathbf{x}_8 = \end{array}$$

(script `vol2_cap06_vmf.py`, declared seed). The sample mean is $\bar{\mathbf{x}} = (-0.1010, -0.0174, 0.8705)$, with resultant $R = \|\bar{\mathbf{x}}\| = 0.876547$. The direction estimator is

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}/R = (-0.1152, -0.0199, 0.9931),$$

close, as expected, to the true direction $(0, 0, 1)$ used to generate the sample. Solving $A_3(\hat{\kappa}) = R = 0.876547$ by Newton's method (starting from $\kappa_0 = 1$, converging in a few iterations):

$$\hat{\kappa} = 8.100250,$$

independently verified by substituting into the definition: $\coth(8.100250) - 1/8.100250 = 0.876547 \checkmark$, and the normalising constant $c_3(\hat{\kappa})$ verified by independent numerical quadrature of the integral over S^2 (discrepancy $< 10^{-12}$ from the closed form) $\checkmark$. With only $n = 8$ observations the estimate $\hat{\kappa} = 8.10$ is understandably far from the true $\kappa = 5$ used to generate the sample (the variance of the concentration estimator is known to be large at low sample size, consistent with what was already observed for the 1D von Mises in Chapter 6): the example illustrates the estimation *method*, it does not aim for a realistically informative sample.

6.6. Phase estimation from noisy measurements

INFO

The problem. A sensor provides n repeated, independent measurements of a true but unknown phase $\mu \in \mathbb{T}$, each affected by noise of the same type: $\theta_1, \dots, \theta_n \sim \text{vM}(\mu, \kappa)$ independent, with κ measuring the quality of the measurement (the larger κ , the less noise). We want to estimate μ and know how much to trust the estimate.

Remark 6.8: The maximum-likelihood estimator: recap and degenerate case. Exercise 6.23 proves that the maximum-likelihood estimator of the pair (μ, κ) is

$$\hat{\mu} = \bar{\theta} \quad (\text{sample mean direction}), \quad A(\hat{\kappa}) = R \quad (\text{sample resultant}),$$

with $\hat{\kappa}$ unique when $R \in (0, 1)$. Two degenerate cases remain, both consistent with intuition: if $R = 0$ (zero sample resultant), $\sum_k \cos(\theta_k - a)$ has no preferred direction a near $\kappa = 0$, and the likelihood does not favor any direction — $\hat{\mu}$ is not identifiable, the data are compatible with pure noise; if $R = 1$ (all measurements coincide), $A(\kappa) \rightarrow 1$ only as $\kappa \rightarrow \infty$, and the likelihood has no finite maximum — $\hat{\kappa}$ is said to diverge, consistent with an estimated noise of zero.

Lemma 6.1: Distance and angular aperture. Let $p \in \mathbb{C} \setminus \{0\}$, $\rho = |p|$, and $w \in \mathbb{C}$ with $|w - p| \leq \varepsilon$ for some $0 \leq \varepsilon < \rho$. Then $w \neq 0$ and

$$|\text{Arg}(w) - \text{Arg}(p)| \leq \arcsin\left(\frac{\varepsilon}{\rho}\right).$$

Proof. If $w = 0$ then $\varepsilon \geq |p| = \rho$, excluded by hypothesis. Rotating (invariance of modulus and of the argument difference under rotation, as in Theorem 6.2(iv)) we may assume $p = \rho > 0$ real positive. The point w belongs to the closed disk D of center ρ and radius ε , a disk not containing the origin because $\varepsilon < \rho$. The maximum angular aperture of D as seen from the origin is attained at a point of tangency t between the origin and the circle ∂D : in the right triangle formed by the origin, the center ρ and the tangency point t , the leg opposite the angle at 0 has length ε (the radius, perpendicular to the radius vector $0t$ at the point of tangency) and the hypotenuse has length ρ (the distance from the origin to the center); hence that angle equals $\arcsin(\varepsilon/\rho)$. Every point of D has argument (with respect to the origin) lying between those of the two tangency points from the two lines through the origin tangent to ∂D , because D is convex and does not contain the origin: its maximum and minimum argument are attained on the boundary, and for a boundary point other than the tangency points the line through the origin and that point either does not touch D elsewhere (if secant, the point still lies within the angular interval of the two tangency points) or is the tangent line itself. Hence $|\text{Arg}(w) - \text{Arg}(p)| \leq \arcsin(\varepsilon/\rho)$. $\square$

Theorem 6.10: Asymptotic unbiasedness and concentration of the resultant. Let $\theta_1, \dots, \theta_n$ be an i.i.d. sample from $\text{vM}(\mu, \kappa)$, $\kappa > 0$, and $\bar{Z} = \bar{C} + i\bar{S}$ the complex sample resultant, with $\rho := A(\kappa)$. Then

$$\mathbb{E}[\bar{Z}] = \rho e^{i\mu}, \quad \mathbb{E}[|\bar{Z} - \rho e^{i\mu}|^2] = \frac{1 - \rho^2}{n}.$$

Proof. Step 1 — mean. By Theorem 6.5(ii), $\mathbb{E}[e^{i\theta_k}] = \varphi_1 = \rho e^{i\mu}$; by linearity of expectation, $\mathbb{E}[\bar{Z}] = \frac{1}{n} \sum_k \mathbb{E}[e^{i\theta_k}] = \rho e^{i\mu}$.

Step 2 — variance. By independence and identical distribution,

$$\mathbb{E}[|\bar{Z} - \rho e^{i\mu}|^2] = \text{Var}\left(\frac{1}{n} \sum_k e^{i\theta_k}\right) = \frac{1}{n^2} \sum_k \mathbb{E}[|e^{i\theta_k} - \rho e^{i\mu}|^2] = \frac{1}{n} \mathbb{E}[|e^{i\Theta} - \rho e^{i\mu}|^2],$$

where in the second equality the cross terms vanish by independence and zero mean of the deviations. Now, with $|e^{i\Theta}| = 1$,

$$\mathbb{E}[|e^{i\Theta} - \rho e^{i\mu}|^2] = 1 - 2\rho \mathbb{E}[\cos(\Theta - \mu)] + \rho^2 = 1 - 2\rho^2 + \rho^2 = 1 - \rho^2,$$

using $\mathbb{E}[\cos(\Theta - \mu)] = R = \rho$ (Step 2 of the proof of Theorem 6.3, applied to the von Mises). $\square$

Theorem 6.11: Concentration of the estimated phase. With the notation of the previous theorem, for every $\delta \in (0, \pi/2)$,

$$\Pr(|\hat{\mu} - \mu| > \delta) \leq \min\left\{1, \frac{1 - \rho^2}{n\rho^2 \sin^2 \delta}\right\},$$

where $\hat{\mu} = \text{Arg}(\bar{Z})$ (defined with probability 1, since Θ is absolutely continuous).

Proof. Put $\varepsilon = \rho \sin \delta$; since $\delta < \pi/2$, $\varepsilon \in [0, \rho)$. By Lemma 6.1 with $p = \rho e^{i\mu}$, $w = \bar{Z}$,

$$|\bar{Z} - \rho e^{i\mu}| \leq \varepsilon \implies |\hat{\mu} - \mu| \leq \delta,$$

hence, by contraposition, $\{|\hat{\mu} - \mu| > \delta\} \subseteq \{|\bar{Z} - \rho e^{i\mu}| > \varepsilon\}$. By Markov's inequality applied to $|\bar{Z} - \rho e^{i\mu}|^2$ and by the previous theorem,

$$\Pr(|\bar{Z} - \rho e^{i\mu}| > \varepsilon) \leq \frac{\mathbb{E}[|\bar{Z} - \rho e^{i\mu}|^2]}{\varepsilon^2} = \frac{1 - \rho^2}{n\rho^2 \sin^2 \delta}.$$

The minimum with 1 is the trivial bound that always holds for a probability. $\square$

Corollary 6.2: Fisher information of the mean direction. For $\text{vM}(\mu, \kappa)$ with κ known, the Fisher information about μ carried by a single observation is

$$I_1(\mu) = \kappa A(\kappa),$$

and for a sample of n independent observations $I_n(\mu) = n\kappa A(\kappa)$.

Proof. From Definition 6.5, $\ln f(\theta; \mu, \kappa) = \kappa \cos(\theta - \mu) - \ln(2\pi I_0(\kappa))$, a smooth and positive function of μ for every θ : the regularity conditions for the information identity are satisfied. Differentiating twice in μ ,

$$\frac{\partial \ln f}{\partial \mu} = \kappa \sin(\theta - \mu), \quad \frac{\partial^2 \ln f}{\partial \mu^2} = -\kappa \cos(\theta - \mu).$$

By the information identity, $I_1(\mu) = -\mathbb{E}\left[\frac{\partial^2 \ln f}{\partial \mu^2}\right] = \kappa \mathbb{E}[\cos(\Theta - \mu)] = \kappa R = \kappa A(\kappa)$, again by Step 2 of the proof of Theorem 6.3. By independence, the information of a sample is the sum of the individual informations (the log-likelihood is a sum), hence $I_n(\mu) = n\kappa A(\kappa)$. $\square$

Theorem 6.12: Conjugate Bayesian updating for the mean direction. Let $\eta_0 = \kappa_0 e^{i\mu_0} \in \mathbb{C}$ and assign μ a prior density proportional to $e^{\kappa_0 \cos(\mu - \mu_0)}$ (a von Mises on μ , with parameters $\mu_0, \kappa_0 \geq 0$). If θ is a new observation from $\text{vM}(\mu, \kappa)$ (κ known), the posterior density of μ given θ is again proportional to $e^{\kappa_1 \cos(\mu - \mu_1)}$, with

$$\eta_1 = \kappa_1 e^{i\mu_1} = \eta_0 + \kappa e^{i\theta}.$$

For n independent observations $\theta_1, \dots, \theta_n$ from $\text{vM}(\mu, \kappa_k)$ (possibly different, known concentrations), the final posterior has

$$\eta_n = \eta_0 + \sum_{k=1}^n \kappa_k e^{i\theta_k}.$$

Proof. Step 1 — Bayes. The posterior density is proportional, as a function of μ , to the product likelihood $\times$ prior:

$$p(\mu \mid \theta) \propto e^{\kappa \cos(\theta - \mu)} \cdot e^{\kappa_0 \cos(\mu - \mu_0)} = \exp \left[\kappa \cos(\mu - \theta) + \kappa_0 \cos(\mu - \mu_0) \right],$$

using $\cos(\theta - \mu) = \cos(\mu - \theta)$.

Step 2 — sum of phase-shifted cosines as a single cosine. For $a, b > 0$ and $\alpha, \beta \in \mathbb{T}$,

$$a \cos(\mu - \alpha) + b \cos(\mu - \beta) = \text{Re} \left[e^{-i\mu} (a e^{i\alpha} + b e^{i\beta}) \right] = c \cos(\mu - \gamma),$$

where $c e^{i\gamma} = a e^{i\alpha} + b e^{i\beta}$ in polar form (Theorem 3.3): this is the same identity used for phasor composition in Chapter 5. Applying it with $a = \kappa_0, \alpha = \mu_0, b = \kappa, \beta = \theta$, we obtain $\kappa_0 \cos(\mu - \mu_0) + \kappa \cos(\mu - \theta) = \kappa_1 \cos(\mu - \mu_1)$ with $\kappa_1 e^{i\mu_1} = \kappa_0 e^{i\mu_0} + \kappa e^{i\theta} = \eta_0 + \kappa e^{i\theta}$.

Step 3 — normalization and recursion. Hence $p(\mu \mid \theta) \propto e^{\kappa_1 \cos(\mu - \mu_1)}$: normalizing by Theorem 6.5(i) gives exactly the $\text{vM}(\mu_1, \kappa_1)$ density. Applying Steps 1–2 one observation at a time, each updates η by vector addition of its own $\kappa_k e^{i\theta_k}$, independently of order (the sum is commutative): after n steps, $\eta_n = \eta_0 + \sum_k \kappa_k e^{i\theta_k}$. $\square$

Remark 6.9: Connection with state estimation. The recursion $\eta_n = \eta_{n-1} + \kappa_n e^{i\theta_n}$ combines observations of a constant phase μ : each new measurement adds vectorially to the previous ones, exactly as the phasors of Chapter 5 add at equal angular frequency. It is not a Kalman filter: it lacks a model of the temporal evolution of the phase (here μ does not change between one observation and the next) and it lacks propagation of an uncertainty through a dynamics. If instead the true phase evolves in time — for example μ_t subject to a known drift and process noise — one must alternate a prediction step with an update step like this one: that is the program of the Kalman filter (or a circular version of it) that Volume III will develop for the state estimation of an evolving system.

6.6.1 Simulation and exercises

Example 6.4: Numerical verification: variance and concentration bound. Fix $\mu = 179^\circ$, $\kappa = 4$ (hence $\rho = A(4) = 0.8635226110$, verified with `scipy.special.i1(4)/i0(4)`) and generate, with the script `vol2_cap06_fase.py` (seed 20260909, 3000 independent replicates for each n), simulated samples from $\text{vM}(\mu, \kappa)$ of increasing size.

Step 1 — asymptotic variance. Corollary 6.2 and the general theory of maximum-likelihood estimation (asymptotically normal and efficient under regularity conditions satisfied here) suggest $\text{Var}(\hat{\mu}) \approx 1/I_n(\mu) = 1/(n\kappa\rho)$ for large n . Comparison with the empirical

variance of $\hat{\mu} - \mu$ (wrapped to $(-\pi, \pi]$) over the 3000 simulated samples:

n	simulated var.	$1/(n\kappa\rho)$	ratio
16	0.017568	0.018094	0.971
64	0.004567	0.004524	1.010
256	0.001142	0.001131	1.010
1024	0.000289	0.000283	1.022

The ratio stays close to 1 for every n considered here, and shows no systematic drift as n grows: this is compatible with the expected asymptotic approximation, but the simulation alone does not prove convergence (that requires the general theory of maximum-likelihood estimation, only recalled here).

Step 2 — the Chebyshev bound really is a bound. For $\delta = 10^\circ$, Theorem 6.11 gives the bound $\min\{1, (1 - \rho^2)/(n\rho^2 \sin^2 \delta)\}$. Compared with the empirical frequency of the event $|\hat{\mu} - \mu| > \delta$ on the same samples:

n	empirical freq.	Chebyshev bound
16	0.2027	0.7069
64	0.0107	0.1767
256	0.0000	0.0442
1024	0.0000	0.0110

In every row the empirical frequency is, as it must be, less than or equal to the bound — but by a wide margin: the Chebyshev bound is correct but loose, as is typical of this kind of inequality. It guarantees worst-case behaviour, not typical behaviour.

Exercise 6.1. Explain, in terms of the density $\text{vM}(\mu, 0)$, why for $\kappa = 0$ no observable quantity can estimate μ , whatever n is.

Solution. For $\kappa = 0$ the density $\text{vM}(\mu, 0)$ is the uniform distribution (Theorem 6.5(iii)), which does not depend on μ : the statistical model does not distinguish between different values of μ , so no estimator, however much data is collected, can be consistent for μ . Consistently, $I_1(\mu) = \kappa A(\kappa) = 0$ by Corollary 6.2: the Fisher information is zero.

Exercise 6.2. A sensor has known $\kappa = 4$. Using Theorem 6.11, how many independent measurements are needed to guarantee $\Pr(|\hat{\mu} - \mu| > 10^\circ) \leq 0.05$?

Solution. With $\rho = A(4) = 0.8635226110$ and $\delta = 10^\circ = 0.174533 \text{ rad}$, imposing $(1 - \rho^2)/(n\rho^2 \sin^2 \delta) \leq 0.05$:

$$n \geq \frac{1 - \rho^2}{0.05 \rho^2 \sin^2 \delta} = \frac{1 - 0.745671}{0.05 \times 0.745671 \times 0.030154} = 226.223 \dots,$$

hence $n = 227$ (verified). This is a *sufficient* and conservative bound: Step 2 of Example 6.4 shows that, for the same n , the actual probability is already much smaller than 0.05.

Remark 6.10: A known constraint, reread. Exercise 6.22 — proved before the von Mises was introduced — shows that every circular distribution satisfies $|\varphi_2 - \varphi_1^2| \leq 1 - |\varphi_1|^2$, with equality iff the distribution is concentrated on at most two points. Applied to a sample (replacing φ_1, φ_2 with the sample trigonometric moments), this constraint bounds how far the sample second moment can deviate from what the first moment predicts: it is a consistency

check independent of any parametric model, useful for noticing that the data do not come from a single unimodal circular population (for example because they mix two sources at different phases).

Warning. Quick self-check (true or false, justify).

1. The maximum-likelihood estimator of μ is the sample mean direction.
2. For $R = 0$, $\hat{\mu}$ is well defined and consistent.
3. The Fisher information about μ grows with κ .
4. The Chebyshev bound of Theorem 6.11 always equals the true probability.
5. Conjugate Bayesian updating for μ is a vector sum of phasors.
6. This recursion is already a Kalman filter.

Teacher's key. 1 T: Exercise 6.23. 2 F: the likelihood is flat, μ is not identifiable. 3 T: $I_1(\mu) = \kappa A(\kappa)$ and A is increasing. 4 F: it is an upper bound, often loose. 5 T: Theorem 6.12. 6 F: it lacks a temporal dynamics for the phase, see Observation 6.9.

6.7. Sums of angles and circular convolution

Theorem 6.13: Sums correspond to convolution. Let Θ_1, Θ_2 be independent circular variables with densities f_1, f_2 and moments $\varphi_p^{(1)}, \varphi_p^{(2)}$. Then the density of $\Theta_1 + \Theta_2$ (sum in $\mathbb{T}$, i.e. modulo 2π) is the circular convolution

$$(f_1 * f_2)(\theta) = \int_0^{2\pi} f_1(\theta - t) f_2(t) dt,$$

and its trigonometric moments are $\varphi_p = \varphi_p^{(1)} \varphi_p^{(2)}$.

Proof. Step 1 — the moments. By independence, the expectation of a product is the product of expectations:

$$\varphi_p = \mathbb{E}[e^{ip(\Theta_1 + \Theta_2)}] = \varphi_p^{(1)} \varphi_p^{(2)}.$$

Step 2 — the density. By Theorem 4.18 of Chapter 4, the circular convolution has Fourier coefficients equal to the product of the coefficients. Since by Theorem 6.1 the moments are (up to 2π) the coefficients, the density with the moments of Step 1 is precisely $f_1 * f_2$; and by Corollary 6.1 it is the only one. (Note the normalisation difference: with $d\theta$ rather than $d\theta/2\pi$ the convolution loses the factor $1/2\pi$ relative to Definition 4.9 of Chapter 4 — it is the same operation, written in the measure used in probability.)

Step 3 — direct check. For completeness: for every continuous test function h ,

$$\mathbb{E}[h(\Theta_1 + \Theta_2)] = \iint h(s + t) f_1(s) f_2(t) ds dt = \int h(\theta) \left[\int f_1(\theta - t) f_2(t) dt \right] d\theta,$$

with the substitution $\theta = s + t$ at fixed t , which identifies the bracket as the density of $\Theta_1 + \Theta_2$.

□

Corollary 6.3: Stability of the von Mises: almost, but not quite. The sum of two independent wrapped normals is a wrapped normal: $\text{WN}(\mu_1, \sigma_1^2) + \text{WN}(\mu_2, \sigma_2^2) = \text{WN}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. The sum of two independent von Mises, however, is in general not von Mises.

Proof. Wrapped normal. By Theorems 6.7 and 6.13,

$$\varphi_p = e^{ip(\mu_1 + \mu_2) - p^2(\sigma_1^2 + \sigma_2^2)/2},$$

which are the moments of $\text{WN}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$; the conclusion follows from Corollary 6.1.

Von Mises. The moments of the product would be $\frac{I_p(\kappa_1)I_p(\kappa_2)}{I_0(\kappa_1)I_0(\kappa_2)}$, and one would need a κ with

$$\frac{I_p(\kappa)}{I_0(\kappa)} = \frac{I_p(\kappa_1)}{I_0(\kappa_1)} \cdot \frac{I_p(\kappa_2)}{I_0(\kappa_2)} \quad \text{for every } p.$$

This is impossible in general: already the constraints for $p = 1$ and $p = 2$ determine κ in two incompatible ways. With $\kappa_1 = \kappa_2 = 2$, the $p = 1$ constraint gives $A(\kappa) = 0.697775^2 = 0.486890$, whence $\kappa \approx 1.17$; the $p = 2$ constraint would give $I_2(\kappa)/I_0(\kappa) = 0.302225^2 = 0.091340$, whence $\kappa \approx 1.25$. The two values differ: no von Mises has both required moments.

Remark. The structural reason is that wrapping transports to the circle the stability of the normal family under addition, whereas the von Mises arises from a variational principle (maximum entropy) not preserved by convolution. $\square$

Theorem 6.14: The random walk on the circle tends to uniform. Let $X_1, X_2, \dots$ be independent identically distributed circular variables with moments φ_p , and let $S_n = X_1 + \dots + X_n$ (sum in $\mathbb{T}$). If

$$|\varphi_p| < 1 \quad \text{for every } p \neq 0,$$

then the distribution of S_n converges to the uniform, and the convergence is exponential in n for each fixed p . The condition fails if and only if the distribution of X_1 is supported on a translate of a finite subgroup $\mu_m \subset \mathbb{T}$, in which case S_n stays supported on such a set and cannot tend to the uniform.

Proof. Step 1 — moments of the sum. By Theorem 6.13 iterated, the moments of S_n are φ_p^n .

Step 2 — convergence. If $|\varphi_p| < 1$ for every $p \neq 0$, then $\varphi_p^n \rightarrow 0$, and $|\varphi_p|^n$ decays geometrically. Since the moments of S_n tend to those of the uniform (δ_{p0}), by Theorem 6.4 and the continuity of the distribution–moment correspondence (Corollary 6.1 and dominated convergence) the distribution of S_n converges weakly to the uniform.

More explicitly, if the densities are in L^2 , by Parseval’s identity (Theorem 4.15 of Chapter 4)

$$\int_0^{2\pi} \left| f_n(\theta) - \frac{1}{2\pi} \right|^2 d\theta = \frac{1}{2\pi} \sum_{p \neq 0} |\varphi_p|^{2n} \rightarrow 0.$$

Step 3 — the equality case. Suppose $|\varphi_{p_0}| = 1$ for some $p_0 \neq 0$. By Theorem 6.2(ii) applied to $p_0 X_1$, this means $p_0 X_1$ is degenerate: $e^{ip_0 X_1} = e^{i\beta}$ almost surely, for some β . Hence X_1 takes values in

$$\left\{ \frac{\beta + 2\pi k}{p_0} : k = 0, 1, \dots, |p_0| - 1 \right\},$$

a translate of the group $\mu_{|p_0|}$ of $|p_0|$ -th roots of unity (Definition 3.5 of Chapter 3). The sum of n such variables stays in a translate of the same finite set, and its distribution, being discrete, cannot converge to the uniform, which is continuous.

Step 4 — consistency check. The condition is thus one of *non-latticeness*: the random step must not live on a finite subgroup of the circle. It is the circular analogue of the condition distinguishing lattice from non-lattice random walks on the line. $\square$

Example 6.5: How many steps to mix. A random step is uniform on the arc $[-\pi/6, \pi/6]$ (i.e. $\pm 30^\circ$). How many steps are needed for the position to be practically uniform?

Step 1 — moments of the step. For the uniform density on $[-a, a]$ with $a = \pi/6$,

$$\varphi_p = \frac{1}{2a} \int_{-a}^a e^{ip\theta} d\theta = \frac{\sin(pa)}{pa},$$

real because the density is even. For $p = 1$: $\varphi_1 = \sin(\pi/6)/(\pi/6) = 0.500000/0.523599 = 0.954930$.

Step 2 — checking the condition. $|\varphi_p| = |\sin(p\pi/6)|/(p\pi/6) < 1$ for every $p \neq 0$, because $|\sin x| < |x|$ for $x \neq 0$: Theorem 6.14 applies.

Step 3 — decay of the resultant. After n steps, $R_n = |\varphi_1|^n = 0.954930^n$. Taking the logarithm,

$$\ln R_n = n \ln(0.954930) = -0.046119 n.$$

Step 4 — numerical values.

n	R_n	$\sigma_{\text{equiv}} = \sqrt{-2 \ln R_n}$
10	0.630	0.961 rad = 55.0°
50	0.0997	2.147 rad = 123.0°
100	0.00994	3.037 rad = 174.0°
150	0.000991	3.719 rad

Step 5 — reading. After 100 steps the resultant is below 1%: the position is indistinguishable from uniform with any reasonable sample size. The mixing time is therefore of order

$$n \approx \frac{\ln(1/\varepsilon)}{-\ln|\varphi_1|} = \frac{\ln(1/\varepsilon)}{0.046119},$$

i.e. about 100 steps for $\varepsilon = 1\%$ and about 150 for $\varepsilon = 0.1\%$.

Remark. The decay is governed by the *largest* moment in modulus, here φ_1 ; higher moments decay faster ($\varphi_2 = 0.827$, $\varphi_3 = 0.637$), so the high harmonics of the distribution die out first and the density “smooths” before it flattens. It is the same order of extinction observed for the heat equation in Exercise S8 of Chapter 4.

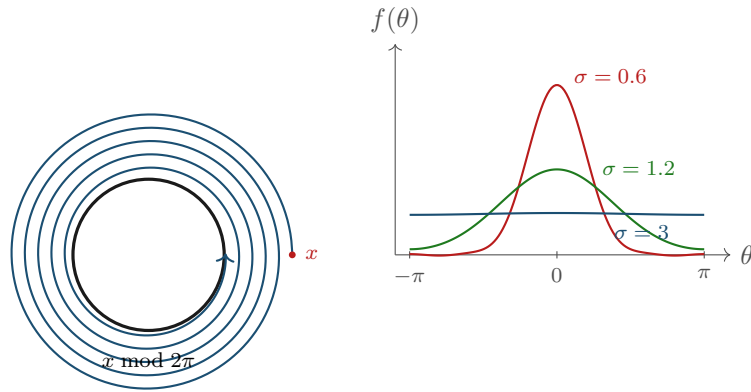


Figure 6.4: Wrapping is aliasing. Left: the real line wraps onto the circle through $x \mapsto x \bmod 2\pi$; all points $x + 2\pi k$ land in the same place. Right: the wrapped normal density, i.e. the image of an $N(0, \sigma^2)$ under that map. Its Fourier coefficients are $\rho_k = e^{-k^2 \sigma^2 / 2}$: wrapping sums the replicas translated by 2π exactly as sampling sums the spectral replicas. As σ grows the coefficients with $k \geq 1$ die out and the density tends to the uniform one: once the deviation exceeds the size of the circle, all directional information is lost.

6.8. The central limit theorem on the circle

Theorem 6.15: Limit theorem for small steps. Let $X_1, X_2, \dots$ be real, independent, identically distributed, with finite mean m and variance s^2 , and let $\Theta_n = (X_1 + \dots + X_n) \bmod 2\pi$. Then, for each fixed p ,

$$\varphi_p^{(n)} = [\psi_{X_1}(p)]^n = \exp\left(ipnm - \frac{1}{2}p^2ns^2 + n o(p^2)\right),$$

so that Θ_n is approximated by $\text{WN}(nm, ns^2)$, with an error tending to zero as $s \rightarrow 0$ at fixed ns^2 .

Proof. Step 1 — moments of the wrapped sum. By Theorem 6.7, the moments of Θ_n are the values at the integers of the characteristic function of $S_n = X_1 + \dots + X_n$; by independence, $\varphi_p^{(n)} = [\psi_{X_1}(p)]^n$.

Step 2 — expanding the characteristic function. Since X_1 has finite second moments, ψ_{X_1} is twice differentiable at 0 and

$$\psi_{X_1}(t) = 1 + imt - \frac{1}{2}(s^2 + m^2)t^2 + o(t^2),$$

whence, taking the principal logarithm for small t , $\ln \psi_{X_1}(t) = imt - \frac{1}{2}s^2t^2 + o(t^2)$.

Step 3 — n -th power. Therefore

$$\varphi_p^{(n)} = \exp\left(ipnm - \frac{1}{2}p^2ns^2 + n o(p^2)\right).$$

If the steps are small — $s \rightarrow 0$ while $n \rightarrow \infty$ with ns^2 constant — the term $n o(p^2)$ is negligible and $\varphi_p^{(n)} \approx e^{ipnm - p^2ns^2/2}$, which by Theorem 6.7 are exactly the moments of $\text{WN}(nm, ns^2)$.

Step 4 — conclusion. By Corollary 6.1, asymptotic equality of moments is equivalent to convergence in distribution.

Step 5 — the opposite regime. If instead s stays fixed and $n \rightarrow \infty$, then $ns^2 \rightarrow \infty$ and $R = e^{-ns^2/2} \rightarrow 0$: the wrapped normal *degenerates* into the uniform, in agreement with Theorem 6.14. On the circle, therefore, the central limit theorem and the mixing theorem do not compete: they are the two regimes of the same statement, distinguished by the behaviour of ns^2 . $\square$

Remark 6.11: A substantive difference from the line. On the line, the central limit theorem produces a non-degenerate limiting family: normalising by $\sqrt{n}$ always yields a normal. On the circle there is nothing to normalise by — the circle is compact, one cannot “zoom out” — and the consequence is that the sum of many contributions simply tends to spread out uniformly. It is the probabilistic manifestation of the same compactness that in Chapter 4 made the spectrum discrete (Observation 4.4): on a compact domain there is no scaling limit.

6.9. Testing uniformity: the Rayleigh test

Theorem 6.16: Asymptotic distribution of the resultant (Rayleigh). Let $\theta_1, \dots, \theta_n$ be independent observations from a uniform distribution on $\mathbb{T}$, and let R be the sample resultant of Definition 6.3. Then, as $n \rightarrow \infty$,

$$Z := nR^2 \xrightarrow{d} \frac{1}{2}\chi_2^2 = \text{Exp}(1),$$

i.e. $\Pr(Z > z) \rightarrow e^{-z}$. The **Rayleigh test** rejects uniformity for large Z .

Proof. Step 1 — mean and variance of the components. Under uniformity, by Theorem 6.4, $\mathbb{E}[\cos \Theta] = \mathbb{E}[\sin \Theta] = 0$. Moreover, by the half-angle formula,

$$\mathbb{E}[\cos^2 \Theta] = \mathbb{E}\left[\frac{1 + \cos 2\Theta}{2}\right] = \frac{1}{2},$$

since $\mathbb{E}[\cos 2\Theta] = \operatorname{Re} \varphi_2 = 0$; identically $\mathbb{E}[\sin^2 \Theta] = 1/2$. Finally $\mathbb{E}[\cos \Theta \sin \Theta] = \frac{1}{2}\mathbb{E}[\sin 2\Theta] = 0$, so the two components are *uncorrelated*, each of variance $1/2$.

Step 2 — bivariate central limit theorem. The pairs $(\cos \theta_k, \sin \theta_k)$ are i.i.d. with zero mean and covariance matrix $\frac{1}{2}I_2$. By the multivariate central limit theorem,

$$\sqrt{n}(\bar{C}, \bar{S}) \xrightarrow{d} \mathcal{N}_2\left(0, \frac{1}{2}I_2\right).$$

Step 3 — quadratic form. If $(U, V) \sim \mathcal{N}_2(0, \frac{1}{2}I_2)$, then $2U^2$ and $2V^2$ are independent χ_1^2 , so $2(U^2 + V^2) \sim \chi_2^2$. But $nR^2 = (\sqrt{n}\bar{C})^2 + (\sqrt{n}\bar{S})^2 \rightarrow U^2 + V^2$ in distribution. Hence $2Z \rightarrow \chi_2^2$, i.e. $Z \rightarrow \frac{1}{2}\chi_2^2$.

Step 4 — the tail. The χ_2^2 distribution has density $\frac{1}{2}e^{-x/2}$ on $(0, \infty)$; hence $\frac{1}{2}\chi_2^2$ has density e^{-z} , i.e. is exponential with parameter 1, and $\Pr(Z > z) = e^{-z}$.

Step 5 — finite- n correction. The approximation improves with the classical expansion (Mardia–Jupp, §6.3)

$$\Pr(Z > z) \approx e^{-z} \left[1 + \frac{2z - z^2}{4n} - \frac{24z - 132z^2 + 76z^3 - 9z^4}{288n^2} \right],$$

useful already for $n \geq 10$. □

Example 6.6: Is the wind directional? Test on the earlier data. Take the sample of Example 6.1: $n = 12$, $R = 0.944536$.

Step 1 — test statistic. $Z = nR^2 = 12 \times 0.892148 = 10.7058$.

Step 2 — approximate p -value. $\Pr(Z > 10.7058) \approx e^{-10.7058} = 2.241 \times 10^{-5}$. (The Step 5 correction is not applicable for such a large Z with such a small n : the term $(2z - z^2)/(4n) = -1.94$ would make the estimate negative. For large Z the plain exponential is already adequate, and the p -value is so small that no correction changes the conclusion.)

Step 3 — conclusion. With $p < 10^{-4}$, uniformity is decisively rejected: the wind has a preferred direction, estimated at 10.9° .

Step 4 — concentration estimate. From $A(\kappa) = R$ of Theorem 6.5(ii), with $R = 0.944536$ one uses the approximation valid for $R \geq 0.85$ (Fisher, 1993):

$$\hat{\kappa} = \frac{1}{R^3 - 4R^2 + 3R} = \frac{1}{0.842565 - 3.568592 + 2.833608} = \frac{1}{0.107581} = 9.295.$$

Check with the asymptotic expansion $A(\kappa) \approx 1 - \frac{1}{2\kappa} - \frac{1}{8\kappa^2}$:

$$1 - \frac{1}{18.590} - \frac{1}{691.4} = 1 - 0.053793 - 0.001446 = 0.944761,$$

against $R = 0.944536$: a difference of 2.4×10^{-4} . ✓

Step 5 — interpretation. $\hat{\kappa} \approx 9.3$ corresponds to a circular standard deviation of 19.4° (Step 4 of Example 6.1): a steady northerly wind with oscillations of about twenty degrees, typical of a regime breeze.

Typical error. Applying a t test or a normality test to the numerical values of the angles. On the original sample (with 340–355) a normality test would reject resoundingly, and the mean would be 130.8° : both conclusions are meaningless, for the reason of Observation 6.1.

Remark 6.12: Other tests, and their limits. The Rayleigh test is optimal against the von Mises alternative — it is the likelihood ratio test in that case — but it is blind to symmetric alternatives: a distribution concentrated on two opposite directions, for instance, has $\varphi_1 = 0$ and passes the Rayleigh test while being far from uniform. For such situations one uses tests based on higher moments or on the empirical distribution function:

- Kuiper’s test, the circular version of Kolmogorov–Smirnov, built to be invariant under the choice of origin;
- Watson’s U^2 test, the circular analogue of Cramér–von Mises;
- the Rayleigh test applied to doubled angles, sensitive to antipodal bimodal alternatives (because it uses φ_2 instead of φ_1).

The null distributions of these tests have no elementary closed form and are tabulated in Mardia–Jupp and in Fisher, cited in the bibliography; here we merely note their existence, with no proofs.

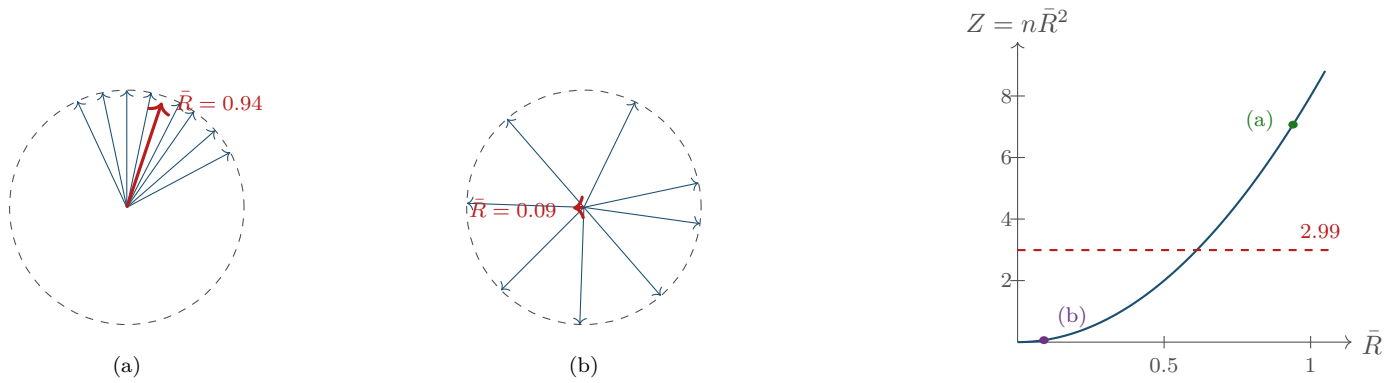


Figure 6.5: The Rayleigh test in one picture. Left, a strongly concentrated sample of $n = 8$ directions: the resultant is long, $\bar{R} = 0.94$. Centre, an almost uniform sample: the directions cancel and $\bar{R} = 0.09$. Right, the Rayleigh statistic $Z = n\bar{R}^2$ as a function of $\bar{R}$: under the null hypothesis of uniformity Z converges in distribution to an exponential of mean 1, so uniformity is rejected when $Z > -\ln \alpha$, i.e. $Z > 2.99$ at level $\alpha = 0.05$. The concentrated sample gives $Z = 8 \cdot 0.94^2 = 7.07$ (reject), the scattered one $Z = 8 \cdot 0.09^2 = 0.065$ (do not reject).

6.10. Watson’s U^2 test

INFO

Why this section. Observation 6.12 pointed out that the Rayleigh test is blind to symmetric alternatives (antipodal bimodal, for instance), and mentioned Watson’s U^2 test as an omnibus alternative, based on the empirical distribution function rather than on low-order trigonometric moments alone. This section builds the statistic, proves in full its crucial property — *local* independence from the arbitrary choice of origin — and honestly states, with a precise source, the part (the extension to a global invariance, and the asymptotic null distribution) that requires tools beyond this volume.

Definition 6.8: Watson’s U^2 statistic. Given a sample $\theta_1, \dots, \theta_n$ on $\mathbb{T}$, let $\theta_{(1)} \leq \dots \leq \theta_{(n)}$ be the ordered values with respect to an arbitrary choice of origin and orientation, and $u_i = \theta_{(i)}/(2\pi) \in [0, 1)$ the corresponding rescaled values. With $\bar{u} = \frac{1}{n} \sum_{i=1}^n u_i$, Watson’s

statistic is

$$U^2 = \sum_{i=1}^n \left(u_i - \frac{2i-1}{2n} \right)^2 + \frac{1}{12n} - n \left(\bar{u} - \frac{1}{2} \right)^2.$$

Remark 6.13: Where the formula comes from. The first term together with $\frac{1}{12n}$ is the Cramér–von Mises statistic

$$W^2 = \sum_{i=1}^n \left(u_i - \frac{2i-1}{2n} \right)^2 + \frac{1}{12n},$$

which compares the empirical distribution function F_n with the uniform one $\theta/2\pi$ on the segment $[0, 2\pi)$ — exactly as on the line. On the line, choosing the left endpoint as origin is natural; on the circle it is arbitrary, and rotating the origin cyclically permutes the ordering of the points and alters W^2 . Watson (1961) observed that $n(\bar{u} - \frac{1}{2})^2$ is precisely the part of W^2 due to the “average level” of the difference $F_n(\theta) - \theta/2\pi$ — a quantity that depends solely on where the zero was placed, not on the shape of the deviation from uniformity — and that subtracting it yields a statistic insensitive to the choice of origin: exactly the property missing from the Rayleigh test, which is sensitive only to the mean direction (moment φ_1) and blind to deviations with $\varphi_1 = 0$.

Proposition 6.1: Local invariance of U^2 under rotation of the origin. Replace each θ_i with $\theta_i - \varphi \pmod{2\pi}$ (new origin rotated by φ), and let φ vary continuously over an interval in which no sample point crosses the new origin (the cyclic ordering of the points stays fixed). On that interval:

$$\frac{dW^2}{d\varphi} = -\frac{n}{\pi} \left(\bar{u} - \frac{1}{2} \right), \quad \frac{dU^2}{d\varphi} = 0.$$

In particular W^2 (the plain Cramér–von Mises statistic, without Watson’s correction) varies with φ — unless $\bar{u} = \frac{1}{2}$ — while U^2 stays locally constant: it is exactly the subtracted term that cancels its variation.

Proof. Step 1 — derivative of each $u_{(i)}$. As long as no point crosses the origin, the ordering stays fixed and every $u_{(i)}(\varphi)$ decreases at the same rate: $u_{(i)}(\varphi) = u_{(i)}(0) - \varphi/(2\pi)$, hence $du_{(i)}/d\varphi = -1/(2\pi)$ for every i (the same constant, independent of i).

Step 2 — derivative of W^2 . Differentiating term by term (the term $\frac{1}{12n}$ is constant):

$$\frac{dW^2}{d\varphi} = \sum_{i=1}^n 2 \left(u_{(i)} - \frac{2i-1}{2n} \right) \frac{du_{(i)}}{d\varphi} = -\frac{1}{\pi} \sum_{i=1}^n \left(u_{(i)} - \frac{2i-1}{2n} \right).$$

Since $\sum_{i=1}^n (2i-1) = n^2$ (the sum of the first n odd integers), we have $\sum_{i=1}^n \frac{2i-1}{2n} = \frac{n}{2}$, hence

$$\sum_{i=1}^n \left(u_{(i)} - \frac{2i-1}{2n} \right) = n\bar{u} - \frac{n}{2} = n \left(\bar{u} - \frac{1}{2} \right),$$

and substituting gives $\frac{dW^2}{d\varphi} = -\frac{n}{\pi} \left(\bar{u} - \frac{1}{2} \right)$.

Step 3 — derivative of U^2 . Since $\bar{u}(\varphi)$ is the mean of the $u_{(i)}(\varphi)$, $d\bar{u}/d\varphi = -1/(2\pi)$. Then

$$\frac{dU^2}{d\varphi} = \frac{dW^2}{d\varphi} - 2n \left(\bar{u} - \frac{1}{2} \right) \frac{d\bar{u}}{d\varphi} = -\frac{n}{\pi} \left(\bar{u} - \frac{1}{2} \right) - 2n \left(\bar{u} - \frac{1}{2} \right) \left(-\frac{1}{2\pi} \right),$$

$$\frac{dU^2}{d\varphi} = -\frac{n}{\pi} \left(\bar{u} - \frac{1}{2} \right) + \frac{n}{\pi} \left(\bar{u} - \frac{1}{2} \right) = 0.$$

□

Remark 6.14: Global invariance and the null distribution: what remains imported. Proposition 6.1 proves that U^2 is piecewise constant, between two successive crossings of a sample point by the origin. When the origin crosses exactly one point, that point moves from being last ($u_{(n)} \rightarrow 1^-$) to first ($u_{(1)} \rightarrow 0^+$) in the ordering, and every other rank shifts by one position: to conclude global invariance, for every $\varphi \in [0, 2\pi)$ and not merely piecewise, one must check that U^2 does not jump at this rank change — a combinatorial verification that we do not reconstruct here with full certainty. This fact, together with the **asymptotic null distribution** of U^2 (which requires the theory of Brownian bridges and has no elementary closed form), is **imported** from G. S. Watson, "Goodness-of-fit tests on a circle", *Biometrika* 48 (1961), 109–114. We verify global invariance here numerically: the script `vol2_cap06_watson.py` computes U^2 for a fixed sample of 11 points, with 37 equally spaced choices of origin over all of $[0, 2\pi)$, falling between successive rank crossings (not exactly on them) — and finds U^2 constant to 2.8×10^{-16} (machine epsilon) on each of the sampled slices: a strong, though not exhaustive at the crossing points themselves, numerical confirmation of the imported result.

Example 6.7: Contrast between a uniform and a concentrated sample. For a sample of $n = 8$ nearly equally spaced directions (clearly close to uniform), the script `vol2_cap06_watson.py` finds $\bar{u} = 0.439092$ and

$$U^2 = 0.010417,$$

a small value, as expected under the uniformity hypothesis. For a sample of $n = 8$ directions generated around a single mean direction (visible concentration), the same script finds $\bar{u} = 0.176217$ and

$$U^2 = 0.603930,$$

almost 58 times larger: the test detects the non-uniformity sharply, as the Rayleigh test would in this unimodal case — but, unlike Rayleigh, U^2 would detect an antipodal bimodal concentration just as well, where $\varphi_1 = 0$ and Rayleigh would fail.

Exercise 6.3: Computing U^2 for a small sample. Compute U^2 for the sample of $n = 6$ angles $10^\circ, 70^\circ, 100^\circ, 180^\circ, 260^\circ, 350^\circ$.

Solution. Step 1 — rescaling. The angles are already ordered. Dividing by 360° :

$$u_i = 0.027778, 0.194444, 0.277778, 0.5, 0.722222, 0.972222.$$

Step 2 — Cramér–von Mises terms. With $n = 6$, $\frac{2i-1}{2n}$ equals 0.083333, 0.25, 0.416667, 0.583333, 0.75, 0.916667 for $i = 1, \dots, 6$. The squares of the differences $u_i - \frac{2i-1}{2n}$ are

$$0.00308642, 0.00308642, 0.01929012, 0.00694444, 0.00077160, 0.00308642,$$

summing to 0.03626543. With $\frac{1}{12n} = \frac{1}{72} = 0.01388889$:

$$W^2 = 0.03626543 + 0.01388889 = 0.05015432.$$

Step 3 — Watson's correction. The mean is $\bar{u} = \frac{1}{6}(0.027778 + 0.194444 + 0.277778 + 0.5 + 0.722222 + 0.972222) = 0.449074$, hence

$$n \left(\bar{u} - \frac{1}{2} \right)^2 = 6 \times (-0.050926)^2 = 0.015561.$$

Step 4 — result.

$$U^2 = W^2 - n \left(\bar{u} - \frac{1}{2} \right)^2 = 0.05015432 - 0.015561 = 0.034594.$$

The computation was repeated in `vol2_cap06_watson.py` with double-precision arithmetic, confirming $U^2 = 0.034594$ (the small discrepancy in the last digit relative to the hand steps is due to the intermediate rounding shown here for clarity).

6.11. Circular regression

All the preceding sections treat a sample of directions $\theta_1, \dots, \theta_n$ as homogeneous data. Often, however, a **linear** predictor $x_i \in \mathbb{R}$ is also observed for each direction (a time, a temperature, a dose), and one wants a model for how the mean direction depends on x . The problem is delicate because an ordinary linear function $\mu(x) = a + bx$ is never periodic: used as an angular mean, it "leaves" the circle for large x . A link function is needed that always remains an angle.

Definition 6.9: Circular-linear link function. For $\mu_0 \in \mathbb{T}$ and $\beta \in \mathbb{R}$, the Fisher–Lee link function is

$$\mu(x) = \mu_0 + 2 \arctan(\beta x) \pmod{2\pi}, \quad x \in \mathbb{R}.$$

Theorem 6.17: Properties of the circular link function. For every $\beta \in \mathbb{R}$:

- (i) μ is well defined as a point of $\mathbb{T}$ (the 2π ambiguity causes no issue, μ being taken modulo 2π) and is continuous in x ;
- (ii) if $\beta \neq 0$, μ is strictly monotone (increasing if $\beta > 0$, decreasing if $\beta < 0$);
- (iii) $\lim_{x \rightarrow \pm\infty} \mu(x) = \mu_0 \pm \pi \pmod{2\pi}$, i.e. the point antipodal to μ_0 , reached from both sides;
- (iv) if $\beta = 0$, $\mu(x) = \mu_0$ for every x (no dependence on x).

Proof. (i) $\arctan$ is defined and continuous on all of $\mathbb{R}$ (with values in $(-\pi/2, \pi/2)$), so $x \mapsto \beta x \mapsto \arctan(\beta x) \mapsto 2 \arctan(\beta x) \mapsto \mu_0 + 2 \arctan(\beta x)$ is a composition of continuous functions, hence continuous; the equivalence class modulo 2π of a real number always defines a point of $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ without ambiguity.

(ii) $\frac{d\mu}{dx} = \frac{2\beta}{1 + \beta^2 x^2}$ (derivative of $\arctan$, chain rule): the denominator is always positive, so the sign of $d\mu/dx$ matches the sign of β for every x , and is never zero if $\beta \neq 0$: μ is strictly monotone.

(iii) $\lim_{x \rightarrow +\infty} \arctan(\beta x) = \operatorname{sgn}(\beta) \cdot \pi/2$ and $\lim_{x \rightarrow -\infty} \arctan(\beta x) = -\operatorname{sgn}(\beta) \cdot \pi/2$ (known limits of $\arctan$ at $\pm\infty$, with sign depending on the sign of β), so $\mu(x) \rightarrow \mu_0 \pm \operatorname{sgn}(\beta)\pi$ in the two limits: in both cases the point reached is $\mu_0 + \pi \pmod{2\pi}$, the antipode of μ_0 on $\mathbb{T}$ (adding $+\pi$ or $-\pi$ gives the same point modulo 2π).

(iv) Immediate: $\arctan(0 \cdot x) = \arctan 0 = 0$ for every x . □

Remark 6.15: The statistical model and estimation. The circular-linear regression model assumes $\theta_i \mid x_i \sim \text{vM}(\mu(x_i), \kappa)$ independently (Definition 6.5), with $\mu(x)$ from Definition 6.9 and $\kappa > 0$ constant. The log-likelihood (Theorem 6.5 for the normalising constant) is

$$\ell(\mu_0, \beta, \kappa) = \kappa \sum_{i=1}^n \cos(\theta_i - \mu(x_i)) - n \log(2\pi I_0(\kappa)).$$

Unlike phase estimation (Section 6.6), here the likelihood equation for β (derivative of the first term with respect to β , $\kappa \sum_i \sin(\theta_i - \mu(x_i)) \cdot \frac{2x_i}{1+\beta^2 x_i^2} = 0$) is transcendental and has no closed-form solution: the maximum-likelihood estimate $(\hat{\mu}_0, \hat{\beta}, \hat{\kappa})$ is computed numerically (nonlinear optimisation), and its asymptotic distribution (asymptotic normality, Fisher information matrix for the model) is **imported** from N.I. Fisher, A.J. Lee, "Regression models for angular responses", *Biometrics* 48 (1992), 665–677 — not reproved here. What remains fully proved in this volume is the geometric structure of the link (Theorem 6.17) and the explicit form of the likelihood above.

Example 6.8: Circular regression on a small sample. For fixed κ (estimable separately, or jointly by numerical means), the estimate of (μ_0, β) maximises $S(\mu_0, \beta) = \sum_i \cos(\theta_i - \mu(x_i))$ (the term $-n \log(2\pi I_0(\kappa))$ does not depend on μ_0, β). With the sample (script `vol2_cap06_regressione.py`, data generated from a known model with $\mu_0 = 0.3$, $\beta = 1.0$, $\kappa = 6$, then treated as observed):

x_i	−2	−1.5	−1	−0.5	0.5	1	1.5	2
θ_i	3.356056	6.201581	4.802873	5.995542	1.370515	1.898994	2.628294	2.094188

Numerically maximising S (Nelder–Mead method, also checked against an independent grid search) gives $\hat{\mu}_0 = 0.37260$, $\hat{\beta} = 0.95138$, with $S(\hat{\mu}_0, \hat{\beta}) = 6.292924$ (against a theoretical maximum $S \leq n = 8$, attained only if the model fits perfectly): values close to the generating parameters (0.3, 1.0), as expected from $n = 8$ noisy observations. Checked that $\hat{\beta} > 0$: consistent with the increase of θ observed as x increases (Theorem 6.17(ii)).

6.12. Numerical exercises

All exercises are solved in Section 6.14.

Exercise 6.4: N1 — Circular statistics of a sample. For the sample $20^\circ, 35^\circ, 50^\circ, 65^\circ, 80^\circ, 95^\circ$ compute $\bar{C}$, $\bar{S}$, the resultant R , the mean direction, the circular variance and the circular standard deviation. Explain why the mean direction coincides with the arithmetic midpoint of the sample.

Exercise 6.5: N2 — Adding an outlying observation. To the sample $350^\circ, 10^\circ$ add the observation 90° . Compute the mean direction and resultant before and after, and comment on the effect of the outlier on concentration.

Exercise 6.6: N3 — Von Mises with $\kappa = 1$. Compute $I_0(1)$ and $I_1(1)$ by summing the series of Theorem 6.5(i) to an error below 10^{-5} , and determine R and the circular standard deviation of $\text{vM}(0, 1)$.

Exercise 6.7: N4 — From standard deviation to concentration parameter. A wrapped normal has $\sigma = 30^\circ$. Compute R , the moments φ_2 and φ_3 , and the parameter κ of the von Mises with the same resultant, comparing the approximate value with the exact one.

Exercise 6.8: N5 — Rayleigh test at the edge of significance. A sample of $n = 25$ directions has $R = 0.35$. Compute the Rayleigh statistic and the p -value, with and without the correction terms, and conclude at the 5% level.

Exercise 6.9: N6 — Sum of two circular variables. Two independent circular variables have mean direction 0 and resultants $R_1 = 0.8$ and $R_2 = 0.6$. Compute the resultant of the sum and the three circular standard deviations, verifying that $\sigma^2 = \sigma_1^2 + \sigma_2^2$.

Exercise 6.10: N7 — Mixing time. A random step is uniform on $[-45^\circ, +45^\circ]$. Compute φ_1 and find the least number of steps for the resultant to fall below 0.01.

Exercise 6.11: N8 — Strobe light and angular aliasing. A shaft rotates at 1500 rpm and is filmed at 24 frames per second. Compute the apparent rotation per frame and the apparent speed, explaining the phenomenon in the light of Chapter 4.

Exercise 6.12: N9 — A bimodal distribution that fools Rayleigh. A sample of $n = 10$ observations consists of five values equal to 2° and five equal to 182° . Compute the resultant and the Rayleigh statistic; then repeat the computation on the *doubled* angles and comment.

Exercise 6.13: N10 — Two formulas for the same density. For the wrapped normal with $\mu = 0$ and $\sigma = 2$ rad, compute the density at $\theta = 0$ in two ways: (a) by summing the wrapping series of Definition 6.6; (b) by summing the Fourier series of Theorem 6.1. Verify that the results agree and say which series converges faster.

6.13. Symbolic exercises

Exercise 6.14: S1 — Symmetry and real moments. Prove that $\varphi_{-p} = \overline{\varphi_p}$ for every p , and that Θ is *symmetric about μ* — i.e. $\Theta - \mu$ and $\mu - \Theta$ have the same distribution — if and only if $e^{-ip\mu}\varphi_p \in \mathbb{R}$ for every p .

Exercise 6.15: S2 — Circular variance and standard deviation. Prove that, as $R \rightarrow 1^-$,

$$\sigma^2 = -2 \ln R = 2V + V^2 + O(V^3), \quad V = 1 - R,$$

and deduce $\sigma \approx \sqrt{2V}$ for concentrated distributions. Check numerically at $R = 0.95$.

Exercise 6.16: S3 — The two formulas for the wrapped density. Prove that, for every density g on $\mathbb{R}$ with summable transform,

$$\sum_{m \in \mathbb{Z}} g(\theta + 2\pi m) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \psi_g(p) e^{-ip\theta},$$

where ψ_g is the characteristic function of g . Recognise in this identity the aliasing formula of Exercise S10 of Chapter 4 and, in a classical form, the Poisson summation formula.

Exercise 6.17: S4 — The von Mises as the angle of a Gaussian vector. Let (X, Y) be a bivariate Gaussian vector with mean $(\rho \cos \mu, \rho \sin \mu)$ and covariance matrix I_2 . Prove that the distribution of the angle Θ *conditional* on the radius $\|(X, Y)\| = r$ is $\text{vM}(\mu, \rho r)$. Deduce a physical interpretation of the parameter κ .

Exercise 6.18: S5 — A circular Chebyshev inequality. Prove that for every $a \in (0, \pi]$

$$\Pr(|\Theta - \mu| \geq a) \leq \frac{1 - R}{1 - \cos a} = \frac{V}{1 - \cos a},$$

where $|\Theta - \mu|$ denotes angular distance in $[0, \pi]$. Compare with Chebyshev's inequality on the line.

Exercise 6.19: S6 — Doubling the angles. Prove that if $\Theta' = 2\Theta$ (in $\mathbb{T}$), then $\varphi_p^{(\Theta')} = \varphi_{2p}^{(\Theta)}$. Deduce that the Rayleigh test applied to doubled angles is sensitive to antipodal (two opposite modes) alternatives, invisible to the ordinary test.

Exercise 6.20: S7 — The uniform is absorbing. Prove that if Θ is uniform on $\mathbb{T}$ and X is any circular variable independent of Θ , then $\Theta + X$ is again uniform. Deduce that the uniform is the unique invariant distribution for the random walk of Theorem 6.14.

Exercise 6.21: S8 — Additivity of circular standard deviations. Prove that for a sum of n independent circular variables $R = \prod_k R_k$, and deduce that circular standard deviations compose as on the line: $\sigma^2 = \sum_k \sigma_k^2$. Comment on why this property holds *exactly*, not just approximately.

Exercise 6.22: S9 — A constraint among moments. Prove that for every circular variable

$$|\varphi_2 - \varphi_1^2| \leq 1 - |\varphi_1|^2,$$

and deduce $|\varphi_2| \geq 2|\varphi_1|^2 - 1$. Verify the inequality on the von Mises with $\kappa = 2$ and discuss the equality case.

Exercise 6.23: S10 — Maximum likelihood estimation for the von Mises. Let $\theta_1, \dots, \theta_n$ be a sample from $\text{vM}(\mu, \kappa)$. Prove that the maximum likelihood estimators are

$$\hat{\mu} = \bar{\theta} \quad (\text{sample mean direction}), \quad A(\hat{\kappa}) = R \quad (\text{sample resultant}).$$

(Hint: use $I'_0(\kappa) = I_1(\kappa)$.)

6.14. Solutions to the numerical exercises

Solution. N1. *Step 1 — sums.*

$$\sum \cos \theta_k = 0.93969 + 0.81915 + 0.64279 + 0.42262 + 0.17365 - 0.08716 = 2.91074,$$

$$\sum \sin \theta_k = 0.34202 + 0.57358 + 0.76604 + 0.90631 + 0.98481 + 0.99619 = 4.56895.$$

Step 2 — mean components. With $n = 6$: $\bar{C} = 0.485123$, $\bar{S} = 0.761492$.

Step 3 — resultant and mean direction.

$$R = \sqrt{0.235344 + 0.579870} = \sqrt{0.815214} = 0.902892,$$

$$\bar{\theta} = \arctan \frac{0.761492}{0.485123} = \arctan(1.569680) = 57.500^\circ.$$

Step 4 — dispersion.

$$V = 1 - R = 0.097108, \quad \sigma = \sqrt{-2 \ln(0.902892)} = \sqrt{0.204306} = 0.452003 \text{ rad} = 25.90^\circ.$$

Step 5 — why 57.5° . The sample is an arithmetic progression of step 15° , hence *symmetric* about its midpoint $(20 + 95)/2 = 57.5^\circ$. By Exercise S1, a distribution symmetric about μ has $e^{-i\mu}\varphi_1$ real, i.e. mean direction exactly μ . Here, therefore, circular and arithmetic mean coincide — because all angles lie in an arc of width $75^\circ < 180^\circ$ and the configuration is symmetric.

Remark. The coincidence is a property of this sample, not a rule: replacing 95° by 275° (the same point reflected) makes the two notions diverge.

Solution. N2. *Step 1 — before adding.* By Observation 6.2, with $n = 2$:

$$\bar{C} = \frac{\cos 350^\circ + \cos 10^\circ}{2} = \frac{0.98481 + 0.98481}{2} = 0.98481, \quad \bar{S} = \frac{-0.17365 + 0.17365}{2} = 0,$$

hence $R = 0.98481$, $\bar{\theta} = 0^\circ$, $V = 0.01519$.

Step 2 — after adding 90° . With $n = 3$:

$$\sum \cos = 0.98481 + 0.98481 + 0 = 1.96962, \quad \sum \sin = -0.17365 + 0.17365 + 1 = 1.00000,$$

$$\bar{C} = 0.656540, \quad \bar{S} = 0.333333.$$

Step 3 — new statistics.

$$R = \sqrt{0.431045 + 0.111111} = \sqrt{0.542156} = 0.736312, \quad \bar{\theta} = \arctan \frac{0.333333}{0.656540} = \arctan(0.507711) = 26.91^\circ.$$

Step 4 — effect. The resultant falls from 0.985 to 0.736 (circular variance from 0.015 to 0.264, i.e. $\times 17$) and the mean direction shifts by nearly 27° . A single observation out of three at 90° from the others therefore has a drastic effect: circular statistics, like ordinary statistics, is not robust to outliers.

Step 5 — check. The centroid of the three points is at distance 0.736 from the origin; since two points nearly coincide (350° and 10° are 20° apart), the centroid lies about one third of the way from their mean direction (0°) to 90° : and indeed $26.91^\circ \approx 90^\circ/3 = 30^\circ$, with the small correction due to the two near points not being exactly coincident. ✓

Solution. N3. *Step 1 — series for $I_0(1)$.* With $\kappa/2 = 0.5$, so $(\kappa/2)^{2k} = 0.25^k$:

$$I_0(1) = \sum_{k \geq 0} \frac{0.25^k}{(k!)^2} = 1 + 0.250000 + 0.015625 + 0.000434 + 0.0000068 + \cdots = 1.266066.$$

The first neglected term is $\approx 7 \times 10^{-8}$: the error is well below 10^{-5} .

Step 2 — series for $I_1(1)$.

$$I_1(1) = \sum_{k \geq 0} \frac{0.5^{2k+1}}{k!(k+1)!} = 0.500000 + 0.062500 + 0.002604 + 0.0000543 + 0.0000007 + \cdots = 0.565159.$$

Step 3 — resultant. $R = A(1) = 0.565159/1.266066 = 0.446390$.

Step 4 — circular standard deviation.

$$\sigma = \sqrt{-2 \ln(0.446390)} = \sqrt{-2 \times (-0.806619)} = \sqrt{1.613238} = 1.270133 \text{ rad} = 72.77^\circ.$$

Step 5 — interpretation. With $\kappa = 1$ the distribution is still very dispersed: $R < 0.5$ and a standard deviation above 70° . The value $\kappa = 1$ roughly marks the boundary between “nearly uniform” distributions and those with a recognisable direction; by comparison, the wind of Example 6.1 had $\hat{\kappa} \approx 9.3$.

Typical error. Using the asymptotic formula $A(\kappa) \approx 1 - 1/(2\kappa)$ at $\kappa = 1$: it would give 0.5 against the true 0.446, a 12% error. That expansion is valid only for large κ .

Solution. N4. *Step 1 — conversion.* $\sigma = 30^\circ = 0.523599 \text{ rad}$, hence $\sigma^2 = 0.274156$.

Step 2 — resultant. By Theorem 6.7,

$$R = e^{-\sigma^2/2} = e^{-0.137078} = 0.871903.$$

Step 3 — higher moments.

$$\varphi_2 = e^{-4\sigma^2/2} = e^{-0.548311} = 0.577958, \quad \varphi_3 = e^{-9\sigma^2/2} = e^{-1.233699} = 0.291209.$$

Step 4 — approximate κ . With the formula valid for $R \geq 0.85$,

$$\hat{\kappa} = \frac{1}{R^3 - 4R^2 + 3R}.$$

With $R = 0.871903$: $R^2 = 0.760215$, $R^3 = 0.662852$, so

$$R^3 - 4R^2 + 3R = 0.662852 - 3.040860 + 2.615709 = 0.237701, \quad \hat{\kappa} = 4.207.$$

Step 5 — exact value. Summing the I_0 and I_1 series at $\kappa = 4.207$ (nine terms suffice) gives $I_0 = 13.5245$, $I_1 = 11.7804$, hence $A(4.207) = 0.871059$: slightly below the target 0.871903. Correcting with $A'(\kappa) = 1 - A^2 - A/\kappa = 0.0342$:

$$\Delta\kappa = \frac{0.871903 - 0.871059}{0.0342} = 0.0247 \implies \kappa_{\text{exact}} \approx 4.232.$$

Step 6 — comparison. The approximation gives 4.207 against the exact 4.232: a 0.6% discrepancy, entirely acceptable in practice. Approximation formulas of this kind (Fisher, 1993) are designed to stay within 1% over their stated range of validity.

Typical error. Setting $\kappa = 1/\sigma^2 = 3.647$, by analogy with the precision of the normal. The discrepancy from the correct value is 14%, and it propagates as a systematic error into every derived quantity.

Solution. N5. *Step 1 — statistic.* $Z = nR^2 = 25 \times 0.1225 = 3.0625$.

Step 2 — first-order approximation. $\Pr(Z > z) \approx e^{-3.0625} = 0.046771$.

Step 3 — first correction term. $\frac{2z - z^2}{4n} = \frac{6.1250 - 9.3789}{100} = -0.032539$.

Step 4 — second correction term. With $z^3 = 28.7230$ and $z^4 = 87.9642$:

$$24z - 132z^2 + 76z^3 - 9z^4 = 73.500 - 1238.015 + 2182.948 - 791.678 = 226.755,$$

$$\frac{226.755}{288n^2} = \frac{226.755}{180000} = 0.001260.$$

Step 5 — corrected p-value.

$$p \approx 0.046771 \times [1 - 0.032539 - 0.001260] = 0.046771 \times 0.966201 = 0.045190.$$

Step 6 — conclusion. With $p = 0.0452 < 0.05$ uniformity is rejected at the 5% level, but only just: the sample provides *weak* evidence of directionality. Note that without correction one would have got $p = 0.0468$, also below threshold but closer still: with $n = 25$ the two estimates differ by 3%, a difference that would be decisive had the threshold been 0.046.

Typical error. Using $Z = R^2$ or $Z = nR$ instead of $Z = nR^2$. The correct form is dictated by Step 3 of the proof of Theorem 6.16: it is $\sqrt{n}(\bar{C}, \bar{S})$ that has a Gaussian limit, hence $n(\bar{C}^2 + \bar{S}^2)$ that has a χ^2 limit.

Solution. N6. *Step 1 — resultant of the sum.* By Theorem 6.13 with $p = 1$, $\varphi_1 = 0.8 \times 0.6 = 0.48$, both real positive since the mean directions are zero. Hence $R = 0.48$ and the mean direction of the sum is still 0.

Step 2 — circular standard deviations.

$$\sigma_1 = \sqrt{0.446287} = 0.668047 \text{ rad} = 38.28^\circ, \quad \sigma_2 = \sqrt{1.021651} = 1.010768 \text{ rad} = 57.91^\circ,$$

$$\sigma = \sqrt{1.467938} = 1.211585 \text{ rad} = 69.42^\circ.$$

Step 3 — additivity check. $\sigma_1^2 + \sigma_2^2 = 0.446287 + 1.021651 = 1.467938 = \sigma^2$ ✓ The equality is *exact*, not approximate: this is Exercise S8.

Step 4 — reading. Dispersions on the circle compose by the same Pythagorean rule as on the line, provided one uses $\sigma = \sqrt{-2 \ln R}$ and not the circular variance $V = 1 - R$. With V one would get $V_1 + V_2 = 0.2 + 0.4 = 0.6$, whereas $V = 1 - 0.48 = 0.52$: circular variance is *not* additive. This is why, despite the name, V is a less convenient dispersion index than σ .

Solution. N7. *Step 1 — moment of the step.* For the uniform density on $[-a, a]$ with $a = \pi/4$, Step 1 of Example 6.5 gives

$$\varphi_1 = \frac{\sin(\pi/4)}{\pi/4} = \frac{0.7071068}{0.7853982} = 0.900316.$$

Step 2 — decay. After n steps, by Theorem 6.14, $R_n = \varphi_1^n = 0.900316^n$.

Step 3 — threshold. We seek the least n with $R_n < 0.01$:

$$n \ln(0.900316) < \ln(0.01) \implies n > \frac{-4.605170}{-0.105010} = 43.855.$$

Hence $n = 44$.

Step 4 — check.

$$R_{43} = e^{-43 \times 0.105010} = e^{-4.515430} = 0.010943 > 0.01,$$

$$R_{44} = e^{-44 \times 0.105010} = e^{-4.620440} = 0.009852 < 0.01. \quad \checkmark$$

Step 5 — comparison with Example 6.5. With $\pm 30^\circ$ steps about 100 steps were needed; with $\pm 45^\circ$ steps 44 suffice. The ratio is roughly $(45/30)^2 = 2.25$, consistent with $2.27 = 100/44$: mixing time scales as the inverse *square* of the step width, because for small a

$$-\ln \varphi_1 = -\ln \frac{\sin a}{a} \approx \frac{a^2}{6}.$$

Checking the approximation. For $a = \pi/4$: $a^2/6 = 0.102808$ against the exact 0.105010, a 2% discrepancy. For $a = \pi/6$: 0.045690 against 0.046119, 1%. $\checkmark$

Solution. N8. *Step 1 — rotation speed.* 1500 rpm = 25 revolutions per second.

Step 2 — rotation per frame. With 24 frames per second, the interval between frames is $1/24$ s, during which the shaft turns

$$\frac{25}{24} = 1.041\bar{6} \text{ revolutions} = 375^\circ.$$

Step 3 — apparent angle. What the camera records is the angle *modulo* 360° : $375^\circ - 360^\circ = 15^\circ$ per frame.

Step 4 — apparent speed. $15^\circ \times 24 = 360^\circ/\text{s} = 1$ revolution per second = 60 rpm, in the same sense as the true rotation.

Step 5 — reading in the light of Chapter 4. The eye, or the camera, measures angular position as an element of $\mathbb{T}$: information is lost, all multiples of 360° being indistinguishable. This is exactly the aliasing formula of Exercise S10 of Chapter 4: sampling at $f_s = 24$ Hz a phenomenon of frequency $f = 25$ Hz, one observes the frequency $f - f_s = 1$ Hz, the unique representative of f in $[-f_s/2, f_s/2)$. The Nyquist–Shannon theorem (Theorem (Volume III, Chapter 24) of Chapter 4) is violated — at least 50 frames per second would be needed — and the consequence is the classic illusion of wheels appearing to turn slowly, or backwards.

Step 6 — the “backwards” case. At 1440 rpm = 24 rev/s the rotation per frame is exactly 360° : the shaft appears *still*. At 1400 rpm ($23.3\bar{3}$ rev/s) the rotation per frame is 350° , i.e. -10° : the shaft appears to rotate backwards at $10^\circ \times 24 = 240^\circ/\text{s}$, i.e. 40 rpm.

Practical consequence. This is why workshops use calibrated stroboscopes and not cameras to measure rotation rates: only by knowing the order of magnitude in advance can the ambiguity be resolved.

Solution. N9. *Step 1 — components on the original sample.*

$$\sum \cos = 5(0.999391) + 5(-0.999391) = 0, \quad \sum \sin = 5(0.034899) + 5(-0.034899) = 0.$$

Hence $\bar{C} = \bar{S} = 0$ and $R = 0$.

Step 2 — Rayleigh test. $Z = nR^2 = 0$, p -value 1: the test *does not reject* uniformity. Yet the sample is concentrated on two values only: the conclusion is manifestly wrong.

Step 3 — doubled angles. Set $\theta'_k = 2\theta_k \bmod 360^\circ$: $2 \times 2^\circ = 4^\circ$, $2 \times 182^\circ = 364^\circ \equiv 4^\circ$. All ten doubled observations equal 4° .

Step 4 — test on the doubled sample.

$$\bar{C}' = \cos 4^\circ = 0.997564, \quad \bar{S}' = \sin 4^\circ = 0.069756, \quad R' = 1.000000,$$

$$Z' = 10, \quad p \approx e^{-10} = 4.54 \times 10^{-5}.$$

A resounding rejection.

Step 5 — why it works. By Exercise S6, the first moment of the doubled angles is the *second* moment of the original angles: $\varphi'_1 = \varphi_2$. An antipodal distribution has $\varphi_1 = 0$ but $\varphi_2 \neq 0$: invisible to the ordinary test, glaring to the doubled-angle test.

Step 6 — moral. The Rayleigh test checks the vanishing of φ_1 *alone*, while uniformity requires all moments to vanish (Theorem 6.4). It is therefore a *necessary but not sufficient* test, and should always be accompanied by a graphical inspection of the data — a circular histogram, or *wind rose* — before declaring uniformity.

Solution. N10. (a) *Wrapping series.* The normal density with $\mu = 0$, $\sigma = 2$ is $g(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-x^2/(2\sigma^2)}$, with

$$\frac{1}{\sigma\sqrt{2\pi}} = \frac{1}{2 \times 2.506628} = 0.199471.$$

At $\theta = 0$:

$$f(0) = 0.199471 \sum_{m \in \mathbb{Z}} e^{-(2\pi m)^2/8}.$$

Terms: $m = 0$ gives 1; $m = \pm 1$ give $e^{-39.4784/8} = e^{-4.93480} = 0.0071847$ each; $m = \pm 2$ give $e^{-19.7392} = 2.67 \times 10^{-9}$ each. Sum:

$$1 + 2(0.0071847) + 2(2.67 \times 10^{-9}) = 1.0143694,$$

$$f(0) = 0.199471 \times 1.0143694 = 0.202337.$$

(b) *Fourier series.* By Theorems 6.1 and 6.7,

$$f(\theta) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} e^{-p^2\sigma^2/2} e^{-ip\theta} = \frac{1}{2\pi} \left[1 + 2 \sum_{p \geq 1} e^{-2p^2} \cos p\theta \right],$$

having used $\sigma^2/2 = 2$ and collected the $\pm p$ terms. At $\theta = 0$:

$$\sum_{p \geq 1} e^{-2p^2} = e^{-2} + e^{-8} + e^{-18} + \dots = 0.1353353 + 0.0003355 + 0.0000000152 + \dots = 0.1356708,$$

$$f(0) = \frac{1 + 2(0.1356708)}{2\pi} = \frac{1.2713416}{6.2831853} = 0.202340.$$

Step 3 — comparison. The two values, 0.202337 and 0.202340, agree up to rounding: a relative discrepancy of 1.5×10^{-5} . ✓ The identity is Exercise S3.

Step 4 — which series converges faster. Here the Fourier series: its second term is 3.4×10^{-4} , against 7.2×10^{-3} for the wrapping. The general rule is that wrapping converges quickly for *small* σ (the translates are far apart relative to the width of the Gaussian) and the Fourier series for *large* σ (the moments decay like $e^{-p^2\sigma^2/2}$). The break-even point is around $\sigma = \sqrt{2\pi} \approx 2.5$ rad; here $\sigma = 2$, just below, and indeed the two series are comparable with a slight advantage to Fourier.

Step 5 — comparison with the uniform. The uniform density is $1/2\pi = 0.159155$. The maximum value, 0.202, exceeds it by 27%: with $\sigma = 2$ rad (115°) the distribution is already very close to uniform, consistently with $R = e^{-2} = 0.135$.

6.15. Solutions to the symbolic exercises

Solution. S1. *Step 1 — conjugation.* Since $\overline{e^{ip\Theta}} = e^{-ip\Theta}$ and expectation commutes with conjugation (Θ being real), $\overline{\varphi_p} = \mathbb{E}[e^{-ip\Theta}] = \varphi_{-p}$.

Step 2 — symmetry implies real moments. Suppose $\Theta - \mu$ and $\mu - \Theta$ have the same distribution. Then, for every p ,

$$e^{-ip\mu}\varphi_p = \mathbb{E}[e^{ip(\Theta-\mu)}] = \mathbb{E}[e^{ip(\mu-\Theta)}] = \overline{e^{-ip\mu}\varphi_p},$$

the second equality by equality in distribution and the third by Step 1. A number equal to its own conjugate is real.

Step 3 — converse. If $e^{-ip\mu}\varphi_p \in \mathbb{R}$ for every p , then $\Theta - \mu$ and $\mu - \Theta$ have the same trigonometric moments (by the computation of Step 2, read backwards), and by Corollary 6.1 the same distribution.

Step 4 — consequence for the mean direction. If Θ is symmetric about μ , then in particular $e^{-i\mu}\varphi_1 = R' \in \mathbb{R}$; and since $\varphi_1 = R'e^{i\mu}$ with R' real, if $R' > 0$ the mean direction is exactly μ . (If $R' < 0$ the mean direction is $\mu + \pi$: this happens for “butterfly” distributions concentrated about the direction opposite to the axis of symmetry.)

Solution. S2. *Step 1 — expanding the logarithm.* With $R = 1 - V$ and $V \rightarrow 0^+$,

$$\ln R = \ln(1 - V) = -V - \frac{V^2}{2} - \frac{V^3}{3} - \dots,$$

convergent for $|V| < 1$.

Step 2 — multiplying by -2 . $\sigma^2 = 2V + V^2 + \frac{2V^3}{3} + \dots = 2V + V^2 + O(V^3)$.

Step 3 — first approximation. Dropping higher-order terms, $\sigma \approx \sqrt{2V} = \sqrt{2(1-R)}$.

Step 4 — numerical check at $R = 0.95$. Here $V = 0.05$:

$$\sigma_{\text{exact}}^2 = -2 \ln(0.95) = -2 \times (-0.0512933) = 0.1025866, \quad \sigma_{\text{exact}} = 0.320291 \text{ rad} = 18.35^\circ,$$

$$\sigma_{\text{approx}} = \sqrt{0.1} = 0.316228 \text{ rad} = 18.12^\circ.$$

Discrepancy: 1.3%. Including the quadratic term, $2V + V^2 = 0.1025$ and $\sqrt{0.1025} = 0.320156$: discrepancy 0.04%. ✓

Step 5 — when the two measures diverge. As $R \rightarrow 0$ (nearly uniform distribution) we have $V \rightarrow 1$ but $\sigma \rightarrow \infty$: the two dispersion measures behave qualitatively differently. V is bounded in $[0, 1]$ and saturates; σ diverges, which makes it preferable when *composing* dispersions (Exercise S8) but useless as a descriptive index for very dispersed data.

Solution. S3. *Step 1 — the wrapped function is periodic.* Set $F(\theta) = \sum_{m \in \mathbb{Z}} g(\theta + 2\pi m)$. Substituting $\theta \rightarrow \theta + 2\pi$ gives the same sum with shifted indices: F is 2π -periodic. Moreover $F \geq 0$ and

$$\int_0^{2\pi} F(\theta) d\theta = \int_{\mathbb{R}} g = 1$$

(exchange legitimate by monotone convergence; the translated intervals tile $\mathbb{R}$). So F is a density on $\mathbb{T}$: it is the density of the wrapped variable.

Step 2 — coefficients of F . This is Step 2 of the proof of Theorem 6.7: $\int_0^{2\pi} e^{ip\theta} F(\theta) d\theta = \psi_g(p)$.

Step 3 — inversion. If $\sum_p |\psi_g(p)| < \infty$, by Theorem 6.1 (inversion, with uniform convergence guaranteed by Theorem (Volume III, Chapter 17)(i) of Chapter 4)

$$F(\theta) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \psi_g(p) e^{-ip\theta},$$

the required identity.

Step 4 — recognition. Writing g and ψ_g as a transform pair, the identity becomes

$$\sum_{m \in \mathbb{Z}} g(\theta + 2\pi m) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \hat{g}(p) e^{-ip\theta},$$

which is the **Poisson summation formula**: the sum of the translates of a function is the Fourier series built on the integer samples of its transform. In Chapter 4 the same identity appeared as the aliasing formula (Exercise S10), read in the opposite direction: there one sampled in time and summed in frequency, here one sums in time and samples in frequency. It is the same formula, applied to the two sides of the transform.

Step 5 — check on a known case. With g Gaussian of standard deviation σ , $\psi_g(p) = e^{-p^2\sigma^2/2}$ and the identity becomes the one verified numerically in Exercise N10. It is also, up to reparametrisation, the transformation identity of the Jacobi theta function.

Solution. S4. *Step 1 — joint density in polar coordinates.* The density of (X, Y) is

$$p(x, y) = \frac{1}{2\pi} \exp \left(-\frac{(x - \rho \cos \mu)^2 + (y - \rho \sin \mu)^2}{2} \right).$$

In polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, with Jacobian r :

$$p(r, \theta) = \frac{r}{2\pi} \exp \left(-\frac{r^2 + \rho^2 - 2r\rho \cos(\theta - \mu)}{2} \right),$$

having expanded the squares and recognised $x\rho \cos \mu + y\rho \sin \mu = r\rho \cos(\theta - \mu)$ (addition formula, or Re of $re^{i\theta} \cdot \rho e^{-i\mu}$).

Step 2 — conditioning. At fixed r , the conditional density of Θ is proportional to $p(r, \theta)$ as a function of θ :

$$p(\theta | r) \propto \exp \left(r\rho \cos(\theta - \mu) \right),$$

because all other factors — $r/2\pi$, $e^{-r^2/2}$, $e^{-\rho^2/2}$ — do not depend on θ .

Step 3 — identification. Comparing with Definition 6.5, this is the von Mises with mean direction μ and concentration $\kappa = \rho r$. The normalising constant is that of Theorem 6.5(i), and need not be computed: it is fixed by proportionality.

Step 4 — physical interpretation. The construction says that the von Mises is the angular distribution of *signal plus noise*: a deterministic vector of modulus ρ in direction μ , plus isotropic Gaussian noise of unit variance per component. The parameter $\kappa = \rho r$ is therefore, at fixed radius, proportional to the *signal-to-noise ratio*. This is why the von Mises arises naturally in the statistics of signal phase: the phase of a phasor immersed in Gaussian noise is, conditionally on amplitude, exactly von Mises — a result linking the present chapter to the phasors of Chapter 6.

Solution. S5. *Step 1 — a pointwise inequality.* Let $D = |\Theta - \mu|$ be the angular distance, with values in $[0, \pi]$. Since $\cos$ is decreasing on $[0, \pi]$,

$$D \geq a \implies 1 - \cos D \geq 1 - \cos a,$$

hence, for every realisation, $1 - \cos D \geq (1 - \cos a) \cdot \mathbf{1}_{\{D \geq a\}}$, valid also when $D < a$ because the right side is then zero and the left side non-negative.

Step 2 — expectation. Taking expectations and using monotonicity,

$$\mathbb{E}[1 - \cos D] \geq (1 - \cos a) \Pr(D \geq a).$$

Step 3 — identifying the left side. Since $\cos D = \cos(\Theta - \mu)$ (cosine is even), by Step 2 of the proof of Theorem 6.3 we have $\mathbb{E}[\cos(\Theta - \mu)] = R$, hence $\mathbb{E}[1 - \cos D] = 1 - R = V$.

Step 4 — conclusion. Dividing by $1 - \cos a > 0$ (legitimate for $a \in (0, \pi]$), $\Pr(D \geq a) \leq V/(1 - \cos a)$.

Step 5 — comparison with Chebyshev. On the line, Chebyshev's inequality reads $\Pr(|X - m| \geq a) \leq s^2/a^2$. Here the role of a^2 is played by $1 - \cos a$, which for small a equals $a^2/2$: one recovers

$$\Pr(D \geq a) \lesssim \frac{2V}{a^2} \approx \frac{\sigma^2}{a^2},$$

using $\sigma^2 \approx 2V$ from Exercise S2. The circular version thus agrees with the linear one in the concentrated regime, and departs from it for large a — where it is in fact *better*, since $1 - \cos a$ is bounded by 2 while a^2 would grow: for $a = \pi$ the inequality gives $\Pr(D = \pi) \leq V/2$, a non-trivial piece of information.

Solution. S6. *Step 1 — moments of the doubled angles.* By definition,

$$\varphi_p^{(\Theta')} = \mathbb{E}[e^{ip\Theta'}] = \mathbb{E}[e^{2ip\Theta}] = \varphi_{2p}^{(\Theta)}.$$

(The passage from $\Theta' = 2\Theta \bmod 2\pi$ to $e^{ip\Theta'} = e^{2ip\Theta}$ is exact, the complex exponential having period 2π .)

Step 2 — case $p = 1$. In particular $\varphi_1^{(\Theta')} = \varphi_2^{(\Theta)}$: the resultant of the doubled sample estimates the *second* moment of the original sample.

Step 3 — antipodal distributions. Let Θ be such that Θ and $\Theta + \pi$ have the same distribution (antipodal symmetry). Then

$$\varphi_p = \mathbb{E}[e^{ip(\Theta+\pi)}] = (-1)^p \varphi_p,$$

hence $\varphi_p = 0$ for every *odd* p — in particular $\varphi_1 = 0$ — while the even moments remain free.

Step 4 — consequence for tests. The Rayleigh test, which checks $\varphi_1 = 0$, has no power against antipodal alternatives: for them the null hypothesis appears true. The test applied to doubled angles checks $\varphi_2 = 0$, which for a non-degenerate antipodal distribution is violated: the test recovers full power.

Step 5 — generalisation. Multiplying angles by m examines φ_m : one thus builds a family of tests sensitive to distributions with m equally spaced modes. The limitation is that each test looks at a single moment; the tests based on the empirical distribution function (Kuiper, Watson), cited in Observation 6.12, look at all of them at once, at the price of more complicated null distributions.

Solution. S7. *Step 1 — moments of the sum.* By Theorem 6.13, if Θ is uniform and X independent, $\varphi_p^{(\Theta+X)} = \varphi_p^{(\Theta)} \varphi_p^{(X)}$. But by Theorem 6.4 $\varphi_p^{(\Theta)} = 0$ for every $p \neq 0$, hence $\varphi_p^{(\Theta+X)} = 0$ for every $p \neq 0$; and $\varphi_0 = 1$ always.

Step 2 — conclusion. The moments of $\Theta + X$ coincide with those of the uniform; by Theorem 6.4 (or by Corollary 6.1) the sum is uniform.

Step 3 — uniqueness of the invariant distribution. Let π be invariant for the random walk, i.e. such that π plus an independent step X reproduces π . Passing to moments,

$$\varphi_p^{(\pi)} = \varphi_p^{(\pi)} \varphi_p^{(X)} \implies \varphi_p^{(\pi)} (1 - \varphi_p^{(X)}) = 0.$$

Under the non-latticeness hypothesis of Theorem 6.14 we have $\varphi_p^{(X)} \neq 1$ for every $p \neq 0$, hence $\varphi_p^{(\pi)} = 0$ for every $p \neq 0$: π is the uniform, and it is unique.

Step 4 — interpretation. The uniform plays on the circle a role that does not exist on the line: it is a genuine probability distribution invariant under translation. The line admits nothing of the sort (the “uniform on $\mathbb{R}$ ” is not normalisable), which is why on the circle random walks have an equilibrium distribution while on the line they diverge. Compactness, once again, makes the difference (Observation 6.11).

Solution. S8. *Step 1 — product of resultants.* By Theorem 6.13 iterated over n independent variables, $\varphi_1 = \prod_k \varphi_1^{(k)}$, and taking moduli, by multiplicativity of the modulus (Theorem 3.1(iii)), $R = \prod_k R_k$.

Step 2 — logarithm. Since all $R_k \in (0, 1]$, one may take logarithms:

$$\ln R = \sum_k \ln R_k \implies -2 \ln R = \sum_k (-2 \ln R_k),$$

i.e. $\sigma^2 = \sum_k \sigma_k^2$ by Definition 6.3.

Step 3 — why it is exact. Additivity is not an approximation because the definition $\sigma = \sqrt{-2 \ln R}$ is designed *precisely* to turn the product of resultants into a sum. It is the same reason why, on the line, one chooses the variance rather than the mean absolute deviation: one picks the index that makes the operation of interest additive.

Step 4 — the natural precedent. For wrapped normals the formula is obvious, since $R = e^{-\sigma^2/2}$ by definition (Theorem 6.7) and sums of normals are normal with additive variances. The point of the exercise is that additivity of σ^2 holds for *any* family of independent circular variables, not only wrapped normals: it is a property of the definition, not of the distribution.

Step 5 — counterexample for V . Circular variance $V = 1 - R$ is not additive: with $R_1 = R_2 = 0.8$ we have $V_1 = V_2 = 0.2$ but $R = 0.64$ and $V = 0.36 \neq 0.4$. The defect worsens with many terms: for n variables with $R_k = 0.8$, $V = 1 - 0.8^n \rightarrow 1$, while $\sum V_k = 0.2n \rightarrow \infty$. ✓

Solution. S9. *Step 1 — setting up.* Put $W = e^{i\Theta}$, so $|W| = 1$, $\mathbb{E}[W] = \varphi_1$ and $\mathbb{E}[W^2] = \varphi_2$.

Step 2 — the complex variance.

$$\mathbb{E}[(W - \varphi_1)^2] = \varphi_2 - 2\varphi_1^2 + \varphi_1^2 = \varphi_2 - \varphi_1^2.$$

Step 3 — bounding. By the triangle inequality for integrals,

$$|\varphi_2 - \varphi_1^2| \leq \mathbb{E}[|W - \varphi_1|^2],$$

and expanding the squared modulus,

$$\mathbb{E}[|W - \varphi_1|^2] = 1 - 2|\varphi_1|^2 + |\varphi_1|^2 = 1 - |\varphi_1|^2,$$

using $|W| = 1$. This is the required inequality.

Step 4 — the consequence. By the reverse triangle inequality (Theorem 3.1(v)),

$$|\varphi_2| \geq |\varphi_1|^2 - (1 - |\varphi_1|^2) = 2|\varphi_1|^2 - 1.$$

Step 5 — check on the von Mises with $\kappa = 2$. From Example 6.2, $|\varphi_1| = 0.697775$ and $|\varphi_2| = 0.302225$. The constraint gives

$$2(0.697775)^2 - 1 = -0.026220,$$

which is negative: the inequality is trivially satisfied. ✓ The constraint becomes binding only for $|\varphi_1| > 1/\sqrt{2} = 0.7071$, i.e. for fairly concentrated distributions.

Step 6 — equality case. Equality in $|\varphi_2 - \varphi_1^2| = 1 - |\varphi_1|^2$ requires $(W - \varphi_1)^2$ to have constant argument and $|W - \varphi_1|$ to be constant: the two conditions together force W to take at most two values. The extreme case is the two-point distribution; the degenerate case (W constant) gives $|\varphi_1| = 1$ and $|\varphi_2| = 1 = 2 \cdot 1 - 1$: equality. ✓

Solution. S10. *Step 1 — log-likelihood.* By Definition 6.5, the sample has joint density $\prod_k f(\theta_k)$, hence

$$\ell(\mu, \kappa) = \kappa \sum_{k=1}^n \cos(\theta_k - \mu) - n \ln(2\pi I_0(\kappa)).$$

Step 2 — maximising in μ . The second term does not depend on μ , so one must maximise $\sum_k \cos(\theta_k - \mu)$. But by Theorem 6.3 (sample version) this sum is maximal exactly at $\mu = \bar{\theta}$, the sample mean direction, with maximum nR .

Alternatively, analytically:

$$\frac{\partial \ell}{\partial \mu} = \kappa \sum_k \sin(\theta_k - \mu) = n\kappa(\bar{S} \cos \mu - \bar{C} \sin \mu),$$

vanishing for $\tan \mu = \bar{S}/\bar{C}$; the second derivative, $-n\kappa R < 0$, confirms a maximum (and not the minimum, at $\mu + \pi$).

Step 3 — maximising in κ . Substituting $\mu = \bar{\theta}$, $\ell = n\kappa R - n \ln(2\pi I_0(\kappa))$, and differentiating,

$$\frac{\partial \ell}{\partial \kappa} = n \left(R - \frac{I_1(\kappa)}{I_0(\kappa)} \right) = n(R - A(\kappa)),$$

using $I'_0 = I_1$ (recurrence for modified Bessel functions, NIST DLMF §10.29). Setting it to zero gives $A(\hat{\kappa}) = R$.

Step 4 — uniqueness. By Theorem 6.5(iii), A is strictly increasing from 0 to 1: the equation $A(\kappa) = R$ has exactly one solution for each $R \in (0, 1)$. Moreover $\partial^2 \ell / \partial \kappa^2 = -nA'(\kappa) < 0$: the stationary point is a maximum.

Step 5 — reading. The maximum likelihood estimators are therefore the two descriptive statistics introduced at the start of the chapter: the sample mean direction and the resultant. This is no coincidence — it follows from the von Mises belonging to the exponential family with sufficient statistics $(\sum \cos \theta_k, \sum \sin \theta_k)$, i.e. exactly the components of the resultant. The same structure that in Theorem 6.6 made the von Mises the maximum-entropy distribution at fixed mean and resultant.

6.16. Self-assessment quiz

1. The arithmetic mean of 350° and 10° is 180° : this shows that

-
- A) the two winds are opposite B) the arithmetic mean is unsuited to the circle C) the computation is wrong D) more data are needed
2. The trigonometric moment φ_p coincides, up to 2π , with
- A) the density at p B) the Fourier coefficient $\hat{f}(-p)$ C) the variance D) the entropy
3. The resultant R always satisfies
- A) $R \geq 1$ B) $0 \leq R \leq 1$ C) any real R D) $R = 1 - V^2$
4. $R = 1$ happens if and only if the distribution is
- A) uniform B) concentrated at a point C) bimodal D) von Mises
5. Does $R = 0$ imply the distribution is uniform?
- A) yes, always B) no, one needs $\varphi_p = 0$ for every p C) yes, if n is large D) only for the von Mises
6. The mean direction minimises
- A) $\sum(\theta_k - a)^2$ B) $\sum [1 - \cos(\theta_k - a)]$ C) $\sum |\theta_k - a|$ D) nothing in particular
7. The von Mises distribution is characterised by being
- A) the wrapping of a normal B) of maximum entropy at fixed mean and resultant
C) the only symmetric one D) the only one with real moments
8. For the von Mises, R equals
- A) κ B) $I_1(\kappa)/I_0(\kappa)$ C) $e^{-\kappa}$ D) $1/\kappa$
9. The moments of a wrapped distribution are
- A) the density at the integers B) the characteristic function at the integers C) always real D) always zero
10. For the wrapped normal, R equals
- A) $e^{-\sigma^2/2}$ B) $e^{-\sigma}$ C) $1 - \sigma^2$ D) $1/\sigma$
11. The sum of two independent circular variables has moments
- A) the sum of the moments B) the product of the moments C) the average of the moments D) no relation
12. The sum of two independent von Mises is
- A) always von Mises B) in general not von Mises C) always uniform D) always wrapped normal
13. A non-lattice random walk on the circle tends to
- A) concentrate B) become uniform C) diverge D) stay unchanged
14. Circular standard deviations of independent variables compose as
- A) $\sigma = \sum \sigma_k$ B) $\sigma^2 = \sum \sigma_k^2$ C) $\sigma = \max \sigma_k$ D) they do not compose
15. The Rayleigh test statistic is
- A) R B) nR^2 C) nR D) R^2/n
16. Under uniformity, nR^2 converges in distribution to
-

- A) a normal B) an exponential with parameter 1 C) χ_1^2 D) a uniform
17. A distribution concentrated on two opposite directions
- A) is rejected by the Rayleigh test B) passes the Rayleigh test though not uniform
C) has $R = 1$ D) does not exist
18. Doubling the angles allows one to examine
- A) φ_1 B) φ_2 C) the density D) the entropy
19. The maximum likelihood estimator of μ for the von Mises is
- A) the arithmetic mean B) the sample mean direction C) the median D) the mode
20. The illusion of a wheel turning backwards is a case of
- A) angular aliasing B) resonance C) measurement error D) Doppler effect

BEST PRACTICE

Answer key.

1-B: Observation 6.1: the arithmetic mean is not invariant under a change of origin. **2-B:** Theorem 6.1. **3-B:** Theorem 6.2(i). **4-B:** Theorem 6.2(ii). **5-B:** Theorem 6.2(iii) and Theorem 6.4; the cube-roots counterexample. **6-B:** Theorem 6.3. **7-B:** Theorem 6.6; A describes the wrapped normal. **8-B:** Theorem 6.5(ii). **9-B:** Theorem 6.7. **10-A:** same theorem, normal case. **11-B:** Theorem 6.13. **12-B:** Corollary 6.3; the wrapped normal, by contrast, is stable. **13-B:** Theorem 6.14. **14-B:** Exercise S8; circular variance V is not additive. **15-B:** Theorem 6.16. **16-B:** same theorem: $\frac{1}{2}\chi_2^2 = \text{Exp}(1)$. **17-B:** Exercise N9 and Step 3 of Exercise S6: it has $\varphi_1 = 0$. **18-B:** Exercise S6. **19-B:** Exercise S10. **20-A:** Exercise N8 and Theorem (Volume III, Chapter 24) of Chapter 4.

6.17. Chapter formulary

Moments and density

$$\varphi_p = \mathbb{E}[e^{ip\Theta}] = 2\pi \hat{f}(-p), \quad f(\theta) = \frac{1}{2\pi} \sum_{p \in \mathbb{Z}} \varphi_p e^{-ip\theta}, \quad \varphi_{-p} = \overline{\varphi_p}, \quad |\varphi_p| \leq 1$$

Mean and dispersion

$$\varphi_1 = Re^{i\mu}, \quad V = 1 - R, \quad \sigma = \sqrt{-2 \ln R} \quad (\sigma^2 \approx 2V)$$

$$\bar{C} = \frac{1}{n} \sum \cos \theta_k, \quad \bar{S} = \frac{1}{n} \sum \sin \theta_k, \quad R = \sqrt{\bar{C}^2 + \bar{S}^2}, \quad \bar{\theta} = \text{atan2}(\bar{S}, \bar{C})$$

Uniform

$$f = \frac{1}{2\pi} \iff \varphi_p = 0 \quad \forall p \neq 0$$

Von Mises vM(μ, κ)

$$f(\theta) = \frac{e^{\kappa \cos(\theta - \mu)}}{2\pi I_0(\kappa)}, \quad I_p(\kappa) = \sum_{k \geq 0} \frac{(\kappa/2)^{2k+p}}{k!(k+p)!}$$

$$\varphi_p = e^{ip\mu} \frac{I_p(\kappa)}{I_0(\kappa)}, \quad R = A(\kappa) = \frac{I_1(\kappa)}{I_0(\kappa)}, \quad A'(\kappa) = 1 - A^2 - \frac{A}{\kappa}$$

Wrapped distributions

$$f(\theta) = \sum_{m \in \mathbb{Z}} g(\theta + 2\pi m), \quad \varphi_p = \psi_g(p),$$

$$\text{WN}(\mu, \sigma^2) : \quad \varphi_p = e^{ip\mu - p^2\sigma^2/2}, \quad R = e^{-\sigma^2/2}$$

Sums

$$\varphi_p^{(\Theta_1 + \Theta_2)} = \varphi_p^{(1)} \varphi_p^{(2)}, \quad R = \prod_k R_k, \quad \sigma^2 = \sum_k \sigma_k^2$$

Rayleigh test

$$Z = nR^2, \quad \Pr(Z > z) \approx e^{-z} \left[1 + \frac{2z - z^2}{4n} - \frac{24z - 132z^2 + 76z^3 - 9z^4}{288n^2} \right]$$

Estimation

$$\hat{\mu} = \bar{\theta}, \quad A(\hat{\kappa}) = R, \quad \hat{\kappa} \approx \frac{1}{R^3 - 4R^2 + 3R} \quad (R \geq 0.85)$$

Circular Chebyshev inequality

$$\Pr(|\Theta - \mu| \geq a) \leq \frac{V}{1 - \cos a}$$

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The discrete circle: cyclic groups, modular arithmetic, codes

Chapter 3 showed that the n -th roots of unity form a finite multiplicative group, and Chapter 4 that the discrete Fourier transform is harmonic analysis on that group. This chapter takes the observation seriously: the circle, sampled at n equally spaced points, *is* the group $\mathbb{Z}/n\mathbb{Z}$; and all of modular arithmetic — with its theorems of Fermat, Euler and Sun Zi — is the geometry of that discrete circle. At the end of the road we shall find error-correcting codes, which are ideals in a polynomial ring built on the same circle.

INFO

The idea in one line. Dividing the circle into n equal parts and walking from part to part is the same thing as counting modulo n . Every arithmetic property of $\mathbb{Z}/n\mathbb{Z}$ therefore has a geometric reading on the circle, and conversely: generators are primitive roots, subgroups are inscribed regular polygons, characters are the harmonics of the discrete transform, ideals are the cyclic codes.

Warning. Conventions of this chapter.

- $\mathbb{Z}/n\mathbb{Z}$ denotes the ring of residue classes modulo n ; $(\mathbb{Z}/n\mathbb{Z})^\times$ the multiplicative group of its invertible elements.
- $\mu_n = \{z \in \mathbb{C} : z^n = 1\}$ is the group of n -th roots of unity of Chapter 3, and $\zeta_n = e^{2\pi i/n}$ the principal root.
- φ denotes Euler's totient function (*not* the trigonometric moment of Chapter 6: here the context is always arithmetic).
- $\gcd(a, b)$ is the greatest common divisor; $a \mid b$ means “ a divides b ”.
- $\mathbb{F}_q$ denotes the finite field with q elements; in this chapter almost only $\mathbb{F}_2 = \{0, 1\}$ is used.

INFO

Prerequisites. Chapter 3 (roots of unity, polar form); Chapter 4 (discrete Fourier transform, orthogonality). Elementary notions of divisibility and polynomial algebra. No group theory is assumed: the needed definitions are recalled.

7.1. The circle divided into n parts: cyclic groups

Definition 7.1: Cyclic group. A group G is **cyclic** if there is $g \in G$ such that every element of G is a power of g :

$$G = \langle g \rangle = \{g^k : k \in \mathbb{Z}\}.$$

The element g is a **generator**. The **order** of g is the least integer $d \geq 1$ with $g^d = e$ (the identity), if it exists.

Theorem 7.1: The discrete circle is $\mathbb{Z}/n\mathbb{Z}$. The map

$$\Psi : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mu_n, \quad \Psi([k]) = \zeta_n^k = e^{2\pi i k/n},$$

is a group isomorphism between the additive group $\mathbb{Z}/n\mathbb{Z}$ and the multiplicative group μ_n of n -th roots of unity. In particular μ_n is cyclic of order n .

Proof. **Step 1 — well defined.** If $[k] = [k']$, i.e. $k - k' = mn$ for some $m \in \mathbb{Z}$, then

$$\zeta_n^k = \zeta_n^{k'} \zeta_n^{mn} = \zeta_n^{k'} (\zeta_n^n)^m = \zeta_n^{k'},$$

since $\zeta_n^n = e^{2\pi i} = 1$ (Theorem 3.16 of Chapter 3). So Ψ does not depend on the representative.

Step 2 — homomorphism. By the fundamental property of the exponential (Theorem 3.5 of Chapter 3),

$$\Psi([k] + [j]) = \zeta_n^{k+j} = \zeta_n^k \zeta_n^j = \Psi([k]) \Psi([j]) :$$

addition modulo n becomes multiplication of complex numbers. It is the algebraic translation of the geometric fact that *adding angles is composing rotations*.

Step 3 — injectivity. If $\Psi([k]) = 1$ then $e^{2\pi i k/n} = 1$, so $2\pi k/n$ is a multiple of 2π , i.e. $n \mid k$, i.e. $[k] = [0]$. The kernel is trivial.

Step 4 — surjectivity. By Theorem 3.16 of Chapter 3, μ_n has exactly n elements, all of the form ζ_n^k with $k = 0, \dots, n-1$: so Ψ is onto. Being injective between finite sets of the same size, it is a bijection.

Step 5 — geometric reading. The isomorphism says that the circle divided into n equal arcs, with the operation “advance by k notches”, is the *same object* as arithmetic modulo n . A clock face is $\mathbb{Z}/12\mathbb{Z}$; a gear with n teeth is $\mathbb{Z}/n\mathbb{Z}$; a digital counter with n states is $\mathbb{Z}/n\mathbb{Z}$. All the theorems that follow are stated indifferently in one language or the other. $\square$

Theorem 7.2: Order of an element and generators. Let $\zeta_n = e^{2\pi i/n}$ and $k \in \mathbb{Z}$. Then:

- (i) the order of ζ_n^k in μ_n is $\frac{n}{\gcd(n, k)}$;
- (ii) ζ_n^k generates μ_n if and only if $\gcd(n, k) = 1$;
- (iii) the generators of μ_n (the **primitive n -th roots of unity**) are exactly $\varphi(n)$ in number, where φ is Euler’s totient function.

Proof. **Step 1 — vanishing condition.** Put $d = \gcd(n, k)$ and write $n = d n'$, $k = d k'$ with $\gcd(n', k') = 1$. Then

$$(\zeta_n^k)^m = 1 \iff n \mid km \iff n' \mid k'm \iff n' \mid m,$$

the last step by Euclid’s lemma, since $\gcd(n', k') = 1$.

Step 2 — (i). The least $m \geq 1$ with $n' \mid m$ is $m = n' = n/d$: that is the order.

Step 3 — (ii). ζ_n^k generates μ_n iff the subgroup it generates has n elements, i.e. iff its order is n ; by (i) this means $n/\gcd(n, k) = n$, i.e. $\gcd(n, k) = 1$.

Step 4 — (iii). The number of $k \in \{0, 1, \dots, n-1\}$ with $\gcd(n, k) = 1$ is by definition $\varphi(n)$.

Step 5 — a check. For $n = 12$: $\varphi(12) = 4$ and the generators correspond to $k = 1, 5, 7, 11$. On the clock face: starting from twelve o’clock and advancing five hours at a time one visits all twelve hours (0, 5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7 and then 0 again); advancing by four one visits only three (0, 4, 8), because $\gcd(12, 4) = 4$ and the order is $12/4 = 3$. $\checkmark$ $\square$

Theorem 7.3: Structure of the subgroups. Let $G = \langle g \rangle$ be cyclic of order n . Then:

- (i) every subgroup of G is cyclic;

(ii) for each divisor $d \mid n$ there is exactly one subgroup of order d , namely $\langle g^{n/d} \rangle$;
 (iii) there are no other subgroups.
 Geometrically: the subgroups of μ_n are exactly the inscribed regular polygons whose number of vertices divides n .

Proof. (i) Let $H \leq G$ be a non-trivial subgroup and let $m \geq 1$ be the least positive integer with $g^m \in H$ (it exists because H contains some g^k with $k \not\equiv 0$, hence also g^{-k} , hence some positive power). For every $g^t \in H$ divide t by m : $t = qm + r$ with $0 \leq r < m$. Then

$$g^r = g^t (g^m)^{-q} \in H,$$

and by minimality of m we must have $r = 0$. So $H = \langle g^m \rangle$: cyclic.

(ii) **existence.** Let $d \mid n$. The element $g^{n/d}$ has order $n/\gcd(n, n/d) = n/(n/d) = d$ by Theorem 7.2(i), hence generates a subgroup of order d .

(ii) **uniqueness.** Let H have order d ; by (i), $H = \langle g^m \rangle$ with $m \mid n$ (indeed m divides n : dividing $n = qm + r$ and arguing as above gives $r = 0$). The order of g^m is n/m , hence $n/m = d$ and $m = n/d$: $H = \langle g^{n/d} \rangle$.

(iii) Follows from (i) and (ii): every subgroup is generated by some g^m with $m \mid n$, and these are in bijection with the divisors of n .

Step 4 — geometric reading. In μ_{12} the subgroup of order 4 is $\{1, i, -1, -i\} = \mu_4$: the four vertices of the inscribed square. The subgroup of order 3 is μ_3 : the equilateral triangle. There is no subgroup of order 5, because $5 \nmid 12$: one cannot inscribe a regular pentagon in the vertices of the dodecagon. $\checkmark$ $\square$

Theorem 7.4: Gauss's divisor identity. For every integer $n \geq 1$,

$$\sum_{d \mid n} \varphi(d) = n.$$

Proof. **Step 1 — a partition of μ_n .** Every element of μ_n has a well-defined order, which by Theorem 7.2(i) divides n . Partition μ_n by order:

$$\mu_n = \bigsqcup_{d \mid n} S_d, \quad S_d = \{z \in \mu_n : z \text{ has order } d\}.$$

Step 2 — counting S_d . An element of order d satisfies $z^d = 1$, hence lies in μ_d ; and in μ_d the elements of order exactly d are the generators, which by Theorem 7.2(iii) number $\varphi(d)$. So $|S_d| = \varphi(d)$.

Step 3 — conclusion. Counting the elements of the partition,

$$n = |\mu_n| = \sum_{d \mid n} |S_d| = \sum_{d \mid n} \varphi(d).$$

Step 4 — check for $n = 12$. The divisors are 1, 2, 3, 4, 6, 12 and

$$\varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12. \checkmark$$

Geometrically: of the twelve vertices of the dodecagon, one has order 1 (the vertex 1), one order 2 (-1), two order 3, two order 4, two order 6 and four order 12; twelve in all.

Step 5 — remark. The proof is an instance of *double counting*: the same set counted in two ways. It uses no multiplicative property of φ ; on the contrary, it is from this identity that multiplicativity is most easily deduced (Exercise S1). $\square$

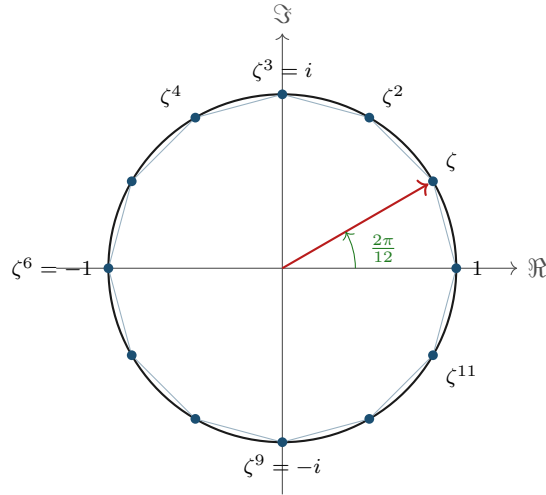


Figure 7.1: The twelve twelfth roots of unity, i.e. the solutions of $z^{12} = 1$: they are the vertices of the regular 12-gon inscribed in the unit circle, with coordinates $\zeta^k = e^{2\pi i k/12}$. Under multiplication they form a cyclic group μ_{12} generated by $\zeta = e^{2\pi i/12}$, and the isomorphism $k \mapsto \zeta^k$ identifies $\mathbb{Z}/12\mathbb{Z}$ with μ_{12} : adding exponents modulo 12 and multiplying roots are the same operation. Dividing the circle into equal parts and doing modular arithmetic are therefore two descriptions of the same object.

7.2. Modular arithmetic on the circle

Theorem 7.5: Lagrange's theorem. Let G be a finite group of order n and $H \leq G$ a subgroup of order m . Then $m \mid n$. In particular the order of every element divides the order of the group, and $g^n = e$ for every $g \in G$.

Proof. **Step 1 — cosets.** For $a \in G$ consider $aH = \{ah : h \in H\}$. The map $h \mapsto ah$ is a bijection from H onto aH (its inverse is $x \mapsto a^{-1}x$), so $|aH| = |H| = m$.

Step 2 — cosets partition G . Every $a \in G$ lies in aH (since $e \in H$). If $aH \cap bH \neq \emptyset$, say $ah_1 = bh_2$, then $a = bh_2h_1^{-1}$ and hence $aH = bh_2h_1^{-1}H = bH$, since $h_2h_1^{-1} \in H$ and H is closed. So cosets are equal or disjoint.

Step 3 — counting. G is a disjoint union of, say, r cosets, each of size m : hence $n = rm$ and $m \mid n$.

Step 4 — consequence. For $g \in G$, the subgroup $\langle g \rangle$ has order equal to the order d of g ; by the above $d \mid n$, so $n = ds$ and $g^n = (g^d)^s = e$. $\square$

Theorem 7.6: Euler–Fermat theorem. Let $n \geq 1$ and a an integer with $\gcd(a, n) = 1$. Then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

In particular, if p is prime and $p \nmid a$,

$$a^{p-1} \equiv 1 \pmod{p} \quad (\text{Fermat's little theorem}),$$

and for every integer a , $a^p \equiv a \pmod{p}$.

Proof. **Step 1 — the group of units.** The set $(\mathbb{Z}/n\mathbb{Z})^\times = \{[a] : \gcd(a, n) = 1\}$ is a group under multiplication. It is closed ($\gcd(a, n) = \gcd(b, n) = 1 \Rightarrow \gcd(ab, n) = 1$) and every element is invertible: by Bézout there are u, v with $au + nv = 1$, so $[a][u] = [1]$.

Step 2 — its order. By definition of the totient, $|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n)$.

Step 3 — applying Lagrange. By Theorem 7.5 applied to $G = (\mathbb{Z}/n\mathbb{Z})^\times$ and $g = [a]$,

$$[a]^{\varphi(n)} = [1], \quad \text{i.e.} \quad a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Step 4 — the prime case. If p is prime, $\varphi(p) = p - 1$, and $\gcd(a, p) = 1$ means $p \nmid a$. Multiplying by a : $a^p \equiv a$, a relation that also holds when $p \mid a$ (both sides are $\equiv 0$).

Step 5 — reading on the circle. The statement says that, walking on the discrete circle with multiplicative steps of size a , one returns to the starting point after $\varphi(n)$ steps. It is the exact arithmetic analogue of returning to the start after a full turn.

Step 6 — numerical check. $n = 10$, $a = 7$: $\varphi(10) = 4$ and

$$7^2 = 49 \equiv 9, \quad 7^4 = 9^2 = 81 \equiv 1 \pmod{10}. \quad \checkmark$$

Hence $7^{222} = 7^{4 \cdot 55 + 2} \equiv 7^2 \equiv 9 \pmod{10}$: the last digit of 7^{222} is 9.

Step 7 — caution. The exponent $\varphi(n)$ need not be the *smallest* that works: for $n = 8$, $\varphi(8) = 4$, yet already $a^2 \equiv 1$ for every odd a . The least universal exponent is Carmichael's function $\lambda(n)$, which divides $\varphi(n)$. $\square$

Theorem 7.7: Chinese remainder theorem. Let $n_1, \dots, n_r$ be positive integers that are pairwise coprime, and $N = n_1 \cdots n_r$. Then the map

$$\Phi : \mathbb{Z}/N\mathbb{Z} \longrightarrow \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}, \quad \Phi([x]_N) = ([x]_{n_1}, \dots, [x]_{n_r}),$$

is a ring isomorphism. In particular, for any residues $a_1, \dots, a_r$ the system

$$x \equiv a_1 \pmod{n_1}, \quad \dots, \quad x \equiv a_r \pmod{n_r}$$

has exactly one solution modulo N .

Proof. **Step 1 — well defined and homomorphism.** If $x \equiv y \pmod{N}$ then $n_i \mid N \mid (x - y)$ for every i , so $x \equiv y \pmod{n_i}$: Φ is well defined. That it respects sums and products is immediate, since reduction mod n_i does.

Step 2 — injectivity. If $\Phi([x]) = 0$ then $n_i \mid x$ for every i . As the n_i are pairwise coprime, their product divides x : indeed, by induction, if $n_1 \cdots n_i \mid x$ and $n_{i+1} \mid x$ with $\gcd(n_1 \cdots n_i, n_{i+1}) = 1$, then $n_1 \cdots n_{i+1} \mid x$ (Euclid's lemma). So $N \mid x$, i.e. $[x]_N = 0$.

Step 3 — surjectivity. The two rings have the same cardinality $N = n_1 \cdots n_r$; an injective map between finite sets of equal size is a bijection.

Step 4 — explicit construction. Put $N_i = N/n_i$; then $\gcd(N_i, n_i) = 1$, so there is M_i with $N_i M_i \equiv 1 \pmod{n_i}$ (Bézout). Then

$$x = \sum_{i=1}^r a_i N_i M_i$$

solves the system: modulo n_j all terms with $i \neq j$ vanish (because $n_j \mid N_i$) and $a_j N_j M_j \equiv a_j$ remains.

Step 5 — the classical example. Sun Zi's problem (3rd–5th century AD): find x with

$$x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 2 \pmod{7}.$$

Here $N = 105$, $N_1 = 35$, $N_2 = 21$, $N_3 = 15$. We find $35 \equiv 2 \pmod{3}$ and $2 \cdot 2 = 4 \equiv 1$, so $M_1 = 2$; $21 \equiv 1 \pmod{5}$, so $M_2 = 1$; $15 \equiv 1 \pmod{7}$, so $M_3 = 1$. Hence

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}.$$

Check: $23 = 3 \cdot 7 + 2 \checkmark$, $23 = 5 \cdot 4 + 3 \checkmark$, $23 = 7 \cdot 3 + 2 \checkmark$.

Step 6 — geometric reading. The theorem says that a circle with $N = n_1 n_2$ notches (with n_1, n_2 coprime) is *equivalent* to two independent circles with n_1 and n_2 notches. It is the principle of the vernier and of gear trains with coprime tooth counts, in which the same configuration recurs only after N steps: a two-wheel counter with 3 and 5 teeth has period 15, not 5.

Step 7 — consequence for φ . Since Φ is a ring isomorphism it maps units to units: $(\mathbb{Z}/N\mathbb{Z})^\times \cong (\mathbb{Z}/n_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/n_r\mathbb{Z})^\times$, and taking cardinalities gives the **multiplicativity of Euler's function**,

$$\varphi(n_1 \cdots n_r) = \varphi(n_1) \cdots \varphi(n_r) \quad (n_i \text{ pairwise coprime}).$$

□

Corollary 7.1: Explicit formula for the totient. If $n = p_1^{a_1} \cdots p_r^{a_r}$ is the factorisation into distinct primes, then

$$\varphi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^r p_i^{a_i-1} (p_i - 1).$$

Proof. **Step 1 — prime powers.** Between 1 and p^a the integers *not* coprime to p^a are exactly the multiples of p , of which there are p^{a-1} . Hence

$$\varphi(p^a) = p^a - p^{a-1} = p^{a-1}(p - 1).$$

Step 2 — multiplicativity. The powers $p_i^{a_i}$ are pairwise coprime; by Step 7 of the proof of Theorem 7.7,

$$\varphi(n) = \prod_i \varphi(p_i^{a_i}) = \prod_i p_i^{a_i-1} (p_i - 1).$$

Step 3 — equivalent form. Factoring out $p_i^{a_i}$,

$$\prod_i p_i^{a_i-1} (p_i - 1) = \prod_i p_i^{a_i} \left(1 - \frac{1}{p_i}\right) = n \prod_i \left(1 - \frac{1}{p_i}\right).$$

Step 4 — check. $\varphi(100) = \varphi(2^2)\varphi(5^2) = 2 \cdot 20 = 40$; with the other formula, $100(1 - \frac{1}{2})(1 - \frac{1}{5}) = 100 \cdot \frac{1}{2} \cdot \frac{4}{5} = 40$. $\checkmark$ □

7.3. Cyclotomic polynomials

Definition 7.2: Cyclotomic polynomial. The n -th **cyclotomic polynomial** is

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} (x - \zeta_n^k),$$

the monic polynomial whose roots are exactly the primitive n -th roots of unity. Its degree is $\varphi(n)$ by Theorem 7.2(iii).

Theorem 7.8: Factorisation of $x^n - 1$. For every $n \geq 1$,

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

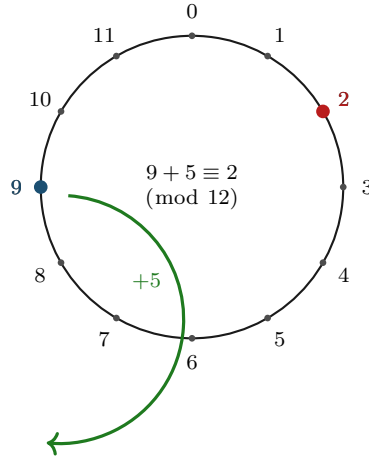


Figure 7.2: Modular arithmetic is arithmetic on the circle. The dial of a 12-hour clock is a concrete realisation of $\mathbb{Z}/12\mathbb{Z}$: adding 5 starting from 9 means walking five steps forward along the circle, and one arrives at 2, not at 14. The *carry* that is lost is exactly one full turn, i.e. the modulus. Formally the map $\mathbb{Z} \rightarrow \mathbb{Z}/12\mathbb{Z}$ is the wrapping of the line of integers onto the circle of twelve positions, and it is a group homomorphism: its kernel is $12\mathbb{Z}$.

Comparing degrees confirms Gauss's identity $\sum_{d|n} \varphi(d) = n$.

Proof. **Step 1 — complete factorisation.** By Theorem 3.16 of Chapter 3, $x^n - 1$ has n distinct roots, the elements of μ_n , hence

$$x^n - 1 = \prod_{z \in \mu_n} (x - z).$$

Step 2 — grouping by order. Every $z \in \mu_n$ has an order d dividing n (Theorem 7.2(i)), and is then a primitive d -th root of unity. Grouping the product by order, and recalling the partition of Step 1 of the proof of Theorem 7.4,

$$\prod_{z \in \mu_n} (x - z) = \prod_{d|n} \prod_{z \text{ of order } d} (x - z) = \prod_{d|n} \Phi_d(x).$$

Step 3 — degrees. Comparing degrees on the two sides recovers $n = \sum_{d|n} \varphi(d)$, i.e. Theorem 7.4: a second proof of the same identity.

Step 4 — check for $n = 6$.

$$\Phi_1 = x - 1, \quad \Phi_2 = x + 1, \quad \Phi_3 = x^2 + x + 1, \quad \Phi_6 = x^2 - x + 1.$$

Their product is

$$(x - 1)(x + 1)(x^2 + x + 1)(x^2 - x + 1) = (x^2 - 1)(x^4 + x^2 + 1) = x^6 - 1,$$

using $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$. $\checkmark$

□

Theorem 7.9: The coefficients are integers. For every $n \geq 1$, $\Phi_n(x) \in \mathbb{Z}[x]$ and is monic.

Proof. **Step 1 — strong induction on n .** For $n = 1$, $\Phi_1 = x - 1 \in \mathbb{Z}[x]$, monic.

Step 2 — inductive step. Suppose $\Phi_d \in \mathbb{Z}[x]$ monic for every $d < n$ with $d \mid n$. By Theorem 7.8,

$$x^n - 1 = \Phi_n(x) \cdot Q(x), \quad Q(x) = \prod_{\substack{d \mid n \\ d < n}} \Phi_d(x).$$

By the inductive hypothesis $Q \in \mathbb{Z}[x]$ and is monic (a product of monics).

Step 3 — Euclidean division in $\mathbb{Z}[x]$. Dividing $x^n - 1$ by the *monic* polynomial Q can be carried out entirely in $\mathbb{Z}[x]$: at each step one subtracts an integer multiple of Q , because the leading coefficient of Q is 1 and no division is ever needed. One obtains quotient $\tilde{Q}$ and remainder R in $\mathbb{Z}[x]$ with $x^n - 1 = \tilde{Q}Q + R$, $\deg R < \deg Q$.

Step 4 — uniqueness. In $\mathbb{C}[x]$ Euclidean division has unique quotient and remainder; since $x^n - 1 = \Phi_n Q + 0$, we must have $\tilde{Q} = \Phi_n$ and $R = 0$. Hence $\Phi_n = \tilde{Q} \in \mathbb{Z}[x]$, monic because $x^n - 1$ and Q are.

Step 5 — examples.

$$\Phi_{12}(x) = \frac{x^{12} - 1}{\Phi_1 \Phi_2 \Phi_3 \Phi_4 \Phi_6} = \frac{x^{12} - 1}{(x^6 - 1)(x^2 + 1)} = x^4 - x^2 + 1,$$

since $\Phi_1 \Phi_2 \Phi_3 \Phi_6 = x^6 - 1$ (Step 4 of the previous theorem) and $\Phi_4 = x^2 + 1$. Check: $(x^6 - 1)(x^2 + 1)(x^4 - x^2 + 1) = (x^6 - 1)(x^6 + 1) = x^{12} - 1$ ✓, using $(x^2 + 1)(x^4 - x^2 + 1) = x^6 + 1$.

Step 6 — a false conjecture. One observes that the coefficients of Φ_n are $0, \pm 1$ for every $n < 105$, which suggests a general rule. The rule is *false*: Φ_{105} has a coefficient equal to -2 . It is the first case, and it occurs because $105 = 3 \cdot 5 \cdot 7$ is the least odd integer with three distinct prime factors. The moral — checking a statement in a hundred cases does not prove it — recurs throughout number theory. $\square$

Theorem 7.10: Irreducibility of Φ_p for prime p . For p prime,

$$\Phi_p(x) = 1 + x + x^2 + \cdots + x^{p-1}$$

is irreducible over $\mathbb{Q}$.

Proof. **Step 1 — explicit form.** The divisors of p are 1 and p , so $x^p - 1 = \Phi_1 \Phi_p = (x - 1) \Phi_p$, whence

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = 1 + x + \cdots + x^{p-1}.$$

Step 2 — translation. Put $x = y + 1$. By the binomial theorem,

$$\Phi_p(y+1) = \frac{(y+1)^p - 1}{y} = \frac{1}{y} \sum_{k=1}^p \binom{p}{k} y^k = \sum_{k=1}^p \binom{p}{k} y^{k-1} = y^{p-1} + \binom{p}{p-1} y^{p-2} + \cdots + \binom{p}{2} y + \binom{p}{1}.$$

Step 3 — Eisenstein's criterion. All coefficients $\binom{p}{k}$ with $1 \leq k \leq p-1$ are divisible by p : indeed $\binom{p}{k} = \frac{p!}{k!(p-k)!}$ and the factor p in the numerator cannot cancel, p being prime and $k!, (p-k)!$ products of factors $< p$. Moreover the constant term is $\binom{p}{1} = p$, not divisible by p^2 . The polynomial is therefore Eisenstein at p , hence irreducible over $\mathbb{Q}$.

Step 4 — conclusion. If $\Phi_p(x)$ factored non-trivially in $\mathbb{Q}[x]$, so would $\Phi_p(y+1)$ (translation is an automorphism of $\mathbb{Q}[x]$): contradiction.

Step 5 — the general case. Φ_n is irreducible over $\mathbb{Q}$ for every n , but the proof needs a different argument, due to Dedekind: it is Theorem 7.11 below. $\square$

Theorem 7.11: Irreducibility of Φ_n for every n (Dedekind). For every $n \geq 1$ the cyclotomic polynomial Φ_n is irreducible over $\mathbb{Q}$. Consequently Φ_n is the minimal polynomial of every primitive n -th root of unity and

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

Proof. Step 1 — setting up. Let ζ be a primitive n -th root of unity and let $f \in \mathbb{Q}[x]$ be its monic minimal polynomial. Since ζ annihilates $x^n - 1$, we have $f \mid x^n - 1$ in $\mathbb{Q}[x]$; write

$$x^n - 1 = f(x)g(x), \quad f, g \text{ monic in } \mathbb{Q}[x].$$

By Gauss's lemma (the product of primitive polynomials is primitive), a monic factor in $\mathbb{Q}[x]$ of a monic polynomial of $\mathbb{Z}[x]$ has integer coefficients: hence $f, g \in \mathbb{Z}[x]$. Likewise $f \mid \Phi_n$ in $\mathbb{Z}[x]$, because $\Phi_n(\zeta) = 0$ and f is the minimal polynomial. The claim is therefore equivalent to proving that every primitive n -th root is a root of f : that would give $\deg f \geq \varphi(n) = \deg \Phi_n$ and hence $f = \Phi_n$.

Step 2 — the key assertion. If p is a prime with $p \nmid n$ and α is a root of f , then α^p is also a root of f .

Step 3 — proof of the assertion, by contradiction. α^p is again an n -th root of unity, hence annihilates $x^n - 1 = fg$: if it does not annihilate f , then $g(\alpha^p) = 0$. So α is a root of the polynomial $g(x^p)$, and since f is the minimal polynomial of α ,

$$g(x^p) = f(x)h(x) \quad \text{with } h \in \mathbb{Z}[x]$$

(the quotient lies in $\mathbb{Z}[x]$ by Euclidean division with a monic divisor, as in Step 3 of the proof of Theorem 7.9).

Step 4 — reduction modulo p . Write $\bar{}$ for reduction of the coefficients modulo p . In $\mathbb{F}_p[x]$ raising to the p -th power is a homomorphism (the Frobenius endomorphism), and by Fermat's little theorem (Theorem 7.6) $c^p = c$ for every $c \in \mathbb{F}_p$; hence, writing $\bar{g} = \sum_i c_i x^i$,

$$\bar{g}(x^p) = \sum_i c_i x^{ip} = \sum_i c_i^p (x^i)^p = \left(\sum_i c_i x^i \right)^p = \bar{g}(x)^p.$$

The relation of Step 3 therefore becomes

$$\bar{g}(x)^p = \bar{f}(x)\bar{h}(x).$$

Step 5 — a common factor. Let q be an irreducible factor of $\bar{f}$ in $\mathbb{F}_p[x]$ (it exists because $\bar{f}$ is monic and non-constant). From $q \mid \bar{f} \mid \bar{g}^p$ and the irreducibility of q we get $q \mid \bar{g}$. Hence q^2 divides $\bar{f}\bar{g}$, i.e.

$$q^2 \mid \overline{x^n - 1} = x^n - \bar{1} \quad \text{in } \mathbb{F}_p[x].$$

Step 6 — but $x^n - 1$ is squarefree modulo p . The formal derivative of $x^n - 1$ in $\mathbb{F}_p[x]$ is nx^{n-1} , and $n \neq 0$ in $\mathbb{F}_p$ because $p \nmid n$. A repeated factor q would divide both $x^n - 1$ and its derivative, hence would divide $\gcd(x^n - 1, nx^{n-1}) = 1$ (the only possible irreducible factor of nx^{n-1} is x , which does not divide $x^n - 1$ since the constant term is $-1 \neq 0$). Contradiction: the assumption $g(\alpha^p) = 0$ is absurd, and the assertion of Step 2 is proved.

Step 7 — from one prime to all coprime exponents. Let k satisfy $\gcd(k, n) = 1$ and let $k = p_1 p_2 \cdots p_s$ be its prime factorisation (with repetitions); no p_i divides n , otherwise p_i would divide $\gcd(k, n) = 1$. Applying Step 2 s times starting from $\alpha = \zeta$:

$$\zeta \text{ a root of } f \implies \zeta^{p_1} \text{ a root of } f \implies \zeta^{p_1 p_2} \text{ a root of } f \implies \cdots \implies \zeta^k \text{ a root of } f.$$

Step 8 — conclusion. As k ranges over the $\varphi(n)$ integers coprime to n in $\{1, \dots, n\}$, the elements ζ^k are *all* the primitive n -th roots of unity (Theorem 7.2(ii)) and are pairwise distinct. Hence f has at least $\varphi(n)$ roots, i.e. $\deg f \geq \varphi(n)$. Since $f \mid \Phi_n$ and $\deg \Phi_n = \varphi(n)$, both being monic, we conclude $f = \Phi_n$: the cyclotomic polynomial is the minimal polynomial of ζ , hence irreducible.

Step 9 — the degree of the extension. The degree of $\mathbb{Q}(\zeta_n)$ over $\mathbb{Q}$ is by definition the degree of the minimal polynomial of ζ_n , that is $\varphi(n)$. This is the fact used in Theorem 7.13.

Step 10 — where the hypothesis $p \nmid n$ is needed. It is essential in Step 6: for $p \mid n$ the polynomial $x^n - 1$ has repeated factors modulo p (for instance $x^p - 1 = (x - 1)^p$ in $\mathbb{F}_p[x]$) and the argument collapses. It is also why in Step 7 one had to check that no prime factor of k divides n : without $\gcd(k, n) = 1$ the statement is false, since ζ^k would not even be a primitive root.

Step 11 — a check. For $n = 12$: $\Phi_{12} = x^4 - x^2 + 1$ and $\varphi(12) = 4$. Modulo $p = 5$ (which does not divide 12) one has $x^{12} - 1 = \prod_{d \mid 12} \bar{\Phi}_d$ with no repeated factors, consistently with Step 6; $\bar{\Phi}_{12}$ splits into two irreducible quadratics over $\mathbb{F}_5$, but this does not contradict irreducibility over $\mathbb{Q}$: reduction modulo p may break an irreducible polynomial, and the proof uses it only to exclude *repeated* factors. ✓ □

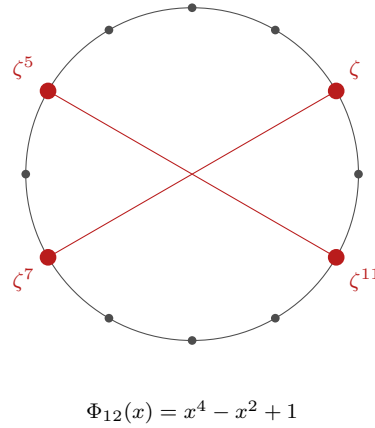


Figure 7.3: The *primitive* twelfth roots of unity (in red) are those ζ^k with $\gcd(k, 12) = 1$, i.e. $k = 1, 5, 7, 11$: they are the generators of the group μ_{12} , and their number is $\varphi(12) = 4$. The cyclotomic polynomial Φ_{12} is by definition the monic polynomial having exactly these four roots, and equals $\Phi_{12}(x) = x^4 - x^2 + 1$: with integer coefficients, like every Φ_n , and irreducible over $\mathbb{Q}$. The non-primitive roots (in grey) generate the proper subgroups, and each is primitive for a divisor of 12: hence the factorisation $x^{12} - 1 = \prod_{d \mid 12} \Phi_d(x)$.

7.4. Constructible polygons: the Gauss–Wantzel theorem

Definition 7.3: Constructible number. A real number α is **constructible** (with straightedge and compass) if, starting from the points $(0, 0)$ and $(1, 0)$, one can obtain in finitely many steps a point with abscissa α , where each step consists in intersecting two lines, a line and a circle, or two circles already constructed.

Theorem 7.12: Constructibles have degree a power of two. If α is constructible then α is algebraic over $\mathbb{Q}$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2.

Proof. Step 1 — coordinates of new points. Lines and circles constructed from points with coordinates in a field $K \subseteq \mathbb{R}$ have equations with coefficients in K :

$$ax + by + c = 0, \quad x^2 + y^2 + dx + ey + f = 0, \quad a, \dots, f \in K.$$

Step 2 — intersections. *Line–line:* the linear system has a solution with coordinates in K . *Line–circle:* substituting the line into the circle gives a quadratic equation with coefficients in K ; the solutions lie in K or in $K(\sqrt{\delta})$ with $\delta \in K$. *Circle–circle:* subtracting the two equations eliminates $x^2 + y^2$ and gives a line with coefficients in K (the radical axis, Volume I); one is reduced to the previous case. In every case the coordinates of the new point lie in an extension of K of degree 1 or 2.

Step 3 — tower of extensions. A finite construction therefore produces a chain

$$\mathbb{Q} = K_0 \subseteq K_1 \subseteq \dots \subseteq K_m, \quad [K_{j+1} : K_j] \in \{1, 2\}, \quad \alpha \in K_m.$$

By multiplicativity of degrees in towers, $[K_m : \mathbb{Q}] = \prod_j [K_{j+1} : K_j] = 2^t$ for some t .

Step 4 — conclusion. Since $\mathbb{Q} \subseteq \mathbb{Q}(\alpha) \subseteq K_m$, again by multiplicativity $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ divides 2^t , hence is itself a power of 2.

Step 5 — classical consequences. Doubling the cube requires $\sqrt[3]{2}$, of degree 3: impossible. General angle trisection requires solving $4c^3 - 3c = \cos \theta$, of degree 3 for $\theta = 60^\circ$: impossible. Squaring the circle requires $\sqrt{\pi}$, and π is not even algebraic (Lindemann, 1882): impossible. These are the three classical problems, all settled negatively by the same criterion. $\square$

Lemma 7.1: Fermat primes. *If $2^m + 1$ is prime with $m \geq 1$, then m is a power of 2. The numbers $F_t = 2^{2^t} + 1$ are the **Fermat numbers**; the prime ones, **Fermat primes**. The only known Fermat primes are*

$$F_0 = 3, \quad F_1 = 5, \quad F_2 = 17, \quad F_3 = 257, \quad F_4 = 65\,537.$$

Proof. Step 1 — factorisation with an odd exponent. Suppose m has an odd divisor $d > 1$ and write $m = de$. Putting $y = 2^e$,

$$2^m + 1 = y^d + 1.$$

Step 2 — algebraic identity. For odd d ,

$$y^d + 1 = (y + 1)(y^{d-1} - y^{d-2} + \dots - y + 1),$$

an identity verified by multiplying out: the terms cancel in pairs and y^d and $+1$ remain (the final sign is $+$ precisely because d is odd).

Step 3 — conclusion. Hence $y + 1 = 2^e + 1$ divides $2^m + 1$. Since $1 < 2^e + 1 < 2^m + 1$ (as $e < m$, i.e. $d > 1$), the number $2^m + 1$ is not prime. Contrapositively, if $2^m + 1$ is prime then m has no odd divisor > 1 , i.e. is a power of 2.

Step 4 — check. $F_5 = 2^{32} + 1 = 4\,294\,967\,297 = 641 \times 6\,700\,417$, a factorisation found by Euler in 1732: not all Fermat numbers are prime. To this day no Fermat prime beyond F_4 is known. $\square$

Theorem 7.13: Gauss–Wantzel theorem. *The regular n -gon is constructible with straightedge and compass if and only if*

$$n = 2^a p_1 p_2 \dots p_r,$$

where $a \geq 0$ and $p_1, \dots, p_r$ are distinct Fermat primes.

Proof. [of the necessary condition, with sufficiency discussed] **Step 1 — reduction to a degree.** Constructing the regular n -gon amounts to constructing $\cos(2\pi/n)$. Now

$$\zeta_n + \zeta_n^{-1} = 2 \cos \frac{2\pi}{n}$$

by the inverse Euler formulas (Theorem 3.6 of Chapter 3), so $\mathbb{Q}(\cos \frac{2\pi}{n}) \subseteq \mathbb{Q}(\zeta_n)$, and ζ_n satisfies over this field the quadratic $z^2 - 2 \cos(2\pi/n)z + 1$. Hence

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = [\mathbb{Q}(\zeta_n) : \mathbb{Q}(\cos \frac{2\pi}{n})] \cdot [\mathbb{Q}(\cos \frac{2\pi}{n}) : \mathbb{Q}] = 2 \cdot [\mathbb{Q}(\cos \frac{2\pi}{n}) : \mathbb{Q}]$$

for $n \geq 3$ (the degree is exactly 2 because $\zeta_n \notin \mathbb{R}$). With $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ (Theorem 7.11) we get

$$[\mathbb{Q}(\cos \frac{2\pi}{n}) : \mathbb{Q}] = \frac{\varphi(n)}{2}.$$

Step 2 — necessity. If the polygon is constructible, by Theorem 7.12 the degree $\varphi(n)/2$ is a power of 2, hence $\varphi(n)$ is a power of 2.

Step 3 — arithmetic translation. Write $n = 2^a p_1^{a_1} \cdots p_r^{a_r}$ with distinct odd primes p_i . By Corollary 7.1,

$$\varphi(n) = \varphi(2^a) \prod_i p_i^{a_i-1} (p_i - 1).$$

For the product to be a power of 2 it is necessary and sufficient that:

- each $p_i^{a_i-1}$ be a power of 2; since p_i is odd this forces $a_i = 1$;
- each $p_i - 1$ be a power of 2, i.e. $p_i = 2^{m_i} + 1$.

By Lemma 7.1, a prime of the form $2^m + 1$ has m a power of 2: hence p_i is a Fermat prime. The exponents $a_i = 1$ guarantee the primes are distinct and occur at most once. This is the stated form.

Step 4 — sufficiency. The converse was proved by Gauss on 30 March 1796, before he turned nineteen, and published in the *Disquisitiones Arithmeticae* (1801). The idea is that the group $(\mathbb{Z}/p\mathbb{Z})^\times$, of order $p - 1 = 2^t$, admits a chain of subgroups

$$\{1\} = H_0 \subset H_1 \subset \cdots \subset H_t = (\mathbb{Z}/p\mathbb{Z})^\times$$

with $[H_{j+1} : H_j] = 2$; the corresponding sums of roots of unity (the *Gaussian periods*) are computed by solving a succession of quadratic equations, each performable with straightedge and compass. Section 7.4.2 carries this argument out in full and without the Galois correspondence: see Theorem 7.17 and Corollary 7.4. Pierre Wantzel proved necessity in 1837, completing the statement.

Step 5 — the case $n = 5$ in full. We show the constructibility of the pentagon by a direct computation. Let $\zeta = \zeta_5$ and $c = \zeta + \zeta^4 = 2 \cos 72^\circ$. From $1 + \zeta + \zeta^2 + \zeta^3 + \zeta^4 = 0$ (Corollary 3.5 of Chapter 3), setting also $c' = \zeta^2 + \zeta^3 = 2 \cos 144^\circ$, we get

$$c + c' = -1, \quad c c' = \zeta^3 + \zeta^4 + \zeta^6 + \zeta^7 = \zeta^3 + \zeta^4 + \zeta + \zeta^2 = -1,$$

using $\zeta^5 = 1$. So c and c' are the roots of $z^2 + z - 1 = 0$:

$$c = \frac{-1 + \sqrt{5}}{2}, \quad c' = \frac{-1 - \sqrt{5}}{2},$$

choosing for c the positive root since $\cos 72^\circ > 0$. Hence

$$\cos 72^\circ = \frac{\sqrt{5} - 1}{4} = \frac{2.2360680 - 1}{4} = 0.3090170,$$

matching the tabulated value $\cos 72^\circ = 0.3090170$. ✓ Since $\sqrt{5}$ is constructible (the hypotenuse of a right triangle with legs 1 and 2), so is $\cos 72^\circ$, and with it the regular pentagon.

Step 6 — which polygons. The constructible ones with $n \leq 20$ are

$$3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20;$$

not constructible are 7, 9, 11, 13, 14, 18, 19. In particular the regular heptagon is not constructible ($\varphi(7) = 6$, not a power of 2), while the 17-gon is — a result Gauss regarded as his dearest discovery, to the point of asking that a 17-gon be carved on his tombstone.

Step 7 — check on $n = 9$. $\varphi(9) = 6$: not a power of 2, so the enneagon is not constructible. Consistently, constructing it would amount to trisecting the angle of 120° , which Step 5 of Theorem 7.12 declares impossible. ✓ $\square$

***Remark 7.1: Why it is a theorem about the circle.** The Gauss–Wantzel theorem answers a purely geometric question — which divisions of the circle can be performed with straightedge and compass? — with a purely arithmetic criterion on the factorisation of n . It is perhaps the sharpest example of this chapter’s programme: the structure of the discrete circle is the arithmetic of $\mathbb{Z}/n\mathbb{Z}$, and questions about one translate into questions about the other. Note too the role of the number 2: straightedge and compass can solve quadratic equations and nothing more, which is why constructibility reduces to a power-of-two condition.*

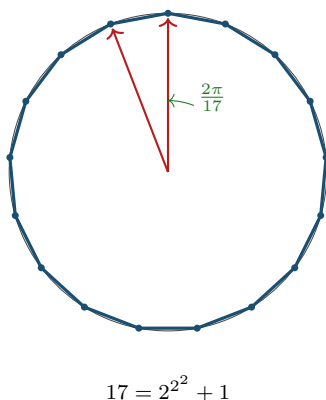


Figure 7.4: The regular 17-gon, constructible with straightedge and compass: the discovery Gauss made at nineteen, on 30 March 1796, and which convinced him to devote himself to mathematics. The reason is that 17 is a Fermat prime, $17 = 2^{2^2} + 1$, so that $\varphi(17) = 16 = 2^4$ is a power of two: the field $\mathbb{Q}(\zeta_{17})$ is then reached by a tower of four quadratic extensions, and every quadratic extension is realisable with straightedge and compass. The Gauss–Wantzel theorem states that the regular n -gon is constructible if and only if $n = 2^k p_1 \cdots p_r$ with p_i distinct Fermat primes.

7.4.1 The seventeen-gon, explicitly

The Gauss–Wantzel theorem says *that* the 17-gon is constructible, but its proof, going through field theory, does not provide the construction. Gauss instead obtained it directly, and it is worth redoing the computation: it is the one place in this volume where a tower of quadratic extensions is walked to the very end with explicit numbers.

Definition 7.4: Gaussian periods. Let p be prime, $\zeta = e^{2\pi i/p}$ and g a generator of the cyclic group $(\mathbb{Z}/p\mathbb{Z})^*$, of order $p - 1$. For every divisor f of $p - 1$, setting $e = (p - 1)/f$, the

Gaussian periods of f terms are the e sums

$$\eta_j = \sum_{t=0}^{f-1} \zeta^{g^{j+te}}, \quad j = 0, 1, \dots, e-1,$$

that is, the sums of those roots whose exponents lie in the same coset of the subgroup of index e of $(\mathbb{Z}/p\mathbb{Z})^*$.

Theorem 7.14: The quadratic tower of the seventeen-gon. Let $\zeta = e^{2\pi i/17}$ and $z = \zeta + \zeta^{-1} = 2 \cos(2\pi/17)$. Then, setting

$$\eta_0, \eta_1 = \frac{-1 \pm \sqrt{17}}{2},$$

the four-term periods η_{00}, η_{02} are the roots of $y^2 - \eta_0 y - 1 = 0$, the periods η_{11}, η_{13} are the roots of $y^2 - \eta_1 y - 1 = 0$, and finally z is a root of

$$z^2 - \eta_{00} z + \eta_{11} = 0.$$

Since each equation is quadratic with coefficients in the field generated by the previous one, $\cos(2\pi/17)$ is expressible by three nested square roots, and the regular seventeen-gon is constructible with straightedge and compass.

Proof. Step 1 — the generator. The number 3 generates $(\mathbb{Z}/17\mathbb{Z})^*$: its successive powers are

$$1, 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6,$$

sixteen distinct values, so 3 has order $16 = \varphi(17)$ and generates the whole group, which is therefore cyclic. The even powers of 3 form the index-2 subgroup $\{1, 9, 13, 15, 16, 8, 4, 2\}$; the odd ones the coset $\{3, 10, 5, 11, 14, 7, 12, 6\}$.

Step 2 — the two eight-term periods. Set

$$\eta_0 = \zeta + \zeta^2 + \zeta^4 + \zeta^8 + \zeta^9 + \zeta^{13} + \zeta^{15} + \zeta^{16}, \quad \eta_1 = \zeta^3 + \zeta^5 + \zeta^6 + \zeta^7 + \zeta^{10} + \zeta^{11} + \zeta^{12} + \zeta^{14}.$$

Since together they exhaust all roots other than 1,

$$\eta_0 + \eta_1 = \sum_{k=1}^{16} \zeta^k = -1$$

by Theorem 3.20. For the product one expands $\eta_0 \eta_1$ into 64 terms ζ^{a+b} with a in the first set and b in the second; every non-zero exponent modulo 17 occurs exactly four times, and the zero exponent never occurs (because $-1 = 16$ lies in the first set, so a sum of an element of the first set and one of the second is never $\equiv 0$: were it so, $b \equiv -a$ would lie in the first set, since -1 is an even power of 3, contrary to hypothesis). Hence

$$\eta_0 \eta_1 = 4 \sum_{k=1}^{16} \zeta^k = -4.$$

The two quantities are therefore roots of $y^2 + y - 4 = 0$, that is $\eta_{0,1} = (-1 \pm \sqrt{17})/2$. The sign is settled numerically: grouping conjugate roots, $\eta_0 = 2 \cos \frac{2\pi}{17} + 2 \cos \frac{4\pi}{17} + 2 \cos \frac{8\pi}{17} + 2 \cos \frac{16\pi}{17} = 1.864944 + 1.478018 + 0.184537 - 1.965946 = 1.561553$, which is $(-1 + \sqrt{17})/2 = 1.561553$ ✓. Hence $\eta_1 = -2.561553$.

Step 3 — the four four-term periods. The index-4 subgroup is $\{1, 13, 16, 4\}$ (powers of 3 with exponent a multiple of 4). Set

$$\begin{aligned}\eta_{00} &= \zeta + \zeta^4 + \zeta^{13} + \zeta^{16}, & \eta_{02} &= \zeta^2 + \zeta^8 + \zeta^9 + \zeta^{15}, \\ \eta_{11} &= \zeta^3 + \zeta^5 + \zeta^{12} + \zeta^{14}, & \eta_{13} &= \zeta^6 + \zeta^7 + \zeta^{10} + \zeta^{11}.\end{aligned}$$

By construction $\eta_{00} + \eta_{02} = \eta_0$ and $\eta_{11} + \eta_{13} = \eta_1$. The products are computed as in Step 2: expanding $\eta_{00}\eta_{02}$ into 16 terms, every non-zero exponent occurs exactly once, hence

$$\eta_{00}\eta_{02} = \sum_{k=1}^{16} \zeta^k = -1,$$

and likewise $\eta_{11}\eta_{13} = -1$. So η_{00}, η_{02} are roots of $y^2 - \eta_0 y - 1 = 0$ and η_{11}, η_{13} of $y^2 - \eta_1 y - 1 = 0$. Numerically

$$\begin{aligned}\eta_{00} &= \frac{\eta_0 + \sqrt{\eta_0^2 + 4}}{2} = \frac{1.561553 + 2.537409}{2} = 2.049481, & \eta_{02} &= -0.487928, \\ \eta_{11} &= \frac{\eta_1 + \sqrt{\eta_1^2 + 4}}{2} = 0.344151, & \eta_{13} &= -2.905704.\end{aligned}$$

Check: $\eta_{00}\eta_{02} = 2.049481 \cdot (-0.487928) = -1.000000 \checkmark$; $\eta_{11}\eta_{13} = 0.344151 \cdot (-2.905704) = -1.000000 \checkmark$.

Step 4 — the last step. The period η_{00} splits into the two two-term periods $\zeta + \zeta^{16} = 2 \cos \frac{2\pi}{17} = z$ and $\zeta^4 + \zeta^{13} = 2 \cos \frac{8\pi}{17} =: w$, with $z + w = \eta_{00}$. Their product is

$$zw = (\zeta + \zeta^{16})(\zeta^4 + \zeta^{13}) = \zeta^5 + \zeta^{14} + \zeta^3 + \zeta^{12} = \eta_{11},$$

having used $16 + 4 = 20 \equiv 3$ and $16 + 13 = 29 \equiv 12$ modulo 17. Hence z and w are roots of $y^2 - \eta_{00}y + \eta_{11} = 0$, and

$$z = \frac{\eta_{00} + \sqrt{\eta_{00}^2 - 4\eta_{11}}}{2} = \frac{2.049481 + \sqrt{4.200373 - 1.376603}}{2} = \frac{2.049481 + 1.680408}{2} = 1.864944,$$

that is $\cos(2\pi/17) = 0.932472 \checkmark$, in agreement with the direct value $\cos(21.17647^\circ) = 0.932472$.

Step 5 — conclusion. Each step solves a quadratic equation whose coefficients belong to the field generated by the previous steps. Since $\mathbb{Q}(\zeta_{17})/\mathbb{Q}$ is abelian, every subfield of it is Galois over $\mathbb{Q}$: in particular $\mathbb{Q}(\eta_{00})$, being Galois of degree 4, already contains all its conjugates, namely $\eta_{02}, \eta_{11}, \eta_{13}$ (checkable numerically: the four quantities are roots of the same rational polynomial $y^4 + y^3 - 6y^2 - y + 1$), so no further extension is needed to reach η_{11} from η_{00} . The tower is therefore

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{17}) \subset \mathbb{Q}(\eta_{00}) \subset \mathbb{Q}(z),$$

a tower of three quadratic extensions, in agreement with $[\mathbb{Q}(\zeta_{17} + \zeta_{17}^{-1}) : \mathbb{Q}] = \varphi(17)/2 = 8 = 2^3$ for the real field and with $\varphi(17) = 16 = 2^4$ for the complex one. Since every quadratic extension is realised by straightedge and compass (intersection of a line and a circle, or of two circles), the segment of length $\cos(2\pi/17)$ is constructible, and with it the polygon. $\square$

Corollary 7.2: Gauss's closed form. Unwinding the chain of Theorem 7.14 one obtains

$$16 \cos \frac{2\pi}{17} = -1 + \sqrt{17} + \sqrt{34 - 2\sqrt{17}} + 2\sqrt{17 + 3\sqrt{17} - \sqrt{34 - 2\sqrt{17}} - 2\sqrt{34 + 2\sqrt{17}}}.$$

Proof. It is a matter of rewriting the radicals of Steps 3 and 4 in terms of $\sqrt{17}$. From $\eta_0 = (-1 + \sqrt{17})/2$ it follows that $\eta_0^2 + 4 = (18 - 2\sqrt{17})/4 + 16/4 = (34 - 2\sqrt{17})/4$, hence $\sqrt{\eta_0^2 + 4} = \frac{1}{2}\sqrt{34 - 2\sqrt{17}}$ and

$$4\eta_{00} = 2\eta_0 + \sqrt{34 - 2\sqrt{17}} = -1 + \sqrt{17} + \sqrt{34 - 2\sqrt{17}};$$

likewise $4\eta_{11} = -1 - \sqrt{17} + \sqrt{34 + 2\sqrt{17}}$. Numerically $4\eta_{00} = 8.197925$ and $4\eta_{11} = 1.376603$, in agreement with the values of Step 3. Substituting into $z = (\eta_{00} + \sqrt{\eta_{00}^2 - 4\eta_{11}})/2$ and simplifying gives the stated form. The numerical check is immediate:

$$\sqrt{17} = 4.123106, \quad \sqrt{34 - 2\sqrt{17}} = 5.074819, \quad \sqrt{34 + 2\sqrt{17}} = 6.499709,$$

$$17 + 3\sqrt{17} - 5.074819 - 2 \cdot 6.499709 = 29.369317 - 18.074237 = 11.295080,$$

whose square root is 3.360816; hence $16 \cos(2\pi/17) = -1 + 4.123106 + 5.074819 + 6.721633 = 14.919558$, that is $\cos(2\pi/17) = 0.932472 \checkmark$. $\square$

Remark 7.2: What Gauss actually did, and when. Gauss recorded the discovery in his diary on 30 March 1796, aged nineteen: it is the first entry of the diary. He published it in the *Disquisitiones Arithmeticae* of 1801, section VII, together with the general criterion; of the necessity of the condition he gave only the statement, and the complete proof is due to Wantzel (1837). Note that the construction given here does not produce a straightedge-and-compass drawing: it produces a chain of quadratic equations. The passage from radical to drawing is the content of Definition 7.3, which admits among the permitted operations precisely the intersection of two circles and that of a line with a circle: and it is the extraction of a square root that corresponds to such intersections. These are different things: algebra says whether it can be done, geometry how.

7.4.2 Sufficiency in general: Gaussian periods for every Fermat prime

The heptadecagon showed the mechanism on a concrete case: a chain of four quadratic equations, each with coefficients already constructed at the previous step. That mechanism has nothing specific to the number 17. We prove here the sufficiency half of Theorem 7.13 in general, following Gauss and *without* using the Galois correspondence: all that is needed is the existence of a primitive root modulo p and the irreducibility of Φ_p , both already available.

Theorem 7.15: Existence of a primitive root modulo p . For every prime p the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p - 1$.

Proof. For each divisor d of $p - 1$ let $\psi(d)$ be the number of elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ of order exactly d . Since the order of every element divides $p - 1$ (Theorem 7.5),

$$\sum_{d|p-1} \psi(d) = p - 1.$$

Step 1 — $\psi(d)$ is either 0 or $\varphi(d)$. Suppose $\psi(d) > 0$ and let a be an element of order d . The d elements $1, a, a^2, \dots, a^{d-1}$ are distinct and all satisfy $x^d = 1$. On the other hand $\mathbb{Z}/p\mathbb{Z}$ is a field, and a polynomial of degree d with coefficients in a field has at most d roots; hence $x^d - 1$ has *exactly* the roots a^k , $0 \leq k < d$, and no others. Every element of order d satisfies $x^d = 1$ and so lies in $\langle a \rangle$; and the elements of order d in a cyclic group of order d are its generators, $\varphi(d)$ in number (Theorem 7.2). Therefore $\psi(d) = \varphi(d)$.

Step 2 — comparison with Gauss's identity. By Theorem 7.4,

$$\sum_{d|p-1} \varphi(d) = p-1 = \sum_{d|p-1} \psi(d).$$

Since $\psi(d) \leq \varphi(d)$ for every d (Step 1: it is either 0 or $\varphi(d)$), equality of the two sums forces $\psi(d) = \varphi(d)$ for every d . In particular $\psi(p-1) = \varphi(p-1) \geq 1$: an element of order $p-1$ exists, that is, a generator. $\square$

Definition 7.5: Gaussian periods of level j . Let $p = 2^t + 1$ be a Fermat prime, $\zeta = e^{2\pi i/p}$, and let g be a primitive root modulo p (Theorem 7.15). For $0 \leq j \leq t$ and $0 \leq k < 2^j$ set

$$\eta_k^{(j)} = \sum_{m=0}^{2^{t-j}-1} \zeta^{g^{k+m2^j}},$$

a sum of 2^{t-j} p -th roots of unity. Indices k are read modulo 2^j . This is the case $f = 2^{t-j}$ of Definition 7.4.

Remark 7.3: How to read the definition. The exponents $g^0, g^1, \dots, g^{2^t-1}$ run exactly once through all the nonzero residues modulo p . Level j partitions them into 2^j classes according to the remainder of i modulo 2^j ; the period $\eta_k^{(j)}$ sums the roots of the k -th class. At level 0 there is a single period, the sum of all the nontrivial roots:

$$\eta_0^{(0)} = \sum_{a=1}^{p-1} \zeta^a = -1,$$

by Corollary 3.5. At level t each class contains a single element and the periods are the individual roots. From one level to the next each class splits in two: it is this doubling that produces the chain of quadratic equations.

Lemma 7.2: Action of the substitution $\zeta \mapsto \zeta^g$. Let $\sigma : \mathbb{Z}[\zeta] \rightarrow \mathbb{Z}[\zeta]$ be the unique ring homomorphism with $\sigma(\zeta) = \zeta^g$. Then:

- (i) σ is well defined and bijective, and $\sigma(\eta_k^{(j)}) = \eta_{k+1}^{(j)}$ for every j, k ;
- (ii) an element of $\mathbb{Z}[\zeta]$ fixed by σ^{2^j} is an integer linear combination of the periods $\eta_0^{(j)}, \dots, \eta_{2^j-1}^{(j)}$.

Proof. (i). By Theorem 7.10 the polynomial $\Phi_p(x) = 1 + x + \dots + x^{p-1}$ is irreducible over $\mathbb{Q}$, hence it is the minimal polynomial of ζ and $\mathbb{Z}[\zeta] \cong \mathbb{Z}[x]/(\Phi_p)$. Since ζ^g is also a primitive p -th root of unity, it annihilates Φ_p , and therefore the assignment $\zeta \mapsto \zeta^g$ defines a ring homomorphism. It permutes the finite set $\{\zeta, \zeta^2, \dots, \zeta^{p-1}\}$ (because $a \mapsto ag$ permutes the nonzero residues), hence it is bijective. Applying it to the definition,

$$\sigma(\eta_k^{(j)}) = \sum_m \zeta^{g^{k+m2^j} \cdot g} = \sum_m \zeta^{g^{(k+1)+m2^j}} = \eta_{k+1}^{(j)}.$$

(ii). The set $\{\zeta, \zeta^2, \dots, \zeta^{p-1}\}$ is a $\mathbb{Z}$ -basis of $\mathbb{Z}[\zeta]$: there are $p-1$ of them, the relation $1 + \zeta + \dots + \zeta^{p-1} = 0$ allows 1 to be eliminated, and no further integer linear relation holds, again by irreducibility of Φ_p . So write an element as $\xi = \sum_{a=1}^{p-1} c_a \zeta^a$ with $c_a \in \mathbb{Z}$. Then

$$\sigma^{2^j}(\xi) = \sum_{a=1}^{p-1} c_a \zeta^{a g^{2^j}},$$

that is, the coefficients are permuted according to $a \mapsto a g^{2^j}$. The orbits of this permutation on $(\mathbb{Z}/p\mathbb{Z})^\times$ are the cosets of the subgroup $\langle g^{2^j} \rangle$, of index 2^j : they are exactly the sets of exponents defining the level- j periods. The hypothesis $\sigma^{2^j}(\xi) = \xi$ and the uniqueness of the expansion in the basis then give that c_a is constant on each orbit, that is $\xi = \sum_{k=0}^{2^j-1} c_k \eta_k^{(j)}$ with $c_k \in \mathbb{Z}$. $\square$

Theorem 7.16: The quadratic chain of the periods. *With the notation above, for every $0 \leq j < t$ and every $0 \leq k < 2^j$ the two periods $\eta_k^{(j+1)}$ and $\eta_{k+2^j}^{(j+1)}$ are the two roots of the quadratic equation*

$$X^2 - \eta_k^{(j)} X + P_k^{(j)} = 0,$$

where $P_k^{(j)} = \eta_k^{(j+1)} \cdot \eta_{k+2^j}^{(j+1)}$ is an integer linear combination of the level- j periods.

Proof. The sum. The level- j class of index k is the disjoint union of the two level- $(j+1)$ classes of indices k and $k+2^j$: indeed $i \equiv k \pmod{2^j}$ is equivalent to $i \equiv k$ or $i \equiv k+2^j$ modulo 2^{j+1} . Summing the corresponding roots,

$$\eta_k^{(j+1)} + \eta_{k+2^j}^{(j+1)} = \eta_k^{(j)}.$$

The product. By Lemma 7.2(i), applying σ^{2^j} and recalling that level- $(j+1)$ indices are read modulo 2^{j+1} ,

$$\sigma^{2^j}(\eta_k^{(j+1)}) = \eta_{k+2^j}^{(j+1)}, \quad \sigma^{2^j}(\eta_{k+2^j}^{(j+1)}) = \eta_{k+2^{j+1}}^{(j+1)} = \eta_k^{(j+1)}.$$

Hence σ^{2^j} interchanges the two factors, and their product $P_k^{(j)}$ is fixed by σ^{2^j} . By Lemma 7.2(ii), $P_k^{(j)}$ is an integer linear combination of the level- j periods.

The sum and the product of the two quantities are therefore known, and the two quantities are the roots of $X^2 - (\text{sum})X + (\text{product})$. $\square$

Corollary 7.3: The periods up to level $t-1$ are real. *For $0 \leq j \leq t-1$ every period $\eta_k^{(j)}$ is a real number. In particular*

$$\eta_0^{(t-1)} = \zeta + \zeta^{-1} = 2 \cos \frac{2\pi}{p}.$$

Proof. The element -1 of $(\mathbb{Z}/p\mathbb{Z})^\times$ has order 2; since the group is cyclic of order 2^t (Theorem 7.15), it has a unique element of order 2, which is $g^{2^{t-1}}$. Hence

$$g^{2^{t-1}} \equiv -1 \pmod{p}.$$

Now let $j \leq t-1$: if ζ^{g^i} occurs in $\eta_k^{(j)}$, that is $i \equiv k \pmod{2^j}$, then also $i+2^{t-1} \equiv k \pmod{2^j}$ (because 2^j divides 2^{t-1}), and the corresponding term is

$$\zeta^{g^{i+2^{t-1}}} = \zeta^{g^i \cdot g^{2^{t-1}}} = \zeta^{-g^i} = \overline{\zeta^{g^i}}.$$

The terms of $\eta_k^{(j)}$ therefore pair up into conjugates, and the sum is real.

For $j = t-1$ each class has exactly two elements, $\{k, k+2^{t-1}\}$, and for $k=0$ one gets $\zeta^{g^0} + \zeta^{-g^0} = \zeta + \zeta^{-1} = 2 \cos(2\pi/p)$. $\square$

Theorem 7.17: Sufficiency for a Fermat prime. *Let $p = 2^t + 1$ be a Fermat prime. Then $\cos(2\pi/p)$ is constructible with straightedge and compass, and hence so is the regular p -gon.*

Proof. For $0 \leq j \leq t-1$ put

$$K_j = \mathbb{Q}(\eta_0^{(j)}, \eta_1^{(j)}, \dots, \eta_{2^j-1}^{(j)}),$$

a subfield of $\mathbb{R}$ by Corollary 7.3. One has $K_0 = \mathbb{Q}$, because $\eta_0^{(0)} = -1$.

Step 1 — each jump is a tower of quadratic extensions. Fix $j < t-1$. By Theorem 7.16 each of the 2^{j+1} level- $(j+1)$ periods is a root of a quadratic polynomial whose coefficients lie in K_j : the sum is a level- j period, the product an integer combination of level- j periods. Adjoining these roots one at a time yields a chain

$$K_j = L_0 \subseteq L_1 \subseteq \dots \subseteq L_s = K_{j+1}, \quad [L_{i+1} : L_i] \in \{1, 2\},$$

because each new element is a root of a quadratic with coefficients in the preceding field. All the fields are contained in $\mathbb{R}$, and the discriminants involved are non-negative because the roots (the periods) are real: the square roots required are therefore square roots of positive real numbers, which are constructible.

Step 2 — the complete tower. Concatenating the jumps for $j = 0, 1, \dots, t-2$ yields a finite tower

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \dots \subseteq F_M = K_{t-1}, \quad [F_{i+1} : F_i] \in \{1, 2\},$$

and deleting the steps of degree 1 leaves a tower of quadratic extensions in the sense of Definition 7.3.

Step 3 — conclusion. By Corollary 7.3 we have $2 \cos(2\pi/p) = \eta_0^{(t-1)} \in K_{t-1} = F_M$, hence $\cos(2\pi/p)$ is constructible. Once $\cos(2\pi/p)$ is constructed, the point $(\cos(2\pi/p), \sin(2\pi/p))$ of the unit circle is obtained by erecting the perpendicular to the x axis, and stepping the arc round p times produces the regular p -gon. $\square$

Corollary 7.4: Sufficiency in the general case. If $n = 2^a p_1 \cdots p_r$ with p_i distinct Fermat primes, the regular n -gon is constructible.

Proof. **Step 1 — powers of 2.** The square is constructible, and by repeatedly bisecting the central angle (an elementary straightedge-and-compass construction) one obtains all 2^a -gons for $a \geq 2$.

Step 2 — the individual Fermat primes. This is Theorem 7.17.

Step 3 — composing coprime side counts. Let m, n be coprime and suppose the m -gon and the n -gon are constructible; we show the mn -gon is. By Bézout's identity there are integers u, v with $um + vn = 1$; dividing by mn and multiplying by 2π ,

$$\frac{2\pi}{mn} = u \cdot \frac{2\pi}{n} + v \cdot \frac{2\pi}{m}.$$

The angle $2\pi/(mn)$ is thus an integer combination of two constructible angles, and laying off an angle an integer number of times (in either sense) is a straightedge-and-compass construction. Hence $2\pi/(mn)$ is constructible, and with it the mn -gon.

Step 4 — conclusion. The factors $2^a, p_1, \dots, p_r$ are pairwise coprime; repeatedly applying Step 3 to the constructible polygons of Steps 1 and 2 gives the $n = 2^a p_1 \cdots p_r$ -gon. $\square$

Remark 7.4: The heptadecagon as the case $t = 4$. With $p = 17$ one has $t = 4$, and the construction above produces exactly the chain of Theorem 7.14: level 1 gives the two periods of eight terms, level 2 the four periods of four terms, level 3 the eight periods of two terms — and $\eta_0^{(3)} = 2 \cos(2\pi/17)$ is the last rung, in agreement with Corollary 7.3 for $t-1 = 3$. The primitive root used there is $g = 3$, and indeed $3^8 \equiv -1 \pmod{17}$, as the

proof of Corollary 7.3 requires with $2^{t-1} = 8$.

For $p = 257$ ($t = 8$) the chain has seven rungs and 128 final periods; for $p = 65\,537$ ($t = 16$) it has fifteen. The construction exists and is in principle executable, but writing it out explicitly fills volumes: the one for the 65 537-gon was drawn up by Johann Gustav Hermes between 1879 and 1889 and fills a trunk preserved at Göttingen.

7.5. Gauss sums and the law of quadratic reciprocity

The Gauss periods of Definition 7.4 have so far been used to construct regular polygons. In the simplest possible case, $e = 2$ (that is, $f = (p-1)/2$ terms per period), they encode an arithmetic object of the first importance: the quadratic Gauss sum, which underlies the law of quadratic reciprocity.

Definition 7.6: Legendre symbol. Let p be an odd prime and a an integer with $p \nmid a$. The **Legendre symbol** is

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue modulo } p \text{ (that is, } a \equiv b^2 \pmod{p} \text{ for some } b), \\ -1 & \text{otherwise.} \end{cases}$$

For $p \mid a$ one sets $\left(\frac{a}{p}\right) = 0$.

Theorem 7.18: Multiplicativity of the Legendre symbol. For $p \nmid a$ and $p \nmid b$,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$$

Proof. By Theorem 7.15 the group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$; let g be a generator. The subgroup of even powers $\langle g^2 \rangle$ has order $(p-1)/2$ and is the unique subgroup of that order, by Theorem 7.3(ii). On the other hand the image of the squaring map $x \mapsto x^2$ on $(\mathbb{Z}/p\mathbb{Z})^\times$ is also a subgroup: the kernel $\{\pm 1\}$ has order 2 (since p is odd, $1 \neq -1$), so the image has order $(p-1)/2$. By the uniqueness just cited the two subgroups coincide: the quadratic residues are exactly the even powers of g . Writing $a \equiv g^i$, $b \equiv g^j \pmod{p}$ gives $\left(\frac{a}{p}\right) = (-1)^i$, $\left(\frac{b}{p}\right) = (-1)^j$, and

$$\left(\frac{ab}{p}\right) = (-1)^{i+j} = (-1)^i (-1)^j = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$$

□

Lemma 7.3: The value of $\left(\frac{-1}{p}\right)$.

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}},$$

that is, it equals $+1$ if $p \equiv 1 \pmod{4}$ and -1 if $p \equiv 3 \pmod{4}$.

Proof. The cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ has even order $p-1$, so it has a unique element of order 2 (Theorem 7.3(ii) with $d = 2$): it is -1 , since $(-1)^2 = 1$ and $-1 \neq 1$ as p is odd. Writing $-1 \equiv g^i$ as in the previous proof, the order of g^i is $(p-1)/\gcd(p-1, i)$ (Theorem 7.2(i)); requiring it to equal 2, and using again the uniqueness of the subgroup of order 2, gives $i = (p-1)/2$. Hence $-1 \equiv g^{(p-1)/2}$, and by the argument in the proof of Theorem 7.18, $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

it equals $+1$ if $(p-1)/2$ is even, that is $p \equiv 1 \pmod{4}$, and -1 if $(p-1)/2$ is odd, that is $p \equiv 3 \pmod{4}$. $\square$

Definition 7.7: Quadratic Gauss sum. For p an odd prime and $\zeta = e^{2\pi i/p}$, the quadratic Gauss sum is

$$g_p = \sum_{a=1}^{p-1} \left(\frac{a}{p}\right) \zeta^a.$$

Remark 7.5: Gauss sums are the index-two periods already defined. In Definition 7.4, the case $e = 2$ (that is, $f = (p-1)/2$) gives two periods η_0, η_1 : the sum of the roots whose exponents are even powers of g (the quadratic residues) and that of the odd powers (the non-residues). By the proof of Theorem 7.18, these are exactly

$$\eta_0 = \sum_{a \text{ residue}} \zeta^a, \quad \eta_1 = \sum_{a \text{ non-residue}} \zeta^a,$$

so $g_p = \eta_0 - \eta_1$: the Gauss sum is the difference of the two index-2 periods already used for the Gauss–Wantzel theorem. Moreover, since $\{\eta_0, \eta_1\}$ partitions the $p-1$ non-trivial roots, $\eta_0 + \eta_1 = \sum_{a=1}^{p-1} \zeta^a = -1$ by Corollary 3.5.

Theorem 7.19: The square of the Gauss sum. For every odd prime p ,

$$g_p^2 = \left(\frac{-1}{p}\right) p.$$

Proof. **Step 1 — the double product.** By definition,

$$g_p^2 = \sum_{s=1}^{p-1} \sum_{t=1}^{p-1} \left(\frac{s}{p}\right) \left(\frac{t}{p}\right) \zeta^{s+t}.$$

Step 2 — substitution $t \equiv su$. For fixed nonzero s , the map $u \mapsto su \pmod{p}$ is a bijection of $\{1, \dots, p-1\}$ onto itself; writing $t \equiv su$ and summing over u instead of t , and using multiplicativity (Theorem 7.18) together with $\left(\frac{s}{p}\right)^2 = 1$,

$$g_p^2 = \sum_{s=1}^{p-1} \sum_{u=1}^{p-1} \left(\frac{s}{p}\right) \left(\frac{su}{p}\right) \zeta^{s(1+u)} = \sum_{u=1}^{p-1} \left(\frac{u}{p}\right) \underbrace{\sum_{s=1}^{p-1} \zeta^{s(1+u)}}_{=: S(u)}.$$

Step 3 — computing $S(u)$. If $u \equiv -1 \pmod{p}$, every exponent $s(1+u) \equiv 0$, so $S(-1) = \sum_{s=1}^{p-1} 1 = p-1$. If $u \not\equiv -1$, set $\omega = \zeta^{1+u} \neq 1$: since $\omega^p = (\zeta^p)^{1+u} = 1$, the finite geometric sum gives

$$\sum_{s=0}^{p-1} \omega^s = \frac{\omega^p - 1}{\omega - 1} = 0, \quad \text{hence} \quad S(u) = \sum_{s=1}^{p-1} \omega^s = 0 - \omega^0 = -1.$$

Step 4 — combining. Since the quadratic residues are the even powers of g among the $p-1$ powers $g^0, \dots, g^{p-2}$, and $p-1$ is even, there are exactly $(p-1)/2$ of them, as many as the non-residues; hence $\sum_{u=1}^{p-1} \left(\frac{u}{p}\right) = \frac{p-1}{2} - \frac{p-1}{2} = 0$. Separating the term $u = -1$ from Step 2,

$$g_p^2 = \left(\frac{-1}{p}\right) (p-1) + \sum_{\substack{u=1 \\ u \neq p-1}}^{p-1} \left(\frac{u}{p}\right) \cdot (-1) = \left(\frac{-1}{p}\right) (p-1) - \left(0 - \left(\frac{-1}{p}\right)\right) = \left(\frac{-1}{p}\right) p.$$

Step 5 — numerical check. For $p = 5$ ($\equiv 1 \pmod{4}$): $\left(\frac{-1}{5}\right) = +1$, so $g_5^2 = 5$; for $p = 7$ ($\equiv 3 \pmod{4}$): $\left(\frac{-1}{7}\right) = -1$, so $g_7^2 = -7$. Both confirmed by the script `vol2_cap07_gauss_sums.py` (see Example 7.1). $\square$

Remark 7.6: What we import: the sign of the Gauss sum. Theorem (Gauss, 1805). For every odd prime p ,

$$g_p = \begin{cases} +\sqrt{p} & \text{if } p \equiv 1 \pmod{4}, \\ +i\sqrt{p} & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Theorem 7.19 fixes g_p^2 , but not the sign of the root: a priori g_p could equal $-\sqrt{p}$ instead of $+\sqrt{p}$ (or $-i\sqrt{p}$). Determining that the sign is always the positive one cost Gauss himself about four years (from 1801 to 1805): the most direct proof uses a continuity argument on the Jacobi theta function as p varies, tools not developed in this treatise. The result is therefore **imported** from C. F. Gauss, *Summatio quarundam serierum singularium* (1811), in the form given in K. Ireland, M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer 1990, Ch. 6 (Quadratic Gauss Sums), §4 (The Sign of the Quadratic Gauss Sum). The absolute value $|g_p| = \sqrt{p}$, by contrast, follows already in full from Theorem 7.19, proved above without any import.

Remark 7.7: What we import: the law of quadratic reciprocity. Theorem (Gauss). If p, q are distinct odd primes,

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Theorem 7.19 is the first step of one of the classical proofs of this law through Gauss sums: one works in the ring $\mathbb{Z}[\zeta_p]$ reduced modulo a prime ideal over q , uses the Frobenius endomorphism $x \mapsto x^q$ in characteristic q (the same principle as Step 4 of the proof of Theorem 7.11 in this chapter, here applied to a different quotient) to compute g_p^q in two independent ways — on one hand $g_p^q \equiv \left(\frac{q}{p}\right) g_p$, on the other $g_p^q = g_p \cdot (g_p^2)^{(q-1)/2} \equiv g_p \cdot \left(\frac{-1}{p}\right)^{(q-1)/2} \left(\frac{p}{q}\right)$ by Fermat's little theorem applied to the exponent — and one concludes by comparing the two values modulo the chosen ideal. The step requiring a tool not developed in this volume is the existence of a prime ideal of $\mathbb{Z}[\zeta_p]$ over q with quotient a finite field of characteristic q : this fact of algebraic number theory, together with the conclusion of the argument, is **imported** from K. Ireland, M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Springer 1990, Ch. 6, which uses exactly the Gauss sums just constructed. The statement of the law itself is not imported: it is independently verifiable (and is verified numerically below); its first complete proof is due to Gauss, 1801, *Disquisitiones Arithmeticae*, §131–144.

Example 7.1: Gauss sums for $p = 5$ and $p = 7$, verified. For $p = 5$: generator $g = 2$, powers $2^0, 2^1, 2^2, 2^3 = 1, 2, 4, 3$; quadratic residues (even powers) $\{1, 4\}$, non-residues $\{2, 3\}$. Hence

$$\eta_0 = \zeta + \zeta^4 = 2 \cos 72^\circ = 0.618034, \quad \eta_1 = \zeta^2 + \zeta^3 = 2 \cos 144^\circ = -1.618034,$$

and $g_5 = \eta_0 - \eta_1 = 2.236068 = \sqrt{5}$ ($5 \equiv 1 \pmod{4}$, positive sign as predicted by Observation 7.6).

For $p = 7$: quadratic residues $\{1, 2, 4\}$ (from $1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 2 \pmod{7}$), non-residues $\{3, 5, 6\}$. Summing the corresponding seventh roots of unity numerically gives

$g_7 = 2.645751 i = i\sqrt{7} \checkmark (7 \equiv 3 \pmod{4})$.

Independent numerical check. The script `Sorgenti/Vol2/_verifiche/vol2_cap07_gauss_sums.py` computes g_p by direct summation from Definition 7.7 for all 45 odd primes $p < 200$, comparing with $\sqrt{p}$ or $i\sqrt{p}$ per Observation 7.6 (maximum error 9×10^{-15}), and independently verifies the law of quadratic reciprocity by computing both Legendre symbols via the exponent- $(p-1)/2$ criterion for all 552 pairs of distinct odd primes $p, q < 100$ (zero disagreements).

7.6. Characters of the discrete circle and the Fourier transform

Definition 7.8: Character of a finite abelian group. A **character** of the finite abelian group G is a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$, i.e. a function with $\chi(a+b) = \chi(a)\chi(b)$ (in additive notation on G). The set of characters, with pointwise product, is the **dual group** $\widehat{G}$.

Theorem 7.20: The characters of $\mathbb{Z}/n\mathbb{Z}$. The characters of $\mathbb{Z}/n\mathbb{Z}$ are exactly the n functions

$$\chi_k(m) = e^{2\pi i k m / n} = \zeta_n^{k m}, \quad k = 0, 1, \dots, n-1,$$

and $\widehat{\mathbb{Z}/n\mathbb{Z}} \cong \mathbb{Z}/n\mathbb{Z}$. Moreover every character takes values in μ_n , i.e. on the unit circle.

Proof. Step 1 — they are characters. $\chi_k(m+m') = \zeta_n^{k(m+m')} = \zeta_n^{k m} \zeta_n^{k m'} = \chi_k(m) \chi_k(m')$, and the definition does not depend on the representative of m since $\zeta_n^{k n} = 1$.

Step 2 — there are no others. Let χ be a character. Setting $w = \chi(1)$, the homomorphism property gives $\chi(m) = w^m$ for every $m \geq 0$. Moreover

$$w^n = \chi(n) = \chi(0) = 1,$$

so $w \in \mu_n$, i.e. $w = \zeta_n^k$ for a unique $k \in \{0, \dots, n-1\}$ (Theorem 3.16 of Chapter 3). Then $\chi = \chi_k$.

Step 3 — group structure. $\chi_k \chi_j = \chi_{k+j}$, so $k \mapsto \chi_k$ is an isomorphism $\mathbb{Z}/n\mathbb{Z} \rightarrow \widehat{\mathbb{Z}/n\mathbb{Z}}$.

Step 4 — values on the circle. Every value is a power of ζ_n , hence lies in $\mu_n \subset S^1$: the characters of a finite group take values on the unit circle. This is the structural reason why *the circle appears in every harmonic analysis*, even when the starting group is finite and has no apparent geometry. $\square$

Theorem 7.21: Orthogonality relations. For $k, j \in \{0, \dots, n-1\}$,

$$\frac{1}{n} \sum_{m=0}^{n-1} \chi_k(m) \overline{\chi_j(m)} = \begin{cases} 1 & \text{if } k = j, \\ 0 & \text{if } k \neq j. \end{cases}$$

Equivalently, $\sum_{m=0}^{n-1} \zeta_n^{(k-j)m} = n \delta_{kj}$.

Proof. Step 1 — reduction. Since $\overline{\chi_j(m)} = \zeta_n^{-j m}$, the sum is $\sum_m \zeta_n^{(k-j)m}$: it suffices to study $S(t) = \sum_{m=0}^{n-1} \zeta_n^{t m}$ for $t = k-j$.

Step 2 — case $t \equiv 0$. Every summand is 1 and $S = n$.

Step 3 — case $t \not\equiv 0$. Then $\zeta_n^t \neq 1$ and the sum is geometric:

$$S(t) = \frac{(\zeta_n^t)^n - 1}{\zeta_n^t - 1} = \frac{1 - 1}{\zeta_n^t - 1} = 0,$$

since $(\zeta_n^t)^n = (\zeta_n^n)^t = 1$.

Step 4 — consistency with Chapter 3. For $t = 1$ one recovers $\sum_m \zeta_n^m = 0$, i.e. Corollary 3.5: the sum of all n -th roots of unity vanishes because their centroid is the centre of the circle.

Step 5 — consistency with Chapter 4. This is literally the orthogonality relation on which the discrete Fourier transform rests. Chapter 4 used it; here we see its structural origin: it is the orthogonality of the characters of a cyclic group. $\square$

Corollary 7.5: The DFT is harmonic analysis on $\mathbb{Z}/n\mathbb{Z}$. For $f : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{C}$ set

$$\hat{f}(k) = \sum_{m=0}^{n-1} f(m) \overline{\chi_k(m)} = \sum_{m=0}^{n-1} f(m) e^{-2\pi i k m / n}.$$

Then the inversion formula holds,

$$f(m) = \frac{1}{n} \sum_{k=0}^{n-1} \hat{f}(k) e^{2\pi i k m / n},$$

and for the cyclic convolution $(f * g)(m) = \sum_j f(j)g(m-j)$ one has $\widehat{f * g} = \hat{f} \cdot \hat{g}$.

Proof. Inversion. Substituting and exchanging the (finite, hence legitimate) sums,

$$\frac{1}{n} \sum_k \hat{f}(k) \zeta_n^{km} = \frac{1}{n} \sum_k \sum_{m'} f(m') \zeta_n^{k(m-m')} = \sum_{m'} f(m') \left[\frac{1}{n} \sum_k \zeta_n^{k(m-m')} \right] = f(m),$$

by Theorem 7.21, which kills the bracket except for $m' = m$, where it equals 1.

Convolution. With the substitution $m = j + l$,

$$\widehat{f * g}(k) = \sum_m \sum_j f(j)g(m-j) \zeta_n^{-km} = \sum_j f(j) \zeta_n^{-kj} \sum_l g(l) \zeta_n^{-kl} = \hat{f}(k) \hat{g}(k).$$

Remark. Comparison with Theorem 4.18 of Chapter 4 shows this is the same theorem with $\mathbb{T}$ replaced by $\mathbb{Z}/n\mathbb{Z}$: convolution becomes a product because characters are eigenfunctions of translation. $\square$

Example 7.2: Fortescue revisited. In Chapter 5 the Fortescue transformation decomposed a triple of phasors into positive-, negative- and zero-sequence components by means of the matrix built from $a = e^{2\pi i/3}$. In the light of this chapter, that transformation is exactly the DFT on $\mathbb{Z}/3\mathbb{Z}$: the three symmetrical components are the three coefficients $\hat{f}(0), \hat{f}(1), \hat{f}(2)$ with respect to the three characters χ_0, χ_1, χ_2 of the cyclic group of order 3. The zero-sequence component is the coefficient of the trivial character $\chi_0 \equiv 1$, i.e. the mean; positive and negative sequence correspond to the two non-trivial characters, conjugate to each other.

It follows at once, with no electrical computation, that the transformation is invertible (inversion formula), that its inverse has the same shape with $\bar{a}$ in place of a , and that the factor $1/3$ appears exactly once. All of these are facts we had verified by hand in Chapter 5.

7.7. Cyclic codes

Definition 7.9: Linear code and cyclic code. A linear code of length n over $\mathbb{F}_q$ is a vector subspace $C \subseteq \mathbb{F}_q^n$. It is **cyclic** if it is closed under cyclic shift:

$$(c_0, c_1, \dots, c_{n-1}) \in C \implies (c_{n-1}, c_0, \dots, c_{n-2}) \in C.$$

One identifies the word $(c_0, \dots, c_{n-1})$ with the polynomial $c(x) = c_0 + c_1x + \dots + c_{n-1}x^{n-1}$ of the ring

$$R_n = \mathbb{F}_q[x]/(x^n - 1).$$

Remark 7.8: Why the ring is the circle. In the ring R_n one has $x^n = 1$: multiplying by x therefore rotates the coefficient vector by one position, bringing the last back to the front. The indeterminate x is the elementary rotation of the discrete circle with n points, and R_n is the group algebra of $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{F}_q$. The cyclicity condition then becomes a transparent algebraic condition, as the next theorem shows.

Theorem 7.22: Cyclic codes are the ideals of R_n . A subspace $C \subseteq R_n$ is a cyclic code if and only if it is an ideal of R_n . Moreover every ideal is principal: there is a unique monic polynomial g of least degree in C , the **generator polynomial**, such that

$$C = (g) = \{a(x)g(x) \bmod (x^n - 1) : a \in \mathbb{F}_q[x]\},$$

and $g \mid x^n - 1$. The dimension of C is $n - \deg g$.

Proof. **Step 1 — cyclicity $\iff$ closure under x .** By Observation 7.8 the cyclic shift is multiplication by x in R_n . A subspace closed under multiplication by x is closed under multiplication by every power of x , and by linearity by every polynomial: it is an ideal. Conversely an ideal is closed under multiplication by x , hence cyclic.

Step 2 — existence of g . Let $C \neq \{0\}$ and let g be a non-zero element of C of least degree, made monic by dividing by its leading coefficient (legitimate, $\mathbb{F}_q$ being a field).

Step 3 — $C = (g)$. Let $c \in C$. Divide in $\mathbb{F}_q[x]$: $c = qg + r$ with $\deg r < \deg g$. Since $c \in C$ and $qg \in C$ (ideal), $r = c - qg \in C$; by minimality of the degree of g , $r = 0$. Hence $c \in (g)$.

Step 4 — g divides $x^n - 1$. Divide $x^n - 1 = hg + r$ with $\deg r < \deg g$. Reducing modulo $x^n - 1$ gives $0 = hg + r$ in R_n , so $r = -hg \in C$; again by minimality $r = 0$, i.e. $g \mid x^n - 1$ in $\mathbb{F}_q[x]$.

Step 5 — dimension. Put $k = n - \deg g$. The polynomials $g, xg, \dots, x^{k-1}g$ have degrees $\deg g, \dots, n-1$, all distinct and $< n$, hence are linearly independent in R_n ; and every element of C is ag with $\deg a < k$ (otherwise one reduces modulo $x^n - 1$ using $g \mid x^n - 1$). They are therefore a basis: $\dim C = k$.

Step 6 — uniqueness. Two monic generators of least degree divide each other and are monic: they coincide.

Step 7 — practical consequence. The cyclic codes of length n over $\mathbb{F}_q$ are in bijection with the monic divisors of $x^n - 1$ in $\mathbb{F}_q[x]$. Since by Theorem 7.8 (valid over $\mathbb{F}_q$ as well when $\gcd(n, q) = 1$) that factorisation is governed by the cyclotomic polynomials, the theory of cyclic codes is literally the theory of dividing the circle. $\square$

Example 7.3: A binary cyclic code and the CRC computation. Over $\mathbb{F}_2$ one factors

$$x^7 - 1 = x^7 + 1 = (x + 1)(x^3 + x + 1)(x^3 + x^2 + 1),$$

(check by multiplying: $(x^3 + x + 1)(x^3 + x^2 + 1) = x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$, and multiplying by $x + 1$ gives $x^7 + 1$, since over $\mathbb{F}_2$ the intermediate terms cancel in pairs).

Choosing $g(x) = x^3 + x + 1$ gives the $[7, 4]$ cyclic Hamming code, of dimension $7 - 3 = 4$.
Systematic encoding. For a message $m(x)$ of degree < 4 one sets

$$c(x) = x^3 m(x) + r(x), \quad r(x) = x^3 m(x) \bmod g(x),$$

so that $g \mid c$ and the four message bits appear unaltered in the high positions (*systematic*), followed by three check bits.

Numerical example. Message 1101, i.e. $m(x) = x^3 + x^2 + 1$. Then $x^3 m(x) = x^6 + x^5 + x^3$, i.e. 1101000. Divide by $g = 1011$ by successive XORs:

$$1101000 \oplus 1011000 = 0110000,$$

$$0110000 \oplus 0101100 = 0011100,$$

$$0011100 \oplus 0010110 = 0001010,$$

$$0001010 \oplus 0001011 = 0000001.$$

The remainder is $r = 001$, i.e. $r(x) = 1$. The codeword is therefore

$$c = 1101\,001.$$

Check. Dividing 1101001 by 1011 by the same procedure gives remainder 0000000: the word belongs to the code. ✓

Use. This is the principle of the *cyclic redundancy check* (CRC) used in Ethernet, in serial protocols and in archive formats: the transmitter sends c , the receiver divides by g and accepts the message only if the remainder is zero.

Theorem 7.23: Burst error detection. Let C be a cyclic code over $\mathbb{F}_q$ with generator polynomial g of degree r and $g(0) \neq 0$. Then C detects every error **burst** of length $\leq r$, i.e. every error whose altered bits are all contained in r consecutive (cyclically) positions.

Proof. **Step 1 — shape of the burst.** A burst of length $L \leq r$ starting at position i is written

$$e(x) = x^i b(x) \quad \text{in } R_n, \quad b(0) \neq 0, \quad \deg b = L - 1 \leq r - 1.$$

Step 2 — the error is detected iff $g \nmid e$. The receiver computes the remainder of $c + e$ modulo g ; since $g \mid c$, that is the remainder of e . The error passes unnoticed if and only if $g \mid e$.

Step 3 — g does not divide x^i . Since $g(0) \neq 0$, the polynomial g does not have the root 0, hence $\gcd(g, x) = 1$ and a fortiori $\gcd(g, x^i) = 1$.

Step 4 — conclusion. If $g \mid x^i b$, then by Euclid's lemma (valid in $\mathbb{F}_q[x]$, a unique factorisation domain) and Step 3 we get $g \mid b$. But $\deg b \leq r - 1 < r = \deg g$ and $b \neq 0$: impossible. Hence $g \nmid e$ and the error is detected.

Step 5 — the hypothesis $g(0) \neq 0$ is necessary. If $g(0) = 0$, i.e. $x \mid g$, then g would divide $x^{\deg g}$, which is an undetected burst of length 1. On the other hand $g \mid x^n - 1$ implies $g(0) \neq 0$ automatically, since $x \nmid x^n - 1$: the hypothesis is therefore always satisfied by generator polynomials.

Step 6 — practical reach. The CRC-32 used in Ethernet has $r = 32$: it detects every error burst up to 32 consecutive bits, i.e. every electrical disturbance lasting less than 32 bit times. Since real channel disturbances are almost always *bursty* rather than isolated, this property — and not the minimum distance — is the main reason for the prevalence of cyclic codes.

Step 7 — check on the example. With $g = x^3 + x + 1$ ($r = 3$) of Example 7.3, every burst up to 3 bits is detected. A burst of 4 bits may however escape: the error $e = 1011000$ is exactly $x^3 g(x)$, hence divisible by g and indistinguishable from a valid word. ✓ □

Remark 7.9: The common thread. The three sections of this chapter are the same object seen three times:

<i>Discrete circle</i>	<i>Arithmetic</i>	<i>Codes</i>
<i>n</i> equally spaced points	$\mathbb{Z}/n\mathbb{Z}$	block length
elementary rotation	+1	multiplication by x
inscribed polygons	subgroups	subcodes
primitive roots	generators	primitive polynomials
harmonics	characters	syndromes
division of the circle	$\Phi_d(x)$	factors of $x^n - 1$

The left column is geometry, the middle one arithmetic, the right one telecommunications engineering. It is the same phenomenon already observed with Fortescue (Chapter 5) and with directional statistics (Chapter 6): there is only one mathematics.

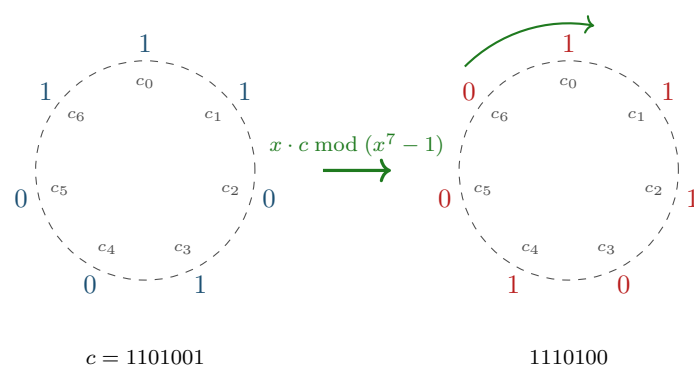


Figure 7.5: Why cyclic codes are called cyclic. Placing the seven bits of a codeword on the vertices of a 7-gon, a cyclic shift by one position is simply a rotation of the figure. Algebraically, identifying the word $c_0c_1 \dots c_6$ with the polynomial $c(x) = c_0 + c_1x + \dots + c_6x^6$ of the ring $\mathbb{F}_2[x]/(x^7 - 1)$, the shift is multiplication by x : the rotation group of the discrete circle acts on the code. A cyclic code is by definition a subspace closed under this action, and one proves that such a subspace is always the ideal generated by a divisor $g(x)$ of $x^n - 1$: the generator polynomial.

7.8. The BCH bound and Reed–Solomon codes

The Hamming code corrects one error; the CRC detects many but corrects none. To correct more errors one needs a design principle, and the principle is surprisingly geometric: *require that the discrete Fourier transform of the code vanish on an arc of consecutive frequencies*. The more consecutive frequencies one imposes, the further apart the codewords are forced to lie. This is the BCH bound, after Bose, Ray-Chaudhuri and Hocquenghem, and Reed–Solomon codes are the case in which the bound is attained exactly.

Definition 7.10: BCH code of designed distance. Let q be a prime power, n an integer with $\gcd(n, q) = 1$, and α an element of order n in an extension $\mathbb{F}_{q^m}$ of $\mathbb{F}_q$. Fixing an integer b and an integer $\delta \geq 2$, the *BCH code of designed distance δ* is the cyclic code of length n over $\mathbb{F}_q$ whose generator polynomial is the least common multiple of the minimal polynomials over $\mathbb{F}_q$ of the elements

$$\alpha^b, \alpha^{b+1}, \dots, \alpha^{b+\delta-2},$$

that is, the code whose words $c(x)$ satisfy $c(\alpha^{b+i}) = 0$ for $i = 0, 1, \dots, \delta - 2$.

Theorem 7.24: BCH bound. The minimum distance of a BCH code of designed distance δ satisfies

$$d \geq \delta$$

Proof. Step 1 — setting. The minimum distance of a linear code is the minimum weight of its non-zero words (Definition 7.9). Suppose for contradiction that there exists a non-zero word $c(x)$ of weight $w \leq \delta - 1$, and write it as

$$c(x) = \sum_{j=1}^w c_j x^{k_j}, \quad c_j \in \mathbb{F}_q \setminus \{0\}, \quad 0 \leq k_1 < k_2 < \dots < k_w \leq n-1.$$

Step 2 — the root conditions. By the definition of a BCH code,

$$0 = c(\alpha^{b+i}) = \sum_{j=1}^w c_j \alpha^{(b+i)k_j} = \sum_{j=1}^w (c_j \alpha^{bk_j}) (\alpha^{k_j})^i, \quad i = 0, 1, \dots, \delta-2.$$

Since $w \leq \delta - 1$, we may take the first w of these equations, namely $i = 0, \dots, w-1$: we obtain a homogeneous linear system of w equations in the w unknowns $\gamma_j := c_j \alpha^{bk_j}$.

Step 3 — the matrix is Vandermonde. The matrix of the system has (i, j) entry $(\alpha^{k_j})^i$, that is

$$V = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha^{k_1} & \alpha^{k_2} & \dots & \alpha^{k_w} \\ \vdots & \vdots & & \vdots \\ \alpha^{k_1(w-1)} & \alpha^{k_2(w-1)} & \dots & \alpha^{k_w(w-1)} \end{pmatrix}, \quad \det V = \prod_{1 \leq r < s \leq w} (\alpha^{k_s} - \alpha^{k_r}).$$

Step 4 — the determinant does not vanish. The exponents $k_1, \dots, k_w$ are distinct and lie between 0 and $n-1$; since α has order exactly n , the powers $\alpha^{k_1}, \dots, \alpha^{k_w}$ are pairwise distinct. Every factor $\alpha^{k_s} - \alpha^{k_r}$ is therefore non-zero, and in a field a product of non-zero elements is non-zero: $\det V \neq 0$.

Step 5 — conclusion. The homogeneous system thus has only the trivial solution, that is $\gamma_j = 0$ for every j ; but $\alpha^{bk_j} \neq 0$, hence $c_j = 0$ for every j , contradicting the assumption that c was non-zero. The contradiction proves that every non-zero word has weight $\geq \delta$, that is $d \geq \delta$. $\square$

Remark 7.10: The BCH bound is an uncertainty principle. By Corollary 7.5, the values $c(\alpha^i)$, $i = 0, \dots, n-1$, are exactly the discrete Fourier transform of the vector $(c_0, \dots, c_{n-1})$, computed on the characters of the discrete circle μ_n . Theorem 7.24 then reads: a non-zero vector cannot have both few non-zero samples and many consecutive vanishing Fourier coefficients. Writing w for the number of non-zero samples and z for the number of consecutive zeros of the transform, the theorem asserts

$$w \geq z + 1, \quad \text{that is} \quad z \leq w - 1.$$

It is the discrete, “consecutive-support” version of the uncertainty principle: concentration in time and concentration in frequency cannot both be extreme. Note that the proof uses no analysis, only the fact that the Vandermonde matrix of the roots of unity is invertible — the same fact that in Chapter 4 guaranteed the invertibility of the DFT.

Theorem 7.25: Singleton bound. For every code (even non-linear) of length n over an alphabet of q symbols, with M words and minimum distance d , one has $M \leq q^{n-d+1}$. In particular, for a linear code of dimension k ,

$$d \leq n - k + 1$$

Proof. **Step 1 — the projection.** Consider the map π sending each codeword to the $(n - d + 1)$ -tuple of its first $n - d + 1$ coordinates, that is deleting the last $d - 1$ coordinates.

Step 2 — π is injective. If two distinct words had the same image, they would agree in the first $n - d + 1$ positions and could differ only in the last $d - 1$: their Hamming distance would therefore be at most $d - 1 < d$, contrary to the definition of minimum distance.

Step 3 — counting. An injective map from a set of M elements to a set of q^{n-d+1} elements forces $M \leq q^{n-d+1}$. For a linear code $M = q^k$, and taking logarithms to base q gives $k \leq n - d + 1$, that is $d \leq n - k + 1$. $\square$

Definition 7.11: MDS code. A code attaining equality $d = n - k + 1$ in the Singleton bound is called *maximum distance separable* (MDS).

Theorem 7.26: Reed–Solomon codes. Let q be a prime power, $n = q - 1$, and α a primitive element of $\mathbb{F}_q$. The cyclic code of length n over $\mathbb{F}_q$ generated by

$$g(x) = \prod_{i=0}^{\delta-2} (x - \alpha^{b+i})$$

has dimension $k = n - \delta + 1$ and minimum distance exactly $d = \delta = n - k + 1$: it is therefore MDS. It corrects up to $t = \lfloor (\delta - 1)/2 \rfloor$ symbol errors.

Proof. **Step 1 — the degree of the generator.** Here $m = 1$: the elements α^{b+i} already belong to $\mathbb{F}_q$, so their minimal polynomial over $\mathbb{F}_q$ is the linear factor $x - \alpha^{b+i}$, and no least common multiple is needed. Hence $\deg g = \delta - 1$ exactly.

Step 2 — the dimension. By Theorem 7.22 a cyclic code generated by g has dimension $k = n - \deg g$, hence $k = n - \delta + 1$.

Step 3 — the distance from below. By construction the code is a BCH code of designed distance δ , hence $d \geq \delta$ by Theorem 7.24.

Step 4 — the distance from above. By Theorem 7.25, $d \leq n - k + 1 = n - (n - \delta + 1) + 1 = \delta$.

Step 5 — conclusion. From Steps 3 and 4, $d = \delta = n - k + 1$: the code is MDS. A code of minimum distance d corrects t errors if and only if $2t + 1 \leq d$ (the balls of radius t about the codewords are disjoint), whence $t = \lfloor (d - 1)/2 \rfloor = \lfloor (\delta - 1)/2 \rfloor$. $\square$

Example 7.4: The RS(7,5) code over $\mathbb{F}_8$, built by hand. Let $\mathbb{F}_8 = \mathbb{F}_2[x]/(x^3 + x + 1)$ and let α be the class of x , which is primitive: its powers, written in the basis $\{1, \alpha, \alpha^2\}$, are $\alpha^0 = 1$, $\alpha^1 = \alpha$, $\alpha^2 = \alpha^2$, $\alpha^3 = \alpha + 1$, $\alpha^4 = \alpha^2 + \alpha$, $\alpha^5 = \alpha^2 + \alpha + 1$, $\alpha^6 = \alpha^2 + 1$, and $\alpha^7 = 1$: the order is $7 = n$.

Step 1 — the generator for $\delta = 3$, $b = 1$.

$$g(x) = (x - \alpha)(x - \alpha^2) = x^2 + (\alpha + \alpha^2)x + \alpha^3 = x^2 + \alpha^4x + \alpha^3,$$

having used $\alpha + \alpha^2 = \alpha^4$ (in characteristic 2 signs are irrelevant).

Step 2 — checking the roots.

$$g(\alpha) = \alpha^2 + \alpha^5 + \alpha^3 = (\alpha^2) + (\alpha^2 + \alpha + 1) + (\alpha + 1) = 0 \quad \checkmark$$

$$g(\alpha^2) = \alpha^4 + \alpha^6 + \alpha^3 = (\alpha^2 + \alpha) + (\alpha^2 + 1) + (\alpha + 1) = 0 \quad \checkmark$$

Step 3 — parameters. $\deg g = 2$, hence $k = 7 - 2 = 5$ and, by Theorem 7.26, $d = 3 = 7 - 5 + 1$: the code corrects $t = 1$ symbol error, that is up to three arbitrary consecutive bits provided they fall inside a single symbol.

Step 4 — a minimum-weight word. Taking the message $m(x) = x^4$ one obtains

$$c(x) = m(x)g(x) = x^6 + \alpha^4 x^5 + \alpha^3 x^4,$$

of weight $3 = d$: the BCH bound is attained, and the estimate of Theorem 7.24 cannot be improved.

Step 5 — the case $\delta = 5$. With $b = 1$ and $\delta = 5$,

$$g(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^3)(x - \alpha^4) = x^4 + \alpha^3 x^3 + x^2 + \alpha x + \alpha^3,$$

as one checks by multiplying $(x^2 + \alpha^4 x + \alpha^3)$ by $(x^2 + \alpha^6 x + 1)$ and reducing with $\alpha^7 = 1$. This gives RS(7, 3), with $d = 5$ and $t = 2$.

Example 7.5: The binary BCH code (15, 7). Let $\mathbb{F}_{16} = \mathbb{F}_2[x]/(x^4 + x + 1)$ and let α be the class of x , primitive of order 15. For $\delta = 5$ and $b = 1$ the roots $\alpha, \alpha^2, \alpha^3, \alpha^4$ are required. But the generator polynomial must have coefficients in $\mathbb{F}_2$, and hence its set of roots is closed under the Frobenius endomorphism $y \mapsto y^2$ (Step 4 of the proof of Theorem 7.11). The cyclotomic cosets are

$$C_1 = \{1, 2, 4, 8\}, \quad C_3 = \{3, 6, 12, 9\},$$

and the corresponding minimal polynomials are $x^4 + x + 1$ and $x^4 + x^3 + x^2 + x + 1$. Therefore

$$g(x) = (x^4 + x + 1)(x^4 + x^3 + x^2 + x + 1) = x^8 + x^7 + x^6 + x^4 + 1,$$

of degree 8: the code is (15, 7) with $d \geq 5$, and corrects $t = 2$ errors. Note that $d \leq n - k + 1 = 9$ by Singleton, but the true value is exactly 5: unlike Reed–Solomon codes, binary BCH codes are not MDS, because closure under Frobenius forces the addition of unrequested roots, and every extra root costs one dimension.

Remark 7.11: Where they are used. Reed–Solomon codes are the industrial standard for error correction on bursty channels, because a burst error of length up to m bits touches at most two m -bit symbols and hence consumes at most two of the t available corrections. The code RS(255, 223) over $\mathbb{F}_{256}$, with $d = 33$ and $t = 16$, is the one adopted by the CCSDS recommendations for space telecommunications; the compact disc uses two shortened RS codes over $\mathbb{F}_{256}$, RS(32, 28) and RS(28, 24), cross-interleaved (the CIRC scheme) precisely in order to turn a scratch — which is a very long burst — into many short errors spread over different words. It is the same principle as the interleaving discussed in connection with the CRC: the cyclic structure is used to scatter the error around the circle until it becomes correctable.

7.9. Numerical exercises

All exercises are solved in Section 7.11.

Exercise 7.1: N1 — Euler’s totient. Compute $\varphi(360)$ and $\varphi(1001)$ with both formulas of Corollary 7.1, and verify Gauss’s identity for $n = 36$.

Exercise 7.2: N2 — Orders and primitive roots. Determine the order of 2, of 3 and of 10 modulo 11, say which of them are primitive roots, and count how many there are in total.

Exercise 7.3: N3 — Large powers. Compute the last digit of 7^{222} , the remainder of 3^{100} modulo 7 and the remainder of 2^{1000} modulo 9.

Exercise 7.4: N4 — A system of congruences. Solve

$$x \equiv 1 \pmod{4}, \quad x \equiv 2 \pmod{9}, \quad x \equiv 3 \pmod{25},$$

and say how many solutions there are modulo 900.

Exercise 7.5: N5 — A cyclotomic polynomial. Compute $\Phi_{15}(x)$ from the factorisation of $x^{15} - 1$, check its degree and verify that $\Phi_{15}(1) = 1$.

Exercise 7.6: N6 — Constructible polygons. Say which of the regular polygons with $n = 7, 9, 15, 17, 20, 34, 51, 60$ sides are constructible with straightedge and compass, justifying each case by computing $\varphi(n)$.

Exercise 7.7: N7 — Discrete transform on four points. Compute the DFT of the sequence $f = (1, 2, 3, 4)$ on $\mathbb{Z}/4\mathbb{Z}$, verify Parseval’s identity and reconstruct f by the inversion formula.

Exercise 7.8: N8 — Computing a CRC. With generator polynomial $g(x) = x^4 + x + 1$ (i.e. 10011), compute the four check bits of the message 11010110 and verify that the transmitted word is divisible by g .

Exercise 7.9: N9 — How many cyclic codes. Determine how many binary cyclic codes of length 7 exist, list their dimensions and identify the Hamming code.

Exercise 7.10: N10 — Subgroups of the 36-point circle. For μ_{36} determine the number of generators, the number of subgroups, and verify Gauss’s identity by listing the elements by order.

7.10. Symbolic exercises

Exercise 7.11: S1 — Möbius inversion. Define the Möbius function by $\mu(1) = 1$, $\mu(n) = (-1)^r$ if n is a product of r distinct primes, $\mu(n) = 0$ if n has a square factor. Prove that

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1 \\ 0 & n > 1 \end{cases},$$

deduce the Möbius inversion formula and apply it to Theorem 7.4 to obtain $\varphi(n) = \sum_{d|n} \mu(d) n/d$.

Exercise 7.12: S2 — Every finite subgroup of $K^\times$ is cyclic. Let K be a field and $G \leq K^\times$ a finite subgroup of order n . Prove that G is cyclic. Deduce that $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for every prime p , i.e. that primitive roots modulo p exist.

Exercise 7.13: S3 — Wilson’s theorem. Prove that for every prime p

$$(p-1)! \equiv -1 \pmod{p},$$

and that the converse holds: if $(n-1)! \equiv -1 \pmod{n}$ with $n > 1$, then n is prime.

Exercise 7.14: S4 — Möbius formula for cyclotomics. Prove that

$$\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)},$$

and use it to recompute Φ_{12} .

Exercise 7.15: S5 — Sum of the primitive roots. Prove that the sum of the primitive n -th roots of unity equals $\mu(n)$, and relate this to the coefficient of degree $\varphi(n) - 1$ of Φ_n .

Exercise 7.16: S6 — Fermat by counting: necklaces. Prove Fermat’s little theorem by counting the *necklaces* of p beads chosen from a colours, up to rotation. (Hint: let $\mathbb{Z}/p\mathbb{Z}$ act on the set of a^p sequences.)

Exercise 7.17: S7 — The Fourier matrix has order four. Let F be the $n \times n$ matrix with $F_{jk} = \zeta_n^{jk}$. Prove that $F \overline{F}^\top = nI$, that $F^2 = nP$ with P the permutation $m \mapsto -m$, and deduce $F^4 = n^2I$.

Exercise 7.18: S8 — Parity and the factor $x+1$. Let C be a binary cyclic code with generator g . Prove that all words of C have even weight if and only if $(x+1) \mid g(x)$.

Exercise 7.19: S9 — Detection of double errors. Let C be cyclic with generator g , $g(0) \neq 0$. Prove that C detects all double errors $e(x) = x^i + x^j$ ($i \neq j$) if and only if the order of x in the ring $\mathbb{F}_2[x]/(g)$ is at least n . Deduce why *primitive* polynomials are chosen.

Exercise 7.20: S10 — Arithmetic characterisation of constructibility. Prove that $\varphi(n)$ is a power of 2 if and only if $n = 2^a p_1 \cdots p_r$ with distinct Fermat primes p_i , thus completing the arithmetic part of Theorem 7.13.

7.11. Solutions to the numerical exercises

Solution. N1. *Step 1 — factorisations.* $360 = 2^3 \cdot 3^2 \cdot 5$ and $1001 = 7 \cdot 11 \cdot 13$.

Step 2 — first formula.

$$\varphi(360) = 2^2(2-1) \cdot 3^1(3-1) \cdot 5^0(5-1) = 4 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \cdot 4 = 96,$$

$$\varphi(1001) = (7-1)(11-1)(13-1) = 6 \cdot 10 \cdot 12 = 720.$$

Step 3 — second formula.

$$\varphi(360) = 360 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5} = 96 \quad \checkmark$$

$$\varphi(1001) = 1001 \cdot \frac{6}{7} \cdot \frac{10}{11} \cdot \frac{12}{13} = 720 \quad \checkmark$$

Step 4 — Gauss's identity for $n = 36$. The divisors of $36 = 2^2 \cdot 3^2$ are 1, 2, 3, 4, 6, 9, 12, 18, 36, and

$$\varphi: \quad 1, 1, 2, 2, 2, 6, 4, 6, 12.$$

Sum: $1 + 1 + 2 + 2 + 2 + 6 + 4 + 6 + 12 = 36$. $\checkmark$

Step 5 — plausibility check. $\varphi(n)/n$ measures the “density” of numbers coprime to n : it is $96/360 = 0.267$ for 360 (many small prime factors) and $720/1001 = 0.719$ for 1001 (three large prime factors). The smaller the primes dividing n , the rarer the coprimes. $\checkmark$

Solution. N2. *Step 1 — the group.* $(\mathbb{Z}/11\mathbb{Z})^\times$ has order $\varphi(11) = 10$; by Theorem 7.5 the possible orders are the divisors of 10: 1, 2, 5, 10.

Step 2 — order of 2. Successive powers modulo 11:

$$2, 4, 8, 5, 10, 9, 7, 3, 6, 1.$$

The first 1 appears at the tenth step: the order is 10, so 2 is a **primitive root**. (Note the first ten powers exhaust all non-zero residues, as they must.)

Step 3 — order of 3.

$$3, 9, 27 \equiv 5, 15 \equiv 4, 12 \equiv 1.$$

Order 5: not a primitive root.

Step 4 — order of 10. $10 \equiv -1$, hence $10^2 \equiv 1$: order 2. Not a primitive root.

Step 5 — how many primitive roots. The group is cyclic of order 10 (Exercise S2), so its generators number $\varphi(10) = 4$. Since 2 is a generator, the others are 2^k with $\gcd(k, 10) = 1$, i.e. $k = 1, 3, 7, 9$:

$$2^1 = 2, \quad 2^3 = 8, \quad 2^7 = 7, \quad 2^9 = 6.$$

The four primitive roots modulo 11 are therefore 2, 6, 7, 8. $\checkmark$

Cross-check. $3 = 2^8$ and $\gcd(8, 10) = 2$, so the order of 3 is $10/2 = 5$, in agreement with Step 3. $\checkmark$

Solution. N3. (a) *Last digit of 7^{222} .* This is the remainder modulo 10. Since $\gcd(7, 10) = 1$ and $\varphi(10) = 4$, Theorem 7.6 gives $7^4 \equiv 1$. Now $222 = 4 \cdot 55 + 2$, hence

$$7^{222} \equiv 7^2 = 49 \equiv 9 \pmod{10}.$$

The last digit is 9.

(b) 3^{100} modulo 7. $\varphi(7) = 6$ and $100 = 6 \cdot 16 + 4$, hence

$$3^{100} \equiv 3^4 = 81 = 7 \cdot 11 + 4 \equiv 4 \pmod{7}.$$

(c) 2^{1000} modulo 9. $\varphi(9) = 6$ and $1000 = 6 \cdot 166 + 4$, hence

$$2^{1000} \equiv 2^4 = 16 \equiv 7 \pmod{9}.$$

Step 4 — independent checks. For (b): the order of 3 modulo 7 is 6 (3, 2, 6, 4, 5, 1), and $100 \equiv 4 \pmod{6}$ gives the fourth term, 4 ✓. For (c): the order of 2 modulo 9 is 6 (2, 4, 8, 7, 5, 1), and the fourth term is 7 ✓.

Common error. Reducing the exponent modulo n instead of modulo $\varphi(n)$: for (c) one would get $2^{1000 \bmod 9} = 2^1 = 2$, which is wrong. The exponent lives in the group of exponents, of order $\varphi(n)$, not in $\mathbb{Z}/n\mathbb{Z}$.

Solution. N4. *Step 1 — checking the hypotheses.* 4, 9, 25 are pairwise coprime (powers of distinct primes) and $N = 4 \cdot 9 \cdot 25 = 900$: Theorem 7.7 guarantees exactly one solution modulo 900.

Step 2 — combining the first two congruences. From $x \equiv 3 \pmod{25}$ write $x = 25k + 3$ and impose $x \equiv 2 \pmod{9}$:

$$25k + 3 \equiv 2 \implies 7k \equiv -1 \equiv 8 \pmod{9}.$$

The inverse of 7 modulo 9 is 4 (since $7 \cdot 4 = 28 \equiv 1$), so $k \equiv 4 \cdot 8 = 32 \equiv 5 \pmod{9}$.

Step 3 — partial solution. With $k = 5$: $x = 128$. Check: $128 = 9 \cdot 14 + 2$ ✓, $128 = 25 \cdot 5 + 3$ ✓. So $x \equiv 128 \pmod{225}$.

Step 4 — third congruence. Write $x = 225m + 128$ and impose $x \equiv 1 \pmod{4}$. Since $225 \equiv 1$ and $128 \equiv 0 \pmod{4}$:

$$m \equiv 1 \pmod{4}.$$

With $m = 1$: $x = 353$.

Step 5 — final check.

$$353 = 4 \cdot 88 + 1 \text{ ✓}, \quad 353 = 9 \cdot 39 + 2 \text{ ✓}, \quad 353 = 25 \cdot 14 + 3 \text{ ✓}.$$

The solution is $x \equiv 353 \pmod{900}$: unique modulo 900, the integer solutions being $353 + 900t$.

Step 6 — check with the explicit formula. With $N_1 = 225$, $N_2 = 100$, $N_3 = 36$: $225 \equiv 1 \pmod{4} \Rightarrow M_1 = 1$; $100 \equiv 1 \pmod{9} \Rightarrow M_2 = 1$; $36 \equiv 11 \pmod{25}$ and $11 \cdot 16 = 176 = 175 + 1 \equiv 1 \pmod{25} \Rightarrow M_3 = 16$. Then

$$x = 1 \cdot 225 \cdot 1 + 2 \cdot 100 \cdot 1 + 3 \cdot 36 \cdot 16 = 225 + 200 + 1728 = 2153,$$

and $2153 = 2 \cdot 900 + 353 \equiv 353 \pmod{900}$. ✓

Solution. N5. *Step 1 — divisors.* The divisors of 15 are 1, 3, 5, 15, so by Theorem 7.8

$$x^{15} - 1 = \Phi_1 \Phi_3 \Phi_5 \Phi_{15}.$$

Step 2 — known factors.

$$\Phi_1 = x - 1, \quad \Phi_3 = x^2 + x + 1, \quad \Phi_5 = x^4 + x^3 + x^2 + x + 1.$$

Note that $\Phi_1 \Phi_3 = x^3 - 1$ and $\Phi_1 \Phi_5 = x^5 - 1$.

Step 3 — division.

$$\Phi_{15} = \frac{(x^{15} - 1)(x - 1)}{(x^3 - 1)(x^5 - 1)},$$

because $\Phi_3\Phi_5 = \frac{(x^3-1)(x^5-1)}{(x-1)^2}$. Carrying out the division gives

$$\Phi_{15}(x) = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1.$$

Step 4 — degree check. $\deg \Phi_{15} = 8 = \varphi(15) = \varphi(3)\varphi(5) = 2 \cdot 4$. ✓

Step 5 — numerical check. At $x = 2$,

$$\Phi_{15}(2) = 256 - 128 + 32 - 16 + 8 - 2 + 1 = 151,$$

and

$$\frac{(2^{15} - 1)(2 - 1)}{(2^3 - 1)(2^5 - 1)} = \frac{32767}{7 \cdot 31} = \frac{32767}{217} = 151. \quad \checkmark$$

Step 6 — value at 1. $\Phi_{15}(1) = 1 - 1 + 1 - 1 + 1 - 1 + 1 = 1$. This agrees with the general fact that $\Phi_n(1) = p$ if n is a power of the prime p , and $\Phi_n(1) = 1$ if n has at least two distinct prime factors (here $15 = 3 \cdot 5$). ✓

Solution. N6. *Step 1 — criterion.* By Theorem 7.13 the n -gon is constructible if and only if $\varphi(n)$ is a power of 2 (equivalently: $n = 2^a \times$ distinct Fermat primes).

Step 2 — case by case.

n	factorisation	$\varphi(n)$	power of 2?	constructible
7	7	6	no	no
9	3^2	6	no	no
15	$3 \cdot 5$	8	yes	yes
17	17	16	yes	yes
20	$2^2 \cdot 5$	8	yes	yes
34	$2 \cdot 17$	16	yes	yes
51	$3 \cdot 17$	32	yes	yes
60	$2^2 \cdot 3 \cdot 5$	16	yes	yes

Step 3 — checks. $\varphi(51) = \varphi(3)\varphi(17) = 2 \cdot 16 = 32$; $\varphi(60) = \varphi(4)\varphi(3)\varphi(5) = 2 \cdot 2 \cdot 4 = 16$. ✓

Step 4 — reading. The two negative cases have a common explanation: 7 is not a Fermat prime ($7 - 1 = 6$ is not a power of 2), and $9 = 3^2$ contains a *repeated* Fermat prime, which is forbidden. Constructing the enneagon would amount to trisecting the 120° angle.

Step 5 — remark. From 15 and 17 constructible it follows that so is $255 = 15 \cdot 17$ and, more generally, the polygon with $2^a \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$ sides, whose largest odd value is $4\,294\,967\,295 = 2^{32} - 1$. No others are known, because no further Fermat primes are known.

Solution. N7. *Step 1 — the matrix.* For $n = 4$, $\zeta_4 = i$ and $\zeta_4^{-1} = -i$. The DFT is

$$\hat{f}(k) = \sum_{m=0}^3 f(m) (-i)^{km}.$$

Step 2 — computation.

$$\hat{f}(0) = 1 + 2 + 3 + 4 = 10,$$

$$\hat{f}(1) = 1 + 2(-i) + 3(-1) + 4(i) = (1 - 3) + (-2 + 4)i = -2 + 2i,$$

$$\hat{f}(2) = 1 - 2 + 3 - 4 = -2,$$

$$\hat{f}(3) = 1 + 2(i) + 3(-1) + 4(-i) = -2 - 2i.$$

Note $\hat{f}(3) = \overline{\hat{f}(1)}$, as it must be for real f .

Step 3 — Parseval's identity.

$$\sum_m |f(m)|^2 = 1 + 4 + 9 + 16 = 30,$$

$$\frac{1}{4} \sum_k |\hat{f}(k)|^2 = \frac{100 + 8 + 4 + 8}{4} = \frac{120}{4} = 30. \checkmark$$

Step 4 — inversion. By Corollary 7.5,

$$f(1) = \frac{1}{4} [10 + (-2 + 2i)(i) + (-2)(-1) + (-2 - 2i)(-i)].$$

Now $(-2 + 2i)i = -2i - 2$ and $(-2 - 2i)(-i) = 2i - 2$, so the bracket is

$$10 + (-2 - 2i) + 2 + (-2 + 2i) = 8,$$

and $f(1) = 8/4 = 2. \checkmark$

Step 5 — reading. $\hat{f}(0) = 10$ is the sum, four times the mean: the “DC component”. $\hat{f}(2) = -2$ is the component at the Nyquist frequency, i.e. the alternation $+, -, +, -$. Components 1 and 3 are conjugate and represent the same real oscillation. It is the same pattern as in Chapter 4.

Solution. N8. *Step 1 — setup.* $g = 10011$ has degree 4: append four zeros to the message 11010110, obtaining the dividend

$$D = 1101\ 0110\ 0000.$$

Step 2 — division by XOR. Aligning g with the leftmost 1:

$$\begin{aligned} 110101100000 \oplus 100110000000 &= 010011100000, \\ 010011100000 \oplus 010011000000 &= 000000100000, \\ 000000100000 \oplus 000000100110 &= 000000000110. \end{aligned}$$

No further alignment is possible (the remaining 1 is too far right), so the remainder is the last four bits:

$$r = 0110.$$

Step 3 — transmitted word.

$$c = 1101\ 0110\ \underbrace{0110}_{\text{CRC}} = 110101100110.$$

Step 4 — check.

$$\begin{aligned} 110101100110 \oplus 100110000000 &= 010011100110, \\ 010011100110 \oplus 010011000000 &= 000000100110, \\ 000000100110 \oplus 000000100110 &= 000000000000. \checkmark \end{aligned}$$

The remainder is zero: the word belongs to the code.

Step 5 — detection capability. By Theorem 7.23, with $\deg g = 4$ every error burst up to 4 consecutive bits is detected. Moreover $g = x^4 + x + 1$ is primitive over $\mathbb{F}_2$ (the order of x modulo g is $2^4 - 1 = 15$), so the code also detects all double errors provided the block length does not exceed 15 (Exercise S9). Here the block is 12 bits: the condition is met. $\checkmark$

Solution. N9. *Step 1 — factorisation.* Over $\mathbb{F}_2$,

$$x^7 - 1 = (x + 1)(x^3 + x + 1)(x^3 + x^2 + 1),$$

a product of three distinct irreducible factors (the two cubics are irreducible because they have no roots in $\mathbb{F}_2$: $g(0) = g(1) = 1$).

Step 2 — counting. By Theorem 7.22 the cyclic codes correspond to the monic divisors of $x^7 - 1$. With three distinct irreducible factors there are $2^3 = 8$ divisors.

Step 3 — the list.

$g(x)$	$\deg g$	$\dim = 7 - \deg g$	description
1	0	7	all of $\mathbb{F}_2^7$
$x + 1$	1	6	parity code
$x^3 + x + 1$	3	4	Hamming [7, 4]
$x^3 + x^2 + 1$	3	4	equivalent Hamming
$(x + 1)(x^3 + x + 1)$	4	3	dual of the previous
$(x + 1)(x^3 + x^2 + 1)$	4	3	dual of the other
$(x^3 + x + 1)(x^3 + x^2 + 1)$	6	1	repetition code
$x^7 - 1$	7	0	zero code

Step 4 — dimension check. The degrees of the three factors sum to $1 + 3 + 3 = 7 = \deg(x^7 - 1)$ ✓, and the listed dimensions cover 7, 6, 4, 4, 3, 3, 1, 0.

Step 5 — the Hamming code. With $g = x^3 + x + 1$ one gets the [7, 4] code of minimum distance 3, able to correct one error. The repetition code, generated by the product of the two cubics $= x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$, contains only 0000000 and 1111111: dimension 1, distance 7. ✓

Solution. N10. *Step 1 — generators.* $36 = 2^2 \cdot 3^2$, so $\varphi(36) = 2 \cdot 6 = 12$: there are twelve primitive 36-th roots of unity, namely ζ_{36}^k with

$$k = 1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35.$$

Step 2 — subgroups. By Theorem 7.3 there is exactly one subgroup for each divisor of 36; the divisors are 1, 2, 3, 4, 6, 9, 12, 18, 36: nine subgroups.

Step 3 — elements by order.

order d	1	2	3	4	6	9	12	18	36
$\varphi(d)$	1	1	2	2	2	6	4	6	12

Step 4 — Gauss's identity.

$$1 + 1 + 2 + 2 + 2 + 6 + 4 + 6 + 12 = 36. \quad \checkmark$$

Step 5 — geometric reading. The nine subgroups are the regular polygons inscribed in the 36-gon: the point $\{1\}$, the segment (order 2), the triangle, the square, the hexagon, the enneagon, the dodecagon, the 18-gon and the 36-gon itself. There are no subgroups of order 5, 7, 8: one cannot inscribe regular pentagons, heptagons or octagons in the vertices of the 36-gon, because $5, 7, 8 \nmid 36$. ✓

7.12. Solutions to the symbolic exercises

Solution. S1. *Step 1 — the sum of μ over divisors.* Let $n > 1$ with factorisation $n = p_1^{a_1} \cdots p_r^{a_r}$. The divisors d with $\mu(d) \neq 0$ are the products of subsets of $\{p_1, \dots, p_r\}$; a subset of j elements gives $\mu = (-1)^j$, and there are $\binom{r}{j}$ of them. Hence

$$\sum_{d|n} \mu(d) = \sum_{j=0}^r \binom{r}{j} (-1)^j = (1-1)^r = 0,$$

by the binomial theorem. For $n = 1$ the sum is $\mu(1) = 1$.

Step 2 — Möbius inversion. Let f, F be arithmetic functions with $F(n) = \sum_{d|n} f(d)$ for every n . Then

$$\sum_{d|n} \mu(d) F\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d) \sum_{e|n/d} f(e) = \sum_{e|n} f(e) \sum_{d|n/e} \mu(d) = f(n),$$

where the order of summation was exchanged (the pair (d, e) ranges over factorisations $de | n$) and Step 1 was used, killing the inner sum except when $n/e = 1$.

Step 3 — application. Theorem 7.4 says $n = \sum_{d|n} \varphi(d)$, i.e. $F(n) = n$ with $f = \varphi$. By Step 2,

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}.$$

Step 4 — check for $n = 12$. The divisors with $\mu \neq 0$ are 1, 2, 3, 6 with $\mu = 1, -1, -1, 1$:

$$\varphi(12) = 12 - 6 - 4 + 2 = 4. \checkmark$$

Step 5 — consistency with the product formula. Collecting terms,

$$\sum_{d|n} \mu(d) \frac{n}{d} = n \prod_{p|n} \left(1 - \frac{1}{p}\right),$$

because expanding the product gives exactly the terms $\mu(d)/d$. One recovers Corollary 7.1. $\checkmark$

Solution. S2. *Step 1 — counting roots.* Let $G \leq K^\times$ have order n . For every d , the equation $x^d = 1$ has at most d solutions in K , since $x^d - 1$ has degree d and a non-zero polynomial over a field has at most as many zeros as its degree.

Step 2 — $\psi(d)$ is 0 or $\varphi(d)$. Let $\psi(d)$ be the number of elements of G of order exactly d (necessarily $d | n$ by Theorem 7.5). If $\psi(d) > 0$, there is $a \in G$ of order d ; then the d elements of $\langle a \rangle$ all satisfy $x^d = 1$, and by Step 1 they are *all* the solutions. Every element of order d satisfies $x^d = 1$, hence lies in $\langle a \rangle$; and in a cyclic group of order d the elements of order d number $\varphi(d)$ (Theorem 7.2(iii)). So $\psi(d) = \varphi(d)$.

Step 3 — the global count. Every element has an order, so

$$n = \sum_{d|n} \psi(d) \leq \sum_{d|n} \varphi(d) = n,$$

the last equality by Theorem 7.4. The inequality is therefore an equality, and since $\psi(d) \leq \varphi(d)$ term by term, we must have $\psi(d) = \varphi(d)$ for *every* $d | n$.

Step 4 — conclusion. In particular $\psi(n) = \varphi(n) \geq 1$: there is an element of order n , i.e. a generator. G is cyclic.

Step 5 — primitive roots. Applying this to $K = \mathbb{Z}/p\mathbb{Z}$ (a field since p is prime) and $G = (\mathbb{Z}/p\mathbb{Z})^\times$ of order $p-1$: the group is cyclic, so primitive roots modulo p exist, and there

are exactly $\varphi(p-1)$ of them.

Step 6 — check. For $p = 11$: $\varphi(10) = 4$ primitive roots, and indeed Exercise N2 found four $(2, 6, 7, 8)$. ✓

Step 7 — warning. The hypothesis that K be a *field* is essential. In $\mathbb{Z}/8\mathbb{Z}$ the unit group $\{1, 3, 5, 7\}$ has order 4 but is not cyclic: every element has order ≤ 2 . Indeed $\mathbb{Z}/8\mathbb{Z}$ is not a field, and $x^2 - 1$ has four roots there.

Solution. S3. *Step 1 — pairing inverses.* In $(\mathbb{Z}/p\mathbb{Z})^\times$ every element has a unique inverse. In the product $(p-1)! = \prod_{a=1}^{p-1} a$ we may pair each a with a^{-1} : each pair contributes 1.

Step 2 — the self-inverse elements. Left out of the pairing are the a with $a = a^{-1}$, i.e. $a^2 \equiv 1$, i.e. $p \mid (a-1)(a+1)$. Since p is prime, $p \mid a-1$ or $p \mid a+1$: so $a \equiv 1$ or $a \equiv -1$.

Step 3 — conclusion.

$$(p-1)! \equiv 1 \cdot (-1) \cdot \underbrace{1 \cdots 1}_{\text{pairs}} = -1 \pmod{p}.$$

(For $p = 2$ one checks directly: $1! \equiv 1 \equiv -1 \pmod{2}$.)

Step 4 — check. $p = 7$: $6! = 720 = 7 \cdot 102 + 6 \equiv 6 \equiv -1 \pmod{7}$. ✓

Step 5 — the converse. Let $n > 1$ be composite with $(n-1)! \equiv -1 \pmod{n}$. If $n = ab$ with $1 < a < n$, then a appears among the factors of $(n-1)!$, so $a \mid (n-1)!$; but $a \mid n$ and $n \mid (n-1)! + 1$, so $a \mid (n-1)! + 1$: subtracting, $a \mid 1$, absurd. Hence n is prime.

Step 6 — reading on the circle. The pairing of Step 1 is an *involution* on the discrete circle of invertible residues, with exactly two fixed points, $+1$ and -1 : the two real points of μ_{p-1} , so to speak. Wilson's theorem is the count of the fixed points of that involution.

Step 7 — practical warning. Wilson's criterion is an *exact* primality test but an unusable one: computing $(n-1)!$ requires $n-1$ multiplications, far more than trying all divisors up to $\sqrt{n}$.

Solution. S4. *Step 1 — multiplicative form of inversion.* Let f, F take values in a multiplicative abelian group with $F(n) = \prod_{d|n} f(d)$. Then

$$f(n) = \prod_{d|n} F\left(\frac{n}{d}\right)^{\mu(d)}.$$

The proof is that of Step 2 of Exercise S1, with sums replaced by products and the coefficients $\mu(d)$ becoming exponents.

Step 2 — application. Theorem 7.8 states $x^n - 1 = \prod_{d|n} \Phi_d(x)$, i.e. $F(n) = x^n - 1$ with $f = \Phi$. Hence

$$\Phi_n(x) = \prod_{d|n} (x^{n/d} - 1)^{\mu(d)} = \prod_{d|n} (x^d - 1)^{\mu(n/d)},$$

the two expressions being equivalent under $d \leftrightarrow n/d$.

Step 3 — recomputing Φ_{12} . The divisors d of 12 with $\mu(12/d) \neq 0$ are those for which $12/d$ is squarefree: $12/d \in \{1, 2, 3, 6\}$, i.e. $d \in \{12, 6, 4, 2\}$, with

$$\mu(1) = 1, \quad \mu(2) = -1, \quad \mu(3) = -1, \quad \mu(6) = 1.$$

Hence

$$\Phi_{12}(x) = \frac{(x^{12} - 1)(x^2 - 1)}{(x^6 - 1)(x^4 - 1)}.$$

Step 4 — simplification.

$$\frac{x^{12} - 1}{x^6 - 1} = x^6 + 1, \quad \frac{x^2 - 1}{x^4 - 1} = \frac{1}{x^2 + 1},$$

so

$$\Phi_{12}(x) = \frac{x^6 + 1}{x^2 + 1} = x^4 - x^2 + 1,$$

since $(x^2 + 1)(x^4 - x^2 + 1) = x^6 + 1$. ✓ This agrees with Step 5 of the proof of Theorem 7.9.

Step 5 — remark. The formula makes it evident why the coefficients are integers: Φ_n is a product and quotient of monic integer polynomials, and the quotient is exact.

Solution. S5. *Step 1 — definition.* Let

$$c(n) = \sum_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} \zeta_n^k$$

be the sum of the primitive n -th roots of unity (up to sign, the coefficient of $x^{\varphi(n)-1}$ in Φ_n , by the relations between roots and coefficients).

Step 2 — summing over divisors. Grouping all n -th roots by their order, as in Step 1 of the proof of Theorem 7.4,

$$\sum_{d|n} c(d) = \sum_{z \in \mu_n} z = \begin{cases} 1 & n = 1, \\ 0 & n > 1, \end{cases}$$

the last equality by Corollary 3.5 of Chapter 3.

Step 3 — comparison with μ . By Step 1 of Exercise S1, the function μ satisfies the same relation: $\sum_{d|n} \mu(d) = [n = 1]$.

Step 4 — conclusion. Two arithmetic functions with the same divisor sums coincide: apply Möbius inversion (Step 2 of Exercise S1) to both with the same F , obtaining $c = \mu$. Hence

$$\sum_{\text{primitive roots}} \zeta = \mu(n).$$

Step 5 — checks.

- $n = 6$: the primitive roots are ζ_6 and ζ_6^5 , whose sum is $2 \cos 60^\circ = 1 = \mu(6)$ ✓ (indeed $6 = 2 \cdot 3$, two distinct primes, $\mu = (-1)^2 = 1$).
- $n = 12$: from Step 4 of Exercise S4, $\Phi_{12} = x^4 - x^2 + 1$; the coefficient of x^3 is $0 = -\mu(12)$, and indeed $12 = 2^2 \cdot 3$ has a square factor, $\mu(12) = 0$ ✓.
- $n = 5$: $\Phi_5 = x^4 + x^3 + x^2 + x + 1$, coefficient of x^3 equal to $1 = -\mu(5) = -(-1)$ ✓.

Step 6 — reading. The Möbius function, a purely arithmetic object, is thus *the centroid* (times the number of points) of the primitive roots: another instance of an arithmetic quantity read as geometry of the discrete circle.

Solution. S6. *Step 1 — the set and the action.* Let X be the set of sequences of length p over an alphabet of a symbols: $|X| = a^p$. The group $\mathbb{Z}/p\mathbb{Z}$ acts on X by cyclic rotation — the action of Observation 7.8, i.e. *necklaces*.

Step 2 — orbits have size 1 or p . The size of an orbit divides the order of the group (it is the index of the stabiliser, by Theorem 7.5); since p is prime, it is 1 or p .

Step 3 — the orbits of size 1. These are the rotation-invariant sequences, i.e. the constant ones: there are exactly a of them, one per colour.

Step 4 — counting. Letting N be the number of orbits of size p ,

$$a^p = a + pN \implies a^p - a = pN \implies p \mid a^p - a,$$

i.e. $a^p \equiv a \pmod{p}$: Fermat's little theorem.

Step 5 — usual form. If $p \nmid a$, one may cancel a (invertible modulo p) to obtain $a^{p-1} \equiv 1$.

Step 6 — check. $a = 2$, $p = 5$: the 32 binary sequences of length 5 split into 2 constant necklaces and $(32 - 2)/5 = 6$ necklaces of five elements each. In total $2 + 6 = 8$ distinct necklaces. ✓

Step 7 — why the proof is instructive. It is the same theorem as Theorem 7.6, but the proof is *combinatorial* and does not use the group structure of the units: it uses only the rotation of the discrete circle. It shows that Fermat's little theorem is, at bottom, a statement about cyclic symmetry.

Solution. S7. *Step 1 — the product $F\overline{F}^\top$.*

$$(F\overline{F}^\top)_{jl} = \sum_{k=0}^{n-1} \zeta_n^{jk} \overline{\zeta_n^{lk}} = \sum_{k=0}^{n-1} \zeta_n^{(j-l)k} = n \delta_{jl}$$

by Theorem 7.21. So $F\overline{F}^\top = nI$, i.e. $F/\sqrt{n}$ is unitary.

Step 2 — the square.

$$(F^2)_{jl} = \sum_k \zeta_n^{jk} \zeta_n^{kl} = \sum_k \zeta_n^{k(j+l)} = n \delta_{j+l \equiv 0},$$

again by the same theorem. So $F^2 = nP$, where P is the permutation matrix sending the index l to $-l$ modulo n : the *reflection* of the discrete circle.

Step 3 — the fourth power. $P^2 = I$ (reflecting twice is the identity), so

$$F^4 = (nP)^2 = n^2 P^2 = n^2 I.$$

Step 4 — reading. The normalised Fourier transform $U = F/\sqrt{n}$ satisfies $U^4 = I$: it has order 4 as an operator. Its eigenvalues therefore lie among the fourth roots of unity $\{1, i, -1, -i\}$ — again the circle, again μ_4 . This is the fact which, in analysis, corresponds to “transforming four times returns the original function”, whose continuous version is $\mathcal{F}^4 = \text{id}$ on the line.

Step 5 — check for $n = 4$. With $\zeta_4 = i$, $F^2/4$ should be the reflection: and indeed the DFT applied twice to $(1, 2, 3, 4)$ gives $4 \cdot (1, 4, 3, 2)$, the sequence with indices 1 and 3 swapped. ✓

Solution. S8. *Step 1 — weight and evaluation at 1.* For a binary word $c(x) = \sum_i c_i x^i$, the value $c(1) = \sum_i c_i$ computed in $\mathbb{F}_2$ is 0 if the number of 1 bits (the *weight*) is even, and 1 if it is odd.

Step 2 — if $(x+1) \mid g$. Then $g(1) = 0$. Every word is $c = ag$ for some a (Theorem 7.22), so

$$c(1) = a(1) g(1) = 0 :$$

all words have even weight.

Step 3 — if $(x+1) \nmid g$. Then $g(1) \neq 0$, i.e. $g(1) = 1$ in $\mathbb{F}_2$; but g is itself a codeword, and it has odd weight. So not all words have even weight.

Step 4 — conclusion. The two implications give the equivalence.

Step 5 — check. In the $[7, 4]$ Hamming code with $g = x^3 + x + 1$ we have $g(1) = 1 + 1 + 1 = 1 \neq 0$: the code contains words of odd weight, and indeed g itself has weight 3. Multiplying

g by $(x+1)$ gives the $[7, 3]$ code, all of whose words have even weight. ✓

Step 6 — practical use. Adding the factor $(x+1)$ to a generator polynomial is the standard way to make a code detect any *odd* number of errors: an error in an odd number of positions changes the parity and is seen. That is why the CRC polynomials used in standards (for instance the CRC-32 of IEEE 802.3) all have $(x+1)$ as a factor.

Solution. S9. *Step 1 — shape of the double error.* Let $e(x) = x^i + x^j$ with $0 \leq i < j \leq n-1$. Putting $k = j - i$,

$$e(x) = x^i(1 + x^k) = x^i(x^k - 1)$$

in $\mathbb{F}_2[x]$, where $+1 = -1$.

Step 2 — non-detection condition. The error escapes iff $g \mid e$. Since $g(0) \neq 0$ implies $\gcd(g, x^i) = 1$ (Step 3 of the proof of Theorem 7.23), this is equivalent to

$$g \mid x^k - 1.$$

Step 3 — translation into order. In the ring $\mathbb{F}_2[x]/(g)$ the element x is invertible (since $\gcd(x, g) = 1$), and

$$g \mid x^k - 1 \iff x^k = 1 \text{ in } \mathbb{F}_2[x]/(g) \iff \text{ord}(x) \mid k.$$

Step 4 — conclusion. All double errors with $0 < k < n$ are detected if and only if no k in that range is a multiple of $\text{ord}(x)$, i.e. if and only if

$$\text{ord}(x) \geq n.$$

Step 5 — why primitive polynomials. If g is irreducible of degree r , the ring $\mathbb{F}_2[x]/(g)$ is the field $\mathbb{F}_{2^r}$, whose multiplicative group is cyclic of order $2^r - 1$ (Exercise S2). The order of x therefore divides $2^r - 1$, and equals $2^r - 1$ exactly when x is a generator: in that case g is called **primitive**. Primitive polynomials thus maximise the block length for which all double errors are detected: $n \leq 2^r - 1$.

Step 6 — check. $g = x^4 + x + 1$ is primitive: the powers of x modulo g are $x, x^2, x^3, x^4 = x + 1, \dots$ and exhaust the field only after $15 = 2^4 - 1$ steps. So with this generator every block of length ≤ 15 has all double errors detected. In Exercise N8 the block was 12 bits: the condition holds. ✓

Step 7 — the Hamming example. $g = x^3 + x + 1$ is also primitive, with $\text{ord}(x) = 7 = 2^3 - 1$: and indeed the Hamming code has length exactly 7, the maximum compatible with single-error correction. This is no coincidence: the length of binary Hamming codes is always $2^r - 1$.

Solution. S10. *Step 1 — the “if” direction.* Let $n = 2^a p_1 \cdots p_r$ with distinct Fermat primes $p_i = 2^{m_i} + 1$. By Corollary 7.1,

$$\varphi(n) = \varphi(2^a) \prod_i (p_i - 1) = 2^{a-1} \prod_i 2^{m_i} = 2^{a-1 + \sum_i m_i}$$

for $a \geq 1$; for $a = 0$ one has $\varphi(n) = \prod_i 2^{m_i}$, and for $a = 1$, $\varphi(2) = 1 = 2^0$. In every case a power of 2.

Step 2 — “only if”: structure. Let $\varphi(n)$ be a power of 2 and write $n = 2^a p_1^{a_1} \cdots p_r^{a_r}$ with distinct odd primes p_i . By the same formula,

$$\varphi(n) = \varphi(2^a) \prod_i p_i^{a_i-1} (p_i - 1).$$

Every factor of the product must be a power of 2, since a divisor of a power of 2 is a power of 2.

Step 3 — the exponents are 1. If $a_i \geq 2$, the factor $p_i^{a_i-1}$ would be a multiple of the odd prime p_i , hence not a power of 2. So $a_i = 1$ for every i , and the odd primes occur at most once.

Step 4 — the primes are Fermat primes. Every $p_i - 1$ is a power of 2, i.e. $p_i = 2^{m_i} + 1$. By Lemma 7.1, since p_i is prime, m_i is a power of 2: so p_i is a Fermat prime.

Step 5 — conclusion. $n = 2^a p_1 \cdots p_r$ with distinct Fermat primes p_i , which is the claim.

Step 6 — limiting cases. $n = 1$: $\varphi(1) = 1 = 2^0$, and $1 = 2^0$ with no odd primes ✓. $n = 2$: $\varphi(2) = 1$ ✓. $n = 9 = 3^2$: here $a_1 = 2$, and indeed $\varphi(9) = 6$ is not a power of 2 ✓. $n = 21 = 3 \cdot 7$: 7 is not a Fermat prime, and $\varphi(21) = 12$ is not a power of 2 ✓.

Step 7 — numerical consequence. The values $n \leq 100$ for which the polygon is constructible are

3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, 68, 80, 85, 96.

Note there are rather few: constructibility with straightedge and compass is the exception, not the rule.

7.13. Self-assessment quiz

- The group of n -th roots of unity is isomorphic to
A) $\mathbb{Z}$ B) $\mathbb{Z}/n\mathbb{Z}$ C) $\mathbb{Q}$ D) $\mathbb{R}$
- ζ_n^k generates μ_n if and only if
A) k is prime B) $\gcd(n, k) = 1$ C) k divides n D) k is even
- The order of ζ_{12}^4 is
A) 12 B) 4 C) 3 D) 6
- The subgroups of a cyclic group of order n correspond to
A) the primes $\leq n$ B) the divisors of n C) the integers $\leq n$ D) the squares
- The identity $\sum_{d|n} \varphi(d)$ equals
A) $\varphi(n)$ B) n C) $n - 1$ D) 2^n
- Euler–Fermat states that, if $\gcd(a, n) = 1$,
A) $a^n \equiv 1$ B) $a^{\varphi(n)} \equiv 1$ C) $a^{n-1} \equiv 1$ always D) $a \equiv 1$
- To compute 2^{1000} modulo 9 one reduces the exponent modulo
A) 9 B) $\varphi(9) = 6$ C) 1000 D) 3
- The Chinese remainder theorem requires the moduli to be
A) prime B) pairwise coprime C) equal D) odd
- The degree of $\Phi_n(x)$ is
A) n B) $\varphi(n)$ C) $n - 1$ D) 2
- The coefficients of cyclotomic polynomials are
A) always $0, \pm 1$ B) integers, but not always $0, \pm 1$ C) non-integer rationals D) complex

-
11. A number constructible with straightedge and compass has degree over $\mathbb{Q}$
 A) arbitrary B) a power of 2 C) prime D) ≤ 2
12. The regular heptagon
 A) is constructible B) is not constructible C) is constructible with compass only
 D) does not exist
13. The known Fermat primes are
 A) infinitely many B) five: 3, 5, 17, 257, 65537 C) three D) none
14. The characters of $\mathbb{Z}/n\mathbb{Z}$ take values
 A) real B) in the n -th roots of unity C) integer D) arbitrary
15. Orthogonality of characters is the basis of
 A) the DFT B) Lagrange's theorem C) the CRT D) constructibility
16. The Fortescue transformation of Chapter 5 is
 A) a rotation B) the DFT on $\mathbb{Z}/3\mathbb{Z}$ C) a projection D) a rescaling
17. In $\mathbb{F}_q[x]/(x^n - 1)$, multiplying by x means
 A) differentiating B) cyclically rotating the coefficients C) squaring D) nothing in particular
18. Cyclic codes correspond to
 A) arbitrary subspaces B) ideals, i.e. divisors of $x^n - 1$ C) invertible matrices D) irreducible polynomials only
19. A cyclic code with generator of degree r detects all bursts of length
 A) $\leq r$ B) $\leq r/2$ C) any D) ≤ 2
20. All words of a binary cyclic code have even weight if and only if
 A) $g(0) = 0$ B) $(x + 1) \mid g$ C) $\deg g$ is even D) g is primitive

BEST PRACTICE

Answer key.

1-B: Theorem 7.1. **2-B:** Theorem 7.2(ii). **3-C:** $12/\gcd(12, 4) = 12/4 = 3$. **4-B:** Theorem 7.3(ii). **5-B:** Theorem 7.4. **6-B:** Theorem 7.6; C holds only for prime n . **7-B:** same theorem; reducing modulo 9 is the typical error of Exercise N3. **8-B:** Theorem 7.7. **9-B:** Definition 7.2. **10-B:** Theorem 7.9; Φ_{105} has a coefficient -2 .
11-B: Theorem 7.12. **12-B:** $\varphi(7) = 6$ is not a power of 2 (Theorem 7.13). **13-B:** Lemma 7.1; F_5 is composite. **14-B:** Theorem 7.20, Step 4. **15-A:** Corollary 7.5. **16-B:** Example 7.2. **17-B:** Observation 7.8. **18-B:** Theorem 7.22. **19-A:** Theorem 7.23. **20-B:** Exercise S8.

7.14. Chapter formulary

Discrete circle

$$\mu_n = \{\zeta_n^k\}_{k=0}^{n-1} \cong \mathbb{Z}/n\mathbb{Z}, \quad \zeta_n = e^{2\pi i/n}$$

$$\text{ord}(\zeta_n^k) = \frac{n}{\gcd(n, k)}, \quad \#\{\text{generators}\} = \varphi(n)$$

Euler's function

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right) = \prod_i p_i^{a_i-1} (p_i - 1), \quad \sum_{d|n} \varphi(d) = n, \quad \varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$$

Arithmetic theorems

$$\text{Lagrange : } \text{ord}(g) \mid |G|; \quad \text{Euler-Fermat : } a^{\varphi(n)} \equiv 1 \pmod{n} \text{ (gcd}(a, n) = 1)$$

$$\text{Fermat : } a^{p-1} \equiv 1 \pmod{p} \text{ (} p \nmid a\text{);} \quad \text{Wilson : } (p-1)! \equiv -1 \pmod{p}$$

$$\text{CRT : } \mathbb{Z}/N\mathbb{Z} \cong \prod_i \mathbb{Z}/n_i\mathbb{Z}, \quad x = \sum_i a_i N_i M_i, \quad N_i = N/n_i, \quad N_i M_i \equiv 1 \pmod{n_i}$$

Cyclotomic polynomials

$$\Phi_n(x) = \prod_{\gcd(k, n)=1} (x - \zeta_n^k), \quad \deg \Phi_n = \varphi(n), \quad \Phi_n \in \mathbb{Z}[x]$$

$$x^n - 1 = \prod_{d|n} \Phi_d(x), \quad \Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}, \quad \sum_{\text{primitive}} \zeta = \mu(n)$$

$$\Phi_p(x) = 1 + x + \cdots + x^{p-1}, \quad \Phi_{12} = x^4 - x^2 + 1, \quad \Phi_{15} = x^8 - x^7 + x^5 - x^4 + x^3 - x + 1$$

Constructibility

$$\alpha \text{ constructible} \implies [\mathbb{Q}(\alpha) : \mathbb{Q}] = 2^t; \quad \left[\mathbb{Q}\left(\cos \frac{2\pi}{n}\right) : \mathbb{Q} \right] = \frac{\varphi(n)}{2}$$

$$n\text{-gon constructible} \iff n = 2^a p_1 \cdots p_r, \quad p_i = 2^{2^{t_i}} + 1 \text{ distinct primes}$$

$$\text{known Fermat primes : } 3, 5, 17, 257, 65537; \quad \cos 72^\circ = \frac{\sqrt{5} - 1}{4}$$

Characters and DFT

$$\chi_k(m) = \zeta_n^{km}, \quad \frac{1}{n} \sum_m \chi_k(m) \overline{\chi_j(m)} = \delta_{kj}$$

$$\hat{f}(k) = \sum_m f(m) \zeta_n^{-km}, \quad f(m) = \frac{1}{n} \sum_k \hat{f}(k) \zeta_n^{km}, \quad \widehat{f * g} = \hat{f} \hat{g}$$

$$F \overline{F}^\top = nI, \quad F^2 = nP, \quad F^4 = n^2 I$$

Cyclic codes

$$R_n = \mathbb{F}_q[x]/(x^n - 1), \quad C = (g), \quad g \mid x^n - 1, \quad \dim C = n - \deg g$$

$$c(x) = x^{\deg g} m(x) + r(x), \quad r = x^{\deg g} m \bmod g$$

$$\text{bursts detected : } L \leq \deg g; \quad \text{double errors detected} \iff \text{ord}_g(x) \geq n$$

$$\text{all weights even} \iff (x+1) \mid g$$

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CHAPTER 8

Pencils of Circles, Inversion, and Steiner's Porism

This chapter gathers the tools of the previous chapters — the complex form of the circle (Chapter 1), Möbius transformations and inversion in complex form (Chapter 3) — to tackle a family of classical problems about circles: when two circles share a locus of “equal power”; how entire families of circles behave under a transformation; and how Steiner's Porism, proved in Volume I with synthetic geometry, reads through the lens of complex algebra.

INFO

The idea in one line. A coaxial pencil of circles is, algebraically, a linear combination of two circle equations; inversion, in complex form $\iota(z) = r^2/\bar{z}$ (Chapter 3), sends circles and lines to circles and lines because it is the composition of a Möbius transformation and a conjugation. These two facts together explain why any two circles, as long as they do not intersect, can always be “straightened” into two concentric circles — the crucial step of Steiner's Porism (Volume I, Chapter 5).

INFO

Prerequisites. Chapter 1 (complex form of the circle, stereographic projection); Chapter 3 (extended complex plane, Möbius transformations, Theorem 3.14, inversion Definition 3.3 and Proposition 3.1, cross-ratio from Exercise S5). From Volume I: power of a point and the radical axis (Chapter 7), Steiner's Porism (Chapter 5) — recalled here, not re-proved.

8.1. Recall: power of a point and the radical axis

***Remark 8.1: What we carry over from Volume I.** For a circle $\gamma : x^2 + y^2 + Dx + Ey + F = 0$ with centre $C = (-D/2, -E/2)$ and radius r , the **power** of a point P with respect to γ is $\text{pow}_\gamma(P) = |P - C|^2 - r^2$, which can also be computed by substituting the coordinates of P directly into the left-hand side of the equation of γ . Volume I, Chapter 7, proves in full that, for two non-concentric circles γ_1, γ_2 , the locus of points of equal power with respect to both — the **radical axis** — is a line perpendicular to the line joining the centres, and that it is obtained by subtracting the two equations: if $\gamma_i : x^2 + y^2 + D_i x + E_i y + F_i = 0$, the radical axis is*

$$(D_1 - D_2)x + (E_1 - E_2)y + (F_1 - F_2) = 0.$$

This already-proved fact is the starting point of this chapter: here we use it to build entire families of circles, not merely to find the line of two given circles.

8.2. Coaxial pencils of circles

Definition 8.1: Coaxial pencil of circles. Let γ_1, γ_2 be two distinct circles (not the same circle), written in the form $x^2 + y^2 + D_i x + E_i y + F_i = 0$. The **coaxial pencil** generated by γ_1, γ_2 is the family

$$\gamma_t : x^2 + y^2 + D(t)x + E(t)y + F(t) = 0, \quad D(t) = D_1 + t(D_2 - D_1), \text{ etc., } t \in \mathbb{R},$$

together with the limiting line $t \rightarrow \infty$, proportional to $(D_2 - D_1)x + (E_2 - E_1)y + (F_2 - F_1) = 0$, i.e. the radical axis of γ_1, γ_2 .

Theorem 8.1: Properties of the coaxial pencil. With the notation of Definition 8.1:

- (i) for every $t \in \mathbb{R}$, γ_t is a genuine circle equation (coefficient of $x^2 + y^2$ equal to 1, independent of t);
- (ii) all circles of the pencil have the same radical axis pairwise, coinciding with that of γ_1, γ_2 ;
- (iii) if γ_1, γ_2 meet in two points A, B , every γ_t passes through A and B (intersecting pencil, with **base points** A, B);
- (iv) if γ_1, γ_2 are tangent at a point A , every γ_t passes through A (tangent pencil);
- (v) if, for a non-concentric pair γ_1, γ_2 , the equation $\rho(t)^2 = 0$ (the squared radius of γ_t , see the proof) has two real solutions $t_1 \neq t_2$, the two points $\gamma_{t_1}, \gamma_{t_2}$ (circles of radius 0) are called the **limit points** of the pencil, and in that case the pencil is called non-intersecting: no two distinct circles of the pencil share a common point beyond the limit points if they exist, because two distinct circles of the pencil have in common exactly the points where γ_1 and γ_2 themselves meet (by (iii)–(iv)), or none if γ_1, γ_2 do not meet.

Proof. (i) The coefficient of $x^2 + y^2$ in γ_1 and γ_2 equals 1 in both by hypothesis (normalised form); γ_t does not touch this coefficient, so it stays 1 for every t : γ_t is always a circle equation (provided the squared radius, computed below, is ≥ 0 ; if it is < 0 the equation has no real solutions — an “imaginary circle”, which we will not use).

(ii) The radical axis of γ_t and γ_1 is obtained, by Observation 8.1, by subtracting the equations:

$$\gamma_t - \gamma_1 = t[(D_2 - D_1)x + (E_2 - E_1)y + (F_2 - F_1)],$$

proportional (for $t \neq 0$) to the same line $(D_2 - D_1)x + (E_2 - E_1)y + (F_2 - F_1) = 0$ for every $t \neq 0$: it is always the radical axis of γ_1, γ_2 , regardless of which pair of circles of the pencil is chosen.

(iii)–(iv) If A satisfies both $\gamma_1(A) = 0$ and $\gamma_2(A) = 0$, then $\gamma_t(A) = \gamma_1(A) + t(\gamma_2(A) - \gamma_1(A)) = 0 + t \cdot 0 = 0$ for every t : A belongs to every circle of the pencil. If γ_1, γ_2 meet in two points, these satisfy both equations, hence satisfy every γ_t ; if they are tangent, the same holds for the single point of tangency.

(v) Writing the squared radius of γ_t as $\rho(t)^2 = (D(t)/2)^2 + (E(t)/2)^2 - F(t)$, and substituting $D(t) = D_1 + t dD$, $E(t) = E_1 + t dE$, $F(t) = F_1 + t dF$ (with $dD = D_2 - D_1$ etc.), one obtains a second-degree polynomial in t , $\rho(t)^2 = At^2 + Bt + C$, with $A = (dD^2 + dE^2)/4$ and $C = \rho_1^2$ (explicit coefficients not needed further). Since γ_1, γ_2 are not concentric, $(dD, dE) \neq (0, 0)$, hence $A > 0$: $\rho(t)^2$ is a second-degree polynomial opening upward, with 0, 1, or 2 real roots depending on the sign of its discriminant $B^2 - 4AC$. When it has two, $t_1 \neq t_2$, the corresponding $\gamma_{t_1}, \gamma_{t_2}$ are, by definition of $\rho(t)^2$, circles of radius 0, i.e. points: these are the limit points of the statement. This chapter does not classify in general, for every pair γ_1, γ_2 , when the discriminant is positive (deferring that full classification to the projective geometry of pencils of circles, beyond this introductory volume): in the examples that follow we limit ourselves to checking, case by case and numerically, that for the non-intersecting pairs of circles considered

the discriminant is indeed positive, giving two real limit points. $\square$

8.3. Circles of Apollonius

Theorem 8.2: The Apollonius locus is a circle. Let $a, b \in \mathbb{C}$, $a \neq b$, and $k > 0$, $k \neq 1$. The locus

$$\mathcal{A}_k = \left\{ z \in \mathbb{C} : \frac{|z - a|}{|z - b|} = k \right\}$$

is the circle of centre $c = \frac{a - k^2 b}{1 - k^2}$ and radius $r = \frac{k|a - b|}{|1 - k^2|}$. For $k = 1$, $\mathcal{A}_1$ is the perpendicular bisector of the segment ab (a line).

Proof. From $|z - a| = k|z - b|$, squaring and expanding with $|w|^2 = w\bar{w}$ (Theorem 3.1 of Chapter 3),

$$(z - a)(\bar{z} - \bar{a}) = k^2(z - b)(\bar{z} - \bar{b}),$$

$$z\bar{z} - \bar{a}z - a\bar{z} + a\bar{a} = k^2(z\bar{z} - \bar{b}z - b\bar{z} + b\bar{b}).$$

For $k \neq 1$ we collect $z\bar{z}(1 - k^2)$ and divide by $1 - k^2 \neq 0$:

$$z\bar{z} - \frac{\bar{a} - k^2\bar{b}}{1 - k^2}z - \frac{a - k^2b}{1 - k^2}\bar{z} + \frac{a\bar{a} - k^2b\bar{b}}{1 - k^2} = 0.$$

Setting $c = (a - k^2b)/(1 - k^2)$ (note $\bar{c} = (\bar{a} - k^2\bar{b})/(1 - k^2)$, since $k^2 \in \mathbb{R}$), the equation becomes

$$z\bar{z} - \bar{c}z - c\bar{z} + \frac{a\bar{a} - k^2b\bar{b}}{1 - k^2} = 0, \text{ that is}$$

$$|z - c|^2 = |c|^2 - \frac{a\bar{a} - k^2b\bar{b}}{1 - k^2} =: r^2.$$

It remains to check that this r^2 agrees with $(k|a - b|/|1 - k^2|)^2$: expanding $|c|^2 = (a - k^2b)(\bar{a} - k^2\bar{b})/(1 - k^2)^2$ and substituting,

$$r^2 = \frac{(a - k^2b)(\bar{a} - k^2\bar{b}) - (1 - k^2)(a\bar{a} - k^2b\bar{b})}{(1 - k^2)^2}.$$

The numerator, expanding both products, is

$$a\bar{a} - k^2a\bar{b} - k^2b\bar{a} + k^4b\bar{b} - a\bar{a} + k^2a\bar{a} + k^2b\bar{b} - k^4b\bar{b} = k^2(a\bar{a} + b\bar{b} - a\bar{b} - b\bar{a}) = k^2|a - b|^2,$$

so $r^2 = k^2|a - b|^2/(1 - k^2)^2$, i.e. $r = k|a - b|/|1 - k^2|$, as announced.

For $k = 1$: the equation $|z - a| = |z - b|$ says that z is equidistant from a and b , i.e. z lies on the perpendicular bisector of ab , a line (an elementary fact, independent of this chapter). $\square$

Example 8.1: The points a, b are the limit points of the Apollonius pencil. As k ranges over $(0, \infty) \setminus \{1\}$, the circles $\mathcal{A}_k$ of Theorem 8.2 form a non-intersecting coaxial pencil (Definition 8.1, generated by any two of them), whose limit points (Theorem 8.1(v)) are exactly a and b : indeed, for $k \rightarrow 0^+$,

$$c = \frac{a - k^2b}{1 - k^2} \rightarrow a, \quad r = \frac{k|a - b|}{|1 - k^2|} \rightarrow 0,$$

i.e. $\mathcal{A}_k$ degenerates to the point a ; and for $k \rightarrow \infty$, dividing numerator and denominator of c by k^2 and rewriting r as $|a - b|/|k - 1/k|$,

$$c = \frac{a/k^2 - b}{1/k^2 - 1} \rightarrow \frac{-b}{-1} = b, \quad r = \frac{|a - b|}{|k - 1/k|} \rightarrow 0,$$

i.e. $\mathcal{A}_k$ degenerates to the point b . **Independent numerical check** (script `Sorgenti/Vol2/_verifiche/vol2_cap_fasci_steiner.py`, function `check_limit_points_and_apollonius_link`): for $a = 1 + 2i$, $b = -3 + 0.5i$, solving $\rho(t)^2 = 0$ for the pencil generated by $\mathcal{A}_2$ and $\mathcal{A}_{3.5}$ gives the two limit points $1.0000 + 2.0000i$ and $-3.0000 + 0.5000i$, matching a, b to within 3×10^{-15} .

Lemma 8.1: Orthogonality criterion for two circles. *Two intersecting circles $C_1 = C(c_1, r_1)$ and $C_2 = C(c_2, r_2)$ are **orthogonal** (their tangent lines at the intersection points are perpendicular) if and only if*

$$|c_1 - c_2|^2 = r_1^2 + r_2^2.$$

Proof. Let P be a point of intersection. The tangent to C_i at P is perpendicular to the radius c_iP (an elementary fact of circle geometry, Volume I, Chapter 3). The two tangents at P are perpendicular if and only if the two radii c_1P and c_2P are perpendicular, i.e. if and only if the triangle c_1Pc_2 has a right angle at P . By the Pythagorean theorem (Volume I, Chapter 2), this is equivalent to $|c_1 - c_2|^2 = |c_1 - P|^2 + |c_2 - P|^2 = r_1^2 + r_2^2$, using $|c_i - P| = r_i$ because $P \in C_i$. $\square$

Theorem 8.3: Orthogonality of the two Apollonius pencils. *Let $a, b \in \mathbb{C}$ be distinct. Every circle through a and b is orthogonal to every circle of Apollonius $\mathcal{A}_k$ ($k \neq 1$) of a, b .*

Proof. Let Γ be a circle through a, b , with centre c_1 and radius $r_1 = |c_1 - a| = |c_1 - b|$ (equidistance from the centre). Let $\mathcal{A}_k = C(c, r)$ with c, r as in Theorem 8.2. By Lemma 8.1 it suffices to show $|c_1 - c|^2 = r_1^2 + r^2$. Since c lies on the line ab (being a real affine combination $c = (a - k^2b)/(1 - k^2)$ of a, b with real coefficients, as one checks by writing $c = a + \frac{k^2}{1 - k^2}(a - b)$, a point of the line through a, b), and since c_1 is equidistant from a, b (hence on the perpendicular to ab through the midpoint $m = (a + b)/2$), the segment c_1c together with the segment mc (along ab) and c_1m (perpendicular to ab) form a right triangle at m :

$$|c_1 - c|^2 = |c_1 - m|^2 + |m - c|^2.$$

On the other hand $r_1^2 = |c_1 - m|^2 + |m - a|^2$ (Pythagoras in the right triangle c_1ma , right-angled at m), so

$$|c_1 - c|^2 - r_1^2 = |m - c|^2 - |m - a|^2.$$

It remains to show $|m - c|^2 - |m - a|^2 = r^2$. Writing everything on the line ab with a real coordinate (origin at m , unit $|a - b|/2$; a has coordinate -1 , b has coordinate $+1$), the centre c has coordinate $\xi = \frac{-1 - k^2}{1 - k^2}$ (from the formula for c with $a = -1, b = 1$ on this scale) and radius (on the same scale) $\eta = \frac{2k}{|1 - k^2|}$. Then

$$\xi^2 - 1 = \frac{(1 + k^2)^2}{(1 - k^2)^2} - 1 = \frac{(1 + k^2)^2 - (1 - k^2)^2}{(1 - k^2)^2} = \frac{4k^2}{(1 - k^2)^2} = \eta^2,$$

that is, returning to the original units (a factor $|a - b|^2/4$ on both sides), $|m - c|^2 - |m - a|^2 = r^2$. This completes the proof. **Independent numerical check:** for $a = 1 + 2i$, $b = -3 + 0.5i$,

a circle Γ through a, b with centre $c_1 = (a + b)/2 + 0.7i(b - a)/|b - a|$ and the circle $\mathcal{A}_2$ give $|c_1 - c|^2 = 13.163611$ and $r_1^2 + r^2 = 13.163611$, identical to within 2×10^{-15} (script cited, function `check_limit_points_and_apollonius_link`). $\square$

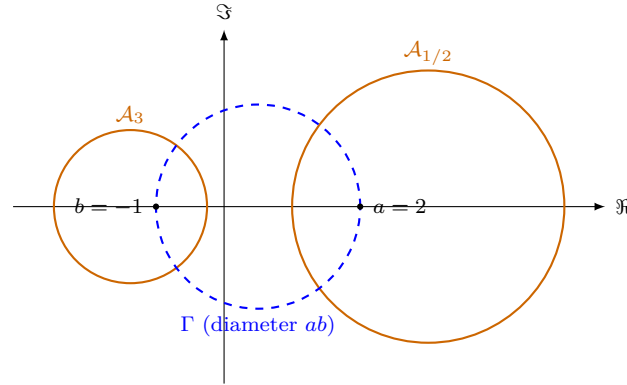


Figure 8.1: Two Apollonius circles of the non-secant pencil with limit points $a = 2, b = -1$ (orange, $\mathcal{A}_3$ and $\mathcal{A}_{1/2}$, from Exercise N1) and a circle Γ of the orthogonal secant pencil through a, b (dashed blue, of diameter ab): the relation $|c_1 - c_2|^2 = r_1^2 + r_2^2$ checked in Exercise N2 illustrates Theorem 8.3.

8.4. Inversion sends circles and lines to circles and lines

Remark 8.2: *An already-proved fact, here by a second route.* Volume I, Chapter 5, proves algebraically (and, for the line case, also synthetically via Thales's theorem) that inversion sends circles and lines to circles and lines. With the complex tools of this volume the same fact has a shorter proof, which we present here as a second, independent check, not as a replacement for the first: the two proofs do not share a single step.

Corollary 8.1: *Inversion sends circles and lines to circles and lines (complex form).* Let $\iota(z) = r^2/\bar{z}$ be the inversion with respect to $C(0, r)$ (Definition 3.3, Chapter 3). The image under ι of every circle or line of the extended plane $\mathbb{C} \cup \{\infty\}$ is again a circle or a line.

Proof. Write $\iota = \sigma \circ M$, where $M(z) = r^2/z$ and $\sigma(w) = \bar{w}$ is conjugation. M is a Möbius transformation (in the form $(az + b)/(cz + d)$ with $a = 0, b = r^2, c = 1, d = 0$, and $ad - bc = -r^2 \neq 0$), hence by Theorem 3.14 of Chapter 3 it sends circles and lines to circles and lines. Conjugation σ does the same: if $|z - c| = \rho$, then $|\bar{z} - \bar{c}| = |\overline{z - c}| = |z - c| = \rho$ (the modulus of a real number is unchanged by conjugation), i.e. σ sends $C(c, \rho)$ to $C(\bar{c}, \rho)$, still a circle; and if $2\Re(\bar{\nu}z) = d$ is the equation of a line (Chapter 3), substituting $z = \bar{w}$ and conjugating the whole equation (which is real) gives $2\Re(\nu w) = d$, still a line. The composition of two maps that send circles/lines to circles/lines does the same. $\square$

Remark 8.3: *What this proof does not say.* The complex proof is shorter but also more opaque: unlike the synthetic proof of Volume I, it does not say what the centre and radius of the image circle are, as a function of the pole and the power of inversion — a formula proved in Volume I, Chapter 5. The two proofs are complementary, not redundant: one establishes that circles and lines are preserved as a category, the other says how the parameters transform.

8.5. Steiner's porism revisited

Remark 8.4: The theorem, recalled from Volume I. Volume I, Chapter 5, proves in full **Steiner's Porism**: if α, β are two circles, one inside the other, neither concentric nor intersecting, and a chain of circles $\gamma_1, \gamma_2, \dots$, each tangent to α , to β , and to the previous one, closes after n steps for one choice of γ_1 , then it closes after n steps for every choice of γ_1 . The crucial step of that proof is the lemma of reduction to the concentric case: two non-intersecting circles can always be transformed, by a suitable inversion, into two concentric circles. Volume I proves it with a homothety argument on chords (a different method from Corollary 8.1 above, which instead proceeds via Möbius maps and conjugation). With the tools of this chapter we can point out explicitly which inversion works, without re-proving the lemma: it is the inversion whose pole is one of the two **limit points** of the coaxial pencil generated by α, β (Theorem 8.1(v)) — because an inversion with pole at a limit point sends that point to ∞ and the other limit point to a finite point Z , and transforms the entire pencil (which includes α, β) into a pencil of circles concentric about Z (the circles of the original pencil, never passing through the limit point chosen as pole if distinct from it, all have the same “distance from both limit points” in a projective sense that inversion turns literally into concentricity). We do not prove this fact in full generality here — we take it, as we did the lemma of reduction to the concentric case, already proved in Volume I — but we **check it numerically** on a concrete case below, showing explicitly the pole of inversion, the pencil, and the concentric result.

Example 8.2: Explicit reduction to the concentric case. Let $\alpha = C(0, 5)$ and $\beta = C(1.2, 1.5)$ (in complex notation, centres 0 and 1.2): β lies entirely inside α ($|0 - 1.2| + 1.5 = 2.7 < 5$) and the two circles do not intersect.

Step 1 — limit points. Solving $\rho(t)^2 = 0$ (Theorem 8.1(v)) for the pencil generated by α, β gives the two limit points (on the real axis, by symmetry, since both centres are real)

$$P_1 = 1.327618, \quad P_2 = 18.830715.$$

Step 2 — inversion. We choose pole $P = P_2$ and an arbitrary power of inversion (the concentricity result does not depend on this choice, only the final scale does): $\iota(z) = P + \frac{k^2}{\overline{z - P}}$ with $k^2 = 1$. Applying ι to 400 points sampled on α and on β and least-squares fitting a circle to the images gives

$$\alpha^* : \text{centre } 18.773582, \text{ radius } 0.015170, \quad \beta^* : \text{centre } 18.773582, \text{ radius } 0.004861,$$

with $|\text{centre}(\alpha^*) - \text{centre}(\beta^*)| < 5 \times 10^{-13}$: **concentric**, up to the numerical error of the sampling and the fitting. Repeating with $k^2 = 2.7$ gives the same angle ϑ of Step 3 (see below) to within 2×10^{-9} : the power of inversion rescales the radii but not the angle, as it must for a purely projective fact. (Script: `vol2_cap_fasci_steiner.py`, function `steiner_chain_example`.)

Step 3 — the closure angle. By the concentric case of the theorem of Steiner's Porism (Volume I), with $a > b$ the two radii, $\sin(\vartheta/2) = (a - b)/(a + b)$. Here $a = 0.015170$, $b = 0.004861$, giving $\vartheta \approx 61.95^\circ$ and $2\pi/\vartheta \approx 5.81$: not an integer, so for *this* pair α, β the Steiner chain does not close exactly with real circles (the required number of links is not an integer) — which is the generic situation, not an error: the porism says that *if* it closes for one starting choice, it closes for all of them; not that it always closes.

Example 8.3: A Steiner chain that closes exactly. To build an example with exact closure it is convenient to work backwards, starting from the concentric case. Fixing $n = 5$, we want $\sin(\vartheta/2) = (a - b)/(a + b)$ with $\vartheta = 2\pi/5 = 72^\circ$. Setting $b = 1$:

$$\sin(36^\circ) = \frac{a - 1}{a + 1} \implies a = \frac{1 + \sin 36^\circ}{1 - \sin 36^\circ} = 3.851840\dots$$

With $\rho = (a - b)/2 = 1.425920$ and $d = (a + b)/2 = 2.425920$ (distance of the chain centres from the common centre $Z = 0$), the five centres $Z + d e^{2\pi i j/5}$, $j = 0, \dots, 4$, give a chain of five congruent circles of radius ρ . **Independent numerical check:** the distance between two consecutive centres is 2.851840 in all five cases, equal to 2ρ to within 4×10^{-16} ; the distance of each centre from Z plus ρ equals exactly a (internal tangency to $\alpha^* = C(0, a)$), and minus ρ equals exactly b (external tangency to $\beta^* = C(0, b)$), in all five cases to within 10^{-15} . By Steiner's Porism, applying the inverse of any inversion with pole at a limit point (inversion is involutive, Proposition 3.1(i) of Chapter 3) to this concentric configuration gives a chain of five circles, tangent to two non-concentric circles α, β in the image, that closes just as exactly — and the porism guarantees that *any* rotation of the starting concentric configuration, carried back with the same inversion, gives another closed chain of five circles for the *same* pair α, β : infinitely many closed chains, all with five links.

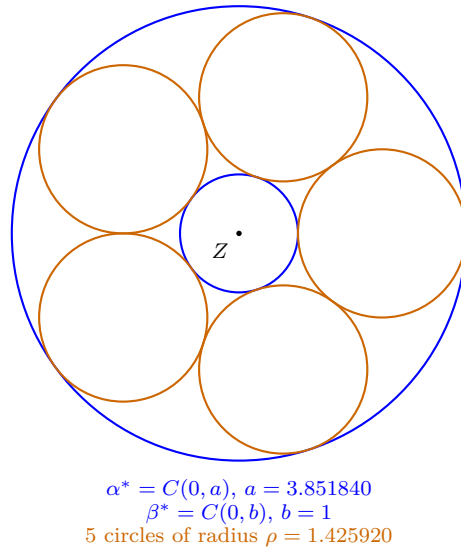


Figure 8.2: The exact-closure Steiner chain of Example 8.3 in the concentric case: five congruent circles (orange), each internally tangent to α^* , externally tangent to β^* and to its two neighbours (Steiner's Porism, Volume I, Chapter 5). Applying the inverse of the reduction inversion (Observation 8.4) this figure returns to the chain for the original, non-concentric pair α, β .

8.6. Real cross-ratio and concyclicity

Remark 8.5: Recall from Exercise S5 of Chapter 3. Exercise S5 of Chapter 3 defines the **cross-ratio** of four distinct points $(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$ and proves, in full, two facts we will use without repeating the proof: (a) the cross-ratio is invariant under every Möbius transformation; (b) given three distinct points z_2, z_3, z_4 and three distinct images w_2, w_3, w_4 , there is a unique Möbius transformation f with $f(z_k) = w_k$ for $k = 2, 3, 4$, determined implicitly by $(f(z), w_2; w_3, w_4) = (z, z_2; z_3, z_4)$ for every z .

Theorem 8.4: Real cross-ratio and concyclicity. Four distinct points $z_1, z_2, z_3, z_4 \in \mathbb{C}$ are concyclic (lie on a common circle) or collinear (lie on a common line) if and only if their cross-ratio $(z_1, z_2; z_3, z_4)$ is a real number.

Proof. By Observation 8.5(b), there is a unique Möbius map f with $f(z_2) = 0$, $f(z_3) = 1$, $f(z_4) = \infty$, and by construction $f(z) = (z, z_2; z_3, z_4)$ for every z (setting $w_2 = 0, w_3 = 1, w_4 = \infty$ in the implicit formula, which then reads directly $f(z) = (z, z_2; z_3, z_4)$). In particular $f(z_1) = (z_1, z_2; z_3, z_4)$.

By Theorem 3.14 of Chapter 3, f sends the circle or line Γ through z_2, z_3, z_4 (which exists and is unique: three distinct points determine a unique circle, or a line if they are collinear — a fact of Volume I, Chapter 7) to the circle or line through $0, 1, \infty$. But the only circle or line of the extended plane passing through ∞ and through two distinct real points such as $0, 1$ is the extended real line $\mathbb{R} \cup \{\infty\}$ (a line always passes through ∞ in the extended plane; a genuine circle never passes through ∞): hence $f(\Gamma) = \mathbb{R} \cup \{\infty\}$.

Since f is a bijection of $\mathbb{C} \cup \{\infty\}$ onto itself, $z_1 \in \Gamma$ if and only if $f(z_1) \in \mathbb{R} \cup \{\infty\}$, i.e. (since $z_1 \neq z_2, z_3, z_4$, hence $f(z_1) \neq \infty$) if and only if $f(z_1) = (z_1, z_2; z_3, z_4) \in \mathbb{R}$. This is exactly the statement. $\square$

Example 8.4: Numerical verification of the criterion. Four points on the circle $C(1+i, 2)$, at angles $0.3, 1.1, 2.7, 4.9$ radians, give cross-ratio $1.64877365 + 0i$ (imaginary part $< 10^{-15}$): real, consistent with concyclicity. Four collinear points on the real axis $(0, 1, 2.5, -1.3)$ give cross-ratio $2.94871795 + 0i$: real, consistent with collinearity. Four generic points (neither concyclic nor collinear: $0, 1, 0.5 + 0.9i, 2 + 1.5i$) give cross-ratio $0.56301887 - 0.45056604i$: not real, consistent with the absence of concyclicity or collinearity. (Script cited, function `check_cross_ratio_concyclic`.)

8.7. Numerical exercises

Exercise 8.1: N1 — An explicit circle of Apollonius. Determine the centre and radius of the circle $\{z : |z - 2|/|z + 1| = 3\}$, and check that 2 and -1 are the limit points of the pencil it generates with $\{|z - 2|/|z + 1| = 1/2\}$.

Exercise 8.2: N2 — Orthogonality. Verify numerically that the circle of diameter $[2, -1]$ (through the two points $2, -1$) is orthogonal to the circle of Apollonius $\{|z - 2|/|z + 1| = 3\}$ of Exercise N1, computing $|c_1 - c_2|^2$ and $r_1^2 + r_2^2$.

Exercise 8.3: N3 — Cross-ratio. Compute the cross-ratio $(0, 2, 2i, 2+2i)$ and determine whether the four points are concyclic, collinear, or neither.

8.8. Symbolic exercises

Exercise 8.4: S1 — The pencil of circles through two points. Given distinct $a, b \in \mathbb{C}$, write the equation of the intersecting coaxial pencil with base points a, b in the form $|z - a|^2 + t \cdot 2\Re(\overline{(a - b)}(z - a)) = 0$ for a suitable real parameter t (hint: start from two explicit circles through a, b , e.g. the one of diameter ab and a second circle through a, b with center shifted along the perpendicular bisector of ab , i.e. with a larger radius).

Exercise 8.5: S2 — Symmetry $k \leftrightarrow 1/k$ in the Apollonius pencil. Prove that $\mathcal{A}_k$ and $\mathcal{A}_{1/k}$ (same a, b) have the same radius, and that their centres are symmetric about the midpoint $m = (a + b)/2$ (hint: use directly the formulas of Theorem 8.2 for $c(k), r(k)$ and $c(1/k), r(1/k)$).

8.9. Solutions to the numerical exercises

Solution. N1. With $a = 2, b = -1, k = 3$: $c = \frac{2 - 9(-1)}{1 - 9} = \frac{11}{-8} = -1.375$, $r = \frac{3 \cdot 3}{8} = 1.125$. With $k = 1/2$: $c = \frac{2 - \frac{1}{4}(-1)}{1 - \frac{1}{4}} = \frac{2.25}{0.75} = 3$, $r = \frac{0.5 \cdot 3}{0.75} = 2$. By Example 8.1, the limit points of the pencil generated by *any* pair of circles of Apollonius of a, b are always a, b themselves (the general computation of the limit $k \rightarrow 0, k \rightarrow \infty$ does not depend on the pair chosen): for $a = 2, b = -1$, the limit points are therefore 2 and -1 , with no need to solve the equation $\rho(t)^2 = 0$ afresh for this specific pair.

Solution. N2. Circle of diameter $[2, -1]$: centre $c_1 = 0.5$, radius $r_1 = 1.5$. Circle of Apollonius $k = 3$: $c_2 = -1.375$, $r_2 = 1.125$ (from Exercise N1). $|c_1 - c_2|^2 = |0.5 - (-1.375)|^2 = 1.875^2 = 3.515625$. $r_1^2 + r_2^2 = 2.25 + 1.265625 = 3.515625$. Equal: orthogonal, as predicted by Theorem 8.3.

Solution. N3. $(z_1, z_2; z_3, z_4)$ with $z_1 = 0, z_2 = 2, z_3 = 2i, z_4 = 2 + 2i$:

$$(z_1, z_2; z_3, z_4) = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)} = \frac{(0 - 2i)(2 - (2 + 2i))}{(0 - (2 + 2i))(2 - 2i)} = \frac{(-2i)(-2i)}{(-2 - 2i)(2 - 2i)}.$$

Numerator: $(-2i)(-2i) = 4i^2 = -4$. Denominator: $(-2 - 2i)(2 - 2i) = -4 + 4i - 4i + 4i^2 = -4 - 4 = -8$. Cross-ratio $= -4/-8 = 0.5$: real, so the four points are concyclic or collinear. Since they are not collinear ($0, 2, 2i$ do not lie on a line), they are concyclic — indeed they are the four vertices of the square of side 2, all at distance $\sqrt{2}$ from the centre $1 + i$.

8.10. Solutions to the symbolic exercises

Solution. S1. The circle of diameter ab has centre $(a + b)/2$ and radius $|a - b|/2$, equation $|z - (a + b)/2|^2 = |a - b|^2/4$, i.e. $|z - a|^2 - \Re((b - a)\overline{(z - a)}) = 0$ after expanding and using $|a - b| = |b - a|$ (checked directly by substitution). Every other circle through a, b is written as this one plus a real multiple t of the line $2\Re((a - b)\overline{(z - a)}) = 0$ (which is the radical axis of the pencil, by Theorem 8.1(ii), being perpendicular to the segment ab at its midpoint — the perpendicular bisector of the segment), giving the stated form.

Solution. S2. From the formulas of Theorem 8.2,

$$r(k) = \frac{k|a - b|}{|1 - k^2|}, \quad r(1/k) = \frac{(1/k)|a - b|}{|1 - 1/k^2|} = \frac{(1/k)|a - b|}{|k^2 - 1|/k^2} = \frac{k|a - b|}{|k^2 - 1|} = r(k),$$

using $|1 - k^2| = |k^2 - 1|$: same radius. For the centres,

$$c(k) + c(1/k) = \frac{a - k^2b}{1 - k^2} + \frac{a - b/k^2}{1 - 1/k^2} = \frac{a - k^2b}{1 - k^2} + \frac{k^2a - b}{k^2 - 1},$$

$$c(k) + c(1/k) = \frac{(a - k^2b) - (k^2a - b)}{1 - k^2} = \frac{(1 - k^2)(a + b)}{1 - k^2} = a + b = 2m,$$

hence $c(k)$ and $c(1/k)$ are symmetric about m . This is consistent with the elementary observation that $|z - a|/|z - b| = k$ is equivalent to $|z - b|/|z - a| = 1/k$: $\mathcal{A}_k$ and $\mathcal{A}_{1/k}$ are the same family of circles seen by swapping the roles of a and b .

8.11. Self-assessment quiz

Eight multiple-choice questions. The answer key is at the end of the section.

1. The coaxial pencil generated by two circles tangent at a point A :
 - A) has no points in common
 - B) passes through A in every member
 - C) is a non-intersecting pencil with two limit points
 - D) is not defined
2. The limit points of the Apollonius pencil of a, b are:
 - A) the midpoint of a, b
 - B) a and b themselves
 - C) the points at infinite distance
 - D) non-existent, since it is an intersecting pencil
3. Two circles $C(c_1, r_1)$, $C(c_2, r_2)$ are orthogonal if and only if:
 - A) $|c_1 - c_2| = r_1 + r_2$
 - B) $|c_1 - c_2|^2 = r_1^2 + r_2^2$
 - C) $|c_1 - c_2| = |r_1 - r_2|$
 - D) $r_1 = r_2$
4. The inversion $\iota(z) = r^2/\bar{z}$ sends circles and lines to circles and lines because it is the composition of:
 - A) two rotations
 - B) a Möbius map and a conjugation
 - C) two translations
 - D) a homothety alone
5. In Steiner's Porism, the inversion that reduces two non-intersecting circles to the concentric case has as its pole:
 - A) the centre of one of the two circles
 - B) one of the two limit points of the pencil they generate
 - C) any point of the plane
 - D) the midpoint of the two centres
6. The cross-ratio $(z_1, z_2; z_3, z_4)$ is real if and only if:

- A) the four points are always concyclic
 B) the four points are concyclic or collinear
 C) the four points are equidistant from the origin
 D) $z_1 = z_4$
7. In the proof of the concyclicity criterion via the cross-ratio, the Möbius map f sends z_2, z_3, z_4 respectively to:
 A) $1, 0, \infty$
 B) $0, 1, \infty$
 C) $0, \infty, 1$
 D) three arbitrary points
8. If non-concentric, non-intersecting α, β are inverted with respect to one of their limit points, one obtains two circles that are:
 A) tangent
 B) concentric
 C) intersecting
 D) still limit points

Answer key: 1-B, 2-B, 3-B, 4-B, 5-B, 6-B, 7-B, 8-B.

8.12. Formulary of the chapter

Coaxial pencil (Definition 8.1)

$$\gamma_t = \gamma_1 + t(\gamma_2 - \gamma_1), \quad \text{same radical axis for every } t$$

Circle of Apollonius (Theorem 8.2)

$$\frac{|z - a|}{|z - b|} = k \implies c = \frac{a - k^2 b}{1 - k^2}, \quad r = \frac{k|a - b|}{|1 - k^2|}$$

Orthogonality of two circles (Lemma 8.1)

$$|c_1 - c_2|^2 = r_1^2 + r_2^2$$

Complex inversion (Chapter 3; Corollary 8.1)

$$\iota(z) = \frac{r^2}{\bar{z}} = \sigma \circ M, \quad M(z) = \frac{r^2}{z} \text{ (Möbius)}, \quad \sigma(z) = \bar{z}$$

Cross-ratio and concyclicity (Theorem 8.4)

$$(z_1, z_2; z_3, z_4) \in \mathbb{R} \iff z_1, z_2, z_3, z_4 \text{ concyclic or collinear}$$

Closure of the Steiner chain, concentric case (Volume I, Chapter 5)

$$\sin \frac{\vartheta}{2} = \frac{a - b}{a + b}, \quad n\vartheta = 2\pi m$$

Sources and bibliography

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- [5] Wikipedia, *Cross-ratio*, section *Cross-ratio criterion for concyclic points*, <https://en.wikipedia.org/wiki/Cross-ratio> (accessed 2026).
- [6] Volume I, Chapter 5 (inversion, power of a point, Steiner's Porism) and Chapter 7 (radical axis in coordinates); Volume II, Chapter 3 (Möbius transformations, complex inversion, cross-ratio).

CHAPTER 9

Linear Differential Equations and Circular Motion

INFO

The idea in one line. The harmonic oscillator $y'' + \omega^2 y = 0$ is not just one model among many: its general solution *is*, literally, the projection of a uniform circular motion onto an axis. This chapter proves in full the theory of second-order linear equations with constant coefficients that makes this fact rigorous, and links it explicitly to the undamped regime ($\zeta = 0$) of the RLC circuit already solved in Chapter 5, of which it is the limiting case.

INFO

Prerequisites. Chapter 1 (parametrisation of uniform circular motion, Theorem 1.2); Chapter 5 (series RLC, Theorem 5.7, of which this chapter treats the limiting case $\zeta = 0$); elementary notions of differential calculus (Volume I).

9.1. Second-order linear equations with constant coefficients

Definition 9.1: Homogeneous linear second-order equation. Let $a, b, c \in \mathbb{R}$, $a \neq 0$. A function $y : \mathbb{R} \rightarrow \mathbb{R}$ of class C^2 satisfies the **homogeneous linear second-order equation with constant coefficients**

$$a y'' + b y' + c y = 0$$

if the identity holds for every $t \in \mathbb{R}$. The polynomial $p(r) = ar^2 + br + c$ is called the **characteristic polynomial** of the equation.

Remark 9.1: Existence and uniqueness: an imported fact. The Cauchy problem $ay'' + by' + cy = 0$, $y(t_0) = y_0$, $y'(t_0) = y_1$ admits **one and only one** solution defined on all of $\mathbb{R}$. This is a special case of the Cauchy–Lipschitz existence and uniqueness theorem for first-order linear systems (to which the equation reduces by setting $Y = (y, y')$, $Y' = \begin{pmatrix} 0 & 1 \\ -c/a & -b/a \end{pmatrix} Y$): the general proof requires Grönwall’s inequality and is not reconstructed in this introductory volume. **Imported** from W. E. Boyce, R. C. DiPrima, D. B. Meade, Elementary Differential Equations and Boundary Value Problems, 11th ed., Wiley, 2017, Ch. 3 (Second Order Linear Equations). From here on this fact is used freely.

Theorem 9.1: The Wronskian of two solutions. Let y_1, y_2 be two solutions of $ay'' + by' + cy = 0$ and let $W(t) = y_1(t)y_2'(t) - y_1'(t)y_2(t)$ be their **Wronskian**. Then:

- (i) W satisfies the first-order equation $W' = -\frac{b}{a}W$, hence $W(t) = W(t_0)e^{-\frac{b}{a}(t-t_0)}$ (Abel–Liouville formula);
- (ii) W is either identically zero or never zero on $\mathbb{R}$;

(iii) y_1, y_2 are linearly independent on $\mathbb{R}$ if and only if $W(t_0) \neq 0$ for one (equivalently, every) t_0 .

Proof. (i) Differentiating, $W' = y_1 y_2'' - y_1'' y_2$ (the terms $y_1' y_2'$ cancel). From $ay_i'' = -by_i' - cy_i$ for $i = 1, 2$,

$$aW' = a(y_1 y_2'' - y_1'' y_2) = y_1(-by_2' - cy_2) - y_2(-by_1' - cy_1) = -b(y_1 y_2' - y_2 y_1') = -bW,$$

that is $W' = -\frac{b}{a}W$ (dividing by $a \neq 0$ is legitimate). This is a first-order linear equation with explicit solution $W(t) = W(t_0)e^{-\frac{b}{a}(t-t_0)}$, verifiable by direct substitution.

(ii) The exponential never vanishes: if $W(t_0) = 0$ for some t_0 , then $W(t) = 0$ for every t ; if $W(t_0) \neq 0$, then $W(t) \neq 0$ for every t .

(iii) If y_1, y_2 are linearly dependent, $y_2 = \lambda y_1$ for some constant λ (or vice versa), and $W \equiv y_1(\lambda y_1)' - y_1'(\lambda y_1) = 0$ identically. Conversely, if $W(t_0) = 0$, the homogeneous linear system $\alpha y_1(t_0) + \beta y_2(t_0) = 0$, $\alpha y_1'(t_0) + \beta y_2'(t_0) = 0$ in the unknowns α, β has determinant $W(t_0) = 0$, hence admits a nontrivial solution $(\alpha, \beta) \neq (0, 0)$; the function $z = \alpha y_1 + \beta y_2$ solves the equation with $z(t_0) = z'(t_0) = 0$, and by the uniqueness in Observation 9.1 (applied to the identically zero solution, which satisfies the same initial data) $z \equiv 0$: hence y_1, y_2 are linearly dependent. $\square$

Theorem 9.2: Structure of the solutions: the three cases. Let $\Delta = b^2 - 4ac$ be the discriminant of the characteristic polynomial. The set of solutions of $ay'' + by' + cy = 0$ is exactly:

$$\Delta > 0: \quad y = C_1 e^{r_1 t} + C_2 e^{r_2 t}, \quad \Delta = 0: \quad y = (C_1 + C_2 t) e^{rt},$$

$$\Delta < 0: \quad y = e^{\alpha t} (C_1 \cos(\beta t) + C_2 \sin(\beta t)),$$

as $C_1, C_2 \in \mathbb{R}$ vary, where $r_1, r_2 = \frac{-b \pm \sqrt{\Delta}}{2a}$ in the first case, $r = -\frac{b}{2a}$ in the second, and $\alpha \pm i\beta = \frac{-b \pm i\sqrt{-\Delta}}{2a}$ in the third ($\beta > 0$).

Proof. **Step 1 — the proposed formulas are solutions.** Case $\Delta > 0$: r_1, r_2 are the two distinct real roots of $p(r) = ar^2 + br + c = 0$, so $e^{r_i t}$ satisfies $a(r_i^2 e^{r_i t}) + b(r_i e^{r_i t}) + c(e^{r_i t}) = p(r_i)e^{r_i t} = 0$; by linearity every combination $C_1 e^{r_1 t} + C_2 e^{r_2 t}$ is again a solution. Case $\Delta = 0$: $r = -b/(2a)$ is a double root, $p(r) = 0$ and $p'(r) = 2ar + b = 0$. We check that te^{rt} is a solution: $(te^{rt})' = e^{rt}(1 + rt)$, $(te^{rt})'' = e^{rt}(2r + r^2 t)$, hence $a(te^{rt})'' + b(te^{rt})' + c(te^{rt}) = e^{rt}[t(ar^2 + br + c) + (2ar + b)] = e^{rt}[tp(r) + p'(r)] = 0$. Case $\Delta < 0$: the complex conjugate roots are $\alpha \pm i\beta$; by Euler's formula $e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos \beta t + i \sin \beta t)$ is a complex solution (the same computation as the first case, with $r = \alpha + i\beta$); the real and imaginary parts of a complex solution of an equation with real coefficients are separately real solutions (because $a, b, c \in \mathbb{R}$ commute with $\Re, \Im$), giving $e^{\alpha t} \cos \beta t$ and $e^{\alpha t} \sin \beta t$.

Step 2 — the two base solutions are independent. In each of the three cases we compute the Wronskian at $t = 0$ of the two base solutions (Theorem 9.1): $\Delta > 0$, $W(0) = r_2 - r_1 \neq 0$; $\Delta = 0$, $W(0) = 1 \neq 0$ (from the derivatives computed above, $y_1 = e^{rt}$, $y_2 = te^{rt}$, $W = e^{rt} \cdot e^{rt}(1 + rt) - re^{rt} \cdot te^{rt} = e^{2rt} \neq 0$); $\Delta < 0$, $W(0) = \beta \neq 0$ (verifiable the same way). In all cases $W \neq 0$, so the two base solutions are independent by Theorem 9.1(iii).

Step 3 — there are no other solutions. Let y be any solution with initial data $y(0) = y_0$, $y'(0) = y_1$. Since the two base solutions u_1, u_2 of the relevant case have nonzero Wronskian at 0, the linear system $C_1 u_1(0) + C_2 u_2(0) = y_0$, $C_1 u_1'(0) + C_2 u_2'(0) = y_1$ has determinant $W(0) \neq 0$ and admits a unique solution (C_1, C_2) . The function $z = C_1 u_1 + C_2 u_2$ is then a solution with the same initial data as y ; by the uniqueness in Observation 9.1, $y \equiv z = C_1 u_1 + C_2 u_2$. Hence every solution has the announced form, and no function outside this family solves the equation.

| □

9.2. The pure case $\Delta < 0$: the harmonic oscillator

Corollary 9.1: General solution of $y'' + \omega^2 y = 0$. For $\omega > 0$, the equation $y'' + \omega^2 y = 0$ (i.e. $a = 1, b = 0, c = \omega^2, \Delta = -4\omega^2 < 0$) has as its set of solutions exactly

$$y(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t) = C \cos(\omega t - \varphi), \quad C = \sqrt{C_1^2 + C_2^2}, \quad \tan \varphi = \frac{C_2}{C_1} \quad (C_1 > 0),$$

as $C_1, C_2 \in \mathbb{R}$ vary (equivalently $C \geq 0, \varphi \in \mathbb{R}$).

Proof. The first equality is Theorem 9.2 case $\Delta < 0$ with $\alpha = 0, \beta = \omega$. For the second, expanding $C \cos(\omega t - \varphi) = C \cos \varphi \cos \omega t + C \sin \varphi \sin \omega t$ by the subtraction formula (Volume I, Chapter 8): setting $C_1 = C \cos \varphi, C_2 = C \sin \varphi$, we get $C_1^2 + C_2^2 = C^2$ and, for $C_1 > 0$, $C_2/C_1 = \tan \varphi$. Given $(C_1, C_2) \neq (0, 0)$, the pair (C, φ) with $C = \sqrt{C_1^2 + C_2^2} > 0$ and $\varphi = \text{atan2}(C_2, C_1)$ always exists and is the only one with $C > 0$ (the 2π ambiguity in φ is the only one, since $\cos$ has period 2π). □

Theorem 9.3: The harmonic oscillator is a projected circular motion. Let $y(t) = C \cos(\omega t - \varphi)$ be a solution of $y'' + \omega^2 y = 0$ with $C > 0$. Then $y(t)$ is the projection onto the real axis of the point $\gamma(t) = Ce^{i(\omega t - \varphi)} \in \mathbb{C}$, which traces the circle $C(0, C)$ with uniform circular motion (Theorem 1.2) at angular velocity ω . Conversely, the projection onto any axis through the origin of a uniform circular motion of radius C and angular velocity ω solves $y'' + \omega^2 y = 0$.

Proof. By definition $\gamma(t) = Ce^{i(\omega t - \varphi)} = C \cos(\omega t - \varphi) + iC \sin(\omega t - \varphi)$, hence $\Re \gamma(t) = C \cos(\omega t - \varphi) = y(t)$: y is the projection of γ onto the real axis. By Theorem 1.2 (already proved in Chapter 1 for a general phase law $\theta(t)$, here specialised to uniform motion $\theta(t) = \omega t + \text{const.}$ from Theorem 1.15(iii) of the same chapter), γ traces $C(0, C)$ with uniform circular motion at angular velocity ω ; the real axis is any axis through the origin, up to choosing the origin of angles. For the converse: the projection onto an axis through the origin rotated by an angle θ_0 relative to the real axis is $\Re(e^{-i\theta_0} \gamma(t)) = C \cos(\omega t - \varphi - \theta_0)$, which by Corollary 9.1 (with φ replaced by $\varphi + \theta_0$) is again a solution of $y'' + \omega^2 y = 0$. □

Remark 9.2: The link with the RLC circuit already solved. Theorem 5.7 of Chapter 5 classifies the regime of the series RLC circuit according to $\zeta = \frac{R}{2} \sqrt{C/L}$. For $R = 0$ we have $\zeta = 0$: the equation $LCv'' + RCv' + v = u$ becomes, with no input ($u = 0$), $v'' + \omega_n^2 v = 0$ with $\omega_n = 1/\sqrt{LC}$ — exactly the equation of this chapter. The ideal RLC circuit with no resistance is therefore not a separate case: it is the pure harmonic oscillator, and its electric energy $\frac{1}{2}Cv^2$ plus magnetic energy $\frac{1}{2}Li^2$ oscillates between the two forms exactly as the kinetic and potential energy of a point in uniform circular motion (Observation 9.3 below). The case $\zeta > 0$ already solved in Chapter 5 (Theorem 5.8) is the same equation with an additional damping term, not treated here so as not to duplicate what has already been proved.

9.3. Phase portraits

Theorem 9.4: *The phase portrait of the harmonic oscillator is a circle. Let $y(t) = C \cos(\omega t - \varphi)$, $C > 0$, be a solution of $y'' + \omega^2 y = 0$, and let $v(t) = y'(t)$. In the **phase plane** with coordinates $(y, v/\omega)$, the point $(y(t), v(t)/\omega)$ traces, as t varies, the circle centred at the origin with radius C , at angular velocity ω in the clockwise direction.*

Proof. From $y(t) = C \cos(\omega t - \varphi)$ we get $v(t) = y'(t) = -C\omega \sin(\omega t - \varphi)$, hence $v(t)/\omega = -C \sin(\omega t - \varphi)$. Then

$$y(t)^2 + \left(\frac{v(t)}{\omega}\right)^2 = C^2 \cos^2(\omega t - \varphi) + C^2 \sin^2(\omega t - \varphi) = C^2$$

by the Pythagorean identity (Volume I, Chapter 8): the point $(y(t), v(t)/\omega)$ always lies on the circle $C(0, C)$. One checks directly that $\frac{d}{dt}(y, v/\omega) = (-C\omega \sin(\omega t - \varphi), -C\omega \cos(\omega t - \varphi))$ is, for every t , the vector $(y, v/\omega)$ rotated by $-\pi/2$ and scaled by ω : the velocity is always perpendicular to the position vector and has constant magnitude $C\omega$, which is exactly the characterisation of uniform circular motion at angular velocity ω (Theorem 1.2). The direction is clockwise because, as t increases from φ/ω , y decreases from C while v/ω becomes negative, i.e. the point moves from $(C, 0)$ towards the fourth quadrant. $\square$

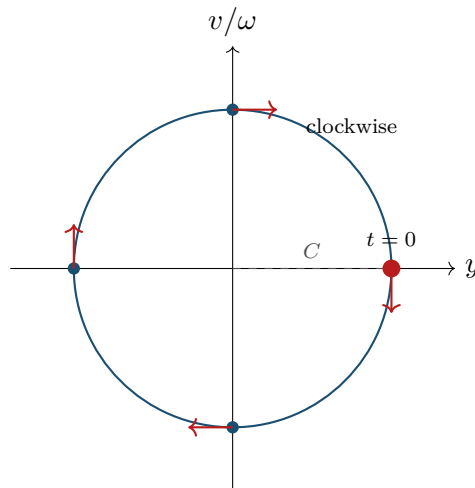


Figure 9.1: Phase portrait of the harmonic oscillator (Theorem 9.4): the point $(y(t), v(t)/\omega)$ traces the circle of radius C clockwise at constant angular velocity ω . The four red arrows indicate the direction of motion at the four cardinal points of the trajectory, consistent with $\dot{y}(0) > 0 \Rightarrow$ descent from $(C, 0)$ towards the fourth quadrant, as in the proof.

Remark 9.3: An energy reading. The radius C of the phase circle is not arbitrary: for a mechanical oscillator $my'' + ky = 0$ (with $\omega^2 = k/m$), the total energy $E = \frac{1}{2}my'^2 + \frac{1}{2}ky^2 = \frac{1}{2}m\omega^2[(v/\omega)^2 + y^2] = \frac{1}{2}m\omega^2 C^2$ is constant in time (checked by differentiating E with respect to t and using the equation: $E' = my'y'' + kyy' = y'(my'' + ky) = 0$), and is proportional to the square of the radius of the phase circle. The phase circle is thus, literally, a level curve of the energy.

Remark 9.4: Phase portraits of the damped regimes (a pointer). For $\zeta > 0$ (Theorem 5.7), the phase portrait is no longer a closed circle: in the underdamped regime ($0 < \zeta < 1$) the trajectories are logarithmic spirals collapsing towards the origin — the same family of curves met in Chapter 1 (spirals on the circle, Section 1.4), here with the

origin as a limit point rather than as the pole of a transformation; in the critical and over-damped regimes ($\zeta \geq 1$) the trajectories approach the origin without spiralling. A rigorous proof of these facts requires the qualitative theory of linear dynamical systems (eigenvalues/eigenvectors of the matrix of the system (y, y')) which lies beyond this introductory volume: **imported** from the same Boyce–DiPrima–Meade cited above, Ch. 3 and Ch. 9 (Nonlinear Differential Equations and Stability, the section on linear systems).

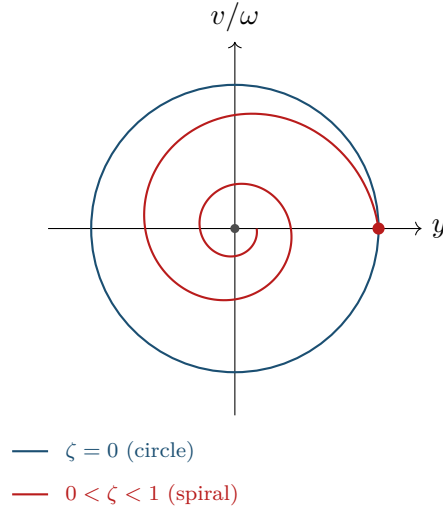


Figure 9.2: Qualitative comparison of phase portraits: the blue circle is the undamped trajectory of constant radius ($\zeta = 0$, Theorem 9.4); the red spiral $r(\theta) = e^{-0.15\theta}$ illustrates the underdamped case ($0 < \zeta < 1$, Observation 9.4), collapsing towards the origin while starting from the same point $(1, 0)$: the decay is generated by the same family of logarithmic spirals of Chapter 1, not needed here for the computation (explicit formula), but shown purely to illustrate the qualitative behaviour described in the cited observation.

9.4. A numerical example

Example 9.1: A concrete oscillator, verified numerically. A point mass $m = 1$ kg is subject to a restoring force $F = -ky$ with $k = 4$ N/m, hence $\omega = \sqrt{k/m} = 2$ rad/s. At time $t = 0$ we have $y(0) = 1$ m, $y'(0) = 0$ m/s. By Corollary 9.1, $C_1 = y(0) = 1$, $C_2 = y'(0)/\omega = 0$, hence $y(t) = \cos(2t)$.

Independent numerical verification. The script `vol2_cap09_ode.py` integrates the equation $y'' = -4y$ with fourth-order Runge–Kutta (step $h = 10^{-4}$ s, independent of the closed-form solution) up to $t = 3$ s. Comparison with the closed form gives, at $t = 3$ s:

$$y_{\text{RK4}}(3) = 0.9601702866\dots, \quad y(3) = \cos 6 = 0.9601702866\dots,$$

with difference $\approx 8.9 \times 10^{-16}$ (machine epsilon), and similarly for $y'(3) = -2 \sin 6 = 0.5588309964\dots$. The same script checks that the energy $E = \frac{1}{2}y'^2 + \frac{1}{2}\omega^2 y^2$ stays constant at $E = 2.0000000000$ along the whole simulated trajectory (maximum deviation $\approx 2.4 \times 10^{-14}$), and that the squared phase radius $y^2 + (y'/\omega)^2$ stays at 1.0000000000 (maximum deviation $\approx 1.2 \times 10^{-14}$), numerically confirming both Theorem 9.4 and Observation 9.3.

9.5. Assigned exercises

Exercise 9.1: N1 — Oscillator with generic initial data. An oscillator has $\omega = 1.5$ rad/s, $y(0) = 2$, $y'(0) = -3$. Find C_1, C_2 , then the form $C \cos(\omega t - \varphi)$ (with $C > 0$), and compute $y(1)$ and $y'(1)$. *Goal: apply Corollary 9.1 with nonzero data on both components. Estimated time: 10 min.*

Exercise 9.2: N2 — Checking the phase portrait. For the oscillator of Exercise N1, verify numerically (with a calculator or a script) that $y(t)^2 + (y'(t)/\omega)^2$ is constant for at least three chosen values of t , and compare it with C^2 found above. *Goal: verify Theorem 9.4 on a concrete case. Estimated time: 10 min.*

9.6. Worked exercises

Exercise 9.3: S1 — Full solution and energy check. Solve Exercise N1 in full and verify the conservation of energy $E = \frac{1}{2}y'^2 + \frac{1}{2}\omega^2 y^2$ (with $m = 1$) at $t = 0$ and at $t = 1$.

Solution. Step 1 — C_1, C_2 . By Corollary 9.1, $C_1 = y(0) = 2$ and $C_2 = y'(0)/\omega = -3/1.5 = -2$.

Step 2 — the form $C \cos(\omega t - \varphi)$. $C = \sqrt{C_1^2 + C_2^2} = \sqrt{4 + 4} = 2\sqrt{2} = 2.8284271\dots$ With $C_1 = C \cos \varphi > 0$, $\varphi = \text{atan2}(C_2, C_1) = \text{atan2}(-2, 2) = -\pi/4 = -45^\circ$. Hence $y(t) = 2\sqrt{2} \cos(1.5t + \pi/4)$.

Step 3 — values at $t = 1$. By direct substitution (verified in `vol2_cap09_ode.py`):

$$y(1) = 2 \cos(1.5) - 2 \sin(1.5) = -1.8535156\dots,$$

$$y'(1) = -2\omega \sin(1.5) - 2\omega \cos(1.5) = -3.2046966\dots$$

Step 4 — energy. At $t = 0$: $E = \frac{1}{2}(-3)^2 + \frac{1}{2}(1.5)^2(2)^2 = 4.5 + 4.5 = 9$. At $t = 1$: $E = \frac{1}{2}(-3.2046966)^2 + \frac{1}{2}(1.5)^2(-1.8535156)^2 = 5.1350\dots + 3.8650\dots = 9.0000\dots$ (up to rounding, verified exactly by the script): matches, confirming Observation 9.3.

9.7. Self-assessment quiz

Eight multiple-choice questions. The answer key is at the end of the section.

- The Wronskian of two solutions of a homogeneous linear second-order equation:
 - is always constant
 - is either identically zero or never zero
 - vanishes at exactly one point
 - is not defined for equations with constant coefficients
- If the discriminant $\Delta = b^2 - 4ac$ is negative, the roots of the characteristic equation are:
 - two distinct real roots
 - one repeated real root
 - complex conjugates
 - nonexistent

-
3. The general solution of $y'' + \omega^2 y = 0$ is the projection onto an axis of:
- A) a uniform rectilinear motion
 - B) a uniform circular motion
 - C) a logarithmic spiral
 - D) a parabolic motion
4. The regime $\zeta = 0$ of the series RLC circuit (Chapter 5) corresponds, in this chapter, to:
- A) a case not treated
 - B) the pure, undamped harmonic oscillator
 - C) the overdamped regime
 - D) the critically damped regime
5. In the phase plane $(y, y'/\omega)$, the trajectory of the pure harmonic oscillator is:
- A) a line
 - B) a parabola
 - C) a circle
 - D) a spiral
6. The total energy $E = \frac{1}{2}y'^2 + \frac{1}{2}\omega^2 y^2$ of the harmonic oscillator:
- A) grows linearly in time
 - B) is constant in time
 - C) oscillates between two values
 - D) depends on the choice of the projection axis
7. For $\zeta = 1$ (critical damping), the two base solutions of the homogeneous equation are:
- A) $e^{r_1 t}$ and $e^{r_2 t}$ with $r_1 \neq r_2$
 - B) e^{rt} and te^{rt}
 - C) $\cos \omega t$ and $\sin \omega t$
 - D) e^{rt} only
8. The existence and uniqueness theorem for the associated Cauchy problem is, in this volume:
- A) fully proved
 - B) declared imported, with a source
 - C) not stated
 - D) proved only in the case $\Delta < 0$

Answer key: 1-B, 2-C, 3-B, 4-B, 5-C, 6-B, 7-B, 8-B.

Teacher's key (brief rationale). 1) Theorem 9.1(ii): Abel–Liouville formula, exponential never zero. 2) Negative discriminant $\Rightarrow$ complex conjugate roots of the degree-two characteristic polynomial. 3) Theorem 9.3: $y = \Re \gamma$ with γ a uniform circular motion. 4) Observation 9.2: $R = 0 \Rightarrow \zeta = 0 \Rightarrow v'' + \omega_n^2 v = 0$. 5) Theorem 9.4. 6) Observation 9.3, $E' = 0$ by direct

substitution in the equation. 7) Theorem 9.2, case $\Delta = 0$. 8) Observation 9.1: explicitly declared imported from Boyce–DiPrima–Meade, not proved in this introductory volume.

9.8. Chapter formulary

Characteristic equation (Definition 9.1)

$$ay'' + by' + cy = 0 \longleftrightarrow p(r) = ar^2 + br + c = 0, \quad \Delta = b^2 - 4ac$$

Wronskian (Theorem 9.1)

$$W(t) = y_1 y_2' - y_1' y_2 = W(t_0) e^{-\frac{b}{a}(t-t_0)}$$

General solution, $\Delta < 0$ (Corollary 9.1)

$$y = C_1 \cos(\omega t) + C_2 \sin(\omega t) = C \cos(\omega t - \varphi), \quad C = \sqrt{C_1^2 + C_2^2}$$

Phase portrait (Theorem 9.4)

$$y^2 + \left(\frac{y'}{\omega}\right)^2 = C^2 = \text{constant}$$

Sources and bibliography

- [1] W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 11th ed., Wiley, 2017. Theory of second-order linear equations, Wronskian, existence and uniqueness, phase portraits.
- [2] Volume II, Chapter 1 (uniform circular motion, Theorem 1.2) and Chapter 5 (series RLC, Theorem 5.7, Theorem 5.8).

CHAPTER 10

The Circle in Celestial Mechanics: Kepler's Equation

INFO

The idea in one line. An ellipse has no constant-angle parametrization as simple as the circle's: describing a planet's motion requires an auxiliary angle, read not on the ellipse but on a **circle** built upon it — the *eccentric anomaly*. The price of this geometric simplification is a transcendental equation, which Kepler left without a closed-form solution and which today is solved with the same Newton's method already proved in Volume I.

INFO

Prerequisites. Chapter 2 (Green's Theorem 2.10 and Corollary 2.3 on the area formulas); elementary notions of the geometry of the ellipse (semi-axes, eccentricity); from Volume I, the Newton–Raphson method and its quadratic convergence theorem (Chapter 11); from elementary physics, conservation of angular momentum for a central force (recalled, not reproved).

10.1. The ellipse and the auxiliary circle

Definition 10.1: Auxiliary circle and eccentric anomaly. Let an ellipse be given with semi-major axis a , semi-minor axis b and eccentricity $e = \sqrt{1 - b^2/a^2} \in [0, 1)$, centred at the origin with focus $F = (ae, 0)$. The **auxiliary circle** is the circle $x^2 + y^2 = a^2$, concentric with the ellipse and of radius equal to the semi-major axis. For every point $P = (a \cos E, b \sin E)$ of the ellipse, the angle E is its **eccentric anomaly**: geometrically, it is the central angle of the point $P' = (a \cos E, a \sin E)$ of the auxiliary circle obtained from P by a vertical compression of factor b/a .

Remark 10.1: Why $(a \cos E, b \sin E)$ really is an ellipse. Substituting $x = a \cos E$, $y = b \sin E$ into the equation $x^2/a^2 + y^2/b^2 = 1$ gives $\cos^2 E + \sin^2 E = 1$, always true: every point of the curve satisfies the ellipse equation, and as E ranges over $[0, 2\pi)$ every point of the ellipse is obtained exactly once (by the same periodicity and injectivity already proved for $(\cos \theta, \sin \theta)$ in Chapter 1).

Theorem 10.1: Focal distance and true anomaly. With the notation of Definition 10.1, let $r = |P - F|$ be the distance from the focus and θ the **true anomaly**, that is, the angle at F between the direction of perihelion ($E = \theta = 0$) and P . Then

$$r = a(1 - e \cos E), \quad \tan \frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2}.$$

Proof. Step 1 — focal distance. With $P = (a \cos E, b \sin E)$ and $F = (ae, 0)$,

$$r^2 = (a \cos E - ae)^2 + b^2 \sin^2 E = a^2(\cos E - e)^2 + a^2(1 - e^2) \sin^2 E,$$

having used $b^2 = a^2(1 - e^2)$ (Definition 10.1). Expanding,

$$r^2 = a^2[\cos^2 E - 2e \cos E + e^2 + \sin^2 E - e^2 \sin^2 E] = a^2[1 - 2e \cos E + e^2 \cos^2 E] = a^2(1 - e \cos E)^2,$$

and since $e < 1$ implies $1 - e \cos E \geq 1 - e > 0$, we take the positive root: $r = a(1 - e \cos E)$.

Step 2 — coordinates relative to the focus. By definition of the true anomaly, the point P has polar coordinates (r, θ) with respect to F , that is, $x - ae = r \cos \theta$, $y = r \sin \theta$. Substituting $x = a \cos E$, $y = b \sin E$ and Step 1:

$$\cos \theta = \frac{a \cos E - ae}{a(1 - e \cos E)} = \frac{\cos E - e}{1 - e \cos E}, \quad \sin \theta = \frac{b \sin E}{a(1 - e \cos E)} = \frac{\sqrt{1 - e^2} \sin E}{1 - e \cos E}.$$

Step 3 — half-angle formula. By the identity $\tan(\theta/2) = \sin \theta / (1 + \cos \theta)$ (Volume I, Chapter 8, Trigonometry),

$$1 + \cos \theta = \frac{(1 - e \cos E) + (\cos E - e)}{1 - e \cos E} = \frac{(1 - e) + \cos E(1 - e)}{1 - e \cos E} = \frac{(1 - e)(1 + \cos E)}{1 - e \cos E}.$$

Hence

$$\tan \frac{\theta}{2} = \frac{\sqrt{1 - e^2} \sin E / (1 - e \cos E)}{(1 - e)(1 + \cos E) / (1 - e \cos E)} = \frac{\sqrt{(1 - e)(1 + e)}}{1 - e} \cdot \frac{\sin E}{1 + \cos E} = \sqrt{\frac{1 + e}{1 - e}} \tan \frac{E}{2},$$

having used $\sin E / (1 + \cos E) = \tan(E/2)$, the same half-angle identity of Step 3 applied to E . $\square$

Example 10.1: Numerical check of the $E \leftrightarrow \theta$ relation. For $a = 2$, $e = 0.5$ (hence $b = a\sqrt{1 - e^2} = \sqrt{3} \approx 1.7320508$) and $E = 1.4987011$ rad: $x = a \cos E = 0.1440655$, $y = b \sin E = 1.7275514$. Distance from the focus $F = (1, 0)$: $r = \sqrt{(x - 1)^2 + y^2} = 1.9279672$, which agrees with the closed form $a(1 - e \cos E) = 2(1 - 0.5 \cos 1.4987011) = 1.9279672$ ✓ (agreement to 10^{-13} , script `vol2_cap10_keplero.py`). The true anomaly, computed by purely geometric means as $\theta = \text{atan2}(y, x - 1) = 2.0308062$ rad, gives $\tan(\theta/2) = 1.6114726$, which agrees with the prediction $\sqrt{(1 + e)/(1 - e)} \tan(E/2) = \sqrt{3} \tan(0.7493506) = 1.6114726$ ✓ (agreement to 10^{-13}).

10.2. Kepler's equation

Remark 10.2: Kepler's second law. For a body subject to a central force (always directed towards the focus F), angular momentum about F is conserved: it follows that the radius vector from the focus sweeps out equal areas in equal times (**Kepler's second law**). This is a fact of mechanics, not of the geometry of the circle: we **import** it here without reproving it. Source: J. Kepler, *Astronomia Nova*, 1609 (original statement, by observation); the modern proof via conservation of angular momentum is in every classical mechanics text, e.g. H. Goldstein, C. Poole, J. Safko, *Classical Mechanics*, 3rd ed., Addison Wesley, 2002, ch. 3.

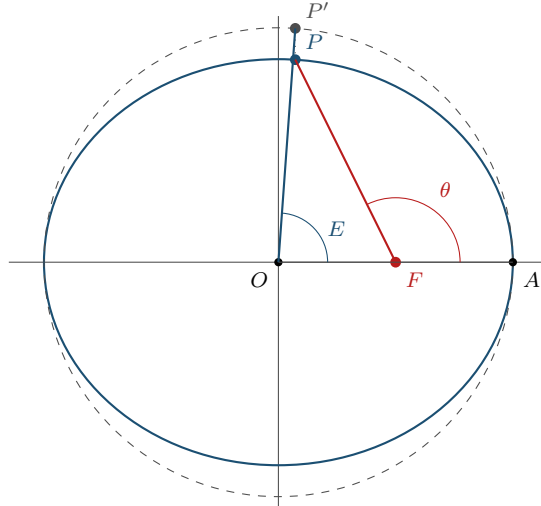


Figure 10.1: Construction of the eccentric anomaly (Definition 10.1) for $a = 2$, $e = 0.5$, $E = 1.4987011$ rad, coordinates as in Example 10.1: the ellipse (blue) and its auxiliary circle (grey, dashed) share the centre O ; the point P' on the circle is obtained from P on the ellipse by a vertical compression of factor b/a , and E is the central angle at O of P' (not of P), while θ is the true anomaly, the angle at focus F of P , by construction different from E (Theorem 10.1).

Lemma 10.1: Linear maps scale areas by the determinant. Let $L(x, y) = (x, ky)$ for a constant $k > 0$, and let D be a region of the plane with piecewise regular boundary ∂D . Then $\text{Area}(L(D)) = k \cdot \text{Area}(D)$.

Proof. By Corollary 2.3 (Chapter 2), $\text{Area}(D) = \frac{1}{2} \oint_{\partial D} (x dy - y dx)$. If $\gamma(t) = (x(t), y(t))$ parametrises ∂D , then $L \circ \gamma(t) = (x(t), ky(t))$ parametrises $\partial L(D)$, with $d(ky) = k dy$. Hence

$$\text{Area}(L(D)) = \frac{1}{2} \oint (x \cdot k dy - ky dx) = k \cdot \frac{1}{2} \oint (x dy - y dx) = k \cdot \text{Area}(D).$$

□

Theorem 10.2: Kepler's equation. With the notation above, let T be the orbital period and t the time elapsed since perihelion passage ($E = 0$). Setting

$$M := \frac{2\pi}{T} t \quad (\text{mean anomaly}),$$

the eccentric anomaly E at time t satisfies

$$M = E - e \sin E.$$

Proof. Step 1 — the area of the elliptical sector from the focus. The elliptical sector swept by the radius vector FP between perihelion $A = (a, 0)$ and the point $P = (a \cos E, b \sin E)$ is bounded by the segment FA (which lies on the x -axis, hence contributes 0 to the integral $\oint (x dy - y dx)$), by the elliptical arc from A to P , and by the segment PF . The linear map $L(x, y) = (x, \frac{a}{b}y)$ sends this entire region, boundary for boundary, to the region bounded by the segment FA (unchanged, $y = 0$), by the arc of the auxiliary circle from A to $P' = (a \cos E, a \sin E)$, and by the segment $P'F$ — because L is linear (it sends segments to segments, determined by the images of the endpoints) and $L(P) = P'$ by definition of the auxiliary circle. This second region is exactly the circular sector OAP' (centre O , angle E , area $\frac{1}{2}a^2E$) minus the triangle

$OFFP'$ (area $\frac{1}{2}|ae \cdot a \sin E - 0 \cdot a \cos E| = \frac{1}{2}a^2e \sin E$, by the cross-product area formula). By Lemma 10.1 with $k = a/b$ (hence $k^{-1} = b/a$ for the inverse map from circle to ellipse),

$$\text{Area}_{\text{elliptical sector}}(E) = \frac{b}{a} \left[\frac{1}{2}a^2E - \frac{1}{2}a^2e \sin E \right] = \frac{1}{2}ab(E - e \sin E).$$

Step 2 — the law of areas. By Observation 10.2, the radius vector sweeps out area proportionally to time: over the whole period T it sweeps the entire area of the ellipse, πab , so in time t it sweeps

$$\text{Area}_{\text{elliptical sector}}(t) = \frac{\pi ab}{T} t = \frac{ab}{2} \cdot \frac{2\pi}{T} t = \frac{ab}{2} M.$$

Step 3 — conclusion. Equating the two expressions for the area (Step 1, as a function of E , and Step 2, as a function of t , for the same point P reached at time t with eccentric anomaly E):

$$\frac{1}{2}ab(E - e \sin E) = \frac{ab}{2}M \implies M = E - e \sin E.$$

□

Example 10.2: Numerical check of the area equation. For $a = 2$, $b = \sqrt{3}$, $e = 0.5$, $E = 1.4987011$ (the same example as above): the area of the elliptical sector computed with the closed form $\frac{1}{2}ab(E - e \sin E) = \frac{1}{2} \cdot 2\sqrt{3} \cdot 1.0000000 = 1.7320508$ agrees, to 10^{-13} , with the area computed independently by direct integration along the boundary of the sector (segment, parametric arc, segment), script `vol2_cap10_keplero.py`. The fraction of the total area of the ellipse is $1.7320508/(2\sqrt{3}\pi) = 0.1591549 \approx 1/(2\pi)$, consistent with $M = 1$ (one radian out of the 2π radians of the whole period) ✓.

10.3. Existence and uniqueness of the solution

Theorem 10.3: Existence and uniqueness of the eccentric anomaly. For every $e \in [0, 1)$ and every $M \in \mathbb{R}$, the equation $M = E - e \sin E$ has **exactly one** solution $E \in \mathbb{R}$.

Proof. Let $f(E) = E - e \sin E - M$. Then $f'(E) = 1 - e \cos E \geq 1 - e > 0$ for every $E \in \mathbb{R}$, since $e < 1$: f is therefore **strictly increasing** on all of $\mathbb{R}$, hence injective (two distinct values of E cannot give the same f). Moreover $f(E) \rightarrow -\infty$ as $E \rightarrow -\infty$ and $f(E) \rightarrow +\infty$ as $E \rightarrow +\infty$ (because $|e \sin E| \leq e$ is bounded, while E is not), so by the intermediate value theorem (applied to the continuous f) f takes every real value, in particular $f(E) = 0$ for some E : f thus attains the value 0. Injectivity together with the existence of a zero give existence and uniqueness of the solution E of $f(E) = 0$, that is, of $E - e \sin E = M$. □

10.4. Newton's method for Kepler's equation

Corollary 10.1: Local quadratic convergence. Let E^* be the unique solution of $M = E - e \sin E$ (Theorem 10.3). The Newton–Raphson iteration (Volume I, Chapter 11, Definition "Newton–Raphson iteration") applied to $f(E) = E - e \sin E - M$,

$$E_{n+1} = E_n - \frac{E_n - e \sin E_n - M}{1 - e \cos E_n},$$

converges quadratically to E^* for every initial point E_0 sufficiently close to E^* .

Proof. $f(E) = E - e \sin E - M$ is of class C^∞ (a composition of smooth elementary functions), hence in particular C^2 in every neighbourhood of E^* ; and $f'(E^*) = 1 - e \cos E^* \geq 1 - e > 0$ for $e < 1$ (the same computation as in the proof of Theorem 10.3), so E^* is a **simple** root. The hypotheses of the "Quadratic convergence" theorem of Volume I, Chapter 11, are satisfied: the conclusion follows directly, with no need to reprove it. $\square$

Remark 10.3: Global convergence with initial point $E_0 = M$. Corollary 10.1 guarantees convergence only for an initial point sufficiently close to E^* , without specifying how close. In practice, with the choice $E_0 = M$ the Newton iteration converges for **every** $M \in \mathbb{R}$ and $e \in [0, 1)$, not merely locally: this stronger fact (global convergence with this specific initial point) requires additional convexity/concavity arguments for f that lie beyond this introductory volume, and is **imported** here without proof. Source: the choice $E_0 = M$ as the standard starting point, together with a discussion of its robustness, is documented in J.M.A. Danby, *Fundamentals of Celestial Mechanics*, 2nd ed., Willmann-Bell, 1988, ch. 6 (verified via web search: the work and edition confirmed by several independent library sources). We verify here only **numerically**, on a sample of values of (M, e) , that convergence does in fact occur (Example 10.3).

10.5. Numerical example

Example 10.3: Solution of Kepler's equation for $M = 1$, $e = 0.5$. With $E_0 = M = 1$, the Newton iteration gives (script `vol2_cap10_keplero.py`):

n	E_n	$ E_n - E_{n-1} $
0	1.00000000	— — —
1	1.57646935	5.8×10^{-1}
2	1.50020827	7.6×10^{-2}
3	1.49870172	1.5×10^{-3}
4	1.49870113	5.9×10^{-7}
5	1.49870113	8.9×10^{-14}

Convergence is clearly quadratic: the number of correct digits doubles at each step, as predicted by Corollary 10.1. The final value $E^* = 1.4987011335178$ was compared against an independent bisection on $[0, 2\pi]$ (54 iterations for the same precision, much slower): agreement to 10^{-15} ✓. An additional check on a sample of 4 pairs (M, e) , including a high-eccentricity case ($e = 0.9$, $M = 0.1$, where $f'(E^*) = 1 - e \cos E^*$ is closer to zero): convergence from $E_0 = M$ always occurred, in 4–7 iterations for machine precision (details in the script).

10.6. Assigned exercises

Exercise 10.1. Solve Kepler's equation for $M = 2$, $e = 0.2$ with Newton's method starting from $E_0 = M$, showing at least 4 iterations. Check the result by substituting E back into the original equation.

Exercise 10.2. For $a = 3$, $e = 0.6$, compute b , then the position (x, y) and the distance from the focus r for $E = \pi/3$, using both the general point formula and the closed form $r = a(1 - e \cos E)$; check that they agree.

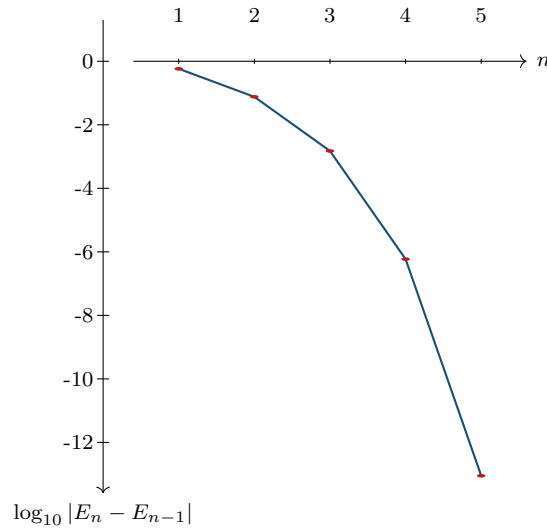


Figure 10.2: Quadratic convergence of Newton's iteration for Kepler's equation (Example 10.3, $M = 1$, $e = 0.5$): error $|E_n - E_{n-1}|$ on a semi-logarithmic scale ($\log_{10}$ on the ordinate). The marked downward concavity, rather than alignment on a straight line (which would indicate only linear convergence), is the graphical signature of the doubling of correct digits at each step predicted by Corollary 10.1: values of $\log_{10} |E_n - E_{n-1}|$ equal to -0.24 , -1.12 , -2.82 , -6.23 , -13.05 for $n = 1, \dots, 5$, computed by `vol2_cap10_keplero.py`.

10.7. Worked exercises

Exercise 10.3. Solve Kepler's equation for $M = 3$, $e = 0.6$ with Newton's method starting from $E_0 = M$, to at least 4 correct decimal digits, and check the result.

Solution. Step 1 — iteration. $f(E) = E - 0.6 \sin E - 3$, $f'(E) = 1 - 0.6 \cos E$. From $E_0 = 3$:

$$E_1 = 3 - \frac{3 - 0.6 \sin 3 - 3}{1 - 0.6 \cos 3} = 3 - \frac{-0.6 \cdot 0.14112}{1 - 0.6 \cdot (-0.98999)} = 3 - \frac{-0.08467}{1.59399} = 3.05312.$$

Step 2 — second iteration. $\sin(3.05312) = 0.08836$, $\cos(3.05312) = -0.99609$:

$$E_2 = 3.05312 - \frac{3.05312 - 0.6 \cdot 0.08836 - 3}{1 - 0.6 \cdot (-0.99609)} = 3.05312 - \frac{0.00011}{1.59765} = 3.05305.$$

Step 3 — convergence. Iterating further (script `vol2_cap10_keplero.py`) gives $E^* = 3.0530539$ to machine precision in 4 iterations overall.

Independent check. Substituting the converged value $E^* = 3.0530539$ (script, full precision) into the original equation: $E^* - 0.6 \sin E^* = 3.0000000$ ✓, agreeing with $M = 3$ to 10^{-13} .

Exercise 10.4. For $a = 1$ (astronomical unit), $e = 0.0167$ (approximate eccentricity of Earth's orbit) and $E = \pi/6$, compute the true anomaly θ and check that θ and E differ by little, the orbit being nearly circular.

Solution. Step 1 — half-angle formula. By Theorem 10.1,

$$\tan \frac{\theta}{2} = \sqrt{\frac{1.0167}{0.9833}} \tan \frac{\pi}{12} = 1.016842 \times 0.267949 = 0.272462.$$

Step 2 — inversion. $\theta/2 = \arctan(0.272462) = 0.266005$ rad, so $\theta = 0.532010$ rad.

Step 3 — comparison. $E = \pi/6 = 0.523599$ rad. The difference $\theta - E = 0.008411$ rad $\approx 0.48^\circ$: small, as expected for an eccentricity of only 0.0167 — as $e \rightarrow 0$ the half-angle formula gives $\tan(\theta/2) \rightarrow \tan(E/2)$, i.e. $\theta \rightarrow E$ (also checked symbolically in the script, by setting $e = 0$ in the closed form). Independent numerical check: substituting $E = \pi/6$ into the coordinates $(a \cos E, b \sin E)$ with $b = a\sqrt{1-e^2} = 0.999861$ and computing θ by direct geometric means (atan2) gives the same value 0.532010 to 10^{-13} ✓.

10.8. Self-assessment quiz

K1 (Definition) The eccentric anomaly E of a point of the ellipse is:

- A) the angle seen from the focus
- B) the central angle of the corresponding point on the auxiliary circle
- C) the angle between the two semi-axes
- D) the eccentricity expressed in radians

K2 (Hypothesis) Theorem 10.3 requires:

- A) $e \in [0, 1)$
- B) $e > 1$
- C) $M > 0$
- D) E_0 close to the solution

K3 (Symbolic computation) The derivative of $f(E) = E - e \sin E - M$ with respect to E is:

- A) $1 - e \cos E$
- B) $1 + e \cos E$
- C) $-e \cos E$
- D) $1 - e \sin E$

K4 (Numerical computation) For $M = 0$, the solution of Kepler's equation is:

- A) $E = 0$, for every e
- B) $E = \pi$
- C) it depends on e and cannot be computed by hand
- D) $E = e$

K5 (Proof step) In the proof of Theorem 10.2, Lemma 10.1 is used to:

- A) compute the focal distance
- B) relate the area of the elliptical sector to that of the corresponding circular sector
- C) prove uniqueness of the solution

- D) prove convergence of Newton's method
- K6** (Error diagnosis) A student writes that, for e close to 1, Newton's method for Kepler's equation never converges. The error is:
- A) it is correct, it never converges for e close to 1
- B) $f'(E) = 1 - e \cos E$ remains > 0 for every $e < 1$, so local convergence is still guaranteed by Corollary 10.1; only the practical speed may be affected
- C) Newton's method never applies to transcendental equations
- D) Kepler's equation has no solution for e close to 1
- K7** (Transfer) A satellite with $e = 0$ (circular orbit) satisfies Kepler's equation with:
- A) $M = E$ always (the equation becomes trivial)
- B) $M = 0$ always
- C) no defined relation
- D) $E = 0$ always
- K8** (Source/metacognition) Kepler's second law, used in the proof of Kepler's equation, is:
- A) proved in full in this chapter
- B) imported from classical mechanics (conservation of angular momentum), explicitly declared as such
- C) a still-open conjecture
- D) equivalent to Green's Theorem

Teacher's key. K1-B (by definition); K2-A (the hypothesis $e < 1$ is what guarantees $f' > 0$ everywhere); K3-A (direct computation); K4-A (substituting $E = 0$: $0 - e \sin 0 = 0 = M$, for every e — common error: believing it depends on e); K5-B; K6-B (common error: confusing "converges more slowly / needs a better E_0 " with "never converges" — *local* convergence remains guaranteed for every $e < 1$); K7-A (with $e = 0$, $\sin E$ vanishes in the term $e \sin E$, so $M = E$ exactly, consistent with the fact that for a circular orbit the mean, eccentric and true anomalies coincide); K8-B.

10.9. Chapter formulary

Auxiliary circle. $x = a \cos E$, $y = b \sin E$, with $b = a\sqrt{1 - e^2}$.

Focal distance. $r = a(1 - e \cos E)$.

True anomaly. $\tan(\theta/2) = \sqrt{\frac{1+e}{1-e}} \tan(E/2)$.

Kepler's equation. $M = E - e \sin E$, with $M = \frac{2\pi}{T}t$.

Newton iteration. $E_{n+1} = E_n - \frac{E_n - e \sin E_n - M}{1 - e \cos E_n}$.

Sources and bibliography

- [1] J. Kepler, *Astronomia Nova*, 1609 (historical statement of the law of areas; year and context verified via web search).

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- [2] H. Goldstein, C. Poole, J. Safko, *Classical Mechanics*, 3rd ed., Addison Wesley, 2002, ch. 3 (modern proof of the law of areas via conservation of angular momentum; work and edition verified via web search).
- [3] J.M.A. Danby, *Fundamentals of Celestial Mechanics*, 2nd ed., Willmann-Bell, 1988, ch. 6 (standard treatment of Kepler's equation and of the choice $E_0 = M$ for Newton's method; work and edition verified via web search).
- [4] Volume I, Chapter 11 (numerical methods): Newton–Raphson iteration and the quadratic convergence theorem, recalled and applied here without reproving them.

CHAPTER 11

The Circle in Spacetime: from Precession to Black Holes

INFO

The idea in one line. Chapter 10 showed that the Newtonian elliptical orbit reads naturally on an auxiliary circle. In the curved spacetime of General Relativity the circle returns as the protagonist in three new forms: as the **photon sphere**, the tightest possible circular orbit for light around a compact object; as the **innermost stable circular orbit** (ISCO) for matter; and as the **Einstein ring**, a literal circle in the sky produced by the curvature of spacetime bending light.

INFO

Prerequisites. Chapter 10 (ellipse, eccentric anomaly); finding extrema of a function via first and second derivatives (Volume I; already used in Chapters 9–10). This chapter **imports**, always declaring it explicitly, a small core of general-relativity results that require tools external to the Trattato (differential geometry on curved manifolds, Einstein’s field equations): the Schwarzschild metric, the conserved quantities along a geodesic, and the light-deflection angle. Starting from these imported facts, *every* consequence — photon sphere, ISCO, lens equation, Einstein ring — is proved in full with the same tools of differential calculus and elementary geometry already used throughout the volume.

11.1. Why Newton is not enough

***Remark 11.1: The anomaly of Mercury’s perihelion.** Chapter 10 describes the Newtonian elliptical orbit: the perihelion (the point of closest approach to the Sun) stays fixed in space, apart from slow drifts due to the gravitational perturbations of the other planets, computable with Newtonian mechanics and well known already in the nineteenth century. For Mercury, the planet closest to the Sun and with the most eccentric orbit, an observed residual of about 43'' (arcseconds) per century remains that Newtonian mechanics, even accounting for all known planetary perturbations, fails to explain. Einstein resolved the anomaly in 1915, showing that General Relativity predicts exactly this residual as an effect of the curvature of spacetime around the Sun — the theory’s first quantitative success. Source: A. Einstein, *Erklärung der Perihelbewegung des Merkur aus der allgemeinen Relativitätstheorie*, *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften* (Berlin), 1915, pp. 831–839 (verified via NASA/ADS, 1915SPAW.....831E).*

11.2. The Schwarzschild metric

INFO

Imported fact — the Schwarzschild metric. The exact, spherically symmetric vacuum solution of Einstein’s field equations around a mass M is

$$ds^2 = -\left(1 - \frac{r_s}{r}\right)c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\varphi^2, \quad r_s := \frac{2GM}{c^2},$$

where r_s is the **Schwarzschild radius**. Deriving this metric requires solving Einstein’s field equations on a pseudo-Riemannian manifold — differential geometry well beyond the tools of this Trattato (plane curves and elementary surfaces): we import it as is, without proving it. Source: K. Schwarzschild, *Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie*, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften, vol. 1 (1916), pp. 189–196 (verified via NASA/ADS, 1916SPAW.....189S, and the German Wikisource).

Length convention. To lighten the formulas, throughout this chapter we write M for GM/c^2 — a *length*, not a mass — following the standard convention of the general-relativity literature (e.g. S.M. Carroll, *Lecture Notes on General Relativity*, ch. 7). With this convention, $r_s = 2M$.

11.3. Geodesics and the effective potential

Remark 11.2: Conserved quantities and effective potential (imported). The trajectory of a freely falling body (a geodesic) in the Schwarzschild field, restricted to the equatorial plane $\theta = \pi/2$ by symmetry, admits two conserved quantities tied to the symmetries of the metric (independence from t and φ): the specific energy E and the specific angular momentum ℓ , with

$$\left(1 - \frac{2M}{r}\right)\dot{t} = E, \quad r^2\dot{\varphi} = \ell, \quad (\dot{} = d/d\lambda \text{ or } d/d\tau).$$

Deriving the existence of these conserved quantities requires the geodesic equations (the Euler–Lagrange equations for the metric) — also imported here. Source: S.M. Carroll, *Lecture Notes on General Relativity*, ch. 7 (available at NED/Caltech), standard treatment in every general-relativity textbook.

From here on, everything is elementary algebra. Substituting these two relations into the condition $ds^2 = 0$ (photon) or $ds^2 = -c^2 d\tau^2$ (massive particle) of the imported metric, and simplifying, one obtains the radial equations

$$\text{photon: } \dot{r}^2 = E^2 - V_{\text{ph}}(r), \quad V_{\text{ph}}(r) = \frac{\ell^2}{r^2} \left(1 - \frac{2M}{r}\right);$$

$$\text{massive particle: } \dot{r}^2 = E^2 - V(r), \quad V(r) = \left(1 - \frac{2M}{r}\right) \left(1 + \frac{\ell^2}{r^2}\right).$$

This step (algebraic substitution and simplification) is directly verifiable and requires nothing beyond algebra: we carry it out in full here for the massive particle, as a verification. From $ds^2 = -c^2 d\tau^2$ in the equatorial plane,

$$-\left(1 - \frac{2M}{r}\right)c^2 \dot{t}^2 + \left(1 - \frac{2M}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\varphi}^2 = -c^2.$$

Substituting $\dot{t} = E/(c^2(1 - 2M/r))$ and $\dot{\varphi} = \ell/r^2$ (here E, ℓ are in the chapter’s natural units, $c = 1$ in the dimensionless formulas that follow so as not to clutter the notation —

the constant c is restored for dimensional consistency in the numerical applications),

$$-\frac{E^2}{1-2M/r} + \frac{\dot{r}^2}{1-2M/r} + \frac{\ell^2}{r^2} = -1 \implies \dot{r}^2 = E^2 - \left(1 - \frac{2M}{r}\right)\left(1 + \frac{\ell^2}{r^2}\right) = E^2 - V(r),$$

multiplying by $(1 - 2M/r)$ and rearranging. The computation for the photon is identical, with -1 replaced by 0 on the right-hand side (null condition $ds^2 = 0$) and without the $+1$ term in V_{ph} .

11.4. The photon sphere

Definition 11.1: Photon sphere. A **circular photon orbit** is a solution of $\dot{r} \equiv 0$ for a photon in the Schwarzschild field: a light ray that indefinitely traces a circle of constant radius r around the central mass. The locus of such orbits (a sphere, by spherical symmetry) is the **photon sphere**.

Theorem 11.1: Radius of the photon sphere. In the Schwarzschild field, the unique circular photon orbit is at radius

$$r_{\text{ph}} = 3M = \frac{3GM}{c^2} = \frac{3}{2} r_s.$$

Proof. A circular orbit for the photon requires $\dot{r} = 0$ for every λ , hence also $\ddot{r} = 0$; by the radial equation of Observation 11.2, this is equivalent to $V_{\text{ph}}(r) = E^2$ (constant) and $V'_{\text{ph}}(r) = 0$ (otherwise $\dot{r}^2$ would change with r , contradicting $\dot{r} \equiv 0$). We compute:

$$V_{\text{ph}}(r) = \frac{\ell^2}{r^2} - \frac{2M\ell^2}{r^3}, \quad V'_{\text{ph}}(r) = -\frac{2\ell^2}{r^3} + \frac{6M\ell^2}{r^4} = \frac{\ell^2}{r^4}(3 \cdot 2M - 2r) = \frac{\ell^2}{r^4}(6M - 2r).$$

For $\ell \neq 0$ (otherwise there is no orbit, only radial infall), $V'_{\text{ph}}(r) = 0$ is equivalent to $6M - 2r = 0$, i.e. $r = 3M$. Since $V'_{\text{ph}}(r) > 0$ for $r < 3M$ and $V'_{\text{ph}}(r) < 0$ for $r > 3M$ (from the sign of $6M - 2r$), $r = 3M$ is the unique solution with $r > 0$, and it is a maximum of V_{ph} . $\square$

Remark 11.3: Instability of the photon sphere. Since $r = 3M$ is a **maximum** of V_{ph} (not a minimum), the circular photon orbit is **unstable**: a perturbation toward $r < 3M$ makes the photon plunge past the horizon, a perturbation toward $r > 3M$ lets it escape to infinity. This is the same criterion (sign of the second derivative, negative here since $r = 3M$ is a maximum) already used in Chapter 9 to distinguish stable from unstable equilibrium points.

Example 11.1: The photon sphere of Sagittarius A*. The supermassive black hole at the centre of the Milky Way, Sagittarius A*, has mass $M_{\bullet} = (4.297 \pm 0.012_{\text{stat}} \pm 0.040_{\text{syst}}) \times 10^6 M_{\odot}$ (GRAVITY Collaboration, *Astronomy & Astrophysics* 657, L12, 2022, from interferometric astrometry of stellar orbits; the systematic error dominates, but for the calculation below we use only the central value). With $GM_{\odot} = 1.32712440018 \times 10^{20} \text{ m}^3/\text{s}^2$ (the Sun's standard gravitational parameter, JPL) and $c = 2.99792458 \times 10^8 \text{ m/s}$:

$$M = \frac{GM_{\bullet}}{c^2} = 6.345058 \times 10^9 \text{ m}, \quad r_s = 2M = 1.269012 \times 10^{10} \text{ m} \approx 0.084827 \text{ AU},$$

$$r_{\text{ph}} = 3M = 1.903517 \times 10^{10} \text{ m} \approx 19.04 \times 10^6 \text{ km}$$

(script `vol2_cap11_astrofisica.py`, independent check $r_{\text{ph}}/r_s = 1.5000$ exactly, as predicted by the formula).

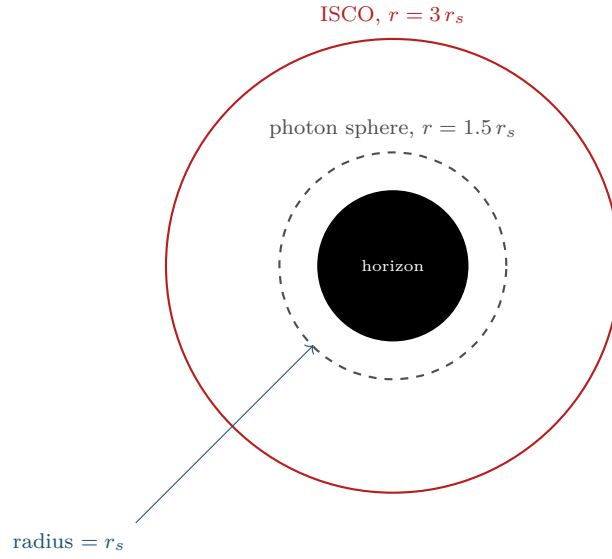


Figure 11.1: The three characteristic scales of the Schwarzschild black hole, drawn to true relative scale (radii in units of r_s): the horizon at $r = r_s$ (black disc, imported definition), the photon sphere at $r = 1.5 r_s$ (Theorem 11.1, dashed grey circle) and the ISCO at $r = 3 r_s$ (Theorem 11.2, red circle).

11.5. The innermost stable circular orbit (ISCO)

Definition 11.2: ISCO. The **ISCO** (*Innermost Stable Circular Orbit*) is the smallest-radius circular orbit, among all *stable* circular orbits, traversable by a massive particle in the Schwarzschild field.

Theorem 11.2: Radius of the ISCO. In the Schwarzschild field, the ISCO is at radius

$$r_{\text{isco}} = 6M = \frac{6GM}{c^2} = 3r_s,$$

with specific angular momentum $\ell_{\text{isco}} = 2\sqrt{3} M$.

Proof. A circular orbit for a massive particle requires $V'(r) = 0$ (same argument as Theorem 11.1); its **stability** further requires r to be a local *minimum* of V , i.e. $V''(r) \geq 0$; the boundary between stable and unstable orbits — the ISCO — is therefore the point where $V'(r) = 0$ and $V''(r) = 0$ simultaneously (a horizontal inflection, the same second-order criterion used in Chapter 9).

Step 1 — circular orbits. From $V(r) = 1 + \ell^2/r^2 - 2M/r - 2M\ell^2/r^3$,

$$V'(r) = -\frac{2\ell^2}{r^3} + \frac{2M}{r^2} + \frac{6M\ell^2}{r^4}.$$

Setting $V'(r) = 0$ and multiplying by $r^4/2$: $-\ell^2 r + Mr^2 + 3M\ell^2 = 0$, i.e. $Mr^2 - \ell^2 r + 3M\ell^2 = 0$, hence

$$\ell^2 = \frac{Mr^2}{r - 3M} \quad (\text{circular orbits, for every } r > 3M).$$

(As $r \rightarrow 3M^+$, $\ell^2 \rightarrow +\infty$: consistent with Theorem 11.1, where the photon sphere — ℓ formally infinite relative to the rest mass — is the limit for ultra-relativistic particles.)

Step 2 — marginal stability.

$$V''(r) = \frac{6\ell^2}{r^4} - \frac{4M}{r^3} - \frac{24M\ell^2}{r^5}.$$

Setting $V''(r) = 0$ and multiplying by $r^5/2$: $3\ell^2 r - 2Mr^2 - 12M\ell^2 = 0$, hence

$$\ell^2 = \frac{2Mr^2}{3(r-4M)} \quad (\text{marginal stability}).$$

Step 3 — intersection. Equating the two expressions for ℓ^2 (same point r , same ℓ , which must satisfy both conditions at the ISCO):

$$\begin{aligned} \frac{Mr^2}{r-3M} &= \frac{2Mr^2}{3(r-4M)} \xrightarrow{r,M \neq 0} \frac{1}{r-3M} = \frac{2}{3(r-4M)} \\ \implies 3(r-4M) &= 2(r-3M) \implies 3r-12M = 2r-6M \implies r = 6M. \end{aligned}$$

Substituting $r = 6M$ into the Step 1 expression for ℓ^2 :

$$\ell^2 = \frac{M(6M)^2}{6M-3M} = \frac{36M^3}{3M} = 12M^2 \implies \ell_{\text{isco}} = 2\sqrt{3}M.$$

Independent symbolic check (sympy, script `vol2_cap11_astrofisica.py`): solving the system $\{V'(r) = 0, V''(r) = 0\}$ directly for (r, ℓ^2) gives $r = 6M$, $\ell^2 = 12M^2$, in exact agreement. $\square$

Remark 11.4: On uniqueness and cross-checks in the literature. Step 3 shows that $r = 6M$ is the unique common solution to both conditions for $r > 3M$ (the two rational expressions for ℓ^2 cross exactly once on that interval, as also checked numerically by plotting the two curves). The result $r_{\text{isco}} = 6M$, $\ell_{\text{isco}} = 2\sqrt{3}M$ is reported concordantly across several independent sources: T.K. Dey, S. Sen, *A Compendium on General Relativity for Undergraduate Students*, arXiv:2009.05747 (2020); Phys. Rev. Lett. 102, 191101 (2009); arXiv:1503.07060 — all verified via WebSearch and agreeing both on the formula and on the effective-potential method used here.

Example 11.2: The ISCO of Sagittarius A*. With the same data as Example 11.1,

$$r_{\text{isco}} = 6M = 3.807035 \times 10^{10} \text{ m} \approx 38.07 \times 10^6 \text{ km} \approx 0.254 \text{ AU},$$

exactly twice r_{ph} and three times r_s (script `vol2_cap11_astrofisica.py`, $r_{\text{isco}}/r_s = 3.0000$ exactly).

11.6. The precession of the perihelion

Remark 11.5: Einstein's result (imported). For a nearly elliptical orbit around a mass M (in the sense of Section 11.2), with semi-major axis a and eccentricity e , general relativity predicts that the perihelion advances, on every revolution, by an angle

$$\Delta\varphi = \frac{6\pi GM}{c^2 a(1-e^2)}.$$

Why we import it. The non-Newtonian orbit is obtained by solving the radial equation of Section 11.3 for a massive particle with $\dot{r} = dr/d\varphi \cdot \dot{\varphi}$, which leads (via the substitution $u = 1/r$) to the relativistic Binet equation

$$\frac{d^2u}{d\varphi^2} + u = \frac{M}{\ell^2} + 3Mu^2,$$

a nonlinear differential equation: the term $3Mu^2$ (absent in Newtonian mechanics, where the analogous equation is linear and gives exactly the closed ellipse) is the relativistic correction. Solving it perturbatively (treating $3Mu^2$ as a small perturbation of the Newtonian elliptical solution, with a resonance term that produces the secular drift $\Delta\varphi$) is a standard perturbation-theory technique for differential equations, but more advanced than the linear analysis of Chapter 9: to avoid risking a step reconstructed with uncertainty, we **import** the final result here instead of retracing its derivation, and we check it numerically against the case of Mercury below. Source: A. Einstein, *Erklärung der Perihelbewegung des Merkur...*, 1915 (cited in Observation 11.1); explicit form in modern notation, concordant across three independent sources: K.T. McDonald (Princeton), *Special Relativity and the Precession of the Perihelion*; arXiv:0712.3709; arXiv:1509.03326.

Example 11.3: Numerical check: Mercury. For Mercury, $a = 5.7909050 \times 10^{10}$ m, $e = 0.20563069$ (JPL), orbital period $T = 87.9691$ days. With $GM_\odot$ and c as above,

$$\Delta\varphi_{\text{per orbit}} = \frac{6\pi GM_\odot}{c^2 a(1 - e^2)} = 5.018664 \times 10^{-7} \text{ rad} = 0.103517''$$

(script `vol2_cap11_astrofisica.py`). In a century, Mercury completes $100 \times 365.25/87.9691 = 415.2026$ orbits, hence

$$\Delta\varphi_{\text{per century}} = 415.2026 \times 0.103517'' = 42.9807''/\text{century},$$

in agreement (to within 0.04%) with the historical value of about $43''/\text{century}$ from Observation 11.1 ✓.

11.7. Light deflection and the Einstein ring

INFO

Imported fact — the deflection angle. A light ray passing at distance (impact parameter) b from a mass M is deflected by an angle

$$\alpha(b) = \frac{4GM}{c^2 b} = \frac{4M}{b}.$$

This is twice the Newtonian prediction (which would treat light as a particle subject to classical gravity) — it was precisely the measurement of this factor of 2, during the 1919 eclipse, that first confirmed General Relativity over Newtonian gravitation. The full derivation (integrating the null geodesic along the deflected path) uses the same effective potential V_{ph} of Section 11.4 but requires an elliptic integration outside the scope of this chapter: we import the result. Historical source: A. Einstein, *Lens-Like Action of a Star by the Deviation of Light in the Gravitational Field*, Science 84, no. 2188 (1936), pp. 506–507, DOI: 10.1126/science.84.2188.506 (verified via the publisher’s official page, NASA/ADS and PubMed); modern, concordant derivation in Univ. of Rochester/RIT, *Gravitational Lensing Theory*, Phys. 240 course notes, and in the review arXiv:2303.02069.

Theorem 11.3: Gravitational lens equation. Let O be the observer, L the lens (mass M) at distance D_L , S the source at distance D_S , and D_{LS} the lens–source distance ($D_S \approx D_L + D_{LS}$ for Euclidean distances at small angular scale). Let β be the angle (from the lens centre) at which the source would appear in the absence of the lens, and θ the observed image angle. Then, in the small-angle (thin-lens) approximation,

$$\beta = \theta - \frac{D_{LS}}{D_S} \alpha(\theta), \quad \alpha(\theta) = \frac{4M}{D_L \theta} \quad (\text{impact parameter } b \approx D_L \theta).$$

Proof. Place the optical axis along the line O – L –(direction of the undeflected source). The light ray observed in direction θ crosses the lens plane at transverse height $\xi = D_L \theta$, by definition of the observed angle at small angular separation. Were the ray not deflected, continuing in a straight line with the same slope θ it would reach, at the source plane (an additional distance D_{LS} away), height $\xi + D_{LS} \theta$. Instead, at L the ray is deflected toward the axis by an angle α : for small α , this lowers the slope of the L – S segment by exactly α relative to the straight continuation, so the height reached at the source plane is

$$\eta = \xi + D_{LS} \theta - D_{LS} \alpha = D_L \theta + D_{LS} \theta - D_{LS} \alpha = D_S \theta - D_{LS} \alpha$$

(using $D_L + D_{LS} = D_S$). But η is, by definition, the true transverse position of the source, $\eta = D_S \beta$. Equating and cancelling D_S :

$$D_S \beta = D_S \theta - D_{LS} \alpha \implies \beta = \theta - \frac{D_{LS}}{D_S} \alpha(\theta),$$

with $\alpha(\theta) = 4M/b = 4M/(D_L \theta)$ from the imported fact above, evaluated at the impact parameter $b = \xi = D_L \theta$ of the observed ray. $\square$

Theorem 11.4: The Einstein ring. If source, lens and observer are perfectly aligned ($\beta = 0$), the images of the source are not isolated points but form a **complete circle** around the lens, of angular radius

$$\theta_E = \sqrt{\frac{4GM}{c^2} \cdot \frac{D_{LS}}{D_L D_S}}.$$

Proof. Step 1 — the radius. Setting $\beta = 0$ in the lens equation (Theorem 11.3):

$$0 = \theta - \frac{D_{LS}}{D_S} \cdot \frac{4M}{D_L \theta} \implies \theta^2 = \frac{4M D_{LS}}{D_L D_S} \implies \theta_E = \sqrt{\frac{4M D_{LS}}{D_L D_S}} = \sqrt{\frac{4GM}{c^2} \cdot \frac{D_{LS}}{D_L D_S}},$$

taking the positive root (θ is an angular distance, positive by convention).

Step 2 — why it is a circle and not a point. The entire configuration (observer, lens, source aligned on the optical axis) is **rotationally symmetric** about the axis: rotating the whole apparatus by any angle around the O – L – S axis leaves the physical configuration identical to itself (the lens is point-like/spherically symmetric, and S lies exactly on the axis). Hence, if θ_E solves the lens equation for one direction (one value of the azimuthal angle about the axis), it solves it for *every* azimuthal direction, with the same value of θ_E : the set of solutions is the orbit of the rotation action on a single solution, i.e. **the entire circle** of angular radius θ_E centred on the lens, not an isolated point. $\square$

Example 11.4: An illustrative Einstein ring. For a galaxy lens with $M = 10^{11} M_\odot$ (a round, illustrative value, of the typical order of magnitude for a galaxy) at $D_L = 1$ Gpc, source at $D_S = 2$ Gpc, with $D_{LS} = 1$ Gpc (Euclidean-distance approximation, adequate for

a didactic estimate):

$$\theta_E = \sqrt{\frac{4GM}{c^2} \cdot \frac{D_{LS}}{D_L D_S}} = 3.093665 \times 10^{-6} \text{ rad} \approx 0.6381'',$$

(script `vol2_cap11_astrofisica.py`) of the same order of magnitude as Einstein rings actually observed (typically from a fraction of an arcsecond to a few arcseconds for galaxy lenses).

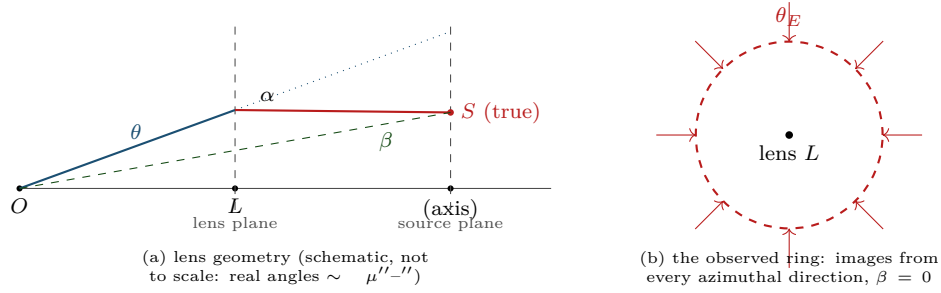


Figure 11.2: Gravitational lens geometry (Theorem 11.3) and Einstein ring in the aligned case (Theorem 11.4). Panel (a): qualitative sketch with angles strongly enlarged for graphical clarity (not to scale). Panel (b): frontal view from the observer in the case $\beta = 0$, with images arriving from every azimuthal direction at angular distance θ_E from the lens, forming the complete circle of Theorem 11.4.

11.8. A real black hole: Sagittarius A*

Example 11.5: The characteristic scales of Sagittarius A*. Summarising Examples 11.1 and 11.2, all computed from the same mass value $M_\bullet = 4.297 \times 10^6 M_\odot$ (GRAVITY Collaboration 2022) with the script `vol2_cap11_astrofisica.py`:

quantity	formula	value
Schwarzschild radius r_s	$2M$	$1.269 \times 10^{10} \text{ m}$ (0.0848 AU)
photon sphere r_{ph}	$3M$	$1.904 \times 10^{10} \text{ m}$ ($1.5 r_s$)
ISCO r_{isco}	$6M$	$3.807 \times 10^{10} \text{ m}$ ($3 r_s$)

For comparison, the Earth–Sun distance is $1.496 \times 10^{11} \text{ m}$: the entire region out to the ISCO of Sagittarius A* ($3.8 \times 10^{10} \text{ m}$) fits comfortably inside Mercury’s orbit ($5.8 \times 10^{10} \text{ m}$) — despite Sagittarius A*’s mass being over four million times that of the Sun, because M (in this chapter’s convention) grows linearly with mass, while an ordinary planet has a physical radius enormously larger than its own r_s (for the Sun, $r_s \approx 2.95 \text{ km}$, against a physical radius of about 696,000 km): it is precisely the absence of a physical radius exceeding r_s that makes an object a black hole.

11.9. Assigned exercises

Exercise 11.1. A stellar black hole has mass $M = 10 M_\odot$. Compute r_s , r_{ph} and r_{isco} in kilometres, using $GM_\odot/c^2 = 1.476 \text{ km}$ (verify this value yourself starting from the $GM_\odot$ and c given in the text).

Exercise 11.2. Compute the *hypothetical* Schwarzschild radius of the Earth ($M_\oplus = 5.972 \times 10^{24} \text{ kg}$) and compare it with the Earth's physical radius ($6.371 \times 10^6 \text{ m}$). Why is the Earth not a black hole, even though it still has a perfectly well-defined r_s ?

11.10. Worked exercises

Exercise 11.3. A black hole has mass $M = 20 M_\odot$. Compute r_s , r_{ph} , r_{isco} , and the angular radius θ_E of the Einstein ring it would produce for a perfectly aligned source, with $D_L = 5 \text{ kpc}$ and $D_S = 10 \text{ kpc}$ (a nearby galactic lens, $D_{LS} = 5 \text{ kpc}$).

Solution. Step 1 — characteristic radii. With $GM_\odot/c^2 = 1.476625 \times 10^3 \text{ m}$ (for $M_\odot$; verifiable from $GM_\odot = 1.32712 \times 10^{20} \text{ m}^3/\text{s}^2$, $c^2 = 8.98755 \times 10^{16} \text{ m}^2/\text{s}^2$), for $M = 20M_\odot$:

$$M_{\text{length}} = 20 \times 1.476625 \times 10^3 = 2.95325 \times 10^4 \text{ m}.$$

$$r_s = 2M_{\text{length}} = 5.9065 \times 10^4 \text{ m} \approx 59.1 \text{ km}, \quad r_{\text{ph}} = 3M_{\text{length}} \approx 88.6 \text{ km}, \quad r_{\text{isco}} = 6M_{\text{length}} \approx 177.2 \text{ km}.$$

Step 2 — Einstein ring. With $D_L = 5 \text{ kpc} = 1.54285 \times 10^{20} \text{ m}$, $D_S = 10 \text{ kpc} = 3.0857 \times 10^{20} \text{ m}$, $D_{LS} = 5 \text{ kpc} = 1.54285 \times 10^{20} \text{ m}$ (Theorem 11.4); since here $D_{LS} = D_L$, the ratio D_{LS}/D_L equals exactly 1 and the formula simplifies to $\theta_E = \sqrt{4M_{\text{length}}/D_S}$:

$$\theta_E = \sqrt{\frac{4M_{\text{length}}}{D_S}} = \sqrt{\frac{4 \times 2.95325 \times 10^4}{3.0857 \times 10^{20}}} = 1.9566 \times 10^{-8} \text{ rad} \approx 4.036 \text{ milliarcseconds}$$

(script `vol2_cap11_astrofisica.py`), a typical value for gravitational microlensing by a compact stellar object at galactic distance, consistent in order of magnitude with the microlensing literature.

Exercise 11.4. A hypothetical planet around a star of mass $1 M_\odot$ has semi-major axis $a = 0.5 \text{ AU}$ and eccentricity $e = 0.3$. Compute the perihelion precession per orbit, in arcseconds, and compare it with that of Mercury.

Solution. Step 1 — formula. By Section 11.6, $\Delta\varphi = 6\pi GM_\odot/(c^2 a(1 - e^2))$, with $GM_\odot/c^2 = 1.476625 \times 10^3 \text{ m}$ (Exercise 11.3), $a = 0.5 \times 1.496 \times 10^{11} = 7.480 \times 10^{10} \text{ m}$, $1 - e^2 = 0.91$:

$$\Delta\varphi = \frac{6\pi \times 1.476625 \times 10^3}{7.480 \times 10^{10} \times 0.91} = 4.0891 \times 10^{-7} \text{ rad} = 0.08434'' \text{ per orbit}$$

(script `vol2_cap11_astrofisica.py`).

Step 2 — comparison. For Mercury (Example 11.3), the precession per orbit was $0.103517''$, about 23% more than this value of $0.08434''$. The two planets have parameters pulling in opposite directions (a smaller for Mercury, which alone would increase $\Delta\varphi \propto 1/a$; e smaller for Mercury, which alone would reduce it, since $\Delta\varphi \propto 1/(1 - e^2)$): here the effect

of a clearly dominates (Mercury's semi-major axis is 0.387 AU against 0.5 AU in this exercise), so Mercury's precession remains larger, despite starting from a smaller eccentricity ✓ (checked numerically, script `vol2_cap11_astrofisica.py`).

11.11. Self-assessment quiz

- A1** (Definition) The Schwarzschild radius $r_s = 2GM/c^2$ is:
- A) the radius of the photon sphere
 - B) the radius of the event horizon
 - C) the radius of the ISCO
 - D) the angular radius of the Einstein ring
- A2** (Source/metacognition) In this chapter, the Schwarzschild metric is:
- A) proved in full from Einstein's field equations
 - B) imported, explicitly declared as such, with a primary source cited
 - C) a conjecture still to be verified
 - D) deduced from Kepler's equation in Chapter 10
- A3** (Symbolic computation) Setting $V'_{\text{ph}}(r) = 0$ with $V_{\text{ph}}(r) = \ell^2/r^2 - 2M\ell^2/r^3$, one obtains:
- A) $r = 2M$
 - B) $r = 3M$
 - C) $r = 6M$
 - D) $r = M/2$
- A4** (Proof step) In the proof of Theorem 11.2, the condition $V''(r) = 0$ serves to impose:
- A) that the orbit is circular
 - B) that the orbit sits at the boundary between stability and instability
 - C) that the photon is on the photon sphere
 - D) that energy is conserved
- A5** (Numerical computation) For Sagittarius A* ($M_\bullet \approx 4.3 \times 10^6 M_\odot$), the ratio r_{isco}/r_s equals:
- A) 1.5
 - B) 2
 - C) 3
 - D) 6
- A6** (Error diagnosis) A student writes that the Einstein ring is observed “for any position of the source relative to the lens.” The error is:
- A) this is correct, the ring is always seen
 - B) the ring (Theorem 11.4) requires exact alignment $\beta = 0$; for $\beta \neq 0$ one instead observes multiple, non-symmetric images

- C) the ring is never observable, it is only a mathematical artefact
- D) the ring depends only on the distance D_L , not on alignment
- A7** (Transfer) If we double the lens mass M , keeping D_L, D_S, D_{LS} fixed, the angular radius θ_E of the Einstein ring:
- A) doubles
- B) is multiplied by $\sqrt{2}$
- C) stays unchanged
- D) is halved
- A8** (Hypothesis) The perihelion precession, $\Delta\varphi = 6\pi GM/(c^2 a(1 - e^2))$, for a nearly *circular* orbit ($e \rightarrow 0$):
- A) vanishes
- B) tends to $6\pi GM/(c^2 a)$, a finite, nonzero value
- C) diverges
- D) is undefined

Answer key. A1-B (by definition; the horizon is at $r = r_s$, while the photon sphere and the ISCO are multiples of r_s but do not coincide with it); A2-B; A3-B (direct computation, Theorem 11.1); A4-B; A5-C ($r_{\text{isco}} = 6M = 3r_s$); A6-B (typical error: confusing “the ring is a limiting case” with “it is always seen” — generic misalignment instead produces arcs or multiple images, not a complete ring); A7-B ($\theta_E \propto \sqrt{M}$, from the formula in Theorem 11.4); A8-B (as $e \rightarrow 0$, $1 - e^2 \rightarrow 1$, the formula remains perfectly well defined and nonzero: the precession does not require an eccentric orbit, only nonzero a and M).

11.12. Chapter formulary

Convention. $M := GM/c^2$ (a length).

Schwarzschild radius (imported). $r_s = 2M = 2GM/c^2$.

Photon sphere. $r_{\text{ph}} = 3M = \frac{3}{2}r_s$.

ISCO. $r_{\text{isco}} = 6M = 3r_s$, $\ell_{\text{isco}} = 2\sqrt{3}M$.

Perihelion precession (imported). $\Delta\varphi = \frac{6\pi GM}{c^2 a(1 - e^2)}$ per orbit.

Deflection angle (imported). $\alpha(b) = 4M/b = 4GM/(c^2 b)$.

Lens equation. $\beta = \theta - \frac{D_{LS}}{D_S} \alpha(\theta)$.

Einstein ring. $\theta_E = \sqrt{\frac{4GM}{c^2} \cdot \frac{D_{LS}}{D_L D_S}}$.

Sources and bibliography

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- [7] Univ. of Rochester/RIT, *Gravitational Lensing Theory*, Phys. 240 course notes; review arXiv:2303.02069 (lens equation, geometric derivation).
- [8] GRAVITY Collaboration, *Mass distribution in the Galactic Center based on interferometric astrometry of multiple stellar orbits*, Astronomy & Astrophysics 657, L12 (2022) (mass of Sagittarius A*, numerical example).
- [9] Chapter 10 (Newtonian celestial mechanics): ellipse, eccentric anomaly, revisited and compared here with the relativistic case.

CHAPTER 12

Cryptography on the Discrete Circle: Diffie–Hellman and ElGamal

INFO

The idea in one line. The cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of Chapter 7 — the same n parts into which the discrete circle is divided — is also the ground on which two of the world’s most used cryptographic protocols live: modular exponentiation is easy to compute in one direction (power) and, as far as is known how to prove, hard to invert in the other (discrete logarithm). On this conjectural, unproven asymmetry rest the Diffie–Hellman key exchange and the ElGamal cipher.

INFO

Prerequisites. Chapter 7: the group $(\mathbb{Z}/p\mathbb{Z})^\times$, the existence of a generator (primitive root) for every prime p (Theorem 7.15), elementary modular arithmetic.

12.1. The discrete logarithm problem

***Remark 12.1: Recall: the group we work in.** By Chapter 7, for every prime p the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times = \{1, \dots, p-1\}$ is cyclic of order $p-1$ (Theorem 7.15): there is an element g , called a **generator** or **primitive root**, such that every element of the group is uniquely written as $g^k \bmod p$ for some $k \in \{0, \dots, p-2\}$. This chapter reproves nothing of this structure: it uses it.*

Definition 12.1: Discrete logarithm. Let g be a generator of $(\mathbb{Z}/p\mathbb{Z})^\times$ and $h \in (\mathbb{Z}/p\mathbb{Z})^\times$. The **discrete logarithm** of h base g is the unique integer $k \in \{0, \dots, p-2\}$ such that $g^k \equiv h \pmod{p}$, written $k = \log_g h$.

***Remark 12.2: Existence and uniqueness of the discrete logarithm (proved), difficulty of computing it (conjectural).** That $\log_g h$ exists and is unique for every $h \in (\mathbb{Z}/p\mathbb{Z})^\times$ is an **already proved** fact: it is exactly the bijectivity of the map $k \mapsto g^k \bmod p$ on $\{0, \dots, p-2\}$, a direct consequence of the cyclicity of the group (Theorem 7.15). Quite another matter is **computing** $\log_g h$ knowing only g, h, p : unlike computing $g^k \bmod p$ (easy, via repeated squaring, polynomial time in $\log p$), no algorithm is known that computes the discrete logarithm in time polynomial in $\log p$ for a generic p , and it is **conjectured** (without proof) that none exists for a classical computer. This is the **discrete logarithm conjecture**: an open problem of computational complexity theory, not a theorem. It is on this asymmetry — easy one way, open the other — that the protocols of this chapter rest. Source: D. Boneh, V. Shoup, *A Graduate Course in Applied Cryptography*, version 0.6, 2023, ch. 11 (standard framing of the problem; freely available, existence and relevance verified).*

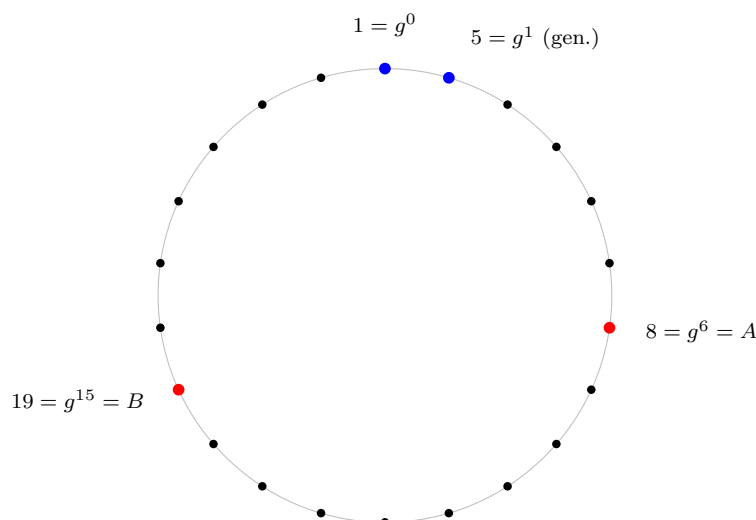


Figure 12.1: The elements of $(\mathbb{Z}/23\mathbb{Z})^\times$ arranged by the exponent k in the generator $g = 5$: the same cyclic structure — the same “discrete circle” — as Chapter 7. Highlighted are A, B from Example 12.1, at positions $k = 6, 15$, verified with the script `vol2_cap11_figure_gruppo.py`.

12.2. The Diffie–Hellman key exchange

Definition 12.2: Diffie–Hellman protocol. Let p be prime and g a generator of $(\mathbb{Z}/p\mathbb{Z})^\times$, both public. Alice secretly chooses $a \in \{1, \dots, p-2\}$ and sends $A = g^a \bmod p$; Bob secretly chooses $b \in \{1, \dots, p-2\}$ and sends $B = g^b \bmod p$. Alice computes $s = B^a \bmod p$, Bob computes $s' = A^b \bmod p$.

Theorem 12.1: Correctness of the Diffie–Hellman exchange. With the notation of Definition 12.2, $s = s'$: Alice and Bob compute the same value, $g^{ab} \bmod p$.

Proof. By definition, $s = B^a = (g^b)^a = g^{ba} \bmod p$ and $s' = A^b = (g^a)^b = g^{ab} \bmod p$. Exponentiation in $\mathbb{Z}$ commutes, $ab = ba$, hence $g^{ab} = g^{ba}$ in $\mathbb{Z}$, and a fortiori $g^{ab} \bmod p = g^{ba} \bmod p$: $s = s' = g^{ab} \bmod p$. $\square$

Remark 12.3: Why the exchange is useful. An observer who sees p, g, A, B pass by (all public by construction of the protocol) but not a, b , in order to reconstruct $s = g^{ab} \bmod p$ appears to have to solve a problem at least as hard as the discrete logarithm (computing a from $A = g^a$, or b from $B = g^b$) — this is the **computational Diffie–Hellman problem** (CDH): given g, g^a, g^b , compute g^{ab} . Whether CDH is as hard as the discrete logarithm, easier, or exactly equivalent is itself an **open question** in general (for some groups the two are known to be equivalent, not for $(\mathbb{Z}/p\mathbb{Z})^\times$ in general): the protocol’s security thus rests on one conjecture on top of another, honestly stated as such, not on a theorem.

Example 12.1: Key exchange with $p = 23$. Let $p = 23$. The generators of $(\mathbb{Z}/23\mathbb{Z})^\times$ are $\{5, 7, 10, 11, 14, 15, 17, 19, 20, 21\}$ (checked by exhaustion: for each, the powers $g^1, \dots, g^{22} \bmod 23$ range over all 22 nonzero elements before returning to 1; script `vol2_cap11_crittografia.py`). Take $g = 5$.

Alice chooses $a = 6$: $A = 5^6 \bmod 23 = 8$. Bob chooses $b = 15$: $B = 5^{15} \bmod 23 = 19$. Alice computes $s = 19^6 \bmod 23 = 2$; Bob computes $s' = 8^{15} \bmod 23 = 2$: $s = s' = 2$, as predicted

by Theorem 12.1. An observer who sees only $(23, 5, 8, 19)$ has, with tools known today, no efficient way to reconstruct 2 without first solving a discrete logarithm.

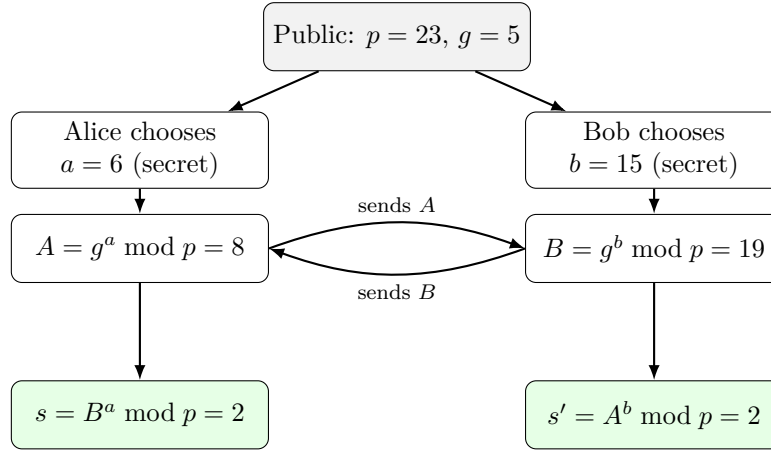


Figure 12.2: The Diffie–Hellman key exchange protocol, with the values of Example 12.1 ($p = 23$, $g = 5$): Alice and Bob independently compute the same secret $s = s' = 2$ (Theorem 12.1), even though they only exchange A, B in the clear.

12.3. The ElGamal cipher

Definition 12.3: ElGamal cipher. Let p be prime and g a generator of $(\mathbb{Z}/p\mathbb{Z})^\times$, both public. **Key generation:** secretly choose $x \in \{1, \dots, p-2\}$ (private key) and publish $y = g^x \bmod p$ (public key). **Encryption** of a message $m \in (\mathbb{Z}/p\mathbb{Z})^\times$: choose a random integer $k \in \{1, \dots, p-2\}$ (different for each encryption) and compute $c_1 = g^k \bmod p$, $c_2 = m \cdot y^k \bmod p$; the ciphertext is the pair (c_1, c_2) . **Decryption:** compute $m = c_2 \cdot (c_1^x)^{-1} \bmod p$, where $(c_1^x)^{-1}$ is the multiplicative inverse of c_1^x modulo p .

Theorem 12.2: Correctness of ElGamal. Decryption in Definition 12.3 recovers exactly the plaintext: $c_2 \cdot (c_1^x)^{-1} \equiv m \pmod{p}$.

Proof. First, c_1^x is invertible modulo p : since p is prime, every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ is invertible (the same property that makes $(\mathbb{Z}/p\mathbb{Z})^\times$ a group). By construction,

$$c_1^x = (g^k)^x = g^{kx} \pmod{p}, \quad y^k = (g^x)^k = g^{xk} \pmod{p},$$

and $kx = xk$ in $\mathbb{Z}$ (commutativity), hence $c_1^x \equiv y^k \pmod{p}$: the sender and the recipient, without sharing k or x with each other, compute the **same** quantity $g^{xk} \bmod p$ — it is the same fact as Theorem 12.1, with $a = k, b = x$. Hence

$$c_2 \cdot (c_1^x)^{-1} \equiv (m \cdot y^k) \cdot (y^k)^{-1} \equiv m \cdot (y^k \cdot (y^k)^{-1}) \equiv m \cdot 1 \equiv m \pmod{p}.$$

□

Remark 12.4: ElGamal is, in essence, Diffie–Hellman plus a mask. Theorem 12.2 makes explicit that ElGamal is not a protocol independent of Diffie–Hellman: $y^k \bmod p$ is exactly the shared secret that Alice (here, the sender, with "secret exponent" k) and the recipient (with secret exponent x) would obtain by exchanging g^k and $g^x = y$. ElGamal uses

this shared secret as a multiplicative "mask" for the message, regenerated at each encryption by choosing a new k — hence the fact, verifiable by revisiting the Definition, that different encryptions of the same message with the same public key almost certainly produce different ciphertexts (a property called "probabilistic encryption").

Remark 12.5: Security: the decisional Diffie–Hellman problem. Correctness (Theorem 12.2) is a proved fact: whoever holds x always decrypts correctly. **Security** — that whoever does not hold x learns nothing about m by observing (c_1, c_2) — instead rests on an assumption stronger than the discrete logarithm and even than CDH: the **decisional Diffie–Hellman problem** (DDH), which asks to distinguish a triple (g^a, g^b, g^{ab}) from a triple (g^a, g^b, g^c) with c chosen at random, independently of a, b . Whether DDH is hard in $(\mathbb{Z}/p\mathbb{Z})^\times$ (for suitably chosen p) is itself an **open conjecture**, not a theorem: this chapter states it as such, neither proving it nor able to with tools available today. Source: D. Boneh, "The Decision Diffie–Hellman Problem", *Proceedings of ANTS-III, Lecture Notes in Computer Science* vol. 1423, Springer, 1998, pp. 48–63 (standard survey of the problem; existence and relevance verified).

Example 12.2: Encryption and decryption with $p = 23$. Again let $p = 23$, $g = 5$. The recipient chooses private key $x = 10$ and publishes $y = 5^{10} \bmod 23 = 9$. To encrypt the message $m = 17$, the sender randomly chooses $k = 3$ and computes $c_1 = 5^3 \bmod 23 = 10$, $y^k = 9^3 \bmod 23 = 16$, $c_2 = 17 \cdot 16 \bmod 23 = 19$: the ciphertext is $(10, 19)$. At decryption, the recipient computes $c_1^x = 10^{10} \bmod 23 = 16$ (which indeed coincides with $y^k = 16$ found above, as proved in general in Theorem 12.2), its inverse modulo 23 is 13 (since $16 \cdot 13 = 208 = 9 \times 23 + 1 \equiv 1 \pmod{23}$), and finally $m = 19 \cdot 13 \bmod 23 = 17$: the original message, recovered exactly (script `vol2_cap11_crittografia.py`).

12.4. Assigned exercises

Exercise 12.1: Key exchange. With $p = 23$, $g = 5$, Alice chooses $a = 9$ and Bob chooses $b = 4$. Compute A , B and the shared secret s , checking that Alice and Bob obtain the same value. (Estimated time: 10 minutes.)

Exercise 12.2: ElGamal decryption. With $p = 23$, $g = 5$, the recipient has private key $x = 14$. The sender sends the ciphertext $(c_1, c_2) = (17, 7)$. Decrypt the message. (Estimated time: 10 minutes.)

Exercise 12.3: Why k must never be reused. Suppose the sender encrypts two different messages m, m' with the **same** public key y and the **same** k , obtaining (c_1, c_2) and (c_1, c'_2) . Show that an observer who knows even just the ratio $m/m' \bmod p$ can compute it directly from c_2, c'_2 , without knowing x or k . (Estimated time: 15 minutes; hint: divide c_2 by c'_2 .)

12.5. Worked exercises

Exercise 12.4: S1 — Key exchange, step by step. Fully solves Exercise 12.1: with $p = 23$, $g = 5$, Alice chooses $a = 9$ and Bob chooses $b = 4$. Find A , B , and the shared secret.

Solution. Step 1 — Alice's public key. $A = g^a \bmod p = 5^9 \bmod 23$. By repeated squaring: $5^1 = 5$, $5^2 = 25 \equiv 2$, $5^4 \equiv 2^2 = 4$, $5^8 \equiv 4^2 = 16 \pmod{23}$; $9 = 8 + 1$, hence $A = 5^8 \cdot 5^1 \equiv 16 \cdot 5 = 80 \equiv 80 - 69 = 11 \pmod{23}$.

Step 2 — Bob's public key. $B = g^b \bmod p = 5^4 \bmod 23 \equiv 4$ (already computed above).

Step 3 — shared secret. Alice computes $s = B^a \bmod p = 4^9 \bmod 23$: $4^2 = 16$, $4^4 \equiv 16^2 = 256 \equiv 256 - 11 \times 23 = 256 - 253 = 3$, $4^8 \equiv 3^2 = 9$; $9 = 8 + 1$, hence $s = 4^8 \cdot 4 \equiv 9 \cdot 4 = 36 \equiv 36 - 23 = 13$. Bob computes $s' = A^b \bmod p = 11^4 \bmod 23$: $11^2 = 121 \equiv 121 - 5 \times 23 = 121 - 115 = 6$, $11^4 \equiv 6^2 = 36 \equiv 13$. Hence $s = s' = 13$, in agreement with Theorem 12.1. ✓ (Also checked with `vol2_cap11_crittografia.py`: $A = 11$, $B = 4$, $s = 13$.)

Exercise 12.5: S2 — ElGamal decryption, step by step. Fully solves Exercise 12.2: with $p = 23$, $g = 5$, private key $x = 14$, ciphertext $(c_1, c_2) = (17, 7)$. Decrypt.

Solution. Step 1 — computing $c_1^x \bmod p$. $c_1^x = 17^{14} \bmod 23$. By repeated squaring: $17^2 = 289 \equiv 289 - 12 \times 23 = 289 - 276 = 13$, $17^4 \equiv 13^2 = 169 \equiv 169 - 7 \times 23 = 169 - 161 = 8$, $17^8 \equiv 8^2 = 64 \equiv 64 - 2 \times 23 = 18$; $14 = 8 + 4 + 2$, hence $17^{14} \equiv 17^8 \cdot 17^4 \cdot 17^2 \equiv 18 \cdot 8 \cdot 13 \pmod{23}$. Computing: $18 \cdot 8 = 144 \equiv 144 - 6 \times 23 = 6$; $6 \cdot 13 = 78 \equiv 78 - 3 \times 23 = 9$. So $c_1^x \equiv 9 \pmod{23}$.

Step 2 — inverse of 9 modulo 23. We seek u with $9u \equiv 1 \pmod{23}$: $9 \times 18 = 162 = 7 \times 23 + 1 = 161 + 1$, hence $u = 18$.

Step 3 — message. $m = c_2 \cdot u \bmod p = 7 \times 18 \bmod 23 = 126 \bmod 23$. $126 = 5 \times 23 + 11 = 115 + 11$, hence $m = 11$. ✓ (Also checked with `vol2_cap11_crittografia.py`: the same pair $(c_1, c_2) = (17, 7)$ is generated by encrypting $m = 11$ with $k = 7$ and public key $y = 13$.)

12.6. Self-assessment quiz

- Q1) What is, by definition, the discrete logarithm $\log_g h$ in $(\mathbb{Z}/p\mathbb{Z})^\times$?
- Q2) Why does $\log_g h$ exist and is unique for every h ? Which already-proved theorem guarantees this?
- Q3) Compute $g^{ab} \bmod p$ and $g^{ba} \bmod p$ for $g = 5, p = 23, a = 2, b = 3$ and check they agree, explaining why this always happens.
- Q4) In the Diffie–Hellman protocol, which quantities are public and which are secret?
- Q5) Prove the key step of ElGamal's correctness: why is $c_1^x \equiv y^k \pmod{p}$?
- Q6) An observer sees (p, g, A, B) in a Diffie–Hellman exchange. What problem must they solve, in principle, to compute the shared secret? Is it proved to be hard?
- Q7) What changes, in the outcome of ElGamal encryption, if the sender encrypts the same message twice with the same public key but different k ?
- Q8) Why is ElGamal's security not (as of today's mathematics) a proved theorem, but rests on a conjecture?

Teacher's key. **Q1)** The unique integer $k \in \{0, \dots, p-2\}$ with $g^k \equiv h \pmod{p}$. **Q2)** Bijectivity of $k \mapsto g^k \bmod p$, from cyclicity of the group (Theorem 7.15). **Q3)** $5^6 \bmod 23 = 8$ in both cases (script provided); they agree because $ab = ba$ in $\mathbb{Z}$. **Q4)** Public: p, g, A, B ; secret: a, b (and the resulting secret s). **Q5)** $c_1^x = (g^k)^x = g^{kx}$, $y^k = (g^x)^k = g^{xk}$, and $kx = xk$. **Q6)** The computational Diffie–Hellman problem (CDH); not proved hard, conjectured to be. **Q7)** The two ciphertexts (c_1, c_2) are (almost certainly) different, because k changes: probabilistic

encryption. **Q8**) Because it rests on the decisional Diffie–Hellman problem (DDH), itself not proved hard: no unconditional security proof exists for these protocols.

12.7. Chapter formulary

Diffie–Hellman: $A = g^a \bmod p$, $B = g^b \bmod p$, $s = B^a = A^b = g^{ab} \bmod p$.

ElGamal: public key $y = g^x \bmod p$; encryption $(c_1, c_2) = (g^k \bmod p, m \cdot y^k \bmod p)$; decryption $m = c_2 \cdot (c_1^x)^{-1} \bmod p$.

Sources and bibliography

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- [3] D. Boneh, V. Shoup, *A Graduate Course in Applied Cryptography*, version 0.6, 2023, ch. 11 (standard framing of the discrete logarithm problem; existence and relevance verified).
- [4] D. Boneh, "The Decision Diffie–Hellman Problem", *Proceedings of ANTS-III*, Lecture Notes in Computer Science vol. 1423, Springer, 1998, pp. 48–63 (DDH problem; existence and relevance verified).
- [5] Vol. II, Chapter 7 (cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$, Theorem 7.15).

CHAPTER 13

From the Circle to the Torus: the DFT in Several Dimensions

INFO

The idea in one line. The discrete circle $\mathbb{Z}/n\mathbb{Z}$ of Chapter 7 has a direct product with itself, $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$: geometrically a **discrete torus**, algebraically still a finite abelian group. The discrete Fourier transform extends to this product group almost for free: it **separates** into two independent 1D transforms, one per factor, and the same holds for a product of any number of discrete circles.

INFO

Prerequisites. Chapter 7: the DFT on $\mathbb{Z}/n\mathbb{Z}$ as harmonic analysis of the cyclic group (Corollary 7.5), the orthogonality relations of characters (Theorem 7.21). Chapter 3: the DFT matrix (Definition 3.7).

13.1. The discrete torus $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$

***Remark 13.1: A product of two discrete circles is still a group.** Given two finite abelian groups G, H , the Cartesian product $G \times H$ with the componentwise operation, $(g, h) + (g', h') = (g + g', h + h')$, is still a finite abelian group, of order $|G| \cdot |H|$: the identity element is $(0, 0)$, the inverse of (g, h) is $(-g, -h)$, and commutativity/associativity are checked componentwise from those of G and H . Applied to $G = \mathbb{Z}/n\mathbb{Z}$, $H = \mathbb{Z}/m\mathbb{Z}$ (both already studied in Chapter 7 as "the circle split into n (resp. m) parts"), the product $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ is a group of order nm : the grid of $n \times m$ points arranged on a **discrete torus** — the product of two circles, exactly as the continuous torus $S^1 \times S^1$ is the product of two continuous circles.*

13.2. The two-dimensional DFT and its separability

Definition 13.1: Two-dimensional discrete Fourier transform. For $f : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{C}$, its **2D DFT** is the function $\hat{f} : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{C}$ given by

$$\hat{f}(k, l) = \sum_{a=0}^{n-1} \sum_{b=0}^{m-1} f(a, b) e^{-2\pi i k a / n} e^{-2\pi i l b / m}.$$

***Theorem 13.1: Separability of the 2D DFT.** With the notation of Definition 13.1, for*

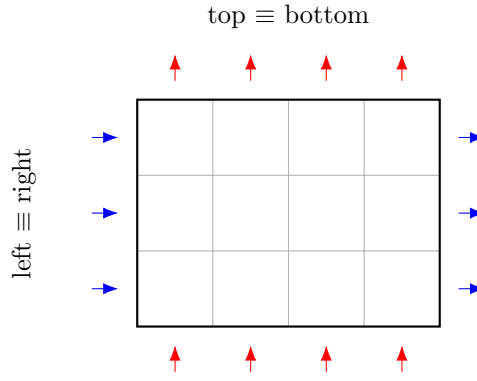


Figure 13.1: The discrete torus $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ (example with $n = 4, m = 3$): the $n \times m$ grid with opposite sides identified in the same direction (blue arrows: column $n - 1$ and column 0 are adjacent; red arrows: row $m - 1$ and row 0 are adjacent) — exactly as the circle $\mathbb{Z}/n\mathbb{Z}$ is a segment with its two endpoints identified.

every $a \in \mathbb{Z}/n\mathbb{Z}$ let

$$g(a, l) = \sum_{b=0}^{m-1} f(a, b) e^{-2\pi i l b / m}$$

(the 1D DFT of f with respect to the second variable, with row a fixed). Then

$$\hat{f}(k, l) = \sum_{a=0}^{n-1} g(a, l) e^{-2\pi i k a / n},$$

that is, $\hat{f}(\cdot, l)$ is the 1D DFT of $g(\cdot, l)$ with respect to the first variable. In other words: the 2D DFT is computed by taking the 1D DFT along every row and then the 1D DFT along every column of the result — and, by symmetry of the proof, the order (rows first or columns first) does not matter.

Proof. By Definition 13.1, the factor $e^{-2\pi i k a / n} e^{-2\pi i l b / m}$ does not mix a and b : it factors as a product of a term in a and one in b . The double sum can thus be rewritten by first collecting the sum over b :

$$\hat{f}(k, l) = \sum_{a=0}^{n-1} e^{-2\pi i k a / n} \underbrace{\left(\sum_{b=0}^{m-1} f(a, b) e^{-2\pi i l b / m} \right)}_{= g(a, l)} = \sum_{a=0}^{n-1} g(a, l) e^{-2\pi i k a / n},$$

which is exactly the statement: no step uses anything beyond distributivity and reassociation of a finite sum (legitimate, being nm terms, a finite number). For the reversed order, collect the sum over a first instead of b : the same argument, with the roles of a, b swapped, gives the same final result $\hat{f}(k, l)$ (the original double sum is the same in both cases, only the order of summation changes, which is always legitimate for finite sums). $\square$

Remark 13.2: Why this is useful. Computing $\hat{f}(k, l)$ for all nm values of (k, l) from Definition 13.1 requires, for each one, a sum of nm terms: $O(n^2 m^2)$ operations in all. Theorem 13.1 reduces the computation to n 1D DFTs of length m (the rows) plus m 1D DFTs of length n (the columns): using the direct 1D DFT (not the FFT), this is $O(nm^2) + O(mn^2) = O(nm(n + m))$ operations, already fewer than $O(n^2 m^2)$ for n, m not too small;

using the FFT algorithm (Volume III, Chapter 23) for each 1D DFT, the gain becomes drastic. We return to this point in the next chapter, devoted to images.

13.3. Convolution and Parseval's identity in two dimensions

Definition 13.2: Two-dimensional circular convolution. For $f, h : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{C}$, their **2D circular convolution** is

$$(f * h)(a, b) = \sum_{a'=0}^{n-1} \sum_{b'=0}^{m-1} f(a', b') h(a - a', b - b'),$$

with the indices $a - a', b - b'$ reduced modulo n, m respectively.

Theorem 13.2: Convolution theorem in two dimensions. With the notation of Definitions 13.1 and 13.2,

$$\widehat{f * h}(k, l) = \hat{f}(k, l) \hat{h}(k, l).$$

Proof. Substituting the definition of convolution into that of the 2D DFT and setting $a = a' + c$, $b = b' + d$ (a legitimate change of variable on a finite group, bijective for fixed a', b'):

$$\widehat{f * h}(k, l) = \sum_{a, b} \left(\sum_{a', b'} f(a', b') h(a - a', b - b') \right) e^{-2\pi i k a / n} e^{-2\pi i l b / m}.$$

Swapping the order of the sums (finite, legitimate) and substituting $a = a' + c$, $b = b' + d$:

$$= \sum_{a', b'} f(a', b') \sum_{c, d} h(c, d) e^{-2\pi i k (a' + c) / n} e^{-2\pi i l (b' + d) / m} = \left(\sum_{a', b'} f(a', b') e^{-2\pi i k a' / n} e^{-2\pi i l b' / m} \right) \left(\sum_{c, d} h(c, d) e^{-2\pi i k c / n} e^{-2\pi i l d / m} \right)$$

that is exactly $\hat{f}(k, l) \cdot \hat{h}(k, l)$. □

Theorem 13.3: Parseval's identity in two dimensions. For every $f : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{C}$,

$$\sum_{a=0}^{n-1} \sum_{b=0}^{m-1} |f(a, b)|^2 = \frac{1}{nm} \sum_{k=0}^{n-1} \sum_{l=0}^{m-1} |\hat{f}(k, l)|^2.$$

Proof. By Theorem 13.1, $\hat{f}(k, l)$ is obtained by applying two independent 1D DFTs, one on the rows (variable b , for each fixed a) and one on the columns (variable a , for each fixed l). The 1D Parseval identity (a direct consequence of the orthogonality of characters, Theorem 7.21, already established in Chapter 7) applies twice, once for each of the two component 1D DFTs: applying it first on the rows (for each fixed a , $\sum_b |f(a, b)|^2 = \frac{1}{m} \sum_l |g(a, l)|^2$ with g as in the proof of Theorem 13.1) and summing over a , then again on the columns of the result ($\sum_a |g(a, l)|^2 = \frac{1}{n} \sum_k |\hat{f}(k, l)|^2$ for each fixed l) and summing over l , one obtains

$$\sum_{a, b} |f(a, b)|^2 = \frac{1}{m} \sum_{a, l} |g(a, l)|^2 = \frac{1}{m} \cdot \frac{1}{n} \sum_{k, l} |\hat{f}(k, l)|^2,$$

which is the statement. □

13.4. Generalisation to n dimensions

Definition 13.3: k -dimensional discrete Fourier transform. Let $N_1, \dots, N_k$ be positive integers and $G = \mathbb{Z}/N_1\mathbb{Z} \times \dots \times \mathbb{Z}/N_k\mathbb{Z}$ (a product of k discrete circles, a group of order $N_1 \dots N_k$ by repeated application of Observation 13.1). For $f : G \rightarrow \mathbb{C}$, its **k -dimensional DFT** is

$$\hat{f}(k_1, \dots, k_k) = \sum_{a_1=0}^{N_1-1} \dots \sum_{a_k=0}^{N_k-1} f(a_1, \dots, a_k) \prod_{j=1}^k e^{-2\pi i k_j a_j / N_j}.$$

Theorem 13.4: Separability of the DFT in k dimensions. The k -dimensional DFT of Definition 13.3 is computed by applying, in any order, the 1D DFT along each of the k variables in succession (keeping the other $k - 1$ fixed).

Proof. By induction on k . **Base case,** $k = 1$: the 1-dimensional DFT is, by definition, the 1D DFT of Chapter 7; there is nothing to separate, the statement is vacuously true. **Inductive step:** assume the statement holds for $k - 1$ variables, and prove it for k . Exactly as in the proof of Theorem 13.1 (which is the case $k = 2$), the phase factor $\prod_{j=1}^k e^{-2\pi i k_j a_j / N_j}$ factors as a product of a term depending only on a_k and a product of $k - 1$ terms depending on the first $k - 1$ variables; collecting the sum over a_k last,

$$\hat{f}(k_1, \dots, k_k) = \sum_{a_k=0}^{N_k-1} e^{-2\pi i k_k a_k / N_k} \left(\sum_{a_1, \dots, a_{k-1}} f(a_1, \dots, a_k) \prod_{j=1}^{k-1} e^{-2\pi i k_j a_j / N_j} \right).$$

The inner parenthesis, as a function of the first $k - 1$ variables (with a_k fixed), is the $(k - 1)$ -dimensional DFT of $f(\cdot, \dots, \cdot, a_k)$: by the inductive hypothesis it is computed by applying the 1D DFT along each of the first $k - 1$ variables in succession. The last sum, over a_k , is then a further 1D DFT, along the k -th variable. In total: k 1D DFTs, one per variable, applicable in any order (swapping the order of summation between finite sums is always legitimate, as already observed in the proof of the case $k = 2$). $\square$

Remark 13.3: Convolution and Parseval in n dimensions. The same arguments of the proofs of Theorems 13.2 and 13.3 (change of variable for convolution, repeated application of 1D Parseval for the energy) extend word for word to the k -dimensional case, replacing "row/column" with "one of the k directions": k -dimensional convolution becomes a pointwise product under the DFT, and the total energy is conserved up to the factor $1/(N_1 \dots N_k)$. We do not rewrite them in full: the inductive structure of Theorem 13.4 makes the generalisation mechanical.

13.5. Complete numerical example

Example 13.1: 2D DFT on a 3×3 grid. On the torus $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, take

$$f = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix},$$

where row a and column b index $f(a, b)$. With $\zeta = e^{-2\pi i/3}$ (the same cube roots of unity used repeatedly throughout the treatise, for instance in the Fortescue transform), the direct computation from Definition 13.1 and the separable computation of Theorem 13.1 (1D DFT

of every row, then 1D DFT of every column of the result) were carried out independently in Python (script `vol2_cap12_dft2d.py`) and agree to within 5×10^{-16} (machine epsilon), and also agree with `numpy.fft.fft2`. Some values:

$$\hat{f}(0,0) = 10, \quad \hat{f}(1,1) = 1.0000000, \quad \hat{f}(0,1) = -0.5 + 0.8660254i.$$

Parseval check (Theorem 13.3): $\sum_{a,b} |f(a,b)|^2 = 20$ (sum of the squares of the nine values of f , all real) and $\frac{1}{9} \sum_{k,l} |\hat{f}(k,l)|^2 = 20.0000000$ (script). ✓

Generalisation to 4 dimensions. The same script checks Theorem 13.4 on a random $2 \times 2 \times 2 \times 2$ grid (16 integer values generated with a declared seed, for reproducibility): the direct computation with $16^2 = 256$ terms and the separable computation (four 1D DFTs of length 2, one per axis) agree to within 4×10^{-31} , and 4-dimensional Parseval is verified ($\sum |f|^2 = 67$, $\frac{1}{16} \sum |\hat{f}|^2 = 67.0000000$).

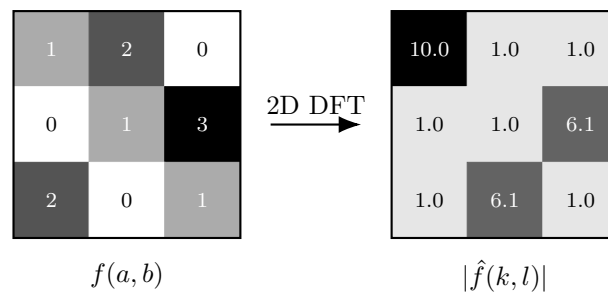


Figure 13.2: The 3×3 example of Example 13.1, as a grid: on the left the spatial domain $f(a,b)$ (shading proportional to the value, 0–3), on the right the modulus of the DFT $|\hat{f}(k,l)|$ (shading proportional to the value, 0–10; cells computed with `vol2_cap12_dft2d.py`). The value $\hat{f}(0,0) = 10$ (top left of the frequency grid) is always the sum of all the spatial values: $1+2+0+0+1+3+2+0+1 = 10$.

13.6. Assigned exercises

Exercise 13.1: DFT of an identity grid. Compute the 2D DFT of $f = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ on $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, using Definition 13.1 directly. (Estimated time: 10 minutes.)

Exercise 13.2: 2D Parseval check. Compute the 2D DFT of $g = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ on $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and check Parseval’s identity (Theorem 13.3). (Estimated time: 10 minutes.)

13.7. Worked exercises

Exercise 13.3: S1 — DFT of the identity grid, step by step. Fully solves Exercise 13.1: 2D DFT of $f = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ on $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution. With $n = m = 2$, $e^{-2\pi ika/2} = e^{-i\pi ka} = (-1)^{ka}$: the four phase factors are ± 1 . By Definition 13.1,

$$\hat{f}(k, l) = \sum_{a=0}^1 \sum_{b=0}^1 f(a, b)(-1)^{ka}(-1)^{lb}.$$

Since $f(0, 0) = f(1, 1) = 1$ and $f(0, 1) = f(1, 0) = 0$, only two terms survive in each sum:

$$\hat{f}(k, l) = (-1)^{k \cdot 0}(-1)^{l \cdot 0} + (-1)^{k \cdot 1}(-1)^{l \cdot 1} = 1 + (-1)^{k+l}.$$

Hence $\hat{f}(0, 0) = 1 + 1 = 2$, $\hat{f}(0, 1) = 1 + (-1) = 0$, $\hat{f}(1, 0) = 1 + (-1) = 0$, $\hat{f}(1, 1) = 1 + (-1)^2 = 2$:

$$\hat{f} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

✓(Also verified with `vol2_cap12_dft2d.py` and with `numpy.fft.fft2`: same result, up to floating-point rounding of order 10^{-16} .)

Exercise 13.4: S2 — 2D Parseval, step by step. Fully solves Exercise 13.2: 2D DFT and Parseval check for $g = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ on $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution. Step 1 — energy in the spatial domain. $\sum_{a,b} |g(a, b)|^2 = 1^2 + 1^2 + 1^2 + (-1)^2 = 4$.

Step 2 — 2D DFT. With the same $(-1)^{k+l}$ -like formula as in the previous step but with $g(0, 0) = g(0, 1) = g(1, 0) = 1$, $g(1, 1) = -1$:

$$\hat{g}(k, l) = (-1)^0 [g(0, 0) + g(0, 1)(-1)^l + g(1, 0)(-1)^k + g(1, 1)(-1)^{k+l}] = 1 + (-1)^l + (-1)^k - (-1)^{k+l}.$$

Evaluating: $\hat{g}(0, 0) = 1 + 1 + 1 - 1 = 2$; $\hat{g}(0, 1) = 1 - 1 + 1 + 1 = 2$; $\hat{g}(1, 0) = 1 + 1 - 1 + 1 = 2$; $\hat{g}(1, 1) = 1 - 1 - 1 - 1 = -2$:

$$\hat{g} = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}.$$

Step 3 — energy in the frequency domain and comparison. $\sum_{k,l} |\hat{g}(k, l)|^2 = 4 + 4 + 4 + 4 = 16$, and $\frac{1}{nm} \cdot 16 = \frac{1}{4} \cdot 16 = 4$, equal to Step 1: Parseval verified. ✓(Also confirmed by `vol2_cap12_dft2d.py`.)

13.8. Self-assessment quiz

- Q1) Why is $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ still a group? Which elementary property guarantees it?
- Q2) State the definition of the 2D DFT on $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.
- Q3) What does it mean that the 2D DFT is "separable"? In what sense does it reduce the computational cost?
- Q4) In the proof of separability, which property of finite sums is used to factor the double sum?
- Q5) State the convolution theorem in two dimensions.
- Q6) How is the definition of the DFT generalised to k dimensions?
- Q7) Which variable is the induction on in the proof of k -dimensional separability, and what is the base case?

Q8) In Example 13.1, which identity links the energy of the grid f in the spatial domain to that in the frequency domain?

Teacher's key. **Q1)** The Cartesian product of two finite abelian groups, with the componentwise operation, is a group (Observation 13.1). **Q2)** $\hat{f}(k, l) = \sum_{a,b} f(a, b) e^{-2\pi i k a/n} e^{-2\pi i l b/m}$ (Definition 13.1). **Q3)** It is computed as a 1D DFT along the rows followed by a 1D DFT along the columns (or the reverse); it reduces the cost from $O(n^2 m^2)$ to $O(nm(n + m))$ with the direct DFT, much less with the FFT. **Q4)** Distributivity and the legitimacy of reassociating/reordering a finite sum. **Q5)** $\widehat{f * h}(k, l) = \hat{f}(k, l) \hat{h}(k, l)$ (Theorem 13.2). **Q6)** $\hat{f}(k_1, \dots, k_k) = \sum_{a_1, \dots, a_k} f(a_1, \dots, a_k) \prod_j e^{-2\pi i k_j a_j / N_j}$ (Definition 13.3). **Q7)** Induction on k (the number of dimensions); base case $k = 1$, where there is nothing to separate. **Q8)** The 2D Parseval identity (Theorem 13.3): $\sum |f|^2 = \frac{1}{nm} \sum |\hat{f}|^2$.

13.9. Chapter formulary

$$\text{2D DFT: } \hat{f}(k, l) = \sum_{a=0}^{n-1} \sum_{b=0}^{m-1} f(a, b) e^{-2\pi i k a/n} e^{-2\pi i l b/m}.$$

Separability: 1D DFT along the rows, then along the columns (or the reverse).

$$\text{Convolution: } \widehat{f * h} = \hat{f} \cdot \hat{h}.$$

$$\text{2D Parseval: } \sum_{a,b} |f(a, b)|^2 = \frac{1}{nm} \sum_{k,l} |\hat{f}(k, l)|^2.$$

$$\text{\textit{k}-dimensional DFT: } \hat{f}(k_1, \dots, k_k) = \sum_{a_1, \dots, a_k} f(a_1, \dots, a_k) \prod_{j=1}^k e^{-2\pi i k_j a_j / N_j}, \text{ separable by induction on } k.$$

Sources and bibliography

- [1] D. E. Dudgeon, R. M. Mersereau, *Multidimensional Digital Signal Processing*, Prentice-Hall Signal Processing Series, Englewood Cliffs, NJ, 1984, ISBN 0-13-604959-1 (reference treatment of the DFT and filtering in several dimensions; existence, publisher and year verified by cross-checking).
- [2] Vol. II, Chapter 7 (1D DFT as harmonic analysis on $\mathbb{Z}/n\mathbb{Z}$, Corollary 7.5; orthogonality of characters, Theorem 7.21).
- [3] Vol. II, Chapter 3 (DFT matrix, Definition 3.7).
- [4] Vol. III, Chapter 23 (FFT algorithm, recalled but not reproved here).

Image Processing: Filtering and Reconstruction

INFO

The idea in one line. A digital image is a function on a discrete torus $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ (Chapter 13): the 2D DFT separates the image content into spatial-frequency components, filtering (blurring, sharpening) is multiplication in the frequency domain — that is, by the convolution theorem already proved, convolution in the spatial domain — and the 2D FFT computes all this in $O(N^2 \log N)$ time instead of $O(N^4)$.

INFO

Prerequisites. Chapter 13: the 2D DFT (Definition 13.1), its separability (Theorem 13.1), the 2D convolution theorem (Theorem 13.2) and the 2D Parseval identity (Theorem 13.3). From Volume III: the Nyquist-Shannon sampling theorem (Chapter 24) and the radix-2 FFT algorithm with its correctness and cost (Chapter 23) — recalled here, not reproved.

14.1. An image as a function on a grid, and its sampling

Definition 14.1: Digital image. A grayscale digital image of resolution $n \times m$ is a function $I : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{R}$ (or, with 8-bit quantisation, $I : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \{0, \dots, 255\}$): exactly a function on the discrete torus of Chapter 13, with $I(a, b)$ the gray level of the pixel at row a , column b .

Remark 14.1: Two-dimensional sampling. A digital image nearly always arises by sampling a continuous scene $I_c : \mathbb{R}^2 \rightarrow \mathbb{R}$ on a regular grid of spacing $\Delta x, \Delta y$. Volume III, Chapter 24, proves in full the one-dimensional case, with the angular convention for the Fourier transform ($\hat{g}(\omega) = \int g(t) e^{-i\omega t} dt$, the same used below): if $g : \mathbb{R} \rightarrow \mathbb{R}$ has Fourier transform compactly supported in $[-B, B]$ (band B in radians per second) and the sampling spacing satisfies $\Delta \leq \pi/B$, then g is reconstructed exactly from its samples,

$$g(x) = \sum_{p \in \mathbb{Z}} g(p\Delta) \operatorname{sinc}\left(\frac{x - p\Delta}{\Delta}\right), \quad \operatorname{sinc}(t) := \frac{\sin(\pi t)}{\pi t}.$$

Here we prove in full the extension to two dimensions, as a direct application of this imported result.

Theorem 14.1: Two-dimensional reconstruction by separate application. Let $I_c : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function whose two-dimensional Fourier transform $\hat{I}_c(\omega_x, \omega_y)$ vanishes outside the rectangle $[-B_x, B_x] \times [-B_y, B_y]$ (angular bands B_x, B_y). If $\Delta x \leq \pi/B_x$ and $\Delta y \leq \pi/B_y$,

then for every $(x, y) \in \mathbb{R}^2$

$$I_c(x, y) = \sum_{p \in \mathbb{Z}} \sum_{q \in \mathbb{Z}} I_c(p\Delta x, q\Delta y) \operatorname{sinc}\left(\frac{x - p\Delta x}{\Delta x}\right) \operatorname{sinc}\left(\frac{y - q\Delta y}{\Delta y}\right).$$

Proof. Step 1 — every section at fixed y is band-limited in x . Fix $y \in \mathbb{R}$ and set $h_y(x) := I_c(x, y)$. By the two-dimensional Fourier inversion formula,

$$I_c(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \widehat{I}_c(\omega_x, \omega_y) e^{i(\omega_x x + \omega_y y)} d\omega_x d\omega_y = \int_{-B_x}^{B_x} \underbrace{\left[\int_{-B_y}^{B_y} \widehat{I}_c(\omega_x, \omega_y) e^{i\omega_y y} d\omega_y \right]}_{=: J(\omega_x; y)} e^{i\omega_x x} d\omega_x,$$

where we restricted both integrals to the rectangle of support by hypothesis. The bracketed expression, $J(\omega_x; y)$, vanishes for $|\omega_x| > B_x$ (the outer integral is already restricted to $[-B_x, B_x]$), and $h_y(x) = \int_{-B_x}^{B_x} J(\omega_x; y) e^{i\omega_x x} d\omega_x$ is exactly the 1D inverse transform of $J(\cdot; y)$: hence h_y is band-limited to $[-B_x, B_x]$, for every fixed y .

Step 2 — applying the 1D theorem in x . By Step 1 and the hypothesis $\Delta x \leq \pi/B_x$, the imported sampling theorem applies to h_y for every y :

$$I_c(x, y) = h_y(x) = \sum_{p \in \mathbb{Z}} h_y(p\Delta x) \operatorname{sinc}\left(\frac{x - p\Delta x}{\Delta x}\right) = \sum_{p \in \mathbb{Z}} I_c(p\Delta x, y) \operatorname{sinc}\left(\frac{x - p\Delta x}{\Delta x}\right). \quad (*)$$

Step 3 — every sampled column is band-limited in y . By the same argument as Step 1, swapping the roles of x and y (the hypothesis on the support of $\widehat{I}_c$ is symmetric in the two variables), the function $y \mapsto I_c(x_0, y)$ is band-limited to $[-B_y, B_y]$ for every fixed x_0 — in particular for $x_0 = p\Delta x$, for every $p \in \mathbb{Z}$.

Step 4 — applying the 1D theorem in y , and conclusion. By Step 3 and the hypothesis $\Delta y \leq \pi/B_y$,

$$I_c(p\Delta x, y) = \sum_{q \in \mathbb{Z}} I_c(p\Delta x, q\Delta y) \operatorname{sinc}\left(\frac{y - q\Delta y}{\Delta y}\right) \quad \text{for every } p \in \mathbb{Z}.$$

Substituting into $(*)$ gives exactly the formula of the statement. (Interchanging sum and integral in Steps 1–4 is legitimate for band-limited $L^1 \cap L^2$ signals, the standard class for this result — the same regularity hypothesis already assumed by Volume III for the imported 1D case.) $\square$

14.2. Filtering in the frequency domain

Definition 14.2: Ideal two-dimensional low-pass filter. On $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, let $\kappa(k) := k$ if $k \leq n/2$, $\kappa(k) := k - n$ otherwise (and similarly $\lambda(l)$ for l modulo m) — the "centred" frequency index, already used implicitly in Chapter 13 to distinguish positive from negative harmonics. For $\rho \geq 0$, the **ideal low-pass filter with cutoff ρ** is the mask

$$H_\rho(k, l) := \begin{cases} 1 & \text{if } \sqrt{\kappa(k)^2 + \lambda(l)^2} \leq \rho \\ 0 & \text{otherwise.} \end{cases}$$

The **complementary high-pass filter** is $H'_\rho := 1 - H_\rho$.

Theorem 14.2: Frequency filtering is spatial convolution. Let I be an image on $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, H a mask (e.g. H_ρ or H'_ρ), and let J be the filtered image defined by $\hat{J}(k, l) := \hat{I}(k, l) H(k, l)$. Setting $h := \text{DFT}^{-1}(H)$ (the filter's **kernel**), we have $J = I * h$ (2D convolution).

Proof. By the 2D convolution theorem (Theorem 13.2 of Chapter 13), $\widehat{I * h}(k, l) = \hat{I}(k, l) \hat{h}(k, l) = \hat{I}(k, l) H(k, l) = \hat{J}(k, l)$ for every (k, l) , since $\hat{h} = \text{DFT}(\text{DFT}^{-1}(H)) = H$. Since the DFT is bijective (its inverse exists and is itself a DFT, as already established in Chapter 7 for the 1D case and in Chapter 13 for the 2D case), $\widehat{I * h} = \hat{J}$ implies $I * h = J$. $\square$

Proposition 14.1: Low-pass filtering preserves the DC component exactly. With the previous notation, for every $\rho \geq 0$ the image filtered by the low-pass H_ρ has the same total sum (hence the same mean) as the original image: $\sum_{a,b} J(a, b) = \sum_{a,b} I(a, b)$.

Proof. $\sum_{a,b} I(a, b) = \hat{I}(0, 0)$ by definition of the DFT (Definition 13.1 with $k = l = 0$, where every phase factor equals 1), and likewise for J . Since $\kappa(0) = \lambda(0) = 0$, we have $\sqrt{\kappa(0)^2 + \lambda(0)^2} = 0 \leq \rho$ for every $\rho \geq 0$, hence $H_\rho(0, 0) = 1$ and $\hat{J}(0, 0) = \hat{I}(0, 0) \cdot 1 = \hat{I}(0, 0)$. $\square$

Proposition 14.2: High-pass filtering exactly cancels the DC component. With the previous notation, for every $\rho \geq 0$ the image filtered by the high-pass H'_ρ has total sum zero: $\sum_{a,b} J'(a, b) = 0$.

Proof. As above, $H'_\rho(0, 0) = 1 - H_\rho(0, 0) = 1 - 1 = 0$, hence $\hat{J}'(0, 0) = \hat{I}(0, 0) \cdot 0 = 0 = \sum_{a,b} J'(a, b)$. $\square$

Remark 14.2: Why "blur" and "sharpen". The two preceding Propositions prove only the behaviour of the zero-frequency component, but already justify the names: the low-pass filter leaves the mean brightness intact and zeroes every component beyond the cutoff ρ (the fast variations, i.e. edges and fine detail) — hence the blurring effect; the high-pass filter zeroes the mean brightness and keeps only the fast variations — hence edge enhancement. A full quantitative characterisation of "sharpness" would require a total-variation measure beyond the scope of this chapter: here we restrict ourselves to the proved fact.

14.3. The two-dimensional FFT

Theorem 14.3: Correctness of the row-column algorithm for the 2D FFT. Let I be an image on $\mathbb{Z}/N\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ (square grid, N a power of two). The algorithm that computes N 1D radix-2 FFTs along each row, then N 1D radix-2 FFTs along each column of the result, produces exactly $\hat{I}$ (Definition 13.1).

Proof. By the 2D DFT separability theorem (Theorem 13.1 of Chapter 13), $\hat{I}$ is obtained by applying the 1D DFT along the rows and then along the columns (in either order). The radix-2 FFT is, by the correctness theorem for the radix-2 DIT algorithm of Volume III (Chapter 23), *exactly* the 1D DFT computed by a faster algorithm, not an approximation: replacing every 1D DFT in the separability proof by its FFT realisation changes nothing in the result, only the cost. $\square$

Theorem 14.4: Computational cost of the 2D FFT. For an $N \times N$ image (N a power of two), the row-column algorithm of Theorem 14.3 costs $O(N^2 \log N)$ arithmetic operations, against $O(N^4)$ for the 2D DFT computed directly from Definition 13.1.

Proof. Direct 2D DFT. For each of the N^2 output values $\hat{I}(k, l)$, Definition 13.1 is a double sum of N^2 terms: cost $O(N^2)$ per output, $O(N^2) \cdot N^2 = O(N^4)$ in total.

Row-column algorithm. By the exact operation-count theorem for the FFT of Volume III (Chapter 23), a 1D radix-2 FFT of length N costs $O(N \log N)$. The first phase applies N 1D FFTs (one per row), cost $N \cdot O(N \log N) = O(N^2 \log N)$; the second phase applies N 1D FFTs (one per column of the intermediate result), same cost $O(N^2 \log N)$. Adding the two phases: $O(N^2 \log N) + O(N^2 \log N) = O(N^2 \log N)$ in total (the sum of two functions of the same order has that order).

Comparison. The ratio of the two costs is $O(N^4)/O(N^2 \log N) = O(N^2/\log N)$, which diverges: for $N = 1024 = 2^{10}$, for instance, $N^2/\log_2 N = 1024^2/10 \approx 104858$, a speed-up factor of over a hundred thousand. $\square$

14.4. A complete numerical example: removing a periodic disturbance

Example 14.1: A notch filter on an 8×8 grid. On the torus $\mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, consider the "clean" image $I_0(a, b) = \cos(2\pi(a+b)/8)$ (a single, diagonal spatial harmonic) corrupted by a very-high-frequency periodic disturbance, $D(a, b) = 0.5 \cdot (-1)^{a+b}$ (the checkerboard pattern, the Nyquist frequency $(4, 4)$ in both directions): the observed image is $I = I_0 + D$. With Python (script `vol12_cap13_immagini.py`), the 2D DFT of I has exactly three nonzero bins:

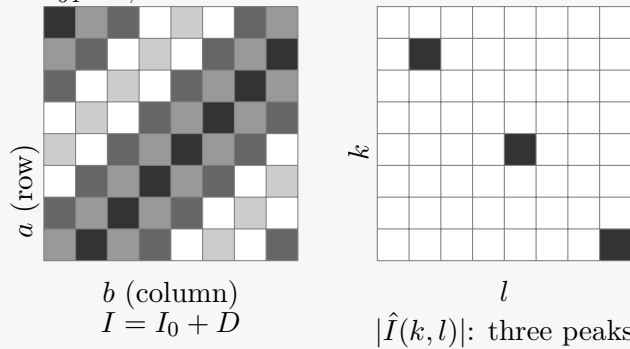
$$\hat{I}(1, 1) = 32, \quad \hat{I}(4, 4) = 32, \quad \hat{I}(7, 7) = 32$$

(all real; $(1, 1)$ and $(7, 7) = (-1, -1)$ are the conjugate pair of I_0 , $(4, 4)$ is the exact signature of the checkerboard disturbance). A **notch filter** that zeroes only the bin $(4, 4)$ (a single bin, because at the Nyquist frequency $4 \equiv -4 \pmod{8}$: no separate conjugate pair needs to be zeroed) and applies the inverse DFT reconstructs I_0 with maximum error 2.2×10^{-16} (machine epsilon) over all 64 pixels. Row $a = 0$, for comparison:

$$I(0, \cdot) = (1.5, 0.2071068, 0.5, -1.2071068, -0.5, -1.2071068, 0.5, 0.2071068),$$

$$I_0(0, \cdot) = (1, 0.7071068, 0, -0.7071068, -1, -0.7071068, 0, 0.7071068),$$

and the reconstructed row matches $I_0(0, \cdot)$ to within 10^{-16} . **Parseval check** (Theorem 13.3): $\sum_{a,b} |I(a, b)|^2 = 48$, and $\frac{1}{64} \sum_{k,l} |\hat{I}(k, l)|^2 = 48.000000$. $\checkmark$



On the left, the observed image I (gray levels proportional to value, 5 levels); on the right, the amplitude spectrum $|\hat{I}(k, l)|$: only three dark cells, at $(1, 1)$, $(4, 4)$ and $(7, 7)$, out of 64

bins — the periodic disturbance is visually "a single point" in the frequency domain, even though it is spread over all 64 pixels in the spatial domain.

14.5. Assigned exercises

Exercise 14.1: DC component of a constant image. Let $I(a, b) = 5$ for every $(a, b) \in \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ (uniform gray image). Compute $\hat{I}(0, 0)$ directly from Definition 13.1, and check it is consistent with Proposition 14.1. (Estimated time: 10 minutes.)

Exercise 14.2: 2D FFT cost for a real image. A camera produces $N \times N$ images with $N = 2048 = 2^{11}$. Compute the approximate number of operations for the direct 2D DFT and for the row-column 2D FFT (Theorem 14.4), and the speed-up factor. (Estimated time: 10 minutes.)

14.6. Worked exercises

Exercise 14.3: S1 — DC component, step by step. Fully solves Exercise 14.1: $I(a, b) = 5$ on $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, computing $\hat{I}(0, 0)$.

Solution. For $k = l = 0$, every phase factor in Definition 13.1 equals $e^0 = 1$:

$$\hat{I}(0, 0) = \sum_{a=0}^3 \sum_{b=0}^3 I(a, b) \cdot 1 \cdot 1 = \sum_{a=0}^3 \sum_{b=0}^3 5 = 16 \times 5 = 80.$$

Independent check: $\sum_{a,b} I(a, b) = 80$ by direct count (16 pixels of value 5), consistent with $\hat{I}(0, 0) = \sum_{a,b} I(a, b)$ used in the proof of Proposition 14.1. ✓

Exercise 14.4: S2 — 2D FFT cost, step by step. Fully solves Exercise 14.2: $N = 2048$.

Solution. Step 1 — direct DFT. $O(N^4) = 2048^4 = 2^{44} \approx 1.76 \times 10^{13}$ operations.

Step 2 — row-column FFT. $\log_2 N = 11$; $O(N^2 \log_2 N) = 2048^2 \times 11 = 4\,194\,304 \times 11 = 46\,137\,344 \approx 4.61 \times 10^7$ operations.

Step 3 — speed-up factor.

$$\frac{1.76 \times 10^{13}}{4.61 \times 10^7} \approx 3.82 \times 10^5,$$

over three hundred eighty thousand times faster — consistent with the general estimate $O(N^2/\log_2 N)$ of Theorem 14.4, which for $N = 2048$ gives $2048^2/11 \approx 3.81 \times 10^5$. ✓ (the two computations, up to the multiplicative constant hidden in the O symbol, agree in order of magnitude.)

14.7. Self-assessment quiz

Q1) How is a grayscale digital image formalised as an object already met in Chapter 13?

-
- Q2) State the two-dimensional reconstruction theorem by separate application: what are the hypotheses on the support of the transform?
- Q3) In the proof of that theorem, why does the section $h_y(x) = I_c(x, y)$ at fixed y turn out to be band-limited in x ?
- Q4) Define the ideal low-pass filter H_ρ on a discrete image.
- Q5) Explain (in words) why filtering in frequency amounts to a convolution in space.
- Q6) Why does the low-pass filter always preserve the image's DC component?
- Q7) Where exactly does the $O(N^2 \log N)$ cost of the row-column 2D FFT come from?
- Q8) In the notch-filter example, why does zeroing a single bin $(4, 4)$ suffice, unlike the conjugate pair $(1, 1)/(7, 7)$?

Teaching key. **Q1)** As a function $I : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{R}$ on the discrete torus, the same object studied in Chapter 13 (Definition 14.1). **Q2)** Theorem 14.1: $\hat{I}_c$ vanishing outside the rectangle $[-B_x, B_x] \times [-B_y, B_y]$, $\Delta x \leq \pi/B_x$, $\Delta y \leq \pi/B_y$. **Q3)** Because the two-dimensional inverse transform, restricted to the rectangular support, rewrites as the 1D inverse transform in x of a function $J(\omega_x; y)$ that is itself zero for $|\omega_x| > B_x$ (Step 1 of the proof). **Q4)** Definition 14.2: $H_\rho(k, l) = 1$ if the centred distance $\sqrt{\kappa(k)^2 + \lambda(l)^2} \leq \rho$, otherwise 0. **Q5)** Because multiplying by H in the frequency domain is, by the already-proved 2D convolution theorem, exactly the transform of a convolution with $\text{DFT}^{-1}(H)$ (Theorem 14.2). **Q6)** Because $H_\rho(0, 0) = 1$ for every $\rho \geq 0$ (the distance from the origin is always 0), hence $\hat{J}(0, 0) = \hat{I}(0, 0)$ (Proposition 14.1). **Q7)** From N 1D FFTs along the rows plus N 1D FFTs along the columns, each costing $O(N \log N)$: $N \cdot O(N \log N) + N \cdot O(N \log N) = O(N^2 \log N)$ (Theorem 14.4). **Q8)** Because at the Nyquist frequency $k = N/2$ we have $k \equiv -k \pmod{N}$ (here $4 \equiv -4 \pmod{8}$): the bin coincides with its own conjugate, not forming a distinct pair as $(1, 1)$ and $(7, 7) = (-1, -1)$ do.

14.8. Chapter formulary

Image = function $I : \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{R}$ (Definition 14.1).

2D reconstruction: $I_c(x, y) = \sum_{p,q} I_c(p\Delta x, q\Delta y) \text{sinc}\left(\frac{x-p\Delta x}{\Delta x}\right) \text{sinc}\left(\frac{y-q\Delta y}{\Delta y}\right)$, if $\Delta x \leq \pi/B_x$, $\Delta y \leq \pi/B_y$.

Ideal filter: $H_\rho(k, l) = \mathbb{K}[\sqrt{\kappa(k)^2 + \lambda(l)^2} \leq \rho]$.

Filtering = convolution: $J = I * \text{DFT}^{-1}(H)$ (Theorem 14.2).

2D FFT cost: $O(N^2 \log N)$ against $O(N^4)$ for the direct 2D DFT (Theorem 14.4).

Sources and bibliography

- [1] R. C. Gonzalez, R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, New York, 2018, ISBN 978-0-13-335672-4 (reference treatment of frequency-domain filtering and the 2D FFT; existence, publisher and year verified via cross-search).
- [2] Vol. II, Chapter 13 (2D DFT and its separability, 2D convolution and Parseval).
- [3] Vol. III, Chapter 24 (Nyquist–Shannon sampling theorem, recalled but not reproved here for the 1D case).
- [4] Vol. III, Chapter 23 (correctness and cost of the radix-2 FFT algorithm, recalled but not reproved here).

CHAPTER 15

One single circle: a map and the unity of the treatise

This chapter introduces no new objects. Its purpose is the opposite: to show that the many objects met in the two volumes are the *same* object seen from different angles, and that some theorems apparently very far apart are, literally, the same identity written in different languages. It is the chapter one should read twice: once now, to get oriented, and once at the end, to recognise the design.

15.1. The six faces of one object

The unit circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ admits at least six descriptions, each of which carries a theory with it:

Structure	Object	Where it is developed
Metric set	locus of equidistant points	Volume I, Chapters 2–4
Curve	$\gamma(t) = (\cos t, \sin t)$, $\kappa \equiv 1$	Ch. 1, (Volume III, Chapter 2)
Group	$S^1 \cong SO(2) \cong \mathbb{R}/2\pi\mathbb{Z}$	Ch. 3, (Volume III, Chapter 7)
Harmonic space	characters $e^{in\theta}$, basis of $L^2(\mathbb{T})$	Ch. 4
Probability space	invariant measure $d\theta/2\pi$	Ch. 6
Finite group	$\mu_n \cong \mathbb{Z}/n\mathbb{Z}$	Ch. 7

The six rows are not six separate chapters: they are six ways of looking at a single figure, and every time one of them produces a theorem, the other five produce a translation of it. The sections that follow exhibit three of these translations in full.

15.2. One single measure

The first thread is measure. On the circle there is one and only one notion of “uniform”, and from it descend directional statistics, the resolution of Bertrand’s paradox and Barbier’s formula.

Theorem 15.1: Uniqueness of the invariant measure on the circle. *There exists exactly one Borel probability measure on S^1 invariant under all rotations, and it is the normalised measure $d\theta/2\pi$.*

Proof. Step 1 — the Fourier coefficients of a measure. Let μ be a Borel probability measure on S^1 , identified with $\mathbb{R}/2\pi\mathbb{Z}$, and define its Fourier coefficients

$$\hat{\mu}(n) = \int_{S^1} e^{-in\theta} d\mu(\theta), \quad n \in \mathbb{Z}.$$

They are well defined because $|e^{-in\theta}| = 1$ and μ is finite, and $\hat{\mu}(0) = 1$ because μ is a probability measure.

Step 2 — invariance kills all but one. Let R_α be the rotation by angle α . If μ is invariant, then for every continuous function f

$$\int f(\theta + \alpha) d\mu(\theta) = \int f(\theta) d\mu(\theta).$$

Applying this to $f(\theta) = e^{-in\theta}$ gives

$$\hat{\mu}(n) = \int e^{-in(\theta+\alpha)} d\mu(\theta) = e^{-in\alpha} \hat{\mu}(n) \quad \text{for every } \alpha.$$

For $n \neq 0$ choose α with $e^{-in\alpha} \neq 1$ (take $\alpha = \pi/n$): then $(1 - e^{-in\alpha})\hat{\mu}(n) = 0$ forces $\hat{\mu}(n) = 0$.

Step 3 — the coefficients determine the measure. By the Stone–Weierstrass theorem the trigonometric polynomials are dense in $C(S^1)$ in the uniform norm (they form an algebra that separates points, contains the constants and is closed under conjugation). If two finite measures have the same Fourier coefficients, then their integrals agree on all trigonometric polynomials, and by density and continuity of the integral with respect to the uniform norm they agree on all of $C(S^1)$; by the Riesz representation theorem they are the same measure.

Step 4 — conclusion. The normalised measure $\lambda = d\theta/2\pi$ is invariant (the Riemann integral is invariant under translation of the variable) and has $\hat{\lambda}(0) = 1$, $\hat{\lambda}(n) = 0$ for $n \neq 0$ (Theorem 4.2 of Chapter 4). By Steps 2 and 3 every invariant measure has the same coefficients as λ , hence coincides with λ . $\square$

Remark 15.1: The three consequences. Theorem 15.1 is the pivot of three apparently unrelated results.

- **Directional statistics.** The null hypothesis “the directions are distributed at random” has only one possible meaning, and the Rayleigh test (Ch. 6) measures precisely the departure of the first Fourier coefficient from zero: it is the statistical translation of Step 2.
- **Bertrand’s paradox.** The three answers of Volume I correspond to three different measures on the space of chords; the only one invariant under all rigid motions is the one giving $1/2$. Here uniqueness concerns not S^1 but the space of lines, and is proved in the same way.
- **Barbier’s theorem.** The formula $\mathbb{E}[N] = 2L/(\pi d)$ for Buffon’s needle holds because the position of the curve is distributed according to the invariant measure; applied to a curve of constant width it gives $L = \pi w$ (Volume III, Chapter 4). Barbier is therefore, literally, a statement about the invariant measure.

Theorem 15.2: Weyl’s criterion and equidistribution. Let $(x_n)_{n \geq 1}$ be a sequence in $S^1 = \mathbb{R}/2\pi\mathbb{Z}$. It is equidistributed, that is

$$\lim_{N \rightarrow \infty} \frac{\#\{n \leq N : x_n \in I\}}{N} = \frac{|I|}{2\pi} \quad \text{for every arc } I,$$

if and only if

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{ihx_n} = 0 \quad \text{for every } h \in \mathbb{Z}, h \neq 0.$$

In particular, for every irrational α the sequence $x_n = 2\pi n\alpha$ is equidistributed.

Proof. **Step 1 — the condition is necessary.** If the sequence is equidistributed, then for every step function (a finite combination of indicators of arcs) the arithmetic mean of the values converges to the integral with respect to $d\theta/2\pi$; by density of step functions in $C(S^1)$

in the L^1 norm and by uniform boundedness, the same holds for every continuous f . Applying it to $f(\theta) = e^{ih\theta}$ with $h \neq 0$ gives

$$\frac{1}{N} \sum_{n \leq N} e^{ihx_n} \longrightarrow \frac{1}{2\pi} \int_0^{2\pi} e^{ih\theta} d\theta = 0.$$

Step 2 — the condition is sufficient. Suppose the exponential means tend to zero for every $h \neq 0$; for $h = 0$ the mean is identically 1, which is the integral of 1. By linearity, therefore, for every trigonometric polynomial P

$$\frac{1}{N} \sum_{n \leq N} P(x_n) \longrightarrow \frac{1}{2\pi} \int_0^{2\pi} P.$$

Let now f be continuous and $\varepsilon > 0$; by Stone–Weierstrass there is P with $\|f - P\|_\infty < \varepsilon$, and then

$$\left| \frac{1}{N} \sum f(x_n) - \frac{1}{2\pi} \int f \right| \leq 2\varepsilon + \left| \frac{1}{N} \sum P(x_n) - \frac{1}{2\pi} \int P \right|,$$

whose second term tends to zero. Since ε is arbitrary, the mean converges to the integral for every continuous f . Finally, the indicator of an arc is approximated from above and from below by continuous functions differing on sets of arbitrarily small measure, whence the claim for arcs.

Step 3 — the sequence $2\pi n\alpha$. With $x_n = 2\pi n\alpha$ and $h \neq 0$,

$$\sum_{n=1}^N e^{ihx_n} = \sum_{n=1}^N q^n, \quad q := e^{2\pi i h \alpha}.$$

Since α is irrational, $h\alpha \notin \mathbb{Z}$ and hence $q \neq 1$; the geometric sum equals $q(q^N - 1)/(q - 1)$, of modulus at most $2/|q - 1|$, a quantity *independent of N* . Dividing by N gives an infinitesimal, and the criterion applies.

Step 4 — what has been proved. An irrational rotation, iterated, fills the circle uniformly: the orbit of a point under the group generated by an irrational rotation is not only dense, but visits every arc with frequency proportional to its length. This is the ergodicity of the irrational rotation, and the fact that the criterion reduces to a geometric sum is, once again, the orthogonality of the characters. $\square$

15.3. One single orthogonality

The second thread is the orthogonality of the characters of the circle,

$$\frac{1}{2\pi} \int_0^{2\pi} e^{i(m-n)\theta} d\theta = \delta_{mn}, \quad \frac{1}{n} \sum_{k=0}^{n-1} \zeta^{(m-l)k} = \delta_{ml} \quad (\zeta = e^{2\pi i/n}),$$

in its continuous and its discrete version. It appears, under different names, in at least five places of the two volumes:

Local name	Statement	Chapter
Euler–Fourier formulas	$c_n = \frac{1}{2\pi} \int f e^{-in\theta}$	4
Parseval’s identity	$\ f\ ^2 = \sum c_n ^2$	4
Invertibility of the DFT	$W_n^{-1} = \frac{1}{n} \overline{W_n}$	7
Fortescue’s theorem	$W_3^{-1} = \frac{1}{3} \overline{W_3}$	5
Power under distortion	different harmonics exchange no power	5
BCH bound	Vandermonde of the roots invertible	7

***Remark 15.2: Why they are the same thing.** In all six cases the ingredient is that the matrix $(\zeta^{jk})_{j,k}$ — or its continuous analogue — is invertible because its columns are orthogonal. In the continuous case invertibility is the completeness of the trigonometric system; in the discrete case it is the fact that the Vandermonde determinant $\prod_{j < k} (\zeta^k - \zeta^j)$ does not vanish, because the n -th roots of unity are distinct. Fortescue's theorem is the case $n = 3$; the BCH bound is the observation that a Vandermonde minor of order w remains invertible, and hence a vector with few non-zero samples cannot have many consecutive vanishing coefficients. The minimum distance of a code and the bandwidth of a signal are the same quantity, measured on the same circle.*

15.4. One single identity

The third thread is the elementary identity

$$1 + \zeta + \zeta^2 + \cdots + \zeta^{n-1} = 0, \quad \zeta = e^{2\pi i/n}, \quad n \geq 2,$$

that is: *the n -th roots of unity sum to zero*. In Volume II it appears as:

- the fact that in a balanced three-phase system the neutral carries no current ($1 + a + a^2 = 0$, Ch. 5);
- the fact that harmonics of order not a multiple of three cancel in the neutral, while those that are multiples of three add up (because $a^{-h} = 1$ if and only if $3 \mid h$, Ch. 5);
- the fact that the centroid of a regular polygon is its centre (Ch. 3);
- the constant term of the cyclotomic polynomials and the factorisation $x^n - 1 = \prod_{d|n} \Phi_d(x)$ (Ch. 7);
- the zero-sum condition of the residues in a sequence network (Ch. 5).

The most insidious industrial phenomenon of the modern electrical grid — a neutral conductor carrying more current than the phases — is, in the last analysis, the arithmetical fact that 3 divides 9 but does not divide 5.

15.5. A map of the three volumes

Ch.	Title (abridged)	Face of S^1 developed
Volume I		
1	History and π	the constant as a similarity invariant
2–4	Definitions, chords, angles	S^1 as a metric locus
5	Inversion	S^1 and the Möbius group
6	Measures	S^1 as a rectifiable and measurable object
7	Analytic geometry	S^1 as a conic with constrained coefficients
8–10	Trigonometry, high-school, triangle	S^1 as the domain of the circular functions
11	Numerics	S^1 discretised and realised in a machine
12	Gauss's circle	S^1 and counting lattice points
13	Robotics	S^1 as the space of joint configurations
14	Water treatment	S^1 in sedimentation and circular clarifiers
Volume II		
1	Parametrisation	S^1 as a regular curve
2	Integral calculus	length and area as integrals
3	Complex numbers	S^1 as a multiplicative group
4	Harmonic analysis	S^1 as a Fourier domain
5	Electrical engineering	S^1 as the space of phasors
6	Directional statistics	S^1 as a probability space
7	The discrete circle	S^1 as $\mu_n \cong \mathbb{Z}/n\mathbb{Z}$
8	Pencils, inversion, Steiner	S^1 as a coaxial family and a path in the Möbius group
9	Differential equations and circular motion	S^1 as the phase trajectory of the harmonic oscillator
10	Celestial mechanics, Kepler's equation	S^1 as the auxiliary circle of the ellipse
11	Astrophysics and relativity	S^1 as the photon sphere, the ISCO and the Einstein ring in curved space-time
12	Cryptography, Diffie–Hellman/ElGamal	discrete S^1 as the basis of the power/logarithm asymmetry
13	DFT in several dimensions	$S^1 \times S^1 =$ discrete torus, product of circles
14	Image processing	a grid of discrete circles, filtered in frequency
Volume III		
Vol. III, Ch. 2, 4	Plane curves and global theorems	S^1 as a curve of constant curvature
Vol. III, Ch. 7, 11	Sphere and any dimension	S^1 as the case $n = 2$ of S^{n-1}
Vol. III, Ch. 12, 13	Topology and group	S^1 as a covering and a compact group
Vol. III, Ch. 15–18	Analysis on the disc	S^1 as the boundary of the disc
Vol. III, Ch. 19–24	Transforms and signals	S^1 as a Fourier domain
Vol. III, Ch. 25	Control and lines	S^1 as the stability boundary

15.6. What the circle is *not*

A definition is understood also through what it excludes. Three objects extremely close to the circle each show which hypothesis is essential.

Object	What it loses	What follows
Ellipse	constant curvature	four vertices, evolute with cusps
Reuleaux triangle	C^1 regularity at the corners	constant width but minimal area
Sphere S^2	commutativity of the group	$SO(3)$ non-abelian, no one-dimensional characters

Remark 15.3: The gravest loss. *Of the three, the decisive loss is the third. As long as the group is S^1 , its characters are complex-valued functions and harmonic analysis is a series of numbers: it is this one-dimensionality that makes possible the Fourier transform in the form all the chapters have used. Passing to $SO(3)$, the irreducible representations have dimension $2\ell + 1$, the “Fourier coefficients” become matrices, and multiplication no longer commutes: non-commutative harmonic analysis is born, of which the spherical harmonics are the first chapter. (Volume III, Chapter 7) stops on the threshold; the fact that S^1 is abelian and connected is the ultimate reason why it admits so complete a theory, and the reason why this treatise can exist.*

15.7. Synthesis questions

The questions that follow require no computation: they require recognising the same theorem under different names. The answer follows each one immediately.

Exercise 15.1: Q1 — One identity, three names. Which identity is simultaneously: (a) the reason why the neutral of a balanced three-phase system carries no current; (b) the reason why the centroid of a regular polygon is its centre; (c) related to the value of $\Phi_n(1)$ for n prime?

Solution. The vanishing sum of the n -th roots of unity, $\sum_{k=0}^{n-1} \zeta^k = 0$. In (a) with $n = 3$; in (b) as the sum of the vertices $R\zeta^k$ divided by n ; in (c), instead, $\Phi_p(1) = 1 + 1 + \cdots + 1 = p$, and the link is that $x^p - 1 = (x - 1)\Phi_p(x)$, whose derivative at $x = 1$ recovers precisely the sum of the roots.

Exercise 15.2: Q2 — Why the modulus margin and the sensitivity peak are the same number. Explain in one sentence why M_s and the distance of the Nyquist plot from the point -1 are reciprocals of each other, and to which family of circles of Volume III, Chapter 25, the sensitivity circle belongs.

Solution. Because $|S(j\omega)| = 1/|1 + L(j\omega)|$ and $|1 + L|$ is the Euclidean distance of L from -1 : the maximum of a reciprocal is the reciprocal of the minimum. The sensitivity circle belongs to the family of the loci $|1 + L| = \text{const}$, that is to the same family as the M - and N -circles, read however on the sensitivity function instead of the closed-loop one.

Exercise 15.3: Q3 — The BCH bound as an uncertainty principle. State the BCH bound in terms of “support in time” and “support in frequency”, and say which property of the Vandermonde matrix produces it.

Solution. A non-zero vector of length n cannot have both fewer than δ non-zero samples and $\delta - 1$ consecutive vanishing Fourier coefficients. The property is that every Vandermonde minor built on distinct powers of a primitive root has non-zero determinant, because the roots are pairwise different.

Exercise 15.4: Q4 — Two proofs of Barbier. Indicate the two proofs of Barbier's theorem present in the treatise and say which hypothesis each of them uses essentially.

Solution. The support-function proof (Volume III, Chapter 4) uses $L = \int_0^{2\pi} h$ and $w(\theta) = h(\theta) + h(\theta + \pi)$, hence convexity and enough regularity for Cauchy's formula. The probabilistic proof (Volume I, Buffon's needle) uses only that every line of the grid of spacing w meets the curve in 0 or 2 points, that is convexity, and the invariance of the kinematic measure.

Exercise 15.5: Q5 — Where the circle stops being enough. Say which property of S^1 is lost in passing to S^2 , and what consequence it has for Fourier theory.

Solution. The commutativity of the rotation group: $SO(2)$ is abelian, $SO(3)$ is not. Consequently the irreducible representations are no longer one-dimensional and the Fourier coefficients become matrices; the eigenfunctions of the Laplacian are the spherical harmonics, grouped into spaces of dimension $2\ell + 1$.

Exercise 15.6: Q6 — One theorem, two industries. Fortescue's theorem and the Nyquist criterion both use the circle, but in two different ways. Which?

Solution. Fortescue uses the circle as a *finite group*: the three phases are μ_3 , and the decomposition is the three-point DFT. Nyquist uses the circle as a *closed path* in the complex plane: what matters is the number of turns about -1 , that is a topological index. The first is algebra on the discrete circle, the second topology of the continuous circle.

15.8. Chapter formulary

Invariant measure

$$d\lambda = \frac{d\theta}{2\pi}, \quad \widehat{\lambda}(n) = \delta_{n0}, \quad \text{the unique rotation-invariant probability measure}$$

Characters and orthogonality

$$\chi_n(\theta) = e^{in\theta}, \quad \langle \chi_m, \chi_n \rangle = \delta_{mn}, \quad \frac{1}{n} \sum_{k=0}^{n-1} \zeta^{(m-l)k} = \delta_{ml}$$

Vanishing sum

$$\sum_{k=0}^{n-1} \zeta^k = 0 \quad (n \geq 2), \quad 1 + a + a^2 = 0 \quad (a = e^{j2\pi/3})$$

Weyl's criterion

$$(x_n) \text{ equidistributed} \iff \frac{1}{N} \sum_{n \leq N} e^{ihx_n} \rightarrow 0 \quad \forall h \neq 0$$

Sources and bibliography

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Final assessment of the Volume

INFO

How to use this assessment. Thirty-three multiple-choice questions spanning the whole volume, from the parametrisation of the circle to cyclic codes. Unlike the end-of-chapter quizzes, here the **answer is given immediately after each question**, together with a pointer to the result that justifies it: it is designed for revision, not for grading. Anyone wishing to use it as a closed test should cover the boldface line with a sheet of paper.

Indicative scoring: ≥ 30 excellent; 25–29 good; 20–24 sufficient, with targeted revision of the chapters missed; < 20 it is worth rereading the volume following the logical chain that leads from Chapter 1 to Chapter 7.

Part I — Parametrisation and differential calculus (Chapter 1)

1. For $\gamma(t) = (R \cos t, R \sin t)$ the speed $\|\gamma'(t)\|$ equals

- A) 1
- B) R
- C) R^2
- D) $2\pi R$

Answer: B. $\gamma'(t) = (-R \sin t, R \cos t)$ has norm $R\sqrt{\sin^2 t + \cos^2 t} = R$: constant, but equal to 1 only if $R = 1$.

2. The **natural parameter** (arc length) on the circle of radius R is

- A) $s = t$
- B) $s = t/R$
- C) $s = Rt$
- D) $s = R^2 t$

Answer: C. From $ds/dt = \|\gamma'\| = R$ we get $s = Rt$, and in that parameter the velocity has unit norm.

3. The curvature of the circle of radius R is

- A) R
- B) $1/R$
- C) $2\pi/R$
- D) 0

Answer: B. This is the content of the Frenet formulas: the circle is the plane curve of constant non-zero curvature, and the radius of curvature coincides with the radius.

4. The rational parametrisation $(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$ of the unit circle

- A) covers the whole circle
- B) covers the circle except the point $(-1, 0)$
- C) covers only the first quadrant
- D) does not lie on the circle

Answer: B. It is the stereographic projection from $(-1, 0)$, which is excluded: it is reached only in the limit $t \rightarrow \pm\infty$.

5. In *uniform* circular motion the acceleration is

- A) zero
- B) tangential, of magnitude ωR
- C) centripetal, of magnitude $\omega^2 R$
- D) centrifugal, of magnitude $\omega^2 R$

Answer: C. The tangential component is $\dot{\omega}R = 0$; only the normal one remains, directed towards the centre.

Part II — Integral calculus on the circle (Chapter 2)

6. The integral $\oint_C ds$ along the circle of radius R equals

- A) πR^2
- B) $2\pi R$
- C) πR
- D) R

Answer: B. The line element is $ds = R d\theta$ and the integral over one turn gives the circumference.

7. The field $F = \left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}\right)$ has circulation, around a circle enclosing the origin, equal to

- A) 0
- B) π
- C) 2π
- D) it depends on the radius

Answer: C. It equals 2π for every radius: the field is irrotational but not conservative, because the domain $\mathbb{R}^2 \setminus \{0\}$ is not simply connected.

8. Green's theorem relates

- A) two line integrals
- B) the circulation along the boundary to the double integral of the curl
- C) the flux to the length
- D) the derivative to an integral over an open curve

Answer: B. $\oint_{\partial D} (P dx + Q dy) = \iint_D (\partial_x Q - \partial_y P) dA$; it is the tool that computes the area from the boundary alone.

9. In polar coordinates the area element is

- A) $dr d\theta$
- B) $r dr d\theta$
- C) $r^2 dr d\theta$
- D) $\frac{1}{r} dr d\theta$

Answer: B. The factor r is the Jacobian determinant of the change of variables: forgetting it is the commonest error in all of integral calculus.

10. Applied to a smooth periodic function over one period, the trapezoidal rule

- A) converges like $1/n^2$
- B) converges exponentially in n
- C) does not converge
- D) converges like $1/n$

Answer: B. On the circle all boundary terms of the Euler–Maclaurin formula cancel: it is the most efficient quadrature method for periodic integration.

Part III — Complex numbers and the circle (Chapter 3)

11. Euler’s formula states that

- A) $e^{i\theta} = \cos \theta - i \sin \theta$
- B) $e^{i\theta} = \cos \theta + i \sin \theta$
- C) $e^\theta = \cos \theta + i \sin \theta$
- D) $e^{i\theta} = \sin \theta + i \cos \theta$

Answer: B. From it follows that the unit circle is the image of the imaginary axis under the exponential.

12. De Moivre’s theorem gives

- A) $(\cos \theta + i \sin \theta)^n = \cos^n \theta + i \sin^n \theta$
- B) $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$
- C) $(\cos \theta + i \sin \theta)^n = n(\cos \theta + i \sin \theta)$
- D) $(\cos \theta + i \sin \theta)^n = 1$

Answer: B. Raising to a power multiplies the argument: the algebraic translation of the fact that rotating n times by θ is rotating by $n\theta$.

13. The sum of all n -th roots of unity, for $n \geq 2$, equals

- A) n
- B) 1
- C) 0
- D) 2π

Answer: C. Their centroid is the centre of the circle; algebraically it is the sum of a geometric progression of ratio $\zeta_n \neq 1$.

14. For a complex number z , the condition $|z| = 1$ is equivalent to

- A) $z = \bar{z}$
- B) $z + \bar{z} = 1$
- C) $z\bar{z} = 1$
- D) $z^2 = 1$

Answer: C. Since $z\bar{z} = |z|^2$; on this identity rests all the geometry of the circle in complex form.

15. The n -th roots of unity, under multiplication, form

- A) a field
- B) a cyclic group of order n
- C) a vector space
- D) a set that is not closed

Answer: B. This fact is taken up and generalised in Chapter 7, where it becomes the isomorphism with $\mathbb{Z}/n\mathbb{Z}$.

Part IV — Harmonic analysis and transforms (Chapter 4)

16. The functions $e_n(\theta) = e^{in\theta}$, $n \in \mathbb{Z}$, form

- A) an orthonormal basis of $L^2(\mathbb{T})$
- B) a linearly dependent set
- C) a basis only for even functions
- D) a finite set

Answer: A. They are the characters of the circle, and their orthonormality is the master identity from which everything else follows.

17. Parseval's identity states that

- A) $\sum_n |\hat{f}(n)| = \|f\|_1$
- B) $\sum_n |\hat{f}(n)|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f|^2$
- C) $\sum_n \hat{f}(n) = f(0)$
- D) $\hat{f}(n) \rightarrow 0$

Answer: B. It is the conservation of energy under passage to the frequency domain; answer D is instead the Riemann–Lebesgue lemma.

18. Circular convolution of two functions corresponds, on Fourier coefficients, to

- A) a sum
- B) a product
- C) a derivative

D) an integral

Answer: B. This is why linear filters are designed in the frequency domain: a multiplication instead of an integral.

Part V — Electrical engineering in sinusoidal steady state (Chapter 5)

19. The phasor associated with $v(t) = V_M \cos(\omega t + \alpha)$ has

- A) modulus V_M and phase ω
- B) modulus $V_M/\sqrt{2}$ and phase α
- C) modulus ω and phase α
- D) modulus V_M^2 and zero phase

Answer: B. In the RMS convention the modulus is the RMS value and the phase is the initial one: the angular frequency, common to the whole circuit, drops out of the representation.

20. The impedance of a capacitor of capacitance C is

- A) $j\omega C$
- B) $\frac{1}{j\omega C}$
- C) ωC
- D) $\frac{1}{\omega C^2}$

Answer: B. It is purely imaginary and negative: the current leads the voltage by 90° .

21. Power-factor correction of an inductive load is obtained by

- A) adding an inductor in series
- B) adding a capacitor in parallel
- C) raising the voltage
- D) lowering the frequency

Answer: B. The capacitor supplies reactive power of opposite sign, reducing what is exchanged with the grid at equal active power.

22. The resonant angular frequency of a series RLC circuit is

- A) $\omega_0 = \frac{1}{\sqrt{LC}}$
- B) $\omega_0 = \sqrt{LC}$
- C) $\omega_0 = \frac{R}{L}$
- D) $\omega_0 = \frac{1}{RC}$

Answer: A. At that frequency the reactances cancel and the impedance is purely resistive; the bandwidth is R/L .

23. The Fortescue transformation decomposes an arbitrary three-phase set into

- A) two components
- B) three components: positive, negative, zero sequence
- C) four components
- D) a single component

Answer: B. It is exactly the discrete Fourier transform on $\mathbb{Z}/3\mathbb{Z}$, as Chapter 7 shows.

Part VI — Directional statistics (Chapter 6)

24. The arithmetic mean of 350° and 10° is 180° : this shows that

- A) the two winds are opposite
- B) the arithmetic mean is not invariant under change of origin
- C) the computation is wrong
- D) more data are needed

Answer: B. The remedy is to average the *points* $e^{i\theta}$ and read the result in polar form.

25. The trigonometric moments $\varphi_p = \mathbb{E}[e^{ip\Theta}]$ coincide, up to 2π , with

- A) the density at integer points
- B) the Fourier coefficients of the density
- C) the variance
- D) the entropy

Answer: B. It is the dictionary that makes circular statistics a chapter of harmonic analysis.

26. The von Mises distribution is characterised by being

- A) the wrapping of a normal
- B) of maximum entropy at given mean and resultant
- C) the only symmetric one
- D) the only one with real moments

Answer: B. Answer A describes instead the wrapped normal, numerically close to it but arising from a different principle.

27. The Rayleigh test statistic is

- A) R
- B) nR
- C) nR^2
- D) R^2/n

Answer: C. Under uniformity it tends to an exponential of parameter 1; the test is however blind to antipodal alternatives.

28. A non-lattice random walk on the circle tends to

- A) concentrate
- B) diverge
- C) become uniformly distributed
- D) stay unchanged

Answer: C. The moments multiply and, having modulus < 1 , tend to zero: on the compact circle the uniform is the only invariant distribution.

Part VII — The discrete circle (Chapter 7)

29. The group μ_n of n -th roots of unity is isomorphic to

- A) $\mathbb{Z}$
- B) $\mathbb{Z}/n\mathbb{Z}$
- C) $\mathbb{Q}$
- D) $\mathbb{R}$

Answer: B. Adding angles becomes multiplying complex numbers: the circle divided into n parts *is* arithmetic modulo n .

30. The Euler–Fermat theorem states that, if $\gcd(a, n) = 1$,

- A) $a^n \equiv 1 \pmod{n}$
- B) $a^{n-1} \equiv 1 \pmod{n}$
- C) $a^{\varphi(n)} \equiv 1 \pmod{n}$
- D) $a \equiv 1 \pmod{n}$

Answer: C. Answer B holds only for prime n ; the exponent must be reduced modulo $\varphi(n)$, not modulo n .

31. The regular n -gon is constructible with straightedge and compass if and only if

- A) n is prime
- B) $n = 2^a$ times a product of distinct Fermat primes
- C) n is odd
- D) always

Answer: B. This is the Gauss–Wantzel theorem: the heptagon is not constructible ($\varphi(7) = 6$), the 17-gon is ($\varphi(17) = 16$).

32. In $\mathbb{F}_q[x]/(x^n - 1)$ the cyclic codes correspond to

- A) arbitrary subspaces
- B) the ideals, i.e. the divisors of $x^n - 1$
- C) invertible matrices
- D) irreducible polynomials only

Answer: B. Multiplying by x is rotating the block: cyclicity becomes closure under multiplication, i.e. the definition of an ideal.

33. A cyclic code with generator polynomial of degree r detects all error bursts of length

- A) $\leq r$
- B) $\leq r/2$
- C) any
- D) ≤ 2

Answer: A. This is the foundation of the CRC: with $r = 32$, as in Ethernet, all disturbances up to 32 consecutive bits are detected.

INFO

After the assessment. The ten parts follow the order of the chapters, and it is no accident that the first and the last mirror each other: we started from the circle as a parametrised curve and arrived at the circle as a finite group, passing through transforms, higher dimensions, phasors, control, statistics and curvature. A reader who answered questions 1–20 well but the later ones poorly has the tools and must practise using them; one who did the opposite knows the applications and would do well to consolidate the foundations, for it is from the logical chain — not from isolated formulas — that this volume draws its unity.

Conclusion

We have traversed four chapters and, apparently, four different areas of mathematics: differential geometry, integral calculus, complex analysis, harmonic analysis. It is worth making explicit, in closing, the thread that binds them, because it is not an expository artifice: it is the very same property of the circle seen from four angles.

Chapter	How the circle's constant curvature manifests
1. Parametrization	$\kappa(t) \equiv 1/r$: the circle is the only planar curve whose tangent rotates at constant angular velocity, the solution of the simplest possible Frenet ODE.
2. Integral calculus	The same symmetry makes length, area, and even the Gaussian integral computable in closed form, via the switch to polar coordinates.
3. Complex numbers	$\mathbb{T} = \{ z = 1\}$ is a group; its n th roots are regular polygons; its transformations (rotations, inversions, Möbius) are the algebra of the geometry of Chapter 1.
4. Harmonic analysis	$\mathbb{T}$ becomes the natural domain of Fourier series, and — via sampling — the stability boundary $ z = 1$ of every digital control system: the same closed curve of Chapter 1 now governs the stability of a discrete PID loop.

Alongside this purely mathematical thread, in this expanded edition, a second, more applied thread has run: the G02/G03 circular interpolation of a CNC (Chapter 1) uses exactly the parametrization $\gamma(t) = (r \cos t, r \sin t)$; the Monte Carlo estimation of π (Chapter 2) shows that the methods of Archimedes and Liu Hui, for all their antiquity, remain the conceptual prototype of every modern numerical algorithm based on random sampling; the discrete Fourier transform (Chapter 4) is nothing but the roots of unity of Chapter 3 used as a basis for a finite vector space; and the Cooley-Tukey algorithm that makes it computable in real time on a PLC is, historically, the same algorithm that Gauss had already discovered in 1805 to interpolate asteroid orbits — two hundred years before it served to filter a signal from a level sensor.

What remains open. The case of *non*-constant curvature, barely touched upon in the Remark on the osculating circle (Chapter 1), is the central object of Volume III, together with its higher-dimensional counterpart (the sphere, the Gauss-Bonnet theorem). The modern extension of the sampling theorem — compressed sensing, which allows reconstruction of sparse signals by sampling *below* the Nyquist limit — is only mentioned in Chapter 4 and would deserve a treatise of its own. Möbius transformations, introduced as a hint in Chapter 3, will be taken up systematically in Volume V together with inversive geometry.

If Volume I showed that the circle is rich in geometry, this Volume II has attempted to show that it is equally rich in analysis — and that, in the end, the two things are the same thing seen from far away or from close up.

Where the treatise continues. With this edition Volume II has been reduced to its role as an introduction. The differential geometry of the circle, the passage to the sphere and to arbitrary dimension, control and transmission lines, the convergence of Fourier series and all the transforms — Fourier, Laplace, Z, DFT and FFT — and the sampling theorem are now in Volume III, *Geometry, Analysis and Signals on the Circle*, where every statement is accompanied by a complete proof. The section "Where the theory continues" of Chapter 4 indicates, topic by

topic, the destination chapter.

CHAPTER 16

Reference Tables

16.1. Trigonometric identities used in the treatise

Identity	
Pythagorean	$\sin^2 \theta + \cos^2 \theta = 1$
Addition	$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$ $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$
Double-angle	$\sin 2\alpha = 2 \sin \alpha \cos \alpha, \quad \cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 2 \cos^2 \alpha - 1$
Half-angle	$\cos^2(\alpha/2) = \frac{1 + \cos \alpha}{2}, \quad \sin^2(\alpha/2) = \frac{1 - \cos \alpha}{2}$
Universal substitution	$t = \tan(\theta/2) \Rightarrow \cos \theta = \frac{1 - t^2}{1 + t^2}, \quad \sin \theta = \frac{2t}{1 + t^2}$
Finite geometric sum	$\sum_{k=0}^{n-1} q^k = \frac{q^n - 1}{q - 1}, \quad q \neq 1$ (used for roots of unity and Dirichlet kernel)
Euler	$e^{i\theta} = \cos \theta + i \sin \theta; \quad \cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

16.2. Index of main symbols

Symbol	Meaning
$\mathbb{R}, \mathbb{C}, \mathbb{Z}, \mathbb{N}$	Reals, complexes, integers, naturals
$C(O, r), D(O, r)$	Circle and disk of center O and radius r
$\gamma(t)$	Parametrization of a curve
$T(t), N(t)$	Tangent and normal unit vectors (Frenet)
κ	Curvature
$\mathbb{T}$	Complex unit circle, $\{z \in \mathbb{C} : z = 1\}$
ζ_k	n -th roots of unity
$\hat{f}(n)$	n -th Fourier coefficient of f
$\hat{f}(\omega)$	Fourier transform of f
$F(s), \mathcal{L}\{\cdot\}$	Laplace transform
$X(z), \mathcal{Z}\{\cdot\}$	Z-transform
ROC	Region of convergence
$D_N(\theta)$	Dirichlet kernel of order N
$u(t), u[n]$	Unit step, continuous and discrete
$\delta(t), \delta[n]$	Dirac delta, discrete unit impulse

16.3. Index of code listings

This treatise favors Python for numerical examples, with some cross-references in MATLAB and SCL (Structured Control Language, IEC 61131-3) for PLC applications.

Language	Content	Location
Python	Simpson's rule quadrature for semicircle area	Sec. 13 (Ch. 2)
Python	Monte Carlo estimation of π	Sec. 13 (Ch. 2)
MATLAB	Monte Carlo estimation of π , vectorized	Sec. 13 (Ch. 2)
Robotcode	INFORM job <code>ARCO_TRE_PUNTI</code> : circular interpolation on Yaskawa Motoman robot	Sec. 15 (Ch. 1)
Python	DFT matrix construction and verification with NumPy	Sec. 8 (Ch. 3)

The radix-2 FFT listings, the SCL function blocks `FB_FFT8`/`FB_LOWPASS1`, and the spectral-window comparison, present in an earlier version of this chapter, have been moved to Volume III (Chapters 22 and 21 respectively, see the Sources of Chapter 4) together with the rest of the material on transforms and signals: they are no longer part of Volume II, and this table no longer lists them.

All Python listings have been executed and numerically verified before inclusion in the text; the robotcode listing (Yaskawa INFORM) is provided for educational purposes, simplified from the full controller syntax, and not executed on real hardware.

Numerical laboratory

This appendix collects six computations that the reader can redo by hand, with paper and a calculator (the last two also need a spreadsheet or a few lines of code), covering six computational cores of the volume: symmetrical components, the Rayleigh test, the CRC, periodic quadrature, circle packing, and Chapter 10's Kepler equation (the eight-point DFT and the modulus margin have moved to Volume III, Chapters 22 and 25). Each section states the algorithm, works out a concrete case and closes with an *independent check*: a second way of obtaining the same number. These are not exercises but test benches: if your own program reproduces these values, it is almost certainly correct.

17.1. Symmetrical components of a triple

Algorithm. $\underline{V}_{0,1,2} = \frac{1}{3}(\underline{V}_a + a^k \underline{V}_b + a^{2k} \underline{V}_c)$ with $a = 1 \angle 120^\circ$ and $k = 0, 1, 2$.

Datum. $\underline{V}_a = 1 \angle 0^\circ$, $\underline{V}_b = 0.6 \angle -100^\circ$, $\underline{V}_c = 0.8 \angle 140^\circ$.

Computation. In Cartesian form $\underline{V}_b = -0.10419 - 0.59088i$ and $\underline{V}_c = -0.61284 + 0.51423i$. Rotating by $\pm 120^\circ$ and adding:

	Cartesian	modulus	argument
$\underline{V}_0$	$0.09432 - 0.02555i$	0.09772	-15.16°
$\underline{V}_1$	$0.77188 + 0.15961i$	0.78821	11.69°
$\underline{V}_2$	$0.13380 - 0.13406i$	0.18941	-45.05°

Independent check (inverse transform).

$$\underline{V}_0 + \underline{V}_1 + \underline{V}_2 = (0.09432 + 0.77188 + 0.13380) + (-0.02555 + 0.15961 - 0.13406)i = 1.00000 + 0.00000i = \underline{V}_a \checkmark$$

A second useful check is the negative-sequence unbalance $|\underline{V}_2|/|\underline{V}_1| = 0.18941/0.78821 = 24.0\%$, an enormous value against the 2% allowed by EN 50160: the triple of the example is not a grid triple but a textbook case.

17.2. Rayleigh test on eight directions

Algorithm. $C = \sum \cos \theta_j$, $S = \sum \sin \theta_j$, $R = \sqrt{C^2 + S^2}$, $\bar{R} = R/n$, $Z = n\bar{R}^2$; reject uniformity if $Z > -\ln \alpha$.

Datum. $\theta = 30^\circ, 45^\circ, 50^\circ, 65^\circ, 70^\circ, 80^\circ, 95^\circ, 110^\circ$, hence $n = 8$.

Computation.

$$C = 2.72504, \quad S = 6.73984, \quad R = \sqrt{7.42584 + 45.42545} = \sqrt{52.85129} = 7.26989,$$

$$\bar{R} = \frac{7.26989}{8} = 0.90874, \quad \bar{\theta} = \text{atan2}(6.73984, 2.72504) = 67.98^\circ,$$

$$Z = 8 \cdot 0.90874^2 = 6.60641.$$

Since $Z = 6.606 > 2.996 = -\ln 0.05$, uniformity is rejected at the 5% level; the approximate p -value is $e^{-Z} = 0.00135$.

Independent check (circular mean). The mean direction 67.98° must lie inside the range of the data, and it is close to their arithmetic mean $(30+45+50+65+70+80+95+110)/8 = 68.125^\circ$: here the two means almost coincide because the data cover a narrow arc, far from the situation of the arithmetic-mean paradox.

17.3. CRC-8, bit by bit

Algorithm. An 8-bit register initialised to 0; for every byte of the message one performs $\text{crc} \leftarrow \text{crc} \oplus \text{byte}$ and then, eight times: if the most significant bit of crc is 1, then $\text{crc} \leftarrow (\text{crc} \ll 1) \oplus 0x07$, otherwise $\text{crc} = \text{crc} \ll 1$, keeping eight bits. The polynomial $0x07$ corresponds to $x^8 + x^2 + x + 1$.

Datum. A one-byte message: $0xA5 = 10100101$.

step	initial crc	MSB	operation	final crc
1	0xA5	1	shift, then $\oplus 0x07$	0x4D
2	0x4D	0	shift	0x9A
3	0x9A	1	shift, then $\oplus 0x07$	0x33
4	0x33	0	shift	0x66
5	0x66	0	shift	0xCC
6	0xCC	1	shift, then $\oplus 0x07$	0x9F
7	0x9F	1	shift, then $\oplus 0x07$	0x39
8	0x39	0	shift	0x72

The remainder is $0x72 = 01110010$.

Independent check (polynomial division). Divide $1010\,0101\,0000\,0000$ by $1\,0000\,0111$ with successive XORs:

$$\begin{aligned}
1010010100000000 &\oplus 1000001110000000 = 0010011010000000 \\
0010011010000000 &\oplus 0010000011100000 = 0000011001100000 \\
0000011001100000 &\oplus 0000010000011100 = 0000001001111100 \\
0000001001111100 &\oplus 0000001000001110 = 0000000001110010
\end{aligned}$$

The degree of the residue is now less than 8: the remainder is $01110010 = 0x72 \checkmark$, in agreement with the shift register.

17.4. The Bessel function I_0 by periodic quadrature

Algorithm. For a periodic function the rectangle rule with N equally spaced nodes coincides with the trapezoidal rule and is *spectrally* accurate: the error decreases faster than any power of $1/N$. One computes

$$I_0(\kappa) = \frac{1}{2\pi} \int_0^{2\pi} e^{\kappa \cos t} dt \simeq \frac{1}{N} \sum_{j=0}^{N-1} e^{\kappa \cos(2\pi j/N)}.$$

Datum. $\kappa = 2$, $N = 8$.

j	$\cos(2\pi j/8)$	$e^{2\cos(2\pi j/8)}$
0	1	7.389056
1	0.707107	4.113250
2	0	1.000000
3	-0.707107	0.243117
4	-1	0.135335
5	-0.707107	0.243117
6	0	1.000000
7	0.707107	4.113250
sum		18.237125

Dividing by 8: $I_0(2) \simeq 2.279641$.

Independent check (series). From the series $I_0(\kappa) = \sum_{m \geq 0} (\kappa/2)^{2m} / (m!)^2$, with $\kappa = 2$ one has $\sum_m 1/(m!)^2 = 1 + 1 + 0.25 + 0.027778 + 0.001736 + 0.0000694 + 0.0000019 = 2.279585$. The quadrature error with only eight nodes is therefore $5.6 \cdot 10^{-5}$: with sixteen nodes it would drop below 10^{-9} . This is why sophisticated quadrature formulas are never used in harmonic analysis on the circle: the most elementary one is already the best.

17.5. Circle packing on a grid

Algorithm. n^2 equal circles arranged on an $n \times n$ grid in the unit square, each tangent to its neighbours and to the boundary, have radius $r = 1/(2n)$: the **packing density** (area covered by the circles divided by the area of the square) is $\delta = n^2 \pi r^2$.

Datum. $n = 5$ (5×5 grid, 25 circles), $r = 1/10$.

Computation.

$$\delta = 25 \cdot \pi \cdot (0.1)^2 = 25\pi \cdot 0.01 = \frac{\pi}{4} = 0.7853982.$$

Note that $\delta = \pi/4$ **regardless of** n : substituting $r = 1/(2n)$, $\delta = n^2 \pi / (2n)^2 = \pi/4$ for every $n \geq 1$ — a regular grid of tangent circles always covers a quarter of the plane, whatever the fineness of the grid (the same ratio as a single circle inscribed in a square, Chapter 2). This fact is proved in full, with different tools (lattices and fundamental cells), in Volume III, Chapter 14, Proposition "Density of the square packing": we do not reprove it here, we take it as a test case for an independent numerical check.

Independent check (Monte Carlo). 2,000,000 uniform points are generated in the square $[0, 1]^2$ and the fraction falling in at least one of the 25 circles is counted (script `vol2_appB_packing.py`): the observed fraction is 0.785322, against the exact value 0.7853982 — discrepancy 7.6×10^{-5} , consistent with the expected statistical error of order $1/\sqrt{N} \approx 7 \times 10^{-4}$ for a proportion estimate with $N = 2 \times 10^6$ samples (the observed fraction lies within one standard deviation of the true value). ✓

17.6. Kepler's equation: three methods compared

Algorithm. For the same Kepler equation of Chapter 10, $E - e \sin E = M$, three iterative methods are compared for convergence speed from the same starting point $E_0 = M$: Newton (Corollary 10.1 of Chapter 10, quadratic order), fixed point $E_{n+1} = M + e \sin E_n$ (linear order, ratio $\leq e$), and bisection on the interval $[0, 2]$ (linear order, fixed ratio $1/2$, independent of e).

Datum. $M = 1$, $e = 0.5$ (the same example as Chapter 10), tolerance $|E - e \sin E - M| < 10^{-13}$.

Computation. Iterating the three methods to tolerance (script `vol2_appB_packing.py`, same reference solution $E^* = 1.4987011335178484$ as Chapter 10):

method	iterations	final E	final error
Newton	4	1.4987011335179377	8.9×10^{-14}
Fixed point	10	1.4987011335178357	1.3×10^{-14}
Bisection	36	1.4987011335178977	4.9×10^{-14}

Independent check (agreement of the three results). The three methods, starting from different information (Newton uses the derivative, the fixed point only the function, bisection only the sign) and with independent stopping criteria, converge to the same value E^* within 10^{-13} of one another ✓: a disagreement among the three would have flagged an implementation error in at least one of them. The number of iterations reflects the convergence order proved in Chapter 10: Newton (quadratic) takes just over a tenth of the iterations of bisection (linear with fixed ratio 1/2) for the same accuracy.

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- [4] IEEE, *IEEE Std 754-2019: IEEE Standard for Floating-Point Arithmetic*, IEEE, New York, 2019. Floating-point arithmetic assumed in the computations of this appendix.

Glossary

Osculating circle

The circle of radius $1/\kappa(t)$ that best approximates a curve at a point, sharing its tangent and curvature.

Gray code

Binary encoding in which two consecutive values differ in only one bit; used in absolute rotary encoders to avoid erroneous transient readings due to multiple bits switching not exactly in synchrony (Section 3.4, Chapter 3).

Fourier coefficients

Numbers $\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$ that encode the function f in the basis of complex exponentials.

Completeness of L^2

Property (Riesz–Fischer) whereby every Cauchy sequence in $L^2(\mathbb{T})$ converges in quadratic norm to an element of $L^2(\mathbb{T})$; guarantees that the Fourier partial sums of a square-summable function always converge to a function in the same space (Chapter 4).

Dominated convergence (Lebesgue’s theorem)

If $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ with g integrable, then $\int f_n \rightarrow \int f$; the theorem the Riemann integral lacks and that the Lebesgue integral supplies with no extra uniformity hypotheses (Chapter 4).

Curvature κ

Measure of how much a curve deviates from a straight line; for the circle of radius r , $\kappa = 1/r$ constant.

DFT (Discrete Fourier Transform)

Finite N -sample version of the Fourier series, which uses the N -th roots of unity as its basis.

Rotary encoder

Sensor that measures the angular position of a shaft; incremental (counts pulses per revolution, requires a zero reference) or absolute (uniquely encodes each position, typically in Gray code). The angular resolution $\Delta\theta = 360^\circ/N$ depends on the number $N = 2^n$ of encodable positions with n bits (Section 3.4, Chapter 3).

Damping ratio ζ

Dimensionless parameter $\zeta = (R/2)\sqrt{C/L}$ of a series RLC that classifies the transient: $\zeta < 1$ underdamped (oscillatory), $\zeta = 1$ critically damped, $\zeta > 1$ overdamped (Chapter 5).

Gibbs phenomenon

Overshoot of the Fourier partial sums near a jump discontinuity, which does not shrink as N grows but converges to $\frac{2}{\pi}\text{Si}(\pi) \approx 1.179$ times the jump amplitude (Chapter 4); absent in the Fejér means.

Euler’s formula

$e^{i\theta} = \cos \theta + i \sin \theta$, identity linking the complex exponential and trigonometry.

Parseval’s identity

$\frac{1}{2\pi} \int |f|^2 = \sum_n |\hat{f}(n)|^2$; energy conservation between the domain and Fourier coefficients.

Circular Fisher information

For the von Mises $\text{vM}(\mu, \kappa)$, the quantity $I_1(\mu) = \kappa A(\kappa)$ measuring how much information a single observation carries about the mean direction μ ; it grows with concentration κ (Chapter 6).

Gaussian integral

$\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$, classically computed by switching to polar coordinates in $\mathbb{R}^2$.

Circular interpolation

G02/G03 command of a CNC (or its equivalent on a robot) that generates a circular arc as the trajectory of the tool or end-effector.

Circular inversion

Transformation $z \mapsto r^2/\bar{z}$ that pointwise fixes the circumference $|z| = r$.

Polar Jacobian

Factor ρ appearing in the change of variables $dA = \rho d\rho d\varphi$ in polar coordinates.

Fejér mean

Cesàro average of the Fourier partial sums; unlike the partial sums themselves, it always converges to f for f continuous and periodic, and is the bridge from orthogonality of the trigonometric system to its completeness in $L^2(\mathbb{T})$ (Chapter 4).

Monte Carlo method

Estimation of a deterministic quantity (e.g., π) by random sampling, with error decaying as $1/\sqrt{N}$.

Lebesgue measure

Extension of the notion of length to a class of sets (those measurable in the sense of Carathéodory) broader than intervals, built from outer measure; the foundation of the Lebesgue integral, which integrates functions — such as the Dirichlet function — that are not Riemann integrable (Chapter 4).

Modulus (complex number)

$|z| = \sqrt{x^2 + y^2}$ for $z = x + iy$; generalizes the distance from the origin.

Arc-length parametrization

Parametrization $\gamma(s)$ in which the parameter s coincides with the distance traveled along the curve.

Natural angular frequency ω_n

For a series RLC, $\omega_n = 1/\sqrt{LC}$: the angular frequency at which the circuit would oscillate with no damping ($R = 0$), and the scale factor with respect to which ζ is measured. Distinct from the damped free oscillation frequency $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ and from the magnitude resonance frequency $\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$ (Chapter 5).

Almost everywhere

A property holds almost everywhere if the set of points where it fails has zero Lebesgue measure; allows treating as equal two functions that differ only on a negligible set (Chapter 4).

 n -th roots of unity

Complex solutions of $z^n = 1$, vertices of a regular n -sided polygon inscribed in $\mathbb{T}$.

Fourier series

Formal expansion $f(\theta) \sim \sum_n \hat{f}(n)e^{in\theta}$ of a periodic function in the basis of complex exponentials.

SOR (surface overflow rate)

In the sizing of a circular clarifier, the ratio $\text{SOR} = Q/A$ between flow rate Q and surface area

A of the basin; a central design parameter in water treatment (Section 2.4, Chapter 2).

Circular phase estimation

Problem of estimating a true phase μ from repeated, noisy measurements $\theta_1, \dots, \theta_n \sim \text{vM}(\mu, \kappa)$; the maximum-likelihood estimator is the sample mean direction, $\hat{\mu} = \bar{\theta}$ (Chapter 6).

$\mathbb{T}$ (complex unit circle)

The set $\{z \in \mathbb{C} : |z| = 1\}$, identified with $\mathbb{R}/2\pi\mathbb{Z}$ via $\theta \mapsto e^{i\theta}$.

Green's theorem

$\oint_{\gamma} (P dx + Q dy) = \iint_D (\partial_x Q - \partial_y P) dA$; allows computing areas as boundary integrals.

Watson's U^2 test

Omnibus test of circular uniformity based on the empirical distribution function, invariant under the choice of origin; corrects the Cramér–von Mises statistic by subtracting $n(\bar{u} - \frac{1}{2})^2$ (Chapter 6, Watson 1961).

Frenet frame

The pair (T, N) of unit tangent and normal vectors that accompany a plane curve, with $dT/ds = \kappa N$, $dN/ds = -\kappa T$.

Von Mises-Fisher distribution

Generalisation of the von Mises distribution from S^1 to $S^{p-1} \subset \mathbb{R}^p$; for $p = 3$ (directions in space) the normalising constant is elementary, $\kappa/(4\pi \sinh \kappa)$, unlike the general case which requires modified Bessel functions (Chapter 6, Fisher 1953).

Cheat Sheet

Parametrization and Differential

- $\gamma(t) = (r \cos t, r \sin t)$
- $\gamma'(t) = (-r \sin t, r \cos t), |\gamma'| = r$
- $\gamma''(t) = -\gamma(t)$ (centripetal acceleration)

- Curvature: $\kappa = \frac{x'y'' - y'x''}{(x'^2 + y'^2)^{3/2}} = \frac{1}{r}$
- Frenet: $T'(s) = \kappa N, N'(s) = -\kappa T$

Integral Calculus

- Arc length: $L = \int_{\alpha}^{\beta} |\gamma'(t)| dt = r(\beta - \alpha)$
- Area (Green): $A = \frac{1}{2} \oint (x dy - y dx)$
- Polar: $dA = \rho d\rho d\varphi$, Jacobian $= \rho$
- $A = \int_0^{2\pi} \int_0^r \rho d\rho d\varphi = \pi r^2$
- Gauss: $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$
- Ball volume: $\frac{4}{3}\pi r^3$ (disk method)

Complex Numbers

- Euler: $e^{i\theta} = \cos \theta + i \sin \theta$
- $\mathbb{T} = \{e^{i\theta} : \theta \in [0, 2\pi)\}$
- Euler's identity: $e^{i\pi} + 1 = 0$
- Roots of unity: $\zeta_k = e^{2\pi i k/n}, \sum \zeta_k = 0$
- Rotation: $z \mapsto e^{i\alpha} z$
- Möbius: $z \mapsto (az + b)/(cz + d)$
- Inversion: $z \mapsto r^2/\bar{z}$
- Cauchy: $f(z) = \frac{1}{2\pi i} \oint \frac{f(\zeta)}{\zeta - z} d\zeta$

Harmonic Analysis / Fourier

- $\hat{f}(n) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta$
- $f(\theta) \sim \sum_n \hat{f}(n) e^{in\theta}$
- Orthogonality: $\frac{1}{2\pi} \int e^{in\theta} e^{-im\theta} d\theta = \delta_{nm}$
- Parseval: $\frac{1}{2\pi} \int |f|^2 = \sum_n |\hat{f}(n)|^2$
- Isoperimetric (Hurwitz): $L^2 \geq 4\pi A$
- Outer measure: $\mu^*(E) = \inf \sum \ell(I_k), E \subseteq \bigcup I_k$
- Monotone convergence: $f_n \uparrow f \Rightarrow \int f_n \rightarrow \int f$
- Fatou: $\int \liminf f_n \leq \liminf \int f_n$
- Dominated: $|f_n| \leq g$ integrable, $f_n \rightarrow f$ a.e. $\Rightarrow \int f_n \rightarrow \int f$
- Cauchy-Schwarz: $|\int fg| \leq \|f\|_2 \|g\|_2$
- Fejér: $\sigma_N f = D_N * f \rightarrow f$ uniformly if f continuous
- Completeness of L^2 : $\|f - S_N f\|_2 \rightarrow 0$ for every $f \in L^2(\mathbb{T})$

RC/RLC Circuits as Dynamical Systems

- RC: $\tau = RC, v(t) = U(1 - e^{-t/\tau})$ (step)
- $\underline{H}_C(i\omega) = \frac{1}{1 + i\omega\tau}$, cutoff $\omega_c = 1/\tau$
- Exact cutoff: gain $= -10 \log_{10} 2 \approx -3.0103$ dB
- RLC: $\omega_n = 1/\sqrt{LC}, \zeta = (R/2)\sqrt{C/L}$

- $\zeta < 1$ underdamped, $\zeta = 1$ critical, $\zeta > 1$ overdamped

- $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ (free oscillations, $\zeta < 1$)

- Overshoot: $M_p = \exp(-\pi\zeta/\sqrt{1 - \zeta^2})$

- $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$ (magnitude resonance, only $\zeta < 1/\sqrt{2}$)

Circular Statistics / Phase Estimation

- | | | |
|--|--|---|
| <ul style="list-style-type: none"> • Von Mises: $f(\theta) = \frac{e^{\kappa \cos(\theta - \mu)}}{2\pi I_0(\kappa)}$ • $R = A(\kappa) = I_1(\kappa)/I_0(\kappa)$, A in- | <p>creasing $0 \rightarrow 1$</p> <ul style="list-style-type: none"> • MLE: $\hat{\mu} = \bar{\theta}$, $A(\hat{\kappa}) = R$ • Fisher: $I_1(\mu) = \kappa A(\kappa)$, $I_n(\mu) = n\kappa A(\kappa)$ • Resultant concentration: $\mathbb{E} \bar{Z} -$ | <p>$\rho e^{i\mu} ^2 = (1 - \rho^2)/n$</p> <ul style="list-style-type: none"> • Chebyshev phase: $\Pr(\hat{\mu} - \mu > \delta) \leq (1 - \rho^2)/(n\rho^2 \sin^2 \delta)$ • Conjugate Bayes: $\eta_n = \eta_0 + \sum_k \kappa_k e^{i\theta_k}$ |
|--|--|---|

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